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ISRO Scientist/Engineer (Scientist-C)

COMPUTER SCIENCE

Solved Papers and Practice Tests

Inside

- Detailed Exam Analysis
- 12 Solved Papers from Previous Years **(2007-2020)**
- Detailed Solutions for All Questions
- 3 Practice Tests on Latest Pattern

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NOTE TO THE ASPIRANTS

Indian Space Research Organization/Department of Space Centres/Units are engaged in Research and Development activities in development of Space Application, Space Science and Technology for the benefit of society at large and for serving the nation by achieving self-reliance and developing capacity to design and build Launch Vehicles and Communication/Remote Sensing Satellites and thereafter launch them. ISRO strives to serve the nation in the areas of television broadcast, location based services, telecommunication, meteorological application and in management of our natural resources. The Indian space programme continues to pursue successful goals on all fronts.

The future ISRO programme envisages development of cutting edge technologies for reusable launch vehicle, development of advanced technologies for Human Spaceflight Programme, advanced high efficiency semi-cryogenic propulsion engine, advanced communication satellite, air breathing propulsion, satellite based navigation system, hyper spectral imaging sensors, lunar and planetary exploration, etc.

ABOUT THE EXAMINATION

ISRO offers challenging opportunities to Scientists/Engineers to undertake development of innovative technologies and establish the advanced technical infrastructure needed for space exploration and beyond. Recruitment to ISRO for Scientists/Engineers 'SC' in Electronics, Mechanical Engineering and Computer Science is done through entrance test conducted by ISRO Centralised Recruitment Board (ICRB).

Eligibility: BE/B.Tech or equivalent qualification in first class with an aggregate minimum of 65% marks or CGPA 6.84/10. Candidates who are going to complete the above course in the current academic year are also eligible to apply, provided final degree is available by the end of August in that year and their aggregate is 65% marks or CGPA 6.84/10 (average of all semesters for which results are available). The qualification prescribed and the benchmark are only the MINIMUM requirement and fulfilling the same does not automatically make candidates eligible for Written Test.

Age Limit: 35 years. Ex-Servicemen and Persons with Disabilities are eligible for age relaxation as per Govt. of India orders.

Selection Process:

- Based on the academic performance and bio-data, initial screening is conducted to short-list candidates for taking-up written test.
- The written test is conducted at twelve venues viz., Ahmedabad, Bengaluru, Bhopal, Chandigarh, Chennai, Guwahati, Hyderabad, Kolkata, Lucknow, Mumbai, New Delhi and Thiruvananthapuram. However, the organization reserves the right to cancel/change the written test venue and re-allot the candidates to any other test centre.
- The call letters for the written test to the short-listed candidates are sent by e-mail.
- The written test paper consists of 80 objective type questions carrying equal marks.
- Based on the performance in the Written Test, candidates are short-listed for interview, the schedule and venue of which is then notified.
- Written test is only a first level screening and written test score is not be considered for final selection process.
- Final selection will be based on the performance of the candidates in the Interview and those who secure minimum 60% marks in the interview will be eligible for consideration for empanelment in the selection panel, in the order of merit.

ABOUT THE BOOK

This book ***ISRO Entrance Test for Scientists/Engineers in Computer Science*** is designed as a must have resource for the students preparing for the post of Scientists / Engineers (Grade: 'SC') in Computer Science. It offers the best-suited pedagogy for the last step in exam preparation, that is, practicing previous year papers and attempting Practice Tests. The book comprises solved papers for ISRO Computer Science for the years 2007–2020 (including May and December 2017 papers). This book also includes three Practice Tests, as per the pattern of recent examination. Detailed solutions are provided for all the questions and questions are tagged topic-wise based on the current syllabus. Practice Tests provided in this book are designed help the students to get more familiar with the pattern of the examination.

ISRO COMPUTER SCIENCE EXAM ANALYSIS 2007–2020

ISRO Entrance Test for Scientists/Engineers in Computer Science is a first level screening and the written test score is used as an eligibility/merit criteria for short-listing the candidates for final interview for the post of Scientist C.

To score well in the written examination, it is very important for the aspirants to get well-versed with the pattern of examination, the level of questions asked and the concept distribution of questions.

The table given below depicts the section-wise analysis of previous years' (2007–2020) papers. This will help students to focus in their preparation on the important and frequently asked topics.

SECTIONS	2007	2008	2009	2011	2013	2014	2015	2016	May-17	Dec-17	April-18	2020
Engineering Mathematics	7	4	22	8	7	9	0	7	4	11	4	4
Digital Logic	12	13	6	9	9	8	11	12	6	6	2	7
Computer Organization and Architecture	6	11	10	13	5	5	3	8	3	6	5	9
Programming and Data Structures	10	13	11	8	9	12	16	6	20	22	20	16
Algorithms	7	2	0	4	2	5	5	4	7	4	7	8
Theory of Computation	1	2	0	3	2	5	3	7	4	6	4	6
Compiler Design	0	3	1	2	1	2	3	3	1	0	3	4
Operating System	19	19	15	10	16	9	11	14	10	10	10	13
Databases	6	2	5	6	6	5	9	8	6	7	12	4
Computer Networks	12	10	8	9	15	12	14	11	12	8	9	7
*** Software Engineering**	0	0	2	5	4	4	2	0	7	0	3	2
***Artificial Intelligence**	0	0	0	1	1	0	0	0	0	0	0	0
***Computer Graphics**	0	1	0	1	2	2	0	0	0	0	1	0
***Web Technologies**	0	0	0	1	0	1	0	0	0	0	0	0
*** Multimedia and Image Processing**	0	0	0	0	1	1	3	0	0	0	0	0

*** Sections which are not covered in B.Tech syllabus.**

CONTENTS

Solved Previous Year Papers

Indian Space Research Organisation Computer Science

2007

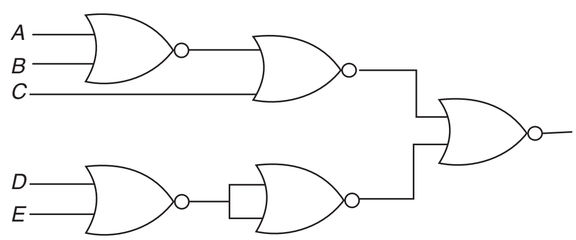
QUESTION PAPER

1. The Boolean expression $Y = \left(A + \overline{B} + \overline{A}B\right)\overline{C}$ is given by

(a) $A\overline{C}$ (b) $B\overline{C}$

(c) $\overline{C}$ (d) AB

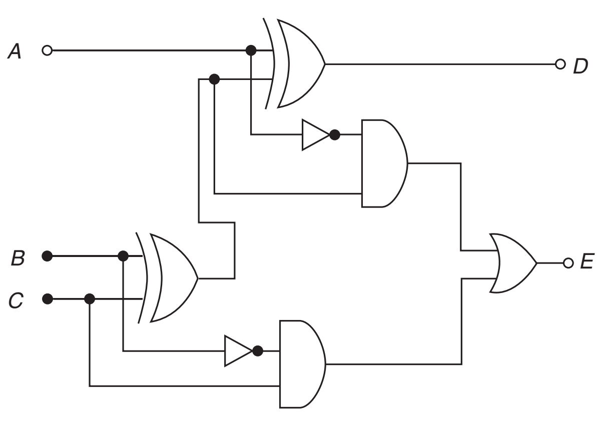
2. The circuit shown in the following figure realizes the function



(a) $\left(\overline{A+B}+C\right)\left(\overline{D}\,\overline{E}\right)$ (b) $\left(\overline{A+B}+C\right)\left(D\overline{E}\right)$

(c) $\left(A+\overline{B+C}\right)\left(\overline{D}E\right)$ (d) $\left(A+B+\overline{C}\right)\left(\overline{D}\,\overline{E}\right)$

3. The circuit shown in the given figure is a



(a) full adder (b) full subtracter

(c) shift register (d) decade counter

4. When two numbers are added in excess-3 code and the sum is less than 9, then in order to get the correct answer it is necessary to

(a) subtract 0011 from the sum

(b) add 0011 to the sum

(c) subtract 0110 from the sum

(d) add 0110 to the sum

5. The characteristic equation of an *SR* flip–flop is given by

(a) $Q_{n+1} = S + RQ_n$ (b) $Q_{n+1} = R\overline{Q}_n + \overline{S}Q_n$

(c) $Q_{n+1} = \overline{S} + RQ_n$ (d) $Q_{n+1} = S + \overline{R}Q_n$

6. A graph with 'n' vertices and '$n-1$' edges that is not a tree is

(a) connected (b) disconnected

(c) Euler (d) a circuit

7. If a graph requires k different colours for its proper colouring, then the chromatic number of the graph is

(a) 1 (b) k

(c) $k-1$ (d) $k/2$

8. A read bit can be read

(a) and written by CPU

(b) and written by peripheral

(c) by peripheral and written by CPU

(d) by CPU and written by the peripheral

9. Eigen vectors of $\begin{bmatrix} 1 & \cos\theta \\ \cos\theta & 1 \end{bmatrix}$ are

(a) $\begin{bmatrix} a^n & 1 \\ 0 & a^n \end{bmatrix}$ (b) $\begin{bmatrix} a^n & n \\ 0 & a^n \end{bmatrix}$

(c) $\begin{bmatrix} a^n & na^{n-1} \\ 0 & a^n \end{bmatrix}$ (d) $\begin{bmatrix} a^n & na^{n-1} \\ -n & a^n \end{bmatrix}$

10. The term "aging" refers to

(a) booting up the priority of a process in multi-level of queue without feedback.

(b) gradually increasing the priority of jobs that wait in the system for a long time to remedy infinite blocking.

(c) keeping track of the following a page has been in memory for the purpose of LRU replacement.

(d) letting job reside in memory for a certain amount of time so that the number of pages required can be estimated accurately.

11. Consider a set of n tasks with known runtimes $r_1, r_2, \ldots, r_n$ to be run on a uniprocessor machine. Which of the following processor scheduling algorithms will result in the maximum throughput?

(a) Round Robin
(b) Shortest job first
(c) Highest response ratio next
(d) First cum first served

12. Consider a job scheduling problem with four jobs J_1, J_2, J_3 and J_4 with corresponding deadlines:

$$(d_1, d_2, d_3, d_4) = (4, 2, 4, 2)$$

Which of the following is not a feasible schedule without violating any job deadline?

(a) J_2, J_4, J_1, J_3 (b) J_4, J_1, J_2, J_3
(c) J_4, J_2, J_1, J_3 (d) J_4, J_2, J_3, J_1

13. By using an eight-bit optical encoder, the degree of resolution that can be obtained is (approximately)

(a) 1.8° (b) 3.4°
(c) 2.8° (d) 1.4°

14. The principal of locality of reference justifies the use of

(a) virtual memory
(b) interrupts
(c) main memory
(d) cache memory

15. Consider the following pseudocode:

```
x : = 1;
i : = 1;
while (x ≤ 1000)
begin
x : = 2^x;
i : = i + 1;
end;
```

What is the value of *i* at the end of the pseudocode?

(a) 4 (b) 5
(c) 6 (d) 7

16. The five items *A*, *B*, *C*, *D* and *E* are pushed in a stack, one after the other starting from *A*. The stack is popped four times and each element is inserted in a queue. Then two elements are deleted from the queue and pushed back on the stack. Now one item is popped from the stack. The popped item is

(a) *A* (b) *B*
(c) *C* (d) *D*

17. Round Robin scheduling is essentially the pre-emptive version of

(a) FIFO
(b) shortest job first
(c) shortest remaining time
(d) longest remaining time

18. The number of digit 1 present in the binary representation of $3\times512+7\times64+5\times8+3$ is

(a) 8 (b) 9
(c) 10 (d) 12

19. Assume that each character code consists of 8 bits. The number of characters that can be transmitted per second through a synchronous serial line at 2400 baud rate and with two stop bits is

(a) 109 (b) 216
(c) 218 (d) 219

20. If the bandwidth of a signal is 5 kHz and the lowest frequency is 52 kHz, what is the highest frequency

(a) 5 kHz (b) 10 kHz
(c) 47 kHz (d) 57 kHz

21. An Ethernet hub

(a) functions as a repeater
(b) connects to a digital PBX
(c) connects to a token-ring network
(d) functions as a gateway

22. Phase transition for each bit are used in

(a) amplitude modulation
(b) carrier modulation
(c) manchester encoding
(d) NRZ encoding

23. Study the following programme:

```
// precondition : x >= 0
public void demo(int x)
{
System.out.print(x %10);
if ((x/10)!=0)
{
demo (x/10);
}
System.out.print(x % 10);
}
```

Which of the following is printed as a result of the call demo (1234)?

(a) 1441 (b) 3443
(c) 12344321 (d) 43211234

24. Bit stuffing refers to

(a) inserting a '0' in user stream to differentiate it with a flag
(b) inserting a '0' in flag stream to avoid ambiguity
(c) appending a nibble to the flag sequence
(d) appending a nibble to the use data stream

25. What is the name of the technique in which the operating system of a computer executes several programs concurrently by switching back and forth between them?

(a) Partitioning (b) Multi-tasking
(c) Windowing (d) Paging

26. If there are five routers and six networks in an intranet using link state routing, how many routing tables are there?

(a) 1 (b) 5
(c) 6 (d) 11

27. Virtual memory is

(a) Part of main memory only used for swapping
(b) A technique to allow a program, of size more than the size of the main memory, to run
(c) Part of secondary storage used in program execution
(d) None of these

28. The level of aggregation of information required for operational control is

(a) detailed (b) aggregate
(c) qualitative (d) none of these

29. The set of all equivalence classes of a set A of cardinality C

(a) is of cardinality 2^C
(b) has the same cardinality as A
(c) forms a partition of A
(d) is of cardinality C^2

30. 0.75 in decimal system is equivalent to ____ in octal system

(a) 0.60 (b) 0.52
(c) 0.54 (d) 0.50

31. In an SR latch made by cross coupling two NAND gates, if both S and R inputs are set to 0, then it will result in

(a) $Q = 0, Q_0 = 1$ (b) $Q = 1, Q_0 = 0$
(c) $Q = 1, Q_0 = 1$ (d) indeterminate states

32. Identify the correct translation into logical notation of the following assertion:

"*Some boys in the class are taller than all the girls*".
Note: taller(x, y) is true if x is taller than y.

(a) $(\exists x)(\text{boy}(x) \rightarrow (\forall x)(\text{girl}(y) \wedge \text{taller}(xy)))$
(b) $(\exists x)(\text{boy}(x) \wedge (\forall y)(\text{girl}(y) \wedge \text{taller}(xy)))$
(c) $(\exists x)(\text{boy}(x) \rightarrow (\forall y)(\text{girl}(y) \rightarrow \text{taller}(xy)))$
(d) $(\exists x)(\text{boy}(x) \wedge (\forall y)(\text{girl}(y) \rightarrow \text{taller}(xy)))$

33. Company X shipped 5 computer chips, 1 of which was defective, and Company Y shipped 4 computer chips, 2 of which were defective. One computer chip is to be chosen uniformly at random from the 9 chips shipped by the companies. If the chosen chip is found to be defective, what is the probability that the chip came from Company Y?

(a) 2/9 (b) 4/9
(c) 2/3 (d) 1/2

34. Ring counter is analogous to

(a) toggle switch (b) latch
(c) stepping switch (d) *SR* flip–flop

35. The output 0 and 1 level for TTL Logic family is approximately

(a) 0.1 and 5 V (b) 0.6 and 3.5 V
(c) 0.9 and 1.75 V (d) −1.75 and 0.9 V

36. Consider a computer system that stores floating-point numbers with 16-bit mantissa and an 8-bit exponent, each in two's complement. The smallest and largest positive values which can be stored are

(a) 1×10^{-128} and $2^{15} \times 10^{128}$
(b) 1×10^{-256} and $2^{15} \times 10^{255}$
(c) 1×10^{-128} and $2^{15} \times 10^{127}$
(d) 1×10^{-128} and $(2^{15} - 1) \times 10^{127}$

37. In comparison with static RAM memory, the dynamic RAM memory has

(a) lower bit density and higher power consumption
(b) higher bit density and higher power consumption
(c) lower bit density and lower power consumption
(d) higher bit density and lower power consumption

38. The hexadecimal equivalent of 01111100110111100011 is

(a) CD73E (b) ABD3F
(c) 7CDE3 (d) FA4CD

39. Disk requests are received by a disk drive for cylinders 5, 25, 18, 3, 39, 8 and 35 in that order. A seek takes 5 ms per cylinder moved. How much seek time is needed to serve these requests for a shortest seek first (SSF) algorithm? Assume that the arm is at cylinder 20 when the last of these requests is made with none of the requests yet served?

(a) 125 ms (b) 295 ms
(c) 575 ms (d) 750 ms

40. Consider a system having 'm' resources of the same type. The resources are shared by 3 processes A, B, C, which have peak time demands of 3, 4, and 6, respectively. The minimum value of 'm' that ensures that deadlock will never occur is

(a) 11 (b) 12
(c) 13 (d) 14

41. A task in a blocked state

(a) is executable
(b) is running
(c) must still be placed in the run queues
(d) is waiting for some temporarily unavailable resources.

42. Semaphores

(a) synchronize critical resources to prevent deadlock
(b) synchronize critical resources to prevent contention
(c) are used to do I/O
(d) are used for memory management

43. On a system using non-preemptive scheduling, processes with expected run times of 5, 18, 9 and 12 are in the ready queue. In what order should they be run to minimize wait time?

(a) 5, 12, 9, 18 (b) 5, 9, 12, 18
(c) 12, 18, 9, 5 (d) 9, 12, 18, 5

44. The number of page frames that must be allocated to a running process in a virtual memory environment is determined by

(a) the instruction set architecture
(b) page size
(c) number of processes in memory
(d) physical memory size

45. A program consists of two modules executed sequentially. Let $f_1(t)$ and $f_2(t)$ respectively denote the probability density functions of time taken to execute the two modules. The probability density function of the overall time taken to execute the program is given by

(a) $f_1(t) + f_2(t)$ (b) $\int_0^t f_1(x) f_2(x) dx$

(c) $\int_0^t f_1(x) f_2(t-x) dx$ (d) $\max(f_1(t) + f_2(t))$

46. Consider a small 2-way set-associative cache memory, consisting of four blocks. For choosing the block to be replaced, use the least recently (LRU) scheme. The number of cache misses for the sequence of block addresses 8, 12, 0, 12, 8 is

(a) 2 (b) 3
(c) 4 (d) 5

47. Which commands are used to control access over objects in relational database?

(a) CASCADE and MVD
(b) GRANT and REVOKE
(c) QUE and QUIST
(d) None of these

48. Which of the following is an aggregate function in SQL?

(a) Avg (b) Select
(c) Ordered by (d) distinct

49. One approach to handle fuzzy logic data might be to design a computer using ternary (base-3) logic so that data could be stored as "true," "false," and "unknown." If each ternary logic element is called a flit, how many flits are required to represent at least 256 different values?

(a) 4 (b) 5
(c) 6 (d) 7

50. A view of a database that appears to an application program is known as

(a) schema (b) subschema
(c) virtual table (d) none of these

51. Armstrong's inference rule does not determine

(a) Reflexivity (b) Augmentation
(c) Transitivity (d) Mutual dependency

52. Which operation is used to extract specified columns from a table?

(a) Project (b) Join
(c) Extract (d) Substitute

53. In the Big-Endian system, the computer stores

(a) MSB of data in the lowest memory address of data unit
(b) LSB of data in the lowest memory address of data unit
(c) MSB of data in the highest memory address of data unit
(d) LSB of data in the highest memory address of data unit

54. BCNF is not used for cases where a relation has

(a) two (or more) candidate keys
(b) two candidate keys and composite

(c) the candidate key overlap
(d) two mutually exclusive foreign keys

55. Selection sort algorithm design technique is an example of
(a) greedy method
(b) divide-and-conquer
(c) dynamic programming
(d) backtracking

56. Which of the following RAID level provides the highest data transfer rate (Read/Write)
(a) RAID 1 (b) RAID 3
(c) RAID 4 (d) RAID 5

57. Which of the following programming language(s) provides garbage collection automatically?
(a) Lisp (b) C++
(c) FORTRAN (d) C

58. The average case and worst case complexities for merge sort algorithm are
(a) $O(n^2)$, $O(n^2)$
(b) $O(n^2)$, $O(n \log_2 n)$
(c) $O(n \log_2 n)$, $O(n^2)$
(d) $O(n \log_2 n)$, $O(n \log_2 n)$

59. The time taken by binary search algorithm to search a key in a sorted array of n elements is
(a) $O(\log_2 n)$ (b) $O(n)$
(c) $O(n \log_2 n)$ (d) $O(n^2)$

60. Which of the following is correct with respect to two-phase commit protocol?
(a) Ensures serializability
(b) Prevents deadlock
(c) Detects deadlock
(d) Recover from deadlock

61. The Fibonacci sequence is the sequence of integers
(a) 1, 3, 5, 7, 9, 11, 13
(b) 0, 1, 1, 2, 3, 5, 8, 13, 21, 54
(c) 0, 1, 3, 4, 7, 11, 18, 29, 47
(d) 0, 1, 3, 7, 15

62. Let X be the adjacency matrix of a graph G with no self loops. The entries along the principal diagonal of X are
(a) all zeros (b) all ones
(c) both zeros and ones (d) different

63. Which of these is not a feature of WAP 2.0?
(a) Push and pull model
(b) Interface to a storage device
(c) Multimedia messaging
(d) Hashing

64. Feedback queues
(a) are very simple to implement
(b) dispatch tasks according to execution characteristics
(c) are used to favour real-time tasks
(d) require manual intervention to implement properly

65. Which of the following is not a UML DIAGRAM?
(a) Use case
(b) Class diagram
(c) Analysis diagram
(d) Swimlane diagram

66. Silly Window Syndrome is related to
(a) Error during transmission
(b) File transfer protocol
(c) Degrade in TCP performance
(d) Interface problem

67. To execute all loops at their boundaries and within their operational bounds is an example of
(a) Black box testing (b) Alpha testing
(c) Recovery testing (d) White box testing

68. SSL is not responsible for
(a) mutual authentication of client and server
(b) secret communication
(c) data integrity protection
(d) error detection and correction

69. A rule in a limited entry decision table is a
(a) row of the table consisting of condition entries
(b) row of the table consisting of action entries
(c) column of the table consisting of condition entries and the corresponding action entries
(d) columns of the table consisting of conditions of the stub

70. The standard for certificates used on Internet is
(a) X.25 (b) X.301
(c) X.409 (d) X.509

71. Hashed message is signed by a sender using
(a) his public key
(b) his private key
(c) receiver's public key
(d) receiver's private key

72. An email contains a textual birthday greeting, a picture of a cake, and a song. The order is not important. What is the content type?
(a) Multipart/mixed
(b) Multipart/parallel
(c) Multipart/digest
(d) Multipart/alternative

73. Range of IP Address from 224.0.0.0 to 239.255.255.255 are
(a) reserved for loopback
(b) reserved for broadcast
(c) used for multicast packets
(d) reserved for future addressing

74. IEEE 802.11 is a standard for
(a) Ethernet
(b) bluetooth
(c) broadband wireless
(d) wireless LANs

75. When a host on network *A* sends a message to a host on network *B*, which address does the router look at?
(a) Port (b) IP
(c) Physical (d) Subnet mask

76. Which of the following is not an approach to Software Process Assessment?
(a) SPICE (ISO/IEC15504)
(b) Standard CMMI Assessment Method for process improvement
(c) ISO 9001:2000
(d) IEEE 2000:2001

77. A physical DFD specifies
(a) what processes will be used
(b) who generates data and who processes it
(c) what each person in an organization does
(d) which data will be generated

78. In UML diagram of a class
(a) state of object cannot be represented
(b) state is irrelevant
(c) state is represented as an attribute
(d) state is represented as a result of an operation

79. Which of the following models is used for software reliability?
(a) Waterfall (b) Musa
(c) COCOMO (d) Rayleigh

80. Djikstra's algorithm is used to
(a) create LSAs
(b) flood an internet with information
(c) calculate the routing tables
(d) create a link state database

ANSWER KEY

1. (c)	**9.** (*)	**17.** (a)	**25.** (b)	**33.** (c)	**41.** (d)	**49.** (c)	**57.** (a)	**65.** (c)	**73.** (c)
2. (a)	**10.** (b)	**18.** (b)	**26.** (b)	**34.** (c)	**42.** (b)	**50.** (b)	**58.** (d)	**66.** (c)	**74.** (d)
3. (b)	**11.** (b)	**19.** (c)	**27.** (b)	**35.** (b)	**43.** (b)	**51.** (d)	**59.** (a)	**67.** (d)	**75.** (b)
4. (a)	**12.** (b)	**20.** (d)	**28.** (a)	**36.** (d)	**44.** (a)	**52.** (a)	**60.** (a)	**68.** (d)	**76.** (d)
5. (d)	**13.** (d)	**21.** (a)	**29.** (c)	**37.** (d)	**45.** (c)	**53.** (a)	**61.** (*)	**69.** (c)	**77.** (b)
6. (b)	**14.** (d)	**22.** (c)	**30.** (a)	**38.** (c)	**46.** (c)	**54.** (b)	**62.** (a)	**70.** (d)	**78.** (c)
7. (b)	**15.** (b)	**23.** (d)	**31.** (d)	**39.** (b)	**47.** (b)	**55.** (b)	**63.** (d)	**71.** (b)	**79.** (d)
8. (d)	**16.** (d)	**24.** (a)	**32.** (d)	**40.** (a)	**48.** (a)	**56.** (b)	**64.** (b)	**72.** (b)	**80.** (c)

*** None of the options is correct.**

ANSWERS WITH EXPLANATION

1. Topic: Digital Logic

(c) Given $Y = (A+\bar{B}+\bar{A}\bar{B})\bar{C}$

Using distributive law $A + (B\cdot C) = (A + B)\cdot(C + A)$, we have

$$Y + (A+\bar{A}\cdot B+\bar{B})\cdot\bar{C} = ((\bar{A}+\bar{A})\cdot(A+B)+\bar{B})\cdot\bar{C}$$

Now $A+\bar{A}=0$ identity. Using this, we get

$$\Rightarrow Y = (1\cdot(A+B)+\bar{B})\cdot\bar{C}$$

Using $1\cdot A = A$, we have

$$Y = (A+B+\bar{B})\cdot\bar{C}$$

Now, using $B+\bar{B}=1$, we have

$$Y = (A+1)\cdot\bar{C}$$

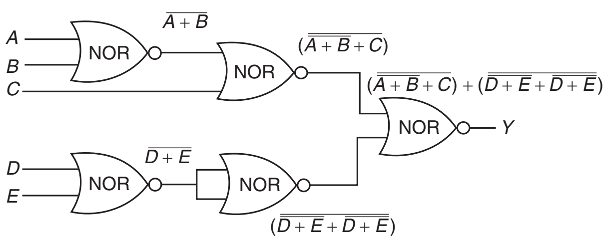
Using $A + 1 = 1$ identity, we have

$$Y = 1\cdot\bar{C} = \bar{C} \qquad \text{(since } 1\cdot A = A)$$

Therefore, $Y = \bar{C}$

2. Topic: Digital Logic

(a) Output of NOR gate with inputs A and B is $\overline{A+B}$:



Solving the resultant $Y = \overline{(\overline{\overline{A+B}+C})+(\overline{\overline{D+E}+\overline{D+E}})}$, using De Morgan's theorem $\overline{A+B}=\bar{A}\cdot\bar{B}$, we have

$$Y\overline{(\overline{\overline{\bar{A}\cdot\bar{B}}+C})+(\overline{\overline{\bar{D}\cdot\bar{E}}+\overline{\bar{D}\cdot\bar{E}}})} = \overline{(\overline{\overline{\bar{A}\cdot\bar{B}}}\cdot\bar{C})+(\overline{\overline{\bar{D}\cdot\bar{E}}}\cdot\overline{\overline{\bar{D}\cdot\bar{E}}})}$$

Using De Morgan's theorem $\overline{A\cdot B}=\bar{A}+\bar{B}$, we get

$$Y = \overline{(\bar{\bar{A}}+\bar{\bar{B}})\cdot\bar{C}+(\bar{\bar{D}}+\bar{\bar{E}})\cdot(\bar{\bar{D}}+\bar{\bar{E}})}$$

Using double negation identity, i.e., $\bar{\bar{A}}=A$, we have

$$Y = \overline{((A+B\cdot\bar{C})+((D+E)\cdot(D+E))}$$

Using De Morgan's theorem,

$$Y = (\overline{(A+B)\cdot\bar{C}})\cdot(\overline{(D+E)(D+E)})$$

$$= ((\overline{A+B})+\bar{\bar{C}})\cdot((\overline{D+E})+(\overline{D+E}))$$

$$= ((\bar{A}\cdot\bar{B})+\bar{\bar{C}})\cdot((\bar{D}\cdot\bar{E})+(\bar{D}\cdot\bar{E}))$$

$$= ((\bar{A}\cdot\bar{B})+C)\cdot(\bar{D}\cdot\bar{E}+\bar{D}\cdot\bar{E}) \qquad (\text{Since } \bar{\bar{C}}=C)$$

$$= (\bar{A}\cdot\bar{B}+C)(\bar{D}\cdot\bar{E}) \qquad (\text{Since } A+A=A)$$

Using De Morgan's law $\overline{A+B}=\bar{A}\cdot\bar{B}$, we have

$$Y = (\overline{A+B}+C)\cdot(\bar{D}\cdot\bar{E})$$

3. Topic: Digital Logic

(b) The final truth table for given circuit as follows:

A	*B*	*C*	*D*	*E*
0	0	0	0	0
0	0	1	1	0
0	1	0	1	1
0	1	1	0	0
1	0	0	1	1
1	0	1	0	0
1	1	0	0	1
1	1	1	1	1

The given Boolean expression is equivalent to full subtracter Boolean expression.

Full subtracter is a combinational circuit capable of performing subtraction on two bits namely minuend and subtrahend. It accepts three inputs: minuend, subtrahend and a borrow bit. It produces two outputs: difference and borrow.

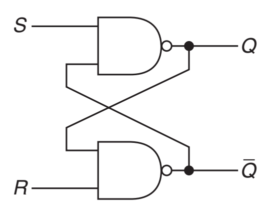
Therefore, given logic circuit is full subtracter.

4. Topic: Digital Logic

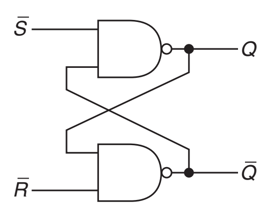
(a) When two numbers are added in excess-3 code and the sum is less than 9, then it does not produce any carry. So, in order to get correct answer, 0011 that is 3 is subtracted from the sum.

5. Topic: Digital Logic

(d) The circuit diagram of *SR* flip–flop is given as follows:



or



Truth table of *SR* flip–flop is as follows:

S	***R***	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	Invalid input

Thus characteristic equation of *SR* flip–flop is

$$Q_{n+1} = \bar{R}Q_n + \bar{S}$$

6. **Topic: Engineering Mathematics**

(b) A graph with n vertices and $n - 1$ edges that is not a tree is disconnected.

7. **Topic: Engineering Mathematics**

(b) Chromatic number is minimum number of colours required to colour the vertices such that no two adjacent vertices are of same colour. Since each pair of vertices is against each other, no two vertices can be assigned the same colour. Thus, chromatic number of graph is k.

8. **Topic: Operating System**

(d) A read bit can be read by CPU and written by the peripheral.

9. **Topic: Engineering Mathematics**

(None) Let λ be the eigen value.
Then, $|A - I\lambda| = 0$

$$\begin{vmatrix} 1-\lambda & \cos\theta \\ \cos\theta & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda)^2 - \cos^2\theta = 0 \Rightarrow (1-\lambda)^2 = \cos^2\theta$$

$$\Rightarrow (1-\lambda) = \cos\,\theta \Rightarrow \lambda = 1 + \cos\,\theta$$

Now, we have $(A - I\lambda)X = 0$

$$\begin{bmatrix} -\cos\theta & \cos\theta \\ \cos\theta & -\cos\theta \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\Rightarrow x_1 = x_2$$

Therefore, eigen vector of given matrix is $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

Now, normalised eigen vector is $\frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

10. **Topic: Programming and Data Structures**

(b) The term **aging** refers to gradually increasing the priority of jobs that wait in the system for a long time to remedy infinite blocking.

11. **Topic: Algorithms**

(b) We know the following:

Throughput = Number of tasks completed per unit time.

For maximum throughput, maximum jobs should be completed in least amount of time.
Shortest job first scheduling, that is, shortest jobs are executed first; thus, maximum tasks are completed.

12. **Topic: Algorithms**

(b) Deadline of job J_1 and J_3 is same, i.e., 4 and of job J_2 and J_4 is same, i.e., 2 since deadline of J_2 and J_4 is less than J_1 and J_3 so it should be scheduled first and J_1 and J_3 should be scheduled after that. J_2 or J_4 can be scheduled in any way since both have same deadline and same is true for J_1 and J_3.
Thus, option (b) is not a feasible schedule.

13. **Topic: Theory of Computation**

(d) Optical encoder converts motion into a sequence of digital pulses, digital resolution is given by

$$\frac{180°}{2^{r-1}} = \frac{360}{2^r}$$

For $r = 8$, digital resolution $= \frac{360}{2^8} = \frac{360}{256} = 1.4$.

14. **Topic: Operating System**

(d) Locality of reference uses related storage locations frequently. Thus the principle of locality of reference justifies use of cache memory.

15. **Topic: Programming and Data Structures**

(b) On execution of the program

Initially $i = 1, x = 1$.
So,
$i = 1, x = 1$
$x = 2^1 = 2$; $i = 1 + 1 = 2$
$x = 2^2 = 4$; $i = 2 + 1 = 3$
$x = 2^4 = 16$; $i = 3 + 1 = 4$
$x = 2^{16} = 65536$; $i = 4 + 1 = 5$
$x > 1000$, loop exists

Value of i at the end of pseudocode = 5.

16. **Topic: Programming and Data Structures**

(d) *ABCDE* are pushed in stack so we have

E
D
C
B
A

- Four items are popped from the stack and inserted in a queue. We have in stack

D
C
B
A

E popped

C
B
A

D popped

B
A

C popped

A

B popped

In queue, we have

B	*C*	*D*	*E*

- Two elements are deleted from queue so in queue we have

E deleted

B	*C*	*D*	

D deleted

B	*C*		

- The deleted elements are pushed back in stack so in stack we have

	D
E	*E*
A	*A*

- One element is popped back

E
A

The popped element is *D*.

17. Topic: Operating System

(a) Round Robin scheduling is essentially the pre-emptive version of FIFO, i.e., first in first out since round robin scheduling assigns equal time to each process without priority in cyclic order, so the process which comes first is executed first.

18. Topic: Digital Logic

(b) The given representation is

$$3 \times 512 + 7 \times 64 + 5 \times 8 + 3$$

Converting above expression in power of 2, we get

$$\begin{aligned}&(2 + 1) \times 2^9 + (4 + 2 + 1)2^6 + (4 + 1)2^3 + (2 + 1)\\ &= 2\cdot 2^9 + 2^9 + 2^2 2^6 + 2\cdot 2^6 + 2^6 + 2^2\cdot 2^3 + 2^3 + 2 + 2^0\\ &= 2^{10} + 2^9 + 2^8 + 2^7 + 2^6 + 2^5 + 2^4 + 2^1 + 2^0\end{aligned}$$

In binary, this can be written as 11111101011.
So, there are nine 1's present in binary representation of the given expression.

19. Topic: ASCII

(c) Character code consists of 8 bits. It requires 1 start bit and 2 stop bit.
So, Total bit characters = 8 + 1 + 2 = 11.
Therefore, the number of characters transmitted per second is

$$\frac{2400}{11} = 218.18 = 218$$

20. Topic: Computer Organization and Architecture

(d) We know that:

Bandwidth = High frequency – low frequency
$\Rightarrow$ 5 kHz = High frequency – 52 kHz
$\Rightarrow$ High frequency = 57 kHz

21. Topic: Computer Networks

(a) An Ethernet hub functions as a repeater. An Ethernet hub, active hub, network hub, repeater hub, multiport repeater, or simply hub is a network hardware device for connecting multiple Ethernet devices together and making them act as a single network segment. A hub works at the physical layer (layer 1) of the OSI model.

22. Topic: Computer Networks

(c) Phase transition for each bit is used in Manchester encoding. In Manchester encoding, logic 0 is indicated by 0 to 1 transition at center of bit and logic 1 is indicated by 1 to 0 transition at center of bit.

23. Topic: Programming and Data Structures

(d) Execution of program gives us the following:

	demo (1234)	
print (4)	demo (123)	print (4)
print (3)	demo (12)	print (3)
print (2)	demo (1)	print (2)
print (1)		print (1)

The result printed is 4 3 2 1 1 2 3 4 because demo (0) will be executed.

24. Topic: Computer Networks

(a) Bit stuffing refers to inserting a '0' in user stream to differentiate it with a flag. Bit stuffing is required when there is a flag of bits. When sender data link layer encounters five consecutive ones in the data stream, it automatically stuffs 0 bit into the outgoing stream. At receiver end, 0 bit is destuffed before sending data to the network layer.

25. Topic: Operating System

(b) Multitasking is a technique in which the operating system of a computer executes several programs concurrently by switching back and forth.

26. Topic: Computer Networks

(b) Number of routing tables is equal to number of routers. Therefore, number of routing tables is 5.

27. Topic: Operating System

(b) Virtual memory is a memory management technique that maps memory address used by a program called virtual address into a physical address in computer memory. Virtual memory is a technique to allow a program, of size more than the size of the main memory to run.

28. Topic: Operating System

(a) The level of aggregation of information required for operational control is detailed and its strategic planning is aggregate.

29. Topic: Programming and Data Structures

(c) The set of all equivalence classes of a set A of cardinality C form a partition of A. An equivalence class is formed by equivalence relation that groups all elements depending on equivalence relation making partitions of given set.

30. Topic: Digital Logic

(a) Given:

$$(0.75)_{10}$$

We know that

$$0.75 \times 8 = 0$$

with a carry of 6.
Therefore, octal equivalent of $(0.75)_{10} = (0.6)_8$.

31. Topic: Digital Logic

(d) When $R = S = 0$, Q and Q_0 are 1.
Now when both inputs are 1, output of NAND gate is zero. Since Q_0 and Q are complement of each other, the logic is not satisfied. There is a race between $Q_0 = 1, Q = 0$ and $Q_0 = 0, Q = 1$ states. Thus, the logic is in intermediate state.

32. Topic: Engineering Mathematics

(d) Suppose boy $= x$ and girl $= y$.
Taller girls in logical notation is written as:

$$(\forall y)\,(\text{girl}(y) \rightarrow \text{taller}(x, y))$$

Thus for assertion, some boys in class are taller than all the girls; the logical notation is

$$(\exists x)(\text{boy}(x) \wedge (\forall y)(\text{girl}(y) \rightarrow \text{taller}(x, y))$$

33. Topic: Engineering Mathematics

(c) Probability that chip is shipped by company $X = 5/9$.
Probability that chip is shipped by company $Y = 4/9$.
Probability that defective piece is shipped by company $X = 1/5$.
Probability that defective piece is shipped by company $Y = 2/4$.
Therefore, Probability that if chosen chip is found to be defective, then chip came from company Y is

$$\frac{\left(\frac{4}{9}\times\frac{2}{4}\right)}{\left(\frac{4}{9}\times\frac{2}{4}\right)+\left(\frac{5}{9}\times\frac{1}{5}\right)} = \frac{\left(\frac{2}{9}\right)}{\left(\frac{2}{9}+\frac{1}{9}\right)} = \frac{\left(\frac{2}{9}\right)}{\left(\frac{3}{9}\right)} = \frac{2}{3}$$

34. Topic: Digital Logic

(c) Ring counter is analogous to stepping switch where each triggering pulse causes on advance of the switch by one step.

35. Topic: Digital Logic

(b) Acceptable input signal voltages range from 0 to 0.8 V for low logic state; and for high logic state, voltages range from 2 to 5 V. Thus, the output 0 and 1 level for TTL logic family is approximately 0.6 and 3.5 V.

36. Topic: Digital Logic

(d) Smallest positive value consists of the following:

Smallest positive mantissa – 0000 0000 0000 0000

and

Largest negative exponent – 1111 1111

These have values in 2's complement as 1 and $-2^7(-128)$
Therefore,

$$\text{Smallest value} = 1 \times 2^{-128}$$

Largest positive value consists of the following:

Largest positive mantissa – 0111 1111 1111 1111

and

Largest positive exponent – 0111 1111

These have values in 2's complement as $2^{15} - 1$ and $2^7 - 1$ (+127), respectively. Therefore, Largest value = $(2^{15} - 1) \times 2^{127}$

37. Topic: Computer Organization and Architecture

(d) In comparison with static RAM memory, the dynamic RAM memory has higher bit density and lower power consumption.

38. Topic: Digital Logic

(c) Given: binary number 0111 1100 1101 1110 0011
Hexadecimal conversion:

$$\underbrace{0111}_{7}\underbrace{1100}_{C}\ \underbrace{1101}_{D}\underbrace{1110}_{E}\ \underbrace{0011}_{3}$$

Therefore, hexadecimal equivalent is $(7CDE3)_{16}$.

39. Topic: Computer Organization and Architecture

(b) Total number of seeks

$= (20 - 18) + (25 - 18) + (39 - 25) + (39 - 8) + (8 - 5) + (5 - 3) = 59$

Each take 5 ms per cylinder.
Total time = 59 × 5 = 295 ms

40. Topic: Operating System

(a) Consider worst-case scenario where one more instance of resource is required by all processes; so, we have 2, 3, 5, which sums to 10. From $m = 10$, deadlock may occur. Therefore, for $m = 10 + 1 = 11$ ensures that deadlock will never occur.

41. Topic: Operating System

(d) A process is blocked if it is waiting for a temporal event or external event. For example, exhausting its CPU time allocation, waiting for an event to occur, semaphore. Thus, a task in a blocked state is waiting for some temporarily unavailable resources.

42. Topic: Operating System

(b) Semaphore is a variable used to solve critical section problem and to achieve synchronization. Semaphore synchronizes critical resources to prevent contention.

43. Topic: Operating System

(b) In non-preemptive scheduling, shortest job first implies less waiting time. Thus, to minimize wait time, order should be 5, 9, 12, 18.

44. Topic: Operating System

(a) The number of page frames that must be allocated to a running process in a virtual memory environment is determined by the instruction set architecture.

45. Topic: Operating System

(c) Let time = t.
$f_1(t)$ executes x units.
Then $f_2(t)$ executes $(t - x)$ units because $f_1(t)$ and $f_2(t)$ are executed sequentially.
Applying convolution on sum of two independent random variables, we get

$$f_1(t) * f_2(t-x) = \int_0^t f_1(x) f_2(t-x)dx$$

46. Topic: Computer Organization and Architecture

(c) Given, 2 blocks in a set and 4 blocks
Block is replaced using least recently scheme (LRU).
For sequence of block addresses 8, 12, 0, 12, 8,
8, 12 and 0 miss in cache when 0 replaces 8 and when 12 comes to cache, 8 is missed again by cache. So total 4 misses are there.

47. Topic: Databases

(b) To control access over objects in relational database – GRANT and REVOKE commands are used.

48. Topic: Databases

(a) SQL aggregate functions return a single value calculated from a column Avg is aggregate function that returns average value.

49. Topic: Operating System

(c) The computer should hold three logics, i.e., true, false and unknown.
We need to represent at least 250 different values.
Now

$3^4 = 81$ (insufficient to hold 256 values)
$3^5 = 243$ (insufficient to hold 256 values)
$3^6 = 729$ (insufficient to hold 256 values)
$3^7 = 2187$ (more than sufficient to hold 256 values)

Therefore, 6 flits are required.

50. Topic: Databases

(b) A view of database that appears to an application program is known as subschema. A subschema describes external view consisting of logical records and relationships.

51. Topic: Databases

(d) There are three Armstrong's inference rule:

(i) Reflexivity – If Y is subset of X, then $X \to Y$
(ii) Augmentation – If $X \to Y$ then $XZ \to YZ$
(iii) Transitivity – If $X \to Y$ and $Y \to Z$ then $X \to Z$

Armstrong's influence rule does not determine mutual dependency.

52. Topic: Databases

(a) Projection operation is used to extract specified columns from a table.

53. Topic: Computer Organization and Architecture

(a) Big-Endian system stores big end of data, i.e., most significant value in sequence at the lowest storage address first.

54. Topic: Databases

(b) If an attribute of composite key depends on attribute of other composite key, then relation is not in BCNF. BCNF is not used for cases where a relation has two candidate keys and composite.

55. Topic: Algorithms

(b) Selection sort algorithm design technique is an example of divide and conquer. It replaces smallest element with first element and second smallest element with second element and so on.

56. Topic: Operating System

(b) In RAID level 3, data is stripped and transferred parallelly; therefore RAID 3 provides highest data transfer rate.

57. Topic: Programming and Data Structures

(a) Lips is a functional programming language that provides garbage collection automatically.

58. Topic: Algorithms

(d) The average case and worst-case complexities for merge sort algorithm are same:

$$O(n \log_2 n)$$

59. Topic: Algorithms

(a) Time taken by binary search algorithm to search a key in a sorted array for n elements is

$$O(\log n)$$

60. Topic: Operating System

(a) Two-phase commit protocol ensures serializability, it does not prevent deadlock neither it detect deadlock nor it recover from deadlock.

61. Topic: Engineering Mathematics

(None) A Fibonacci sequence is a sequence of integers in which each integer is the sum of the previous two integers i.e., 0, 1, 1, 2, 3, 4, 8, 13, 21, 54.

62. Topic: Engineering Mathematics

(a) Since graph G has no self-loops, all elements of the principle diagonal will be zero.

63. Topic: Algorithms

(d) Hashing is not a feature of WAP 2.0.

64. Topic: Programming and Data Structures

(b) Feedback queues dispatch tasks according to execution characteristics.

65. Topic: Computer Organization and Architecture

(c) Analysis diagram is not a UML diagram. Use case is a unified modeling language that defines interactions between a role and a system to achieve goal. Class diagram describes the structure of system by showing the systems classes, their attributes, operations and the relationships among objects. Swim-Lane Diagram distinguishes job sharing and responsibilities for sub-processes of a business process.

66. Topic: Computer Networks

(c) Silly window syndrome is a problem in computer networking caused by poorly implemented TCP flow control. Thus, silly window syndrome is related to degrade in TCP performance.

67. Topic: Programming and Data Structures

(d) To execute all loops at their boundaries and within their operational bounds is an example of white box testing, because white box testing has access to loops.

68. Topic: Computer Networks

(d) SSL – secure rockets layer protocol works at transport layer of TCP/IP. It is an open-standard protocol and is supported by both client and server. It is responsible for authentication, data integrity and data confidentiality through encryption. It is not responsible for error detection and correction.

69. Topic: Operating System

(c) A rule in a limited entry decision table is a column of the table consisting of condition entries and the corresponding action entries. The condition entries are Boolean values and action entries are check-marks.

70. Topic: Computer Networks

(d) X509 is a widely used standard for defining digital certificates.

71. Topic: Computer Networks

(b) Hashed message is signed by a sender using his private key. Only private key can generate a signature and that is verified by corresponding public key.

72. Topic: Computer Networks

(b) A mail containing textual birthday greeting, a picture of cake and a song has multipart/parallel content type. In this, all the parts are intended to be presented in parallel.

73. Topic: Computer Networks

(c) Range of IP address from 224.0.0.0 to 239.255.255.255 is reserved for multicast and therefore used for multicast packets.

74. Topic: Computer Networks

(d) IEEE 802.11 is standard for wireless LANs. Also, IEEE 802.11 is set of media access control and physical layer specifications for implementing WLAN.

75. Topic: Computer Networks

(b) When a host on network A sends a message to a host on network B, the router locks at the IP address. The router operates at third layer of OSI model and is used to connect two different networks based on IP address.

76. Topic: Operating System

(d) The approaches given in first three options, namely, SPICE (ISO/IEC 15504), standard CMMI Assessment Method for process improvement and ISO 9001:2000 are approaches to Software Process Assessment. However, IEEE 2000:2001 is not an approach to Software Process Assessment.

77. Topic: Operating System

(b) A physical DFD specifies who generates data and who process it and how the system is actually implemented.

78. Topic: Programming and Data Structures

(c) In UML diagram of a class state is represented as an attribute. It is a static structure diagram that describes the structure of a system by showing the system classes, their attributes, operations and relationships among objects.

79. Topic: Operating System

(d) Rayleigh model is used for software reliability. It is used for estimating number of defects logged throughout testing process.

80. Topic: Algorithms

(c) Dijkstra's algorithm is used to find the shortest path from one source node to all other nodes in a graph. Thus, Dijkstra's algorithm is used to calculate the routing tables.

Indian Space Research Organisation Computer Science

2008

QUESTION PAPER

1. Which of the following is an illegal array definition?

(a) type COLOGNE: (LIME, PINE, MUSK, MENTHOL); var *a*: array [COLOGNE] of REAL;
(b) var *a*: array [REAL] of REAL;
(c) var *a*: array [‘*A*’…‘*Z*’] of REAL;
(d) var *a*: array [BOOLEAN] of REAL

2. The term Phong is associated with

(a) ray tracing (b) shading
(c) hidden line removal (d) a game

3. The subnet mask 255.255.255.192

(a) extends the network portion to 16 bits
(b) extends the network portion to 26 bits
(c) extends the network portion to 36 bits
(d) has no effect on the network portion of an IP address

4. On a LAN, where are IP datagrams transported?

(a) In the LAN header
(b) In the application field
(c) In the information field of the LAN frame
(d) After the TCP header

5. In Ethernet, the source address field in the MAC frame is the ________ address.

(a) original sender’s physical
(b) previous station’s physical
(c) next destination’s physical
(d) original sender’s service port

6. Which of the following transmission media is not readily suitable to CSMA operation?

(a) Radio (b) Optical fibres
(c) Coaxial cable (d) Twisted pair

7. Consider the grammar

$S \to ABCc|bc$
$BA \to AB$
$Bb \to bb$
$Ab \to ab$
$Aa \to aa$

Which of the following sentences can be derived by this grammar?

(a) *abc* (b) *aab*
(c) *abcc* (d) *abbc*

8. The TCP sliding window

(a) can be used to control the flow of information
(b) always occurs when the field value is 0
(c) always occurs when the field value is 1
(d) occurs horizontally

9. What is the bandwidth of a signal that ranges from 40 kHz to 4 MHz?

(a) 36 MHz (b) 360 kHz
(c) 3.96 MHz (d) 396 kHz

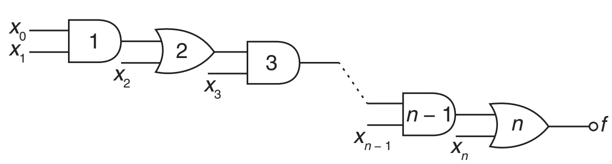
10. Which Project 802 standard provides for a collision-free protocol?

(a) 802.2 (b) 802.3
(c) 802.5 (d) 802.6

11. The Boolean theorem $AB + \bar{A}C + BC = AB + \bar{A}C$ corresponds to

(a) $(A+B)\cdot(\bar{A}+C)\cdot(B+C) = (A+B)\cdot(\bar{A}+C)$
(b) $AB + \bar{A}C + BC = AB + BC$
(c) $AB + \bar{A}C + BC = (A+B)\cdot(\bar{A}+C)\cdot(B+C)$
(d) $(A+B)\cdot(\bar{A}+C)\cdot(B+C) = AB + \bar{A}C$

12. In the given network of AND OR gates, *f* can be written as



(a) $x_0x_1x_2\ldots x_n + x_1x_2\ldots x_n + x_2x_3\ldots x_n + \cdots + x_n$
(b) $x_0x_1 + x_2x_3 + \cdots + x_{n-1}x_n$
(c) $x_0 + x_1 + x_2 + \cdots + x_n$
(d) $x_0x_1 + x_3\ldots x_{n-1} + x_2x_3 + x_5\ldots x_{n-1} + \cdots + x_{n-2}x_{n-1} + x_n$

13. If $N^2 = (7601)_8$ where N is a positive integer, then the value of N is

(a) $(241)_5$ (b) $(143)_6$
(c) $(165)_7$ (d) $(39)_{16}$

14. Assume that each character code consists of 8 bits. The number of characters that can be transmitted per second through a synchronous serial line at 2400 baud rate, and with two stop bits is

(a) 109 (b) 216
(c) 218 (d) 219

15. Four jobs to be executed on a single processor system arrive at time 0 in the order *A, B, C, D*. The burst CPU time requirements are 4, 1, 8, 1, time units, respectively. The completion time of *A* under robin round scheduling with time slice of one time unit is

(a) 10 (b) 4
(c) 8 (d) 9

16. Which one of the following algorithm design techniques is used in finding all pairs of shortest distances in a graph?

(a) Dynamic programming
(b) Backtracking
(c) Greedy
(d) Divide and Conquer

17. The address space of 8086 CPU is

(a) 1 MB (b) 256 KB
(c) 1 K MB (d) 64 KB

18. More than one word are put in one cache block to

(a) exploit the temporal locality of reference in a program
(b) exploit the spatial locality of reference in a program
(c) reduce the miss penalty
(d) none of these

19. The performance of a pipelined processor suffers if

(a) the pipeline stages have different delays
(b) consecutive instructions are dependent on each other
(c) the pipeline stages share hardware resources
(d) all of these

20. If $(12x)_3 = (123)_x$, then the value of x is

(a) 3 (b) 3 or 4
(c) 2 (d) none of the above

21. The advantage of MOS devices over bipolar devices is that

(a) it allows higher bit densities and also cost effective
(b) it is easy to fabricate
(c) its higher-impedance and operational speed
(d) All of these

22. How many 2-input multiplexers are required to construct a 2^{10}-input multiplexer?

(a) 1023 (b) 31
(c) 10 (d) 127

23. A computer uses 8-digit mantissa and 2-digit exponent. If $a = 0.052$ and $b = 28E + 11$, then $b + a - b$ will

(a) result in an overflow error
(b) result in an underflow error
(c) be 0
(d) be $5.28E + 11$

24. The Boolean expression $(A+\overline{C})(\overline{B}+\overline{C})$ simplifies to

(a) $\overline{C}+A\overline{B}$ (b) $\overline{C}(\overline{A}+B)$
(c) $\overline{B}\overline{C}+A\overline{B}$ (d) None of these

25. In the expression $\overline{A}(\overline{A}+\overline{B})$ by writing the first term A as $A + 0$, the expression is best simplified as

(a) $A + AB$ (b) AB
(c) A (d) $A + B$

26. The logic operations of two combinational circuits given in Figure I and Figure II are

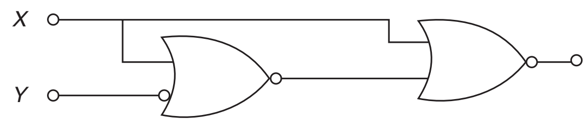


Figure I

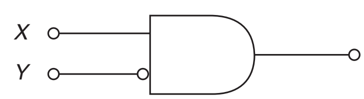
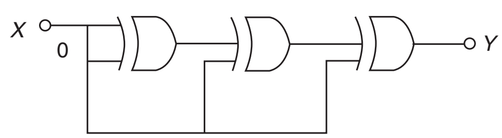


Figure II

(a) entirely different
(b) identical
(c) complementary
(d) dual

27. The output Y of the given circuit is



(a) 1 (b) zero
(c) X (d) $\overline{X}$

28. Which of the following is not a valid rule for XOR?

(a) 0 XOR 0 = 0 (b) 1 XOR 1 = 1
(c) 1 XOR 0 = 1 (d) B XOR B = 0

29. The number of distinct simple graphs with up to three nodes is

(a) 15 (b) 10
(c) 7 (d) 9

30. Maximum number of edges in a n-node undirected graph without self-loops is

(a) n^2 (b) $\frac{n(n-1)}{2}$
(c) $n-1$ (d) $\frac{n(n+1)}{2}$

31. If the two matrices $\begin{bmatrix} 1 & 0 & x \\ 0 & x & 1 \\ 0 & 1 & x \end{bmatrix}$ and $\begin{bmatrix} x & 1 & 0 \\ x & 0 & 1 \\ 0 & x & 1 \end{bmatrix}$ have the same determinant, then the value of x is

(a) 1/2 (b) $\sqrt{2}$
(c) ±1/2 (d) $\pm 1/\sqrt{2}$

32. The network 198.78.41.0 is a

(a) Class A network (b) Class B network
(c) Class C network (d) Class D network

33. The join operation can be defined as

(a) a Cartesian product of two relations followed by a selection
(b) a Cartesian product of two relations
(c) a union of two relations followed by Cartesian product of the two relations
(d) a union of two relations

34. If a square matrix A satisfies $A^TA = I$, then the matrix A is

(a) idempotent (b) symmetric
(c) orthogonal (d) Hermitian

35. Embedded pointer provides

(a) an inverted index
(b) a secondary access path
(c) a physical record key
(d) a primary key

36. An interrupt in which the external device supplies its address as well as the interrupt requests is known as

(a) vectored interrupt
(b) maskable interrupt
(c) non-maskable interrupt
(d) designated interrupt

37. The ability to temporarily halt the CPU and use this time to send information on buses is called

(a) direct memory access (b) vectoring the interrupt
(c) polling (d) cycle stealing

38. Relative to the program translated by a compiler, the same program when interpreted runs

(a) faster
(b) slower
(c) at the same speed
(d) may be faster or slower

39. Consider the following Assembly language program:

```
MVIA  30 H
ACI   30 H
XRA   A
POP   H
```

After the execution of the above program, the contents of the accumulator will be

(a) 30 H (b) 60 H
(c) 00 H (d) contents of stack

40. Consider the following *C* function:

```
int f(int n)
{static int i = 1;
if (n >=5) return n;
n = n + i;
i++;
return f(n);
}
```

The value returned by *f*(1) is

(a) 5 (b) 6
(c) 7 (d) 8

41. In a resident-OS computer, which of the following systems must reside in the main memory under all situations?

(a) Assembler (b) Linker
(c) Loader (d) Compiler

42. Which of the following architecture is/are not suitable for realizing SIMD?

(a) Vector processor (b) Array processor
(c) Von Neumann (d) All of these

43. Consider the following code segment.

```
for (int k = 0; k < 20; k = k + 2)
{
if (k % 3 = = 1)
System. out. print (k + " ");
}
```

What is printed as a result of executing the code segment?

(a) 4 16

(b) 4 10 16

(c) 0 6 12 18

(d) 1 4 7 10 13 16 19

44. The device which is used to connect a peripheral to bus is known as

(a) control register (b) interface

(c) communication protocol (d) none of these

45. The TRAP is one of the interrupts available in INTEL 8085. Which one of the following statements is true of TRAP?

(a) It is level triggered

(b) It is negative edge triggered

(c) It is the positive edge triggered

(d) It is both positive and negative edges triggered

46. Raid configurations of disks are used to provide

(a) fault-tolerance (b) high speed

(c) high data density (d) none of these

47. Which of the following need not necessarily be saved on a context switch between processes?

(a) General-purpose registers

(b) Translation look aside buffer

(c) Program counter

(d) All of these

48. Which of the following is termed as minimum error code?

(a) Binary code (b) Gray code

(c) Excess 3 code (d) Octal code

49. The total time to prepare a disk drive mechanism for a block of data to be read from it is

(a) seek time (b) latency

(c) latency plus seek time (d) transmission time

50. Feedback queues

(a) are very simple to implement

(b) dispatch tasks according to execution characteristics

(c) are used to favour real-time tasks

(d) require manual intervention to implement properly

51. With Round-Robin CPU scheduling in a time shared system

(a) using very large time slices (quantas) degenerates into First-Come First-Served (FCFS) algorithm

(b) using extremely small time slices improves performance

(c) using very small time slices degenerates into Last-ln First-Out (LIFO) algorithm

(d) using medium-sized time slices leads to Shortest Request Time First (SRTF) algorithm

52. Dynamic Address translation

(a) is part of the operating system paging algorithm

(b) is useless when swapping is used

(c) is the hardware necessary to implement paging

(d) storage pages at a specific location on disk

53. Thrashing

(a) always occurs on large computers

(b) is a natural consequence of virtual memory systems

(c) can always be avoided by swapping

(d) can be caused by poor paging algorithms

54. What is the name of the operating system that reads and reacts in terms of actual time?

(a) Batch system

(b) Quick response system

(c) Real time system

(d) Time sharing system

55. The memory address register

(a) is a hardware memory device which denotes the location of the current instruction being executed

(b) is a group of electrical circuit that performs the intent of instructions fetched from memory

(c) contains the address of the memory location that is to be read from or stored into

(d) contains a copy of the designated memory location specified by the MAR after a "read" or the new contents of the memory prior to a "write".

56. An example of spooled device is a

(a) line printer used to print the output of a number of jobs

(b) terminal used to enter input data to a running program

(c) secondary storage device in a virtual memory system

(d) graphic display device

57. Dirty bit for a page in a page table

(a) helps avoid unnecessary writes on a paging device

(b) helps maintain LRU information

(c) allows only read on a page

(d) none of these

58. Checkpointing a job
(a) allows it to be completed successfully
(b) allows it to continue executing later
(c) prepares it for finishing
(d) occurs only when there is an error in it

59. A public key encryption system
(a) allows anyone to decode the transmissions
(b) allows only the correct sender to decode the data
(c) allows only the correct receiver to decode the data
(d) does not encode the data before transmitting it

60. Overlaying
(a) requires use of a loader
(b) allows larger programs, but requires more effort
(c) is most used on large computers
(d) is transparent to the user

61. A critical section is a program segment
(a) which should run in a certain specified amount of time
(b) which avoids deadlock
(c) where shared resources are accessed
(d) which must be endorsed by a pair of semaphore operations, P & U

62. In which of the following four necessary conditions for deadlock processes claim exclusive control of the resources they require?
(a) No preemption (b) Mutual exclusion
(c) Circular wait (d) Hold and wait

63. Fork is
(a) the creation of a new job
(b) the dispatching of a task
(c) increasing the priority of a task
(d) the creation of a new process

64. Which of the following need not necessarily be saved on a Context Switch between processes?
(a) General-purpose register
(b) Translation look aside buffer
(c) Program counter
(d) Stack pointer

65. Consider a logical address space of 8 pages of 1024 words mapped into memory of 32 frames. How many bits are there in the logical address?
(a) 13 bits (b) 15 bits
(c) 14 bits (d) 12 bits

66. The performance of Round Robin algorithm depends heavily on
(a) size of the process
(b) the I/O bursts of the process
(c) the CPU bursts of the process
(d) the size of the time quantum

67. The page replacement algorithm which gives the lowest page fault rate is
(a) LRU
(b) FIFO
(c) Optional page replacement
(d) Second chance algorithm

68. Which of the following class of statement usually produces no executable code when compiled?
(a) Declaration
(b) Assignment statements
(c) Input and output statements
(d) Structural statements

69. What is the value of $F(4)$ using the following procedure?

```
function F(k : integer):
integer;
begin
if(k < 3) then F := k else F: = F(k − 1)
* F(k − 2) + F(k − 3)
end;
```

(a) 5 (b) 6
(c) 7 (d) 8

70. Stack A has the entries a, b, c (with a on top). Stack B is empty. An entry popped out of stack A can be printed immediately or pushed to stack B. An entry popped out of the stack B can only be printed. In this arrangement, which of the following permutations of a, b, c are not possible?
(a) $b\ a\ c$ (b) $b\ c\ a$
(c) $c\ a\ b$ (d) $a\ b\ c$

71. The time required to search an element in a linked list of length n is
(a) $O(\log_2 n)$ (b) $O(n)$
(c) $O(1)$ (d) $O(n^2)$

72. Which of the following operations is performed more efficiently by doubly linked list than by linear linked list?
(a) Deleting a node whose location is given
(b) Searching an unsorted list for a given item
(c) Inserting a node after the node with a given location
(d) Traversing the list to process each node

73. We can make a class abstract by
- (a) Declaring it abstract using the virtual keyword
- (b) Making at least one member function as virtual function
- (c) Making at least one member function as pure virtual function
- (d) Making all member function const

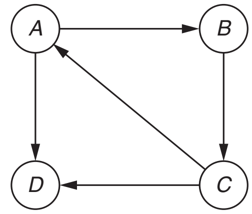
74. A Steiner patch is
- (a) Biquadratic Bezier patch
- (b) Bicubic patch
- (c) Circular patch only
- (d) Bilinear Bezier patch

75. A complete binary tree with the property that the value at each node is at least as large as the values at its children is known as
- (a) binary search tree (b) AVL tree
- (c) completely balanced tree (d) Heap

76. The minimum number of fields with each node of doubly linked list is
- (a) 1 (b) 2
- (c) 3 (d) 4

77. How many comparisons are needed to sort an array of length 5 if a straight selection sort is used and array is already in the opposite order?
- (a) 1 (b) 10
- (c) 15 (d) 20

78. Consider the graph shown in the following figure:



Which of the following is a valid strong component?
- (a) *a*, *c*, *d* (b) *a*, *b*, *d*
- (c) *b*, *c*, *d* (d) *a*, *b*, *c*

79. Repeated execution of simple computation may cause compounding of
- (a) round-off errors (b) syntax errors
- (c) run-time errors (d) logic errors

80. In C, what is the effect of a negative number in a field width specifier?
- (a) The values are displayed right justified.
- (b) The values are displayed centered.
- (c) The values are displayed left justified.
- (d) The values are displayed as negative numbers.

ANSWER KEY

1. (b)	**9.** (c)	**17.** (a)	**25.** (*)	**33.** (a)	**41.** (c)	**49.** (c)	**57.** (a)	**65.** (a)	**73.** (c)
2. (b)	**10.** (c)	**18.** (b)	**26.** (a)	**34.** (c)	**42.** (c)	**50.** (b)	**58.** (b)	**66.** (d)	**74.** (a)
3. (b)	**11.** (a)	**19.** (d)	**27.** (b)	**35.** (b)	**43.** (b)	**51.** (a)	**59.** (c)	**67.** (c)	**75.** (d)
4. (d)	**12.** (*)	**20.** (d)	**28.** (b)	**36.** (a)	**44.** (b)	**52.** (c)	**60.** (b)	**68.** (c)	**76.** (b)
5. (b)	**13.** (b)	**21.** (d)	**29.** (c)	**37.** (d)	**45.** (d)	**53.** (d)	**61.** (c)	**69.** (a)	**77.** (b)
6. (a)	**14.** (c)	**22.** (a)	**30.** (b)	**38.** (b)	**46.** (a)	**54.** (c)	**62.** (b)	**70.** (c)	**78.** (d)
7. (*)	**15.** (d)	**23.** (c)	**31.** (a)	**39.** (c)	**47.** (b)	**55.** (c)	**63.** (d)	**71.** (b)	**79.** (a)
8. (a)	**16.** (a)	**24.** (a)	**32.** (c)	**40.** (c)	**48.** (b)	**56.** (a)	**64.** (b)	**72.** (a)	**80.** (c)

*** None of the options is correct.**

ANSWERS WITH EXPLANATION

1. **Topic: Programming and Data Structures**

(b) An array is a kind of data structure that holds a fixed number of values of a single type. An array is declared as

type arrayname [arraysize];

Array size must be an integer constant greater than zero. We can also use enum, char and boolien in the place of integer to define array size but arraysize cannot be real. Therefore, option (b) var *a*:array[REAL] of REAL is an illegal array definition.

2. **Topic: Computer Graphics**

(b) Phong is an interpolation technique for surface shading in 3D computers. It is also called phong interpolation. It is surface for smoothing out multi-surface shapes and creating more sophisticated computer modelled images. So, the term "Phong" is associated with shading. Therefore, option (b) is correct.

3. **Topic: Computer Networks**

(b) A subnet mask is a 32- or 128-bit number that segments an existing IP address in TCP/IP network into discrete network and host address. A subnet mask can increase the number IP addresses and improve network security and performance. A class C address has 3 bytes for network and 1 byte for host.

3 bytes = 24 bits

Now,

192 in binary = 11000000

That is, first two bits of host thereby extending the network to 26 bits.

4. **Topic: Computer Networks**

(d) LAN or local area network is a computer network that connects computers within a limited area.
Data transmitted over an Internet using Internet Protocol is carried out in messages called IP datagrams. An IP datagram consists of an IP header, top header and data.

IP HEADER	TCP HEADER	DATA

Then on a LAN, the data is transferred after the TCP header. Therefore, option (d) is correct.

5. **Topic: Computer Networks**

(b) (In Ethernet, the source address field in the MAC frame is the previous station's physical address. The only field that changes through the path of an IP packet is source and destination MAC at data link layer. The source address is MAC of the station that sends it to next station.

6. **Topic: Computer Networks**

(a) CSMA or carrier sense multiple access is the protocol for carrier transmission access in Ethernet networks. It determines how network device responds when two devices attempt to use a data channel simultaneously called collisions. Wireless trans-receivers cannot send and receive on the same channel at the same time so collisions cannot be detected. Therefore, radio transmission media is not readily suitable to CSMA operation.

7. **Topic: Theory of Computation**

(None) A grammar is a set of production rules that describes all possible strings in a formal language from the given grammar, the only string that can be generated is *bc*.
S $\rightarrow$ ABCc cannot be further evaluated, so none of the given options is correct.

8. **Topic: Computer Networks**

(a) TCP or transmission control protocol uses a sliding window for flow control. It determines the number of unacknowledged bytes. The value of these bytes is determined by size of send buffer and size and space of receive buffer. Thus, TCP sliding window protocol is used for controlling information flow only. Therefore, option (a) is correct.

9. **Topic: Computer Networks**

(c) Bandwidth refers to amount of data that can be transmitted in a fixed amount of time. It is also defines the range within a bound of frequencies.
Bandwidth = highest frequency – lowest frequency:

$$= 4 \text{ MHz} - 40 \text{ kHz}$$
$$= 4 \times 10^6 - 40 \times 10^3 = 4 \times 10^6 - 4 \times 10^4$$
$$= (400 - 4) \times 10^4 = 396 \times 10^4 \text{ Hz}$$
$$= 3.96 \times 10^6 \text{ Hz} = 3.96 \text{ MHz}$$

10. **Topic: Computer Networks**

(c) (Project 802 standards describe a network's physical elements: NICs, cables, connectors, signaling technologies, media access control, etc.
802.2 is logical control link that covers error control and flow control over data frames.
802.3 is Ethernet LAN that covers all forms of Ethernet media and interfaces.
802.5 is token ring LAN that covers all forms of token ring media and interfaces. It uses CSMA/CD which is a collision free protocol.
802.6 is metropolitan area network that covers MAN technologies, addressing and services.
Therefore, 802.5 standard provides for a collision-free protocol.

11. Topic: Digital Logic

(a) It is given that

$$AB + \bar{A}C + BC = AB + \bar{A}C \quad (1)$$

Now according to consensus theorem, a Boolean expression can be simplified by eliminating the redundant term in the expression to form an equivalent expression. The redundant term is called redundant term or dual form. This is obtained by simply replacing ANDs win ORs and ORs with ANDs. The complements remain unaffected. So, applying this on Eq. (1), we get

$$(A+B)\cdot(\bar{A}+C)\cdot(B+C) = (A+B)\cdot(\bar{A}+C)$$

12. Topic: Digital Logic

(None) We have:
Output of gate 1 = x_0x_1
Output of gate 2 = $x_0x_1 + x_2$
Output of gate 3 = $(x_0x_1 + x_2)x_3$
Output of gate 4 = $(x_0x_1 + x_2)x_3 + x_4$
Output of gate 5 = $((x_0x_1 + x_2)x_3 + x_4)x_5$
Output of gate $n - 1$ = $((x_0x_1 + x_2)x_3 + x_4)x_5 + \ldots + x_{n-2})x_{n-1}$
$f = ((((x_0x_1 + x_2)x_3 + x_4)x_5 + x_6)x_7 + \ldots + x_{n-2})x_{n-1} + x_n$
$= (((x_0x_1x_3 + x_2x_3 + x_4)x_5 + x_6)x_7 + \ldots + x_{n-2})x_{n-1} + x_n$
$= ((x_0x_1x_3x_5 + x_2x_3x_5 + x_4x_5 + x_6)x_7 + \ldots + x_{n-2})\, x_{n-1} + x_n$
$= (x_0x_1x_3x_5x_7 + x_2x_3x_5x_7 + x_4x_5x_7 + x_6x_7 + \ldots\, x_{n-2})x_{n-1} + x_n$
$= x_0x_1x_3x_5x_7\ldots\, x_{n-1} + x_2x_3x_5x_7\ldots\, x_{n-1} + \ldots + x_n$

Therefore,

$$f = x_0x_1x_3x_5\ldots\, x_{n-1} + x_2x_3x_5\ldots\, x_{n-1} + x_4x_5x_7 + x_{n-1}\ldots + x_6\ldots x_{n-1} + x_n$$

So, none of the options is correct

13. Topic: Digital Logic

(b) Given:

$$N^2 = (7601)_8$$
$$\Rightarrow N^2 = 7 \times 8^3 + 6 \times 8^2 + 0 \times 8^1 + 1 \times 8^0 = 3969$$
$$\Rightarrow N = \sqrt{3969} = 63$$

Now,

$$(143)_6 = 1 \times 6^2 + 4 \times 6^1 + 3 \times 6^0 = 63$$

Therefore,

$$N = (143)_6$$

14. Topic: Computer Networks

(c) Given: character code consists of 8 bits.

Band rate = 2400
Stop bits = 2

$$\text{Number of characters transmitted} = \frac{\text{Band Rate}}{\text{Bits per character}}$$

Now,

Bits per character = Data bits + Start bit + Stop bit
= 8 + 2 + 1 = 11 bits

Therefore, number of characters transmitted is

$$\frac{2400}{11} = 218.18$$

So, the number of characters that can be transmitted per second = 218

15. Topic: Operating System

(d) Given, four jobs are to be executed on a single processor system in order A, B, C, D.
First CPU time of these jobs are 4, 1, 8, 1 time units.
Robin round scheduling with time dice of one time unit is used. Robin round scheduling is essentially the pre-emptive version of FIFO, that is, First-In First-Out since robin round scheduling assigns equal time to each process without priority in cyclic order so what process comes first is executed first.
So, the jobs will be executed as follows:

A	*B*	*C*	*D*	*A*	*C*	*A*	*C*	*A*	*C*	…	*C*

0 1 2 3 4 5 6 7 8 9 10 14

Therefore, completion of job *A* requires 9 units of time.

16. Topic: Algorithms

(a) Floyd–Warshall algorithm is an algorithm used to find all pairs of shortest path. Floyd–Warshall algorithm is a dynamic programming algorithm. It is a method for solving complex problem by breaking it into simple subproblems.
Hence, dynamic programming algorithm design techniques is used in finding all pairs of shortest distances in a graph.

17. Topic: Computer Organisation and Architecture

(a) 8086 CPU is a 16-bit processor. It has 20-bit address line; therefore, address space is

$$2^{20} = 1 \text{ MB}$$

It ranges from 0000016 to FFFFF16.
The 1 MB of address space is partitioned into 16 segments size of each segment is 64 KB. All the segments are not active at same time. Only four segments are active at a time.

18. Topic: Computer Organization and Architecture

(b) More than one word is put in one cache block to exploit the spatial locality of reference in a program.
Locality of reference is a phenomenon in which same or related values or storage locations are accessed frequently depending on memory access pattern.

Spatial locality of reference is a concept that if a resource is referenced then likelihood of referencing a resource near it is high.

19. Topic: Computer Organization and Architecture

(d) Pipelining is a process in which hardware elements of CPU are arranged such that overall performance is increased. The performance of a pipelined processor suffers if

- the pipeline stages have different delays.
- consecutive instructions are dependent on each other.
- the pipeline stages share hardware resources.

Therefore, all of the given options are correct.

20. Topic: Digital Logic

(d) Given:

$$(12x)_3 = (123)_x$$

Now,
$(123)_x$ implies a radix greater than 3
$(12x)_3$ implies a digit less than 3.
Therefore, the value of x cannot be possible.

21. Topic: Computer Organization and Architecture

(d) MOS devices contain three level of conducting materials which are separated by insulating material. The three levels are metal, polysilicon and diffusion levels, respectively.
Bipolar devices contain two mutually connected *pn* junctions with layer sequence of *pnp* or *npn*. The fusion produces a three-layer device. The three layers are emitter, bas and collector's levels, respectively.
The advantages of MOS device over bipolar devices are as follows:

1. MOS device has input impedance higher than BJTs.
2. MOS devices are much smaller than BJTs in size.
3. MOS devices are less noisier than BJTs.
4. MOS device can be easily fabricated and cost-effective.
5. MOS device allows higher bit densities.

Therefore, all the given options are correct.

22. Topic: Digital Logic

(a) Multiplexer is a device having multiple inputs and a single output. A multiplexer can operate on both digital and analog signals. Multiplexers are used in communication system, computer memory, telephone network, etc.
To construct a 2^n input multiplexers, $2^n - 1$ two input multiplexes are required.
So, for 2^{10} inputs multiplexers, number of two input multiplexers required are

$$2^{10} - 1 = 1024 - 1 = 1023$$

23. Topic: Digital Logic

(c) Given, mantissa is 8 digits and exponents are 2 digits.
For $a = 0.052$, mantissa = 0.52 and exponent = –1
For $b = 28E + 11$, mantissa = 0.28 and exponent = 13
For comparing $a + b$, both a and b should have same exponent.

$$\Rightarrow b = 28000000000000E - 1$$
$$\Rightarrow a = 0.52\,E - 1$$

Now, a is very small compared to b, so $a + b$ will evaluate to b

$$\Rightarrow a + b = b$$
$$\Rightarrow a + b - b = b - b = 0$$

24. Topic: Digital Logic

(a) Given:
$(A+\bar{C})(\bar{B}+\bar{C}) = A\cdot\bar{B} + A\cdot\bar{C} + \bar{B}\cdot\bar{C} + \bar{C}\cdot\bar{C}$

$= A\bar{B} + A\bar{C} + \bar{B}\bar{C} + \bar{C}$ (since $C\cdot C = C$)

$= A\bar{B} + A\bar{C} + (\bar{B}+1)\bar{C}$

$= A\bar{B} + A\bar{C} + \bar{C}.1$ (since $\bar{B}+1 = 1$, annulment law)

$= A\bar{B} + A\bar{C} + \bar{C}$ (since $\bar{C}\cdot 1 = \bar{C}$, identity law)

$= A\bar{B} + (A+1)\bar{C} = A\bar{B} + 1\cdot\bar{C}$ (since $A+1 = 1$)

$= A\bar{B} + \bar{C}$

Therefore,

$$(A+\bar{C})(\bar{B}+\bar{C}) = \bar{C} + A\bar{B}$$

25. Topic: Digital Logic

(None) Given:

$$\bar{A}(\bar{A}+\bar{B})$$

Writing first term A as $A + 0$, we have

$$(\overline{A+0})(\bar{A}+\bar{B}) = (\bar{A}\cdot\bar{0})(\bar{A}+\bar{B})$$

(De Morgan's law $\overline{A+B} = \bar{A}.\bar{B}$)

$\Rightarrow (\bar{A}\cdot 1)(\bar{A}+\bar{B})$ (since $\bar{0} = 1$)

$= \bar{A}(\bar{A}+\bar{B})$ (since $\bar{A}\cdot 1 = \bar{A}$, Identity law)

$= \bar{A}\cdot\bar{A} + \bar{A}\cdot\bar{B} = \bar{A} + \bar{A}\cdot\bar{B}$ (since $\bar{A}\cdot\bar{A} = \bar{A}$, Idempotent law)

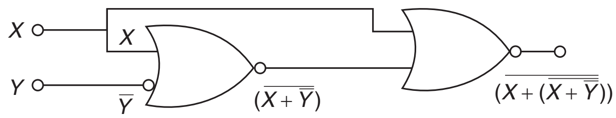
$= \bar{A}(1+\bar{B}) = \bar{A}\cdot 1$ (since $1+\bar{B} = 1$, Null law)

$= \bar{A}$ (since $\bar{A}.1 = \bar{A}$, Identity law)

Therefore, the expression simplifies to $\bar{A}$.
So, none of the options is correct.

26. Topic: Digital Logic

(a) Consider combinational circuit in Figure I:



The output of circuit is $(\overline{X+(\overline{\overline{X}+\overline{Y}})})$.

Solving this Boolean expression, we have

$$\overline{X+(\overline{\overline{X}+\overline{Y}})}=\overline{X}\cdot(\overline{\overline{X+\overline{Y}}})$$

(using De Morgan's law $\overline{A+B}=\overline{A}\cdot\overline{B}$)

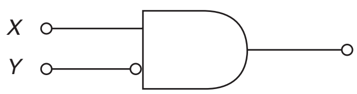
$=\overline{X}\cdot(X+\overline{Y})$ (since $\overline{\overline{A}}=A$)

$=\overline{X}\cdot X+\overline{X}\cdot\overline{Y}=0+\overline{X}\overline{Y}$ (since $\overline{X}\cdot X=0$ by Inverse law)

$=\overline{X}\cdot\overline{Y}$ (since $A+0=A$ by Identity law)

Therefore, output is $\overline{X}\cdot\overline{Y}$.

Now, consider combinational circuit in Figure II.

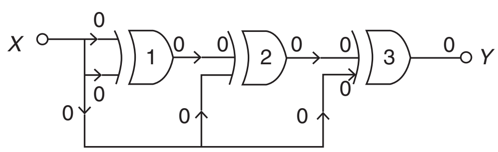


The output is $X\cdot\overline{Y}$.

Therefore, the two combinational circuits are entirely different.

27. Topic: Digital Logic

(b) The given figure consists of three XOR gates:



The truth table for XOR gate is as follows:

X	*Y*	**Output**
0	0	0
0	1	1
1	0	1
1	1	0

XOR gate 1: inputs are 0, 0 as output is 0
XOR gate 2: inputs are 0, 0 so output will be 0
XOR gate 3: inputs are 0, 0 so output will be 0
Therefore, output y will be 0.

28. Topic: Digital Logic

(b) The truth table of XOR gate is as follows:

A	*B*	***A* XOR *B***
0	0	0
0	1	1
1	0	1
1	1	0

Therefore, 1 XOR 1 = 1 is not a valid rule for XOR.

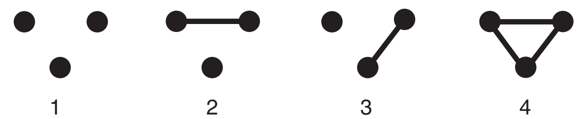
29. Topic: Engineering Mathematics

(c) A simple graph is also called strict graph. It is an undirected graph containing no graph loops and multiple edges.

Number of distinct graphs with one node is 1:

Number of distinct graphs with two nodes is 2:

Number of distinct graphs with three nodes is 4:



Therefore, total number of distinct graphs with up to three nodes is 1 + 2 + 4 = 7.

30. Topic: Engineering Mathematics

(b) An undirected graph is a set of nodes that are connected together such that all edges are bidirectional. It is also called undirected network. For n nodes, each node can have $(n-1)$ edges. So, n nodes will have $n \times (n-1)$ edges.

Since all edges are bidirectional, they are counted twice.

Therefore, number of edges in n node undirected graph is $\dfrac{n(n-1)}{2}$.

31. Topic: Engineering Mathematics

(a) Given two matrices $\begin{bmatrix}1 & 0 & x\\ 0 & x & 1\\ 0 & 1 & x\end{bmatrix}$ and $\begin{bmatrix}x & 1 & 0\\ x & 0 & 1\\ 0 & x & 1\end{bmatrix}$ have same determinant.

Determinant of matrix $\begin{bmatrix}1 & 0 & x\\ 0 & x & 1\\ 0 & 1 & x\end{bmatrix}$ is

$$D_1=1[x^2-1]-0[0]+x[0-0]=x^2-1 \qquad (1)$$

Determinant of matrix $\begin{bmatrix}x & 1 & 0\\ x & 0 & 1\\ 0 & x & 1\end{bmatrix}$ is

$$D_2=x[-x]-1[x]+0[x^2]=-x^2-x \qquad (2)$$

Now, we know that $D_1=D_2$

Using Eqs. (1) and (2) we have

$$x^2 - 1 = -x^2 - x$$
$$\Rightarrow 2x^2 + x - 1 = 0$$
$$\Rightarrow x = \frac{1}{2}$$

Therefore, value of x is 1/2.

32. Topic: Computer Networks

(c) IPv4 is current mainstream addressing protocol for the internet and other TCP/IP network. IPv4 contains three main categories of networks, namely Class A, Class B and Class C. These are defined by first three bits in IP address. The number of bits used to identify the network and host vary according to the network class of the address.
For class A: First octet in address is between 0 and 127, i.e., 0.0.0.0 to 127.255.255.255
For class B: First octet in address is between 128 and 191, i.e., 128.0.0.0 to 191.255.255.255
For class C: First octet in address is between 192 and 223, i.e., 192.0.0.0 to 223.255.255.255
Therefore, the network 198.78.41.0 is a class C network.

33. Topic: Databases

(a) Join operation is a combination of Cartesian product followed by selection process. This operation pairs two types from different relations if the given join operation satisfies. Therefore, option (a) is correct definition of join operation.

34. Topic: Engineering Mathematics

(c) If a square matrix A satisfies $A^TA = 1$, then the matrix A is orthogonal.
A matrix is orthogonal if its determinant is either 1 or –1.
Given:

$$A^TA = I$$

Taking determinant on both sides

$$\det(A^TA) = \det I$$
$$\Rightarrow \det A \times \det A^T = \det I$$

We know that $\det I = 1$.
So,

$$\det A \times \det A^T = 1 \quad (1)$$

Also, if A^T is transpose of matrix A then determinant of A^T is same as determinant of A.
That is,

$$\det A = \det A^T$$

Using this from Eq. (1), we have

$$(\det A)^2 = 1$$
$$\Rightarrow \det A = \pm 1$$

Hence, A is an orthogonal matrix.

35. Topic: Databases

(b) Embedded pointers are embedded in data structures, such as arrays, unions and structures. Embedded pointers are null on input and only write output to a buffer. Embedded pointer is a set of pointers in a dataset which points to answer data set. Embedded pointer provides a secondary access path. Therefore, option (b) is correct.

36. Topic: Computer Organization and Architecture

(a) An interrupt in which the external device supplies its address as well as the interrupt request is known as vectored interrupt. Vectored interrupt is an input and output interrupt that informs part of the computer that handles input-output interrupt that a request from an input and output device has been received. It also identifies the device that has sent the request.

37. Topic: Computer Organization and Architecture

(d) The ability to temporarily halt the CPU and use this time to send information on buses is called cycle stealing. Cycle stealing is a direct memory access method. It is a method of accessing computer memory or bus without interfering with the CPU the data is transferred in cycle stealing mode at a rate of one byte at a time. It is a design technique that periodically grabs machine cycles from the main processor usually by same peripheral control unit.

38. Topic: Compiler Design

(b) Compiler scans entire program and translates it as a whole into machine language. Programming language like C and C++ use compiler. The time taken by compiler to analyze source code is very large; however, overall execution time is very fast. Interpreter on other hand translates program one statement at a time. Programming languages such as python and ruby use interpreters. The time taken by interpreter to analyze source code is very small; however, overall execution time is very slow.
Therefore, relative to the program translated by a compiler, the same program when interpreted runs slower.

39. Topic: Theory of Computation

(c) For the given assembly language program, MVI implies move immediate data to register or memory location.
Therefore, MVI A 30H means 30H is stored in register A.
ACI implies add immediate to accumulator with carry.
Therefore, AC1 30H means add 30H to the contents of accumulator.
XRA implies the contents of register specified are XORed with the contents of accumulator and the result is stored in the accumulator.
Therefore, XRA A means XOR the contents of register A with contents of accumulator.

Now, register A has 30H and accumulator also has 30H. When XOR operation is carried out, resultant will be 00H. This result is stored in accumulator POP and implies POP contents of accumulator to specified register.
Therefore, POP H means POP the contents of accumulator to H.
After the execution of the program, the contents of accumulator will be 00H.

40. Topic: Programming and Data Structures

(c) Static variables in *C* preserve their value even after they are not in the scope. When the function is called

Call 1: $i = 1, n = 1$
Call 2: $n = n + i = 1 + 1 \Rightarrow n = 2$
$i{+}{+} \Rightarrow i = 2$
Call 3: $n = n + i = 2 + 2 \Rightarrow n = 4$
$i{+}{+} \Rightarrow i = 3$
Call 4: $n = n + i = 4 + 3 \Rightarrow n = 7$
$i{+}{+} \Rightarrow i = 4$
Call 5: If $(n \geq 5)$ condition satisfied

Therefore, n is returned.
Thus, value returned by $f(1)$ is 7.

41. Topic: Compiler Design

(c) Assembler: It is a program that converts the basic computer instructions into a pattern of bits that the computer's processor can use to perform its basic operations.
Linker: A linker is a program combines one or more object files generated by the compiler into one executable program or a unified program.
Loader: It is a major component of an operating system. It ensures that all necessary programs which are essential during the start-up phase of running a program and libraries are loaded and prepared for execution by the OS.
Compiler: It is a software program that transforms high-level source code into a low-level object code in machine language so that it can be understood by the processor.
From all these systems, loader must reside in the main memory under all situations.

42. Topic: Computer Organization and Architecture

(c) SIMD or Single Instruction Multiple Data implies that all parallel units share same instruction but carry out on different data elements. SIMD processors are highly specialized computers.
From the given architecture vector processor and array processor are suitable for realizing SIMD. Whereas, Von Neumann architecture is SISD or Single Instruction Single Data. In this, a single processor executes a single instruction stream to operate on data stored in a single memory. Therefore, Von Neumann is not suitable for realizing SIMD.

43. Topic: Programming and Data Structures

(b) For loop for k, runs over

$$k = 0, 2, 4, 6, 8, 10, 12, 14, 16, 18$$

The program prints the k values with remainder 1 when k is divided by 3.

For $k = 2$ $\quad k \% 3 \neq 1$
For $k = 4$ $\quad k \% 3 == 1$, prints 4
For $k = 6$ $\quad k \% 3 == 0$
For $k = 8$ $\quad k \% 3 == 2$
For $k = 10$ $\quad k \% 3 == 1$, prints 10
For $k = 12$ $\quad k \% 3 == 0$
For $k = 14$, $\quad k \% 3 == 2$
For $k = 16$, $\quad k \% 3 == 1$, prints 16
For $k = 18$, $\quad k \% 3 == 0$

Therefore, 4 10 16 is printed as a result.

44. Topic: Computer Organization and Architecture

(b) Control register: It is a processor register that controls the general behavior of CPU or other digital device.
Interface: It is a device which is used to connect a peripheral to bus.
Communication protocol: It is formal descriptions of digital message formats and rules. It exchanges messages between computer systems and is therefore required in telecommunications.
Therefore, option (b) is correct.

45. Topic: Computer Organization and Architecture

(d) The microprocessor of INTEL 8085 has five interrupt pins. These are used for hardware interrupts, namely TRAP, RST7.5, RST6.5, RST5.5, INTR.
TRAP is a non-maskable edge and level triggered interrupt. This means that TRAP goes high and remain high until it is acknowledged. In case of a failure, TRAP executes ISR and sends data from main memory to backup memory. TRAP has highest priority and vectors interrupt, i.e., the interrupt address is known to the processor. It has both positive and negative edges triggered. Therefore, option (d) is true.

46. Topic: Operating System

(a) RAID or redundant array of independent disks configurations is for storing data redundantly in order to recover in case of disk fail. It is a way of storing some data in different places on multiple hard disks. Therefore, RAID configurations of disks are used to provide fault-tolerance.

47. Topic: Operating System

(b) A context switch is a procedure that a computer's CPU follows to change from one task to another while ensuring that the tasks do not conflict. Translation look aside buffer is not necessarily saved on a context switch between processes. Translation lookaside buffer is present only to speed up operations by caching virtual address to physical address mappings.

48. Topic: Digital Logic

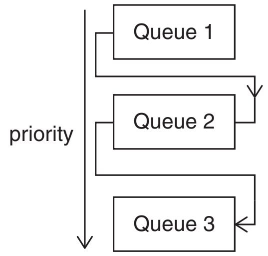
(b) A binary code is simplest form of computer code that is represented entirely by binary system of digits.
Gray code represents numbers using binary. It groups a sequence of bits so that one bit in the group changes from the number before and after.
Excess 3 code is a non-weighted code used to express decimal number. Excess 3 code for a given number is determined by adding 3 to each decimal digit in a given number and then replacing each digit of newly found decimal number by its four-bit binary equivalent. Octal code is a base-8 code. It is obtained from binary numerals by grouping consecutive binary digits into group of three.
From all these, gray code is termed as a minimum error code because the hamming distance between successive number in gray code is 1 which results in minimum error.

49. Topic: Operating System

(c) Seek time is the time taken by a disk drive to locate the area where the data is to be read is stored on the disk.
Latency is the amount of time a message takes to traverse a system.
Transmission time is the time to transmit first bit to last bit of message in transmission node.
Therefore, total time to prepare a disk drive mechanism for a block of data to be read from it is latency plus seek time.

50. Topic: Programming and Data Structures

(b) Feedback queues analyse the behavior or execution characteristics of processor and according to the characteristics it changes its priority. It is an enhancement of queue scheduling where processes can more between the queues based on execution characteristics. Hence, feedback queues dispatch tasks according to execution characteristics.



51. Topic: Operating System

(a) Round robin scheduling is essentially the pre-emptive version of FIFO, i.e., First-In First-Out since round robin scheduling assigns equal time to each process without priority in cyclic order so what process comes first is executed first. Therefore, option (a) is correct. With round-robin CPU scheduling in a time-shared system, using very large time slices (quanta's) degenerates into first come first served algorithm.

52. Topic: Operating System

(c) Dynamic address translation is a process of translating a virtual address into the corresponding real address during a storage reference. Dynamic address translation is the hardware necessary to implement paging.

53. Topic: Operating System

(d) Thrashing is a computer activity that makes little or no progress. It happens because of limited resource or exhaustion of memory. It happens when a page fault occurs. The effect of thrashing is: CPU becomes idle and degree of multiprogramming increases which increasing thrashing exponentially. Thus, thrashing can be caused by poor paging algorithms.

54. Topic: Operating System

(c) A real-time operating system reads and reacts in terms of actual time. Real-time operating system is intended to serve real time applications that process data without any buffer delays. It provides a deterministic execution pattern which is of interest to embedded systems.

55. Topic: Computer Organization and Architecture

(c) Registers are memory locations which can be accessed quickly for rapid retrieval of data. Memory address register is a register located in the computer's processor. It is also known as a memory address translator or decoder. It contains the address of the memory location that is to be read from or stored into. It translates the data to the read from memory or written to memory into an actual location on the memory, assigning the space on RAM to be used by the CPU.

56. Topic: Operating System

(a) Spooling is a process of putting jobs in a buffer, special area of memory where it can be accessed by a device. An example of spooled device is a line printer used to print the output of a number of jobs. In print spooling, document is loaded into a buffer and then the printer pulls them at its own rate.

57. Topic: Operating System

(a) Dirty bit allows performance optimization. Dirty bit for a page in a page table helps avoid unnecessary writes on a paging device. Before performing the write operation, the dirty bit status is checked.

58. Topic: Operating System

(b) Checkpointing a job is method of periodically saving the state of a job so that if job is not completed due to some reason, it can be restarted from the saved state. Checkpoints can be taken either under the control of the user application or external to the application. Thus, checkpointing a job allows it to continue executing later.

59. Topic: Computer Networks

(c) A public key encryption is a scheme that uses two mathematically related, non-identical keys, namely a public key and a private key. Each key performs a unique function. The public key is used to encrypt and the private key is used to decrypt. A public key encryption system allows only the correct receiver to decode the data.

60. Topic: Operating System

(b) Overlaying is the process of transferring a block of program code or data into internal memory replacing what is already stored in the memory. Overlaying is a programming method that allows programs to be larger than the computer's main memory, but it requires more effort. Overlaying is transparent to the user.

61. Topic: Operating System

(c) Critical section is a set of instructions that need to be executed atomically. This means that in a group of cooperating process at a given point of time, only one process must be executing its critical section. A critical section is a program segment where shared resources are accessed. Therefore, option (c) is correct.

62. Topic: Operating System

(b) Mutual exclusion claims exclusive control of the resources they require.
The four conditions for dead lock processes are as follows:

(i) No preemption: Resources already allocated to a process cannot be preempted or removed from the processes.
(ii) Mutual exclusion: The resources involved are non-sharable. At least one of the resource should be non-sharable so that only one process at a time claims exclusion control of the resource.
(iii) Circular wait: The processes from a circular list where each process in the list waits for the resource held by the next process in the list.
(iv) Hold and wait: There exists a process that is holding a resource and waiting for an additional resource being held by another process.

63. Topic: Operating System

(d) Fork () system call is used to create processes. The new process is called child process of the caller. After the new processes is created, both processes will execute the next instruction following the fork () call. Thus, fork is the creation of a new process. Option (4) is correct.

64. Topic: Operating System

(b) Context switch is a process followed by CPU to move from one task to another in order to ensure that the tasks do not conflict. In a CPU, the term context refers to the data in register and program counter at a specific moment in time. Translation look aside buffer is not necessarily saved on a context switch between processes.

65. Topic: Operating System

(a) Given:

$$\begin{aligned}
\text{Number of pages} &= 8 \\
\text{Number of words} &= 1024 = \text{Page offset} \\
\text{Frames in memory} &= 32 \\
\text{Now logical address space} &= \text{Number of pages} \times \text{Page offset} \\
&= 8 \times 1024 \\
&= 2^3 \times 2^{10} \\
&= 2^{13}
\end{aligned}$$

Therefore, number of bits in logical address = 13.

66. Topic: Operating System

(d) Round robin scheduling is pre-emptive version of First-In First-Out, i.e., FIFO. Round robin scheduling assigns equal time to each process without priority in cyclic order, therefore what process comes in first is executed first. Since the scheduling process assigns equal time to each process, the performance of round robin algorithm depends heavily on the size of the time quantum.

67. Topic: Operating System

(c) Page replacement algorithm which gives the lowest page fault rate is optimal page replacement algorithm. In this algorithm, pages are replaced which are not used for the longest duration of time in the future. This is the best possible page replacement algorithm but is impossible to implement, i.e., it is unrealizable because operating system cannot predict future requests. However, this algorithm is used as a reference for other page replacement algorithms.

68. Topic: Compiler Design

(c) Input and output statements do not produce executable code when compiled. Input statements provide means of transferring data from external media to internal storage. This process is called reading. Output statements provide means of transferring data from internal storage to external media. This process is called writing. Input and output statements pertain to gather information from an input device or sending information to an output device.

69. Topic: Programming and Data Structures

(a) Given instruction:

```
if (k < 3)      F: = k
else F: = F(k - 1) * F(k - 2) + F(k - 3)
```

$F(0) = 0$
$F(1) = 1$
$F(2) = 2$
$F(3) = F(3-1) * F(3-2) + F(3-3) = F(2) * F(1) + F(0)$

$= 2 \times 1 + 0 = 2$

So, $F(3) = 2$

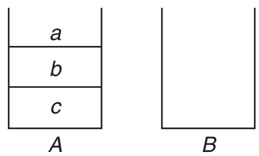
Now, $F(4) = F(4-1) * F(4-2) + F(4-3) = F(3) * F(2) + F(1)$

So, $F(4) = 2 * 2 + 1 = 4 + 1 = 5$

Therefore, the correct value of $F(4)$ is 5.

70. Topic: Programming and Data Structures

(c) Given, stack A has entries a, b, c and stack B is empty. Entry popped out from A can be printed immediately or pushed to stack B.
Entry popped out from B can be printed only.
In this arrangement, $c\ a\ b$ is not possible.



If a is popped and pushed to stack B.
Then b is popped and pushed to stack B.
Then c is popped and printed.
Then b will be popped from B and printed and then a will be popped.
These give $c\ b\ a$, therefore $c\ a\ b$ is not possible.

71. Topic: Programming and Data Structures

(c) The time required to search an element in a linked list of length n is of $O(n)$. In worst-case scenario, the element to be searched has to be compared with all the elements present in the linked list, thus the time complexity will be $O(n)$.

72. Topic: Programming and Data Structures

(a) The operation of 'deleting a node whose location is given' will be performed more efficiently by doubly linked rest than by linear linked list.
A linear linked list contains linear collection of data elements called nodes. Each node contains information of the element and the address of the next node (next pointer).
A doubly linked list contains an extra pointer called previous pointer along with the next pointer and data information which are present in linear linked list.

73. Topic: Programming and Data Structures

(c) An abstract class is similar to interface. An abstract class cannot be instantiated and it may contain a mix of methods declared with or without an implementation. An abstract class may be subclassed. A class abstract can be made by making at least one member function as pure virtual function.

74. Topic: Computer Graphics

(a) A Steiner patch is a biquadratic Bezier path. The Steiner patch is the simplest free-form surface. A Steiner patch is a triangular surface patch. The Cartesian coordinates of point on the patch are defined parametrically by quadratic polynomial function of two variables defined by functions $x(s, t)$, $y(s, t)$ and $z(s, t)$.

75. Topic: Programming and Data Structures

(d) A complete binary tree with the property that the value at each node is at least as large as the value at its children is known as heap. Heap is a special case of balanced binary tree data structure where the root node key is compared with its children and arranged accordingly:

$$\text{key (a)} \geq \text{key } (b)$$

where a has child node b.

76. Topic: Programming and Data Structures

(b) A doubly linked list contains linear collection of data which are called nodes. Each node contains information of the element, i.e., data, the address of the next node, i.e., next pointer and address of the previous node, i.e., previous pointer. The minimum number of fields with each node of doubly linked list is 2, namely, the next pointer and the previous pointer, data may or may not be present.

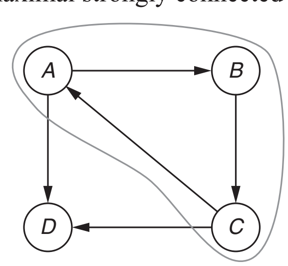
77. Topic: Algorithms

(b) The selection sort algorithm sorts an array by repeatedly finding the smallest element from the unsorted part of array and putting it at the beginning selecting the lowest element requires scanning all n elements and doing $(n - 1)$ comparisons and then swapping the element to first position.
In order to sort an array of length 5 that is in opposite order, number of comparisons required is obtained follows:

$$\frac{n(n-1)}{2} = \frac{5(5-1)}{2} = \frac{5 \times 4}{2} = 5 \times 2 = 10$$

78. Topic: Programming and Data Structures

(d) The given graph is a directed graph. A directed graph is a graph that is connected together and all edges are directed from one vertex to another. A directed graph is strongly connected if there is a path between all pairs of vertices. A strongly connected component of a directed graph is a maximal strongly connected subgraph.



Note that a, b, c are valid strong components of the graph.

79. Topic: Digital Logic

(a) Repeated execution of simple computation may cause compounding of round-off errors. Round-off error is the difference between an approximation of a number used in computation and its exact value. Round-off error can be magnified as any initial errors are carried through one or more intermediate steps.

80. Topic: Programming and Data Structures

(c) If there is a negative number in a field width specifier, then the values are displayed lift justified. Field width is most useful when printing multiple lines of output which are to be lined up in a table format. To left justify a negative number is used in field width as follows:

```
Printf ("data to be" %d, data);
```

Indian Space Research Organisation
Computer Science

2009

QUESTION PAPER

1. The subnet mask for a particular network is 255.255.31.0. which of the following pairs of IP addresses could belong to this network?

(a) 172.57.88.62 and 172.56.87.23
(b) 10.35.28.2 and 10.35.29.4
(c) 191.203.31.87 and 191.234.31.88
(d) 128.8.129.43 and 128.8.161.55

2. In networking, UTP stands for

(a) Unshielded T-connector port
(b) Unshielded twisted pair
(c) Unshielded terminating pair
(d) Universal transmission process

3. The address resolution protocol (ARP) is used for

(a) Finding the IP address from the DNS
(b) Finding the IP address of the default gateway
(c) Finding the IP address that corresponds to a MAC address
(d) Finding the MAC address that corresponds to an IP address

4. Which of the following is a MAC address?

(a) 192.166.200.50
(b) 00056A:01A5CCA7FF60
(c) 568, Airport Road
(d) 01:A5:BB:A7:FF:60

5. What is the primary purpose of a VLAN?

(a) Demonstrating the proper layout for a network
(b) Simulating a network
(c) To create a virtual private network
(d) Segmenting a network inside a switch or device

6. SHA-1 is a

(a) Encryption algorithm
(b) Decryption algorithm
(c) Key exchange algorithm
(d) Message digest function

7. Advanced Encryption Standard (AES) is based on

(a) Asymmetric key algorithm
(b) Symmetric key algorithm
(c) Public key algorithm
(d) Key exchange

8. The primary purpose of an operating system is

(a) To make the most efficient use of the computer hardware
(b) To allow people to use the computer
(c) To keep systems programmers employed
(d) To make computers easier to use

9. Which is the correct definition of a valid process transition in an operating system?

(a) Wake up: ready $\rightarrow$ running
(b) Dispatch: ready $\rightarrow$ running
(c) Block: ready $\rightarrow$ running
(d) Timer runout: $\rightarrow$ ready $\rightarrow$ blocked

10. The correct matching for the following pairs is

(A) Disk check	(1) Round Robin
(B) Batch processing	(2) Scan
(C) Time sharing	(3) LIFO
(D) Stack operation	(4) FIFO

	A	B	C	D
(a)	3	4	2	1
(b)	4	3	2	1
(c)	3	4	1	2
(d)	2	4	1	3

11. A page fault

(a) occurs when a program accesses an available page of memory
(b) is an error in a specific page
(c) is a reference to a page belonging to another program
(d) occurs when a program accesses a page not currently in memory

12. Using larger block size in a fixed block size file system leads to

(a) Better disk throughput but poorer disk space utilization

(b) Better disk throughput and better disk space utilization

(c) Poorer disk throughput but better disk space utilization

(d) Poorer disk throughput and poorer disk space utilization

13. Which of the following statements about synchronous and asynchronous I/O is NOT true?

(a) An ISR is invoked on completion of I/O in synchronous I/O but not in asynchronous I/O

(b) In both synchronous and asynchronous I/O an ISR (Interrupt Service Routine) is invoked after completion of the I/O

(c) A process making a synchronous I/O call waits until I/O is complete, but a process making an asynchronous I/O call does not wait for completion of the I/O

(d) In the case of synchronous I/O, the process waiting for the completion of I/O is woken up by the ISR that is invoked after the completion of I/O

14. Consider three CPU-intensive processes, which require 10, 20 and 30 time units and arrive at times 0, 2, and 6, respectively. How many context switches are needed if the operating system implements a shortest remaining time first scheduling algorithm? Do not count the context switches at time zero and at the end.

(a) 1 (b) 2

(c) 3 (d) 4

15. The performance of Round Robin algorithm depends heavily on

(a) Size of the process

(b) The I/O bursts of the process

(c) The CPU bursts of the process

(d) The size of the time quantum

16. Consider a system having "n" resources of same type. These resources are shared by three processes A, B, C. These have peak demands of 3, 4 and 6, respectively. For what value of "n" deadlock won't occur

(a) 15 (b) 9

(c) 10 (d) 13

17. Consider a set of five processes whose arrival time, CPU time needed, and the priority are given in the following table:

Process Priority	Arrival Time (in ms)	CPU Time Needed (in ms)	Priority
P1	0	10	5
P2	0	5	2
P3	2	3	1
P4	5	20	4
P5	10	2	3

(Smaller the number, higher the priority.)
If the CPU scheduling policy is priority scheduling without pre-emption, the average waiting time will be

(a) 12.8 ms (b) 11.8 ms

(c) 10.8 ms (d) 09.8 ms

18. The range of integers that can be represented by an n-bit 2's complement number system is

(a) -2^{n-1} to $(2^{n-1} - 1)$

(b) $-(2^{n-1} - 1)$ to $(2^{n-1} - 1)$

(c) -2^{n-1} to 2^{n-1}

(d) $-(2^{n-1} + 1)$ to $(2^{n-1} - 1)$

19. The switching expression corresponding to $f(A, B, C, D) = \Sigma(1, 4, 5, 9, 11, 12)$ is

(a) $B\bar{C}\bar{D} + \bar{A}\bar{C}D + A\bar{B}D$

(b) $AB\bar{C} + ACD + \bar{B}\bar{C}D$

(c) $AC\bar{D} + \bar{A}B\bar{C} + A\bar{C}\bar{D}$

(d) $\bar{A}BD + AC\bar{D} + BC\bar{D}$

20. Consider the boolean function of four variables $f(w, x, y, z) = \Sigma(1, 3, 4, 6, 9, 11, 12, 14)$. The function is

(a) Independent of one variable

(b) Independent of two variables

(c) Independent of three variables

(d) Dependent on all the variables

21. In which addressing mode, the effectives address of the operand is generated by adding a constant value to the content of a register?

(a) Absolute mode (b) Indirect mode

(c) Immediate mode (d) Index mode

22. A certain microprocessor requires 4.5 microseconds to respond to an interrupt. Assuming that the three interrupts I_1, I_2 and I_3 require the following execution time after the interrupt is recognized:

(i) I_1 requires 25 microseconds
(ii) I_2 requires 35 microseconds
(iii) I_3 requires 20 microseconds

I_1 has the highest priority and I_3 has the lowest. What is the possible range of time for I_3 to be executed assuming that it may or may not occur simultaneously with other interrupts?

(a) 24.5 microseconds to 39.5 microseconds
(b) 24.5 microseconds to 93.5 microseconds
(c) 4.5 microseconds to 24.5 microseconds
(d) 29.5 microseconds to 93.5 microseconds

23. The process of organizing the memory into two banks to allow 8- and 16-bit data operation is called

(a) Bank switching
(b) Indexed mapping
(c) Two-way memory interleaving
(d) Memory segmentation

24. Suppose the numbers 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are inserted in that order into an initially empty binary search tree. The binary search tree uses the usual ordering on natural numbers. What is the in order traversal sequence of the resultant tree?

(a) 7 5 1 0 3 2 4 6 8 9 (b) 0 2 4 3 1 6 5 9 8 7
(c) 0 1 2 3 4 5 6 7 8 9 (d) 9 8 6 4 2 3 0 1 5 7

25. A data structure is required for storing a set of integers such that each of the following operations can be done in (log n) time, where n is the number of elements in the set.

1. Deletion of the smallest element.
2. Insertion of an element if it is not already present in the set.

Which of the following data structures can be used for this purpose?

(a) A heap can be used but not a balanced binary search tree
(b) A balanced binary search tree can be used but not a heap
(c) Both balanced binary search tree and heap can be used
(d) Neither balanced binary search tree nor heap can be used

26. The following numbers are inserted into an empty binary search tree in the given order: 10, 1, 3, 5, 15, 12, 16. What is the height of the binary search tree (the height is the maximum distance of a leaf node from the root)?

(a) 2 (b) 3
(c) 4 (d) 6

27. Assume that the operators +, −, × are left associative and ^ is right associative. The order of precedence (from highest to lowest) is ^, ×, +, −. The postfix expression corresponding to the infix expression $a + b \times c - d \wedge e \wedge f$ is

(a) $abc \times + def \wedge\wedge -$ (b) $abc \times + de \wedge f \wedge$
(c) $ab + c \times d - e \wedge f \wedge$ (d) $- + a \times bc \wedge\wedge def$

28. The infix expression $A + (B - C) * D$ is correctly represented in prefix notation as

(a) $A + B - C * D$ (b) $+ A * - BCD$
(c) $ABC - D * +$ (d) $A + BC - D *$

29. A one-dimensional array A has indices 1…75. Each element is a string and takes up three memory words. The array is stored at location 1120 decimal. The starting address of A [49] is

(a) 1267 (b) 1164
(c) 1264 (d) 1169

30. The five items A, B, C, D, and E are pushed in a stack, one after the other starting from A. The stack is popped four items and each element is inserted in a queue. Then two elements are deleted from the queue and pushed back on the stack. Now one item is popped from the stack. The popped item is

(a) A (b) B
(c) C (d) D

31. A full binary tree with n leaves contains

(a) n nodes (b) $\log_2 n$ nodes
(c) $2n - 1$ (d) 2^n nodes

32. The expression 1 * 2 ^ 3 * 4 ^ 5 * 6 will be evaluated as

(a) 32^{30} (b) 162^{30}
(c) 49152 (d) 173458

33. The feature in object-oriented programming that allows the same operation to be carried out differently, depending on the object, is

(a) inheritance (b) polymorphism
(c) overfunctioning (d) overriding

34. The microinstructions stored in the control memory of a processor have a width of 26 bits. Each microinstruction is divided into three fields: micro-operation fields of 13 bits, a next address field (X), and a MUX select field (Y). There are 8 status bits in the inputs of the MUX. How many bits are there in the X and Y fields, and what is the size of the control memory in number of words?

(a) 10, 3, 1024 (b) 8, 5, 256
(c) 5, 8, 2048 (d) 10, 3, 512

35. A CPU has 24-bit instructions. A program starts at address 300 (in decimal). Which one of the following is a legal program counter (all values in decimal)?

(a) 400 (b) 500
(c) 600 (d) 700

36. Consider a disk pack with 16 surfaces, 128 tracks per surface and 256 sectors per track. In total, 512 bytes of data are stored in a bit serial manner in a sector. The capacity of the disk pack and the number of bits required to specify a particular sector in the disk are respectively

(a) 256 MB, 19 bits (b) 256 MB, 28 bits
(c) 512 MB, 20 bits (d) 64 GB, 28 bits

37. Consider a pipelined processor with the following four stages

IF: Instruction Fetch
ID: Instruction Decode and Operand Fetch
EX: Execute WB: Write Back
The IF, ID and WB stages take one clock cycle each to complete the operation. The ADD and SUB instructions need 1 clock cycle and the MUL instruction need three clock cycles in the EX stage. Operand forwarding is used in the pipelined processor. What is the number of clock cycles taken to complete the following sequence of instructions?

ADD $R2$, $R1$, $R0$ $R2 \leftarrow R1 + R0$
MUL $R4$, $R3$, $R2$ $R4 \leftarrow R3 * R2$
SUB $R6$, $R5$, $R4$ $R6 \leftarrow R5 - R4$

(a) 7 (b) 8
(c) 10 (d) 14

38. The use of multiple register windows with overlap causes a reduction in the number of memory accesses for

1. Function locals and parameters
2. Register saves and restores
3. Instruction fetches

(a) 1 only (b) 2 only
(c) 3 only (d) 1, 2 and 3

39. A processor that has carry, overflow and sign flag bits as part of its program status word (PSW) performs addition of the following two 2's complement numbers 01001101 and 11101001. After the execution of this addition operation, the status of the carry, overflow and sign flags, respectively, will be

(a) 1, 1, 0 (b) 1, 0, 0
(c) 0, 1, 0 (d) 1, 0, 1

40. The two numbers given below are multiplied using the Booth's algorithm.

Multiplicand: 0101 1010 1110 1110
Multiplier: 0111 0111 1011 1101
How many additions/subtractions are required for the multiplication of the above two numbers?

(a) 6 (b) 8
(c) 10 (d) 12

41. The addition of 4-bit, two's complement, binary numbers 1101 and 0100 results in

(a) 0001 and an overflow
(b) 1001 and no overflow
(c) 0001 and no overflow
(d) 1001 and an overflow

42. Which of the flowing statements about relative addressing mode is FALSE?

(a) It enables reduced instruction size
(b) It allows indexing of array element with same instruction
(c) It enables easy relocation of data
(d) It enables faster address calculation than absolute addressing

43. Substitution of values for names (whose values are constants) is done in

(a) Local optimization (b) Loop optimization
(c) Constant folding (d) Strength reduction

44. A root α of equation $f(x) = 0$ can be computed to any degree of accuracy if a 'good' initial approximation x_0 is chosen for which

(a) $f(x_0) > 0$ (b) $f(x_0)f''(x_0) > 0$
(c) $f(x)_0 f''(x_0) < 0$ (d) $f''(x_0) > 0$

45. Which of the following statement is correct?

(a) $\Delta(u_k v_k) = u_k \Delta v_k + v_k \Delta u_k$
(b) $\Delta(u_k v_k) = u_{k+1} \Delta v_k + v_{k+1} \Delta u_k$
(c) $\Delta(u_k v_k) = v_{k+1} \Delta u_k + u_k \Delta v_k$
(d) $\Delta(u_k v_k) = u_{k+1} \Delta v_k + v_k \Delta u_k$

46. The shift operator E is defined as $E[f(x_i)] = f(x_i + h)$ and $E^{-1}[f(x_i)] = f(x_i - h)$. Then Δ (forward difference) in terms of E is

(a) E^{-1} (b) E
(c) $1-E^{-1}$ (d) $1-E$

47. The formula

$$\int_{x0}^{xn} x \simeq h/2\left(y_0 + 2y_1 + \cdots + 2y_{n-1} + y_n\right)$$
$$-h/12\left(\nabla y_n - \Delta y_0\right) - h/24\left(\nabla^2 y_n + \Delta^2 y_0\right)$$
$$-19h/720\left(\nabla^3 y_n - \Delta^3 y_0\right)\ldots \text{ is called}$$

(a) Simpson rule (b) Trapezoidal rule
(c) Romberg's rule (d) Gregory's formula

48. The cubic polynomial $y(x)$ which takes the following values:
$y(0) = 1$, $y(1) = 0$, $y(2) = 1$ and $y(3) = 10$ is

(a) $x^3 + 2x^2 + 1$ (b) $x^3 + 3x^2 - 1$
(c) $x^3 + 1$ (d) $x^3 - 2x^2 + 1$

49. $x = a\cos(t)$, $y = b\sin(t)$ is the parametric form of

(a) Ellipse (b) Hyperbola
(c) Circle (d) Parabola

50. The value of x at which y is minimum for $y = x^2 - 3x + 1$ is

(a) –3/2 (b) 3/2
(c) 0 (d) –5/4

51. The formula

$$p_k = y_0 + k\nabla y_0 + \frac{k(k+1)}{2}\nabla^2 y_0 + \cdots + \frac{k\ldots(k+n-1)}{n!}\nabla^n y_0 \text{ is}$$

(a) Newton's backward formula
(b) Gauss forward formula
(c) Gauss backward formula
(d) Stirling's formula

52. If G is a graph with e edges and n vertices, the sum of the degrees of all vertices in G is

(a) e (b) $e/2$
(c) e^2 (d) $2e$

53. Let G be an arbitrary graph with n nodes and k components. If a vertex is removed from G, the number of components in the resultant graph must necessarily lie between

(a) k and n (b) $k-1$ and $k+1$
(c) $k-1$ and $n-1$ (d) $k+1$ and $n-k$

54. A graph in which all nodes are of equal degree is known as

(a) Multigraph (b) Non-regular graph
(c) Regular graph (d) Complete graph

55. If in a graph G there is one and only one path between every pair of vertices, then G is a

(a) Path (b) Walk
(c) Tree (d) Circuit

56. A simple graph (a graph without parallel edge or loops) with n vertices and k components can have at most

(a) n edges
(b) $(n-k)$ edges
(c) $(n-k)(n-k+1)$ edges
(d) $(n-k)(n-k+1)/2$ edges

57. Consider the polynomial $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$, where $a_i \neq 0, \forall i$. The minimum number of multiplications needed to evaluate ρ on an input x is

(a) 3 (b) 4
(c) 6 (d) 9

58. Consider the following code written in a pass-by-reference language like FORTRAN.

Subroutine swap (ix, iy)

```
it = ix
L1:     ix = iy
L2:     iy = it
end
ia = 3
ib = 8
call swap (ia, ib +5)
print*, ia, ib
end
```

S_1: The compiler will generate code to allocate a temporary nameless cell, initialize it to 13, and pass the address of the cell to swap
S_2: On execution the code will generate a runtime error on line $L1$
S_3: On execution the code will generate a runtime error on line $L2$
S_4: The program will print 13 and 8
S_5: The program will print 13 and −2
Exactly the following set of statement(s) is correct:

(a) S_1 and S_2 (b) S_1 and S_4
(c) S_3 (d) S_1 and S_5

59. A square matrix A is called orthogonal if $A'A =$

(a) I (b) A
(c) $-A$ (d) $-I$

60. If two adjacent rows of a determinant are interchanged, the value of the determinant

(a) becomes zero
(b) remains unaltered
(c) becomes infinite
(d) becomes negative of its original value

61. If $\begin{pmatrix} 3 & 3 \\ x & 5 \end{pmatrix} = 3$, then the value of x is

(a) 2 (b) 3
(c) 4 (d) 5

62. If A, B, C are any three matrices, then $A' + B' + C'$ is equal to

(a) a null matrix (b) $A + B + C$
(c) $(A + B + C)'$ (d) $-(A + B + C)$

63. $\begin{vmatrix} 265 & 240 & 219 \\ 240 & 225 & 198 \\ 219 & 198 & 181 \end{vmatrix} =$

(a) 779 (b) 679
(c) 0 (d) 256

64. Let $f(x)$ be the continuous probability density function of a random variable x. The probability that $a < x \le b$ is

(a) $f(b - a)$ (b) $f(b) - f(a)$

(c) $\int_a^b f(x)dx$ (d) $\int_a^b xf(x)\,dx$

65. If the mean of a normal frequency distribution of 1000 items is 25 and its standard deviation is 2.5, then its maximum ordinate is

(a) $\frac{1000}{\sqrt{2\pi}}e^{-25}$ (b) $\frac{1000}{\sqrt{2\pi}}$

(c) $\frac{1000}{\sqrt{2\pi}}e^{-2.5}$ (d) $\frac{400}{\sqrt{2\pi}}$

66. If the pdf of a Poisson distribution is given by $f(x) = \frac{e^{-2}2^x}{x!}$, then its mean is

(a) 2^x (b) 2

(c) –2 (d) 1

67. Activities which ensure that the software that has been built is traceable to customer requirement is covered as part of

(a) Verification (b) Validation

(c) Maintenance (d) Modeling

68. A testing method, which is normally used as the acceptance test for a software system, is

(a) Regression Testing

(b) Integration Testing

(c) Unit Testing

(d) System Testing

69. The 'command' used to change contents of one database using the contents of another database by linking them on a common key field is called

(a) Replace (b) Join

(c) Change (d) Update

70. A locked database file can be

(a) Accessed by only one user

(b) Modified by users with the correct password

(c) Used to hide sensitive information

(d) Updated by more than one user

71. Which of the following contains complete record of all activity that affected the contents of a database during a certain period of time?

(a) Transaction log

(b) Query language

(c) Report writer

(d) Data manipulation language

72. Purpose of 'Foreign Key' in a table is to ensure

(a) Null Integrity

(b) Referential Integrity

(c) Domain Integrity

(d) Null & Domain Integrity

73. Which of the following scenarios may lead to an irrecoverable error in a database system?

(a) A transaction writes a data item after it is read by an uncommitted transaction

(b) A transaction reads a data item after it is read by an uncommitted transaction

(c) A transaction reads a data item after it is written by a committed transaction

(d) A transaction reads a data item after it is written by an uncommitted transaction

74. Use of IPSEC in tunnel mode results in

(a) IP packet with same header

(b) IP packet with new header

(c) IP packet without header

(d) No changes in IP packet

75. Special software to create a job queue is called a

(a) Driver (b) Spooler

(c) Interpreter (d) Linkage editor

76. Process is

(a) a program in high level language kept on disk

(b) contents of main memory

(c) a program in execution

(d) a job in secondary memory

77. When a process is rolled back as a result of deadlock, the difficulty which arises is

(a) Starvation (b) System throughput

(c) Low device utilization (d) Cycle stealing

78. On receiving an interrupt from an I/O device, the CPU

(a) Halts for a predetermined time

(b) Branches off to the interrupt service routine after completion of the current instruction

(c) Branches off to the interrupt service routine immediately

(d) Hands over control of address bus and data bus to the interrupting device

79. Compared to CISC processors, RISC processors contain

(a) more register and smaller instruction set

(b) larger instruction set and less registers

(c) less registers and smaller instruction set

(d) more transistor elements

80. Which of the following is/are true of the auto-increment addressing mode?

1. It is useful in creating self-relocating code
2. If it is included in an instruction set architecture, then an additional ALU is required for effective address calculation
3. The amount of increment depends on the size of the data item accessed

(a) 1 only (b) 2 only
(c) 3 only (d) 2 and 3 only

ANSWER KEY

1. (d)	**9.** (b)	**17.** (c)	**25.** (b)	**33.** (b)	**41.** (c)	**49.** (a)	**57.** (a)	**65.** (d)	**73.** (a)
2. (b)	**10.** (d)	**18.** (a)	**26.** (b)	**34.** (a)	**42.** (d)	**50.** (b)	**58.** (b)	**66.** (b)	**74.** (b)
3. (d)	**11.** (d)	**19.** (a)	**27.** (a)	**35.** (c)	**43.** (c)	**51.** (a)	**59.** (a)	**67.** (b)	**75.** (b)
4. (d)	**12.** (a)	**20.** (b)	**28.** (b)	**36.** (a)	**44.** (*)	**52.** (d)	**60.** (d)	**68.** (d)	**76.** (c)
5. (d)	**13.** (c)	**21.** (d)	**29.** (c)	**37.** (b)	**45.** (b)	**53.** (c)	**61.** (c)	**69.** (d)	**77.** (a)
6. (d)	**14.** (b)	**22.** (b)	**30.** (d)	**38.** (a)	**46.** (a)	**54.** (c)	**62.** (c)	**70.** (a)	**78.** (b)
7. (b)	**15.** (d)	**23.** (c)	**31.** (c)	**39.** (b)	**47.** (b)	**55.** (c)	**63.** (c)	**71.** (a)	**79.** (a)
8. (a)	**16.** (*)	**24.** (c)	**32.** (c)	**40.** (b)	**48.** (d)	**56.** (d)	**64.** (c)	**72.** (b)	**80.** (c)

*** None of the options is correct.**

ANSWERS WITH EXPLANATION

1. Topic: Computer Networks

(d) Given, subnet mask = 255.255.31.0. If on applying AND operation to the subnet mask and the pair of IP addresses, the resultant is same for both IP addresses, then that pair belongs to this network.
Since the second segment of IP address in pair (a) and (c) is different, therefore they cannot belong to this network.

Consider option (b):
255.255.31.0 && 10.35.28.2 = 10.35.28.0
255.255.31.0 && 10.35.29.4 = 10.35.29.0
Consider option (d):
255.255.31.0 && 128.8.129.43 = 128.8.1.0
255.255.31.0 && 128.8.161.55 = 128.8.1.0

In option (d), the resultant number matches; therefore this pair of IP addresses could belong to this network.

2. Topic: Computer Networks

(b) In networking, UTP stands for unshielded twisted pair. UTP is a type of copper cabling used in LANs. In this, conductors are twisted around each other to cancel out EM interference from external sources and no other shielding like meshes or aluminum foil are used. These are used in networking such as Ethernet for short-to-medium distances, because they are cheaper than coaxial cables and optical fiber.

3. Topic: Computer Networks

(d) ARP or address resolution protocol is used to get MAC address of a node by providing IP address. It is a protocol that maps on IP address to a physical machine address that is recognized in the local network. When an incoming packet destined for a host machine on a particular LAN arrives at gateway, then the ARP is asked to find physical machine address or MAC address that matches the IP address so that the packet can be converted to right packet length and format and sent to the machine.

4. Topic: Computer Networks

(d) MAC address or media access control is a unique identifier that is assigned to network interfaces for communications on the physical network. According to IEEE standards, MAC address is a group of six groups of two hexadecimal digits separated by colons.
Among the given options, only option (d) follows the format of MAC address.

5. Topic: Computer Networks

(d) VLAN or Virtual LAN comprises of a subset of ports on a single switch or subset of ports on multiple switches. It allows network administrator to partition their networks in order to match the functional and security requirements of their system. Therefore, the primary purpose of a VLAN is segmenting a network inside a switch or device.

6. **Topic: Computer Networks**

(d) SHA-1 or secure hash algorithm 1 is a cryptographic computer security algorithm. It takes an input data which requires encryption and produces a 160-bit or 20-byte hash value or message digest. SHA-1 is used in applications where high data integrity is required. It is also used to index hash functions and identity data corruption and checksum errors.

7. **Topic: Computer Networks**

(b) AES or advanced encryption standard is a symmetric encryption algorithm use to protect classified information. It is implemented in software and hardware throughout the world to encrypt sensitive data. It is stronger and faster than triple DES. Provides full specification and design details. It is implemented in C and Java. Therefore, option (b) is correct. AES is based on symmetric key algorithm.

8. **Topic: Operating System**

(a) Operating system is the most important program that runs of the computer. Operating system manages all other programs in a computer called the application programs. In multitasking, operating system determines which applications should run in what order and time that should be allocated for each application. It manages sharing of internal memory among applications. It handles input/output processes from the attached hardware devices. Thus, the primary purpose of an operating system is to make the most efficient use of the computer hardware.

9. **Topic: Operating System**

(b) In operating system, moving from one process state to another process state is called process transition. Process transition occurs if a process is waiting for input/output operation or waiting for data from disk or if the process execution is completed. The correct definition of a valid process transition in an operating system is Dispatch: ready → running.

10. **Topic: Operating System**

(d) Scan algorithm is used to read from magnetic disks. Batch processing is a processing method where items are processes in some order they come in. This is also called first in first out or FIFO.

Round Robin approach is used to divide available CPU time among multiple waiting processes. This reduces waiting time. Stack operations are last in first out or LIFO. This is because, in a stock only the last entered item is available. Consider this analogues to a stack of plates. Hence, correct matching is $A - 2$, $B - 4$, $C - 1$, $D - 3$.

11. **Topic: Operating System**

(d) A page fault is a type of exception or on interrupt. It is raised by computer hardware when a running program accesses a page that is not currently in memory. In order to handle page fault, the operating system generally trees to make the required page accessible at the location in physical memory or terminate the program in case of on illegal memory access.

12. **Topic: Operating System**

(a) Using a larger block size in a fixed block size file system leads to better disk throughput, but poorer disk space utilization. By using a larger block size, the seek time gets less since. There are fewer blocks to seek. Also, the disk performance is improved. But using larger block size causes wastage of disk space, leading to poorer disk space utilization.

13. **Topic: Operating System**

(c) Synchronous input/output is when the issuer waits for the input/output operation to complete. Synchronous input/output is also called blocking input/output. On the other hand, asynchronous input/output does not block the issuer. Rather a callback is issued on completion of the input/output operation notifying the issuer. Thus statement (c) is correct that A process making synchronous I/O call waits until I/O is complete, but a process making an asynchronous I/O call does not wait for completion of the I/O.

14. **Topic: Operating System**

(b) In shortest remaining time first scheduling algorithm, the process with least time remaining to complete is executed. Thus, for currently executing processes, the process will run until they are completed or a new process which requires less amount of time is added. Given, 3 processes arrive at times 0, 2 and 6 respectively and these require 10, 20 and 30-time units.

At $t = 0$, 1st process arrive and is the only process available and time units required = 10.

At $t = 2$, 2nd process arrives; but since 1st process has shortest remaining time; therefore process 1 continues.

At $t = 6$, 3rd process arrives; but since process 1 has shortest remaining time among all, therefore process 1 continues.

At $t = 10$, process 1 completes and process 2 is scheduled because it has shortest remaining time.

At $t = 30$, process 2 completes and process 3 is scheduled.

Therefore, two context switches are needed, process 1 to process 2 and process 2 to process 3.

15. Topic: Operating System

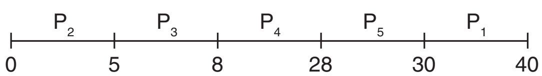
(d) Round Robin scheduling is a pre-emptive version of first in first out, i.e., FIFO. Round Robin scheduling assigns equal time to each process without priority in cyclic order, therefore what process come in first are executed first. Since the scheduling process assigns equal time to each process, therefore the performance of round Robin algorithm depends heavily on the size of the time quantum.

16. Topic: Operating System

(None) A deadlock occurs when a group of processes, with shared resources, have the resources allocated in such a way that none of the deadlocked processes have enough resources to complete execution and the system has no more resources. In this example, if A holds 2 resources, B holds 3 resources and C holds 5 resources, the system will be deadlocked. That is, $n = 2 + 3 + 5 = 10$. But if $n = 11$, and extra resource can be used by either process, deadlock will never occur. None of the answer is correct.

17. Topic: Operating System

(c) Since the CPU scheduling policy is priority scheduling without pre-emption, therefore, the scheduling order will be



Therefore, waiting time of process $P_1 = W_1 = 30 - 0 = 30$

Waiting time of process $P_2 = W_2 = 0$

Waiting time of process $P_3 = W_3 = 5 - 2 = 3$

Waiting time of process $P_4 = W_4 = 8 - 5 = 3$

Waiting time of process $P_5 = W_5 = 28 - 10 = 18$

$$\text{Average waiting time} = \frac{W_1 + W_2 + W_3 + W_4 + W_5}{\text{Total number of processes}}$$

$$\text{Average waiting time} = \frac{30 + 0 + 3 + 3 + 18}{5} = \frac{54}{5} = 10.8$$

Therefore, average waiting time = 10.8 ms.

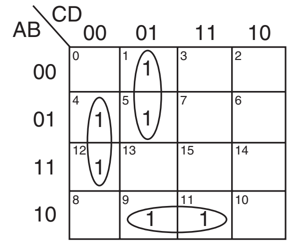
18. Topic: Digital Logic

(a) 2's complement is a binary operation on binary numbers. n-bit 2's complement is defined as complement of n-bit number with respect to 2^n. For a given set of n-bit values, the lower-half of the set can be assigned integers from 0 to $(2^{n-1} - 1)$ and upper-half of the site can be assigned integers from -2^{n-1} to -1. Thus, the range of integers that can be represented by n-bit 2's complement number system is -2^{n-1} to $(2^{n-1} - 1)$.

19. Topic: Digital Logic

(a) Using Karnaugh map to simplify Boolean algebra expression. It deals with technique of inserting values of output variable in cells within a grid according to a definite pattern. The number of cells is determined by number of input variables n as 2^n.

Given, $f(A, B, C, D) = \Sigma(1, 4, 5, 9, 11, 12)$

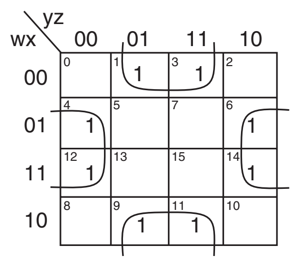


Therefore, $f(A, B, C, D) = BC'D' + A'C'D + AB'D$.

20. Topic: Digital Logic

(b) Using Karnaugh map to simplify Boolean algebra expression. It is the technique of inserting the values of the output variable in cells within a grid according to a definite pattern. The number of cells is determined by number of input variables n as 2^n.

Given, $f(w, x, y, z) = \Sigma(1, 3, 4, 6, 9, 11, 12, 14)$



So, $f(w, x, y, z) = x'z + xz'$.

Therefore, the function is independent of two variables.

21. Topic: Computer Organization and Architecture

(d) In Index addressing mode, the effective address of the operand is generated by adding a constant value to the content of a register. The address can be in a register used specially for this purpose or any of the general-purpose register called index register. The effective address is computed as:

(Index * Scalar Factor) + Signed Displacement

Indexed addressing mode is useful to access elements of an array particularly if the element size is 2, 4 or 8 bytes.

22. Topic: Computer Organization and Architecture

(b) Minimum time required is processing one instruction at a time.

Minimum Time = Response + Execution Time
= 4.5 + 20 = 24.5 μs

Maximum time required is processing all instructions

$$\text{Maximum time} = I_1 + I_2 + I_3 = (25 + 4.5) + (35 + 4.5) + (20 + 4.5) = 93.5\ \mu s$$

Therefore, range of time is 24.5 microseconds to 93.5 microseconds.

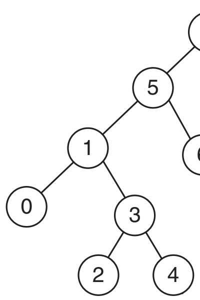
23. Topic: Computer Organization and Architecture

(c) The process of organizing the memory into two banks to allow 8- and 16-bit data operation is called two-way memory interleaving. In memory interleaving, the main memory is divided into two or more sections and the CPU can access alternate sections immediately, without waiting for memory to catch up. It is used to compensate for the slow speed of dynamic RAM.

24. Topic: Programming and Data Structures

(c) A binary search tree is a node-based binary tree data structure. In this, left subtree of a node contains nodes with keys less than node's key and the right subtree of a node contains nodes with key greater than node's key. In in order traversal, nodes in left subtree are visited first, then the root node and nodes in right subtree in the last.

Given that the numbers 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are inserted in an initially empty binary tree. Thus, the resultant tree structure will be



Then the in-order traversal sequence of the resultant tree will be 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.

25. Topic: Programming and Data Structures

(b) A balanced binary search tree is a tree that keeps its height small for a sequence of insertions and deletions. A balanced search tree has height log n.

Thus, deletion of smallest element will take time –O(log n) insertion operation of element will take time –O(log n).

Heap is a special case of balanced binary tree where the root node is compared with its children and arranged accordingly:

$$\text{key } (a) \geq \text{key } (b)$$

where, a is parent node and b is child node,

height of heap is log n,

deletion of smallest element = O(log n), and

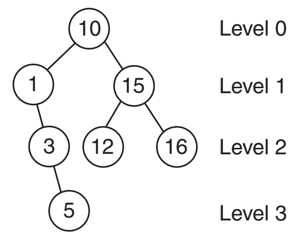
insertion operation of element = O(n).

Thus option (b) is correct. A balanced binary search tree can be used but not a heap.

26. Topic: Programming and Data Structures

(b) Given, numbers 10, 1, 3, 5, 15, 12, 16 are inserted into an empty binary search tree.

A binary search tree is a node-based binary tree data structure. In this, left subtree of a node has all nodes with key less than the root node's key and the right subtree of a node contains nodes with key greater than the root node's key. Thus, the resultant tree structure will be



Thus, the height of the binary search tree is 3.

27. Topic: Programming and Data Structures

(a) Infix Expression: Operators are written in-between their operands. For example, $a + b$.

Post-Fix Expression: Operators are written after their operands for example $ab+$. The order of evaluation is always from left to right and operators act on values immediately to the left of them.

Given: Infix expression $a + b \times c - d \wedge e \wedge f$

The order of precedence is ^, ×, +, −. Therefore, using brackets according to precedence, we have the following expression:

$$((a + (b \times c)) - (a \wedge (e \wedge f)))$$

Thus, postfix expression is $((a + (bc\times)) - (d \wedge (ef \wedge))$

$$= ((abc\times+) - (def \wedge\wedge)) = abc\times+def \wedge \wedge -$$

Hence expression (a) is postfix expression corresponding to the infix expression given above.

28. Topic: Programming and Data Structures

(b) In infix expression, operators are written in between their operands like $a + b$. In prefix expression, the operators are written before their operands for example $+ab$. The order of elevation is always from left to right and operators act on values immediately to the right of them given infix expression is

$$A + (B - C) * D$$

The prefix expression is evaluated as follows:

$$A + (B - C) * D = A + ((-BC) * D) = A + (* - BCD)$$
$$= +A * - BCD$$

Hence the prefix expression of the given infix expression is $+A * -BCD$.

29. Topic: Programming and Data Structures

(c) If index begins at zero,

Element Address = Base Address + Index × Element Size for the given problem

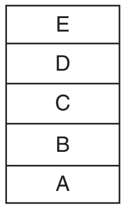
Element Address = Base Address + (Index – 1) × Element Size

$= 1120 + (49 - 1) \times 3 = 1264$

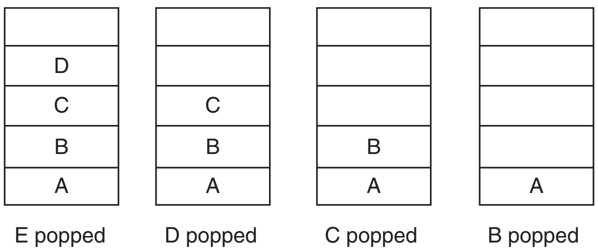
Therefore, starting address of A [49] is 1264.

30. Topic: Programming and Data Structures

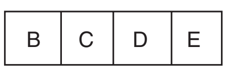
(d) ABCDE are pushed in stack, so we have



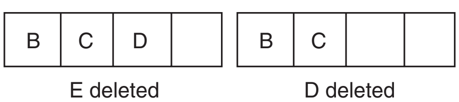
Next, four items are popped from the stack and inserted in a queue and we have in stack



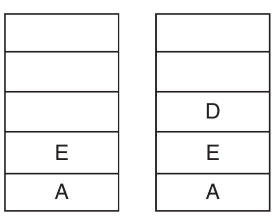
and we have in queue:



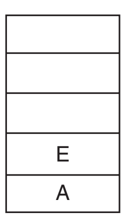
Then two elements are deleted from queue, so in queue we have:



Next the deleted elements are pushed back in stack, so in stack we have:



then one element is popped:



So, the popped element is 'D'.

31. Topic: Programming and Data Structures

(c) A full binary tree is a special tree data structure. In this structure, every node can have a maximum of two children. One child is called left child and another child is called right child. In a full binary tree, every node can have either 0 child or 2 children, but not more than that. A full binary tree with n leaves contain $2n - 1$ nodes. Thus, option (c) is correct.

32. Topic: Programming and Data Structures

(c) Given, expression is 1 * 2 ^ 3 * 4 ^ 5 * 6

The order of precedence is ^ *

Using brackets according to precedence we have the following expression:

$$1 * (2 \wedge 3) * (4 \wedge 5) * 6$$
$$\Rightarrow 1 * 8 * 1024 * 6 \Rightarrow 49152$$

Thus, the expression 1 * 2 ^ 3 * 4 ^ 5 * 6 will be evaluated as 49152.

33. Topic: Programming and Data Structures

(b) The feature in object-oriented programming that allows the same operation to be carried out differently, depending on the object is polymorphism. Polymorphism is the ability of an object to take many forms. The most common use of polymorphism is when a parent class reference is used to refer to a child class object.

34. Topic: Computer Organization and Architecture

(a) Given:

Microinstruction has a width of 26 bits

Microinstruction is divided into 3 fields:

Micro-operation field of 13 bits, a next address field (X) and a Mux select field (Y).

Number of status bits in input of Mux = 8.

Since,

$$\text{Mux has 8 inputs} \Rightarrow 2^3 \text{ inputs}$$

So,

$$Y = 3 \text{ bits}$$

Number of bits in next address field = 26 – 13 – 3 = 10.

10-bit address field yields $2^{10} = 1024$ memory size

Therefore, X, Y = 10, 3 and size = 1024.

35. Topic: Computer Organization and Architecture

(c) Given, CPU has 24-bit instructions program starts at address 300.

Therefore, size of instruction $= \dfrac{24}{8} = 3$ bytes

Thus, program counter can shift 3 bytes at a time to jump to next instruction. Thus, the program counter must be a multiple of 3. Thus, 600 must be a legal program counter.

36. Topic: Operating System

(a) Given:

Number of surfaces = 16
Number of tracks per surface = 128
Number of sectors per track = 256
Data byte stored = 512 bytes
Capacity of Disk Pack = Number of Surfaces × Number of Tracks per Surface × Number of Sectors per Track × Data Byte Stored
Capacity of Disk Pack = 16 × 128 × 256 × 512 = 256 MB

Now,

Total Number of Sectors = Number of Surfaces × Number of Tracks per Surface × Number of Sectors per Track
$= 16 \times 128 \times 256 = 2^{19}$
Number of bits required to specify a particular sector in the disk = 19.

Thus option (a) is correct.

37. Topic: Computer Organization and Architecture

(b)

	1	2	3	4	5	6	7	8
ADD	IF	ID	EK	WB				
MUL		IF	ID	EK	EK	EK	WB	
SUB		IF	ID				EK	WB

Therefore, total of eight clock cycles are taken to complete the given sequence of instructions. So, the correct option is (b).

38. Topic: Computer Organization and Architecture

(a) The use of multiple register windows with overlap causes a reduction in the number of memory access for the function locals and parameters because by using multiple register, the need of accessing variables again and again from memory is eliminated. The accessing variables are stored in the registers. However, the memory access for register saves and stores and instruction fetches are not affected by using multiple register windows. Thus, only option (a) is correct.

39. Topic: Digital Logic

(b) Addition of the given two numbers gives

$$\begin{array}{r} 0\,1\,0\,0\,1\,1\,0\,1 \\ \underline{1\,1\,1\,0\,1\,0\,0\,1} \\ \underline{1}\,1\,0\,1\,1\,0\,1\,1\,0 \end{array}$$

Since, there is an extra bit out of MSB, therefore carry flag = 1.
If two binary numbers added have same sign and the resultant sum has different sign, then overflow is possible. Therefore, overflow flag = 0.
Most significant bit determines the sign of the binary number.
Therefore, sign bit = 0.

Thus, the status of carry, overflow and sign flags are 1, 0, 0 respectively.

40. Topic: Digital Logic

(b) Booth's algorithm is a multiplication algorithm that multiplies two signed binary numbers in 2's compliment notation. The algorithm is as follows:

1. Initialize the partial product to 0.
2. Add a dummy 0 bit to LSB of multiplier.
3. Starting from LSB make pairs of bits and check:
   - if two bits are 00, do nothing
   - if two bits are 01, add multiplicand to partial product
   - if two bits are 10, subtract multiplicand to partial product.
   - if two bits are 11, do nothing
4. Multiply the multiplicand by 2 at each pair checking.
5. Partial product is the full product.

Here, given multiplicand: 0101 1010 1110 1110
Multiplier: 0111 0111 1011 1101
Adding dummy 0 bit to LSB of multiplier, we have

0111 0111 1011 1101 $\underline{0}$

Making pairs of bits from LSB, we have
Total (01) pairs = 5
Total (10) pairs = 3
Thus, total of eight additions and subtractions are required for the multiplication of above two numbers.

41. Topic: Digital Logic

(c) Sum of two 2's complement binary numbers overflows, if sum of two positive numbers results in a negative number or if sum of two negative numbers results in a positive number. The sum of two 4-bit numbers given here will be

$$\begin{array}{r} 1\,1\,0\,1 \\ \underline{0\,1\,0\,0} \\ \underline{1}\,0\,0\,0\,1 \end{array}$$

Thus, addition will be 0001 with a carry bit and there is no overflow.

42. Topic: Computer Organization and Architecture

(d) In relative addressing mode, the address is specified by indicating its distance from another address called base address. It enables reduced instruction size. It also allows indexing of array element with same instruction. It also enables easy relocation of data by just changing the base address. However, relative addressing mode does not enable faster address calculation than absolute addressing as absolute address must be calculated from relative address. By using specialized hardware unit, relative address can function as good as absolute addressing but not faster than that.

43. Topic: Compiler Design

(c) Substitution of values for names is done in constant folding. Constant folding is the process of simplifying constant expression at compilation time. In some compiler, constant folding is done early so that statements can accept simple arithmetic expression, for example, array initializers. The terms in constant expressions are typically simple literals, such as the integer literals.

44. Topic: Engineering Mathematics

(None) Given, a root α of equation $f(x) = 0$ can be computed to any degree of accuracy if a 'good' initial approximation x_0 is chosen.
Since there are many approximations and it is not specified which approximation to use, therefore any of the options cannot be chosen.

45. Topic: Engineering Mathematics

(b) Among the given statements

$\Delta(u_k v_k) = u_k\, \Delta v_k + v_k\, \Delta u_k$ is incorrect
$\Delta(u_k v_k) = v_{k+1}\, \Delta u_k + u_k\, \Delta v_k$ is incorrect
$\Delta(u_k v_k) = u_{k+1}\, \Delta v_k + v_k\, \Delta u_k$ is incorrect
$\Delta(u_k v_k) = u_{k+1}\, \Delta v_k + v_{k+1}\, \Delta u_k$ is correct

Hence option (b) is correct.

46. Topic: Engineering Mathematics

(a) Shift operator E is defined as

$$E[f(x_i)] = f(x_i + h) \qquad (1)$$

$$E^{-1}[f(x_i)] = f(x_i - h) \qquad (2)$$

Then Δ (forward difference) is given as

$$\Delta f(x) = f(x+h) - f(x) \qquad (3)$$

Using Eq. (1) $E[f(x)] = f(x+h)$ adding and subtracting $f(x)$, we get

$$E[f(x)] = f(x+h) - f(x) + f(x) = \Delta f(x) + f(x)$$
$$= (\Delta + 1) f(x)$$
$$\Rightarrow E = \Delta + 1 \Rightarrow \Delta = E - 1$$

47. Topic: Engineering Mathematics

(b) The formula

$$\int_{x_0}^{x_a} y(n)dx \simeq \frac{h}{2}(y_0 + 2y_1 + \cdots + 2y_{n-1} + y_n)$$
$$-\frac{h}{12}(\nabla y_n - \Delta y_0) - \frac{h}{2y}(\nabla^2 y_n + \Delta^2 y_0)$$
$$-\frac{19h}{720}(\nabla^3 y_n - \Delta^3 y_0)$$

is called Trapezoidal rule. It is an approximate method for calculating definite integrals using linear approximations. It approximate area under the curve by adding area of all the trapezoids.

48. Topic: Engineering Mathematics

(d) Consider option (1):

$$y(x) = x^3 + 2x^2 + 1$$
$$y(0) = 0^3 + 2 \times 0^2 + 1 = 1$$
$$\Rightarrow y(0) = 1$$
$$y(1) = 1^3 + 2(1)^2 + 1 = 4$$
$$\Rightarrow y(1) = 4$$

Therefore, option (a) is incorrect.
Consider option (b):

$$y(x) = x^3 + 3x^2 - 1$$
$$y(0) = 0^3 + 3 \times 0^2 - 1 = -1$$
$$\Rightarrow y(0) = -1$$
$$y(1) = 1^3 + 3 \times 1^2 - 1 = 3$$
$$\Rightarrow y(0) = 3$$

Therefore, option (b) is incorrect.
Consider option (c):

$$y(x) = x^3 + 1$$
$$y(0) = 0^3 + 1 = 1$$
$$\Rightarrow y(0) = 1$$
$$y(1) = 1^3 + 1 = 2$$
$$\Rightarrow y(1) = 2$$

Therefore, option (c) is incorrect.
Consider option (d):

$$y(x) = x^3 - 2x^2 + 1$$
$$y(0) = 0^3 - 2 \times 0^2 + 1 = 1$$
$$\Rightarrow y(0) = 1$$
$$y(1) = 1^3 - 2 \times 1^2 + 1 = 0$$
$$\Rightarrow y(1) = 0$$
$$y(2) = 2^3 - 2 \times 2^2 + 1$$
$$\Rightarrow y(2) = 1$$
$$y(3) = 3^3 - 2 \times 3^2 + 1$$
$$\Rightarrow y(3) = 10$$

Thus, option (d) is correct.

49. Topic: Engineering Mathematics

(a) Given:

$$x = a \cos(t) \qquad (1)$$
$$y = b \sin(t) \qquad (2)$$

Consider Eq. (1). Squaring both sides, we get

$$x^2 = a^2 \cos^2 t \Rightarrow \frac{x^2}{a^2} = \cos^2 t \qquad (3)$$

Consider Eq. (2). Squaring both sides, we get

$$y^2 = b^2 \sin^2 t \Rightarrow \frac{y^2}{b^2} = \sin^2 t \qquad (4)$$

Adding Eq. (3) and Eq. (4), we get:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \sin^2 t + \cos^2 t$$

Now, we know $\sin^2 t + \cos^2 t = 1$.
Therefore,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

This is equation of ellipse.
Thus, $x = a \cos t$, $y = b \sin t$ is the parametric form of ellipse.

50. Topic: Engineering Mathematics

(b) Given:

$$y = x^2 - 3x + 1 \qquad (1)$$

Differentiating it w.r.t. 'x', we get

$$\frac{dy}{dx} = 2x - 3 \qquad (2)$$

Now, putting $\frac{dy}{dx} = 0$, we get

$$2x - 3 = 0$$
$$\Rightarrow x = 3/2 \qquad (3)$$

Differentiate Eq. (2) again w.r.t. 'x', we get
$\frac{d^2y}{dx^2} = 2$ since $\frac{d^2y}{dx^2} > 0$; therefore the function y has a minimum at $x = 3/2$.

51. Topic: Engineering Mathematics

(a) The following formula is Newton's backward formula:

$$P_k = y_0 + k\nabla y_0 + \frac{k(k+1)}{2}\nabla^2 y_0 + \cdots + \frac{k\cdots(k+n-1)}{n!}\nabla^n y_0$$

It is another way of approximating a function with n^{th} degree polynomial passing through $(n + 1)$ equally spaced points. It is used for interpolating the value of a function $yf(x)$ near the end values and to extrapolate value of the function a short distance forward from y_n.

52. Topic: Engineering Mathematics

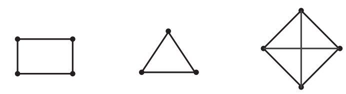
(d) Given, graph G has e edges and n vertices; then, sum of the degrees of all the vertices = $2e$. This theorem is called handshaking theorem. Since, in a graph, each edge contributes twice to sum of degrees. Therefore, the sum of degrees of all vertices = $2e$.

53. Topic: Engineering Mathematics

(c) Given, graph G has n nodes and k components.
If a vertex is removed, then number of components decreases by one, that is, components = $k - 1$.
If we consider a vertex connected to all other vertices in a component and all other vertices, this component being isolated. If a vertex is removed, the vertex we have considered makes $(n - 1)$ vertices isolated, thus the lower bound will be $n - 1$.
Therefore, the number of components in resultant graph must lie between $k - 1$ and $n - 1$.

54. Topic: Engineering Mathematics

(c) A graph in which all nodes are of equal degree is known as a regular graph. A regular graph is a graph where each vertex has the same number of neighbors. A regular graph satisfies condition that indegree and outdegree of each vertex are equal to each other. Some regular graphs are as follows:



55. Topic: Engineering Mathematics

(c) If in a graph G there is one and only one path between every pair of vertices, then G is a tree. Trees belong to the simplest class of graphs. Trees represent hierarchical structure in a graphical form and contain not even a single cycle. In other words, a connected graph with no cycles is called a tree. The edges of a tree are called branches and elements of the tree are called nodes.

56. Topic: Engineering Mathematics

(d) A simple graph with n vertices and k components will have the most edges if every vertex is connected to other vertex within the component. For 1 component, $n(n - 1)$ edges will be there, but each edge is counted twice.

So, for 1 component, edges = $\frac{n(n-1)}{2}$

For k components, number of edges = $\frac{(n-k)(n-k+1)}{2}$

Hence option (d) is correct.

57. Topic: Engineering Mathematics

(a) Given:

$$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$$

If λ_1, λ_2 and λ_3 are roots of the equation, then the given equation can be factorized as:

$$(x + \lambda_1)(x + \lambda_2)(x + \lambda_3)$$

Therefore, three multiplications are needed to evaluate p on an input x.

58. Topic: Programming and Data Structures

(b) The given code swaps two numbers pass by reference is used in function call.
Statement S_1 will be executed: the compiler will generate code to allocate a temporary nameless all, initialize it to 13, and pass the address of the cell to swap.
Statement S_2 is incorrect: There will not be any runtime error on line $L1$ during execution.

Statement S_3 is incorrect: There will not be any runtime error on line *L*2 during execution.
Statement S_4 is correct: The program will print 13&8 which follows that statement S_5 is incorrect.
Thus, only Statement S_1 and S_4 are correct.

59. Topic: Engineering Mathematics

(a) A square matrix A is called orthogonal if $A'A = I$
We know that for a square matrix, that is orthogonal,

$$A^T = A^{-1} \text{ or } A' = A^{-1} \qquad (1)$$

Now we know that

$$A \cdot A^{-1} = I$$

Using Eq. (1),

$$A \cdot A' = I$$

60. Topic: Engineering Mathematics

(d) If two adjacent rows of a determinant are interchanged, the value of the determinant becomes negative of its original value.
Determinant of a matrix changes its sign if we interchange any two rows or columns present in a matrix.
Let:

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$|A| = a(ei - fh) - b(di - fg) + c(dh - eg)$$

$$|A| = aei - afh + bfg - bdi + chd - ceg$$

Now, interchanging two rows, we get

$$B = \begin{bmatrix} g & h & i \\ d & e & f \\ a & b & c \end{bmatrix}$$

$$|B| = \mathrm{g}(ec - fb) - h(dc - fa) + \mathrm{i}c(db - ea)$$

$$|B| = ceg - bfg - chd + afh - bdi - aei$$

$$|B| = -|A|$$

Therefore, $|B| = -|A|$ when we exchange any two rows or columns of a matrix the sign of determinant changes.

61. Topic: Engineering Mathematics

(c) Given:

$$\begin{vmatrix} 3 & 3 \\ x & 5 \end{vmatrix} = 3$$

$$\Rightarrow 3 \times 5 - 3 \times x = 3$$

$$\Rightarrow 15 - 3x = 33(5 - x) = 3$$

$$\Rightarrow 5 - x = 1 \Rightarrow x = 5 - 1$$

$$\Rightarrow x = 4$$

Therefore, value of x is 4 for $\begin{vmatrix} 3 & 3 \\ x & 5 \end{vmatrix} = 3$.

62. Topic: Engineering Mathematics

(c) According to properties of transpose of a matrix, the transpose of two added matrices is the same as the addition of the two transpose matrices.
That is,

$$(A + B)^T = A^T + B^T$$

Therefore, if A, B, C are any three matrices, then

$$A' + B' + C' = (A + B + C)'$$

63. Topic: Engineering Mathematics

(c) Let:

$$\begin{vmatrix} 265 & 240 & 219 \\ 240 & 225 & 198 \\ 219 & 198 & 181 \end{vmatrix} = A$$

Using elementary row operations to solve the above determinant,

$$C_1 \to C_1 - C_3$$

$$\Rightarrow A = \begin{vmatrix} 46 & 240 & 219 \\ 42 & 225 & 198 \\ 38 & 198 & 181 \end{vmatrix}$$

$$C_2 \to C_2 - C_3$$

$$\Rightarrow A = \begin{vmatrix} 46 & 21 & 219 \\ 42 & 27 & 198 \\ 38 & 17 & 181 \end{vmatrix}$$

$$C_1 \to C_1 - 2C_2$$

$$\Rightarrow A = \begin{vmatrix} 4 & 21 & 219 \\ -12 & 27 & 198 \\ 4 & 17 & 181 \end{vmatrix}$$

$$C_3 - 10C_2$$

$$\Rightarrow A = \begin{vmatrix} 4 & 21 & 9 \\ -12 & 27 & -72 \\ 4 & 17 & 11 \end{vmatrix}$$

$$R_2 \to R_2 + 3R_3$$

$$\Rightarrow A = \begin{vmatrix} 4 & 21 & 9 \\ 0 & 78 & -39 \\ 4 & 17 & 11 \end{vmatrix}$$

$$R_1 \to R_1 - R_3$$

$$\Rightarrow A = \begin{vmatrix} 0 & 4 & -2 \\ 0 & 78 & -39 \\ 4 & 17 & 11 \end{vmatrix} = 2\begin{vmatrix} 0 & 2 & -1 \\ 0 & 78 & -39 \\ 4 & 17 & 11 \end{vmatrix}$$

$$\Rightarrow A = 2\times 39\begin{vmatrix} 0 & 2 & -1 \\ 0 & 2 & -1 \\ 4 & 17 & 11 \end{vmatrix}$$

$$= 2\times 39[0(22+17)-2(0+4)-1(0-8)]$$

$$A = 2\times 39[-8+8] = 2\times 39\times 0 = 0$$

Therefore,

$$\begin{vmatrix} 265 & 240 & 219 \\ 240 & 225 & 198 \\ 219 & 198 & 181 \end{vmatrix} = 0$$

64. Topic: Engineering Mathematics

(c) A probability density function of a random variable can be interpreted as the probability of a random variable falling within a particular range of values. If $f(x)$ is continuous probability density function of a random variable x, the probability that $a < x \le b$ is

$$\int_a^b f(x)dx$$

65. Topic: Engineering Mathematics

(d) Given, mean of normal frequency distribution of 1000 items is $\bar{x} = 25$.
Standard deviation, $\sigma = 2.5$.
Probability density function is given as

$$f(x) = \frac{e^{\left[-\frac{(x-\bar{x})}{(2\sigma)^2}\right]}}{\sqrt{2\pi}\times\sigma}$$

At $x = \bar{x}$, maximum value of $f(x) = \dfrac{1}{\sqrt{2\pi}\times\sigma}$.

Therefore, maximum value of $f(x) = \dfrac{1000}{\sqrt{2\pi}\times 2.5} = \dfrac{400}{\sqrt{2\pi}}$.

So, maximum ordinate is $\dfrac{400}{\sqrt{2\pi}}$.

66. Topic: Engineering Mathematics

(b) Poisson distribution is a discrete function.
It is a statistical distribution which shows frequency probability of specific events when the average probability of single occurrence is known. In other words, it shows probability of a number of independent events that occur in a fixed time interval. If the probability distribution function of a Poisson distribution is given by

$$f(x) = \frac{e^{-2}2^x}{x!}$$

On comparing with $f(x) = \dfrac{e^{-\lambda}\lambda^x}{x!}$

λ = mean number of events per interval = 2
Therefore, mean = 2.

67. Topic: Software Engineering

(b) Activities which ensure that the software that has been built, is traceable to customer requirement is covered as part of validation. Validation is the process of evaluating the final product of check whether the software meets customer expectations and requirements. Validation is done at the end of the development process after all verifications have been completed.

68. Topic: Software Engineering

(d) A testing method, which is normally used as the acceptance test for a software system is system testing. Unit testing is used to test independent modules and integration testing is used to test integrated modules while regression testing is done after finding defects to check dependent modules. System testing is done by the customer to test a system for acceptance.

69. Topic: Databases

(d) Update command is used to change contents of one database using the contents of another database by linking them on a common key field. Replace command is used to return char with every occurrence of search – string replaced with replacement string.
Join command is used to fetch data from two or more tables, join the data to form a single set of data. Change command is used to change the structure of tables.

70. Topic: Databases

(a) A database file is locked to present inconsistency. During locked state, the values cannot be read by anyone except the user for which the database is locked. Thus, a locked database file can be accessed by only one user.

71. Topic: Databases

(a) Transaction log in any SQL database records all activity taking place in a database. It is the most important component when disaster recovery is done. Thus, transaction log contains complete record of all activity that affected the contents of a database during a certain period of time.

72. Topic: Databases

(b) Foreign key is the primary key of one table that is referred in another table to establish relation among the data. Purpose of 'Foreign Key' in a table is to ensure referential integrity. Thus, foreign key of one table can only have the values that are in some primary key of referenced table. A foreign key cannot have unmatched values.

73. Topic: Databases

(a) A scenario where a transaction writes a data item after it is read by an uncommitted transaction leads to an irrecoverable error in a database system. This scenario is called Blind write. When a transaction writes without reading a value, this is equivalent to writing without reading the updated value.

74. Topic: Computer Networks

(b) IPSEC tunnel mode is usually found between site-to-site virtual private networks. It provides security for packets in network layer. It takes an IP packet, including the header, then applies IPSEC security methods to the entire packet and then adds a new IP header. Thus, use of IPSEC in tunnel mode results in IP packet with new header. It protects the original IP header as well.

75. Topic: Operating System

(b) Special software to create a job queue is called a spooler. Spooler maintains an orderly queue of jobs for peripheral devices and feeds it data at a specific rate thereby controlling spooling. Spooling is a process in which data is temporarily held to be used and executed by a device, program or the system.

76. Topic: Operating System

(c) Process is a program in execution. Process is an instance of a program in execution. It contains program code and its current activity. Depending on the operating system, a process may hold different data. Several processes may be associated with the same program.

77. Topic: Operating System

(a) Starvation is a major problem associated with deadlock rollback because in cases where the same process is rolled back every time the process may strive for resources and may never complete. Therefore, when a process is rolled back as a result of deadlock, the difficulty which arises is called starvation. Hence option (a) is correct.

78. Topic: Computer Organization and Architecture

(b) Interrupt is a signal send by hardware or software to the processor indicating the need of attention. When an interrupt occurs, the current instruction is completed and then the CPU is branched off to the interrupt service. Thus, on receiving an interrupt from an I/O device, the CPU branches off to the interrupt service routine after completion of the current instruction.
Hence option (b) is correct.

79. Topic: Operating System

(a) RICS stands for reduced instruction set computer and CISC stands for complex instruction set computer. CISC provides a large instruction set compared to RISC, but a lot of CISC instructions execute multiple low-level instructions. This means RISC has fewer instruction set than CISC. Since registers are the fastest accessible memory type, to help speed up operations RISC has more registers than CISC. Thus, compared to CISC processors, RISC processors contain more registers and smaller instruction set; hence, option (a) is correct.

80. Topic: Computer Organization and Architecture

(c) In auto-increment addressing mode, the effective address of the operand is the contents of a register specified in the instruction. After accessing the operand, the contents of this register are automatically incremented to point to the next consecutive memory location. Thus, auto-increment addressing mode is not useful in creating self-relocation code, and it does not require additional ALU for effective address calculation. However, the amount of increment depends on the size of the data item accessed. Hence only statement 3 is true of the auto-increment addressing mode.

Indian Space Research Organisation Computer Science

2011

QUESTION PAPER

1. The encoding technique used to transmit the signal in giga ethernet technology over fiber-optic medium is

(a) differential Manchester encoding
(b) non-return to zero
(c) 4B/5B encoding
(d) 8B/10B encoding

2. Which of the following is an unsupervised neural network?

(a) RBS (b) Hopfield
(c) Back propagation (d) Kohonen

3. In compiler terminology, reduction in strength means

(a) replacing run time computation by compile time computation
(b) removing loop invariant computation
(c) removing common subexpressions
(d) replacing a costly operation by a relatively cheaper one

4. The following table shows the processes in the ready queue and time required for each process for completing its job.

Process	Time (ms)
P_1	10
P_2	5
P_3	20
P_4	8
P_5	15

If Round Robin scheduling with 5 ms is used, what is the average waiting time of the processes in the queue?

(a) 27 ms (b) 26.2 ms
(c) 27.5 ms (d) 27.2 ms

5. MOV [BX], AL type of data addressing is called

(a) register addressing
(b) immediate addressing
(c) register indirect addressing
(d) register relative

6. Evaluate (X XOR Y) XOR Y

(a) All 1's (b) All 0's
(c) X (d) Y

7. Which of the following is true about the z-buffer algorithm?

(a) It is a depth sort algorithm
(b) No limitation on total number of objects in the scene
(c) Comparison of objects is done
(d) z-buffer is initialized to background colour at start of algorithm

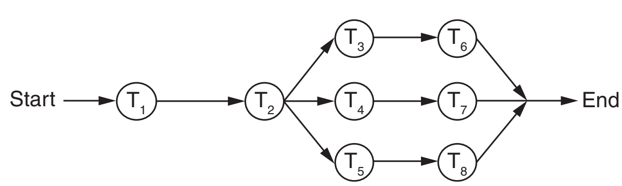
8. What is the decimal value of the floating-point number C1D00000 (hexadecimal notation)? (Assume 32-bit, single precision floating point IEEE representation)

(a) 28 (b) –15
(c) –26 (d) –28

9. What is the raw throughput of USB 2.0 technology?

(a) 480 Mbps (b) 400 Mbps
(c) 200 Mbps (d) 12 Mbps

10. Following is the precedence graph for a set of tasks to be executed on a parallel processing system S.



What is the efficiency of this precedence graph on S if each of the tasks $T_1, \ldots, T_8$ takes the same time and the system S has five processors?

(a) 25% (b) 40%
(c) 50% (d) 90%

11. How many distinct binary search trees can be created out of four distinct keys?

(a) 5 (b) 14
(c) 24 (d) 35

12. The network protocol which is used to get MAC address of a node by providing IP address is

(a) SMTP (b) ARP
(c) RIP (d) BOOTP

13. Which of the following statements about peephole optimizations is False?

(a) It is applied to a small part of the code.
(b) It can be used to optimize intermediate code.
(c) To get the best out of this, it has to be applied repeatedly.
(d) It can be applied to a portion of the code that is not contiguous.

14. Which one of the following in place sorting algorithms needs the minimum number of swaps?

(a) Quick-sort (b) Insertion sort
(c) Selection sort (d) Heap sort

15. What is the equivalent serial schedule for the following transactions?

Transaction:

T_1	T_2	T_3
		R(Y)
		R(Z)
R(X)		
W(X)		
		W(Y)
		W(Z)
	W(Z)	
R(Y)		
W(Y)		
	R(Y)	
	W(Y)	
	R(Y)	
	W(X)	

(a) $T_1 - T_2 - T_3$ (b) $T_3 - T_1 - T_2$
(c) $T_2 - T_1 - T_3$ (d) $T_1 - T_3 - T_2$

16. Consider a direct mapped cache with 64 blocks and a block size of 16 bytes. To what block number does the byte address 1206 map to?

(a) Does not map (b) 6
(c) 11 (d) 54

17. A context model of a software system can be shown by drawing a

(a) LEVEL–0 DFD (b) LEVEL–1 DFD
(c) LEVEL–2 DFD (d) LEVEL–3 DFD

18. An example of poly-alphabetic substitution is

(a) *P*-box (b) *S*-box
(c) Caesar cipher (d) Vigenere cipher

19. If node A has three siblings and B is parent of A, what is the degree of A?

(a) 0 (b) 3
(c) 4 (d) 5

20. The IEEE standard for WiMax technology is

(a) IEEE 802.16 (b) IEEE 802.36
(c) IEEE 812.16 (d) IEEE 806.16

21. Which type of DBMS provides support for maintaining several versions of the same entity?

(a) Relational Data Base Management Systems
(b) Hierarchical
(c) Object Oriented Data Base Management Systems
(d) Network

22. A system is having 8 M bytes of video memory for bit-mapped graphics with 64-bit colour. What is the maximum resolution it can support?

(a) 800×600 (b) 1024×768
(c) 1280×1024 (d) 1920×1440

23. What is the meaning of $\overline{RD}$ signal in Intel 8151 A?

(a) Read (when it is low)
(b) Read (when it is high)
(c) Write (when it is low)
(d) Read and Write (when it is high)

24. If the page size in a 32-bit machine is 4 K bytes, then the size of page table is

(a) 1 M bytes (b) 2 M bytes
(c) 4 M bytes (d) 4 K bytes

25. A processor takes 12 cycles to complete an instruction I. The corresponding pipelined processor uses six stages with the execution times of 3, 2, 5, 4, 6 and 2 cycles, respectively. What is the asymptotic speedup assuming that a very large number speedup assuming that a very large number of instructions are to be executed?

(a) 1.83 (b) 2
(c) 3 (d) 6

26. The in-order traversal of a tree resulted in FBGADCE. Then the pre-order traversal of that tree would result in

(a) FGBDECA
(b) ABFGCDE
(c) BFGCDEA
(d) AFGBDEC

27. Which one of the following is 'true'?

(a) $R \cap S = (R \cup S) - [(R - S) \cup (S - R)]$

(b) $R \cup S = (R \cap S) - [(R - S) \cup (S - R)]$

(c) $R \cap S = (R \cup S) - [(R - S) \cap (S - R)]$

(d) $R \cap S = (R \cup S) \cup (R - S)$

28. 

The above figure represents which one of the following UML diagram for a single send session of an online chat system?

(a) Package Diagram (b) Activity Diagram
(c) Class Diagram (d) Sequence Diagram

29. Which 'Normal Form' is based on the concept of 'full functional dependency'?

(a) First Normal Form
(b) Second Normal Form
(c) Third Normal Form
(d) Fourth Normal Form

30. In Boolean algebra, rule $(X + Y)(X + Z) = ?$

(a) $Y + XZ$ (b) $X + YZ$
(c) $XY + Z$ (d) $XZ + Y$

31. How many 3-to-8-line decoders with a chip having enable pin are needed to construct a 6-to-64-line decoder without using any other logic gates?

(a) 7 (b) 8
(c) 9 (d) 10

32. In which layer of network architecture, the secured socket layer (SSL) is used?

(a) Physical layer (a) Session layer
(c) Application layer (d) Presentation layer

33. What is the bit rate of a video terminal unit with 80 character/line, 8 bits/character and horizontal sweep time of 100 μs (including 20 μs of retrace time)?

(a) 8 Mbps (b) 6.4 Mbps
(c) 0.8 Mbps (d) 0.64 Mbps

34. Black Box software testing method focuses on the

(a) boundary condition of the software
(b) control structure of the software
(c) functional requirement of the software
(d) independent paths of the software

35. How many edges are there in a forest with v vertices and k components?

(a) $(v + 1) - k$ (b) $(v + 1)/2 - k$
(c) $v - k$ (d) $v + k$

36. If A and B are square matrices of the same order and A is symmetric, then B^TAB is

(a) skew symmetric (b) symmetric
(c) orthogonal (d) idempotent

37. Find the memory address of the next instruction executed by the microprocessor (8086), when operated in real mode for CS = 1000 and IP = E000.

(a) 10E00 (b) 1E000
(c) F000 (d) 1000E

38. A fast wide SCSI-II disk drive spins at 7200 RPM, has a sector size of 512 bytes, and holds 160 sectors per track. Estimate the sustained transfer rate of this drive.

(a) 576000 kilobytes/sec
(b) 9600 kilobytes/sec
(c) 4800 kilobytes/sec
(d) 19200 kilobytes/sec

39. Two control signals in microprocessor which are related to Direct Memory Access (DMA) are

(a) INTR and INTA
(b) RD and WR
(c) S0 and S1
(d) HOLD and HLDA

40. Consider the following pseudocode.

```
x : = 1;
i : = 1;
while ` (x ≤ 500)
begin
  x : = 2^x;
  i : = i + 1;
end;
```

What is the value of i at the end of the pseudocode?

(a) 4 (b) 5
(c) 6 (d) 7

41. If a microcomputer operates at 5 MHz with an 8-bit bus and a newer version operates at 20 MHz with a 32-bit bus, the maximum speed-up possible approximately will be

(a) 2 (b) 4
(c) 8 (d) 16

42. The search concept used in associative memory is

(a) Parallel search
(b) Sequential search
(c) Binary search
(d) Selection search

43. Which variable does not drive a terminal string in the grammar

$$S \to AB$$
$$A \to a$$
$$B \to b$$
$$B \to C$$

(a) A (b) B
(c) C (d) S

44. In Java, after executing the following code what are the values of x, y and z?

```
int x, y = 10, z = 12;
x = y++ + z++;
```

(a) $x = 22, y = 10, z = 12$
(b) $x = 24, y = 10, z = 12$
(c) $x = 24, y = 11, z = 13$
(d) $x = 22, y = 11, z = 13$

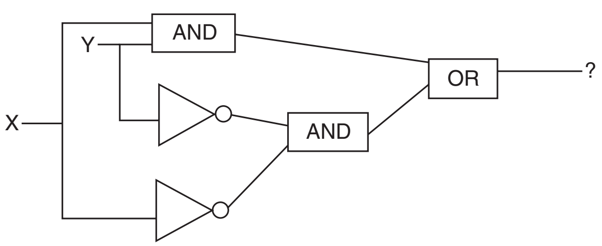
45. The broadcast address for IP network 172.16.0.0 with subnet mask 255.255.0.0 is

(a) 172.16.0.255
(b) 172.16.255.255
(c) 255.255.255.255
(d) 172.255.255.255

46. Which RAID level gives block-level striping with double distributed parity?

(a) RAID 10 (b) RAID 2
(c) RAID 6 (d) RAID 5

47. The output expression of the following gate network is



(a) $X \cdot Y + \overline{X} \cdot \overline{Y}$ (b) $X{\cdot}Y + X{\cdot}Y$
(c) $X{\cdot}Y$ (d) $X + Y$

48. The Hamming distance between the octets of $0 \times AA$ and 0×55 is

(a) 7 (b) 5
(c) 8 (d) 6

49. Consider a 32-bit machine where four-level paging scheme is used. If the hit ratio to TLB is 98%, and it takes 20 nanoseconds to search the TLB and 100 nanoseconds to access the main memory, what is effective memory access time in nanoseconds?

(a) 126 (b) 128
(c) 122 (d) 120

50. Data is transmitted continuously at 2.048 Mbps rate for 10 hours and received 512 bit errors. What is the bit error rate?

(a) 6.9 e – 9 (b) 6.9 e – 6
(c) 69 e – 9 (d) 4 e – 9

51. Warnier Diagram enables the analyst to represent

(a) Class Structure
(b) Information Hierarchy
(c) Data Flow
(d) State Transition

52. Given:

X:	0	10	16
Y:	6	16	28

The interpolated value at $X = 4$ using piecewise linear interpolation is

(a) 11 (b) 4
(c) 22 (d) 10

53. In functional dependency, Armstrong's inference rules refer to

(a) Reflexive, Augmentation and Decomposition
(b) Transitive, Augmentation and Reflexive
(c) Augmentation, Transitive, Reflexive and Decomposition
(d) Reflexive, Transitive and Decomposition

54. Number of chips (128 × 8 RAM) needed to provide a memory capacity of 2048 bytes is

(a) 2 (b) 4
(c) 8 (d) 16

55. There are three processes in the ready queue. When the currently running process requests for I/O, how many process switches take place?

(a) 1 (b) 2
(c) 3 (d) 4

56. Let $T(n)$ be defined by $T(1) = 10$ and $T(n + 1) = 2n + T(n)$ for all integers $n \geq 1$. Which of the following represents the order of growth of $T(n)$ as a function of n?

(a) $O(n)$ (b) $O(n \log n)$
(c) $O(n^2)$ (d) $O(n^3)$

57. Which of the following UNIX command allows scheduling a program to be executed at the specified time?

(a) Cron (b) Nice
(c) Date and time (d) Schedule

58. In DMA transfer scheme, the transfer scheme other than burst mode is

(a) Cycle technique
(b) Stealing technique
(c) Cycle stealing technique
(d) Cycle bypass technique

59. n^{th} derivative of x^n is

(a) nx^{n-1} (b) $n^n\, n!$
(c) $nx^n!$ (d) $n!$

60. A total of nine units of a resource type are available, and given the safe state as shown, which of the following sequence will be a safe state?

Process	Used	Max
P_1	2	7
P_2	1	6
P_3	2	5
P_4	1	4

(a) $<P_4, P_1, P_3, P_2>$ (b) $<P_4, P_2, P_1, P_3>$
(c) $<P_4, P_2, P_3, P_1>$ (d) $<P_3, P_1, P_2, P_4>$

61. Three coins are tossed simultaneously. The probability that they will fall two heads and one tail is

(a) 5/8 (b) 1/8
(c) 2/3 (d) 3/8

62. The average depth of a binary search tree is

(a) O($n^{0.5}$) to (b) O(n)
(c) O(log n) (d) O(n log n)

63. What is the output of the following C code?

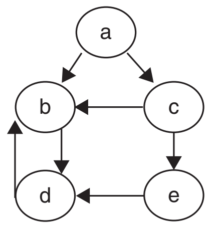
```
#include <stdio.h>
#include <conio.h>
void main()
{
    int index;
    for(index=1; index<=5;i++)
    {
        printf("%d", index);
        if(i == 3)
            continue;
    }
}
```

(a) 1245 (b) 12345
(c) 12245 (d) 12354

64. When n-type semiconductor is heated,

(a) Number of electrons increases while that of holes decreases
(b) Number of holes increases while that of electrons decreases
(c) Number of electrons and holes remain same
(d) Number of electron and holes increases equally

65. The Cyclomatic Complexity metric $V(G)$ of the following control flow graph is



(a) 3 (b) 4
(c) 5 (d) 6

66. Which of the following algorithm design techniques is used in merge sort?

(a) Greedy method
(b) Backtracking
(c) Dynamic programming
(d) Divide and Conquer

67. The arithmetic mean of attendance of 49 students of class A is 40% and that of 53 students of class B is 35%. Then the % of arithmetic mean of attendance of class A and B is

(a) 27.2% (b) 50.25%
(c) 51.13% (d) 37.4%

68. Which of the following sentences can be generated by

$S \rightarrow aS|bA$
$A \rightarrow d|cA$

(a) *bccdd* (b) *abbcca*
(c) *abcabc* (d) *abcd*

69. Lightweight Directory Access Protocol is used for

(a) Routing the packet
(b) Authentication
(c) Obtaining IP address
(d) Domain name resolving

70. Number of comparisons required for an unsuccessful search of an element in a sequential search organized, fixed length, symbol table of length L is

(a) L (b) $L/2$
(c) $(L + 1)/2$ (d) $2L$

71. One SAN switch has 24 ports. All 24 port supports 8 Gbps Fiber Channel technology. What is the aggregate bandwidth of that SAN switch?

(a) 96 Gbps (b) 192 Mbps
(c) 512 Gbps (d) 192 Gbps

72. Find the output of the following Java code line:

```
System.out.println(math.floor(– 7.4))
```

(a) – 7 (b) – 8
(c) – 7.4 (d) – 7.0

73. Belady's anomaly means
(a) Page fault rate is constant even on increasing the number of allocated frames.
(b) Pages fault rate may increase on increasing the number of allocated frames.
(c) Pages fault rate may increase on decreasing the number of allocated frames.
(d) Pages fault rate may decrease on increasing the number of allocated frames.

74. In an *RS* flip-flop, if the *S* line (Set line) is set high (1) and the *R* line (Reset line) is set low (0), then the state of the flip-flop is
(a) Set to 1 (b) Set to 0
(c) No change in state (d) Forbidden

75. In HTML, which of the following can be considered a container?
(a) <SELECT> (b) <Value>
(c) <INPUT> (d) <BODY>

76. What is the matrix that represents rotation of an object by $\theta°$ about the origin in 2D?
(a) $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ (b) $\begin{bmatrix} \sin\theta & -\cos\theta \\ \cos\theta & \sin\theta \end{bmatrix}$
(c) $\begin{bmatrix} \cos\theta & -\sin\theta \\ \cos\theta & \sin\theta \end{bmatrix}$ (d) $\begin{bmatrix} \sin\theta & -\cos\theta \\ \cos\theta & \sin\theta \end{bmatrix}$

77. In a system having a single processor, a new process arrives at the rate of six processes per minute and each such process requires seven seconds of service time. What is the CPU utilization?
(a) 70% (b) 30%
(c) 60% (d) 64%

78. A symbol table of length 152 is possessing 25 entries at any instant. What is occupation density?
(a) 0.164 (b) 127
(c) 8.06 (d) 6.08

79. A problem whose language is recursion is called
(a) Unified problem
(b) Boolean function
(c) Recursive problem
(d) Decidable

80. Logic family popular for low power dissipation
(a) CMOS (b) ECL
(c) TTL (d) DTL

ANSWER KEY

1. (d)	**9.** (a)	**17.** (a)	**25.** (b)	**33.** (b)	**41.** (d)	**49.** (b)	**57.** (a)	**65.** (b)	**73.** (b)
2. (d)	**10.** (b)	**18.** (d)	**26.** (b)	**34.** (c)	**42.** (a)	**50.** (a)	**58.** (c)	**66.** (d)	**74.** (a)
3. (d)	**11.** (b)	**19.** (*)	**27.** (a)	**35.** (c)	**43.** (c)	**51.** (b)	**59.** (d)	**67.** (d)	**75.** (d)
4. (b)	**12.** (b)	**20.** (a)	**28.** (b)	**36.** (b)	**44.** (d)	**52.** (d)	**60.** (d)	**68.** (d)	**76.** (a)
5. (c)	**13.** (d)	**21.** (c)	**29.** (b)	**37.** (b)	**45.** (b)	**53.** (b)	**61.** (d)	**69.** (b)	**77.** (a)
6. (c)	**14.** (c)	**22.** (b)	**30.** (b)	**38.** (b)	**46.** (c)	**54.** (d)	**62.** (c)	**70.** (a)	**78.** (a)
7. (d)	**15.** (b)	**23.** (a)	**31.** (c)	**39.** (d)	**47.** (a)	**55.** (b)	**63.** (b)	**71.** (d)	**79.** (d)
8. (c)	**16.** (c)	**24.** (c)	**32.** (d)	**40.** (b)	**48.** (c)	**56.** (c)	**64.** (b)	**72.** (b)	**80.** (a)

*** None of the options is correct.**

ANSWERS WITH EXPLANATION

1. Topic: Computer Networks

(d) The encoding technique used to transmit the signal in giga ethernet technology over fiber-optic medium is 8B/10B encoding. In 8B/10B encoding, each byte of data is assigned a 10-bit code. The byte is split up into the three most significant bits and five least significant bits, which is then represented as two decimal numbers with least significant bits first. 8B/10B encoding enables long transmission distances and more effective error detection.

2. Topic: Artificial Intelligence

(d) In unsupervised neural network, the network is provided with inputs but not with desired outputs. Kohonen network is one of the basic types of self-organizing neural network. It is an unsupervised neural network.

3. Topic: Compiler Design

(d) Reduction in strength is a technique in which expensive operations are replaced with equivalent but less-expensive operations. Example of this compiler optimization technique is replacing multiplication which a loop with addition, replacing exponentiation within a loop with multiplication. Hence option (d) is correct.

4. Topic: Operating System

(b) Round Robin scheduling is pre-emptive version of first in first out, i.e., FIFO. Round Robin Scheduling assigns equal time to each process without priority in cyclin order. Therefore, what process comes in first is executed first.

The processes are executed as follows:

P_1	P_2	P_3	P_4	P_5	P_1	P_3	P_4	P_5	P_3	P_5	P_3	
0 ms	5 ms	10 ms	15 ms	20 ms	25 ms	30 ms	35 ms	38 ms	43 ms	48 ms	53 ms	58 ms

Waiting time of P_1 = 25 ms – 5 ms = 20 ms
Waiting time of P_2 = 5 ms
Waiting time of P_3 = 53 ms –15 ms = 38 ms
Waiting time of P_4 = 35 ms – 5 ms = 30 ms
Waiting time of P_5 = 48 ms – 10 ms = 38 ms
Therefore,

Average Waiting Time

$$= \frac{\text{Waiting Time of } P_1 + P_2 + P_3 + P_4 + P_5}{\text{Total Processes}}$$

$$= \frac{20\text{ ms} + 5\text{ ms} + 38\text{ ms} + 30\text{ ms} + 38\text{ ms}}{5}$$

$$= \frac{131\text{ms}}{5}$$

$$= 26.2\text{ ms}$$

5. Topic: Computer Organization and Architecture

(c) The given instruction moves the contents of AL to memory location stored in BX. Thus, the instruction MOV[BX], AL type of addressing is called register indirect addressing. In this, a second register, called base register, is used to hold the actual memory address that the program refers to.

6. Topic: Digital Logic

(c) Given:

$$(X \text{ XOR } Y) \text{ XOR } Y$$

We know that

$$A \text{ XOR } B = A\bar{B} + \bar{A}B$$

Let us consider

$$X \text{ XOR } Y = W$$

We get

$$\begin{aligned}
(X \text{ XOR } Y) \text{ XOR } Y = W \text{ XOR } Y &= \bar{W}Y + \bar{Y}W \\
&= \overline{(X \text{ XOR } Y)}Y + (X \text{ XOR } Y)\bar{Y} \\
&= \overline{(\bar{X}Y + X\bar{Y})}Y + (\bar{X}Y + X\bar{Y})\bar{Y} \\
&= (\overline{\bar{X}Y} \cdot \overline{X\bar{Y}})Y + \bar{X}Y\bar{Y} + X\bar{Y}\bar{Y}
\end{aligned}$$

(Since $\overline{A+B} = \bar{A}\cdot\bar{B}$ De Morgan's Law)

$$= (\bar{\bar{X}} + \bar{Y})\cdot(\bar{X} + \bar{\bar{Y}})\cdot Y + \bar{X}Y\bar{Y} + X\bar{\bar{Y}}\bar{Y}$$
$$= (X + \bar{Y})\cdot(\bar{X} + Y)\cdot Y + \bar{X}\cdot 0 + X\bar{Y}$$

(Since $Y\bar{Y} = 0, \bar{Y}\ \bar{Y} = \bar{Y}$)

$$= (X\bar{X} + XY + \bar{Y}\ \bar{Y} + Y\bar{Y})Y + 0 + X\bar{Y})$$

(Since $A\cdot 0 = 0$)

$$\begin{aligned}
&= (0 + XY + \bar{X}\bar{Y} + 0)Y + X\bar{Y} \\
&= (XY + \bar{X}\bar{Y})Y + X\bar{Y} = XYY + \bar{X}\bar{Y}Y + X\bar{Y} \\
&= XY + \bar{X}\cdot 0 + X\bar{Y} = XY + X\bar{Y} = X(Y + \bar{Y}) \\
&= X\cdot 1
\end{aligned}$$

(Since $Y + \bar{Y} = 1$)

Therefore, $(X \text{ XOR } Y) \text{ XOR } Y = X$.

7. Topic: Computer Graphics

(d) z buffering is used to determine which parts of object are visible and which are hidden. z buffering is simple and easy to implement in object an image space. It can be executed quickly with many polygons. It assigns z-value to each polygon and then display the one (pixel by pixel) that has smallest value. z buffer is initialized to background colour at start of algorithm. Hence option (d) is correct.

8. **Topic: Digital Logic**

(c) Given floating-point number is C1D00000 (in hexadecimal notation).
In binary representation, the number can be written as:

$$\underline{1}\underline{100\ 0001\ 1}101\ 0000\ 0000\ 0000\ 0000\ 0000$$

Now, according to single precision floating point IEEE representation, first bit is sign bit (0 for positive and 1 for negative), next 8 bits are for exponent and last 23 bits are for mantissa.
Here,

Sign bit = 1 (this implies number is negative)
Exponent = 131 – 127 = 4

and

$$\text{Significant} = 1.101\ 0000\ 0000\ 0000\ 0000\ 0000$$
$$= 1\times 2^0 + 1\times\frac{1}{2} + 0\times\frac{1}{2^2} + 1\times\frac{1}{2^3} + 0\cdots+0$$
$$= 1+\frac{1}{2}+\frac{1}{8}=\frac{13}{8}$$

$$\text{Therefore, Decimal value} = \text{Sign}\times 2^{\text{exponent}}\times\text{Significant}$$
$$= (-1)\times 2^4\times\frac{13}{8} = -26$$

9. **Topic: Computer Oganization and Architecture**

(a) USB or universal serials bus is a plug-and-play interface. USB allows a computer to communicate with peripheral and other devices. USB 2.0 technology is a high-speed USB. It can support a transfer rate of up to 480 megabits per second (Mbps) or 60 megabytes per second (MBps). Thus, the saw throughput of USB 2.0 technology is 480 Mbps (option a).

10. **Topic: Operating System**

(b) A precedence graph is a directed, acyclic graph where nodes represent sequential activities which could be a software task or statements within a program.
Given, a system has five processors and tasks T_1 to T_8 takes same time. And according to the graph,
One processor executes T_1 while other four processors are idle. Then one processor executes T_2 and remaining four processors are idle; then, three processors execute T_3, T_4, T_5 and remaining two processors are idle. These processors are seen parallelly. Then, three processors T_6, T_7, T_8 execute parallely and the remaining two processors are idle.
Thus, 4 unit of time are required from start to end. Maximum number of tasks that can be executed in 4 units of time using 5 processors is given as

$$4 \times 5 = 20 \text{ tasks}$$

Now, number of tasks executed in 4 units of time is 1 + 1 + 3 +3 = 8.
Therefore,

$$\text{Efficiency} = \frac{\text{(Number of Tasks Executed)}}{\text{(Maximum Number of Tasks that can be Executed)}}\times 100$$
$$= \frac{8}{20}\times 100 = 40\%$$

11. **Topic: Programming and Data Structures**

(b) Binary search tree is a node-based data structure. In a binary search tree, the left subtree of a node contains nodes less than the nodes key and the right subtree of a node contains nodes greater than the node's key. It provides ordering among keys. For N distinct keys, there are $\frac{^{2n}C_n}{(n+1)}$ unlabeled trees possible.
Thus for n = 4, numbers of distinct binary search trees that can be created are

$$\frac{^{2n}C_n}{n+1} = \frac{^8C_4}{4+1} = \frac{1}{5}\times\frac{8!}{4!(8-4)!} = \frac{8!}{5\times 4\times 4!}$$
$$= \frac{8\times 7\times 6\times 5\times 4!}{5\times 4\times 3\times 2\times 1\times 4!} = \frac{8\times 7\times 6\times 5}{5\times 4\times 3\times 2\times 1} = 7\times 2$$
$$= 14$$

Thus, 14 distinct binary search trees can be created.

12. **Topic: Computer Networks**

(b) ARP or address resolution protocol is used to get MAC address of a node by providing IP address. It is a protocol that maps an IP address to a physical machine address that is recognized in the local network. When an incoming packet destined for a host machine on a particular LAN arrives at gateway, then the ARP is asked to find physical machine address or MAC address that matches the IP address so that the packet can be converted to right packet length and format and sent to the machine.

13. **Topic: Compiler Design**

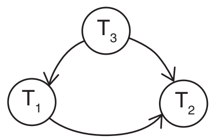
(d) Peephole optimization is performed over a small set of instructions in a piece of code. The set of instruction is called peephole or window. This optimization is performed to reduce code size, improve performance and reduce memory footprint. This method can be applied on intermediate codes as well as target codes. From the given statements, statement (d) is false about peephole optimization.

14. Topic: Algorithms

(c) Section sort requires minimum number of swaps. It is of the order of n. Selection sort algorithm sorts by finding minimum element in an unsorted array and putting it in its correct position in sorted array. Thus, the algorithms maintain two subarrays in a given array: sorted subarray and unsorted subarray.

15. Topic: Databases

(b) Here each transaction T_i in schedule. Create a node labelled T_i in precedence graph.



Consider transaction T_1 it is doing R and W operations using X and Y variables operations $R(X)$ and $W(X)$ are not used in T_3 transaction after T_1 and some is case for $R(Y)$ and $W(Y)$. There is no outgoing edge from T_1 to T_3. While they are being used in transaction T_2 so there is an edge from T_1 to T_2. Moreover, operation $R(Y)$ $R(Z)$ and $W(Y)$ $W(Z)$ are being used in T_1 and T_2 transactions after T_3, therefore there is an outgoing edge from T_3 to T_1 and T_3 to T_2, thus equivalent serial schedule will be $T_3 - T_1 - T_2$.

16. Topic: Computer Organization and Architecture

(c) Given, cache has 64 blocks:

$$\text{Size of block} = 16 \text{ bytes}$$
$$\text{Byte address} = 1206$$

Now,

$$\text{Memory Block Number} = \frac{\text{Byte Address}}{\text{Size of Block}}$$
$$= \frac{1206}{16} = 75.376 \sim 75$$

Now,

$$\text{Cache Block Number} = (\text{Memory Block Number}) \bmod (\text{Number of Blocks in Cache})$$

Therefore,

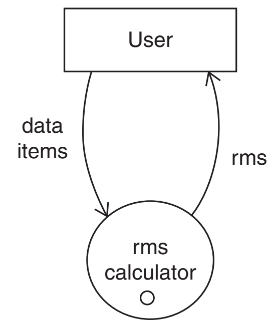
$$\text{Cache Block Number} = 75 \bmod 64 = 11$$

Thus, Byte address 1206 map to 11 block number.

17. Topic: Software Engineering

(a) Context model of a software system shows the system as a single high-level process and shows the relationship of system with other external entities. It is a specialized version of data flow diagram. Context model is also called context-level data flow diagram or level 0 data flow diagram.

Context model example for an RMS calculating software:



18. Topic: Computer Networks

(d) Poly-alphabetic substitution is simply a substitution cipher with an alphabet that changes. In this, a mixed alphabet is used to encrypt a plaintext. Polyalphabetic substitution is useful because they are less easily broken by frequency analysis. The number of letters encrypted before a polyalphabetic substitution cipher returns to its first cipher alphabet is called its period. Larger the period, stronger the cipher. Vigenere cipher is an example of polyalphabetic substitution.

19. Topic: Programming and Data Structures

(None) Given, node A has three siblings and B is parent of A.
Now we know that degree of a node is the total number of children it has, i.e., total number of nodes originating from it.
From the given information, it is not possible to determine number of children node A has or whether A is the leaf thus degree of A cannot be determined.

20. Topic: Computer Networks

(a) WiMAX or wireless microwave access technology is a broadband wireless technology. It is a standardized wireless version of ethernet used as an alternative to wire technologies to provide broadband access to customer premises. It is a scalable wireless platform for constructing alternative and complementary broadband networks. The IEEE standard for WiMAX technology is IEEE 802.16.

21. Topic: Databases

(c) Object-oriented database management systems provide support for maintaining several versions of the same entity. An object-oriented database management system is a database management system that supports the modelling and creation of data as objects. It is also called 00DBMS. It also includes support for classes of objects and inheritance of class properties. It also incorporates methods, subclasses and their objects.

22. Topic: Computer Organizational and Architecture

(b) Given, video memory = 8 M bytes

$$\text{Pixel requires} = 64\text{-bit color}$$

Therefore,

$$\text{Total number of pixels} = \frac{8\text{ MB}}{64\text{ bits}} = \frac{8\text{ MB}}{8\text{ bytes}}$$
$$= 1\text{ M}$$

Out of the given options, only option (b) can support these number of pixels 1024 × 768.

23. Topic: Computer Organization and Architecture

(a) $\bar{R}D$ signal in Intel 8151 A stands for: Read (when it is low).

24. Topic: Operating System

(c) Given:

$$\text{Page size} = 4\text{ K Bytes} = 4\times 2^{10} = 2^{12}\text{ bytes}$$
$$\text{Address space} = 32\text{ bit}$$

Therefore,

$$\text{Number of page table entries} = \frac{2^{32}}{2^{12}} = 2^{20}$$

Since each entry in page table has address of size 4 bytes, therefore size of page table is

Size of page table

= size of each entry × number of page table entries

$= 4\times 2^{20}$ bytes $= 4\,\text{M Bytes}$

25. Topic: Computer Organization and Architecture

(b) Given, to complete an instruction 12 cycles are required.
With pipelining, processor uses six stages with execution times of 3, 2, 5, 4, 6 and 2 cycles, respectively.

Therefore,

$$\text{Asymptotic speed-up} = \frac{\text{Time without Pipeline}}{\text{Time with Pipeline}}$$

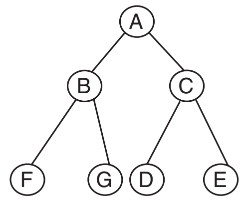
Now, since for non-pipeline processor 12 cycles are required therefore time = 12 cycles
For pipeline processor consider maximum execution of 6 cycles therefore time = 6 cycles
Therefore,

$$\text{Asymptatic speed-up} = \frac{12}{6} = 2$$

26. Topic: Databases

(b) In order traversal, all the nodes in the left subtree are visited first, then the root node and finally all the nodes in the right subtree are visited.

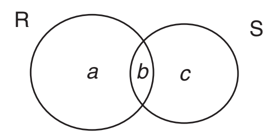
In preorder traversal, root node is visited first, then all nodes in left subtree and finally all nodes in right subtree.
Given, in-order traversal of tree resulted in FBGADCE.
Generating tree from given order:



Therefore, preorder traversal would result in ABFGCDE.

27. Topic: Engineering Mathematics

(a) Given:



Then,

$$R\cap S = b \text{ and } (R\cup S)-[(R-S)\cup(S-R)]$$
$$= \{a,b,c\}-[a\cup c] = b$$

Therefore,

$$R\cap S = (R\cup S)-[(R-S)\cup(S-R)]$$
$$R\cup S = \{a,b,c\} = (R\cap S)-[(R-S)\cup(S-R)]$$
$$= \{b\}-\{a\cup c\}$$

Therefore,

$$R\cup S \neq (R\cap S)-[(R-S)\cup(S-R)]$$

Now,

$$(R\cup S)-[(R-S)\cap(S-R)] = \{a,b,c\}-[a\cap c] \neq R\cap S$$

and

$$(R\cup S)\cup(R-S) = \{a,b,c\}\cup\{a\} = \{a,b,c\} \neq R\cap S$$

Therefore option (a) is correct.

28. Topic: Software Engineering

(b) In UML, an activity diagram is a flow chart. It is a subset of behaviour diagram. It captures the dynamic behaviour of the system. It not only visualizes the dynamic nature of a system, but also constructs the executable system by using forward and backward engineering techniques. An activity diagram describes the sequence from one activity to another. The given UML diagram for a single send session of an online chat system represents activity diagram.

29. Topic: Databases

(b) The biggest problem in database is data redundancy. Normalization is the process of removing redundant data to improve storage efficiency, data integrity and scalability. The process involves splitting of tables into multiple tables which can be rejoined or linked whenever a query is issued.

First Normal Form: A relational table is in first normal form or 1NF if each attribute name is unique and is single valued. Each row is unique and there are no repeating groups.

Second Normal Form: A relational table is in second normal form or 2NF if the table is in 1NF. All attributes are fully dependent on the primary key. There are no partial dependencies. 2NF is based on the concept of full functional dependency.

Third Normal Form: A relational table is in third normal form or 3NF if the table is already in 2NF. Non-primary key attributes do not depend on other non-primary key attributes. There is no transitive dependency.

Fourth Normal Form: A relational table is in fourth normal form or 4NF if the table is already in BCNF and the table does not contain multivalued dependency.

Thus, second normal form is based the concept of full functional dependency.

30. Topic: Digital Logic

(b) Given:

$$(X+Y)(X+Z)$$
$$(X+Y)(X+Z) = X \cdot X + Y \cdot X + X \cdot Z + Y \cdot Z$$

Rearranging it, we get

$$= X \cdot X + X \cdot Z + Y \cdot X + Y \cdot Z$$

Using identity law $X \cdot X = X$, thus we have:

$$(X+Y)(X+Z) = X + X \cdot Z + Y \cdot X + Y \cdot Z$$
$$= X(1+Z) + Y \cdot X + Y \cdot Z$$

Using identity law $1 + Z = 1$, we have

$$(X+Y)(X+Z) = X \cdot 1 + Y \cdot X + Y \cdot Z$$

Using identity law $X \cdot 1 = X$, we have

$$(X+Y)(X+Z) = X + Y \cdot X + Y \cdot Z = X(1+Y) + Y \cdot Z$$

Again, using identity law $1+Y=1$, we have

$$(X+Y)(X+Z) = X \cdot 1 + Y \cdot Z$$

Again, using identity law $X \cdot 1 = X$, we have

$$(X+Y)(X+Z) = X + YZ$$

This law is called Distributive law.

31. Topic: Digital Logic

(c) A decoder is a combinational logic circuit. It is used to change code into a set of signals. A 3- to 8-line decoder circuit gives eight logical output for three inputs. And 6 to 64 lines decode circuit gives 64 logical outputs for 6 inputs. In general, a decoder circuit gives 2^n logical unique outputs for n inputs.

Since 3- to 8-line decoder gives 8 outputs and 6- to 64-line decoder gives 64 outputs, thus to construct a 6- to 64-line decoder from 3 to 8 line decoder.

$$\frac{64}{8} = 8 \text{ decoders are required}$$

An additional decoder is required to select anyone of these eight decoders. Thus, nine decoders are required.

32. Topic: Computer Networks

(d) Secured socket layer or SSL is a standard security technology used for establishing an encrypted link between a web server and a browser, so that all the data that passes through web server and browsers remain private. In OSI or open system interconnection model, SSL is used in presentation layer of the network architecture.

33. Topic: Operating System

(b) Given, horizontal sweep time = 100 μs

Retrace time = 20 μs

Data = 80 characters/line and 8 bits/character

Total number of bits transmitted = 80 × 8 = 640 bits

Therefore,

$$\text{Bit rate} = \frac{\text{Total number of bits transmitted}}{\text{Horizontal sweep time}}$$
$$= \frac{640 \text{ bits}}{100\ \mu\text{s}} = \frac{640}{100 \times 10^{-6}} \text{ bits/sec}$$
$$= 6.4 \times 10^6 \text{ bits / sec} = 6.4 \text{ Mbps}$$

34. Topic: Software Engineering

(c) Black box software testing is a software testing method in which internal structure, design and implementation of the item being tested is not known to the tester. Thus, the software program to be tested is like a black box to the tester inside which one cannot see. This method is used to find errors, like initialization and termination errors, behaviour or performance errors, interface errors, incorrect or missing functions, errors in data structures or external database access. Thus, black box software testing method focusses on the functional requirement of the software.

35. Topic: Engineering Mathematics

(c) A forest is a graph comprising collection of trees. Every connected component in a forest is a tree. Mathematically, forest is a disjoint union of trees. Given, forest has v vertices and k components.

$$\text{Number of edges} = \sum_{i=1}^{k} v_i - 1$$
$$= (v_1 - 1) + (v_2 - 1) + (v_3 - 1) + \cdots + (v_k - 1)$$
$$= (v_1 + v_2 + v_3 + \cdots + v_k) - k = v - k$$

Thus edges in the forest = $v - k$.

36. Topic: Engineering Mathematics

(b) A symmetric matrix is a square matrix that is equal to its transpose that is:

$$A^T = A$$

Given, A and B are square matrices of same order A is symmetric.
Considering B^TAB take transpose, we have

$$(B^TAB)^T = B^TA^TB$$

(Since $(AB)^T = B^TA^T$)

Now, we know A is symmetric. Therefore,

$$A^T = A$$

Therefore,

$$(B^TAB)^T = B^TAB$$

Thus, B^TAB is symmetric.

37. Topic: Computer Organization and Architecture

(b) In 8086, the address generated using offset and segment base is given as:

Address = Segment × 0 × 10 + Offset

In the given problem, it is given that
CS = 1000 which is segment and IP = E000 which is offset. Therefore,

$$\text{Address} = 0 \times 1000 * 0 \times 10 + 0 \times \text{E000}$$
$$= 0 \times 1\text{E000}$$

Thus, memory address of next instruction = 1E000.

38. Topic: Operating System

(b) Given, SCSI-II disk drive spins at 7200 RPM sector size = 512 bytes

Capacity = 160 Sectors per track

Rotation = 7200 RPM

$$\Rightarrow 1 \text{ rotation} = \frac{60}{7200}\text{sec} = \frac{1}{120}\text{sec}$$

Now in 1 rotation, 1 track is read.

Therefore, in 1 rotation. 160 ×512 bytes will be read since 1 rotation takes $\frac{1}{120}$ sec.

So, in $\frac{1}{120}$sec 160 × 512 Bytes will be read

$$\text{Bytes read in 1 sec} = \frac{160 \times 512}{1/120} = 9830400 \text{ bytes}$$
$$= \frac{9830400}{1024}\text{K8} = 9600 \text{ kB}$$

Thus, transfer rate = 9600 kilobytes/sec.

39. Topic: Computer Organization and Architecture

(d) DMA or direct memory access is a method that allows an input/output device to send or receive data directly to or from the main memory. It is a method of transferring data from the computer RAM to another part of the computer without processing it using the CPU.
HOLD and HLDA are two control signals that are related to direct memory access.
HOLD is an input signal to the processor used to request a DMA action.
HLDA is an output signal that acknowledges the DMA action.

40. Topic: Algorithms

(b) Given:

$x = 1$ & $i = 1$, loop begins
$i = 1$; $x = 2^1 = 2$, $i = 1 + 1 = 2$
$i = 2$; $x = 2^2 = 4$, $i = 2 + 1 = 3$
$i = 3$; $x = 2^4 = 16$, $i = 3 + 1 = 4$
$i = 4$; $x = 2^{16} = 65536$, $i = 4 + 1 = 5$
$x > 500$

Therefore, control exists the loop. Thus, value of i at the end of the pseudocode is 5.

41. Topic: Computer Organization and Architecture

(d) Given, old version of microcomputer operates at 5 MHz with 8-bit bus.
New version of microcomputer operates at 20 MHz with 32 bit bus. By using newer version of microcomputer, the bus expansion increases data flow by factor of $\frac{32}{8} = 4$.

Similarly, operating speed will increase by factor $\frac{20}{5} = 4$.

Thus maximum speed up possible = 4 × 4 = 16.

42. Topic: Computer Organization and Architecture

(a) Associative memory is a set of storage locations that are accessed through its contents rather than through an address. Search concept used in associative memory is parallel search. For K-way associativity we can have K

compares which compared the tag bits of K blocks in parallel. An associative memory is also known as content addressable memory.

43. Topic: Theory of Computation

(c) Given:

$$S \to AB$$
$$A \to a$$
$$B \to b$$
$$B \to C$$

Here, $S \to AB$ and $A \to a$ and $B \to b$ or C. If $B \to C$, then there is no termination string, thus C is a useless symbol or variable that does not drive a terminal string in the grammar.

44. Topic: Programming and Data Structures

(d) In the following code:

$$x = 10, y = 10, z = 12$$

Now in expression $x = y{+}{+} + z{+}{+}$
Since post-increment is used for y and z variables, thus their value will be incremented after their values have been used in expression of x.
Therefore,

$$x = 10 + 12 = 22$$

Then, y and z are incremented:

$$\Rightarrow y = 11 \text{ and } z = 13 \text{ and } x = 22$$

45. Topic: Computer Networks

(b) Given:

IP network = 172.16.0.0
Subnet mask = 255.255.0.0

First, we will find network address. It is obtained by applying ANO operation between IP network and subset mask.

172.16.0.0 && 255.255.0.0 = 172.16.0.0

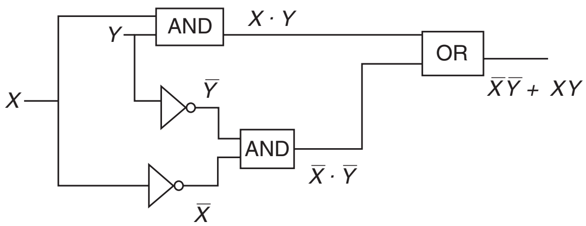
Now broadcast address can be obtained by setting the host portion of network address to 1's. Therefore, broadcast address = 172.16.255.255.

46. Topic: Databases

(c) RAID or redundant array of independent disks configurations employ techniques of striping, mirroring or parity to create reliable data stores. It enables better storage performance and higher availability.
RAID 0 – Splits data evenly across two or more disks without parity information, redundancy or fault tolerance.
RAID 1 – Consists of an exact copy of a set of data on two or more disks.
RAID 2 – Stripes data at the bit level and uses a hamming code for error correction.
RAID 3 – Consists of byte level striping with a dedicated parity disk.
RAID 4 – Consists of block level striping with a dedicated parity desk.
RAID 5 – Consists of block-level striping with distributed parity.
RAID 6 – Used block level striping with two parity block distributed across all disks.
Thus correct option is (c) RAID 6 gives block level striping with double distributed parity.

47. Topic: Digital Logic

(a) Given:



The output expression of the given gate network is $X \cdot Y + \overline{X} \cdot \overline{Y}$.

48. Topic: Digital Logic

(c) Hamming distance between two strings is the number of positions at which the corresponding character are different. It is obtained by performing XOR operation between the two strings. The number of 1's in the exultant string indicate hamming distance.
Now,

$$0xAA = 1010\ 1010$$
$$0x55 = 0101\ 0101$$

Since, at all positions the character are different, therefore hamming distance = 8.
Moreover, $0xAA$ XOR $0x55$ = (10101010) XOR (01010101)

$$= 11111111$$

Total 8 1's, thus hamming distance = 8.

49. Topic: Operating System

(b) We know that:

Effective Memory Access Time = Hit Ratio (TLB Access Time + Main Memory Access Time) + (1– Hit Ratio) (TLB Access Time + (Number of Levels + 1) Main Memory Access Time)

Here, Hit Ratio = 0.98.
TLB Access Time = 20 ns.
Main Memory Access Time = 100 ns.
Number of Levels = 4.
Therefore, effective memory access time

$$= 0.98\,(20 + 100) + (1–0.98) \times (20 + 5\times100)$$
$$= 0.98 \times 120 + 0.02 \times 520 = 128 \text{ ns}$$

50. Topic: Computer Networks

(a) Given, data is transmitted at 2.48 Mbps for 10 hours.

$$\text{Error Rate} = \frac{\text{Error Bits}}{\text{Time}} = \frac{512}{36000}$$

$$\text{Bit Error Rate} = \frac{\text{Error Rate}}{\text{Data Transmission Bandwidth}}$$

$$= \frac{512}{36000} \times \frac{1}{2.048 \quad 10^6}$$

$$\text{Bit Error Rate} = 6.94 \times 10^{-9} = 6.9e-9$$

51. Topic: Software Engineering

(b) Warnier diagram is a hierarchical flowchart that allows the description of the organization of data and procedures. It is also called Warnier diagram. It enables representation of information hierarchy in a compact manner. It is used in software engineering for system analysis and design. Hence, option (b) is correct.

52. Topic: Engineering Mathematics

(d) Given,

$$X: 0 \quad 10 \quad 16$$
$$Y: 6 \quad 16 \quad 28$$

Using ($x_1 = 0, y_1 = 6$) and ($x_2 = 10, y_2 = 16$)
Then,

$$P(x) = y_1 \frac{(x_2 - x)}{(x_2 - x_1)} + y_2 \frac{(x - x_1)}{(x_2 - x_1)}$$

$$P(4) = 6\frac{(10-4)}{(10-0)} + 16\frac{(4-0)}{(10-0)} = \frac{6 \times 6}{10} + \frac{16 \times 4}{10}$$

$$= 3.6 + 6.4 = 10$$

Thus, interpolated value at $X = 4$ is 10.

53. Topic: Databases

(b) Armstrong's inference rules are a set of rules used to infer all functional dependencies on a relational database. These are the most basic inference rules. The Armstrong's inference rules are: let A, B and C be sets of attributes in relation to R, then

1. If $B \subseteq A$ then $A \rightarrow B$. This axiom is called reflexivity rule.
2. If $A \rightarrow B$ then $AC \rightarrow BC$. This axiom is called augmentation rule.
3. If $A \rightarrow B$ and $B \rightarrow C$ then $A \rightarrow C$. This axiom is called transitivity rule.

Thus, in functional dependency, Armstrong's inference rules refers to transitive, augmentation and reflexive.

54. Topic: Computer Organization and Architecture

(d) Given, chip size = 128 × 8 RAM

Memory capacity = 2048 bytes

Now,

128 × 8 RAM, ⇒ there are 128 locations in chip of storing capacity of 8 bits

Therefore,

$$\text{Total Number of Chips required}$$
$$= \frac{\text{Memory Capacity}}{\text{Number of locations in Chips}}$$
$$= \frac{2048}{128} = 16$$

Thus, number of chips needed = 16.

55. Topic: Operating System

(b) When a process requests for I/O, then two process switches takes place. In the first switch, the process to be switched is taken out and the scheduler starts executing. Then the next process is brought to execution. Thus, the currently running process requests for I/O, two process switches take place. Option (b) is correct.

56. Topic: Algorithms

(c) Given $T(n)$ is defined as

$$T(1) = 10$$
$$T(n+1) = 2n + T(n)$$

Using recursive definition of T here, we have

$$T(n+1) = 2n + 2(n-1) + T(n-1)$$
$$= 2n + 2(n-1) + 2(n-2) + T(n-2)$$
$$= 2n + 2(n-1) + 2(n-2) + 2(n-3) + T(n-3)$$
$$= 2n + 2(n-1) + 2(n-2) + 2(n-3) + \cdots$$
$$+ 2(1) + T(1)$$
$$= 2(n + (n-1) + (n-2) + (n-3) + \cdots + 1) + T(1)$$
$$= 2 \times \frac{n(n+1)}{2} + T(1)$$
$$= n(n+1) + T(1)$$
$$= n^2 + n + 10$$

Thus, growth of $T(n)$ as a function of n is of the order of n^2.

57. Topic: Computer Organization and Architecture

(a) Unix is a computer operating system.
It handles activities from multiple uses at the same time.
Cron is a long running process that executes commands at specific dates and times.
Nice is a program used to invoke a shell script with a particular priority thereby giving a process more or less CPU time than other processes.
Date and time is used to set date and time in various formats.

Schedule is used to asynchronously execute tasks at any desired time of the day. It is made possible by cron clock daemon.
Thus, UNIX command cron allows scheduling a program to be executed at the specified time.

58. Topic: Computer Organization and Architecture

(c) In DMA transfer scheme, the transfer scheme other than burst mode is cycle stealing technique.
DMA or direct memory access is a method that allows transferring data from the computer's RAM to another past of the computer without processing it using the CPU. Cycle stealing and burst mode are two methods for DMA. In cycle stealing technique, data is not transferred at once like in burst mode; rather, it is transferred 1 byte at a time hence correct option is (c).

59. Topic: Engineering Mathematics

(d) Given x^n
First derivative:

$$\frac{d}{dx}x^n = nx^{n-1}$$

Second derivative:

$$\frac{d}{dx}(nx^{n-1}) = n(n-1)x^{n-2}$$

Third derivative:

$$\frac{d}{dx}(n(n-1)x^{n-2}) = n(n-1)(n-2)x^{n-3}$$

Thus, n^{th} derivative:

$$\begin{aligned}\frac{d^n}{dx^n}x^n &= n(n-1)(n-2)\cdots(n-n-2)(n-n-1)\\ &= n(n-1)(n-2)\cdots 2.1\\ &= n!\end{aligned}$$

Thus, n^{th} derivative of $x^n = n!$

60. Topic: Operating System

(d) Given, resource type available = 9.
Process P_1, P_2, P_3 and P_4 are using 2, 1, 2, 1 resources respectively
$\Rightarrow$ 6 of 9 resources are already allocated and 3 resources are available.
Now P_1 is using two resources and can use seven resources at max.
Therefore,

$$7 - 2 = 5 \text{ resources can be used}$$

Similarly,

$$\begin{aligned}P_2 &\Rightarrow 6 - 1 = 5 \text{ resources}\\ P_3 &\Rightarrow 5 - 2 = 3 \text{ resources}\\ P_4 &\Rightarrow 4 - 1 = 3 \text{ resources}\end{aligned}$$

Thus, P_3 can use remaining 3 resources after that new resource available = 3 + 2 = 5 so P_1 can use, then P_2 and P_4. So safe sequence will be $<P_3, P_1, P_2, P_4>$.

61. Topic: Engineering Mathematics

(d) When three coins are tossed simultaneously, then possible combinations are

HHH, HHT, HTH, THH, HTT, THT, TTH, TTT

Out of these, the possible combinations of two heads and one tail is HTH, THH and HHT.
Therefore,

$$\text{Probability} = \frac{\text{Favourable Combinations}}{\text{All posible Combinations}} = \frac{3}{8}$$

62. Topic: Programming and Data Structures

(c) A binary tree is a special tree data structure. In this structure, every node can have a maximum of two children. One child is called left child and other child is called right child. In a binary tree, every node can have either 0 child, 1 child or 2 children, but not more than that. The average depth of a binary search tree is $O(\log n)$. Thus, correct option is (c).

63. Topic: Programming and Data Structures

(b) Continue Statement in C programming forces the next iteration of the loop to take place skipping any code in between. Since continue statement is in the last statement; therefore, it will not skip any code and output of the C code will be 12345.

64. Topic: Semiconductors

(b) n-type semiconductor is created by adding pentavalent impurities to pure or intrinsic semiconductor. In n-type semiconductor, electrons are majority carriers and holes are minority carriers. When n-type semiconductor is heated, the number of holes increases while number of electrons decreases.

65. Topic: Software Engineering

(b) Cyclomatic complexity is a software metric. It is used to indicate the complexity of a program. It measures independent paths through program source code. Cyclomatic complexity can be calculated with respect to functions, modules, methods or classes within a program.

$$\text{Cyclomatic complexity} = E - N + 2$$

where, E is number of edges and N is number of nodes.
In the given flow graph, number of nodes = 5

$$\text{Number of edges} = 7$$

Therefore, cyclomatic complexity = 7 – 5 + 2 = 4

66. Topic: Algorithms

(d) Merge sort is a divide and conquer algorithm. The algorithm involves during the consorted list into N sub-lists, each containing one element. Then, these sub-lists are merged such that the resultant list is sorted. The algorithm works as follows:

1. Divide the comported list into N sub lists each containing one element.
2. The adjacent pairs of two singleton lists are merged to form a list of two elements. Thus, N sub-lists are converted into N/2 sub-lists containing two elements each.
3. This process is repeated until a single sorted list is produced.

67. Topic: Engineering Mathematics

(d) Given, arithmetic mean of attendance of 49 students of class A is 40%.
Arithmetic mean of 53 students of attendance of class B is 35%.
Arithmetic mean is average of set of numerical values. It is obtained by adding numerical values of the set and dividing by total elements in the set.

So, for class A, attendence $= \dfrac{40}{100} \times 49 = 19.6$

For class B, attendence $= \dfrac{35}{100} \times 53 = 18.55$

Therefore, % of arithmetic mean of attendance of class A and B is

$$= \frac{(19.6 + 18.55)}{49 + 53} \times 100 = \frac{38.15 \times 100}{102} = 37.40\%$$

68. Topic: Theory of Computation

(d) Given:

$$S \to aS/bA$$
$$A \to d/cA$$

So, we have,

$$S \to aS$$
$$S \to abA$$
$$S \to abcA$$
$$S \to abcd$$

Therefore, from the given grammar *abcd* can be generated.

69. Topic: Computer Networks

(b) Lightweight Directory Access Protocol is a client/server protocol that is used to access and manage directory information. It reads and edits directories over IP networks and runs directly over TCP/IP using simple string formats for data transfer. Lightweight directly access protocol is used to keep all the information of a user in one place. Thus, it is used for authentication.

70. Topic: Programming and Data Structures

(a) Given, symbol table of length L is organized using sequential search. Therefore, the entire table needs to be searched every time for an unsuccessful search of an element. Thus, number of comparisons required is L.

71. Topic: Computer Networks

(d) Given, number of ports in one SAN switch = 24.
All ports support 8 Gbps fiber technology SAN or storage area network is a device that connects servers and shaded pools of storage devices. It is a fibre channel switch compatible with fibre channel protocol.
Aggregate bandwidth of SAN switch = Number of ports × Operating frequency

$$= 24 \times 8 \text{ Gbps}$$

72. Topic: Programming and Data Structures

(b) System.out.println(math.floor(−7.4)) floor function. Floor () returns the largest integer which is not greater than the argument value thus floor (−7.4) will return lower limit, i.e., −8 thus output of the java code will be −8.

73. Topic: Operating System

(b) Belady, anomaly is an unusual behaviour that occurs when the number of frames in first in first out page replacement policy increases and so does the number of page faults increases. Thus Belady's anomaly means page fault rate may increase on increasing the number of allocated frames. Hence, option (b) is correct.

74. Topic: Digital Logic

(a) The truth table of *RS* flip-flop is

S	R	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	Invalid input

Therefore, if S line (set line) is set high (l) and the R line (Reset line) is set low (0), then the state of the flip-flop is set to 1.

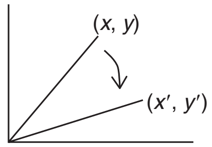
75. Topic: Web Technologies

(d) HTML or hypertext markup language is used on web to develop web pages.
<select> is used to create a drop-down list.
<value> specifies value of an input element
<input> specifies input field where user can enter data
<body> this element contains all the contents of an HTML document.

Therefore, <body> is considered as a container.

76. Topic: Engineering Mathematics

(a) The matrix that represents rotation of an object by θ° about the origin in 2D is



$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

It is represented as

$$x' = x\cos\theta - y\sin\theta$$
$$y' = x\sin\theta + y\cos\theta$$
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$$

77. Topic: Operating System

(a) Given, in single processor, a new process arrives at rate of six processes per minute each process requires 7 seconds.
Now,

$$\text{CPU utilisation} = \frac{\text{Useful time}}{\text{Total time}} \times 100\%$$

Since each process requires 7 seconds, total time required by 6 processes arriving in 1 minute

$$= 7 \times 6 = 42 \text{ seconds}$$

So,

Useful time = 42 seconds
Total time = 1 minute = 60 seconds

Therefore,

$$\text{CPU utilisation} = \frac{42}{60} \times 100 = 70\%$$

78. Topic: Programming and Data structure

(a) Given:

Table length = 152
Number of entries = 25

So,

$$\text{Occupation Density} = \frac{\text{Number of Entries}}{\text{Table Length}} = \frac{25}{152} = 0.164$$

Therefore, occupation density = 0.164.

79. Topic: Theory of Computation

(d) A language L is decidable it is a recursive language. All decidable languages are recursive languages and vice versa. Thus, a problem whose language is recursion is called decidable. Hence, option (d) is correct.

80. Topic: Digital Logic

(a) Logic family popular for low power dissipation is CMOS. CMOS or complementary metal-oxide semiconductor is used to describe small amount of memory on computer motherboard that stores the BIOS settings. In CMOS, both n- and p-type transistors are used in complementary way to form a current gate that forms an effective means of electric control. These transistors use almost no power when not needed and are low-power dissipation transistors.

Indian Space Research Organisation Computer Science 2013

QUESTION PAPER

1. Let A(1:8, –5:5, –10:5) be a three-dimensional array. How many elements are there in the array A?

(a) 1200 (b) 1408
(c) 33 (d) 1050

2. The number of rotations required to insert a sequence of elements 9, 6, 5, 8, 7, 10 into an empty AVL tree is

(a) 0 (b) 1
(c) 2 (d) 3

3. Opportunistic reasoning is addressed by which of the following knowledge representation?

(a) Script (b) Blackboard
(c) Production Rules (d) Fuzzy Logic

4. The following steps in a linked list

```
p = getnode()
info (p) = 10
next (p) = list
list = p
```

result in which type of operation?

(a) Pop operation in stack
(b) Removal of a node
(c) Inserting a node
(d) Modifying an existing node

5. Shift reduce parsing belongs to a class of

(a) bottom up parsing (b) top down parsing
(c) recursive parsing (d) predictive parsing

6. Which of the following productions eliminate left recursion in the productions given below:

$$S \to Aa|b$$
$$A \to Ac|Sd|\varepsilon$$

(a) $S \to Aa \mid b,\ \ A \to bdA',\ \ A' \to A'c \mid A'ba \mid A \mid \varepsilon$
(b) $S \to Aa \mid b,\ \ A \to A' \mid bdA',\ \ A' \to cA' \mid adA' \mid \varepsilon$
(c) $S \to Aa \mid b,\ \ A \to A'c \mid A'd,\ \ A' \to bdA' \mid cA \mid \varepsilon$
(d) $S \to Aa \mid b,\ \ A \to cA' \mid adA' \mid bdA',\ \ A' \to A \mid \varepsilon$

7. Consider the following psuedocode:

```
x : integer:= 1
y : integer: = 2
procedure add
    x := x + y
procedure second (P: procedure)
    x : integer := 2
    P()
procedure first
    y : integer := 3
    second(add)
first()
write_integer (x)
```

What does it print if the language uses dynamic scoping with deep binding?

(a) 2 (b) 3
(c) 4 (d) 5

8. Which logic gate is used to detect overflow in 2's complement arithmetic?

(a) OR gate (b) AND gate
(c) NAND gate (d) XOR gate

9. In an array of $2N$ elements that is both 2-ordered and 3-ordered, what is the maximum number of positions that an element can be from its position if the array were 1-ordered?

(a) 1 (b) 2
(c) $N/2$ (d) $2N - 1$

10. If the frame buffer has 8 bits per pixel and 8 bits are allocated for each of the R, G, B components, what would be the size of the lookup table?

(a) 24 bytes (b) 1024 bytes
(c) 768 bytes (d) 256 bytes

11. When two BCD numbers 0 × 14 and 0 × 08 are added what is the binary representation of the resultant number?

(a) 0 × 22 (b) 0 ×1c
(c) 0 × 16 (d) results in overflow

12. Which of the following sorting algorithms has the minimum running time complexity in the best and average case?

(a) Insertion sort, Quick sort
(b) Quick sort, Quick sort
(c) Quick sort, Insertion sort
(d) Insertion sort, Insertion sort

13. The number 1102 in base 3 is equivalent to 123 in which base system?

(a) 4 (b) 5
(c) 6 (d) 8

14. A processor is fetching instructions at the rate of 1 MIPS. A DMA module is used to transfer characters to RAM from a device transmitting at 9600 bps. How much time will the processor be slowed down due to DMA activity?

(a) 9.6 ms (b) 4.8 ms
(c) 2.4 ms (d) 1.2 ms

15. A pipeline P operating at 400 MHz has a speedup factor of 6 and operating at 70% efficiency. How many stages are there in the pipeline?

(a) 5 (b) 6
(c) 8 (d) 9

16. How much speed do we gain by using the cache, when cache is used 80% of the time? Assume cache is faster than main memory.

(a) 5.27 (b) 2.00
(c) 4.16 (d) 6.09

17. Two eight bit bytes 1100 0011 and 0100 1100 are added. What are the values of the overflow, carry and zero flags, respectively, if the arithmetic unit of the CPU uses 2's complement form?

(a) 0, 1, 1 (b) 1, 1, 0
(c) 1, 0, 1 (d) 0, 1, 0

18. How many check bits are required for 16 bit data word to detect 2 bit errors and single bit correction using hamming code?

(a) 5 (b) 6
(c) 7 (d) 8

19. What is the maximum number of characters (7 bits + parity) that can be transmitted in a second on a 19.2 kbps line. This asynchronous transmission requires 1 start bit and 1 stop bit.

(a) 192 (b) 240
(c) 1920 (d) 1966

20. IEEE 1394 is related to

(a) RS-232 (b) USB
(c) Firewire (d) PCI

21. What will be the cipher text produced by the following cipher function for the plain text ISRO with key $k = 7$. [Consider 'A' = 0, 'B' = 1,…,'Z'= 25]

$C_k(M) = (kM + 13) \bmod 26$

(a) *RJCH* (b) *QIBG*
(c) *GQPM* (d) *XPIN*

22. Any set of Boolean operators that is sufficient to represent all boolean expressions is said to be complete. Which of the following is not complete?

(a) {NOT, OR} (b) {NOR}
(c) {AND, OR} (d) {AND, NOT}

23. Which of the following is the highest isolation level in transaction management?

(a) Serializable
(b) Repeated Read
(c) Committed Read
(d) Uncommitted Read

24. Consider the following relational schema:

Suppliers (<u>sid:integer</u>, sname:string, saddress:string)
Parts (<u>pid:integer</u>, pname:string, pcolor:string)
Catalog (<u>sid:integer, pid:integer</u>, pcost:real)

What is the result of the following query?

(SELECT Catalog.pid from Suppliers, Catalog
WHERE Suppliers.sid = Catalog.pid)
MINUS
(SELECT Catalog.pid from Suppliers, Catalog
WHERE Suppliers.sname <> 'sachin' and Suppliers.sid = Catalog.sid)

(a) pid of Parts supplied by all except sachin
(b) pid of Parts supplied only by sachin
(c) pid of Parts available in catalog supplied by sachin
(d) pid of Parts available in catalogs supplied by all except sachin

25. Consider the following dependencies and the BOOK table in a relational database design. Determine the normal form of the given relation.

ISBN → Title
ISBN → Publisher
Publisher → Address

(a) First Normal Form
(b) Second Normal Form
(c) Third Normal Form
(d) BCNF

26. Calculate the order of leaf (p_{leaf}) and non-leaf (p) nodes of a B^+ tree based on the information given below:

Search key field = 12 bytes
Record pointer = 10 bytes
Block pointer = 8 bytes
Block size = 1 KB

(a) $p_{leaf} = 51$ and $p = 46$ (b) $p_{leaf} = 47$ and $p = 52$
(c) $p_{leaf} = 46$ and $p = 51$ (d) $p_{leaf} = 52$ and $p = 47$

27. The physical location of a record determined by a formula that transforms a file key into a record location is

(a) hashed file (b) B-tree file
(c) indexed file (d) sequential file

28. The most simplified form of the Boolean function

$x(A, B, C, D) = \Sigma(7, 8, 9, 10, 11, 12, 13, 14, 15)$

(expressed in sum of minterms) is

(a) $A + \bar{A}BCD$ (b) $AB + CD$
(c) $A + BCD$ (d) $ABC + D$

29. How many programmable fuses are required in a PLA which takes 16 inputs and gives 8 outputs? It has to use 8 OR gates and 32 AND gates.

(a) 1032 (b) 776
(c) 1284 (d) 1536

30. In a three-stage counter, using *RS* flip–flops what will be the value of counter after giving 9 pulses to its input? Assume that the value counter before giving any pulses is 1.

(a) 1 (b) 2
(c) 9 (d) 10

31. In which of the following shading models of polygons, the interpolation of intensity values is done along the scan line?

(a) Gourard shading (b) Phong shading
(c) Constant shading (d) Flat shading

32. Which of the following number of nodes can form a full binary tree?

(a) 8 (b) 15
(c) 14 (d) 13

33. What is the matrix transformation which takes the independent vectors $\begin{pmatrix}1\\2\end{pmatrix}$ and $\begin{pmatrix}2\\5\end{pmatrix}$ and transforms them to $\begin{pmatrix}1\\1\end{pmatrix}$ and $\begin{pmatrix}3\\2\end{pmatrix}$, respectively?

(a) $\begin{pmatrix}1 & -1\\1 & 0\end{pmatrix}$ (b) $\begin{pmatrix}0 & 0\\0.5 & 0.5\end{pmatrix}$

(c) $\begin{pmatrix}-1 & 0\\1 & 1\end{pmatrix}$ (d) $\begin{pmatrix}-1 & 1\\1 & 0\end{pmatrix}$

34. In 8086, the jump condition for the instruction JNBE is?

(a) CF = 0 or ZF = 0
(b) ZF = 0 and SF = 1
(c) CF = 0 and ZF = 0
(d) CF = 0

35. How many number of times the instruction sequence below will loop before coming out of the loop?

```
      MOV AL, 00H
A1:   INC AL
      JNZ A1
```

(a) 1
(b) 255
(c) 256
(d) Will not come out of the loop

36. In 8085 microprocessor, the ISR for handling trap interrupt is at location?

(a) 3CH (b) 34H
(c) 74H (d) 24H

37. The voltage ranges for a logic high and a logic low in RS-232 C standard is

(a) Low is 0.0 V to 1.8 V, High is 2.0 V to 5.0 V
(b) Low is –15.0 V to –3.0 V, High is 3.0 V to 15.0 V
(c) Low is 3.0 V to 15.0 V, High is –3.0 V to –15.0 V
(d) Low is 2.0 V to 5.0 V, High is 0.0 V to 1.8 V

38. In the Ethernet, which field is actually added at the physical layer and is not part of the frame?

(a) Preamble (b) CRC
(c) Address (d) Location

39. Ethernet layer-2 switch is a network element type which gives

(a) different collision domain and same broadcast domain
(b) different collision domain and different broadcast domain
(c) same collision domain and same broadcast domain
(d) same collision domain and different broadcast domain

40. If the frame to be transmitted is 1101011011 and the CRC polynomial to be used for generating checksum is $x^4 + x + 1$, then what is the transmitted frame?

(a) 11010110111011 (b) 11010110111101
(c) 11010110111110 (d) 11010110111001

41. What will be the efficiency of a Stop and Wait protocol, if the transmission time for a frame is 20 ns and the propagation time is 30 ns?

(a) 20% (b) 25%
(c) 40% (d) 66%

42. IPv6 does not support which of the following addressing modes?

(a) Unicast addressing
(b) Multicast addressing
(c) Broadcast addressing
(d) Anycast addressing

43. What is IP class and number of sub-networks if the subnet mask is 255.224.0.0?

(a) Class A, 3 (b) Class A, 8
(c) Class B, 3 (d) Class B, 32

44. Which algorithm is used to shape the bursty traffic into a fixed rate traffic by averaging the data rate?

(a) Solid bucket algorithm
(b) Spanning tree algorithm
(c) Hocken helm algorithm
(d) Leaky bucket algorithm

45. A packet filtering firewall can

(a) deny certain users from accessing a service
(b) block worms and viruses from entering the network
(c) disallow some files from being accessed through FTP
(d) block some hosts from accessing the network

46. Which of the following encryption algorithms is based on the Fiestal structure?

(a) Advanced encryption standard
(b) RSA public key cryptographic algorithm
(c) Data encryption standard
(d) RC4

47. The protocol data unit for the transport layer in the internet stack is

(a) segment (b) message
(c) datagram (d) frame

48. The Guass–Seidal iterative method can be used to solve which of the following sets?

(a) Linear algebraic equations
(b) Linear and non-linear algebraic equations
(c) Linear differential equations
(d) Linear and non-linear differential equations

49. What is the least value of the function $f(x) = 2x^2 - 8x - 3$ in the interval [0, 5]?

(a) -15 (b) 7
(c) -11 (d) -3

50. Consider the following set of processes, with arrival times and the required CPU-burst times given in milliseconds.

Process	Arrival Time	Burst Time
$P1$	0	4
$P2$	2	2
$P3$	3	1

What is the sequence in which the processes are completed? Assume Round Robin scheduling with a time quantum of 2 milliseconds.

(a) $P1, P2, P3$ (b) $P2, P1, P3$
(c) $P3, P2, P1$ (d) $P2, P3, P1$

51. In case of a DVD, the speed of data transfer is mentioned in multiples of

(a) 150 KB s^{-1} (b) 1.38 MB s^{-1}
(c) 300 KB s^{-1} (d) 2.40 MB s^{-1}

52. Suppose we have variable logical records of lengths of 5 bytes, 10 bytes, and 25 bytes while the physical block size in disk is 15 bytes. What is the maximum and minimum fragmentation seen in bytes?

(a) 25 and 5 (b) 15 and 5
(c) 15 and 0 (d) 10 and 5

53. A CPU scheduling algorithm determines an order for the execution of its scheduled processes. Given 'n' processes to be scheduled one processor, how many possible different schedules are there?

(a) n (b) n^2
(c) $n!$ (d) 2^n

54. Which of the following are the likely causes of thrashing?

(a) Page size was very small
(b) There are too many users connected to the system
(c) Least recently used policy is used for page replacement
(d) First-in First-out policy is used for page replacement

55. Consider a logical address space of 8 pages of 1024 words each, mapped onto a physical memory of 32 frames. How many bits are there in the physical address and logical address, respectively?

(a) 5, 3 (b) 10, 10
(c) 15, 13 (d) 15, 15

56. In a 64-bit machine, with 2 GB RAM, and 8 KB page size, how many entries will be there in the page table if it is inverted?

(a) 2^{18} (b) 2^{20}
(c) 2^{33} (d) 2^{51}

57. Which of the following is not a necessary condition for deadlock?

(a) Mutual exclusion (b) Reentrancy
(c) Hold and wait (d) No pre-emption

58. Consider the following process and resource requirement of each process.

Process	Type 1		Type 2	
	Used	Max	Used	Max
*P*1	1	2	1	3
*P*2	1	3	1	2
*P*3	2	4	1	4

Predict the state of this system, assuming that there are a total of 5 instances of resource type 1 and 4 instances of resource type 2.

(a) Can go to safe or unsafe state based on sequence
(b) Safe state
(c) Unsafe state
(d) Deadlock state

59. A starvation free job scheduling policy guarantees that no job indefinitely waits for a service. Which of the following job scheduling policies is starvation free?

(a) Priority queuing (b) Shortest job first
(c) Youngest job first (d) Round Robin

60. The state of a process after it encounters an I/O instruction is

(a) ready (b) blocked
(c) idle (d) running

61. Embedded pointer provides

(a) a secondary access path
(b) a physical record key
(c) an inverted index
(d) a prime key

62. A particular parallel program computation requires 100 seconds when executed on a single CPU. If 20% of this computation is strictly sequential, then theoretically the best possible elapsed times for this program running on 2 CPUs and 4 CPUs, respectively, are

(a) 55 and 45 seconds (b) 80 and 20 seconds
(c) 75 and 25 seconds (d) 60 and 40 seconds

63. Consider the following C code.

```
#include <stdio.h>
#include <math.h>
void main()
{
    double pi = 3.1415926535;
    int a = 1;
    int i;
    for(i = 0; i < 3; i++)
            if(a = cos(pi * i/2))
                    printf("%d", 1);
            else printf("%d", 0);
}
```

What would the program print?

(a) 0 0 0 (b) 0 1 0
(c) 1 0 1 (d) 1 1 1

64. What is the output of the following Java program?

```
Class Test
{
    public static void main (String []
    args)
    {
            int x = 0;
            int y = 0;
            for (int z = 0; z < 5; z++)
            {
                if((++x > 2) || (++y > 2))
                {
                        x++;
                }
            }
            System.out.println(x + " " +
            y);
    }
}
```

(a) 8 2 (b) 8 5
(c) 8 3 (d) 5 3

65. Consider the list of page references in the time line as below:

9 6 2 3 4 4 4 4 3 4 4 2 5 8 6 8 5 5 3 2 3 3 9 6 2 7

What is the working set at the penultimate page reference if Δ is 5?

(a) {8, 5, 3, 2, 9, 6} (b) {4, 3, 6, 2, 5}
(c) {3, 9, 6, 2, 7} (d) {3, 9, 6, 2}

66. What is the cyclomatic complexity of a module which has seventeen edges and thirteen nodes?

(a) 4 (b) 5
(c) 6 (d) 7

67. Which of the following types of coupling has the weakest coupling?

(a) Pathological coupling (b) Control coupling
(c) Data coupling (d) Message coupling

68. Which of the following testing methods uses fault simulation technique?

(a) Unit testing (b) Beta testing
(c) Stress testing (d) Mutation testing

69. If a program *P* calls two subprograms *P*1 and *P*2 and *P*1 can fail 50% of the time and *P*2 can fail 40% of the time, what is the failure rate of program *P*?

(a) 50% (b) 60%
(c) 70% (d) 10%

70. Which of the following strategy is employed for overcoming the priority inversion problem?

(a) Temporarily raise the priority of lower priority level process
(b) Have a fixed priority level scheme
(c) Implement kernel pre-emption scheme
(d) Allow lower priority process to complete its job

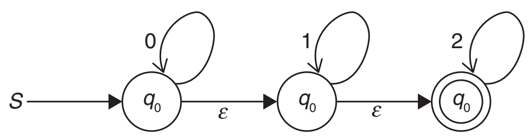
71. Let $P(E)$ denote the probability of the occurrence of event E. If $P(A) = 0.5$ and $P(B) = 1$, then the values of $P(A/B)$ and $P(B/A)$, respectively, are

(a) 0.5, 0.25 (b) 0.25, 0.5
(c) 0.5, 1 (d) 1, 0.5

72. How many diagonals can be drawn by joining the angular points of an octagon?

(a) 14 (b) 20
(c) 21 (d) 28

73. What are the final states of the DFA generated from the following NFA?



(a) q_0, q_1, q_2 (b) $[q_0, q_1], [q_0, q_2], []$
(c) $q_0, [q_1, q_2]$ (d) $[q_0, q_1], q_2$

74. The number of elements in the power set of the set $\{\{A, B\}, C\}$ is

(a) 7 (b) 8
(c) 3 (d) 4

75. What is the right way to declare a copy constructor of a class if the name of the class is MyClass?

(a) MyClass (constant MyClass * arg)
(b) MyClass (constant MyClass & arg)
(c) MyClass (MyClass arg)
(d) MyClass (MyClass * arg)

76. The number of edges in a '*n*' vertex complete graph is?

(a) $n * (n - 1)/2$ (b) n^2
(c) $n * (n + 1)/2$ (d) $n * (n + 1)$

77. The binary equivalent of the decimal number 42.75 is

(a) 101010.110 (b) 100110.101
(c) 101010.101 (d) 100110.110

78. Which of the following is not provided as a service in cloud computing?

(a) Infrastructure as a service
(b) Architecture as a service
(c) Software as a service
(d) Platform as a service

79. The built-in base class in Java, which is used to handle all exceptions, is

(a) raise (b) exception
(c) error (d) throwable

80. In graphics, the number of vanishing points depends on

(a) the number of axes cut by the projection plane
(b) the centre of projection
(c) the number of axes which are parallel to the projection plane
(d) the perspective projections of any set of parallel lines that are not parallel to the projection plane

ANSWER KEY

1. (b)	**9.** (a)	**17.** (d)	**25.** (b)	**33.** (d)	**41.** (b)	**49.** (c)	**57.** (b)	**65.** (d)	**73.** (*)
2. (d)	**10.** (b)	**18.** (b)	**26.** (c)	**34.** (c)	**42.** (c)	**50.** (b)	**58.** (c)	**66.** (c)	**74.** (d)
3. (b)	**11.** (a)	**19.** (c)	**27.** (a)	**35.** (c)	**43.** (b)	**51.** (b)	**59.** (d)	**67.** (d)	**75.** (b)
4. (b)	**12.** (a)	**20.** (c)	**28.** (c)	**36.** (d)	**44.** (d)	**52.** (d)	**60.** (b)	**68.** (d)	**76.** (a)
5. (a)	**13.** (b)	**21.** (a)	**29.** (b)	**37.** (c)	**45.** (d)	**53.** (c)	**61.** (a)	**69.** (c)	**77.** (a)
6. (b)	**14.** (d)	**22.** (c)	**30.** (b)	**38.** (a)	**46.** (c)	**54.** (a)	**62.** (d)	**70.** (a)	**78.** (b)
7. (c)	**15.** (d)	**23.** (a)	**31.** (a)	**39.** (a)	**47.** (a)	**55.** (c)	**63.** (c)	**71.** (b)	**79.** (d)
8. (d)	**16.** (c)	**24.** (*)	**32.** (b)	**40.** (c)	**48.** (a)	**56.** (a)	**64.** (a)	**72.** (b)	**80.** (d)

*** None of the options is correct.**

ANSWERS WITH EXPLANATION

1. **Topic: Programming and Data Structures**

   (b) If the three-dimensional array is denoted as

   $$(l_1{:}u_1,\ l_2{:}u_2,\ l_3{:}u_3)$$

   then number of elements in 3D array is given by

   $$(u_3 - l_3 + 1) \times (u_2 - l_2 + 1) \times (u_1 - l_1 + 1)$$

   Given:

   $$A = (1{:}8, -5{:}5, -10{:}5)$$

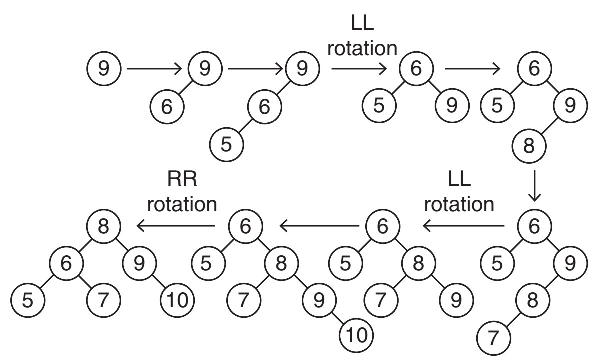
   Therefore, the number of elements is

   $$(8 - 1 + 1) \times (5 + 5 + 1) \times (5 + 10 + 1)$$
   $$= 8 \times 11 \times 16 = 1408$$

2. **Topic: Programming and Data Structures**

   (d) AVL trees are height-balanced binary search tree. An AVL tree has balance factor calculated at every node. Balance factor of a node is the difference of height of left subtree and height of right subtree. For every node, heights of left subtree and right subtree can differ by no more than one. Insert operation may cause balance factor to become more than one. In that case, tree is adjusted by rotation around the node.

   To insert a sequence of elements 9, 6, 5, 8, 7, 10:

   Therefore, the total number of rotations required is 3.

3. **Topic: Artificial Intelligence**

   (b) Opportunistic reasoning is a type of artificial intelligence. It is a method of selecting a suitable logical inference strategy within artificial intelligence applications. Opportunistic reasoning uses either forward chaining or backward chaining depending on the situation. Opportunistic reasoning is an integral part of blackboard knowledge representation. The blackboard holds the data gathered so far and also the current set of partial solutions which have been pursued simultaneously.

4. **Topic: Programming and Data Structures**

   (b) Given:

   $$p = \text{get node}();$$

   It implies that space is being allocated for a new node to be created.

info(p) = 10; setting info value to 10
next(p) = list; pointing to the next node the next node of p contains address of the existing list.
list = p; storing address of p as the first node of the list.
Therefore, the cited list steps in a linked list is inserting a new node at the beginning of the existing list.

5. **Topic: Compiler Design**

(a) Shift-reduce parsing constructs a parse tree for an input string beginning at the leaves and working up towards the root. Shift-reduce parsing belongs to a class of bottom-up parsing. In this, we start from a sentence and then apply production rules in reverse manner in order to reach the start symbol.

Bottom up
↓
Shift reduce
↓
LR parsing
↙ ↓ ↘
SLR parsing LR parsing LA LR parser

Above figure depicts the bottom-up parsers available.

6. **Topic: Theory of Computation**

(b) A grammar G is left recursive if it has at least one left recursive nonterminal. A production of G is said left recursive, if it has the form

$$A \to A\alpha$$

Given:

$$S \to Aa|b$$
$$A \to Ac|Sd|E$$

So,

$$A \to Ac|Aad|bd|E$$

or

$$A \to bdA'|A'$$

where

$$A' \to cA'|adA'|E$$

Therefore, the final grammar is

$$S \to Aa|b$$
$$A \to bdA'|A'$$
$$A' \to cA'|adA'|E$$

Thus, correct option is (b).

7. **Topic: Algorithms**

(c) In dynamic scoping, global identifiers refer to the identifier associated with the most recent environment. Each time a new function is executed, a new scope is pushed onto the stack.
Deep binding: It binds the environment at the time a procedure is passed as an argument.
Thus, for dynamic scoping and deep binding, in second environment when add is passed, $x = 1$ and $y = 3$.

$$\Rightarrow x = 1 + 3 = 4$$

Therefore, the pseudocode prints 4.

8. **Topic: Digital Logic**

(d) If sum of two positive numbers yield a negative number, a sum of two negative numbers yields a positive number then the sum has overflowed. In order to detect overflow in 2's complement, XOR gate is used. It is detected by applying XOR operation between carry in and carry out.

9. **Topic: Programming and Data Structures**

(a) An array $A[1, 2, \ldots, n]$ is said to be of k-order if
$A[i - k] < A[t] < A[i + k]$ for all i such that $k < i < n - k$.
In an array of $2N$ elements, for array to be 2-ordered the first element should be less than 3rd element and for array to be 3-ordered. The first element should be less than 4th element. If the array is both 2-ordered and 3-ordered, then first element should be less than 4th element, 6th element and so on. If the array were 1-ordered, the maximum number of positions that an element can be from its position will be 1.

10. **Topic: Image Processing**

(b) A frame buffer stores color representation for every pixel on the screen and its size depends on the screen resolution. Lookup table is used to achieve greater color depth by using lesser amount of memory.
Given, frame buffer has 8 bits per pixel ⇒ Number of entries = 2^8 each R, G, B component has 8 bits.
R, G, B components having 8 bits implies each entry in lookup table has 24 bits.
So, size of lookup table = Size of entry × Number of entries

$$= 24 \text{ bits} \times 2^8 = 24 \times 256 \text{ bits}$$
$$= \frac{24 \times 256}{8} \text{ bytes} = 768 \text{ bytes}$$

Therefore, size of lookup table is 768 bytes.

11. **Topic: Digital Logic**

(a) In order to convert BCD to binary, first convert BCD to decimal and then convert decimal to binary:

0 × 14 in binary representation is 0001 0100
0 × 08 in binary representation is 000 1000

Sum of these numbers is greater than 9 therefore adding a correction of 6. Thus, we get

$$\begin{array}{rl} 0001\ 1100 & \text{(sum of two BCD numbers)} \\ \underline{0110} & \text{(correction of 6)} \\ 0010\ 0010 & \end{array}$$

Therefore, the resultant number in BCD is 0 × 22.

12. Topic: Algorithms

(a) Insertion sort is a simple sorting algorithm that builds final sorted list one at a time. In this a sublist is maintained which is always sorted. The array is searched and unsorted elements are moved and inserted into sorted sublist. In the best-case scenario, time complexity of insertion sort is $O(n)$ and in average case scenario time complexity is $O(n^2)$.

Quick sort is a divide and conquer algorithm. It selects an element called pivot and partitions the array around the pivot. Elements on left side of pivot are less than the pivot element and all elements on the right side are greater than pivot. In the best-case scenario, time complexity of quick sort is $O(n\log n)$ and in average-case scenario, time complexity is $O(n\log n)$.

Hence, insertion sort has minimum time complexity in best case and quick sort has minimum time complexity in average case.

13. Topic: Digital Logic

(b) Given:

$$\begin{aligned}(1102)_3 &= 1\times 3^3 + 1\times 3^2 + 0\times 3^1 + 2\times 3^0\\ &= 1\times 27 + 1\times 9 + 0\times 3 + 2\times 1\\ &= 27+9+0+2 = 38\end{aligned}$$

So,

$$(1102)_3 = 38$$

Now, suppose this is equivalent to 123 in x base system, then

$$(123)_x = 1\times x^2 + 2\times x^1 + 3\times x^0 = x^2+2x+3$$

Now,

$$\begin{aligned}&(1102)_3 = (123)_x\\ &\Rightarrow 38 = x^2+2x+3\\ &\Rightarrow x^2+2x-35=0\\ &\Rightarrow x^2+7x-5x-35=0\\ &\Rightarrow x(x+7)-5(x+7)=0\\ &\Rightarrow (x-5)(x+7)=0\\ &\Rightarrow x=5 \text{ or } x=-7\end{aligned}$$

However, x cannot be negative. Therefore,

$$x=5$$
$$\Rightarrow (1102)_3 = (123)_5$$

14. Topic: Operating System

(4) DMA or direct memory access is a method that allows on input/output device to send and receive data directly from main memory. This speeds up memory operations.

Given, the processor fetches instruction at rate of 1 MIPS.

1 MIPS implies 1 million instructions per second.

So, fetching rate = 10^6

Characters from a device to RAM are transmitting at 9600 bps implies bits per second.

Therefore, transmitting rate is

$$\frac{9600}{8} = 1200 \text{ bps}$$

Time by which processor will be slowed down due to this DMA activity is

$$\frac{1200}{10^6} = 0.0012 \text{ s}$$
$$= 1.2\times 10^{-3} \text{ sec} = 1.2 \text{ ms}$$

15. Topic: Computer Organization and Architecture

(d) Given, pipeline P is operating at 400 MHz

Speedup factor = 6

Operating efficiency = 70%

Now,

$$\text{Operating efficiency} = \frac{\text{Speedup factor}}{\text{Number of stages in pipeline}}$$

So,

$$\text{Number of stages in pipeline} = \frac{\text{Speedup factor}}{\text{Operating efficiency}}$$
$$= \frac{6}{70\%} = \frac{6\times 100}{70} = 8.57 \approx 9$$

Therefore, number of stages in pipeline = 9

16. Topic: Computer Organization and Architecture

(c) We know that:

$$\text{Speed gain} = \frac{\text{Memory access time without cache}}{\text{Memory access time with cache}}$$

Given, cache is used 80% of time.

Let time taken by memory be x.

Then, the time taken by cache memory = $\frac{x}{20}$

So, the memory access time without cache = x

Memory access time with cache is

$$\frac{80}{100}\frac{x}{20} + x\times\frac{20}{100} = 0.04x + 0.2x = 0.24x$$

Therefore, speed gain = $\frac{x}{0.24x} = 4.16$

17. Topic: Digital Logic

(d) Given, two 8-bit bytes added.

$$\begin{array}{cc} & 1100\,0011\\ & 0100\,1100\\ \hline \boxed{1} & 0000\,1111\\ \downarrow & \downarrow\\ \text{Carry} & \text{Sign}\end{array}$$

Thus, carry flag = 1; sign flag = 0

Now, overflow flag is obtained by applying XOR operation of input carry and output carry.
$\Rightarrow$ (Input carry) XOR (Output carry) = 1 XOR 1 = 0
Therefore, over flow, carry and zero flanges are 0, 1, 0, respectively.

18. Topic: Computer Networks

(b) Hamming code is a set of error correction code used to detect and correct bit errors that can occur when data is being moved or stored.
Now, for detecting d bit error, $d + 1$ bits are required and For correcting d bit error, $2d + 1$ bits are required.
Here, it is given that we are required to detect 2-bit errors and correct single bit.
So, for detecting 2-bit error, 2 + 1 = 3 bits are required.
For correcting single bit, 2 × 1 + 1 = 3 bits are required.
Therefore, total check bits required = 3 + 3 = 6.

19. Topic: Computer Networks

(c) Given, characters that are transmitted on 19.2 kbps line is

$$19.2 \times 10^3 \text{ bps} = 19200 \text{ bps}$$

Now, one character includes 7 bits + parity = 8 bits.
Also, the transmission requires 1 start bit + 1 stop bit.
Thus, the total number of bits required to transmit one character will be 8 + 1 + 1 = 10 bits.
Therefore, the maximum number of characters that can be transmitted is $\frac{19200}{10} = 1920$ characters.

20. Topic: Computer Networks

(c) IEEE 1394 is related to Firewire. This refers to the type of cables, parts and connectors used to connect external devices to computer. It provides a single plug and socket connection on which up to 63 devices can be attached. The standard describes a serial bus or pathway between one or more peripheral devices and computer's microprocessor.

21. Topic: Computer Networks

(a) Cipher text is encrypted text. Cipher is an algorithm which is applied to plain text to get ciphertext. Cipher text is unreadable and has to be converted into plain text using a key to make it understandable.
Here, a cipher text is to be produced for plain that ISRO with key $k = 7$.
Consider $A = 0, B = 1, \ldots, Z = 25$
and $C_k(M) = (kM + 13) \bmod 26$
for $I = 8$, $C(J) = (7 \times 8 + 13) \bmod 26 = 17$, which is R
for $S = 18$, $C(S) = (7 \times 18 + 13) \bmod 26 = 9$, which is J
for $R = 17$, $C(R) = (17 \times 7 + 13) \bmod 26 = 2$, which is C
for $O = 14$, $C(O) = (7 \times 14 + 13) \bmod 14 = 7$, which is H
Therefore, cipher text for plain text ISRO is $RJCH$.

22. Topic: Digital Logic

(c) NAND gate and NOR gate are said to be universal gates. With the help of these gates, any Boolean function can be implemented without the use of any other gate.

- NOT and OR gate together can be used as NOR gate; therefore, this is functionally complete.
- NOR is universal gate; therefore, it is functionally complete.
- AND and OR together are not functionally complete, because none of these is universal gate.
- AND and NOT gate together can be used as NAND gate, so these are functionally complete.

Therefore, option (c) is not complete.

23. Topic: Databases

(a) Transaction is a logical unit of work that contains one or more SQL statements. Isolation levels defines the degree to which a transaction must be isolated from the data modifications made by any other transaction in the database system. The isolation levels defined by SQL standard are listed as follows:

(i) Serializable
(ii) Repeatable reads
(iii) Read committed
(iv) Read uncommitted
(v) Dirty reads
(vi) Non-repeatable reads
(vii) Phantom reads

Therefore, the highest isolation level in transaction management is serializable.

24. Topic: Databases

(None) Query 1. (SELECT catalog.pid from suppliers, catalog WHERE suppliers.sid = catalog.pid)
This gives pid's of all parts that are being supplied.
Query 2. (SELECT catalog.pid from suppliers, catalog WHERE suppliers.sname <> 'sachin' and suppliers.sid = catalog.sid)
This gives pid's of all parts supplied by all except those supplied by Sachin.
However, if a part is supplied by Sachin as well as some other supplier, then it will also be excluded. Therefore, no option is correct.

25. Topic: Databases

(b) Normalization is a process of organizing data in database to avoid data redundancy, insertion deletion and update anomaly.
First normal form: An attribute of a table cannot hold multiple values. It should hold only atomic values second normal form: The relation is in first normal form and all non-key attributes are fully functional dependent on the primary key.

Third normal form: The relation is in second normal form and there is no transitive functional dependency.
BCNF or Boyce-Codd normal form: There is no non-trivial functional dependencies of attributes.
The given relation is in second normal form since there is transitive functional dependency, ISBN $\rightarrow$ address.

26. Topic: Databases

(c) Given:

Search key field = 12 bytes
Record pointer = 10 bytes
Block pointer = 8 bytes
Block size = 1 KB = 2^{10} B = 1024 B

Order of leaf (p_{leaf}) is given as follows:

(p_{leaf}) × (Key field + Record pointer) + Block pointer $\leq$ Block size

$(p_{\text{leaf}}) \times (12 + 10) + 8 \leq 1024$

$$p_{\text{leaf}} \leq \frac{1024-8}{22} \Rightarrow p_{\text{leaf}} \leq 46$$

Order of non-leaf (p) is given as follows:

$$p \times \text{Block pointer} + (p - 1) \times \text{Key field} \leq \text{Block size}$$
$$p \times 8 + (p - 1) \times 12 \leq 1024$$
$$8p + 12p - 12 \leq 1024$$
$$20p - 12 \leq 1024 \Rightarrow p \leq \frac{1024+12}{20} \leq 51$$

Therefore, $p_{\text{leaf}} = 46$ and $p = 51$.

27. Topic: Databases

(a) The physical location of a record determined by a formula that transforms a file key into a record location is hashed file. Hashing is the transformation of a string of characters into a usually shorter fixed-length value or key that represents the original string. It is used to index and retrieve items in a database.

28. Topic: Digital Logic

(c) Let as use Karnaugh map to simplify Boolean algebra expression. It deals with technique of inserting values of output variable in cells within a grid according to a definite pattern. The number of cells is determined by number of input variables n as 2^n.
Given:

$$X(A, B, C, D) = \Sigma\,(7, 8, 9, 10, 11, 12, 13, 14, 15)$$

AB \ CD	00	01	11	10
00				
01			1	
11	1	1	1	1
10	1	1	1	1

Therefore, $X(A, B, C, D) = A + BCD$.

29. Topic: Digital Logic

(b) Total programmable fuses required = Fuses required by AND gates + Fuses required by OR gates
Now,
Fuses required by AND gates = 2 × No. of inputs × no. of AND gate

$$= 2 \times 16 \times 32 = 1024$$

Fuses required by OR gates = No. of outputs × no. of AND gates

$$= 8 \times 32 = 256$$

Total programmable fuses required = 1024 + 256 = 1280.

30. Topic: Digital Logic

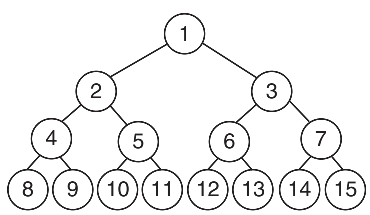
(b) Given, a three-stage counter using *RS* flip–flops; thus, it is a 3-bit counter.
Thus, the total number of possible states = $2^n = 2^3 = 8$.
Given, initial value of counter is 1.
State after 8 clock pulses will be 1, then it will return to the initial state.
Therefore, the value of counter after giving 9 pulses to its input is 2.

31. Topic: Computer Graphics

(a) Shading is referred as the implementation of illumination model at the pixel points or polygon surfaces of the graphics objects. In Gourard sharing, the interpolation of intensity values is done along the second line. It renders a polygon surface by linearly interpolating intensity values across the surface. Intensity values for each polygon are matched with the values of adjacent polygons along the common edges and eliminating the intensity discontinuities that can occur in flat shading.

32. Topic: Programming and Data Structures

(b) A full binary tree is a tree where every node has two children or no children. A full binary tree is also called a proper binary tree or 2 trees.
In a full binary tree, $L = I + 1$, where
L is number of leaf nodes
and I is number of internal nodes



Therefore, 15 nodes can form a full binary tree.

33. Topic: Engineering Mathematics

(d) Consider option (a):

$$\begin{bmatrix}1 & -1\\ 1 & 0\end{bmatrix}\begin{bmatrix}1\\ 2\end{bmatrix} = \begin{bmatrix}1 & -2\\ 1 & +0\end{bmatrix} = \begin{bmatrix}-1\\ 1\end{bmatrix}$$

Therefore, this transformation matrix does not transform $\begin{bmatrix}1\\2\end{bmatrix}$ to $\begin{bmatrix}1\\1\end{bmatrix}$

Now, consider transformation matrix in option (b):

$$\begin{bmatrix}0 & 0\\0.5 & 0.5\end{bmatrix}\begin{bmatrix}1\\2\end{bmatrix}=\begin{bmatrix}0\\0.5+0.5\end{bmatrix}=\begin{bmatrix}0\\3.5\end{bmatrix}$$

Therefore, this transformation matrix does not transform $\begin{bmatrix}1\\2\end{bmatrix}$ to $\begin{bmatrix}1\\1\end{bmatrix}$

Now, consider transformation matrix in option (c):

$$\begin{bmatrix}-1 & 0\\1 & 1\end{bmatrix}\begin{bmatrix}1\\2\end{bmatrix}=\begin{bmatrix}-1\\1+2\end{bmatrix}=\begin{bmatrix}-1\\3\end{bmatrix}$$

Therefore, this transformation matrix does not transform $\begin{bmatrix}1\\2\end{bmatrix}$ to $\begin{bmatrix}1\\1\end{bmatrix}$

Now, consider transformation matrix is option (d):

$$\begin{bmatrix}-1 & 1\\1 & 0\end{bmatrix}\begin{bmatrix}1\\2\end{bmatrix}=\begin{bmatrix}-1+2\\1+0\end{bmatrix}=\begin{bmatrix}1\\1\end{bmatrix}$$

$$\begin{bmatrix}-1 & 1\\1 & 0\end{bmatrix}\begin{bmatrix}2\\5\end{bmatrix}=\begin{bmatrix}-2+5\\2+0\end{bmatrix}=\begin{bmatrix}3\\2\end{bmatrix}$$

Therefore, this transformation matrix transforms independent vectors $\begin{bmatrix}1\\2\end{bmatrix}$ and $\begin{bmatrix}2\\5\end{bmatrix}$ to $\begin{bmatrix}1\\1\end{bmatrix}$ and $\begin{bmatrix}3\\2\end{bmatrix}$, respectively.

Hence option (d) is correct.

34. Topic: Computer Organization and Architecture

(c) In 8086, the jump condition JNBE is short jump if first operand is not below and not equal to second operand. The instruction is given as follows:

If (CF = 0) and (ZF = 0), then jump, where CF and ZF are CPU flags. Therefore option (c) is correct.

35. Topic: Computer Organization and Architecture

(c) The instruction sequence will loop 256 times before coming out of the loop. At each iteration, AL is incremented by 1 till 256 since AL is an 8-bit register. Then, AL becomes zero and the iterations stop.

36. Topic: Computer Organization and Architecture

(d) TRAP is one edge and level triggered interrupt which implies that the TRAP must go high and remain high until it is acknowledged. It is a non-maskable interrupt which remains unaffected by any mask or interrupt enable.

In 8085, there are five hardware interrupts. These interrupts and their locations are as follows:

(i) TRAP – 0024H
(ii) RST5.5 – 002CH
(iii) RST6.5 – 0034H
(iv) RST7.5 – 003CH

Therefore, ISR for handling trap interrupt is at 24H location. Thus option (d) is correct.

37. Topic: Operating System

(c) RS-232 C standard is an asynchronous serial communication method. It defines the voltage levels that correspond to logical one and logical zero levels for the data transmission and the control signal lines. Valid signals are either in the range of +3 to +15 V or –3 to –15 V.

Thus, voltage range for a high logic is –3.0 to –15.0 V and for a low logic is +3.0 to +15.0 V.

Therefore option (c) is correct.

38. Topic: Computer Networks

(a) Ethernet is a protocol in TCP/IP which describes how networked devices can format data for transmission to other network devices on the same network segment and how to put that data out on the network connection. Preamble is added at the physical layer and is not part of the frame. It consists of 56-bit pattern of alternating 0's and 1's. It allows devices on the network to easily synchronize their receiver clocks.

39. Topic: Computer Networks

(a) Switches are used as network connection point for hosts at the edge of a network layer 2 switches are used for high-speed connectivity between and stations at the data link layer. It is a network element type which gives different collision domain and same broadcast domain.

A collision domain: It is a network segment connected by shared medium where data packets may collide with one another while being sent.

Broadcast domain: It is a logical division of network where all nodes can reach each other by broadcast.

40. Topic: Computer Networks

(c) Given, frames to be transmitted is 1101011011

$$\text{Generating checksum} = x^4 + x + 1$$

Since, generating checksum is of order 4, therefore, appending 4 0's to the frame.

$$\text{CRC polynomial} = 10011$$

$$\begin{array}{r} 10011\overline{)11010110110000} \\ \underline{10011} \\ 10011 \\ \underline{10011} \\ 10110 \\ \underline{10011} \\ 10100 \\ \underline{10011} \\ 1110 \end{array}$$

Transmitting frame = 1101 0110 111110

41. Topic: Computer Networks

(b) Given, transmission time for frame = 20 ns
Propagation time = 30 ns

$$\text{Efficiency} = \frac{1}{1+2\times\left(\dfrac{\text{Propagation time}}{\text{Transmission time}}\right)}$$

$$= \frac{1}{1+2\times\left(\dfrac{30\text{ ns}}{20\text{ ns}}\right)}$$

$$= \frac{1}{(1+3)} = \frac{1}{4} = 0.25$$

Therefore, efficiency of stop and wait protocol in percentage is 25%.

42. Topic: Computer Networks

(c) IPv6 or internet protocol version 6 is a set of specifications that provides identification and location system of computers on networks and routes traffic across the internet. IPv6 supports following addressing modes:

Unicast address – Packet is delivered to one interface.
Multicast addresses – Packet is delivered to multiple interface.
Anycast address – Packet is delivered to nearest of multiple interface.

Therefore, broadcast addressing mode is not supported by IPv6.

43. Topic: Computer Networks

(b) Given, subnet mask is 255.224.0.0 converting to binary equivalent we have the following:
11 11 11 11: 11 100 000 : 00000000 : 00000000
First segment of 8 bits corresponds to network. Class since all 8 bits are 1's, this corresponds to class A. Now, the next segment of 8 bits corresponds to number of sub-networks. Since 3 bits are 1's, the number of sub network = $2^3 = 8$.

Therefore, IP class and number of sub-networks for given subnet mask is class A and B, respectively.

44. Topic: Computer Networks

(d) Leaky bucket algorithm is used to shape the bursty into a fixed rate traffic by averaging the data rate.
The leaky bucket algorithm is used to control rate in a network. In this, the input rate can vary but the output rate remains constant. This is implemented as a single server queue with constant service time. If the bucket overflows, then pockets are discarded.

45. Topic: Computer Networks

(d) Packet filtering is a firewall technique. It is used to control network access by monitoring outgoing and incoming packets and allowing them to pass or halt based on source and destination IP addresses and protocols. Thus, a packet filtering firewall disallows some files from being accessed through FTP.

46. Topic: Computer Networks

(c) Feistel structure is a symmetric key cryptography. It is used in the construction of block ciphers. Data encryption standard is based on the Festal structure. It uses same key to encrypt and decrypt a message so that both the sender and the receiver must know and use the same private key.

47. Topic: Computer Networks

(a) The protocol data unit for the transport layer in the internet stack is segment. It is a specific block of information transferred over a network. It describes different types of data that are transferred from each layer. For transport layer, it is a segment that includes TCP header and data.

48. Topic: Engineering Mathematics

(a) Gauss–Seidel method is also known as Liebman method or method of successive displacement. It is an iterative method used to solve a linear system of equation. Hence, option (a) is correct, i.e., Gauss–Seidel method can be used to solve linear algebraic equations of the form $AX = b$.

49. Topic: Engineering Mathematics

(c) Given:

$$f(x) = 2x^2 - 8x - 3 \qquad (1)$$

Now, differentiating Eq. (1) with respect to x, we get

$$f'(x) = 4x - 8 \qquad (2)$$

Now, putting $f'(x) = 0$, we get

$$4x - 8 = 0 \Rightarrow x = 2 \qquad (3)$$

Now, differentiating Eq. (2) again with respect to x, we get

$$f''(x) = 4x$$

Since

$$f''(x) > 0$$

So the function has minima at point $x = 2$.
Minimum value of function is

$$f(2) = 2 \times 2^2 - 8 \times 2 - 3 = 8 - 16 - 3 = 8 - 19 = -11$$

Therefore, the least value of the function is -11.

50. Topic: Operating System

(b) Round Robin algorithm is a CPU scheduling algorithm. In this algorithm, each process is assigned a fixed time slot in a cyclic way. Therefore, the process that comes first is executed first. In this set of process, Round Robin scheduling with time quantum 2 ms is employed:

$P1$	$P2$	$P1$	$P3$

$t = 0$ $t = 2$ $t = 4$ $t = 6$ $t = 7$

Therefore, processes are completed in the sequence $P2, P1, P3$.

51. Topic: Operating System

(b) DVD or digital video disk is an optical disk technology with 4.7 gigabytes storage capacity on a single-sided, one-layered disk. DVDs are commonly used as a medium for digital representation of movies and other multimedia presentations that combine sound with graphics. The speed of data transfer for a DVD is mentioned in multiples of 1.38 MB s^{-1}.

52. Topic: Operating System

(d) Given logical records have variable length of 5 bytes, 10 bytes and 25 bytes.
Physical block size is 15 bytes.
Since physical block is of size 15 bytes, maximum fragmentation that is observed is of record length 10 bytes and minimum fragmentation that is observed is of record length 5 bytes.

15 bytes		5	10

Physical block

53. Topic: Operating System

(c) CPU scheduling is a process which allows one process to use the CPU while the execution of another process is on hold due to unavailability of any resource. CPU scheduling is used to make system fast and efficient. Given n processes to be scheduled on one processor, then the number of possible different schedules are $n!$.

54. Topic: Operating System

(a) Thrashing is a computer activity that makes no processes or very little progress because memory or other resources have become exhausted or limited to perform operation. Thrashing can be due to page fault when page size was very small. Thus, option (a) is correct.

55. Topic: Operating System

(c) Given, logical address space is of 8 pages of 1024 words each.
Physical memory has 32 frames.
Physical Address: Since physical memory has 32 frames, that is, 2^5 and size of word = 1024 = 2^{10}, the bits in physical address = 5 + 10 = 15 bits.
Logical address: Since logical address space is 8, we have the pages $\Rightarrow 2^3$ and size of word = 1024 = 2^{10}, the bits in logical address = 10 + 3 = 13 bits.
Therefore, the bits in physical address and logical address are 15 and 13, respectively.

56. Topic: Operating System

(a) Given:

$$\text{RAM size} = 2\text{ GB}$$
$$\text{Page size} = 8\text{ KB}$$

Inverted page table is maintained by operating system. There is one table for entire system which stores one entry per physical frame. In inverted page table, for each occupied physical memory frame, there is a virtual page associated to it.

$$\text{Number of entries in page table} = \frac{\text{RAM size}}{\text{Page size}}$$
$$= \frac{2\text{ GB}}{8\text{ kB}} = \frac{2 \times 2^{30} B}{2^3 \times 2^{10} B}$$
$$= \frac{2^{31} B}{2^{13} B} = 2^{18}$$

57. Topic: Operating System

(b) Deadlock is a blocked process holding a resource and waiting for another resource held by another process, and deadlock can be avoided by avoiding one of the four conditions. These four conditions are essential to cause deadlock.

(i) **Mutual exclusion:** The resources involved are non-sharable. At least one of the resource should be non-sharable so that only one process at a time claims exclusive control of the resource.
(ii) **Hold and wait:** There exists a process that is holding a resource and waiting for on additional resource being held by another process.
(iii) **No pre-emption:** Resources already allocated to a process cannot be precomputed or removed from the processes.

(iv) **Circular wait:** The processes form a circular list where each process in the list waits for the resource held by the next process in the list.

Hence, Re-entrancy is not a necessary condition for deadlock.

58. Topic: Operating System

(c) Given, the total number of instances of resource type 1 is 5.
Total number of instances of resource type 2 is 4.
Resource type 1 allocated to process P1, P2 and P3 is

$$1 + 1 + 2 = 4$$

Therefore, instance available of resource type 1 is

$$5 - 4 = 1$$

Now, the total number of instances of resource type 2 allocated to process P1, P2 and P3 is $1 + 1 + 1 = 3$ the and instance available of resource type 2 is $4 - 3 = 1$.
These available resources 1 and 1 of type 1 and type 2, respectively, cannot handle any of the processes.
Therefore, the state of system is unsafe state.

59. Topic: Operating System

(d) Given, a starvation-free job scheduling policy guarantees that no job indefinitely waits for a service. From the given scheduling policies, Round Robin is starvation free. Round Robin scheduling is pre-emptive version of First-In First-Out, i.e., FIFO. Round Robin scheduling assigns equal time to each process without priority in cyclic order.
Therefore, what process comes in first is executed first. Since the scheduling process assigns equal, no job indefinitely awaits for a service.

60. Topic: Operating System

(b) The state of a process after it encounters on I/O instruction is blocked. A process that is blocked is one that is waiting for some event after completion of an I/O operation. A process is unable to run until the external event has happened.

61. Topic: Databases

(a) Embedded pointers are embedded in data structures, such as arrays, unions and structures. Embedded pointers are null on input and only write output to a buffer. Embedded pointer is a set of pointers in a dataset which points to another data set. Embedded pointer provides a secondary access path.

62. Topic: Operating System

(d) Given, parallel program computation requires 100 seconds when executed on single CPU. 20% of computation is strictly sequential.
For program running on two CPU's:
Since 20% of computation is sequential; therefore, CPU spends 20 secs for sequential computation.
The remaining 80% parallel computation is divided in CPU1 and CPU2 such that CPU1 works for 40 s and CPU2 works for 40 s.
Then the total time elapsed = 20 + 40 = 60 s
For program reconnecting on four CPU's:
Since 20% of sequential computation runs on CPU 1, therefore, CPU 1 spends 20 s for sequential computation.
The remaining 80% parallel computation is divided among four CPUs such that CPU 1 works for 20 s, CPU 2 works for 20 s, CPU 3 works for 20 s and CPU 4 works for 20 s.
Then total time elapsed = 20 + 20 = 40 s
Therefore, the best possible elapsed times for this program running on two CPUs and four CPUs, respectively, are 60 and 40 s.

63. Topic: Programming and Data Structures

(c) Given:

$$a = 1; \pi = 3.1415926535$$

For loop runs for $i = 0$, $i = 1$ and $i = 2$:
If $a = \cos(\text{pi} * i/2)$ condition is true print 1.
if condition fails print 0.
for $i = 0$, $\cos(0) = 1 = a$, prints 1
for $i = 1$, $\cos(\text{pi}/2) = 0 \neq a$; prints 0
for $i = 2$, $\cos(\text{pi}) = -1 = a$; prints 1
Therefore, the program will print 101.

64. Topic: Programming and Data Structures

(a) Given:

$$-x = 0, y = 0$$

$z = 0$; $x = 1$ and $y = 1$ if condition fails.
$z = 1$; $x = 2$ and $y = 2$ if condition fails.
$z = 2$; $x = 3$ if condition true.

So,

$$x = 4 \text{ and } y = 2$$

$$z = 3; x = 5 \text{ if condition true}$$

So,

$$x = 6 \text{ and } y = 2$$

$$z = 4; x = 7 \text{ if condition true}$$

So,

$$x = 8 \text{ and } y = 2$$

Therefore, the output of program will be 8 and 2.

65. Topic: Operating System

(d) Working sets are
$9 \to \{9\}$, $6 \to \{9, 6\}$, $2 \to \{9, 6, 2\}$, $3 \to \{9, 6, 2, 3\}$,
$4 \to \{9, 6, 2, 3, 4\}$, $4 \to \{6, 2, 3, 4\}$, $4 \to \{2, 3, 4\}$
$4 \to \{3, 4\}$, $3 \to \{3, 4\}$, $4 \to \{3, 4\}$, $4 \to \{3, 4\}$,
$2 \to \{3, 4, 2\}$, $5 \to \{3, 4, 2, 5\}$, $8 \to \{2, 4, 5, 8\}$,
$6 \to \{2, 4, 5, 8, 6\}$, $8 \to \{2, 5, 8, 6\}$, $5 \to \{5, 8, 6\}$,
$5 \to \{5, 8, 6\}$, $3 \to \{3, 5, 8, 6\}$, $2 \to \{2, 3, 5, 8\}$,
$3 \to \{2, 3, 5\}$, $9 \to \{2, 3, 9\}$, $6 \to \{2, 3, 9, 6\}$,

$2 \to \{3, 9, 6, 2\}$, $7 \to \{3, 9, 6, 2, 7\}$
Therefore, the penultimate page reference working set is

$$\{3, 9, 6, 2\}$$

66. Topic: Software Engineering

(c) Cyclomatic complexity is a software metric. It is used to indicate the complexity of a program. It measures independent path through program source code. Cyclomatic complexity can be calculated with respect to functions, modules, method or classes within a program.

$$\text{Cyclomatic complexity} = E - N + 2$$

where E is number of edges and N is number of nodes.
Therefore, cyclomatic complexity is $17 - 13 + 2 = 6$.

67. Topic: Software Engineering

(d) Pathological coupling: This is also known as content coupling occurs when one module modifies or relies on the internal workings of another module.
Control coupling: One module controls the flow of another by passing it information on what to do, e.g., flag, switches, etc.
Data coupling: When module shares data it is called data coupling. It occurs between two modules when data are passed by parameters using a simple argument list and every item in the list is used.
Message coupling: It is the weakest form of coupling. Message coupling is obtained by state decentralization.
Therefore, option (d) is correct.

68. Topic: Software Engineering

(d) Fault simulation technique is used to evaluate the quality of a test set and to construct fault dictionary. I consists of simulating a circuit in the presence of faults. Mutation testing uses fault simulation. In mutation testing, the source code is mutated by changing certain statements. Then it is tested for errors. Mutation testing is mainly used for unit testing.

69. Topic: Software Engineering

(c) Given, P calls $P1$ and $P2$
$P1$ fails 50% of the time
$P2$ fails 40% of the time
Program P fails if either $P1$ fails or $P2$ fails or both fails
Thus, failure rate of P is given as follows:
Fail (P) = Failure of $P1$ + Failure of $P2$ – (Failure of $P1 \cap$ Failure of $P2$)

$$= \frac{50}{100} + \frac{40}{100} - \left(\frac{50}{100} \times \frac{40}{100}\right)$$

$$\frac{90}{100} - \frac{2000}{10000} = \frac{9}{10} - \frac{2}{10} = \frac{7}{10}$$

Therefore, the failure rate of $P = 70\%$.

70. Topic: Operating System

(a) Priority inversion problem occurs in scheduling. In this, a high-priority task is indirectly pre-empted by a low-priority task thereby inverting the relative priority of the two tasks. To overcome the priority inversion problem, temporarily raise the priority of lower priority-level process.

71. Topic: Engineering Mathematics

(b) Given:

$$P(A) = 0.5$$
$$P(B) = 1$$

Then, $P(A/B)$ = Probability of occurrence of A when B has already occurred. Therefore,

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{0.5 \times 1}{1.0} = 0.5$$

Also, $P(B/A)$ = Probability of occurrence of B when A has already occurred. Therefore,

$$P(B/A) = \frac{P(B \cap A)}{P(A)} = \frac{1 \times 0.5}{0.5} = 1$$

Therefore, the values of $P(A/B)$ and $P(B/A)$ are 0.5 and 1, respectively.

72. Topic: Engineering Mathematics

(b) For a polygon of n vertices, corresponding to each vertex there are total to $(n - 3)$ diagonals.

So, the total number of diagonals is $\frac{n(n-3)}{2}$.

The total number is divided by 2 because of duplicate diagonals.
For octagon, $n = 8$. Therefore, total number diagonals

$$\frac{8(8-3)}{2} = 4 \times 5 = 20$$

73. Topic: Theory of Computation

(None) NFA stands for non-deterministic finite automata. DFA stands for deterministic finite automata. In NFA, each pair of state and input symbol can have many possible next states, while in DFA the set of next possible states are distinctly set. Thus, for the given NFA the final states of DFA should be q_0, q_2, $[q_1, q_2]$ and $[q_0, q_1, q_2]$
So, none of the options is correct.

74. Topic: Engineering Mathematics

(d) For n elements in a set:
Number of elements of power set are 2^n.
Given, set $\{\{A, B\}, C\}$
Here, number of elements = 2
Therefore, number of elements of power set = $2^2 = 4$

$$\{\phi\}, \{A, B\}, \{C\}, \{\{A, B,\}C\}$$

75. Topic: Programming and Data Structures

(b) Copy constructor creates on object by initializing it with an object of the same class which has been created previously. It is a member function which initializes an object using another object of the same class. A copy constructor has prototype.

Class Name (const classname &old-obj);

Thus, the right way to declare a copy constructor of a class named Myclass is

Myclass (constant Myclass &arg)

76. Topic: Engineering Mathematics

(a) A complete graph is a graph in which each pair of graph vertices is connected by an edge. For *n* vertices, a complete graph is denoted by following number of edges:

$$^{n}C_{2} = \frac{n(n-1)}{2}$$

These are undirected edges and thus, the number of edges are denoted by $\frac{n(n-1)}{2}$.

77. Topic: Digital Logic

(a) Binary equivalent of decimal 42 is

$1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$

So,

$$42 = (101010)_2$$

Also, binary equivalent of fraction part of decimal number is

$$75 = \frac{1 \times 1}{2^2} + 1 \times \frac{1}{2^1} + 0 \times \frac{1}{2^0}$$

Then,

$$0.75 = (0.110)_2$$

Therefore,

$$42.75 = (101010.110)_2$$

78. Topic: Computer Networks

(b) Cloud computing is a general term for the delivery of hosted services over the Internet.

The main benefits of cloud computing are self-service provisioning, elasticity, pay per use, workload resistance and migration flexibility. Cloud computing is divided into three service categories:

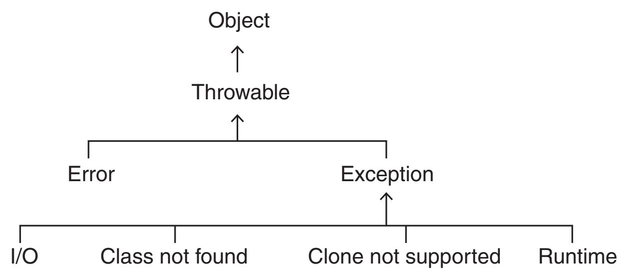
(i) Software as a Service (SaaS)
(ii) Platform as a service (PaaS)
(iii) Infrastructure as a Service (IaaS)

Therefore, Architecture, as a service, is not provided as a service in cloud computing.

79. Topic: Programming and Data Structures

(d) In build-in base class in Java, Throwable is used to handle all exceptions.

Throwable class is the superclass of all errors and exceptions in the Java Language.



80. Topic: Computer Graphics

(d) Vanishing point is the perspective image of the point where a bundle of parallel lines seems to converge. In graphics, the number of vanishing points depends on the perspective projections of any set of parallel lines that are not parallel to the projection plane. These can be found by shooting a ray in the direction of the parallel bundle and finding its intercept with the imaging plane.

Indian Space Research Organisation Computer Science 2014

QUESTION PAPER

1. Consider a 33 MHz CPU based system. What is the number of wait states required if it is interfaced with a 60 ns memory? Assume a maximum of 10 ns delay for additional circuitry like buffering and decoding.

(a) 0 (b) 1
(c) 2 (d) 3

2. The number of states required by a Finite State Machine, to simulate the behaviour of a computer with a memory capable of storing '*m*' words, with each word being '*n*' bits long is

(a) $m \times 2^n$ (b) 2^{m+n}
(c) 2^{mn} (d) $m + n$

3. What is the output of the following C program?

```
#include <stdio.h>
#define SQR(x) (x * x)
int main()
{
   int a;
   int b = 4;
   a = SQR(b + 2);
   printf("%d\n",a);
   return 0;
}
```

(a) 14 (b) 36
(c) 18 (d) 20

4. Consider the following pseudo-code

```
   while (m < n)
         if (x > y) and (a < b) then
               a = a + 1
               y = y - 1
         end if
         m = m + 1
   end while
```

What is the cyclomatic complexity of the above pseudo-code?

(a) 2 (b) 3
(c) 4 (d) 5

5. What is the number of steps required to derive the string ((() ()) ()) for the following grammar.

$$S \rightarrow SS$$
$$S \rightarrow (S)$$
$$S \rightarrow \varepsilon$$

(a) 10 (b) 15
(c) 12 (d) 16

6. The process of modifying IP address information in IP packet headers while in transit across a traffic routing device is called

(a) port address translation (PAT)
(b) network address translation (NAT)
(c) address mapping
(d) port mapping

7. What does a pixel mask mean?

(a) String containing only l's
(b) String containing only 0's
(c) String containing two 0's
(d) String containing l's and 0's

8. In the standard IEEE 754 single precision floating point representation, there is 1 bit for sign, 23 bits for fraction and 8 bits for exponent. What is the precision in terms of the number of decimal digits?

(a) 5 (b) 6
(c) 7 (d) 8

9. Let R be the radius of a circle. What is the angle subtended by an arc of length R at the centre of the circle?

(a) 1 degree (b) 1 radian
(c) 90 degrees (d) π radians

10. The number of logical CPUs in a computer having two physical quad-core chips with hyper-threading enabled is _________.

(a) 1 (b) 2
(c) 8 (d) 16

11. An aggregation association is drawn using which symbol?

(a) A line which loops back on to the same table

(b) A small open diamond at the end of a line connecting two tables
(c) A small closed diamond at the end of a line connecting two tables
(d) A small closed triangle at the end of a line connecting two tables

12. How many states are there in a minimum state deterministic finite automaton accepting the language $L = \{w \mid w \in (0,1)^*$, number of 0's is divisible by 2 and number of 1's is divisible by 5, respectively$\}$?

(a) 7 (b) 9
(c) 10 (d) 11

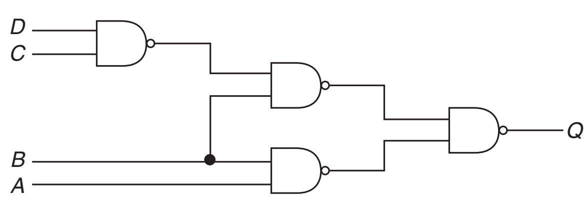
13. Which of the following is TRUE with respect to a Reference?

(a) A reference can never be NULL
(b) A reference needs an explicit dereferencing mechanism
(c) A reference can be reassigned after it is established
(d) A reference and pointer are synonymous

14. There are 200 tracks on a disk platter and the pending requests have come in the order– 36, 69, 167, 76, 42, 51, 126, 12, and 199. Assume the arm is located at the 100th track and moving towards track 200. If sequence of disc access is 126, 167, 199, 12, 36, 42, 51, 69, and 76 then which disc access scheduling policy is used?

(a) Elevator
(b) Shortest Seek-time First
(c) C-SCAN
(d) First Come First Served

15. Consider the logic circuit given below:



$Q =$ ————?

(a) $\overline{A}C + B\overline{C} + CD$
(b) $ABC + \overline{C}D$
(c) $AB + B\overline{C} + B\overline{D}$
(d) $A\overline{B} + \overline{A}C + \overline{C}D$

16. What is routing algorithm used by OSPF routing protocol?

(a) Distance vector (b) Flooding
(c) Path vector (d) Link state

17. If each address space represents one byte of storage space, how many address lines are needed to access RAM chips arranged in a 4×6 array, where each chip is $8K \times 4$ bits?

(a) 13 (b) 15
(c) 16 (d) 17

18. Consider the following segment table in segmentation scheme:

Segment Id	Base	Limit
0	200	200
1	5000	1210
2	1527	498
3	2500	50

What happens if the logical address requested is – Segment Id 2 and Offset 1000?

(a) Fetches the entry at the physical address 2527 for Segment Id 2
(b) A trap is generated
(c) Deadlock
(d) Fetches the entry at offset 27 in Segment Id 3

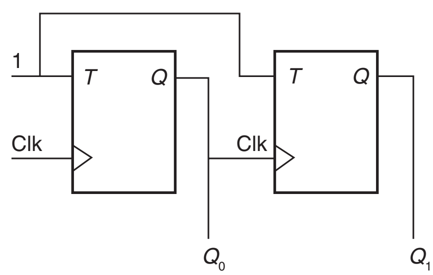
19. The number of bit strings of length 8 that will either start with 1 or end with 00 is

(a) 32 (b) 128
(c) 160 (d) 192

20. Which of the following is not a maturity level as per Capability Maturity Model?

(a) Initial (b) Measurable
(c) Repeatable (d) Optimized

21. Consider the following sequential circuit.



What are values of Q_0 and Q_1 after 4 clock cycles if the initial values are 00?

(a) 11 (b) 01
(c) 10 (d) 00

22. Consider the schema R (A, B, C, D) and the functional dependencies $A \to B$ and $C \to D$. If the decomposition is made as R_1 (A, B) and R_2 (C, D), then which of the following is TRUE?

(a) Preserves dependency but cannot perform lossless join
(b) Preserves dependency and performs lossless join
(c) Does not preserve dependency and cannot perform lossless join
(d) Does not preserve dependency but performs lossless join

23. The test suite (set of test input) used to perform unit testing on a module could cover 70% of the code. What is the reliability of the module if the probability of success is 0.95 during above testing?

(a) 0.665 to 0.95 (b) At the most 0.665
(c) At the most 0.95 (d) At least 0.665

24. In a system an RSA algorithm with $p = 5$ and $q = 11$ is implemented for data security. What is the value of the decryption key if the value of the encryption key is 27?

(a) 3 (b) 7
(c) 27 (d) 40

25. Suppose you want to build a memory with 4 byte words and a capacity of 2^{21} bits. What is type of decoder required if the memory is built using 2K × 8 RAM chips?

(a) 5 to 32 (b) 6 to 64
(c) 4 to 16 (d) 7 to 128

26. The output of a tristate buffer when the enable input in 0 is

(a) always 0
(b) always 1
(c) retains the last value when enable input was high
(d) disconnected state

27. How many different BCD numbers can be stored in 12 switches? (Assume two position or on-off switches).

(a) 2^{12} (b) $2^{12} - 1$
(c) 10^{12} (d) 10^{3}

28. Suppose there are eleven items in sorted order in an array. How many searches are required on the average, if binary search is employed and all searches are successful in finding the item?

(a) 3.00 (b) 3.46
(c) 2.81 (d) 3.33

29. Consider the following Java code fragment. Which of the following statement is true?

Line No. **Code Statement**
```
1   public class While
2   {
3     public void loop()
4     {
5       int x = 0;
6       while (1)
7         {
8         System.out.println("x
          plus one is" + (x +1));
9         }
10      }
11  }
```

(a) There is syntax error in line no. 1
(b) There are syntax errors in line nos. 1 and 6
(c) There is syntax error in line no. 8
(d) There is syntax error in line no. 6

30. Every time the attribute A appears, it is matched with the same value of attribute B but not the same value of attribute C. Which of the following is true?

(a) $A \to (B, C)$ (b) $A \to B, A \to\to C$
(c) $A \to B, C \to\to A$ (d) $A \to\to B, B \to C$

31. A IP packet has arrived in which the fragmentation offset value is 100, the value of HLEN is 5 and the value of total length field is 200. What is the number of the last byte?

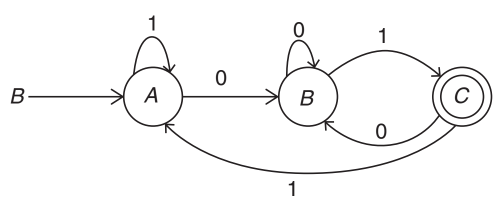
(a) 194 (b) 394
(c) 979 (d) 1179

32. What is the output of the following C program?

```
#include <stdio.h>
void main (void)
{
   int shifty;
   shifty = 0570;
   shifty = shifty >> 4;
   shifty = shifty >> 6;
   printf("The value of shifty is %o
   \n", shifty);
}
```

(a) The value of shifty is 15c0
(b) The value of shifty is 4300
(c) The value of shifty is 5700
(d) The value of shifty is 2700

33. The following Finite Automaton recognizes which of the given languages?



(a) {1, 0} * {0 1} (b) {1, 0} * {1}
(c) {1} {1, 0} * {1} (d) 1 * 0 * {0, 1}

34. How much memory is required to implement z-buffer algorithm for a 512 × 512 × 24 bit-plane image?

(a) 768 KB (b) 1 MB
(c) 1.5 MB (d) 2 MB

35. Using the page table shown below, translate the physical address 25 to virtual address. The address length is 16 bits and page size is 2048 words while the size of the physical memory is four frames.

Page	Present (1-In, 0-Out)	Frame
0	1	3
1	1	2
2	1	0
3	0	–

(a) 25 (b) 6169
(c) 2073 (d) 4121

36. Consider a standard circular queue 'q' implementation (which has same condition for queue full and queue empty) whose size is 11 and the elements of the queue are q[0], q[1],..., q[10]. The front and rear pointers are initialized to point at q[2]. In which position will the ninth element be added?

(a) q[0] (b) q[l]
(c) q[9] (d) q[10]

37. The probability that two friends are born in the same month is ——?

(a) 1/6 (b) 1/12
(c) 1/144 (d) 1/24

38. How many lines of output does the following C code produce?

```
#include <stdio.h>
float i = 2.0;
float j = 1.0;
float sum = 0.0;
main()
{
   while (i/j > 0.001)
   {
         j+=j;
         sum = sum + (i/j);
         printf("%f\n", sum);
   }
}
```

(a) 8 (b) 9
(c) 10 (d) 11

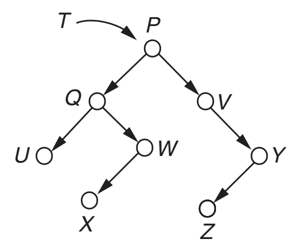
39. If only one memory location is to be reserved for a class variable, no matter how many objects are instantiated, then the variable should be declared as ________.

(a) extern (b) static
(c) volatile (d) const

40. Assume that a 16 bit CPU is trying to access a double word starting at an odd address. How many memory operations are required to access the data?

(a) 1 (b) 2
(c) 3 (d) 4

41. Consider the following binary search tree T given below. Which node contains the fourth smallest element in T?



(a) Q (b) V
(c) W (d) X

42. Let x, y, z, a, b, c be the attributes of an entity set E. If $\{x\}$, $\{x, y\}$, $\{a, b\}$, $\{a, b, c\}$, $\{x, y, z\}$ are superkeys, then which of the following are the candidate keys?

(a) $\{x, y\}$ and $\{a, b\}$ (b) $\{x\}$ and $\{a, b\}$
(c) $\{x, y, z\}$ and $\{a, b, c\}$ (d) $\{z\}$ and $\{c\}$

43. The five items: A, B, C, D, and E are pushed in a stack, one after the other starting from A. The stack is popped four times and each element is inserted in a queue. Then two elements are deleted from the queue and pushed back on the stack. Now, one item is popped from the stack. The popped item is ________.

(a) A (b) B
(c) C (d) D

44. A computer has sixteen pages of virtual address space but the size of main memory is only four frames. Initially the memory is empty. A program references the virtual pages in the order of 0, 2, 4, 5, 2, 4, 3, 11, 2, 10. How many page faults occur if LRU page replacement algorithm is used?

(a) 3 (b) 5
(c) 7 (d) 8

45. Consider a 50 kbps satellite channel with a 500 milliseconds round trip propagation delay. If the sender wants to transmit 1000 bit frames, how much time will it take for the receiver to receive the frame?

(a) 250 milliseconds (b) 20 milliseconds
(c) 520 milliseconds (d) 270 milliseconds

46. If the maximum output voltage of a DAC is V volts and if the resolution is R bits, then the weight of the most significant bit is ________.

(a) $V/(2^{R-1})$ (b) $(2^{R-1})\cdot V/(2^{R-1})$
(c) $(2^{R-1})\cdot V$ (d) $V/(2^{R-1})$

47. The following three 'C' language statements is equivalent to which single statement?

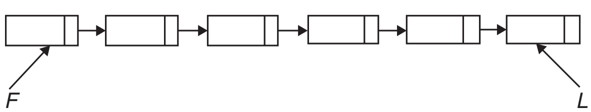
$$y = y + 1;$$
$$z = x + y;$$
$$x = x + 1;$$

(a) $z = x + y + 2;$
(b) $z = (x++) + (++y);$
(c) $z = (++x) + (y++);$
(d) $z = (x++) + (++y) + 1$

48. A frame buffer array is addressed in row-major order for a monitor with pixel locations starting from (0, 0) and ending with (100, 100). What is address of the pixel (6, 10)? Assume one bit storage per pixel and starting pixel location is at 0.

(a) 1016 (b) 1006
(c) 610 (d) 616

49. Consider a single linked list where F and L are pointers to the first and last elements respectively of the linked list. The time for performing which of the given operations depends on the length of the linked list?



(a) Delete the first element of the list
(b) Interchange the first two elements of the list
(c) Delete the last element of the list
(d) Add an element at the end of the list

50. Let A be a finite set having x elements and let B be a finite set having y elements. What is the number of distinct functions mapping B into A.

(a) x^y (b) $2^{(x+y)}$
(c) y^x (d) $y!/(y-x)!$

51. Which of the following is NOT represented in a subroutine's activation record frame for a stack-based programming language?

(a) Values of local variables
(b) Return address
(c) Heap area
(d) Information needed to access non local variables

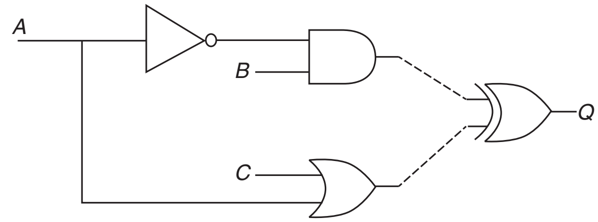
52. Consider the following grammar.

$$S \to AB$$
$$A \to a$$
$$A \to BaB$$
$$B \to bbA$$

Which of the following statements is FALSE?

(a) The length of every string produced by this grammar is even
(b) No string produced by this grammar has three consecutive a's
(c) The length of substring produced by B is always odd.
(d) No string produced by this grammar has four consecutive b's

53. Consider the logic circuit given below.



The inverter, AND and OR gates have delays of 6, 10 and 11 nanoseconds respectively. Assuming that wire delays are negligible, what is the duration of glitch for Q before it becomes stable?

(a) 5 (b) 11
(c) 16 (d) 27

54. The conic section that is obtained when a right circular cone is cut through a plane that is parallel to the side of the cone is called

(a) parabola (b) hyperbola
(c) circle (d) ellipse

55. An IP packet has arrived with the first 8 bits as 0100 0010. Which of the following is correct?

(a) The number of hops this packet can travel is 2.
(b) The total number of bytes in header is 16 bytes.
(c) The upper layer protocol is ICMP.
(d) The receiver rejects the packet.

56. Which of the following is not a valid Boolean algebra rule?

(a) $X \cdot X = X$
(b) $(X + Y)\cdot X = X$
(c) $X + XY = Y$
(d) $(X + Y)\cdot (X + Z) = X + YZ$

57. A supernet has a first address of 205.16.32.0 and a supernet mask of 255.255.248.0. A router receives four packets with the following destination addresses. Which packet belongs to this supernet?

(a) 205.16.42.56 (b) 205.17.32.76
(c) 205.16.31.10 (d) 205.16.39.44

58. Assume the following information.

Original timestamp value = 46
Receive timestamp value = 59
Transmit timestamp value = 60
Timestamp at arrival of packet = 69
Which of the following statements is correct?

(a) Receive clock should go back by 3 milliseconds
(b) Transmit and receive clocks are synchronized
(c) Transmit clock should go back by 3 milliseconds
(d) Receive clock should go ahead by 1 millisecond

59. Which of the following is FALSE with respect to possible outcomes of executing a Turing Machine over a given input?

(a) It may halt and accept the input
(b) It may halt by changing the input
(c) It may halt and reject the input
(d) It may never halt

60. Suppose you are browsing the world wide web using a web browser and trying to access the web servers. What is the underlying protocol and port number that are being used?

(a) UDP, 80 (b) TCP, 80
(c) TCP, 25 (d) UDP, 25

61. A mechanism or technology used in Ethernet by which two connected devices choose common transmission parameters such as speed, duplex mode and flow control is called

(a) Autosense (b) Synchronization
(c) Pinging (d) Autonegotiation

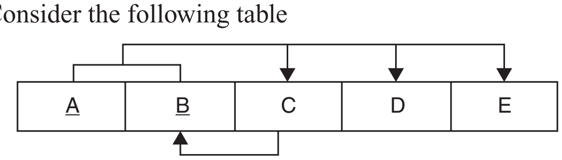
62. Consider the following sorting algorithms.

I. Quicksort
II. Heapsort
III. Mergesort

Which of them perform in least time in the worst case?

(a) I and II only (b) II and III only
(c) III only (d) I, II and III

63. Consider the following table



The table is in which normal form?

(a) First Normal Form
(b) Second Normal Form
(c) Third Normal Form but not BCNF
(d) Third Normal Form and BCNF

64. Consider a 13 element hash table for which f(key) = key mod 13 is used with integer keys. Assuming linear probing is used for collision resolution, at which location would the key 103 be inserted, if the keys 661, 182, 24 and 103 are inserted in that order?

(a) 0 (b) 1
(c) 11 (d) 12

65. A cube of side 1 unit is placed in such a way that the origin coincides with one of its top vertices and the three axes run along three of its edges. What are the coordinates of the vertex which is diagonally opposite to the vertex whose coordinates are (1, 0, 1)?

(a) (0, 0, 0) (b) (0, –1, 0)
(c) (0, 1, 0) (d) (1, 1, 1)

66. Consider a system where each file is associated with a 16-bit number. For each file, each user should have the read and write capability. How much memory is needed to store each user's access data?

(a) 16 KB (b) 32 KB
(c) 64 KB (d) 128 KB

67. What is the time complexity for the following C module? Assume that $n > 0$;

```
int module(int n)
{
   if(n = = 1)
         return 1;
   else
         return (n+module(n-1));
}
```

(a) $O(n)$ (b) $O(\log n)$
(c) $O(n^2)$ (d) $O(n!)$

68. What is the minimum number of resources required to ensure that deadlock will never occur, if there are currently three processes P_1, P_2, and P_3 running in a system whose maximum demand for the resources of same type are 3, 4 and 5, respectively.

(a) 3 (b) 7
(c) 9 (d) 10

69. For a software project, the spiral model was employed. When will the spiral stop?

(a) When the software product is retired

(b) When the software product is released after Beta testing
(c) When the risk analysis is completed
(d) After completing five loops

70. Dirty bit is used to indicate which of the following?
(a) A page fault has occurred
(b) A page has corrupted data
(c) A page has been modified after being loaded into cache
(d) An illegal access of page

71. Which of the following is not a valid multicast MAC address?
(a) 01:00:5E:00:00:00
(b) 01:00:5E:00:00:FF
(c) 01:00:5E:00:FF:FF
(d) 01:00:5E:FF:FF:FF

72. The rank of the matrix $A = \begin{bmatrix} 1 & 2 & 1 & -1 \\ 9 & 5 & 2 & 2 \\ 7 & 1 & 0 & 4 \end{bmatrix}$ is
(a) 0 (b) 1
(c) 2 (d) 3

73. How many different trees are there with four nodes A, B, C and D?
(a) 30 (b) 60
(c) 90 (d) 120

74. What is the median of the data if its mode is 15 and the mean is 30?
(a) 20 (b) 25
(c) 22.5 (d) 27.5

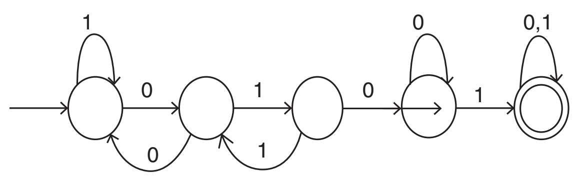
75. An organization is granted the block 130.34.12.64/26. It needs to have four subnets. Which of the following is not an address of this organization?
(a) 130.34.12.124 (b) 130.34.12.89
(c) 130.34.12.70 (d) 130.34.12.132

76. Consider the following scenario.
A web client sends a request to a web server. The web server transmits a program to that client and is executed at client. It creates a web document. What are such web documents called?
(a) Active (b) Static
(c) Dynamic (d) Passive

77. What is the size of the physical address space in a paging system which has a page table containing 64 entries of 11 bit each (including valid/invalid bit) and a page size of 512 bytes?
(a) 2^{11} (b) 2^{15}
(c) 2^{19} (d) 2^{20}

78. Which of the following is not an optimization criterion in the design of a CPU scheduling algorithm?
(a) Minimum CPU utilization
(b) Maximum throughput
(c) Minimum turnaround time
(d) Minimum waiting time

79. Consider the following Deterministic Finite Automaton M.



Let S denote the set of eight bit strings whose second, third, sixth and seventh bits are 1. The number of strings in S that are accepted by M is
(a) 0 (b) 1
(c) 2 (d) 3

80. A computing architecture, which allows the user to use computers from multiple administrative domains to reach a common goal is called as
(a) grid computing (b) neural networks
(c) parallel processing (d) cluster computing

ANSWER KEY

1. (c)	**5.** (a)	**9.** (b)	**13.** (a)	**17.** (d)	**21.** (d)	**25.** (a)	**29.** (d)	**33.** (a)	**37.** (b)
2. (c)	**6.** (b)	**10.** (d)	**14.** (c)	**18.** (b)	**22.** (a)	**26.** (d)	**30.** (b)	**34.** (c)	**38.** (d)
3. (a)	**7.** (d)	**11.** (b)	**15.** (c)	**19.** (c)	**23.** (b)	**27.** (d)	**31.** (c)	**35.** (d)	**39.** (b)
4. (c)	**8.** (c)	**12.** (c)	**16.** (d)	**20.** (b)	**24.** (a)	**28.** (a)	**32.** (d)	**36.** (b)	**40.** (c)

41. (c)	**45.** (d)	**49.** (c)	**53.** (a)	**57.** (d)	**61.** (d)	**65.** (b)	**69.** (a)	**73.** (d)	**77.** (c)
42. (b)	**46.** (b)	**50.** (a)	**54.** (a)	**58.** (a)	**62.** (b)	**66.** (a)	**70.** (c)	**74.** (b)	**78.** (a)
43. (d)	**47.** (b)	**51.** (c)	**55.** (d)	**59.** (b)	**63.** (c)	**67.** (a)	**71.** (d)	**75.** (d)	**79.** (c)
44. (c)	**48.** (d)	**52.** (d)	**56.** (c)	**60.** (b)	**64.** (b)	**68.** (d)	**72.** (c)	**76.** (a)	**80.** (a)

*** None of the options is correct.**

ANSWERS WITH EXPLANATION

1. Topic: Computer Organization and Architecture

(c) A wait state is a delay which occurs when computer processor is accessing another device that is slow to respond.
Given:

$$\text{CPU speed} = 33 \text{ MHz}$$

This implies 1 clock time $= \dfrac{1}{33 \text{ MHz}} = 30.30 \times 10^{-9}$ s

Now, the given access time = 60 ns and 10 ns delay
So, the total access time = 60 ns + 10 ns = 70 ns

$$\text{Therefore, number of wait states} = \frac{\text{Access time}}{\text{Clock time}} = \frac{70 \times 10^{-9}}{30.30 \times 10^{-9}} = 2.31 \approx 2$$

2. Topic: Theory of Computation

(c) A finite state machine (FSM) is a device that has limited or finite number of possible states. At any time, memory will be in one configuration and in next instance it may remain in same configuration or goes to a different configuration.
Here, we need to find number of configurations possible. Given, memory is capable of storing 'm' words and each word is n bits long therefore, total number of bits = $m \times n$
Number of possible configuration = $2^{m \times n} = 2^{mn}$

3. Topic: Programming and Data Structures

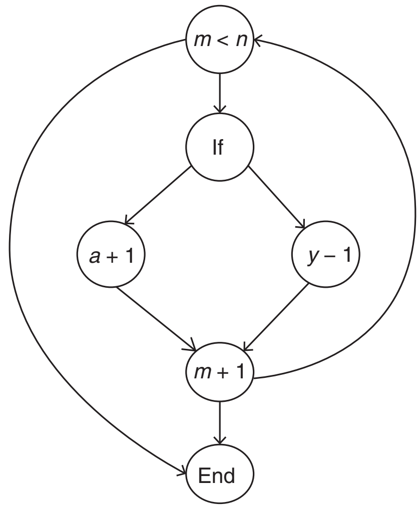
(a) The program is executed as follows:

$$\text{int } a, \text{ int } b = 4 \text{(initialize)}$$
$$a = \text{sqr } (b+2) \text{ (macro function call)}$$
$$\Rightarrow a = b+2*b+2 = 4+2*4+2 = 4+8+2$$
$$a = 14$$

Therefore, output of program is 14.

4. Topic: Software Engineering

(c) Cyclomatic complexity is a quantitative measure of number of linearly independent paths:



Cyclomatic complexity is 8 – 6 + 2 = 4.

5. Topic: Compiler Design

(a) In order to derive the string ((() ()) ()), 10 steps are required:

Step 1: (S)
Step 2: (SS)
Step 3: ((S)S)
Step 4: ((SS)S)
Step 5: (((S)S)S)
Step 6: ((()S)S)
Step 7: ((()(S))S)
Step 8: ((()())S)
Step 9: ((()())(S))
Step 10: ((()())())

6. Topic: Computer Networks

(b) The process of modifying IP address information in IP packet headers while in transit across a traffic routing device is called network address translation (NAT). NAT is the process where a network device, usually a firewall, assigns a public address to a computer inside a private network. It operates on a router, connecting two networks together.

7. **Topic: Image Processing**

   (d) A pixel mask is a special type of non-negative integer image used to select a small set of pixels or regions in an image. Pixel mask is formed by digits 0 and 1. It is used to determine positions to create a plot within the line path. Hence, option (d) is correct. Pixel mask is string containing 1's and 0's.

8. **Topic: Digital Logic**

   (c) Format of IEEE –754 is

$$(-1)^S 1\cdot M^{E-127}$$

   where S is sign, M is mantissa and E is exponent.
   Given: $S = 1$ bit, $M = 23$ bits, $E = 8$ bits
   Precision is represented as 1.M so total bits = 24
   Now, these 24 bits can be represented by 2^{24} number. In decimal, this is represented as follows:

$$2^{24} = 10^x$$

   Taking log on both sides, we have

$$\log_2 2^{24} = \log_2 10^x \Rightarrow 24 = x\log_2 10 \Rightarrow x = 7.22$$

   Therefore, $x \simeq 7$ or precision in decimal is 7.

9. **Topic: Engineering Mathematics**

   (b) We know that

$$l = r\theta \qquad (1)$$

   where l is length of the arc, θ is the angle subtended by arc and r is the radius of the circle.
   In the given question, it is given that radius of circle = R; length of the arc = R.
   So, using Eq. (1), we get

$$R = R\theta \Rightarrow \theta = 1$$

   Therefore, the angle subtended by the arc is 1 radian.

10. **Topic: Computer Organization and Architecture**

   (d) Hyper-threading is a technology that allows a single microprocessor to act like two separate processors to the operating system. It is given that computer is having two physical quad-core chips.
   Therefore, the total number of physical CPU = 2 × 4 = 8.
   Thus, the number of logical CPUs = 2 × 8 = 16.

11. **Topic: Databases**

   (b) There are three primaries inter-object relationships in database management systems, namelyassociation, aggregation and composition. Aggregation is a feature of object relationship model that allows a relationship set to participate in another relationship set. It is a special case of association. An aggregation association is drawn using a small open diamond at the end of a line connecting two tables.

12. **Topic: Theory of Computation**

   (c) In automata theory, deterministic finite automaton minimization transforms a given deterministic finite automaton into an equivalent deterministic finite automaton that has minimum number of states.
   Here, it is given that minimum state deterministic finite automaton accepts language: $L = \{\omega/\omega \in \{0,1\}^*$, number of 0's is divisible by 2 and numbers of 1's divisible by 5, respectively$\}$.
   Now, if number of 0's is divisible by x and number of 1's divisible by y, then number of states is given by $x \times y$.
   Here, $x = 2$ and $y = 5$.
   Therefore, number of states is 2 × 5 = 10.

13. **Topic: Programming and Data Structures**

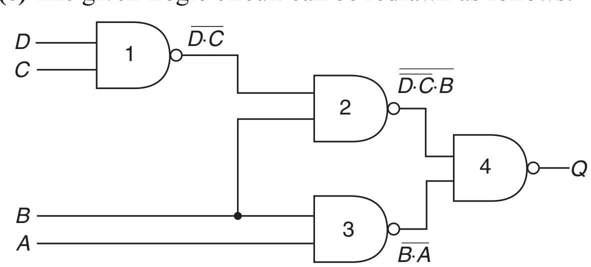
   (a) A reference is a special datatype used in C++ to refer an object. It cannot be reassigned after it is established. A reference cannot be null.
   A pointer is used in C to refer an address. It can be reassigned after it is established. A pointer can be null.
   Therefore, from the given statements only statement given in option (a) is true that a reference can never be NULL.

14. **Topic: Operating System**

   (c) For the given problem, C-SCAN scheduling policy is used. C-SCAN is a variant of SCAN scheduling. In this policy, head moves from one end of the disk to other end, serving requests along the way. When head reaches the end, it returns to the beginning without servicing requests.

15. **Topic: Digital Logic**

   (c) The given Logic circuit can be redrawn as follows:



   Output from logic gate 1, which is NAND gate, is $\overline{D\cdot C}$
   Output from logic gate 2, which is a NAND gate, is

$$\overline{(\overline{D\cdot C})\cdot B}$$

   Output from logic gate 3, which is a NAND gate, is $\overline{B\cdot A}$
   Output from logic gate 4, which is a NAND gate, is

$$Q = \overline{(\overline{B\cdot A})\cdot(\overline{\overline{D\cdot C}\cdot B})}$$

Now simplifying the expression:

$$Q = \overline{(\overline{B \cdot A}) \cdot (\overline{\overline{D \cdot C} \cdot B})} = \overline{(\overline{B \cdot A})} + \overline{\overline{\overline{D \cdot C} \cdot B}}$$

(using De Morgan's law $\overline{A \cdot B} = \overline{A} + \overline{B}$)

$$= B \cdot A + \overline{D \cdot C} \cdot B \qquad (\because \overline{\overline{A}} = A)$$

$$= B \cdot A + (\overline{D} + \overline{C}) \cdot B \qquad (\because \overline{A \cdot B} = \overline{A} + \overline{B})$$

$$= B \cdot A + \overline{D} \cdot B + \overline{C} \cdot B$$

$$= AB + B\overline{C} + B\overline{D}$$

16. Topic: Computer Networks

(d) OSPF or open shortest path first is a router protocol used to find best path for packets that pass through connected networks. It uses link state routing algorithm. In link state routing algorithm, every node constructs a map of the connecting to the network showing which nodes connected to each other. Then, each node calculates best logical path to every possible destination network.

17. Topic: Computer Organization and Architecture

(d) Given, RAM chips are arranged in 4 × 6 array:

$$4 \times 6 = 24 \text{ chips}$$

Now, $2^5 = 32 > 24$, so 5 bits are required to address this array.Also each chip is $8k \times 4$ bits.

So, the number of bytes in RAM $= \dfrac{8 \times 10^3 \times 4}{8} = 4 \times 10^5$

and the number of bytes in RAM $= 4k = 2^{12}$
Therefore, 12 bits are required to represent this.
Hence, the total address lines required is

$$12 + 5 \text{ bits} = 17 \text{ bits.}$$

18. Topic: Operating System

(b) The logical address requested is segment Id2 and offset 1000 then a trap is generated. A trap is an exception or fault that is caused by an exceptional condition.
Here, offset is 1000 while size of segment is 498. This causes an exception and hence a trap gets generated.

19. Topic: Engineering Mathematics

(c) Given, bit string is of length 8.
If the string starts with 1: First place is fixed and seven places can have two choices of '0' or '1'. So, the total number of bit strings can be $2^7 = 128$.
If the string ends with 00: Two places are fixed at the end and six places can have two choices of '0' or '1'. So, the total number of bit strings can be $2^6 = 64$.
Now, some of the bit strings will be common to both the cases. Those can be found by fixing 1 place for 1 at start and 2 places for 00 at end. So, three places are fixed and five can have two choices of '0' or '1'. So, such bit strings are $2^5 = 32$.
Therefore, the total number is 128 + 64 – 32 = 160.

20. Topic: Software Engineering

(b) Capability Maturity Model (CMM) is a methodology used for software development and for refinement of software development process. It specifies an increasing series of levels of software development. Higher the level, better is the software development process. The different levels of capability maturity model are listed as follows:

Level 1: Initial
Level 2: Repeatable
Level 3: Defined
Level 4: Managed
Level 5: Optimizing

Therefore, the measurable is not a maturity level as per capability maturity model.

21. Topic: Digital Logic

(d) The given sequential circuit has T flip–flop. A T flip–flop is a clocked flip–flop, which toggles the state if input is 1 and does not change state if input is 0.

Initially state is 00 $\Rightarrow Q_0 = 0$ and $Q_1 = 0$
Clock 1: input T is 1 $\Rightarrow Q_0 = 0$ and $Q_1 = 0$
Clock 2: input $T = 1$ and clock $= 0 \Rightarrow Q_0 = 0$ and $Q_1 = 1$
Clock 3: input $T = 1 \Rightarrow Q_0 = 1$ and $Q_1 = 0$
Clock 4: input $T = 1$ and clock $= 0 \Rightarrow Q_0 = 0$ and $Q_1 = 0$

The value of Q_0 and Q_1 after 4 clocks will be 00.

22. Topic: Databases

(a) Given, schema R (A, B, C, D) and functional dependencies $A \to B$ and $C \to D$.
A decomposition $R_1(A, B)$ and $R_2(C, D)$ preserves dependcy but cannot perform lossless join because $R_1 \cap R_2$ is empty and we know that decomposition is lossless if there is at least one of the favoring dependencies is

$$R_1 \cap R_2 \to R_1 \text{ or } R_1 \cap R_2 \to R_2$$

23. Topic: Software Engineering

(b) Given, test suit used for testing could cover 70% of the code $\Rightarrow \dfrac{70}{100} = 0.7$

Probability of success = 0.95

Therefore, reliability = 0.70 × 0.95 = 0.665
For remaining 30% of the code, reliability cannot be calculated without testing. Thus, the reliability of the module during testing can be at most 0.665.

24. Topic: Computer Networks

(a) RSA is asymmetric cryptography algorithm. It is one of the most popular and secure public-key encryption method. Using an encryption key (e, n), the message is

encrypted by raising e^{th} power modulo n. This is called cipher text message c. To decrypt message c, raise it to power d modulo n.
Relation between e, d and n are as follows:

$$n = p \times q$$

where p and q are prime numbers.

$$\text{GCD}(d, (p-1)*(q-1)) = 1$$

and

$$e*d \bmod((p-1)*(q-1)) = 1$$

Given:

$$p = 5 \text{ and } q = 11$$

and

$$\text{encryption key } e = 27$$

So,

$$n = p*q = 5\times 11 = 55$$
$$(d*e)\%((p-1)\times(q-1)) = 1$$
$$\Rightarrow (d\times 27)\%\ 4\times 10 = 1$$
$$\Rightarrow (d\times 27)\%\ 40 = 1$$

For $d = 3$,

$$(3 \times 27)\ \%\ 40 = 81\%\ 40 = 1$$

Therefore, decryption key $d = 3$.

25. Topic: Digital Logic

(a) Given:

$$\text{Capacity} = 2^{21}\text{ bits} = \frac{2^{21}}{2^3}\text{ bytes}$$
$$= 2^{18}\text{ bytes}$$

and size of word is

$$4 \text{ byte} = 2^2 \text{ bytes}$$

So, the number of words is

$$\frac{2^{18}}{2^2} = 2^{16}\text{ words}$$

Now, it is given that memory is built using 2k × 8 RAM chips $\Rightarrow$ RAM capacity $= 2\times 2^{10} = 2^{11}$ words

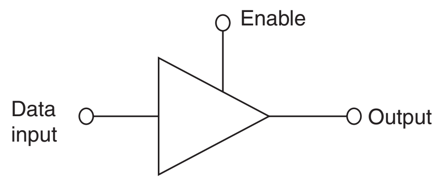
$$\text{So, number of decoder required} = \frac{\text{Number of words}}{\text{RAM capacity}}$$

$$= \frac{2^{16}}{2^{11}} = 2^5 = 32$$

Therefore, 5 to 32 decoders are required.

26. Topic: Digital Logic

(d) A tristate buffer is an input-controlled switch. The output of this can be controlled and turned ON or OFF by means of external control or enable input signal.



Truth table of tristate buffer is

Enable	Input	Output
0	0	Disconnects
0	1	Disconnects
1	0	0
1	1	1

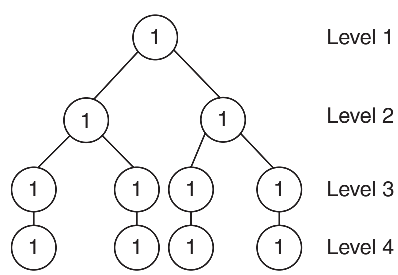
Therefore, the output of tristate buffer is disconnected state when enable input is 0.

27. Topic: Digital Logic

(d) We know that 1 switch is equivalent to 1 bit. It is given that there are 12 switches which means 12 bits. Now, we know that BCD code or binary coded decimal is represented by 4 bit binary number. Thus, the given 12 bits can be grouped into 3 groups of 4 bits each.
Now, each group can store 10 numbers.
Thus, a total of 10 × 10 × 10 BCD numbers can be stored in 12 switches. Thus, correct option is 10^3.

28. Topic: Algorithms

(a) Binary search is used to search a sorted array by dividing the search interval in half. In order to sort an array of 11 items, a binary search tree is constructed. Thus at level 1, one comparison is required, at level 2, two comparisons and so on.



Also, there are 1, 2, 4 and 4 items at level 1, 2, 3 and 4, respectively. Therefore, Average number of comparisons is

$$\frac{1\times 1 + 2\times 2 + 4\times 3 + 4\times 4}{11}$$
$$= \frac{1+4+12+16}{11} = \frac{33}{11} = 3$$

29. Topic: Programming and Data Structures

(d) In the given code fragment, line 6 will give a syntax error. "While (1)" will give a compile time error because in Java, there will be a type mismatch in conversion of int to bool. Therefore, among the given statements, option (d) is true.

30. Topic: Databases

(b) It is given that attribute A is matched with same value of attribute B but not same value of attribute C. Thus, it is multivalued dependency service there is more than one independent multivalued attributes. Therefore, $A \rightarrow B, A \rightarrow\rightarrow C$ is true among all given options.

31. Topic: Computer Networks

(c) Given:

Fragmentation offset value = 100

HLEN value = 5

Total length field = 200

Now, data field after excluding headed will be

$$200 - 4 \times 5 = 200 - 20 = 180$$

So, data field is 0 to 179.
Also, number of bytes ahead from the given packet is

$$100 \times 8 = 800$$

Therefore, last byte = 800 + 179 = 979.

32. Topic: Programming and Data Structures

(d) On program execution, we have

$$\text{shifty} = 0570$$

In binary representation, shifty is given as

$$000\ 101\ 111\ 000$$

Now,

$$\text{shifty} = \text{shifty} \gg 4; \text{ right shifts the bits by 4}$$

So,

$$\text{shifty} = 0\ 1\ 0\ 1\ 1\ 1$$

Next,

$$\text{shifty} = \text{shifty} \ll 6; \text{ left shifts the bits by 6}$$

So,

$$\text{shifty} = 010\ 111\ 000\ 000$$

$\Rightarrow$ shifty = 2700
Therefore, the value of shifty is 2700.

33. Topic: Theory of Computation

(a) Finite automaton is a simple machine that recognizes patterns from given options {1, 0} * {1}. This will generate on invalid string whose output will be rejected.

{1} {1, 0}*{1}, this will be an invalid string whose output will be rejected.
1 * 0 * {0, 1} is invalid, so output is rejected.
{1, 0} * {0, 1} will generate strings like 1001 whose output is accepted.

34. Topic: Computer Graphics

(c) The *z* buffering algorithm is used in 3D graphics. It ensures that perspective works the same way in the virtual world as it does in the real world. It is known as visual determination algorithm or VSD algorithm.
Given:

Plane image has $512 \times 512 \times 24$ bits

$$\text{Frame buffer size} = \frac{512 \times 512 \times 24}{8} = 786432 \text{ bytes}$$

$$\begin{aligned}\text{Memory required} &= 2 \times \text{Frame buffer size}\\ &= 2 \times 786432 = 1572864 \text{ bytes}\\ &= 1.572864 \text{ MB} \approx 1.5 \text{ MB}\end{aligned}$$

35. Topic: Operating System

(d) We have to translate physical address 25 to virtual address.
Given:

Address length = 16 bits
Page size = 2048
Size of physical memory = 4 frames

Now, we know that 4 frames = 2048 words = 11 bits.
Also, physical address 25 in binary is represented as 0000 0000 11001 have 2 bits represents frame number and 11 bits represent word. That is, 00 00000011001 from table, frame 0 corresponds to page 2. In binary, page 2 can be represented as 00010. Thus, physical address (13 bits) in virtual address (16 bit) is represented as 00010 000000 11001 = 4121

36. Topic: Programming and Data Structures

(b) Given circular queue of size 11 with elements *q*[0], *q*[1], …, *q*[10] the front and rear pointer are initialized at *q*[2]:

0	1	2	3	4	5	6	7	8	9	10
		↑								

Since front points at *q*[2], 0th element will be added at *q*[3], 1st element at *q*[4], and so on 7th element will be added at *q*[10], 8th at *q*[0] and 9th at *q*[1].
So, correct option is (b).

37. Topic: Engineering Mathematics

(b) First friend could have born in any of the 12 months.

So, probability is $\frac{12}{12}$.

The second friend can also have born in any of the 12 months but only one case is favorable in which first friend is also born.

So, probability is $\frac{1}{12}$.

Therefore, probability that two friends are born in same month is

$$= \frac{12}{12} \times \frac{1}{12} = \frac{1}{12} \text{ [option (b)]}$$

38. Topic: Programming and Data Structures

(b) Here, $i = 2.0$ and $j = 1.0$. While the $\frac{i}{j} > 0.001$ condition is true, j increases are 2^n for the condition $\frac{i}{j} > 0.001$ to be false: $j > 2000$. Then,

$$2^n > 2000 \Rightarrow \text{for } n = 11, 2048 > 2000$$

So, till $n = 11$ while condition will be true.
Therefore, the given C code produces 11 lines of output.

39. Topic: Programming and Data Structures

(b) If only one memory location is to be reserved for a class variable, no matter how many objects are instantiated, the variable should be declared as static. Static members of a class share same memory. Static member variables are shared by all objects of the class.

40. Topic: Computer Organization and Architecture

(c) For a 16-bit CPU, to access a single word beginning at odd address, two memory operations are required.
To access a single word beginning at even address, one memory operation is required.
To access a double word beginning at odd address, three memory operation is required.
To access a double word beginning at even address, two memory operation is required.
Therefore, correct option is (c).

41. Topic: Programming and Data Structures

(c) A binary search tree is a node-based tree data structure. The left subtree of node has nodes with key less than the node's key and the right subtree was nodes with key more than the node's key. The traversal of binary search tree given sorts the elements as follows:

$$U\ Q\ X\ W\ P\ V\ Z\ Y$$

Therefore, the fourth smallest element is T is W.

42. Topic: Databases

(b) A superkey is a set of attributes whose value uniquely identifies a topic. A candidate key is a minimal set of attributes necessary to identify a tuple. A candidate key is also called a minimal superkey.
Here, the entity set:

$$E = x, y, z, a, b, c$$

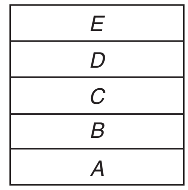
and superkeys are

$$\{x\}, \{x, y\}, \{a, b\}, \{a, b, c\}, \{x, y, z\}$$

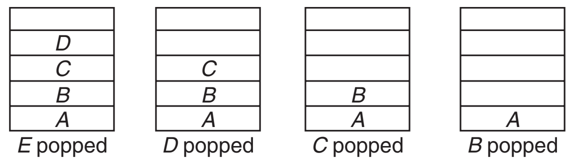
Then, $\{x\}$ and $\{a, b\}$ are the candidate keys.

43. Topic: Programming and Data Structures

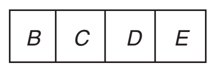
(d) *ABCDE* are pushed in stack, so we have the following:



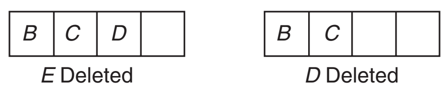
Next, four times are popped from the stack and inserted in a queue, we have the following in stack:



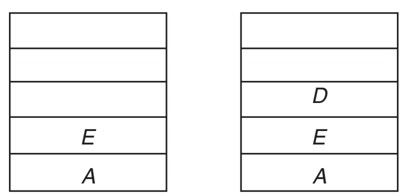
and in queue, we have the following:



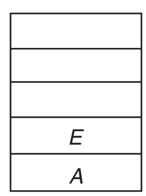
Then, two elements are deleted from queue; so, in queue, we have the following:



Next, the deleted elements are pushed back in stack; so, in stack we have the following:



Then, one element is popped, which is shown as below:



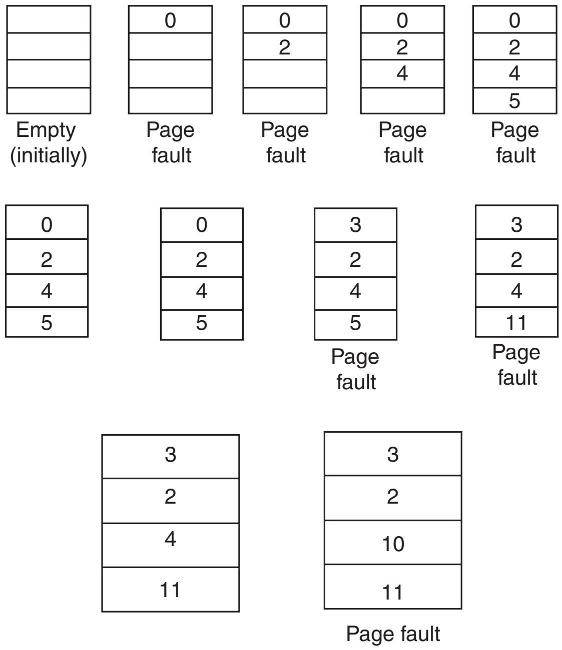
So, the popped element is '*D*'.

44. Topic: Operating System

(c) Given, memory has four frames and program references the virtual pages in order of

$$0, 2, 4, 5, 2, 4, 3, 11, 2, 10$$

So, we have the following:



So, 7-page fault occurs.

45. Topic: Computer Networks

(d) Given bandwidth of satellite channel = 50 kbps

Round trip time = 500 m sec
Size of packet to be transmitted = 1000 bit

Now,

Round trip time = 2 × Propagation time

$$\Rightarrow \text{Propagation time} = \frac{1}{2} \times \text{Round trip time}$$
$$= \frac{1}{2} \times 5000 \text{ ms} = 250 \text{ ms}$$

and the transmission time is

$$\frac{\text{Packet size}}{\text{Bandwidth of channel}} = \frac{1000 \text{ bits}}{50 \text{ kbps}} = 20 \text{ ms}$$

Time to receive the frame is

Propagation time + Transmission time
= 250 ms + 20 ms = 270 ms

46. Topic: Computer Organization and Architecture

(b) In Digital to analog converter (DAC), every input has a weight and the output voltage is the sum of the weights of 1's in the input.
Given, maximum output voltage is V volts.

Resolution = R bits

Then,

$$\text{Weight of least significant bit} = \frac{V}{2^R - 1}$$

$$\text{Weight of second least significant bit} = \frac{2V}{2^R - 1}$$

$$\text{Weight of third least significant bit} = \frac{4V}{2^R - 1}$$

Therefore, the weight of the most significant bit is

$$\frac{(2^R - 1)V}{2^R - 1}$$

47. Topic: Programming and Data Structures

(b) Given:

$y = y + 1$; increments y by 1

Then, x and y are added as z in the statement:

$$z = x + y;$$

Then, x is incremented by 1 in the statement:

$$x = x + 1;$$

Therefore, y is pre-incremented end x is post-incremented and added together as z so a single statement equivalent to all the statement will be

$$z = (x++) + (++y)$$

48. Topic: Computer Graphics

(d) Address of location in row major is

$$A[i][j] = B + w * [N * (i - L_r) + (j - L_c)]$$

where B is base address, w is storage size of 1 element, N is number of row, i is row subscript, j is column subscript, L_r is lower limit of row and L_c is lower limit of column.
Therefore,

$$A = 0 + 1[101 \times (6 - 0) + (10 - 0)] = 616$$

49. Topic: Programming and Data Structures

(c) To delete the first element of the list, pointer F that is pointing to the first element of linked list will be set to the next node of the linked list. This operation is independent of the length of linked list.
To interchange the first two elements of the list, a temporary node is required to perform swapping. This operation is independent of the length of linked list.
To add an element at the end of the list, the pointer l that is pointing at the last element of linked list will be set to the next node of the linked list. This operation is independent of the length of linked list.
To delete the last element of the list, the address of the last node is required to be deleted which requires linear

search of the list so it depends on length of the linked list.

50. Topic: Engineering Mathematics

(a) Given: Set A has x elements; Set B has y elements.
Now, in order to map B onto set A, each element in B has x choices.
Therefore, the total number of functions that map set B into set A is

$$\underbrace{x \times x \times x \times \cdots \times x}_{y\text{ times}} = x^y$$

So, the correct option is (a).

51. Topic: Programming and Data Structures

(c) Activation record frame has same or all of the following:

(i) Arguments
(ii) Link to caller's stack frame
(iii) Return address
(iv) Local variables
(v) Copies of called-save registers
(vi) Spilled registers
(vii) Information needed to access non-local variables hence option (c), heap area is not represented in activation record frame.

52. Topic: Compiler Design

(d) From the given grammar, we have following:

$$S \rightarrow AB$$
$$A \rightarrow a$$
$$A \rightarrow BaB$$
$$B \rightarrow bbA$$

We get

$$S \rightarrow AB$$
$$S \rightarrow aB$$
$$S \rightarrow abbA$$
$$S \rightarrow abbBaB$$
$$S \rightarrow abbbbAabbA$$
$$S \rightarrow abbbbaabba$$

So, the length of the string produced by this grammar is even. This statement is true.
The string does not have three consecutive a's; this statement is true.
The length of substring produced by B is always odd; this statement is true.
The string has no 4-consecutive b's; this statement is false as the string has 4 consecutive b's.
So, the correct option is (d).

53. Topic: Digital Logic

(a) Given, delays of inverter, AND and OR are 6, 10 and 11, respectively.
Inverter and OR operations start at some time and AND operation waits for the output from inverter.
So, the day time required for inverter and AND operation to complete is

$$6 + 10 = 16 \text{ ns}$$

and the delay time of OR is 11 ns.
So, the duration of glitch is 16 ns – 11 ns = 5 ns.

54. Topic: Engineering Mathematics

(a) A circle cuts the axis at right angles.
An ellipse cuts the axis at angle $\theta < 90°$.
A parabola cuts the plane parallel to the side of the cone.
A hyperbola cuts the axis at angle $\theta = 0°$, i.e., parallel.
Therefore, option (a) is correct.

55. Topic: Computer Networks

(d) Given, IP packet has arrived and the first eight bits are 0100 0010.
The receiver will reject the packet since the information at the beginning of IP packet is called header information and Internet header length is four bits. The first eight bits arrived is a mismatch; therefore, the packet will be rejected.

56. Topic: Digital Logic

(c) Option (c) is not a valid Boolean algebra rule:

$$\overline{X} + XY = (\overline{X} + X)(\overline{X} + Y) = \overline{X} + Y$$

So, this is not a rule whereas $XX = X$ is idempotent law.

$$(X+Y)\cdot X = X\cdot X + Y\cdot X = X + Y\cdot X$$
$$= (1+Y)\cdot X$$
$$= X$$

So,

$$(X+Y)\cdot X = X$$

and

$$(X+Y)\cdot(X+Z) = X\cdot X + Y\cdot X + X\cdot Z + Y\cdot Z$$
$$= X + Y\cdot X + X\cdot Z + Y\cdot Z$$
$$= X(1+Y) + X\cdot Z + Y\cdot Z$$
$$= X + X\cdot Z + Y\cdot Z$$
$$= X(1+Z) + Y\cdot Z$$
$$= X + Y\cdot Z$$

Therefore,

$$(X+Y), (X+Z) = X + Y\cdot Z$$

57. Topic: Computer Networks

(d) Given:

First address is 205.16.32.0
Supernet mask is 255.255.248.0

The binary equivalent of the first address and supernet mask is

First address: 11001101.00010000.00100000.00000000
Supernet mask: 11111111.11111111.11111000.00000000

Now, AND operation of IP address and supernet mask gives first address
So, consider option (d) 205·16·39·44
The binary equivalent of this is

11001101.00010000.00100111.00101100

AND operation of this address with supernet mask gives

11001101.00010000.00100000.00000000

This is the first address given in the question so correct option is (d).

58. Topic: Computer Networks

(a) Given:

Original time stamp value = 46
Receive timestamp value = 59
Transmit timestamp value = 60
Timestamp at arrival of packet = 69

Now, sending time = Receive timestamp value – Original timestamp value = 59 – 46 = 13

$$\begin{aligned}\text{Receiving time} &= \text{Timestamp at arrival of packet} \\ &\quad - \text{Transmit timestamp value} \\ &= 69 - 60 = 9\end{aligned}$$

$$\text{Round trim time} = 13 + 9 = 22$$

$$\begin{aligned}\text{Time difference} &= \text{Transmit timestamp value} \\ &\quad - (\text{Original timestamp value} \\ &\quad + \text{one-way time}) \\ &= \text{Transmit timestamp value} \\ &\quad - \left(\text{Original timestamp value} \right. \\ &\quad \left. + \frac{\text{Round trip time}}{2}\right) \\ &= 60 - \left(46 + \frac{22}{2}\right) = 60 - (46 + 11) \\ &= 60 - 57 = 3\end{aligned}$$

Therefore, the receive clock should go back by 3 ms.

59. Topic: Theory of Computation

(b) A turing machine is an idealized computing device. It consists of a read or write head with a paper tape passing through it. The paper tape is divided into squares bearing ‘0’ or ‘1’. The turing machine reads the symbol in the tape that is under the head or writes a symbol on the tape under the head, then moves the tape to left or right by one square and finally changes the state and halts.
A turing machine cannot halt by changing the input, therefore this outcome is false.

60. Topic: Computer Networks

(b) TCP protocol is used for accessing the web servers. TCP or transmission control protocol is a protocol that defines a set of rules. These rules describe the movement of data between source and destination or the internet and by default port number for a web server is 80. Thus, TCP, 80 is being used.

61. Topic: Computer Networks

(d) A mechanism or technology used in Ethernet by which two connected devices choose common transmission parameters such as speed, duplex mode and flow control is called autonegotiation. Autonegotiation permits devices to exchange information over line segments and to perform automatic configuration to achieve best modes of operations over links.

62. Topic: Algorithms

(b) Quick sort is based on divide and conquer approach. An element called pivot element is chosen and array is portioned around it. Left side of pivot has all elements less than it and right side has all elements greater than pivot element. In worst case, complexity of quick sort is $O(N^2)$.
Heapsort is based on comparison approach where the largest element is found first and placed at the end. Same process is repeated for all the remaining elements. In worst case, complexity of heapsort is $O(N \log N)$.
Merge sort is based on divide and conquer approach. The algorithm divides the array into two halves and further dividing the subarrays until there is single element in each subarray. Then the subarrays are merged together in sorted manner. In worst case, complexity of merge sort is $O(N \log N)$.
Therefore, heapsort and merge sort will perform in least time in worst case.

63. Topic: Databases

(c) In first normal form, all attributes are atomic and there are no repeating groups.
In second normal form, the database is in first normal form. All non-key attributes are fully functional dependent on the primary key.
In third normal form, the database is in second normal form and there is no transitive functional dependency.
In Boyce-Codd normal form (BCNF), there is no non-trivial functional dependency of attributes.
For the given database table, following dependencies exist:

$$AB \to CDE$$
$$C \to B$$

Therefore, the database is in third normal form but not BCNF because C is not a superkey.

64. Topic: Algorithms

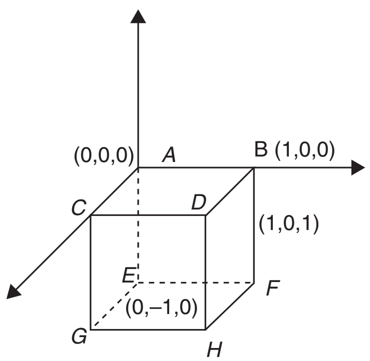
(b) Given, hash table has 13 elements and $f(\text{key})$ = key mod 13

Then for key = 661, $f(\text{key})$ = 661 mod 13 = 11
So, key 661 will be inserted in 11th position.

for key = 182, f(key) = 182 mod 13 = 0
So, key 182 will be inserted in 0th position.
for key = 24, f(key) = 24 mod 13 = 11
Since, 11th position is occupied by 661,
So, key 24 will be inserted in 12th position.
For key = 103, f(key) = 103 mod 13 = 12
Since, 12th position is occupied, so next free space available 1 so 103 will be inserted in 1st position.

65. Topic: Engineering Mathematics

(b) Given, cube of side 1 unit is placed such that the origin coincides with one of its top vertices (say A). If D is vertex whose coordinates are (1, 0, 1), then vertex diagonally opposite to D is E. Coordinates of E are (0, –1, 0).



66. Topic: Operating System

(a) Given, each file is associated with 16-bit number. Each user should have the read and write capability. Thus, memory needed to store user's access data is

$$2\times 2^{16} = 128 \text{ k bits} = \frac{128 \text{ k bits}}{8} = 16 \text{ k bytes}$$

67. Topic: Algorithms

(a) There is a recursive function call of module $(n-1)$ and there is a constant addition n + module $(n-1)$
So, the recursion relation is

$$\begin{aligned} M(n) &= M(n-1) + c \\ &= M(n-2) + c + c \\ &= M(n-2) + 2c \\ &= M(n-3) + c + 2c \\ &= M(n-3) + 3c \end{aligned}$$

or

$$M(n) = M(n-k) + kc$$

$$\text{For } k = n-1: \quad \begin{aligned} M(n) &= M(n-n+1) + (n-1)c \\ &= M(1) + (n-1)c \end{aligned}$$

Therefore, time complexity of module is $O(n)$.

68. Topic: Operating System

(d) In worst-case scenario, when all processes are holding one resource less than the maximum demand. Therefore,

$$P_1 = 2, P_2 = 3, P_3 = 4$$

So,

$$\text{Total resources} = 2 + 3 + 4 = 9$$

If one new resource enters, then it will be occupied by one of these processes. Thus, minimum resources required is

$$9 + 1 = 10$$

Deadlock is a situation where each process is holding a resource and the set of process are blocked.

69. Topic: Software Engineering

(a) In software engineering, spiral model is a risk-driven process model where more emphasis is given on risk analysis. The spiral model has four phases:

(i) Planning
(ii) Risk Analysis
(iii) Engineering
(iv) Evaluation

The software process, repeatedly passes through these phases.
The spiral model stops when the software product is retired.

70. Topic: Operating System

(c) Dirty bit is a specific bit in computer memory that is used to indicate a modification. Dirty bit is also known as modified bit. Dirty bit can also be used as markers in memory-specific processes.
Therefore, option (c) is correct; dirty bit is used to indicate a page that has been modified after being loaded into cache.

71. Topic: Computer Networks

(d) MAC or media access control address is a globally unique identifier assigned to network devices. MAC address is 6 byte in length and has following format.
$MM : MM : MM : SS : SS : SS$
where MM bytes are 10 numbers of the manufacturer, SS bytes are serial numbers assigned by manufacturer.
Multicast MAC address has range of

$$01 : 00 : 5E : 00 : 00 : 00 \text{ to } 01 : 00 : 5E : 7F : FF : FF$$

Thus, 01 : 00 : 5E : FF : FF : FF is not a valid multicast MAC address.

72. Topic: Engineering Mathematics

(c) Given:

$$A = \begin{bmatrix} 1 & 2 & 1 & -1 \\ 9 & 5 & 2 & 2 \\ 7 & 1 & 0 & 4 \end{bmatrix}$$

Using elementary row operations:

$$R_2 \to R_2 - 9R_1$$
$$R_3 \to R_3 - 7R_1$$

we get

$$A = \begin{bmatrix} 1 & 2 & 1 & -1 \\ 0 & -13 & -7 & 11 \\ 0 & -13 & -7 & 11 \end{bmatrix}$$

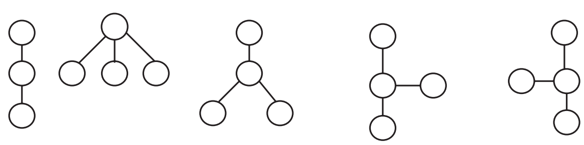
and using $R_3 \to R_3 - R_2$:

$$A = \begin{bmatrix} 1 & 2 & 1 & -1 \\ 0 & -13 & -7 & 11 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Number of linearly independent rows are 2. Hence, the rank of matrix $A = 2$.

73. Topic: Engineering Mathematics

(d) If there are n nodes, then there exists five ways in which tree can be formed:



The labels A, B, C, D can be permuted in 4! ways
So, the numbers of different trees possible is

$$4! \times 5 = 4 \times 3 \times 2 \times 1 \times 5 = 120 \text{ ways}$$

74. Topic: Engineering Mathematics

(b) Given:

$$\text{Mode} = 15, \text{Mean} = 30$$

Mode is set of data value that appears most often.
Mean is arithmetic average of the values.
Median is the middle of a sorted list of numbers.
Now, that we know that

$$3 \times \text{Median} = \text{Mode} + 2 \times \text{Mean}$$
$$3 \times \text{Median} = 15 + 2 \times 30$$

Therefore, median is

$$\frac{15+60}{3} = \frac{75}{3} = 25$$

75. Topic: Computer Networks

(d) Given, block given to the organization

130.34.12.64/26

The subnet mask has address 255.255.255.192
Taking binary of last segment of the address we have
Subnet mask: 255.255.255.11000000
Organization start address: 130.34.12.01000000
On complementing the mask, the range of 1 p's available are 130.34.12.64 to 130.34.12.127

So, 130.34.12.132 is not an address of organization.

76. Topic: Web Technologies

(a) There are three basic types of web documents:

(i) **Static:** A static web document is in the file which is associated with web server. The contents of static file are determined while writing each request has exactly same response.
(ii) **Dynamic:** A dynamic web document is not predefined, it is created when a request arrives at web server. A fresh document is created for each request.
(iii) **Active:** A active web document consists of a computer program that the server sends to browser. It is seen locally and the program interacts with user and change the display continuously.

The given scenario in the question implies that the web document is active.

77. Topic: Operating System

(c) Given, page table consists of 64 entries.
Each entry is of 11 bits.
Page size = 512 bytes
Physical address space = Number of frames × Frame size
Now, number of frames = 2^{11}
Frame size = 512 byte
Therefore, physical address space = $2^{11} \times 512 = 2^{20}$ bytes
Since, the valid/invalid bit is included, 1 bit is used to identify if page is valid or invalid.
Thus, physical address space = 2^{19} bytes.

78. Topic: Operating System

(a) The various optimization criteria in the design of a CPU scheduling algorithm is as listed below:

(i) Maximum CPU utilization
(ii) Minimum TAT (Turn Around Time)
(iii) Minimum waiting time
(iv) Minimum response time
(v) Maximum throughout

Therefore, option (a) minimum CPU utilization is not an optimization criterion in design of a CPU scheduling algorithm.

79. Topic: Theory of Computation

(c) In deterministic finite automaton, machine goes to one state only for a particular input character. Deterministic finite automaton can change state only for an input character.

If S is set of 8-bit strings whose second, third, sixth and seventh bits are 1, then the number of 8-bit strings that are accepted by *M* are as follows:

01110111
01110110

Therefore, the correct option is (c), i.e., 2.

80. Topic: Algorithms

(a) Grid computing: it is a collection of computer resources from multiple domains to reach a common goal.

Neural networks: It is an interconnected group of nodes inspired by biological neural networks. Such networks lean and improve their performance progressively.

Parallel processing: It is a type of computation in which many processes are carried out simultaneously by dividing the instructions among multiple processors.

Cluster computing: It is a set of computers that are connected together and work together such that they are viewed as a single system.

Therefore, the correct option is (a).

Indian Space Research Organisation Computer Science

2015

QUESTION PAPER

1. Which of the given number has its IEEE-754 32-bit floating-point representation as

(0 10000000 110 0000 0000 0000 0000 0000)

(a) 2.5 (b) 3.0
(c) 3.5 (d) 4.5

2. The range of integers that can be represented by an n-bit 2's complement number system is

(a) -2^{n-1} to $(2^{n-1}-1)$
(b) $-2(2^{n-1}-1)$ to $(2^{-n-1}-1)$
(c) -2^{n-1} to 2^{n-1}
(d) $-2(2^{n-1}+1)$ to $(2^{n-1}-1)$

3. How many 32 K × 1 RAM chips are needed to provide a memory capacity of 256 K-bytes?

(a) 8 (b) 32
(c) 64 (d) 128

4. A modulus-12 ring counter requires a minimum of

(a) 10 flip-flops (b) 12 flip-flops
(c) 8 flip-flops (d) 6 flip-flops

5. The complement of the Boolean expression $AB(\overline{B}C + AC)$ is

(a) $\left(\overline{A}+\overline{B}\right)+\left(B+\overline{C}\right)\cdot\left(\overline{A}+\overline{C}\right)$
(b) $\left(\overline{A}\cdot\overline{B}\right)+\left(B\overline{C}+\overline{A}\ \overline{C}\right)$
(c) $\left(\overline{A}+\overline{B}\right)\cdot\left(B+\overline{C}\right)+\left(A+\overline{C}\right)$
(d) $\left(A+B\right)\cdot\left(\overline{B}+C\right)\left(A+C\right)$

6. The code which uses 7 bits to represent a character is

(a) ASCII (b) BCD
(c) EBCDIC (d) Gray

7. If half adders and full adders are implements using gates, then for the addition of two 17 bit numbers (using minimum gates) the number of half adders and full adders required will be

(a) 0, 17 (b) 16, 1
(c) 1, 16 (d) 8, 8

8. Minimum number of 2 × 1 multiplexers required to realize the following function, $f = \overline{A}\overline{B}C + \overline{A}\overline{B}C$.

Assume that inputs are available only in true form and Boolean a constant 1 and 0 are available.

(a) 1 (b) 2
(c) 3 (d) 7

9. The number of 1s in the binary representation of $(3*4096+15*256+5*16+3)$ are

(a) 8 (b) 9
(c) 10 (d) 12

10. The Boolean expression $AB + A\overline{B} + \overline{A}C + AC$ is independent of the Boolean variable

(a) A (b) B
(c) C (d) None of these

11. If the sequence of operations – push (1), push (2), pop, push (1), push (2), pop, pop, pop, push (2), pop are performed on a stack, the sequence of popped out values

(a) 2, 2, 1, 1, 2 (b) 2, 2, 1, 2, 2
(c) 2, 1, 2, 2, 1 (d) 2, 1, 2, 2, 2

12. A machine needs a minimum of 100 s to sort 1000 names by quick sort. The minimum time needed to sort 100 names will be approximately

(a) 50.2 s (b) 6.7 s
(c) 72.7 s (d) 11.2 s

13. Six files $F1$, $F2$, $F3$, $F4$, $F5$ and $F6$ have 100, 200, 50, 80, 120, 150 records respectively. In what order should they be stored so as to optimize act. Assume each file is accessed with the same frequency

(a) $F3$, $F4$, $F1$, $F5$, $F6$, $F2$
(b) $F2$, $F6$, $F5$, $F1$, $F4$, $F3$
(c) $F1$, $F2$, $F3$, $F4$, $F5$, $F6$
(d) Ordering is immaterial as all files are accessed with the same frequency

14. A hash table with 10 buckets with one slot per bucket is depicted in fig. The symbols, $S1$ and $S7$ are initially entered using a hashing function with linear probing. The maximum number of comparisons needed in searching an item that is not present is

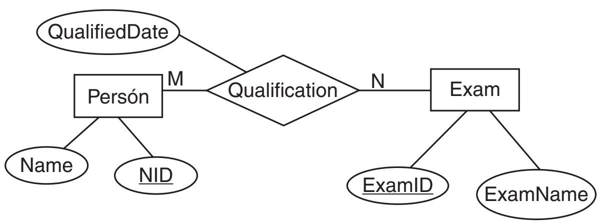
0	$S7$
1	$S1$
2	
3	$S4$
4	$S2$
5	
6	$S5$
7	
8	$S6$
9	$S3$

(a) 4 (b) 5
(c) 6 (d) 3

15. The queue data structure is to be realized by using stack. The number of stacks needed would be

(a) It cannot be implemented
(b) 2 stacks
(c) 4 stacks
(d) 1 stack

16. Consider the following Entity Relationship Diagram (ERD)



Which of the following possible relations will not hold if the above ERD is mapped into a relation model?

(a) Person (NID, Name)
(b) Qualification (<u>NID</u>, <u>ExamID</u>, QualifiedDate)
(c) Exam (<u>ExamID</u>, <u>NID</u>, ExamName)
(d) Exam (<u>ExamID</u>, ExamName)

17. Consider the following log sequence of two transactions on a bank account, with initial balance 12000, that transfer 2000 to a mortgage payment and, then apply a 5% interest

(i) T 1 start
(ii) T 1 B old = 12000 new = 10000
(iii) T 1 M old = 0 new = 2000
(iv) T 1 commit
(v) T 2 start
(vi) T 2 B old = 10000 new = 10500
(vii) T 2 commit

Suppose the database system crashed just before log record 7 is written. When the system is restarted, which one statement is true of the recovery procedure?

(a) We must redo log record 6 set B to 10500
(b) We must undo log record 6 to set B to 10000 and then redo log record 2 and 3
(c) We need not redo log records 2 and 3 because transaction $T1$ has committed
(d) We can apply redo and undo operations in arbitrary order because they are idempotent

18. Given a block can hold either 3 records or 10 key pointers. A database contains n records, then how many blocks do we need to hold the data file and the dense index

(a) $\frac{13n}{30}$ (b) $\frac{n}{3}$
(c) $\frac{n}{10}$ (d) $\frac{n}{30}$

19. The maximum length of an attribute of type text is

(a) 127 (b) 255
(c) 256 (d) It is variable

20. Let $R = (A, B, C, D, E, F)$ be a relation scheme with the following dependencies $C \rightarrow F$, $E \rightarrow A$, $EC \rightarrow D$, $A \rightarrow B$. Which of the following is a key for R?

(a) CD (b) EC
(c) AE (d) AC

21. If $D_1, D_2, \ldots, D_n$ are domains in a relational model, then the relation is a table, which is a subset of

(a) $D_1 \oplus D_2 \oplus \ldots \oplus D_n$
(b) $D_1 \times D_2 \times \ldots \times D_n$
(c) $D_1 \cup D_2 \cup \ldots \cup D_n$
(d) $D_1 \cap D_2 \cap \ldots \cap D_n$

22. Consider the following relational query on the above database:

```
SELECT S.sname
FROM Suppliers S
WHERE S.sid NOT IN (SELECT C.sid
    FROM Catalog C
    WHERE C.pid NOT IN (SELECT P.pid
            FROM Parts P
            WHERE P.color < > 'blue'))
```

Assume that relations corresponding to the above schema are not empty. Which of the following is the correct interpretation of the above query?

(a) Find the names of all suppliers who have supplied a non-blue part
(b) Find the names of all suppliers who have not supplied a non-blue part
(c) Find the names of all suppliers who have supplied only non-blue parts
(d) Find the names of all suppliers who have not supplied only non-blue parts

23. Consider the following schema:

Emp (Empcode, Name, Sex, Salary, Deptt)
A simple SQL query is executed as follows:
SELECT Deptt FROM Emp
WHERE sex = 'M'
GROUP by Dept
Having avg (Salary) > {select avg (Salary) from Emp}
The output will be

(a) Average salary of male employee is the average salary of the organization
(b) Average salary of male employee is less than the average salary of the organization
(c) Average salary of male employee is equal to the average salary of the organization
(d) Average salary of male employees is more than the average salary of the organization

24. Given the following expression grammar:

$$E \rightarrow E * F \,|F + E|\, F$$
$$F \rightarrow F - F \mid id$$

Which of the following is true?
(a) * has higher precedence than +
(b) – has higher precedence than *
(c) + and – have same precedence
(d) + has higher precedence than *

25. The number of token the following C statement is

```
printf ("i = %d, &i =%x", i&i);
```

(a) 13 (b) 6
(c) 10 (d) 11

26. Which grammar rules violate the requirement of the operator grammar? A, B, C are variables and a, b, c are terminals

(i) $A \rightarrow BC$
(ii) $A \rightarrow CcBb$
(iii) $A \rightarrow BaC$
(iv) $A \rightarrow \varepsilon$

(a) (i) only (b) (i) and (ii)
(c) (i) and (iii) (d) (i) and (iv)

27. Which one of the following is a top-down parser?
(a) Recursive descent parser
(b) Shift left associative parser
(c) SLR (k) parser
(d) LR (k) parser

28. Yacc stands for
(a) yet accept compiler constructs
(b) yet accept compiler compiler
(c) yet another compiler constructs
(d) yet another compiler compiler

29. Which statement is true?
(a) LALR parser is most powerful and costly as compare to other parsers
(b) All CFG's are LP and not all grammars are uniquely defined
(c) Every SLR grammar is unambiguous but not every unambiguous grammar is SLR
(d) LR(K) is the most general back tracking shift reduce parsing method

30. Semaphores are used to solve the problem of

(i) race condition
(ii) process synchronization
(iii) mutual exclusion
(iv) None of the above

(a) (i) and (ii) (b) (ii) and (iii)
(c) All of the above (d) None of the above

31. If there are 32 segments, each size 1 k bytes, then the logical address should have
(a) 13 bits (b) 14 bits
(c) 15 bits (d) 16 bits

32. In a lottery scheduler with 40 tickets, how we will distribute the tickets among 4 processes P_1, P_2, P_3 and P_4 such that each process gets 10%, 5%, 60% and 25% respectively?

	P_1	P_2	P_3	P_4
(a)	12	4	70	30
(b)	7	5	20	10
(c)	4	2	24	10
(d)	8	5	40	30

33. Suppose a system contains n processes and system uses the round-robin algorithm for CPU scheduling then which data structure is best suited ready queue of the processes
(a) stack (b) queue
(c) circular queue (d) tree

34. A hard disk system has the following parameters:

- Number of track = 500
- Number of sectors/track =100
- Number of bytes/sector = 500
- Time taken by the head to move from one track to adjacent track = 1 ms
- Rotation speed = 600 rpm

What is the average time taken for transferring 250 bytes from the disk?

(a) 300.5 ms (b) 255.5 ms
(c) 255 ms (d) 300 ms

35. At a particular time of computation the value of a counting semaphore is 7. Then 20 P operations and 15 V operations were completed on this semaphore. The resulting value of the semaphore is

(a) 42 (b) 2
(c) 7 (d) 12

36. Increasing the RAM of a computer typically improves performance because

(a) Virtual memory increases
(b) Larger RAMs are faster
(c) Fewer page faults occur
(d) Fewer segmentation faults occur

37. Consider the following program

```
main()
{
   fork();
   fork();
   fork();
}
```

How many new processes will be created?

(a) 9 (b) 6
(c) 7 (d) 5

38. Suppose two jobs, each of which needs 10 min of CPU time, start simultaneously. Assume 50% I/O wait time. How long will it take for both to complete if they run sequentially?

(a) 10 (b) 20
(c) 30 (d) 40

39. If a node has K children in B tree, then the node contains exactly ______________ keys.

(a) K^2 (b) $K - 1$
(c) $K + 1$ (d) $\sqrt{K}$

40. The time complexity of the following C function is (assume $n > 0$):

```
int recursive (int n)
{
   if (n == 1)
   return (1);
   else
   return (recursive (n-1) + recursive
   (n-1));
}
```

(a) $O(n)$ (b) $O(n \log n)$
(c) $O(n^2)$ (d) $O(2^n)$

41. The number of spanning trees for a complete graph with seven vertices is

(a) 2^5 (b) 7^5
(c) 3^5 (d) $2^{2 \times 5}$

42. If one uses straight two-way merge sort algorithm to sort the following elements in ascending order: 20, 47, 25, 8, 9, 4, 40, 30, 12, 17, and then the order of these elements after second pass of the algorithms is

(a) 8, 9, 15, 20, 47, 4, 12, 7, 30, 30
(b) 8, 15, 20, 47, 4, 9, 30, 40, 12, 17
(c) 15, 20, 47, 4, 8, 9, 12, 30, 40, 17
(d) 4, 8, 9, 15, 20, 47, 12, 17, 30, 40

43. Let $R1$ and $R2$ be regular sets defined over the alphabet, then

(a) $R_1 \cap R_2$ is not regular
(b) $R_1 \cup R_2$ is not regular
(c) $\Sigma^* - R_1$ is regular
(d) R_1^* is not regular

44. The DNS maps the IP addresses to

(a) a binary address as strings
(b) an alphanumeric address
(c) a hierarchy of domain names
(d) a hexadecimal address

45. To add a background color for all <*h*l> elements, which of the following HTML syntax is used?

(a) *h*1 {background-color: #FFFFFF}
(b) {background-color: #FFFFFF}. *h*1
(c) {background-color: #FFFFFF}. *h*l(all)
(d) *h*1. all{bgcolor= #FFFFFF}

46. The correct syntax to write “Hi There” in Javascript is

(a) jscript. write (“Hi There”)
(b) response. write (“Hi There”)
(c) print (“Hi There”)
(d) print. jscript (“Hi There”)

47. To declare the version of XML, the correct syntax is

(a) <?xml version=‘1.0’/>
(b) <*xml version=‘1.0’/>
(c) <?xml version=“1.0”/>
(d) </xml version=’1.0’/>

48. A T-switch is used to

(a) control how messages are passed between computers
(b) echo every character that is received
(c) transmit characters one at a time
(d) rearrange the connections between computing equipments

49. What frequency range is used for microwave communications, satellite and radar?

(a) Low frequency: 30 kHz to 300 kHz
(b) Medium frequency: 300 kHz to 3 MHz
(c) Super high frequency: 3000 MHz to 30000 MHz
(d) Extremely high frequency: 30000 kHz

50. How many bits internet address is assigned to each host on a TCP/IP internet which is used in all communication with the host?

(a) 16 bits (b) 32 bits
(c) 48 bits (d) 64 bits

51. How many characters per sec (7 bits + 1 parity) can be transmitted over a 2400 bps line if the transfer is synchronous (1 start and 1 stop bit)?

(a) 300 (b) 240
(c) 250 (d) 275

52. In CRC if the data unit is 100111001 and the divisor is 1011 then what is dividend at the receiver?

(a) 100111001101 (b) 100111001011
(c) 100111001 (d) 100111001110

53. An ACK number of 1000 in TCP always means that

(a) 999 bytes have been successfully received
(b) 1000 bytes have been successfully received
(c) 1001 bytes have been successfully received
(d) None of the above

54. In a class B subnet, we know the IP address of one host and the mask as given below:

IP address = 125.134.112.66
Mask = 255.255.224.0
What is the first address (Network address)?

(a) 125.134.96.0 (b) 125.134.112.0
(c) 125.134.112.66 (d) 125.134.0.0

55. A certain population of ALOHA users manages to generate 70 request sec^{-1}. If the time is slotted in units of 50 ms, then channel load would be

(a) 4.25 (b) 3.5
(c) 450 (d) 350

56. Which statement is false?

(a) PING is a TCP/IP application that sends datagrams once every second in the hope of an echo response from the machine being PINGED
(b) If the machine is connected and running a TCP/IP protocol stack, it should respond to the PING datagram with a datagram of its own
(c) If PING encounters an error condition, an ICMP message is not returned
(d) PING display the time of the return response in milliseconds or one of several error message

57. A router uses the following routing table:

Destination	Mask	Interface
144.16.0.0	255.255.0.0	eth0
144.16.64.0	255.255.224.0	ethl
144.16.68.0	255.255.255.0	eth2
144.16.68.64	255.255.255.224	eth3

A packet bearing an estimation address 144.16.68.117 arrives at the router. On which interface will it be forwarded?

(a) eth0 (b) ethl
(c) eth2 (d) eth3

58. Which layers of the OSI reference model are host-to-host layers?

(a) Transport, session, presentation, application
(b) Session, presentation, application
(c) Datalink, transport, presentation, application
(d) Physical, datalink, network, transport

59. Alpha and beta testing are forms of

(a) Acceptance testing (b) Integration testing
(c) System testing (d) Unit testing

60. If in a software project the number of user input, user output, enquiries, files and external interfaces are (15, 50, 24, 12, 8), respectively, with complexity average weighing factor. The productivity if effort = 70 person-month is

(a) 110.54 (b) 408.74
(c) 304.78 (d) 220.14

61. The contents of the flag register after execution of the following program by 8085 microprocessor will be

```
Program
SUB A
MVI B, (01)H
DCR B
HLT
```

(a) $(54)_H$ (b) $(00)_H$
(c) $(01)_H$ (d) $(45)_H$

62. The minimum time delay between the initiations of two independent memory operations is called

(a) access time (b) cycle time
(c) rotational time (d) latency time

63. Which of the following compression algorithms is used to generate a .png file?

(a) LZ78 (b) Deflate
(c) LZW (d) Huffman

64. Dirty bit for a page in a page table

(a) helps avoid unnecessary writes on a paging device
(b) helps maintain LRU information
(c) allows only read on a page
(d) None of these

65. Which of the following is not an image type used in MPEG?

(a) A frame (b) B frame
(c) D frame (d) P frame

66. Consider an uncompressed stereo audio signal of CD quality which is sampled at 44.1 kHz and quantized using 16 bits. What is required storage space if a compression ratio of 0.5 is achieved for 10 seconds of this audio?

(a) 172 KB (b) 430 KB
(c) 860 KB (d) 1720 KB

67. What is the compression ratio in typical *mp*3 audio file?

(a) 4:1 (b) 6:1
(c) 8:1 (d) 10:1

68. Consider the following program fragment

```
if (a > b)
if (b > c)
   s1;
else s2;
```

s2 will be executed if

(a) $a <= b$ (b) $b > c$
(c) $b >= c$ and $a <= b$ (d) $a > b$ and $b <= c$

69. If n has the value 3, then the statement $a[++n] = n++;$

(a) assigns 3 to a [5]
(b) assigns 4 to a [5]
(c) assigns 4 to a [4]
(d) what is assigned is compiler dependent

70. The following program

```
main()
{
   inc(); inc(); inc();
}
inc()
{
   static int x;
   printf("%d", ++x);
}
```

(a) prints 012
(b) prints 123
(c) prints 3 consecutive, but unpredictable numbers
(d) prints 111

71. Consider the following program fragment

```
i=6720; j=4;
while((i%j)==0)
{
   i = i/j;
   j=j+1;
}
```

on termination j will have the value

(a) 4 (b) 8
(c) 9 (d) 6720

72. Consider the following declaration,

```
int a, *b = &a, **c = &b;
```

the following program fragment

```
a = 4;
**c =5;
```

(a) does not change the value of *a*
(b) assigns address of *c* to *a*
(c) assigns the value of *b* to *a*
(d) assigns 5 to *a*

73. The output of the following program is

```
main()
{
   static int x[] = {1, 2, 3, 4, 5, 6,
   7, 8};
   int i;
   for (i=2; i<6; ++i)
   x[x[i]] = x[i];
   for (i=0; i<8; ++i)
   printf("%d", x[i]);
}
```

(a) 1 2 3 3 5 5 7 8 (b) 1 2 3 4 5 6 7 8
(c) 8 7 6 5 4 3 2 1 (d) 1 2 3 5 4 6 7 8

74. Which of the following has the compilation error in C?

(a) int n = 17;

(b) char c = 99;

(c) float f = (float) 99.32;

(d) include <stdio.h>

75. The for loop

```
for(i = 0; i < 10; ++i)
printf(%d”, i&1);
```

prints

(a) 0101010101 (b) 0111111111

(c) 0000000000 (d) 1111111111

76. Consider the following statements

define hypotenuse (a, b) sqrt (a * a + b * b);

The macro call hypotenuse (a + 2, b + 3);

(a) Finds the hypotenuse of a triangle with sides $a + 2$ and $b + 3$

(b) Finds the square root of $(a + 2)^2 + (b + 3)^2$

(c) Is invalid

(d) Find the square root of 3 * 2 + 4 * b + 5

77. In $X = (M + N \times O)/(P \times Q)$, how many one-address instructions are required to evaluate it?

(a) 4 (b) 6

(c) 8 (d) 10

78. A decimal number has 64 digits. The number of bits needed for its equivalent binary representation is

(a) 200 (b) 213

(c) 246 (d) 277

79. Consider the following C declaration

```
struct
{  short s[5];
   union
   {
         float y;
         long z;
   } u;
} t;
```

Assume that objects of type short, float and long occupy 2 bytes, 4 bytes and 8 bytes, respectively. The memory requirement for variable t, ignoring alignment considerations, is

(a) 22 bytes (b) 18 bytes

(c) 14 bytes (d) 10 bytes

80. Consider the following code segment

```
void foo(int x, int y)
{
   x+=y;
   y+=x;
}
main()
{
   int x = 5.5;
   foo(x, x);
}
```

What is the final value of x in both call by value and call by reference, respectively?

(a) 5 and 16 (b) 5 and 12

(c) 5 and 20 (d) 12 and 20

ANSWER KEY

1. (c)	**9.** (c)	**17.** (c)	**25.** (*)	**33.** (c)	**41.** (b)	**49.** (d)	**57.** (c)	**65.** (a)	**73.** (a)
2. (a)	**10.** (b)	**18.** (a)	**26.** (d)	**34.** (d)	**42.** (*)	**50.** (b)	**58.** (a)	**66.** (c)	**74.** (*)
3. (c)	**11.** (a)	**19.** (c)	**27.** (a)	**35.** (b)	**43.** (c)	**51.** (a)	**59.** (a)	**67.** (d)	**75.** (a)
4. (b)	**12.** (b)	**20.** (b)	**28.** (d)	**36.** (c)	**44.** (c)	**52.** (b)	**60.** (*)	**68.** (d)	**76.** (d)
5. (a)	**13.** (a)	**21.** (b)	**29.** (c)	**37.** (c)	**45.** (a)	**53.** (d)	**61.** (a)	**69.** (d)	**77.** (c)
6. (a)	**14.** (b)	**22.** (a, c)	**30.** (b)	**38.** (d)	**46.** (*)	**54.** (a)	**62.** (b)	**70.** (b)	**78.** (b)
7. (c)	**15.** (b)	**23.** (d)	**31.** (c)	**39.** (b)	**47.** (c)	**55.** (b)	**63.** (b)	**71.** (c)	**79.** (b)
8. (b)	**16.** (c)	**24.** (b)	**32.** (c)	**40.** (d)	**48.** (d)	**56.** (c)	**64.** (a)	**72.** (d)	**80.** (c)

*** None of the options is correct.**

ANSWERS WITH EXPLANATION

1. Topic: Digital Logic

(c) Floating point representation is based on scientific notation. It has two parts, that is, the mantissa (or significand) and the exponent. In floating point representation, the values of mantissa bits 'float' along with the decimal point based on the exponent is given value.
Given, the floating point representation is given as 0 10000000 1100000 0000 0000 0000 0000
Here,

$$\text{Sign} = 0 \text{ (positive)}$$
$$\text{Exponent} = 10000000$$
$$\text{Mantissa} = 110\ 0000\ 0000\ 0000\ 0000\ 0000$$

Now,

$$\text{Actual exponent } (p) = \text{exponent} - \text{bias}$$
$$= 128 - 127 = 1$$

Therefore,

$$\text{Value} = (-1)^{\text{sign}} \times (1.11000000000000000000000) \times 2^1$$
$$= (-1)^0 \times (1\times 2^0 + 1\times 2^{-1} + 1\times 2^{-2} + 0\times 2^{-3} + 0\times 2^{-4} + \cdots)\times 2^1$$
$$= \left(1 + \frac{1}{2} + \frac{1}{4}\right)\times 2 = \frac{(4+2+1)}{4}\times 2 = \frac{7}{2} = 3.5$$

2. Topic: Digital Logic

(a) 2's complement is a binary operation on binary numbers. n-bit 2's complement is defined as complement of n-bit number with respect to 2^n. For a given set of n-bit values, the lower half of the set can be assigned integers from 0 to $(2^{n-1} - 1)$ and upper half of the set can be assigned integers from -2^{n-1} to -1 thus the range of integers that can be represented by n-bit 2's complement number system is -2^{n-1} to $(2^{n-1} - 1)$.

3. Topic: Digital Logic

(c) Total number of RAM chips is given by:

$$\text{Number of RAM chips} = \frac{\text{Total size}}{\text{1 RAM size}}$$
$$= \frac{256\text{ K bytes}}{32\text{K}\times 1\text{ RAM}} = \frac{2^8\times 2^{10}\times 2^3}{2^5\times 2^{10}}$$
$$= \frac{2^{11}}{2^5} = 2^6 = 64$$

Therefore, number of RAM chips = 64

4. Topic: Digital Logic

(b) A ring counter is a shift register with output of last flip–flop connected to the input of first flip–flop. The modulus of a counter is number of states the counter counts before repeating itself. A modulus-n ring counter will require n flip–flops to circulate a single bit. Therefore, a modulus-12 ring counter requires 12 flip–flops.

5. Topic: Digital Logic

(a) Given:

$$AB(\bar{B}C + AC)$$

Taking complement

$$\overline{AB(\bar{B}C + AC)}$$

Using De Morgan's law $\overline{XY} = \bar{X} + \bar{Y}$

$$\overline{AB(\bar{B}C + AC)} = \overline{AB} + \overline{(\bar{B}C + AC)}$$
$$= \bar{A} + \bar{B} + \overline{(\bar{B}C + AC)}$$

Using De Morgan's law $\overline{X+Y} = \bar{X}\bar{Y}$
So,

$$\overline{AB(\bar{B}C + AC)} = \bar{A} + \bar{B} + \overline{\bar{B}C}\cdot\overline{AC}$$
$$= (\bar{A} + \bar{B}) + (\bar{\bar{B}} + \bar{C})\cdot(\bar{A} + \bar{C})$$

Now,

$$\bar{\bar{B}} = B$$

Therefore,

$$\overline{AB(\bar{B}C + AC)} = (\bar{A} + \bar{B}) + (B + \bar{C})\cdot(\bar{A} + \bar{C})$$

6. Topic: Digital Logic

(a) ASCII (American standard code for information interchange). ASCII code is a numerical representation for character set such as letters, digit, punctuation marks, special characters and central characters. ASCII table has 128 characters, with values from 0 to 127. Thus, 7 bits are sufficient to represent a character in ASCII.

7. Topic: Digital Logic

(c) Adders are digital circuits that carry out addition. Half adder adds two binary digits and produces two outputs – sum and carry. The full adder adds three one bit numbers and produces two bit output – carry and sum. Hence for addition of two 17 bit numbers only one half adder is required while $(n - 1)$ that is, 16 full adders are required.

8. Topic: Digital Logic

(b) Given:

$$f = \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C}$$

Simplifying the given expression

$$f = \overline{A}\overline{B}(C+\overline{C})$$
$$= \overline{A}\overline{B}\cdot 1 \qquad \text{(Since, } C+\overline{C}=1\text{)}$$
$$= \overline{A}\overline{B} \qquad \text{(Since, } \overline{B}\cdot 1=\overline{B}\text{)}$$
$$= \overline{A+B} \qquad \text{(Since, } \overline{A+B}=\overline{A}\overline{B} \text{ De Morgan's law)}$$

The final function represents XOR gate.
To implement this function two 2 × 1 multiplexer is required.

9. Topic: Digital Logic

(c) Given:

$$3 * 4096 + 15 * 256 + 5 * 16 + 3$$

Converting into binary representation

$(2 + 1)4096 + (8 + 4 + 2 + 1)256 + (4 + 1) * 16 + (2 + 1)$
$(2^1 + 2^0)2^{12} + (2^3 + 2^2 + 2^1 + 2^0)2^8 + (2^2 + 2^0)2^4 + (2^1 + 2^0)$
$2^{13} + 2^{12} + 2^{11} + 2^{10} + 2^9 + 2^8 + 2^6 + 2^4 + 2^1 + 2^0$

Therefore, there will be 10 1's in the binary representation of the given expression (because binary equivalent of 2 is 10).

10. Topic: Digital Logic

(b) Given Boolean expression is

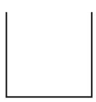
$$AB + A\overline{B} + \overline{A}C + AC$$

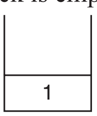
Simplifying this we get

$$A(B+\overline{B}) + C(\overline{A}+A) = A + C$$
$$\text{(Since, } B+\overline{B}=1,\ \overline{A}+A=1\text{)}$$

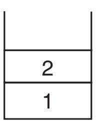
Therefore, the expression is independent of B.

11. Topic: Programming and Data Structures

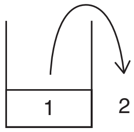
(a) Initially the stack is empty 

operation push (1) 

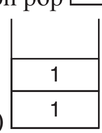
'1' is pushed to stack

operation push (2) 

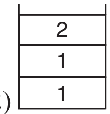
'2' is pushed to stack

operation pop 

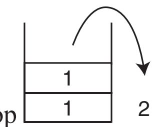
'2's is popped out from stack

push (1) 

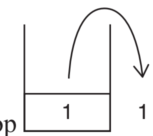
(1) is pushed to stack

push (2) 

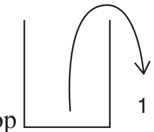
'2' is pushed to stack

pop 

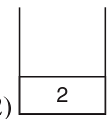
'2' is popped out from stack

operation pop 

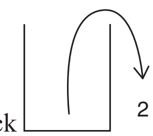
'1' is popped out from stack

operation pop 

'1' is popped out from stack

operation push (2) 

'2' is pushed to stack

pop operation will pop out '2' from stack 

Therefore, sequence of popped out values are 22112.

12. Topic: Algorithms

(b) Quick sort is a sorting algorithm that uses divide and conquer approach. One element called pivot element is chosen and array is portioned around it. The elements on left side are less than pivot element and elements on right are greater than the pivot element.
Now, running time of quick sort = $C\, n \log n$
Given, for n = 1000, running time = 100 s

$$\Rightarrow 100 = C1000 \log 1000$$
$$\Rightarrow C = \frac{1}{10 \log 1000} = 0.0333$$

Now for n = 100

Running time = 0.0333 × 100 × log 100 = 6.66

Therefore, running time ~ 6.7 s

13. Topic: Operating System

(a) This problem can be solved using greedy approach where solution is obtained piece by piece such that the next piece has most obvious and immediate benefit. Here the files should be stored such that the access of file is done at minimum cost. Thus the files should be stored in order of increasing length that is, $F3$, $F4$, $F1$, $F5$, $F6$, $F2$.

14. Topic: Algorithms

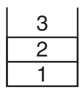
(b) In worst-case scenario, number of comparisons needed to reach an item not in hash table is given as:

Size of largest cluster + 1

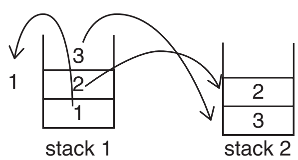
In the given hash table with 10 buckets, the largest cluster is ($S6$, $S3$, $S7$, $S1$) i.e. size of largest cluster is 4. Thus, number of comparisons needed to search an item that is not present = 4 + 1 = 5

15. Topic: Programming and Data Structures

(b) Stack is an ordered list that follows last in first out structure while queue is an abstract data structure that is open at both ends and follows first in first out structure. To implement a queue using stack, 2 stacks are required. In first stack, items will be pushed



to perform queue operation '1' should come out first, so second stack is used to push the items popped out from stack 1



and the last element i.e., '1' is popped at the last.

16. Topic: Databases

(c) An entity relationship diagram (ERD) is structural diagram used in database design. It contains information about the major entities within the system scope and the inter-relationships among these entities.
Relation data model is model that structures data using relation into grid like form consisting of rows and columns.
There are several processes & algorithms to correct ER diagrams into Relational Schema. In the given ERD, there is relationship with m:n cardinality
To express such type of relationships tables are used.
From given possible relations Exam (<u>Exam10</u>, <u>NID</u>, ExamName) relation will not hold since it has already been represented in Exam relation. Exam (<u>Exam10</u>, ExamName)

17. Topic: Databases

(c) For recovery, two approaches are used deferred database modification and immediate database modification. In deferred database-permanent database update happens when transaction commits & only new value logs are needed. In immediate database modification-permanent database update happens immediately and both old value & new value logs are needed. If deferred modification is used there is no need of redo operations.
Therefore, option (c) will be true.

18. Topic: Databases

(a) Given, database contains n records

If block holds 3 records then,

$$\text{Number of data blocks} = \frac{n}{3}$$

If block holds 10 key pointers then,

$$\text{Number of index blocks} = \frac{n}{10}$$

Therefore, total number of blocks needed to hold data file and dense index $= \frac{n}{3} + \frac{n}{10} = \frac{13n}{30}$

19. Topic: Database

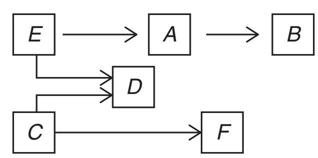
(c) Text is a datatype of MySQL database. Text is a variable length string type. A text is stored using a length prefix which requires 1 to 4 bytes depending on the datatype plus data. The maximum attribute of type text is 256.

20. Topic: Databases

(b) Given $R = (A, B, C, D, E, F)$ is a relation scheme and following dependencies are given

$$C \to F$$
$$E \to A$$
$$EC \to D$$
$$A \to B$$

Using pictorial representation to show the dependencies



From the diagram it is clear that E and C both are required to reach all attributes of relation R. Thus, EC is key for R.

21. Topic: Databases

(b) A relation model structures data using relation into grid like form consisting of rows and columns. So if D_1, D_2, …, D_n are domains in a relational model, then relation can be at most $D_1 \times D_2 \times \ldots \times D_n$.
Therefore, option (b) is correct.

22. Topic: Databases

(a, c) Query "SELECT C.Sid FROM catalog C WHERE C.pid NOT IN (SELECT P.pid FROM Parts P where P.color <> 'blue')" gives sids of all suppliers who supply blue parts and query" SELECT P.pid FROM P where P.color <> 'blue')" gives pids of parts which are not blue thus output gives names of all suppliers who have supplied non-blue part.

23. Topic: Databases

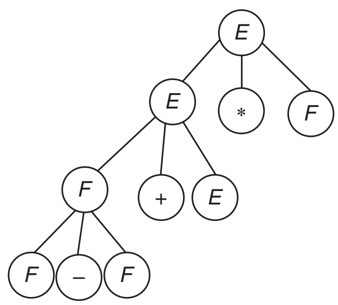
(d) For given schema:

Emp (Empcode, Name, Sex, Salary, Dept)

From this department is selected an all-male employees are grouped for each department. Then average salary of male employees whose salary is more than average salary of the organization is calculated. Thus, the output of the SQL query will be average salary of male employee is more than the average salary of the organization.

24. Topic: Theory of Computation

(b) Grammar describes the syntax of the programming languages. It is a set of production rules for strings operator which is at lower level in grammar is said to have higher precedence. In order to find the precedence of given expression, construct a tree



Therefore ‘–’ has higher precedence than *

25. Topic: Programming and Data Structures

(None) Tokens are building blocks that are constructed together to wire a C program. Tokens are classified into: keywords, identifiers, constants, strings, special symbols and operators.

For the given C statement:

```
printf(“i =%d, &i = %x”, i&i);
```

number of token are:

1. printf
2. (
3. “i = %d, &i = %d”
4. ,
5. i
6. &
7. i
8.)
9. ;

Therefore, there are 9 tokens in the statement. So, none of the options is correct.

26. Topic: Theory of Computation

(d) Grammar describes the syntax of the programming languages. According to operator precedence (i) right hand side cannot be empty i.e. no nullable variable like E should be there.

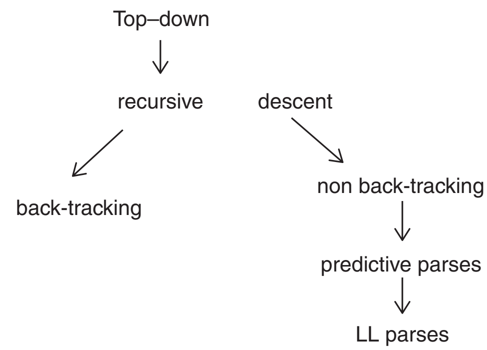
(ii) two productions cannot be adjacent to each other. Thus rule (i) and (iv) violates the requirement of the operator grammar.

$A \rightarrow BC$ (two productions are adjacent)

$A \rightarrow \varepsilon$ (nullable variable)

27. Topic: Compiler Design

(a) Top–down parsing technique parses the input and constructs a parse tree from root node gradually moving to the leaf nodes. Types of top–down parsing are as follows



Therefore, from the given options – recursive descent is a top–down parser.

28. Topic: Compiler Design

(d) Yacc stands for yet another compiler. It is standard parser generator for Unix Operating System. It analyses the structure of input stream and operate of the ‘big picture’. This kind of analysis of text files is normally associated with writing compilers.

29. Topic: Compiler Design

(c) (a) LALR parser is most powerful and costly as compare to other parser is incorrect statement. The order of parser are as follows

$$LR(0) < SLR\ (1) < LALR\ (1) < CLR\ (1)$$

(b) All CFG’s are LP and not all grammars are uniquely defined – is incorrect statement. The grammar generated by LP are CFG but this does not imply that all CFG are LP.

(c) Every SLR grammar is unambiguous but not every unambiguous grammar is SLR – this statement is TRUE.

(d) LR(k) is the most general back tracking shift reduce parsing method – this statement is incorrect. LR(k) is the most general non-back tracking shift reduce parsing method.

30. Topic: Operating System

(b) Semaphore is a variable used to solve critical section problem and to achieve synchronization.

Mutual exclusion is a program object that prevents simultaneous access to a shared resource. Process synchronization means sharing of system resources by processes such that inconsistency is minimum. Thus semaphore is used to solve the problem of process synchronization and mutual exclusion.

31. Topic: Operating System

(c) Given, there are 32 segments of size 1 k bytes
So, 32 segments = 2^5
Thus 5 bits are required to specify a particular segment.
Also, 10 bits are required to select a particular byte after selecting a page because 1 k byte = 2^{10}
Therefore, total bits required = 5 + 10 = 15 bits.

32. Topic: Operating System

(c) We have to distribute 40 tickets among 4 processes.

$$\text{Process } P_1 \text{ gets } 10\% \Rightarrow 40 \times \frac{10}{100} = 4$$
$$\text{Process } P_2 \text{ gets } 5\% \Rightarrow 40 \times \frac{5}{100} = 2$$
$$\text{Process } P_3 \text{ gets } 60\% \Rightarrow 40 \times \frac{60}{100} = 24$$
$$\text{Process } P_4 \text{ gets } 25\% \Rightarrow 40 \times \frac{25}{100} = 10$$

Therefore, the 4 processes P_1, P_2, P_3, P_4 get 4, 2, 24, 10 tickets respectively.

33. Topic: Operating System

(c) Round-Robin algorithm is a CPU scheduling algorithm. In this algorithm each process is assigned a fixed time slot in a cyclic way. Therefore circular queue is the best suited ready queue of the processes.

34. Topic: Operating System

(d) We know that:
Average time taken = Average seek time + Average rotational delay + Data transfer time average seek time is the time taken to move from 1st track to 2nd track, 1st track to 3rd track and so on.
So,

$$\text{Average seek time} = \frac{1+2+3+\cdots+499}{500} = 249.5 \text{ ms}$$

Average rotational delay – Given rotational speed = 600 rpm.

$$\text{Therefore, time of one rotation } = \frac{60 \text{ s}}{600} = 0.1 \text{ s}$$

$$\Rightarrow \text{average rotational delay} = \frac{0.1}{2} = 50 \text{ ms}$$

Data transfer time is time to transfer 250 bytes from disk data read on one track in one rotation is

$$100 \times 500 = 50000 \text{ B data}$$

Then,

$$\text{Data transfer time} = \frac{0.1 \times 250}{50000} = 0.5 \text{ ms}$$

Therefore,

$$\text{Average time to transfer} = 249.5 + 0.5 + 50 = 300 \text{ ms}$$

35. Topic: Operating System

(b) Each P operation decreases semaphore value by 1 and each V operation increase the semaphore value by 1. If the value of semaphore at particular time is 7, then after 20 P and 15 V operations the resulting value of semaphore will be:

$$7 - 20 + 15 = 2$$

36. Topic: Operating System

(c) Increasing RAM of a computer improves performance because of fewer page faults. Virtual memory of computer is independent of RAM, speed of RAM does not depend on its size and segmentation fault occurs because of invalid access by a program or illegal memory address which is not related to size of memory. Therefore option (c) is correct.

37. Topic: Operating System

(c) Fork () function creates new process by duplicating the existing process. The existing process is called the parent process and newly created process is called child process. The number of child process created by n fork () function calls = $2^n - 1$.
In the given program fork () function is called 3 times; therefore, the number of new child process created is

$$2^3 - 1 = 7$$

38. Topic: Operating System

(d) Given:

$$50\% \text{ I/O wait time suppose CPU time} = 2x$$

So,

$$\text{I/O operation time} = x$$
$$\text{Process seen time} = x$$

Given, two jobs need 10 min CPU time
Then,

$$x = 10 \Rightarrow 2x = 20$$
$$\text{Time for completion of both jobs} = 20 + 20 = 40$$

39. Topic: Databases

(b) A B-tree is a self-balancing tree. A B-tree is defined by minimum degree K. Every node of B-tree except root must contain at least $(K - 1)$ keys and all nodes including is the root may contain $(2K - 1)$ keys. Therefore, if a node has K children then node contains exactly $(K - 1)$ keys

40. Topic: Algorithms

(d) The recurrence relation found from the given program is

$$T(n) = 2T(n-1) + a$$

using back substitution method

$$T(n-1) = 2T(n-2) + a$$
$$T(n-2) = 2T(n-3) + a$$

and so on
So,

$$T(n) = 2[2T(n-2) + a] + a = 4T(n-2) + 3a$$
$$\Rightarrow T(n) = 4[2T(n-3) + a] + 3a = 8T(n-3) + 7a$$
$$= 2^3 T(n-3) + (2^3 - 1)a$$

So,

$$T(n) = 2^m T(n-m) + (2^m - 1)a$$

Using condition $T(1) = 1 \Rightarrow n - m = 1 \Rightarrow m = n - 1$
Then,

$$T(n) = 2^{n-1} T(1) + (2^{n-1} - 1)a \sim O(2^n)$$

Therefore, time complexity of the given *C* function is $O(2^n)$.

41. Topic: Programming and Data Structures

(b) A spanning tree is a subset of a graph. It has minimum number of edges possible on all the vertices. If a graph is a complete graph with *n* vertices then total number of spanning trees is n^{n-2}. Where *n* is number of nodes in graph. So for a graph with 7 vertices number of spanning trees = $7^{7-2} = 7^5$.

42. Topic: Algorithms

(None) A two-way merge sort, sorts a data stream using repeated merges. It distributes input into two streams, sorting it and then writing it to the next stream. It is them merged into an output stream. It is then repeatedly distributed to two streams and merged until there is single sorted output.

	20 47	25 8	9 4	40 30	12 17
1st Pass:	20 47	8 25	4 9	30 40	12 17
2nd Pass:	8 25	20 47	4 9	30 40	12 17

43. Topic: Theory of Computation

(c) Regular languages are closed under the following:-

1. intersection
2. complement
3. union
4. kleene – closure

$\Sigma^* - R_1$ implies complement of regular set R_1. Therefore, $\Sigma^* - R_1$ is regular and the correct option.

44. Topic: Computer Networks

(c) DNS or domain name system is the way that internet domain names are located and translated into internet protocol addresses. If maps the name people use to locate website to IP address. Therefore DNS maps the IP address to a hierarchy of domain names.

45. Topic: Computer Networks

(a) HTML or hypertext markup language is a standard language for creating web pages and web applications. To add a background color for all <*h*l> elements following syntax is used

*h*1{background-color: # FFFFFF}

46. Topic: Programming and Data Structures

(None) Correct syntax to write 'Hi there' in Javascript is document. Write ("hithere")
Therefore none of the options is correct.

47. Topic: Computer Networks

(c) XML or extensible markup language is a markup language like HTML designed to store and transport data and to be self-descriptive. To declare the version of XML the correct syntax is

<?xml version="1.0"/>

48. Topic: Computer Networks

(d) A switch is a device in a computer network that connects devices together on a network. A T– switch is a switch used to rearrange the connections between computing equipment's.

49. Topic: Computer Networks

(d) Microwave communications, satellite and radar all use microwaves because this frequency can easily penetrate through the Earth's atmosphere with less interference. Microwave has frequency range 3 – 300 GHz/100 – 1 m. Therefore extremely high frequency is used for microwave communications satellite and radar.

50. Topic: Computer Networks

(b) TCP/IP or Transmission control protocol and internet protocol is network model used in internet architecture. These are set of rules or protocol that describe the movement of data between the source & destination or the internet. The IP standards specify that each host is assigned a unique 32 bit number called internet address. Therefore, correct option is (b).

51. Topic: Computer Networks

(a) In synchronous transfer, data is transferred as a continuous stream of data in form of signals which are accompanied by regular timing signals that are generated by external clocking system. In synchronous transfer start and stop bit is not required.
So,

$$\text{Total bits transmitted} = 7 + 1 = 8 \text{ bits}$$

$$\text{Number of characters transmitted} = \frac{2400}{8} = 300 \text{ bps}$$

52. Topic: Computer Networks

(b) CRC or cyclic redundancy check is error detecting code used in digital networks and storage devices. It is used to detect accidental changes in raw data. In CRC, a dividend is generated at sender that is send to the receiver. Dividend that is required to send is data unit padded by remainder bits at sender. So,

Data unit = 100111001
Divisor = 1011

```
1011 | 100111001000      | 101000001
     | 1011
     |  1011001000
     |  1011
     |     1000
     |     1011
     |      011           Remainder
```

So, the remainder = 011
Therefore, dividend = 100111001011

53. Topic: Computer Networks

(d) ACK number in TCP tells which is next expected to be received. It depends on the initial random sequence number. ACK number of 1000 in TCP means that TCP is expecting a packet numbered 1000 to be delivered next. If sequence number is 1, then ACK number of 1000 means 999 bytes successfully received. If sequence number is 0 then ACK of 1000 means 1000 bytes received correctly. Here no such information is given therefore none of the options is correct.

54. Topic: Computer Networks

(a) The binary equivalent of given IP address and mask is

IP address: 01111101.10000110.01110000.01000010
mark: 11111111.11111111.11100000.00000000

The first address (network address) is obtained by using AND operation on IP address and mark. So,

Network Address:
01111101.10000110.01100000.00000000

The decimal equivalent of above binary is 125.134.96.00
Therefore, the network address is 125.134.96.0

55. Topic: Computer Networks

(b) In this case, time is divided into slots and request can be send only at the beginning of the time slot.
Given:

Number of requests = 70 requests/s

Time slot = 50 ms

$$\text{Number of slots per second} = \frac{1}{50\times 10^{-3}} = 20 \text{ slots}$$

$$\text{Channel load} = \frac{\text{number of requests}}{\text{number of slots}} = \frac{70}{20}$$

$$= 3.5$$

56. Topic: Computer Networks

(c) PING is a TCP/IP application that sends datagrams once every second in the hope of an echo response from the machine being PINGED-this statement is true.
If a machine is connected & running a TCP/IP protocol stack, it should respond to the PING datagram with a datagram of its own-this statement is true.
If PING in counters an error condition, an ICMP message is not returned-this statement is false an ICMP message is returned if PING encounters an error condition.
PING display the time of the return response in milliseconds or one of several errors messages–this statement is true.

57. Topic: Computer Networks

(c) To find out the interface the estimated address should be forwarded we need to perform bitwise AND operation of the estimated address with the mask. If the output obtained matched with the destination address, then the packet will be forwarded to the corresponding interface

Estimated address = 144.16.68.117

In binary representation it will be written as

10010000.00010000.01000100.01110101
Mask = 255.255.255.0

In binary representation it is written as

11111111.11111111.11111111.00000000

Applying bitwise AND operation, we get:

10010000.00010000.01000100.00000000

The decimal equivalent is 144.16.68.0
Which matches with the destination mask hence the packet will be forwarded to eth2.

58. Topic: Computer Networks

(a) The OSI or open system inter connection model characterizes the communication function of computing system. The model defines seven layers.

7.	Application layer	host layers
6.	Presentation layer	
5.	Session layer	
4.	Transport layer	
3.	Network layer	media layers
2.	Data link layer	
1.	Physical layer	

Therefore, transport, session, presentation, application are host-to-host layers of OSI model.

59. Topic: Software Engineering

(a) Acceptance testing is a level of software testing where system is tested for acceptability. It consists of two parts: alpha testing & beta testing, alpha testing is a system testing performed by development team and beta testing is a system testing performed by a friendly set of customers.

60. Topic: Software Engineering

(None) Weighing factors of the given functional units are: - user input (4), user output (5), enquiries (4), files (10), external interfaces (7)
So,

$$\text{Average complexity} = 15 \times 4 + 50 \times 5 + 24 \times 4 + 12 \times 10 + 8 \times 7 = 582$$

Given:

$$\text{Effort} = 70 \text{ person-month}$$

Therefore,

$$\text{Productivity} = 582 \times \frac{70}{100} = 407.4$$

61. Topic: Computer Organization and Architecture

(a) 8085 microprocessor has 8-bit flag register that indicates conditions arising after arithmetic & logical operations.

Here SUB A subtracts content of accumulator A from contents of accumulator A
MVI B,$(01)_H$ stores 01 is register B
DCR B. decreases content of register B
Hex code of SUB A = 97
Hex code of MV1 B = 06
Hex code of DCR B = 05
Hex code of HLT = 76
Content of flag register will be $(54)_H$.

62. Topic: Computer Organization and Architecture

(b) Access time is the time that lapses between the initiation of an operation and the completion of the operation. Cycle time is the minimum time delay between the initiations of two independent memory operations rotational time is the amount of time taken for the disk to rotate till the required location is reached. Latency time is overhead of finding the right memory location and preparing to access it.

63. Topic: Algorithms

(b) .png stand for portable graphics format. It is a bitmap format on the internet .png uses deflate algorithm which is the most commonly used file compression algorithm. It has following characteristics high reliability, good compression, good encoding speed, excellent decoding speed, minimal overhead on compression.

64. Topic: Operating System

(a) Dirty bit allows performance optimization. Dirty bit for a page in a page table helps avoid unnecessary writes on a paging device. Dirty bit status is checked before the write operation is performed.

65. Topic: Image Processing

(a) A flame is a single still image within the sequence of images that comprises the video. To compress video signals, MPEG categorizes frames into different formats. These formats vary from frame types to frame types. MPEG frame types include I frame, P frame, B frame, D frame. Thus, A frame is not an image type used in MPEG.

66. Topic: Multimedia

(c) Given:

Sampling frequency = 44.1 kHz
$= 44.1 \times 10^3$ bps

This is quantized using 16 bits.
Since sampling frequency is stereo audio signal, two channels will be used.
So,

bit rate $= 2 \times 16 \times 44.1 \times 10^3 = 1411.2 \times 10^3$ bps
Compression ratio = 0.5

Then,

Compressed bit rate = bit rate × compression ratio
$= 1411.2 \times 10^3 \times 0.5 = 705.6 \times 10^3$

Therefore,

$$\text{Data in 10 s} = 705.6 \times 10^3 \text{ bits}$$
$$= 705.6 \times \frac{10^3 \times 1024}{8} \text{ kB}$$
$$= 861.33 \text{ kB} \sim 860 \text{ kB}$$

67. Topic: Multimedia

(d) The compression ratio in typical mp3 audio file is 10:1 This implies that 1 min of CD quality music compression requires 10 MB of storage.

68. Topic: Programming and Data Structures

(d) The program fragment is as follows:

```
if (a > b)
{
   if (b > c)
   {
          s1;
   }
   else
   {
          s2;
   }
}
```

Therefore, $s2$ will be executed when 1st if condition is satisfied while the second if condition is not satisfied, that is, $a > b$ and $b \le c$.

69. Topic: Programming and Data Structures

(d) The value assigned will depend on complier. This semantic operation is defined as "undefined". It's result is not prescribed by the language specification rather the compiler is allowed to give a compile time diagnostic in this case.

70. Topic: Programming and Data Structures

(b) Static variable preserve its value even out of their scope. So in main function, inc() function is called 3 times and in each function call 'x' variable which is static is incremented and printed. And the incremented value is preserved till the next function call which increments 'x' again. So the output will be 1 2 3.

71. Topic: Programming and Data Structures

(c) Given:

$$i = 6720, j = 4$$

Iteration 1:

$$i\%j = 6720\%4 == 0$$
$$i = 6720/4 = 1680$$
$$j = 4 + 1 = 5$$

Iteration 2:

$$i\%j = 1680\%5 == 0$$
$$i = 1680/5 = 336$$
$$j = 5 + 1 = 6$$

Iteration 3:

$$i\%j = 336\%6 == 0$$
$$i = 336/6 = 56$$
$$j = 6 + 1 = 7$$

Iteration 4:

$$i\%j = 56\%7 == 0$$
$$i = 56/7 = 8$$
$$j = 7 + 1 = 8$$

Iteration 5:

$$i\%j = 8\%8 == 0$$
$$i = 8/8 = 1$$
$$j = 8 + 1 = 9$$

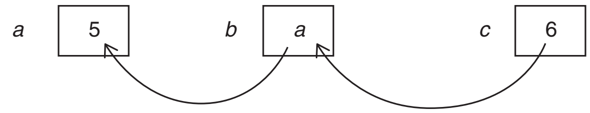
Iteration 6:

$$i\%j = 1\%9 \text{ condition fails}$$

Therefore, j will have value 9.

72. Topic: Programming and Data Structures

(d) In given fragment, C saves address of b and b saves address of a, assigning 5 to 0 assigns 5 to a.



73. Topic: Programming and Data Structures

(a) We have:

Initially: $x[] = \{1, 2, 3, 4, 5, 6, 7, 8\};$

for $i = 2$, $x[x[2]] = x[2] \Rightarrow x[3] = x[2] \Rightarrow x[3] = 3$
So, $x[] = \{1, 2, 3, 3, 5, 6, 7, 8\}$
for $i = 3$, $x[x[3]] = x[3] \Rightarrow x[3] = x[3]$
So, $x[] = \{1, 2, 3, 3, 5, 6, 7, 8\}$
for $i = 4$, $x[x[4]] = x[4] \Rightarrow x[5] = x[4] \Rightarrow x[5] = 5$
So, $x[] = \{1, 2, 3, 3, 5, 5, 7, 8\}$
for $i = 5$, $x[x[5]] = x[5] \Rightarrow x[5] = x[5]$
So, $x[] = \{1, 2, 3, 3, 5, 5, 7, 8\}$
for $i = 6$, loop exists

Therefore, output of program is 1 2 3 3 5 5 7 8

74. Topic: Programming and Data Structures

(None) None of the given statements have compilation error in C.

75. Topic: Programming and Data Structures

(a) The following loop is applying bitwise AND operation, i & 1. Now we know that bitwise operation AND between 1 and an odd number gives 1 and between an even number and 1 gives 0. Therefore there will be alternate 0 and 1 in the output.

76. Topic: Programming and Data Structures

(d) Macros are inclined directly into a program. The macro call hypotenuse $(a + 2, b + 3)$ evaluates

$$\text{sqrt}\ (a + 2 * a + 2 + b + 3 * b + 3)$$
$$\text{sqrt}(a + 2a + 2 + b + 3b + 3) = \text{sqrt}(3a + 4b + 5)$$

Therefore, option (d) is correct sqrt$(3 * a + 4 * b + 5)$

77. Topic: Computer Organization and Architecture

(c) Accumulator is one address instruction, so 8 one address instructions are required to evaluate:

$$X = (M + N \times O)/(P \times Q)$$

1. Load P: ACC $\leftarrow$ M[P]
2. Mul Q: ACC $\leftarrow$ ACC*M[Q]
3. Store R: m[R] $\leftarrow$ ACC
4. Load N: ACC $\leftarrow$ M[N]
5. Mul O: ACC $\leftarrow$ ACC*M[O]
6. Add M: ACC $\leftarrow$ ACC + M[M]
7. Div R: ACC $\leftarrow$ ACC/M[R]
8. store X: M[X] $\leftarrow$ ACC

78. Topic: Digital Logic

(b) The number of bits is the exponent of the smallest power of two, mathematically greater than the number:

$$d > n \log_{10} 2$$

where d is the digits and n is number of bits

So,

$$64 = n\log_{10} 2$$

$$\Rightarrow n = \frac{64}{\log_{10} 2} = \frac{64}{0.3010} \approx 213$$

79. Topic: Programming and Data Structures

(b) From the C declaration structure creates memory for array S and union and union creates memory for z.

So,

$$\text{short } s[5] \Rightarrow 5 \times 2 = 10 \text{ bytes}$$

and

$$\text{long } z \Rightarrow 8 \text{ bytes}$$

Therefore, total memory = 18 bytes

80. Topic: Programming and Data Structures

(c) In call by reference, arguments to a function copies the address of an argument into the formal parameter.

In call by value, arguments to a function copies actual values of an argument into the formal parameter of the function.

If call by value is used to pass arguments to a function; whatever changes happen in function foo will not affect the actual value of x. Since x is integer, final value of x will be 5.

If call by reference is used to pass arguments to a function, x and y in foo function are aliases of x in main so final value of x will change. So,

$$x += y \Rightarrow 5 + 5 = 10$$
$$y += x \Rightarrow 10 + 10 = 20$$

Therefore, final value of $x = 20$.

Indian Space Research Organisation
Computer Science

2016

QUESTION PAPER

1. Which of the following is true?

(a) $\sqrt{3}+\sqrt{7}=\sqrt{10}$ (b) $\sqrt{3}+\sqrt{7}\le\sqrt{10}$

(c) $\sqrt{3}+\sqrt{7}<\sqrt{10}$ (d) $\sqrt{3}+\sqrt{7}>\sqrt{10}$

2. What is the sum to infinity of the series, $3 + 6x^2 + 9x^4 + 12x^6 + \cdots$ given $|x| < 1$?

(a) $\dfrac{3}{(1+x^2)}$ (b) $\dfrac{3}{(1+x^2)^2}$

(c) $\dfrac{3}{(1-x^2)^2}$ (d) $\dfrac{3}{(1-x^2)}$

3. $\lim\limits_{x\to 0}\dfrac{\sqrt{1+x}-\sqrt{1-x}}{x}$ is given by

(a) 0 (b) -1

(c) 1 (d) $\dfrac{1}{2}$

4. If $(G,\cdot)$ is a group such that $(ab)^{-1} = a^{-1}\,b^{-1}$, $\forall\ a, b \in G$ then G is a/an

(a) Commutative semi group (b) Abelian group

(c) Non-abelian group (d) None of these

5. A given connected graph G is an Euler Graph if and only if all vertices of G are of

(a) Same degree (b) Even degree

(c) Odd degree (d) Different degree

6. Maximum number of edges in a n – node undirected graph without self-loops is

(a) n^2 (b) $\dfrac{n(n-1)}{2}$

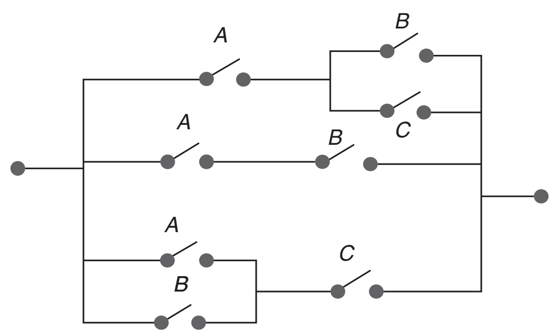
(c) $n-1$ (d) $\dfrac{n(n+1)}{2}$

7. The minimum number of NAND gates required to implement the Boolean function $A+A\overline{B}+A\overline{B}C$ is equal to

(a) 0 (Zero) (b) 1

(c) 4 (d) 7

8. The minimum Boolean expression for the following circuit is



(a) $AB + AC + BC$ (b) $A + BC$

(c) $A + B$ (d) $A + B + C$

9. For a binary half-subtractor having two inputs A and B, the correct set of logical expression for the outputs D (= A minus B) and X (= borrow) are

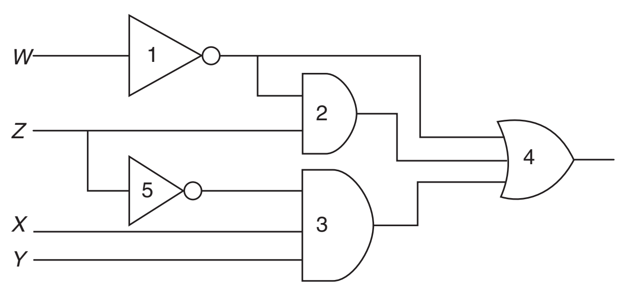
(a) $D = AB+\overline{A}B,\ X = \overline{A}B$

(b) $D = \overline{A}B+A\overline{B},\ X = A\overline{B}$

(c) $D = \overline{A}B+A\overline{B},\ X = \overline{A}B$

(d) $D = AB+\overline{A}B,\ X = A\overline{B}$

10. Consider the following gate network



Which one of the following gates is redundant?

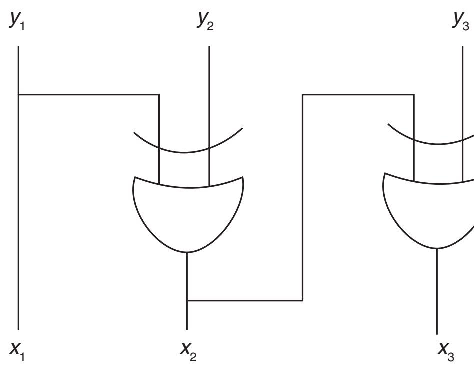
(a) Gate No. 1 (b) Gate No. 2

(c) Gate No. 3 (d) Gate No. 4

11. The dynamic hazard problem occurs in

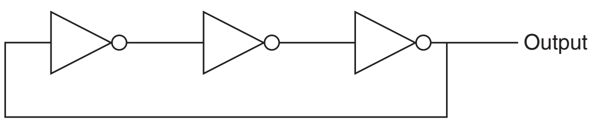
(a) Combinational circuit alone
(b) Sequential Circuit only
(c) Both (a) and (b)
(d) None of the above

12. The logic circuit given below converts a binary code y_1, y_2, y_3 into



(a) Excess-3 code (b) Gray code
(c) BCD code (d) Hamming Code

13. The circuit shown in the figure below is



(a) An oscillating circuit and its output is square wave
(b) The one whose output remains stable in '1' state
(c) The one having output remains stable in '0' state
(d) Has a single pulse of three times propagation delay

14. If $12A7C_{16} = X_8$ then the value of X is

(a) 224174 (b) 425174
(c) 6173 (d) 225174

15. The Excess-3 Code is also called

(a) Cyclic Redundancy Code
(b) Weighted Code
(c) Self-Complementing Code
(d) Algebraic Code

16. The simplified Sum of Product form of the following Boolean expression

$(P+\overline{Q}+R)(P+Q+R)\ (P+Q+\overline{R})$

(a) $(\overline{P}Q+R)$ (b) $(P+Q\overline{R})$
(c) $(P\overline{Q}+R)$ (d) $(PQ+R)$

17. Which of the following binary number is the same as its 2's complement?

(a) 1010 (b) 0101
(c) 1000 (d) 1001

18. The functional difference between *SR* flip-flop and *JK* flip-flop is that

(a) *JK* flip-flop is faster than *SR* flip-flop
(b) *JK* flip-flop has a feedback path
(c) *JK* flip-flop accepts both inputs 1
(d) None of the above

19. Consider a non-pipelined processor with a clock rate of 2.5 GHz and average cycles per instruction of four. The same processor is upgraded to a pipelined processor with five stages; but due to the internal pipeline delay, the clock speed is reduced to 2 GHz. Assume that there are no stalls in the pipeline. The speed up achieved in this pipelined processor is

(a) 3.2 (b) 3.0
(c) 2.2 (d) 2.0

20. What is the output of this C code?

```
#include <stdio.h>
void main0
{
    int k = 5;
    int *p = &k;
    int **m = &p;
    printf("%d %d %d\n", k, *p, **m);
}
```

(a) 5 5 5 (b) 5 5 junk
(c) 5 junk junk (d) Compile time error

21. Consider a disk pack with 16 surfaces, 128 tracks per surface and 256 sectors per track. 512 bytes of data are stored in a bit serial fashion in a sector. The capacity of the disk pack and the number of bits required to specify a particular sector in the disk respectively

(a) 256 MB, 19 bits (b) 256 MB, 28 bits
(c) 512 MB, 20 bits (d) 64 GB, 28 bits

22. Let the page fault service time be 10 ms in a computer with average memory access time being 20 ns. If one page fault is generated for every 10^6 memory accesses, what is the effective access time for the memory?

(a) 21.4 ns (b) 29.9 ns
(c) 23.5 ns (d) 35.1 ns

23. Register renaming is done in pipelined processors

(a) As an alternative to register allocation at compile time

(b) For efficient access to function parameters and local variables

(c) To eliminate certain kind of hazards

(d) As part of address translations

24. In which class of Flynn's taxonomy, Von Neumann architecture belongs to?

(a) SISD (b) SIMD

(c) MIMD (d) MISD

25. What will be output of following program? Assume that you are running this program in little-endian processor.

```
#include<stdio.h>
int main()
{
    short a = 320;
    char *ptr;
    ptr = (char *)&a;
    printf("%d",*ptr);
    return 0;
}
```

(a) 1 (b) 320

(c) 64 (d) Compilation Error

26. Consider the following segment of C code

```
int j, n;
j =1;
while (j< = n)
    j = j*2;
```

The number of comparisons made in the execution of the loop for any $n > 0$ is

(a) $[\log_2 n]*n$ (b) n

(c) $[\log_2 n]$ (d) $[\log_2 n] + 1$

27. The following postfix expression with single digit operands is evaluated using a stack

8 2 3 ^/2 3 * + 5 1* – (Note that ^ is the exponential operator)

The top two elements of the stack after the first * operator is evaluated are

(a) 6, 1 (b) 5, 7

(c) 3, 2 (d) 1, 5

28. Average number of comparison required for a successful search for sequential search on 'n' items is

(a) $\frac{n}{2}$ (b) $\frac{(n-1)}{2}$

(c) $\frac{(n+1)}{2}$ (d) None of the above

29. A Hash Function f is defined as f(key) = key mod 7. With linear probing, while inserting the keys 37, 38, 72, 48, 98, 11, 56 into a table indexed from 0, in which location the key 11 will be stored (count table Index 0 as 0^{th} location)?

(a) 3 (b) 4

(c) 5 (d) 6

30. A complete binary tree with n non-leaf nodes contains

(a) $\log_2 n$ nodes (b) $n + 1$ nodes

(c) $2n$ nodes (d) $2n + 1$ nodes

31. Algorithm design technique used in quick sort algorithm is

(a) Dynamic programming (b) Backtracking

(c) Divide and Conquer (d) Greedy Method

32. An FSM (Finite State Machine) can be considered to be a Turning Machine of finite tape length

(a) Without rewinding capability and unidirectional tape movement

(b) Rewinding capability and unidirectional tape movement

(c) Without rewinding capability and bidirectional tape movement

(d) Rewinding capability and bidirectional tape movement

33. Let $L = \{\omega \in (0+1)^* \mid \omega \text{ has even number of 1's}\}$, i.e., L is the set of all bit strings with even number of 1s. Which one of the regular expression below represents L?

(a) (0*10*1)* (b) 0*(10*10*)*

(c) 0*(10*1*)*0* (d) 0*1(10*1)*10*

34. Consider the following recurrence

$$T(n) = 2T(\sqrt{n}) + 1$$
$$T(1) = 1$$

Which of the following is true?

(a) $T(n) = O(\log \log n)$ (b) $T(n) = O(\log n)$

(c) $T(n) = O(\sqrt{n})$ (d) $T(n) = O(n)$

35. Consider the following statements about the context-free grammar:

$$G = \{S \to SS, S \to ab, S \to ba, S \to \wedge\}$$

I. G is ambiguous

II. G produces all strings with equal number of a's and b's

III. G can be accepted by a deterministic PDA

Which combinations below expresses all the true statements about G?

(a) I only (b) I and III only
(c) II and III only (d) I, II and III

36. If L and $\overline{L}$ are recursively enumerable, then L is

(a) Regular (b) Context-free
(c) Context-sensitive (d) Recursive

37. $S \rightarrow a\ Sa \mid bSb \mid a \mid b$

The language generated by the above grammar over the alphabets $\{a, b\}$ is the set of

(a) All palindromes
(b) All odd length palindromes
(c) Strings that begin and end with same symbol
(d) All even length palindromes

38. What is the highest type number that can be assigned to this following grammar:

$$S \rightarrow Aa, A \rightarrow Ba, B \rightarrow abc$$

(a) Type 0 (b) Type 1
(c) Type 2 (d) Type 3

39. Access time of the symbol table will be logarithmic, if it is implemented by

(a) Linear list
(b) Search Tree
(c) Hash Table
(d) Self-organization list

40. Recursive descent parsing is an example of

(a) Top-down parsers (b) Bottom-up parsers
(c) Predictive parsers (d) None of these

41. A top-down parser generates

(a) Rightmost derivation
(b) Rightmost derivation in reverse
(c) Leftmost derivation
(d) Leftmost derivation in reverse

42. Relative mode of addressing is most relevant to writing

(a) Co-routines
(b) Position-independent code
(c) Sharable code
(d) Interrupt Handlers

43. A simple two-pass assembler does which of the following in the first pass?

(a) Checks to see if the instructions are legal in the current assembly mode
(b) It allocates space for the literals
(c) It builds the symbol table for the symbols and their values
(d) All of these

44. Peephole optimization is a form of

(a) Loop optimization (b) Local optimization
(c) Constant folding (d) Data flow analysis

45. At a particular time of computation, the value of a counting semaphore is 7. Then 20 P operation and xV operations were completed on this semaphore. If the final value of the semaphore is 5, x will be

(a) 18 (b) 22
(c) 15 (d) 13

46. With single resource, deadlock occurs

(a) If there are more than two processor competing for that resource
(b) If there are only two processes competing for that resource
(c) If there is a single process competing for that resource
(d) None of these

47. A system has 3 processes sharing 4 resources. If each process needs a maximum of 2 units, then

(a) Deadlock can never occur
(b) Deadlock may occur
(c) Deadlock has to occur
(d) None of these

48. Determine the number of page faults when references to pages occur in the following order: 1, 2, 4, 5, 2, 1, 2, 4. Assume that the main memory can accommodate 3 pages and the main memory already has the pages 1 and 2, with page one having brought earlier than page 2. (LRU page replacement algorithm is used)

(a) 3 (b) 5
(c) 4 (d) None of these

49. Working Set (t, k) at an instant of time t is

(a) The set of k future references that the Operating System (OS) will make
(b) The set of future references that the OS will make in next t unit of time
(c) The set of k references with high frequency
(d) The k set of pages that have been referenced in the last t time units

50. A CPU generates 32-bit virtual addresses. The page size is 4 KB. The processor has a translation look-aside buffer (TLB) which can hold a total of 128 page table entries and is 4-way set associative. The minimum size of the TLB tag is

(a) 11 bits (b) 13 bits
(c) 15 bits (d) 20 bits

51. The real-time operating system, which of the following is the most suitable scheduling scheme?

(a) Round robin
(b) First come first serve
(c) Pre-emptive
(d) Random scheduling

52. In which one of the following page replacement policies, Balady's anomaly may occur?

(a) FIFO (b) Optimal
(c) LRU (d) MRU

53. Consider Join of a relation R with a relation S. If R has m tuples and S has n tuples, then maximum and minimum sizes of the Join respectively are

(a) $m + n$ and 0 (b) mn and 0
(c) $m + n$ and $|m - n|$ (d) mn and $m + n$

54. Let R (a, b, c) and S (d, e, f) be two relations in which d is the foreign key of S that refers to the primary key of R

Consider the following four operations in R and S

I. Insert into R
II. Insert into S
III. Deletion from R
IV. Deletion from S

Which of the following can cause violation of the relational integrity constraint above?

(a) Both I and IV (b) Both II and III
(c) All of these (d) None of these

55. The relation book (<u>title</u>, <u>price</u>) contains the titles and prices of different books. Assuming that no two books have the same price, what does the following SQL query list?

[select title from book as B where (select count(*) from book as T where T.price > B.price) < 5]

(a) Titles of the four most expensive books
(b) Title of the fifth most inexpensive book
(c) Title of the fifth most expensive book
(d) Titles of the five most expensive books

56. Goals for the design of the logical schema include

(a) Avoiding data inconsistency
(b) Being able to construct queries easily
(c) Being able to access data efficiently
(d) All of these

57. Given the relations

***employee (name, salary, deptno)* and**
department (deptno, deptname, address)

Which of the following queries cannot be expressed using the basic relational algebra operations (U,– x, π, σ, p)?

(a) Department address of every employee
(b) Employees whose name is the same as their department name
(c) The sum of all employees' salaries
(d) All employees of a given department

58. Trigger is

(a) Statement that enables to start any DBMS
(b) Statement that is executed by the user when debugging an application program
(c) The condition that the system tests for the validity of the database user
(d) Statement that is executed automatically by the system as a side effect of a modification to the database

59. The order of a leaf node in a B+ tree is the maximum number of (value, data record pointer) pairs it can hold. Given that the block size is 1 K bytes, data record pointer is 7 bytes long, the value field is 9 bytes long and a block pointer is 6 bytes long, what is the order of the leaf node?

(a) 63 (b) 64
(c) 67 (d) 68

60. A clustering index is defined on the fields which are of type _________.

(a) Non-key and ordering
(b) Non-key and non-ordering
(c) Key and ordering
(d) Key and non-ordering

61. The Extent to which the software can continue to operate correctly despite the introduction of invalid inputs is called as

(a) Reliability (b) Robustness
(c) Fault tolerance (d) Portability

62. Which one of the following is a functional requirement?

(a) Maintainability
(b) Portability
(c) Robustness
(d) None of the mentioned

63. Configuration management is not concerned with

(a) Controlling changes to the source code
(b) Choice of hardware configuration for an application

(c) Controlling documentation changes
(d) Maintaining versions of software

64. A company needs to develop a strategy for software product development for which it has a choice of two programming languages *L*1 and *L*2. The number of lines of code (LOC) developed using *L*2 is estimated to be twice the LOC developed with *L*1. The product will have to be maintained for five years. Various parameters for the company are given in the table below.

Parameter	**Language *L*1**	**Language *L*2**
Man years needed for development	LOC/10000	LOC/10000
Development cost per man year	Rs. 10,00,000	Rs. 7,50,000
Maintenance time	5 years	5 years
Cost of maintenance per year	Rs. 1,00,000	Rs. 50,000

Total cost of the project includes cost of development and maintenance. What is the LOC for *L*1 for which the cost of the project using *L*1 is equal to the cost of the project using *L*2?

(a) 10,000 (b) 5,000
(c) 7,500 (d) 75,000

65. A company needs to develop digital signal processing software for one of its newest inventions. The software is expected to have 20000 lines of code. The company needs to determine the effort in person-months needed to develop this software using the basic COCOMO model. The multiplicative factor for this model is given as 2.2 for the software development on embedded systems, while the exponentiation factor is given as 1.5. What is the estimated effort in person-months?

(a) 196.77 (b) 206.56
(c) 199.56 (d) 210.68

66. In the spiral model of software development, the primary determinant in selecting activities in each iteration is

(a) Iteration size
(b) Cost
(c) Adopted process such as Rational Unified Process or Extreme Programming
(d) Risk

67. Bit stuffing refers to

(a) Inserting a '0' in user stream to differentiate it with a flag
(b) Inserting a '0' in flag stream to avoid ambiguity
(c) Appending a nibble to the flag sequence
(d) Appending a nibble to the user data stream

68. Dynamic routing protocol enable routers to:

(a) Dynamically discover and maintain routes
(b) Distribute routing updates to other routers
(c) Reach agreement with other routers about the network topology
(d) All of the above

69. In Ethernet CSMA/CD, the special bit sequence is transmitted by media access management to handle collision is called

(a) Preamble (b) Postamble
(c) Jam (d) None of these

70. Which network protocol Allows hosts to dynamically get a unique IP number on each bootup?

(a) DHCP (b) BOOTP
(c) RARP (d) ARP

71. In a token ring network, the transmission speed is 10^7 bps and the propagation speed is 200 m μs^{-1}. Then 1 bit delay in this network is equivalent to

(a) 500 m of cable (b) 200 m of cable
(c) 20 m of cable (d) 50 m of cable

72. The address of a class B host is to be split into subnets with a 6-bit subnet number. What is the maximum number of subnets and the maximum number of hosts in each subnet?

(a) 62 subnets and 262142 hosts
(b) 64 subnets and 262142 hosts
(c) 62 subnets and 1022 hosts
(d) 64 subnets and 1024 hosts

73. The message 11001001 is to be transmitted using the CRC polynomial $x^3 + 1$ to protect it from errors. The message that should be transmitted is

(a) 11001001000 (b) 11001001011
(c) 11001010 (d) 110010010011

74. What is the maximum size of data that application layer can pass on to the TCP layer below?

(a) Any size
(b) (2^{16} bytes-the size of TCP header)
(c) 2^{16} bytes
(d) 1500 bytes

75. Frames of 1000 bits are sent over a 10^6 bps duplex link between two hosts. The propagation time is 25 ms. Frames are to be transmitted into this link to maximally pack them in transit (within the link).

What is the minimum number of bits (*i*) that will be required to represent the sequence numbers distinctly?

Assume that no time gap needs to be given between transmission of two frames.

(a) $i = 2$ (b) $i = 3$
(c) $i = 4$ (d) $i = 5$

76. Which of the following is TRUE only for XML, but not for HTML?

(a) It is derived from SGML
(b) It describes content and layout
(c) It allows user defined tags
(d) It is restricted only to be used with web browsers

77. Consider a system with 2 level caches. Access times of Level 1 cache, Level 2 cache and main memory are 1 ns, 10 ns, and 500 ns, respectively. The hit rates of Level 1 and Level 2 caches are 0.8 and 0.9, respectively. What is the average access time of the system ignoring the search time within the cache?

(a) 13.0 ns (b) 12.8 ns
(c) 12.6 ns (d) 12.4 ns

78. If a class C is derived from class B, which is derived from class A, all through public inheritance, then a class C member function can access

(a) Only protected and public data of C and B
(b) Only protected and public data of C
(c) All data of C and private data of A and B
(d) Public and protected data of A and B and all data of C

79. Which one of the following is correct about the statements given below?

I. All function calls are resolved at compile-time in C language.
II. All function calls are resolved at compile-time in C++.

(a) Only II is correct
(b) Both I and II are correct
(c) Only I is correct
(d) Both I and II are incorrect

80. When a DNS server accepts and uses incorrect information from a host that has no authority giving that information, then it is called

(a) DNS lookup
(b) DNS hijacking
(c) DNS spoofing
(d) None of the mentioned

ANSWER KEY

1. (d)	**9.** (c)	**17.** (c)	**25.** (c)	**33.** (d)	**41.** (c)	**49.** (d)	**57.** (c)	**65.** (a)	**73.** (b)
2. (c)	**10.** (b)	**18.** (c)	**26.** (d)	**34.** (b)	**42.** (b)	**50.** (c)	**58.** (d)	**66.** (d)	**74.** (a)
3. (c)	**11.** (c)	**19.** (a)	**27.** (a)	**35.** (b)	**43.** (d)	**51.** (c)	**59.** (a)	**67.** (a)	**75.** (d)
4. (b)	**12.** (d)	**20.** (a)	**28.** (c)	**36.** (d)	**44.** (b)	**52.** (a)	**60.** (a)	**68.** (d)	**76.** (c)
5. (b)	**13.** (a)	**21.** (a)	**29.** (c)	**37.** (b)	**45.** (a)	**53.** (b)	**61.** (b)	**69.** (c)	**77.** (a, c)
6. (b)	**14.** (d)	**22.** (b)	**30.** (d)	**38.** (c)	**46.** (d)	**54.** (b)	**62.** (d)	**70.** (a)	**78.** (d)
7. (a)	**15.** (c)	**23.** (c)	**31.** (c)	**39.** (b)	**47.** (a)	**55.** (d)	**63.** (b)	**71.** (c)	**79.** (c)
8. (a)	**16.** (b)	**24.** (a)	**32.** (a)	**40.** (a)	**48.** (c)	**56.** (d)	**64.** (b)	**72.** (c)	**80.** (c)

*** None of the options is correct.**

ANSWERS WITH EXPLANATION

1. **Topic: Engineering Mathematics**

(d) We know that:

$$\begin{aligned}(\sqrt{3}+\sqrt{7})^2 &= (\sqrt{3})^2+(\sqrt{7})^2+2\sqrt{3}\cdot\sqrt{7}\\ &= 3+7+2\sqrt{3}\cdot\sqrt{7}\\ &= 10+2\sqrt{21}\end{aligned}$$

So,

$$(\sqrt{3}+\sqrt{7})^2 = 10+2\sqrt{21} \quad (1)$$

also

$$(\sqrt{10})^2 = 10 \quad (2)$$

from Eq. (1) and Eq. (2), we have

$$10+2\sqrt{21} > 10$$
$$\Rightarrow (\sqrt{3}+\sqrt{7})^2 > (\sqrt{10})^2 \Rightarrow \sqrt{3}+\sqrt{7} > \sqrt{10}$$

2. **Topic: Engineering Mathematics**

(c) Given:

$$S = 3 + 6x^2 + 9x^4 + 12x^6 + \cdots \quad (1)$$

Multiply both sides by x^2, we get:

$$Sx^2 = 3x^2 + 6x^4 + 9x^6 + 12x^8 + \cdots \quad (2)$$

Subtract Eq. (2) from Eq. (1), we get

$$\begin{aligned}S(1-x^2) &= 3 + 3x^2 + 3x^4 + 3x^6 + \cdots\\ &= 3(1 + x^2 + x^4 + x^6 + \cdots)\end{aligned}$$

Now, consider $1 + x^2 + x^4 + x^6 + \ldots$, this forms geometric progression with initial term $a = 1$ and common ratio $r = x^2$ So,

$$S_\infty = \frac{a}{1-r} = \frac{1}{1-x^2}$$

Therefore,

$$S(1-x^2) = 3\times\frac{1}{1-x^2} \Rightarrow S = 3\times\frac{1}{(1-x^2)^2}$$

3. **Topic: Engineering Mathematics**

(c) Given:

$$\lim_{x\to 0}\frac{\sqrt{1+x}-\sqrt{1-x}}{a}$$

$$= \lim_{x\to 0}\frac{(1+x)-(1-x)}{x(\sqrt{1+x}+\sqrt{1-x})} \quad \text{(By rationalizing)}$$

$$= \lim_{x\to 0}\frac{2x}{x(\sqrt{1+x}+\sqrt{1-x})} = \lim_{x\to 0}\frac{2}{\sqrt{1+x}+\sqrt{1-x}} = \frac{2}{2} = 1$$

4. **Topic: Engineering Mathematics**

(b) In an obtain group, operation applied on the group elements is independent of the order in which they are written. Hence for G if $(ab)^{-1} = a^{-1}b^{-1}$ then G is on abelian group.

5. **Topic: Engineering Mathematics**

(b) An Euler graph is a graph where every vertex has even degree. Thus G is a Euler Graph if and only if all vertices of G are of even degrees.

6. **Topic: Engineering Mathematics**

(b) For n nodes, each node can have $(n - 1)$ edges have n nodes can have n $(n - 1)$ edges. Since each edge is counted twice, total edges are $\frac{n(n-1)}{2}$

7. **Topic: Digital Logic**

(a) Given:

$$A + A\overline{B} + A\overline{B}C$$

Simplifying the boolean function we get

$$A + A\overline{B} + A\overline{B}C = A + A\overline{B}(1+C)$$

Using identity $1 + C = 1$ we get

$$A + A\overline{B}\cdot 1 = A + A\overline{B} \quad (\text{Since, } A\cdot 1 = A)$$
$$A + A\overline{B} = A(1+\overline{B}) = A\cdot 1 \quad (\text{Since, } 1+\overline{B} = 1)$$

So, the simplified boolean function is A which requires no NAND gate.

8. **Topic: Digital Logic**

(a) The boolean expression for the given circuit is:

$$A\,(B + C) + AB + (A + B)C$$

Simplifying this we get

$$AB + AC + AB + AC + BC = (AB + AB) + (AC + C) + BC$$

Using identity, $A + A = A$ we get

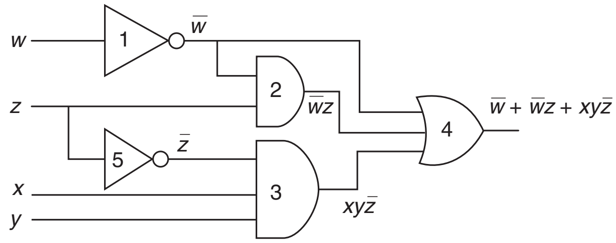
$$\Rightarrow AB + AC + BC$$

9. **Topic: Digital Logic**

(c) Half subtractor is XOR operation and borrow is AND operation with complement of 1 of the inputs. So, correct option is (c).

10. **Topic: Digital Logic**

(b) Given:



Output of given gate network is $y=\bar{w}+\bar{w}z+xy\bar{z}$.
Simplifying this expression we get:

$$y=\bar{w}+\bar{w}z+xy\bar{z}=\bar{w}(1+z)+xy\bar{z}$$

Using identity $1+z=1$ we get:

$$y=\bar{w}\cdot 1+xy\bar{z}=\bar{w}+xy\bar{z} \quad \text{(Since, } A\cdot 1=1 \text{ is on identity)}$$

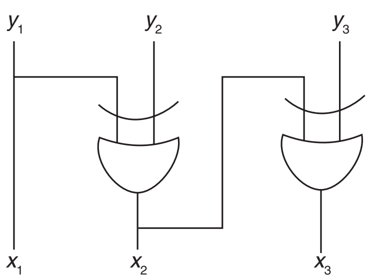
If gate 2 was not present the output of gate network would have been same that is, $y=\bar{w}+xy\bar{z}$. This gate 2 is redundant.

11. Topic: Digital Logic

(c) Dynamic hazard is possibility of output changing more than once for a single input change. It can occur in large circuits where multiple paths to output exists.

12. Topic: Digital Logic

(d) From figure given below, we have:



$$x_1=y_1$$
$$x_2=x_1\oplus y_2=y_1\oplus y_2$$
$$x_3=x_2\oplus y_3=y_1\oplus y_2\oplus y_3$$

The given logic circuit converts binary code y_1, y_2, y_3 into a hamming code.

13. Topic: Digital Logic

(a) The given circuit is having NOT gates in sequence. NOT gate changes state from 0 to 1 and 1 to 0. So, it is clear that given circuit is oscillating circuit.

14. Topic: Digital Logic

(d) Given:

$$(12\text{A}7\text{C})_{16}=X_8$$

Converting to binary digits

$$(0001\ 0010\ 1010\ 0111\ 1100)_2$$

Rearranging the bits as

$$(000\ 010\ 010\ 101\ 001\ 111\ 100)_2$$

Converting to octal we have $(0225174)_8$.

15. Topic: Digital Logic

(c) Excess-3 code is a BCD code. It is a 4-bit code. It is used along with BCD to convert decimal numbers to excess 3 forms. Excess-3 code is also called self-complementing code.

16. Topic: Digital Logic

(b) Given:

$$(P+\bar{Q}+\bar{R})(P+Q+R)(P+Q+\bar{R})$$
$$=(PP+PQ+PR+\bar{Q}P+\bar{Q}Q+\bar{Q}R+\bar{R}P+\bar{R}Q+\bar{R}R)(P+Q+\bar{R})$$
$$=(P+PQ+PR+\bar{Q}P+0+\bar{Q}R+\bar{R}P+\bar{R}Q+0)(P+Q+\bar{R})$$
$$\text{(Since, } PP=P, \bar{Q}Q=0, \bar{R}R=0)$$
$$=(P+PQ+PR+\bar{Q}P+\bar{Q}R+\bar{R}P+\bar{R}Q)(P+Q+\bar{R})$$
$$=(P(1+Q+R+\bar{Q}+\bar{R})+\bar{Q}R+\bar{R}Q)(P+Q+\bar{R})$$
$$=(P(1+R+\bar{Q}+\bar{R})+\bar{Q}R+\bar{R}Q)(P+Q+\bar{R}) \quad \text{(Since, } 1+Q=1)$$
$$=(P(1+\bar{Q}+\bar{R})+\bar{Q}R+\bar{R}Q)(P+Q+\bar{R}) \quad \text{(Since, } 1+R=1)$$
$$=(P(1+\bar{R})+\bar{Q}R+\bar{R}Q)(P+Q+\bar{R}) \quad \text{(Since, } 1+\bar{Q}=1)$$
$$=(P\cdot 1+\bar{Q}R+\bar{R}Q)(P+Q+\bar{R}) \quad \text{(Since, } 1+\bar{R}=1)$$
$$=(P+\bar{Q}R+\bar{R}Q)(P+Q+\bar{R}) \quad \text{(Since, } P\cdot 1=P)$$
$$=PP+PQ+P\bar{R}+\bar{Q}RP+Q\bar{Q}R+\bar{Q}R\bar{R}+P\bar{R}Q+\bar{R}QQ+\bar{R}\bar{R}Q$$
$$=P+PQ+P\bar{R}+P\bar{Q}R+P\bar{R}Q+O.R+\bar{Q}.O+\bar{R}Q+\bar{R}Q$$
$$\text{(Since, } Q\bar{Q}=R\bar{R}=0, QQ=Q \text{ and } \bar{R}\bar{R}=\bar{R})$$
$$=P+PQ+P\bar{R}+P\bar{R}Q+\bar{R}Q$$
$$\text{(Since, } 0\cdot R=\bar{Q}\cdot 0=0, \bar{R}Q+\bar{R}Q=\bar{R}Q)$$
$$=P(1+Q+\bar{R}+\bar{Q}R+\bar{R}Q)+\bar{R}Q$$
$$=P(1+\bar{R}+\bar{Q}R+\bar{R}Q)+\bar{R}Q \quad \text{(Since, } 1+Q=1)$$
$$=P(1+\bar{Q}R+\bar{R}Q)+\bar{R}Q \quad \text{(Since, } 1+\bar{R}=1)$$
$$=P(1+\bar{R}Q)+\bar{R}Q \quad \text{(Since, } 1+R\bar{Q}=1)$$
$$=P\cdot 1+\bar{R}Q \quad \text{(Since, } 1+\bar{R}Q=1)$$
$$=P+\bar{R}Q \quad \text{(Since, } P\cdot 1=P)$$

17. Topic: Digital Logic

(c) To get 2's complement, first invert the binary digits and then add one to the result.
2's complement of 1000 is

$$\begin{array}{r} 0\,1\,1\,1 \\ +\quad 1 \\ \hline 1000 \end{array}$$

Therefore, 1000 is same as its 2's complement

18. Topic: Digital Logic

(c) In *SR* flip-flop, when both inputs are 1 output is invalid state whereas in *JK* flip-flop output is toggled thus the functional difference between *SR* flip-flop and *JK* flip-flop is that *JK* flip-flop accepts both inputs 1.

19. Topic: Computer Organization and Architecture

(a) We know that:

$$\text{Speed up} = \frac{\text{Clock cycle without pipeline}}{\text{Clock cycle with pipeline}}$$

$$\text{Clock cycle without pipeline} = \frac{4}{2.5\text{ GHz}} = \frac{4}{2.5\times10^9\text{ Hz}}$$

$$\text{Clock cycle with pipeline} = \frac{1}{2.0\text{ GHz}} = \frac{1}{2.0\times10^7\text{ Hz}}$$

$$\text{Speed up} = \frac{4}{2.5\times10^9}\times2.0\times10^9$$

$$= \frac{8}{2.5} = 3.2$$

20. Topic: Programming and Data Structures

(a) On executing the program

$$k = 5$$
$$*p = \&k = 5$$
$$**m = \&p = 5$$

Therefore, all 3 variables point to the same value thus output is 5 5 5.

21. Topic: Computer Organization and Architecture

(a) Given:

Surfaces = 7 bits, Track = 4 bits, sector = 8 bits
Number of bits required = 7 + 4 + 8 = 19 bits
Capacity of disk pack = $2^{(7+4+8+9)} = 2^{28}$ = 256 MB

22. Topic: Computer Organization and Architecture

(b) Given:

Fault service time = 10 ms = 10×10^{-3}
Memory access time = 20 ns = 20×10^{-9}

Page fault is generated for every 10^6 memory accesses then, effective access time for memory is

$$0.000001 \times 10 \times 10^{-3} + 0.999999 \times 20 \times 10^{-9}$$
$$= 10 \times 10^{-9} + 19.9 \times 10^{-9} = 29.9 \times 10^{-9}\text{ s} = 29.9\text{ ns}$$

23. Topic: Computer Organization and Architecture

(c) Register renaming is done in pipelined processors to eliminate write after read and write after write dependency between instructions. Hence, register renaming is done to eliminate certain kind of hazards.

24. Topic: Computer Organization and Architecture

(a) Van Neumann architecture belongs to single instruction single data (SISD).

25. Topic: Programming and Data Structures

(c) Given

short a = 320

In little endian processor, least significant byte is at lowest address.
In binary system a is represented as:

00000001 01000000

When a is type casted to char, least significant byte value is at lowest address.
So,

*ptr = 01000000

Therefore, output = 01000000 = 64

26. Topic: Programming and Data Structures

(d) This code requires $[\log_2 n]$ comparison within the loop and 1 comparison on termination,
Therefore, the number of comparisons made in the execution of the loop for any $n > 0$ is:

$$[\log_2 n] + 1$$

27. Topic: Programming and Data Structures

(a) Given expression:

8 2 3 ^ / 2 3 * + 5 1 * –

1. Push 8 to stack. So, stack = 8
2. Push 2 to stack. So, stack = 8 2
3. Push 3 to stack. So, stack = 8 2 3
4. Push pop 3 and 2 perform 2^3 = 8
5. Push 2^3 to stack. So, stack = 8 8
6. Push pop 8 and 8 perform 8/8 = 1
7. Push 8/8 to stack. So, stack = 1
8. Push 2 to stack. So, stack = 1 2
9. Push 3 to stack. So, stack = 1 2 3
10. Pop 3 and 2 perform 2 × 3 = 6
11. Put 2 × 3 to stack. So, stack = 1 6

Therefore, two elements in stack after first * is evaluated is 6, 1.

28. Topic: Algorithms

(c) Suppose we are searching element e

Number of comparisons = 1 × probability of 1st element be e +

$2 \times$ probability of 2^{nd} element be $+ \cdots + n \times$ probability of 3^{rd} element be e

$$= \frac{1}{n} + \frac{2}{n} + \frac{3}{n} + \cdots \frac{n}{n} = \frac{1}{n}(1+2+3+\cdots+n) = \frac{1}{n}\left[\frac{n(n+1)}{2}\right]$$

Therefore,

$$\text{Number of comparison} = \frac{n+1}{2}$$

29. Topic: Engineering Mathematics

(c) We have:

	Location
37 mod 7 = 2	0
38 mod 7 = 3	1
72 mod 7 = 2	2
48 mod 7 = 6	3
98 mod 7 = 0	4
11 mod 7 = 4	5

Thus key 11 will be stored at 5.

30. Topic: Algorithms

(d) In binary tree each non-leaf node has exactly two sons thus number of leaves = $n + 1$.
Therefore, total number of nodes = $2n + 1$

31. Topic: Algorithms

(c) Quick sort algorithm is divide and conquer algorithm which divides array into two smaller sub arrays which are then sorted.

32. Topic: Theory of Computation

(a) An FSM can be considered to be a Turning Machine of finite tape length without rewinding capability and unidirectional tape movement.

33. Topic: Theory of Computation

(d) L can be 0*(10*10*)* since (0*10*1)* will not generate all bit strings with even numbers of 1's, (0*10*1*)*0* will have some strings like 010 that will not have even number of 1's and 0*1(10*1)*10* will have generate all bit strings with only 0's.

34. Topic: Theory of Computation

(b) Given:

$$T(n) = 2T(\sqrt{n}) + 1$$

Put $n = 2^m$,

$$T(2^m) = 2T(\sqrt{2^m}) + 1 = 2T(2^{m/2}) + 1$$

Put $T(2^m) = s(m)$,

$$\Rightarrow s(m) = 2s(m/2) + 1$$

Now, using master method

$$O(m) = O(\log n) \Rightarrow T(n) = O(\log n)$$

35. Topic: Theory of Computation

(b) Given:

$$G = \{S \to ss, S \to ab, S \to ba, S \to \wedge\}$$

I. G is ambiguous. The statement is true since string ab has multiple derivation trees.
II. G produces all strings with equal number of a's and b's. This statement is false. G cannot produce all strings with equal number of a's and b's, $aabb$ cannot be produced.
III. G can be accepted by a deterministic PDA. This statement is true a deterministic PDA can be designed for G.

36. Topic: Theory of Computation

(d) If L and $\overline{L}$ are recursively enumerable, then L is recursive because for L Turing machines accept all strings in L and for $\overline{L}$ Turing machine accepts all strings in $\overline{L}$ so we can check if string is L or not.

37. Topic: Theory of Computation

(b) Given:

$$S \to a\, s\, a\ 1\ b\, s\, b\ 1\ a\ 1\ b$$

The given grammar generates strings like $\{a, b, aaa, bbb \ldots\}$. Thus, all strings generated are set of odd length palindromes.

38. Topic: Theory of Computation

(c) Given:

$$S \to Aa, A \to Ba, B \to abc$$

Type 2 grammar must have production of form $A \to V$
Therefore, $A \in N$ (non-terminals) and $V \in (T \cup N)$ string of terminals and non-terminals

39. Topic: Algorithms

(b) Search tree has search time of O(log$n$) while linear list has search time of O(n/2) and Hash table has search time of O(1).

40. Topic: Compiler Design

(a) Recursive descent parsing is an example of top-down parsers.

41. Topic: Compiler Design

(c) Top down parser is a LL parser which means left to right parsing performing left most derivation.

42. Topic: Computer Organization and Architecture

(b) In relative mode of addressing the addresses are not absolute but instead is displacement against a base address. Relative mode of addressing is most relevant to writing position independent code.

43. Topic: Computer Organization and Architecture

(d) Two passes assemblers, in the first pass checks to see if the instructions are legal in the current assembly mode, allocates space for the literals and builds symbol table for the symbols and their values.

44. Topic: Compiler Design

(b) Peephole optimization is a form of local optimization, which is done on segment of code that has been generated.

45. Topic: Operating System

(a) Given:
P is wait that decreases the value of semaphore by 1 and V is signal that increases the value of semaphore by 1.
Given, at particular time value of semaphore = 7
After this $20P$ and xV operations are completed.
Final value of semaphore = 5

$$7 - 20 + x = 5 \Rightarrow x = 18$$

46. Topic: Operating System

(d) With a single resource, deadlock cannot occur.

47. Topic: Operating System

(a) Given, 3 processes are sharing 4 resources. If each process has one resource allocated, even then one resource is available so deadlock can never occur.

48. Topic: Operating System

(c) Given main memory already has pages 1 and 2.

1	2	

Page fault occurs:

1	2	4

Page fault occurs:

5	2	4

5	2	4

Page fault occurs:

5	2	1

5	2	1

Page fault occurs:

4	2	1

There are total 4-page faults.

49. Topic: Operating System

(d) Working set (t, k) is set of unique pages referenced within t unit of time. So, working set (t, k) at on instant of time t is the k set of pages that have been referenced in the last t time units.

50. Topic: Operating System

(c) Given:

Size of page = 4 kB = 2^{12} bits

Bit needed to address a page frame = 32 – 12 = 20 bits

Page table entries TLB can hold = 128 = 2^7

Number of sets in cache = $\frac{2^7}{4} = 2^5$

Therefore,

$$\text{Minimum size of TLB tag} = 20 - 5 = 15 \text{ bits}$$

51. Topic: Operating System

(c) Real time operating system serves real time application requests without buffering delays. Pre-emptive scheduling scheme is most suitable for real time operating systems because it allows processes running to be temporarily suspended.

52. Topic: Operating System

(a) Belady's anomaly occurs when page fault increases despite increase in page frames. Belady's anomaly occurs while FIFO (first in first out) page replacement policy.

53. Topic: Databases

(b) Maximum tuples is the product of m and n as every tuple in relation R can be joined to every tuple in relation s. Minimum tuples are 0 when no match is made. Hence maximum tuples of join = mn and minimum tuples of join = 0

54. Topic: Databases

(b) Relational integrity constraint is violated if the foreign key referenced is not existing in the reference table. This can happen when inserting into S and deleting from R.

55. Topic: Databases

(d) The SQL query lists titles of the five most expensive books.
Outer query select all titles from book table.
sub query returns count of books which are most expensive from the selected book and where condition of outer query will be true for 5 count that is 5 most expensive book.

56. Topic: Databases

(d) Goods for the design of the logical schema should include all of the given options i.e. Goals for schema should be to avoid data inconsistency, easy construction of queries, efficient data access.

57. Topic: Databases

(c) The basic relational algebra operations are 'U' – union, '–' – difference, 'x' – Cartesian product, 'π' – projection, 'σ' – selection and 'p' – rename.

Since, from these operations aggregation is not possible. So, the sum of all employee's salaries cannot be expressed using basic relational algebra operations.

58. Topic: Databases

(d) Trigger is a statement that is executed automatically by the system as a side effect of a modification to the database.

59. Topic: Databases

(a) Let there be n value fields, n data record pointer and 1 block pointer to point next block given value field is a byte long, data record pointer is 7 bytes long and block pointer is
6 byte long then,

$$9n + 7n + 6 \leq 1024$$
$$\Rightarrow 16n \leq 1018$$
$$\Rightarrow n \leq 63.625 \Rightarrow n \approx 63$$

60. Topic: Databases

(a) A clustering index is defined on the fields which are of type non-key and ordering. Clustering index is an ordered data file. The data file is ordered on non-key field.

61. Topic: Operating System

(b) The extent to which the software can continue to operate correctly despite the introduction of invalid inputs is called as Robustness. Robustness is ability of computer system to cope with errors during execution and erroneous input.

62. Topic: Operating System

(d) Functional requirements are related to functionality for example set of inputs, behavior and outputs. All the given options are non-functional requirements.

63. Topic: Operating System

(b) Configuration management is concerned with tracking and controlling changes in the software. Option (b) is not concerned with software.

64. Topic: Operating System

(b) Let $L_1 = x$ and $L_2 = 2x$
Now,

$$\text{LOC}(L_1) = \text{LOC}(L_2)$$
$$\Rightarrow \left(\frac{x}{10,000} \times 10,00,000\right) + (5 \times 1,00,000)$$
$$= \left(\frac{2x}{10,000} \times 7,50,000\right) + (5 \times 50,000)$$
$$\Rightarrow 100x + 5,00,000 = 150x + 2,50,000$$
$$\Rightarrow 50x = 2,50,000$$
$$\Rightarrow x = 5000$$

65. Topic: Operating System

(a) Given:

$$\text{Lines of code (LOC)} = 20,000$$
$$\Rightarrow \text{KLOC} = 20$$

Now,

$$\text{Person months} = a\,(\text{KLOC})^b$$

where $a = 2.2$ and $b = 1.5$
Therefore,

$$\text{Person months} = (2.2) \times (20)^{1.5} = 196.77$$

66. Topic: Operating System

(d) In spiral model of software development, the primary determinant in selecting activities in each iteration is risk since spiral model is a risk-driven process model generator for software projects.

67. Topic: Computer Networks

(a) Bit stuffing refers to inserting a '0' in user stream to differentiate it with a flag.

68. Topic: Computer Networks

(d) All the given options are handled by the dynamic routing protocols.
If a router on the route goes down the destination may become unreachable. Dynamic routing allows routing tables in routers to change as the possible routes change. There are several protocols used to support dynamic routing including RIP and OSPF.

69. Topic: Computer Networks

(c) In Ethernet CSMA/CD, the special bit sequence is transmitted by media access management to handle collision is called Jam. Jam is sent to handle collision. It is a 32 bit pattern sent by the data station to inform other station about collision.

70. Topic: Computer Networks

(a) DHCP stands for dynamic host configuration protocol. It is used to assign IP addresses dynamically to devices

71. Topic: Computer Networks

(c) We know that:

$$\text{Transmission delay} = \frac{\text{Length}}{\text{Band width}} = \frac{1}{10^7} = 10^7\,\text{s}$$
$$\text{Propagation speed} = 200\ \text{m}\ \mu\text{s}^{-1}$$

Therefore,

$$\text{1-bit delay in network} = \frac{200\ \text{m}}{\mu\text{s}} \times 10^{-7}\,\text{s}$$
$$= \frac{200\ \text{m}}{\mu\text{s}} \times 0.1 \times 10^{-6}\,\text{s}$$

$$= \frac{200\text{ m}}{\mu\text{s}} \times 10^{-1}\,\mu\text{s} = 20\text{ m}$$

72. Topic: Computer Networks

(c) First 2 octets are reserved for Network 10 and remaining 2 octets are reserved for host 10. Thus first 6 bits of 3rd octet are used for subnet and remaining 10 bits for hosts.

$\Rightarrow$ Maximum number of subnets = $2^6 - 2 = 62$
Number of hosts = $2^{10} - 2 = 1022$

73. Topic: Computer Networks

(b) Given:

Message = 11001001

and

CRC polynomial = $x^3 + 1$

So,

```
1001)11001001000(11010011
     1001
      1011
      1001
       0100
       0000
        1000
        1001
         0011
         0000
          0110
          0000
           1100
           1001
            1010
            1001
             011
```

Divisor = 1001

and

Remainder = 011

Thus, CRC sends dividend to receiver generated by CRC procedure

Dividend = data unit padded by remainder bits at sender
= number of divisor bits – 1
= 1100 1001 011

Therefore,

Message transmitted = 11001001011

74. Topic: Computer Networks

(a) There is no limit on data size that application layer can pass on to the TCP layer.

75. Topic: Computer Networks

(d) Given, frames of 1000 bits are sent over a 10^6 bps duplex link between two hosts.
Propagation time is 25 ms

$$\text{Number of frames} = \frac{25 \times 10^{-3} \times 10^{6}}{1000} = 25$$

So,

$$\text{Minimum bits required} = \log_2 25 = 5$$

Therefore,

$$i = 5$$

76. Topic: Computer Networks

(c) HTML does not allow user defined tags.

77. Topic: Computer Organization and Architecture

(a, c) Assuming simultaneous access, we have:

$$\text{Average access time} = (h_1 t_1) + (1 - h_1)h_2 t_2 + (1 - h_1)(1 - h_2)\,t_m$$
$$= (0.8 \times 1 \times 10^{-9}) + (1 - 0.8)\,0.9 \times 10 \times 10^{-9} + (1 - 0.8)(1 - 0.9) \times 500 \times 10^{-9}$$
$$= 0.8 \times 10^{-9} + 1.8 \times 10^{-9} + 10 \times 10^{-9}$$
$$= 12.6 \times 10^{-9} = 12.6\text{ ns}$$

Assuming hierarchical access, we have:

$$\text{Average access time} = h_1 t_1 + (1 - h_1)h_1(t_1 + t_2) + (1 - h_1)(1 - h_2)(t_1 + t_2 + t_1)$$
$$= 0.8 \times 1 + (0.2)(0.9)(11) + (0.2 \times 0.1)(511)$$
$$= 13\text{ ns}$$

78. Topic: Programming and Data Structures

(d) Private members of a class are accessed only inside the class declared. Protected members of a class can be accessed in 1st and 2nd child after inheritance as well. Public members of a class are available to all. Therefore, class C member function can access public and protected data of A and B and all data of class C.

79. Topic: Programming and Data Structures

(c) C supports early binding which means compile time binding. C++ supports both early binding and late binding which means run time binding. Therefore, only statement I is correct.

80. Topic: Computer Networks

(c) DNS spoofing or DNS cache poisoning is when corrupt DNS data is introduced into the DNS resolver cache. Therefore option (c) is correct.

Indian Space Research Organisation
Computer Science

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QUESTION PAPER

1. If A is a skew symmetric matrix, then A^T

(a) diagonal matrix (b) A
(c) 0 (d) $-A$

2. If A and B be two arbitrary events, then

(a) $P(A \cap B) = P(A)P(B)$
(b) $P(A \cup B) = P(A) + P(B)$
(c) $P(A/B) = P(A \cap B) + P(B)$
(d) $P(A \cup B) \leq P(A) + P(B)$

3. Using Newton–Raphson method, a root correct to 3 decimal places of the equation $x^3 - 3x - 5 = 0$

(a) 2.222 (b) 2.275
(c) 2.279 (d) None of the above

4. What does the data dictionary identify?

(a) Field names (b) Field formats
(c) Field Types (d) All of these

5. Which of the following concurrency control protocol ensures both conflict serializability and free from deadlock?

(a) Time stamp ordering (b) 2 Phase locking
(c) Both (a) and (b) (d) None of the above

6. ACID properties of a transactions are

(a) Atomicity, consistency, isolation, database
(b) Atomicity, consistency, isolation, durability
(c) Atomicity, consistency, integrity, durability
(d) Atomicity, consistency, integrity, database

7. Database table by name Overtime_allowance is given below:

Employee	**Department**	**OT_allowance**
RAMA	Mechanical	5000
GOPI	Electrical	2000
SINDHU	Computer	4000
MAHESH	Civil	1500

What is the output of the following SQL query?
select count(*) from ((select Employee, Department from Overtime_allowance) as S natural join (select Department, OT_allowance from Overtime_allowance) as T);

(a) 16 (b) 4
(c) 8 (d) None of the above

8. Which symbol denote derived attributes in ER Model?

(a) Double ellipse
(b) Dashed ellipse
(c) Squared ellipse
(d) Ellipse with attribute name underlined

9. The symmetric difference of sets $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$ and $B = \{1, 3, 5, 6, 7, 8, 9\}$ is

(a) $\{1, 3, 5, 6, 7, 8\}$
(b) $\{2, 4, 9\}$
(c) $\{2, 4\}$
(d) $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$

10. The problems 3-SAT and 2-SAT are

(a) Both NP-complete
(b) Both in P
(c) NP-complete and in P, respectively
(d) Undecidable and NP-complete, respectively

11. Given the following statements

S1: Every context-sensitive language L is recursive
S2: There exists a recursive language that is not context-sensitive
Which statements are true?

(a) Only S1 is correct
(b) Only S2 is correct
(c) Both S1 and S2 are not correct
(d) Both S1 and S2 are correct

12. Which one of the following is FALSE?

(a) There is a unique minimal DFA for every regular language
(b) Every NFA can be converted to an equivalent PDA
(c) Compliment of every context-free language is recursive
(d) Every non-deterministic PDA can be converted to an equivalent deterministic PDA

13. In some programming languages, an identifier is permitted to be a letter followed by any number of letters or digits. If L and D denotes the set of letters and digit, respectively. Which of the following expression defines an identifier?

(a) $(L + D)^*$ (b) $(L \cdot D)^*$
(c) $L(L + D)^*$ (d) $L(L \cdot D)^*$

14. The recurrence relation that arises in relation with the complexity of binary search is

(a) $T(n) = 2T(n/2) + k$, where k is constant
(b) $T(n) = T(n/2) + k$, where k is constant
(c) $T(n) = T(n/2) + \log n$
(d) $T(n) = T(n/2) + n$

15. Which one of the following in-place sorting algorithms needs the minimum number of swaps?

(a) Insertion Sort (b) Quick Sort
(c) Heap Sort (d) Selection Sort

16. Given two statements

(i) Insertion of an element should be done at the last node in a circular list
(ii) Deletion of an element should be done at the last node of the circular list

(a) Both are true
(b) Both are false
(c) First is false and second is true
(d) None of the above

17. Which of the following data structure is useful in traversing a given graph by breadth first search?

(a) Stack (b) Queue
(c) List (d) None of the above

18. How many 128 × 8 bit RAMs are required to design 32 K × 32 bit RAM?

(a) 512 (b) 1024
(c) 128 (d) 32

19. The most appropriate matching for the following pairs:

X: Indirect Addressing	1. Loop
Y: Immediate Addressing	2. Pointers
Z: Auto Decrement Addressing	3. Constants

(a) X – 3, Y – 2, Z – 1 (b) X – 2, Y – 3, Z – 1
(c) X – 3, Y – 1, Z – 2 (d) X – 2, Y – 1, Z – 3

20. Which interrupt in 8085 Microprocessor is unmaskable?

(a) RST 5.5 (b) RST 7.5
(c) TRAP (d) Both (a) and (b)

21. A cache memory needs an access time of 30 ns and main memory 150 ns, what is the average access time of CPU (assume hit ratio = 80%)?

(a) 60 ns (b) 30 ns
(c) 150 ns (d) 70 ns

22. Which one of the following Boolean expressions is NOT a tautology?

(a) $((a \to b) \wedge (b \to c)) \to (a \to c)$
(b) $(a \leftrightarrow c) \to (\sim b \to (a \wedge c))$
(c) $(a \wedge b \wedge c) \to (c \vee a)$
(d) $a \to (b \to a)$

23. What is the minimum number of two-input NAND gates used to perform the function of two input OR gate?

(a) One (b) Two
(c) Three (d) Four

24. When two n-bit binary numbers are added the sum will contain at the most

(a) n bits (b) $(n + 3)$ bits
(c) $(n + 2)$ bits (d) $(n + 1)$ bits

25. The 2-input XOR has a high output only when the input values are

(a) low (b) high
(c) same (d) different

26. Advantage of synchronous sequential circuits over asynchronous one is

(a) Lower hardware requirement
(b) Better noise immunity
(c) Faster operation
(d) All of the above

27. Physical topology of FDDI is?

(a) Bus (b) Ring
(c) Star (d) None of the above

28. In networking terminology UTP means

(a) Uniquitous teflon port
(b) Uniformly terminating port
(c) Unshielded twisted pair
(d) Unshielded T-connector port

29. The default subnet mask for a class B network can be

(a) 255.255.255.0 (b) 255.0.0.0
(c) 255.255.192.0 (d) 255.255.0.0

30. If there are n devices (nodes) in a network, what is the number of cable links required for a fully connected mesh and a star topology respectively

(a) $n(n-1)/2, n-1$ (b) $n, n-1$
(c) $n-1, n$ (d) $n-1, n(n-1)/2$

31. Which of the following protocol is used for transferring electronic mail messages from one machine to another?

(a) TELNET (b) FTP
(c) SNMP (d) SMTP

32. Which media access control protocol is used by IEEE 802.11 wireless LAN?

(a) CDMA (b) CSMA/CA
(c) ALOHA (d) None of the above

33. An Ethernet frame that is less than the IEEE 802.3 minimum length of 64 octets is called

(a) Short frame (b) Small frame
(c) Mini frame (d) Runt frame

34. Match with the suitable one:

List-I	List-II
(1) Multicast group membership	1. Distance vector routing
(2) Interior gateway protocol	2. IGMP
(3) Exterior gateway protocol	3. OSPF
(4) RIP	4. BGP

(a) A-2, B-3, C-4, D-1 (b) A-2, B-4, C-3, D-1
(c) A-3, B-4, C-1, D-2 (d) A-3, B-1, C-4, D-2

35. MD5 is a widely used hash function for producing hash value of

(a) 64 bits (b) 128 bits
(c) 512 bits (d) 1024 bits

36. Which protocol suite designed by IETF to provide security for a packet at the Internet layer?

(a) IPsec (b) NetSec
(c) PacketSec (d) SSL

37. Pretty Good Privacy (PGP) is used in

(a) Browser security (b) FTP security
(c) Email security (d) None of the above

38. What is WPA?

(a) Wired Protected Access
(b) Wi-Fi Protected Access
(c) Wired Process Access
(d) Wi-Fi Process Access

39. Estimation of software development effort for organic software in basic COCOMO is

(a) $E = 2.0(KLOC)^{1.05}$PM
(b) $E = 3.4(KLOC)^{1.06}$PM
(c) $E = 2.4(KLOC)^{1.05}$PM
(d) $E = 2.4(KLOC)^{1.07}$PM

40. XPath is used to navigate through elements and attributes in

(a) XSL document (b) XML document
(c) XHTML document (d) XQuery document

41. What is the output of this C++ program?

```
#include <iostream>
using namespace std;
void square (int *x)
{
   *x = (*x)++ * (*x);
}
void square (int *x, int *y)
{
   *x = (*x) * -- (*y);
}
int main ()
{
   int number = 30;
   square(&number, &number);
   cout << number;
   return 0;
}
```

(a) 910 (b) 920
(c) 870 (d) 900

42. Which of the following operator(s) cannot be overloaded?

(a) . (Member Access or Dot operator)
(b) ?: (Ternary or Conditional Operator)
(c) :: (Scope Resolution Operator)
(d) All of the above

43. Which of the following UML 2.0 diagrams capture behavioural aspects of a system?

(a) Use Case Diagram, Object Diagram, Activity Diagram, and State Machine Diagram
(b) Use Case Diagram, Activity Diagram, and State Machine Diagram
(c) Object Diagram, Communication Diagram, Timing Diagram, and Interaction diagram
(d) Object Diagram, Composite Structure Diagram, Package Diagram, and Deployment Diagram

44. Which of the following is associated with objects?

(a) State (b) Behavior
(c) Identity (d) All of the above

45. Which one of these is characteristic of RAID 5?

(a) Dedicated parity (b) Double parity
(c) Hamming code parity (d) Distributed parity

46. SATA is the abbreviation of

(a) Serial Advanced Technology Attachment
(b) Serial Advanced Technology Architecture
(c) Serial Advanced Technology Adapter
(d) Serial Advanced Technology Array

47. Capability Maturity Model (CMM) is a methodology to

(a) develop and refine an organization's software development process
(b) develop the software
(c) test the software
(d) all of the above

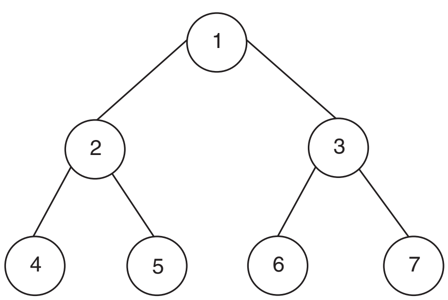
48. What problem is solved by Dijkstra banker's algorithm?

(a) Mutual exclusion (b) Deadlock recovery
(c) Deadlock avoidance (d) Cache coherence

49. The number of swappings needed to sort the numbers 8, 22, 7, 9, 31, 5, 13 in ascending order, using bubble sort is

(a) 11 (b) 12
(c) 13 (d) 10

50. Consider the following tree



If the post order traversal gives $ab - cd* +$ then the label of the nodes 1, 2, 3, ... will be

(a) +, –, *, *a*, *b*, *c*, *d* (b) *a*, –, *b*, +, *c*, *, *d*
(c) *a*, *b*, *c*, *d*, –, *, + (d) –, *a*, *b*, +, *, *c*, *d*

51. What is the output of the following program?

```
main()
{
   int a = 10;
   if ((fork() == 0))
      a++;
      printf ("%d\n", a);
}
```

(a) 10 and 11 (b) 10
(c) 11 (d) 11 and 11

52. Given reference to the following pages by a program

0, 9, 0, 1, 8, 1, 8, 7, 8, 7, 1, 2, 8, 2, 7, 8, 2, 3, 8, 3

How many page faults will occur if the program has three page frames available to it and uses an optimal replacement?

(a) 7 (b) 8
(c) 9 (d) None of these

53. In a doubly linked list, the number of pointers affected for an insertion operation will be

(a) 4
(b) 0
(c) 1
(d) Depends upon the nodes of the doubly linked list

54. Consider the following C function

```
void swap (int x, int y)
{
   int tmp;
   tmp = x;
   x = y;
   y = tmp;
}
```

In order to exchange the values of two variables *a* and *b*:

(a) Call swap (*a*, *b*)
(b) Call swap (&*a*, &*b*)
(c) swap (*a*, *b*) cannot be used as it does not return any value
(d) swap (*a*, *b*) cannot be used as the parameters passed by value

55. What does the following C-statement declare?

int (**f*) (int*);

(a) A function that takes an integer pointer as argument and returns an integer
(b) A function that takes an integer as argument and returns an integer pointer
(c) A pointer to a function that takes an integer pointer as argument and returns an integer
(d) A function that takes an integer pointer as argument and returns a function pointer

56. Mutual exclusion problem occurs

(a) Between two disjoint processes that do not interact
(b) Among processes that share resources
(c) Among processes that do not use the same resource
(d) Between two processes that uses different resources of different machine

57. $(1217)_8$ is equivalent to

(a) $(1217)_{16}$ (b) $(028F)_{16}$
(c) $(2297)_{10}$ (d) $(0B17)_{16}$

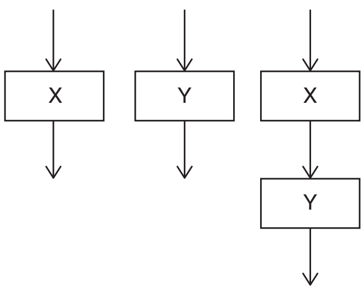
58. Which of the following is not a life cycle model?

(a) Spiral model
(b) Prototyping model
(c) Waterfall model
(d) Capability maturity model

59. The best data structure to check whether an arithmetic expression has balanced parenthesis is a

(a) Queue (b) Stack
(c) Tree (d) List

60. The cyclomatic complexity of each of the modules *X* and *Y* shown below is 10. What is the cyclomatic complexity of the sequential integration shown on the right hand side?



(a) 21 (b) 19
(c) 20 (d) 10

61. In software maintenance tackling the changes in the hardware or software environment where the software works, is

(a) Corrective maintenance
(b) Perfective maintenance
(c) Adaptive maintenance
(d) Preventive maintenance

62. What will be the output of the following C code?

```
#include <stdio.h>
main()
{
int i;
for(i = 0; i < 5; i++)
{
    int i = 10;
    printf("%d", i);
    i++;
}
return 0;
}
```

(a) 10 11 12 13 14 (b) 10 10 10 10 10
(c) 0 1 2 3 4 (d) Compilation error

63. What does the following program do when the input is unsigned 16-bit integer?

```
#include <stdio.h>
main()
{
    unsigned int num;
    int i;
    scanf("%u", &num);
    for(i = 0; i < 16; i++)
    {
    printf ("%d", (num << i & 1 << 15)?
    1:0);
    }
}
```

(a) It prints all even bits from num
(b) It prints all odd bits from num
(c) It prints binary equivalent of num
(d) None of the above

64. What is the output of the following program?

```
#include <stdio.h>
int tmp=20;
main()
{
    printf("%d ",tmp);
    func();
    printf("%d ",tmp);
}
func()
{
    static int tmp=10;
    printf("%d ",tmp);
}
```

(a) 20 10 10 (b) 20 10 20
(c) 20 20 20 (d) 10 10 10

65. Which product metric gives the measure of the average length of words and sentence in documents?

(a) SCI number
(b) Cyclomatic complexity
(c) LOC
(d) Fog index

66. Consider a disk system with 100 cylinders. The request to access the cylinders occur in the following sequences

4, 37, 10, 7, 19, 73, 2, 15, 6, 20

Assuming the head is currently at cylinder 50, what is the time taken to satisfy all requests if it takes 1 ms to move from one cylinder to adjacent one and shortest seek time first algorithm is used

(a) 95 ms (b) 119 ms
(c) 233 ms (d) 276 ms

67. A B-Tree used as an index for a large database table has four levels including the root node. If a new key is inserted in this index, then the maximum number of nodes that could be newly created in the process are

(a) 5 (b) 4
(c) 1 (d) 2

68. A critical region

(a) is a piece of code which only one process executes at a time
(b) is a region prone to deadlock
(c) is a piece of code which only a finite number of processes execute
(d) is found only in windows NT operating system

69. Choose the equivalent prefix form of the following expression

$$(a + (b - c)) * ((d - e)/(f + g - h))$$

(a) $* + a - bc/- de - + fgh$
(b) $* + a - bc -/de - + fgh$
(c) $* + a - bc/- ed + - fgh$
(d) $* + ab - c/- ed + - fgh$

70. We use malloc and calloc for

(a) Dynamic memory allocation
(b) Static memory allocation
(c) Both dynamic and static memory allocation
(d) None of the above

71. At particular time, the value of a counting semaphore is 10, it will become 7 after

(a) 3 V operations
(b) 3 P operations
(c) 5 V operations and 2 P operations
(d) 2 V operations and 5 P operations

72. The linux command "mknod myfifo *b* 4 16"

(a) Will create a character device if the user is root
(b) Will create a named pipe FIFO if the user is root
(c) Will create a block device if the user is root
(d) None of the above

73. Which of the following statement is true?

(a) Hard real time OS has less jitter than soft real time OS
(b) Hard real time OS has more jitter than soft real time OS
(c) Hard real time OS has equal jitter as soft real time OS
(d) None of the above

74. Which of these is a super class of all errors and exceptions in the Java language?

(a) RunTimeExceptions
(b) Throwable
(c) Catchable
(d) None of the above

75. Choose the most appropriate HTML tag in the following to create a numbered lists.

(a) <dl> (b) <list>
(c) (d)

76. Which of the following algorithm solves the all-pair shortest path problem?

(a) Prim's algorithm
(b) Dijikstra's algorithm
(c) Bellman–Ford's algorithm
(d) Floyd–Warshall's algorithm

77. If L and P are two recursively enumerable languages, then they are not closed under

(a) Kleene Star L^* of L
(b) Intersection $L \cap P$
(c) Union $L \cup P$
(d) Set difference

78. In the context of modular software design, which one of the following combinations is desirable?

(a) High cohesion and high coupling
(b) High cohesion and low coupling
(c) Low cohesion and high coupling
(d) Low cohesion and low coupling

79. The output of a lexical analyzer is

(a) A parse tree (b) Intermediate code
(c) Machine code (d) A stream of tokens

80. The time complexity of computing the transitive closure of a binary relation on a set of *n* elements is known to be

(a) $O(n \log n)$ (b) $O(n^{3/2})$
(c) $O(n^3)$ (d) $O(n)$

ANSWER KEY

1. (d)	**9.** (b)	**17.** (b)	**25.** (d)	**33.** (d)	**41.** (c)	**49.** (d)	**57.** (b)	**65.** (d)	**73.** (a)
2. (d)	**10.** (c)	**18.** (b)	**26.** (*)	**34.** (a)	**42.** (d)	**50.** (a)	**58.** (d)	**66.** (b)	**74.** (b)
3. (c)	**11.** (d)	**19.** (b)	**27.** (b)	**35.** (b)	**43.** (b)	**51.** (a)	**59.** (b)	**67.** (a)	**75.** (d)
4. (d)	**12.** (d)	**20.** (c)	**28.** (c)	**36.** (a)	**44.** (d)	**52.** (a)	**60.** (b)	**68.** (a)	**76.** (d)
5. (a)	**13.** (c)	**21.** (a)	**29.** (d)	**37.** (c)	**45.** (d)	**53.** (*)	**61.** (c)	**69.** (a)	**77.** (d)
6. (b)	**14.** (b)	**22.** (b)	**30.** (a)	**38.** (b)	**46.** (a)	**54.** (d)	**62.** (b)	**70.** (a)	**78.** (b)
7. (b)	**15.** (d)	**23.** (c)	**31.** (d)	**39.** (c)	**47.** (a)	**55.** (c)	**63.** (c)	**71.** (b, d)	**79.** (d)
8. (b)	**16.** (b)	**24.** (d)	**32.** (b)	**40.** (b)	**48.** (c)	**56.** (b)	**64.** (b)	**72.** (c)	**80.** (c)

*** None of the options are correct.**

ANSWERS WITH EXPLANATION

1. Topic: Engineering Mathematics

(d) If A is a skew symmetric matrix.
Then,

$$A^T = -A$$

For skew symmetric matrix the elements of matrix $a_{ij} = -a_{ji}$ where i is the row & j is column.

2. Topic: Engineering Mathematics

(d) If A and B are two arbitrary events then,

$$P(A \cup B) = P(A) + P(B) - P(A \cup B)$$

If events are mutually exclusive then $P(A \cap B) = 0$ and in general $P(A \cap B) \geq 0$

$$\Rightarrow P(A \cup B) \leq P(A) + P(B)$$

3. Topic: Engineering Mathematics

(c) Given:

$$f(x) = x^3 - 3x - 5$$
$$f'(x) = 3x^2 - 3$$

Iteration	x_n	$f(x_n)$	$f'(x_n)$	x_{n+1}
1.	$x_0 = 3$	13	24	2.45833
2.	2.45833	2.48165	15.13016	2.29431
3.	2.29431	0.19399	12.79158	2.27914
4.	2.27914	0.00153	12.58344	2.27902
5.	2.27902	0.00001	12.581796	2.27902

Therefore, root of equation is 2.27902

4. Topic: Programming and Data Structures

(d) The data dictionary identifies field names, field formats and field types. A data dictionary is collection of descriptions of data items in data model. This is done for the benefits of other programmers who need to refer to them, so that they can understand where the data fits in the structure and what values it may contain.

5. Topic: Databases

(a) Time stamp ordering is a non-lock concurrency control method. This ensures both conflict serializability and free from deadlock. This used in some databases to handle transactions safely using timestamps.

6. Topic: Databases

(b) Transaction is a very small unit of program. It contains several low-level tasks. The acronym ACID refers to 4 properties. ACID properties of a transaction are atomicity, consistency, isolation, durability.
Atomicity: are changes to data set a performed as a single operation.

Consistency: data is consistent between start and end of a transaction.
Isolation: Intermediate states of a transaction are invisible to other transaction.
Durability: data changes persist after completion of transaction.

7. **Topic: Databases**

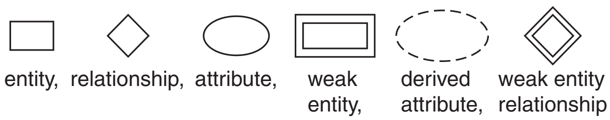
(b) The query counts employee, department from overtime allowance and natural join of department, overtime allowance.

S =	Employee	Department		T =	Department	OT_ Allowance
	RAMA	Mechanical	→		Mechanical	5000
	GOPI	Electrical	→		Electrical	2000
	SINDHU	Computer	→		Computer	4000
	MAHESH	Civil	→		Civil	1500

Count function will return 4.

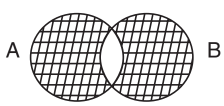
8. **Topic: Databases**

(b) Dashed ellipse denote derived attributes in ER model. ER model refers to visual representation of different entities within a system and this relation with each other various ER model symbols & notations are given as



9. **Topic: Engineering Mathematics**

(b) Symmetric difference is set fall elements which belong to either A or to B but not both.



Shaded part represents symmetric difference of A and B

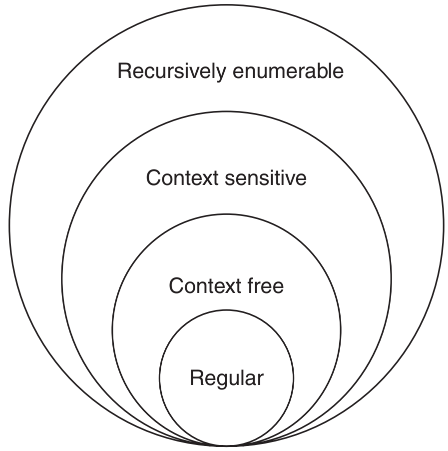
$$\begin{aligned}\text{Symmetric difference} &= (A - B) \cup (B - A)\\ &= \{2, 4\} \cup \{9\}\\ &= \{2, 4, 9\}\end{aligned}$$

10. **Topic: Programming and Data Structures**

(c) 3-SAT is in P and 2-SAT is in NP complete.

11. **Topic: Theory of Computation**

(d) We can explain the condition with the figure given



Every context sensitive language L is recursive – this statement is true.
There exists a recursive language that is not context sensitive – this statement is true
Hence both S1 and S2 are correct.

12. **Topic: Theory of Computation**

(d) Non-deterministic PDA is more powerful than deterministic PDA. Deterministic PDA cannot handle languages with ambiguity whereas non-deterministic PDA can handle languages with ambiguity, deterministic PDA is a subset of non-deterministic PDA. Therefore, non-deterministic PDA cannot be converted to an equivalent deterministic PDA. Therefore, option (d) is false.

13. **Topic: Theory of Computation**

(c) If L is set of letters and D is digit then identifier that should be a letter followed by any number of letters or digits is defined as

$$L(L + D)^*$$

14. **Topic: Algorithms**

(b) Binary search does researching operation only for half of the list therefore the recurrence relation will be

$$T(n) = T(n/2) + k,$$

where k is constant

15. **Topic: Algorithms**

(d) In worst case scenario, insertion sort requires $O(n^2)$ swaps, quick sort requires $O(n^2)$ swaps, heap sort requires $O(n \log n)$ swaps and selection sort requires $O(n)$ swaps. Therefore, selection sort needs minimum number of swaps

16. **Topic: Programming and Data Structures**

(b) Both the statements are false since this is no insertion and deletion rule in a circular list. There is a rule of insertion of an element to be done at the last node and deletion of element from the front node in a queue.

17. Topic: Programming and Data Structures

(b) Breadth first search uses queue in traversing a given graph. Breadth first search starts from the source node one traverse the graph longer wise exploring the neighbours.

18. Topic: Operating System

(b) We have:

$$\text{Number of RAM} = \frac{32\,\text{K} \times 32}{128 \times 8}$$
$$= \frac{2^5 \times 2^{10} \times 2^5}{2^7 \times 2^3} = \frac{2^{20}}{2^{10}}$$
$$= 2^{10} = 1024$$

19. Topic: Computer Organization and Architecture

(b) Indirect addressing does not have the address of data; it points where address is stored
So,

$$\text{Indirect addressing} \rightarrow \text{pointers}$$

Immediate addressing stores the data to be operated
So,
Immediate addressing $\rightarrow$ constants auto decrement addressing decreases the value in base register by size of the data which is to be operated.
Therefore,

$$\text{Auto decrement addressing} \rightarrow \text{loop}$$

20. Topic: Computer Organization and Architecture

(c) TRAP is a non-maskable edge and level triggered interrupt. It is an exception in user process that is caused by division by zero an invalid memory access. A trop can be used to call operating system routines or to catch arithmetic errors.

21. Topic: Computer Organization and Architecture

(a) We have:
Access time

$$= (\text{Cache hit}) \times (\text{Cache access time}) + (\text{Cache miss})$$
$$\times (\text{Cache miss penalty} + \text{Memory access time})$$
$$= \frac{80}{100} \times 30 \text{ ns} + \left(1 - \frac{80}{100}\right) \times (30 + 150) \text{ ns}$$
$$= 24 \text{ ns} + 0.2 \times 180 \text{ ns} = 245 \text{ ns} + 36 \text{ ns}$$
$$= 60 \text{ ns}$$

22. Topic: Digital Logic

(b) We know that, $(a \leftrightarrow c) \rightarrow (\sim b \rightarrow (a \wedge c))$ is not a tautology.
A tautology is a formula that satisfies all possible interpretation that is, it is always TRUE. $a \leftrightarrow c$ is TRUE for same values of a and c assuming a and b as FALSE, $\sim b$ is TRUE so RHS is false which is a contradiction.

23. Topic: Digital Logic

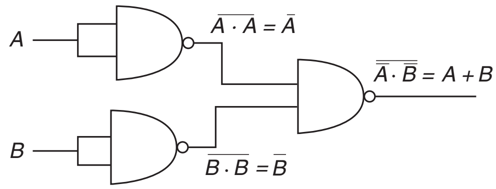
(c) We know that:
Two input OR gate have truth table:

Input		Output
A	**B**	**Y**
0	0	0
0	1	1
1	0	1
1	1	1

Two input NAND gate have truth table:

Input		Output
A	**B**	**Y**
0	0	1
0	1	1
1	0	1
1	1	0

So, 3 NAND gates are required to perform a function of OR gate.



24. Topic: Digital Logic

(d) If two n-bit binary numbers are added and summation results a carry then result will contain
$(n + 1)$ bits, thus the sum will contain at most $(n + 1)$ bits.

25. Topic: Digital Logic

(d) Truth table of two input XOR gate is:

Inputs		Outputs
A	**B**	**C**
0	0	0
0	1	1
1	0	1
1	1	0

Therefore, XOR has a high output only when the input values are different.

26. Topic: Digital Logic

(None) Advantages of Synchronous over asynchronous circuits are:

1. Performance is much better
2. Circuit is liable and portable
3. Easy to design.

Therefore, none of the given options are correct.

27. Topic: Computer Networks

(b) Physical topology of FDDI is ring. In ring network, cabling link loops from one node to another to form a closed loop. Fiber distributed data interface or FDDI is set of ANSI and ISO standard for data transmission on fiber optic lines in LAN. FDDI protocol is based on token ring protocol.

28. Topic: Computer Networks

(c) UTP stands for unshielded twisted pair. UTP is a type of copper cubing used in LANs. In this, conductors are twisted around each other to cancel out EM interference from external sources and no other shielding like mashes or aluminum foil are used.

29. Topic: Computer Networks

(d) Subnet masks are used to divide network address into two or more subnets. The default subnet mask for class B network is 255.255.0.0.

30. Topic: Computer Networks

(a) A fully connected mesh has every computer network connected to each of the other computers in the network thus number of cable links required

$$= \frac{n(n-1)}{2}$$

A star topology has 1 computer network connected to the remaining computers in the network thus number of cable links required

$$= (n - 1)$$

31. Topic: Computer Networks

(d) SMTP is used for transferring electronic mail messages from one machine to another. SMTP stands for simple mail transfer protocol. It is an internet standard for electronic mail transmission across internet protocol networks.

32. Topic: Computer Networks

(b) IEEE 802.11 standard defines the physical layer and media access control (MAC) layer for wireless local area network (LAN). The stations in IEEE 802.11 wireless LAN should coordinate their access and use of the shared communication media which is done by MAC protocol. The MAC protocol is a carrier sense multiple access protocol with collision avoidance (CSMA/CA).

33. Topic: Computer Networks

(d) IEEE 802.3 is widely used for computer networking and general data communications. It defines physical layer and media access control (MAC) of the data link layer for wired Ethernet networks. An Ethernet frame is encapsulated data defined by Network Access layer. The Ethernet frame structure is defined in the IEEE 802.3 standard. An Ethernet frame that is less than the IEEE 802.3's minimum lengths of 64 octets is called a Runt frame.

34. Topic: Computer Networks

(a) The correct match is:

Multicast group membership $\rightarrow$ IGMP
Interior gateway protocol $\rightarrow$ OSPF
Exterior gateway protocol $\rightarrow$ BGP
RIP $\rightarrow$ Distance vector routing

IGMP (internet group management protocol): is a multicast group membership protocol. IGMP allows a network host to inform a router that it is interested in receiving a particular multicast stream.

OSPF (open shortest path first): is an Interior gateway protocol. OSPF allows routers to dynamically learn routes from other routers and to advertise routs to other routers. OSPF selects the best routes that has lowest cost paths to the destination.

BGP (Border Gateway Protocol): is an exterior gateway protocol. BGP is a routing protocol that calculates loop-free paths across the internet. It is dynamic and handled outages and link failures fairly gracefully.

RIP (routing information protocol): is a distance vector routing protocol. It allows routers to dynamically adopt to changes in network connections by communicating information about which networks each router can reach and its distance from the networks.

35. Topic: Algorithms

(b) MD5 (message digest) algorithm is a widely used hash function that produces a hash value of 128 bits. The hash generated by MD5 is a 32-digit hexadecimal number. In order to map data sets of variable length to data sets of fixed length, input message is split into 512-bit blocks. These blocks are processed by MD5 which operates in 128-bit state and the result will be 128-bit hash value.

36. Topic: Computer Networks

(a) IPsec provide security for a packet at the internet layer. IPsec is an end to end security scheme that authenticates and encrypts the packets of data sent over a network. IPsec is the only security system that protects all application traffic over an IP network.

37. Topic: Computer Networks

(c) Pretty Good Privacy (PGP) is used in email security. PGP provides cryptographic privacy and authentication for data communication. It is used for signing, encrypting and decrypting emails over the internet.

38. Topic: Computer Networks

(b) WPA stands for wi-fi protected access. It is a security standard for users of computing devices equipped with wi-fi. WPA provides data encryption using temporal key integrity protocol and advanced encryption standard.

39. Topic: Software Engineering

(c) The basic COCOMO (constructive cost estimation model) gives an approximate estimate of the project parameters. The basic COCOMO model is given by expression:

$$E = a_b(KLOC)^{b_b}$$

where E is effort applied in person months (PM), $KLOC$ is estimated number of delivered lines of code for the project, a_b is coefficient and b_b is exponent.
Organic software project deals with developing a well understood application program developing team size is reasonably small and experienced in developing similar type of projects.
For organic software project $a_b = 2.4$ and $b_b = 1.05$
Therefore,

$$E = 2.4\ (KLOC)^{1.05}\ \text{PM}$$

40. Topic: Software Engineering

(b) XPath is used to navigate through elements and attributes in XML document. XPath is a major element in XSLT. It is a syntax for defining parts of a XML document. It contains a library of standard functions and uses path expressions to select nodes or node sets in on XML document.

41. Topic: Programming and Data Structures

(c) Given:

$$\text{Number} = 30$$

Square function with two arguments is called function will compute:

$$*x = (*x) * --(*y)$$
$$\Rightarrow *x = 30 \times 29$$
$$\Rightarrow *x = 870$$

Therefore, output of the program will be 870.

42. Topic: Programming and Data Structures

(d) Overloading is a concept in C++. It is a type of polymorphism in which operator is overloaded to give it a user defined meaning. These overloaded operators are operated on user defined data type. From the given operators none can be overloaded. Since, it is syntactically not possible to overload these operators.

43. Topic: Programming and Data Structures

(b) UML 2.0 diagrams capture behavioral aspects of a system; use case diagram, activity diagram and state machine diagram.
Use case diagram: captures use cases and relationship among actors and the system. It describes functional requirements, system boundary interaction and response of the system.
Activity diagram: models the behavior of the system and it's relation with overall flow of the system.
State machine diagram: illustrate movement of an element between states.

44. Topic: Programming and Data Structures

(d) Object is a physical as well as logical entity that has state and behavior.
An object has three characteristics:

1. State: it represents data of an object
2. Behavior: it represents functionality of an object.
3. Identity: it is implemented using unique ID.

45. Topic: Databases

(d) RAID level 5 is a duster: level implementation of data striping with distributed parity for enhanced performance. The dusters and parity are evenly distributed across multiple hard drives; this provides better performance than those where single drive for parity is used.

46. Topic: Operating System

(a) SATA stands for serial advanced technology attachment. It is technology that connects hard drive or SSD to the rest of the computer. It is a computer bus interface that connects host bus adapters to mass storage devices.

47. Topic: Software Engineering

(a) The Capability Maturity Model (CMM) is a methodology used to develop and refine an organization's software development process. It is a process improvement approach that is based on a process model. The model describes a five-level evolutionary path of increasingly organized and systematically more mature processes.

48. Topic: Operating System

(c) Dijkstra banker's algorithm solves deadlock avoidance problem. The Dijkstra banker's algorithm pretends to have allocated the required resources to the process and if the system does not lead to deadlock, it actually allocates the resources.

49. Topic: Algorithms

(d) Bubble sort is a simple sorting algorithm that repeatedly steps through the list to be sorted. It compares the pair of adjacent elements in the list and if they are in wrong order then it swaps them.

List:	8, 22, 7, 9, 31, 5, 13
Step 1:	8, 7, 22, 9, 31, 5, 13
Step 2:	8, 7, 9, 22, 31, 5, 13
Step 3:	8, 7, 9, 22, 5, 31, 13
Step 4:	8, 7, 9, 22, 5, 13, 31
Step 5:	7, 8, 9, 22, 5, 13, 31
Step 6:	7, 8, 9, 5, 22, 13, 31
Step 7:	7, 8, 9, 5, 13, 22, 31
Step 8:	7, 8, 5, 9, 13, 22, 31
Step 9:	7, 5, 8, 9, 13, 22, 31
Step 10:	5, 7, 8, 9, 13, 22, 31

Therefore, number of swapping needed to sort the numbers 8, 22, 7, 9, 31, 5, 13 in ascending order is 10 swaps.

50. Topic: Algorithms

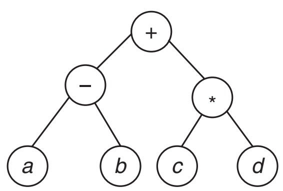
(a) Tree traversal refers to process of visiting nodes in a tree data structure. In post order traversal left subtree is visited first then the right subtree and the root is visited at last. The post order traversal of the given tree will be:

4 5 2 6 7 3 1

Comparing with the given order → $ab - cd* +$. We have

node 1 = +	node 3 = *	node 5 = b	node 7 = d
node 2 = –	node 4 = a	node 6 = c	

So, we have



51. Topic: Programming and Data Structures

(a) Output of given program is 10 and 11. Fork () is used to create new process which becomes child process of the caller. After creation of new child, both process will execute the next instructions. So, parent process will print value 10 and child process will execute a++ and will therefore print 11 because a++ command is associated with if condition that is calling fork function to create the child.

52. Topic: Operating System

(a) Optional page replacement algorithm replace the page in memory that will not be used for the longest period of time. It gives less number of page faults as compared to any other algorithm

Therefore, 7 page fault will occur

53. Topic: Programming and Data Structures

(None) There is no answer to the question. It requires the number of nodes in doubly linked list and the location of the insertion.

54. Topic: Programming and Data Structures

(d) Swap (a, b) cannot be used as the parameters are passed by value without pointers. This is illustrated as follows:

$$\text{swap } (a, b) \Rightarrow x = a, y = b$$
$$\text{tmp} = x \Rightarrow x = a, y = b, \text{tmp} = a$$
$$x = y \Rightarrow x = b, y = b, \text{tmp} = a$$
$$y = \text{tmp} \Rightarrow x = b, y = a, \text{tmp} = a$$

Therefore, the function has swapped value of x and y however a still holds value of a and b holds value of b.

55. Topic: Programming and Data Structures

(c) We have:

int f (int *) is a function that takes integer pointer as an argument and returns integer.

int * f (int) is a function that takes an integer as an argument and returns an integer pointer.

∴ int (*f) (int *) is a pointer to a function that takes integer pointer as an argument and returns an integer.

56. Topic: Operating System

(b) Mutual exclusion problem is an object that prevents simultaneous access to a shared resource. It is used in concurrent programming with a critical section. Thus mutual exclusion problem occurs among processes that share resources.

57. Topic: Digital Logic

(b) Converting $(1217)_8$ to binary equivalent:

$$(1217)_8 = (001\ 010\ 001\ 111)_2$$

Rearranging the binary equivalent in group of four we have:

$$(001\ 010\ 001\ 111)_2 = (0010\ 1000\ 1111)_2$$

Now, converting to hexadecimal equivalent:

$$(0010\ 1000\ 1111)_2 = (28\text{F})_{16}$$

Therefore,

$$(1217)_8 \text{ is equivalent to } (028\text{F})_{16}$$

58. Topic: Software Engineering

(d) Capability maturity model is a strategy for improving software process. It is not a life cycle model. It specifies an increasing series of levels of a software development organization the higher the level, the better the software development process.

On other hand life cycle model is a process for planning, creating, testing & deploying an information system. Spiral model, prototyping model and waterfall model all are life cycle models.

59. Topic: Programming and Data Structures

(b) The best data structure to check whether an arithmetic expression has balanced parenthesis is a stack. Stack scans the expression from left to right. Whenever left parenthesis is encountered, it is pushed into the stack and whenever a right parenthesis is encountered, it is popped out of stack. If at the end of expression, the stack is empty then expression has a balanced parenthesis.

60. Topic: Algorithms

(b) Cyclomatic complexity of module is given as number of decisions points + 1

Given, cyclomatic complexity of each module x and y is 10

Therefore, number of decisions points of x = cyclomatic complexity of module $x - 1$
Number of decisions points of $x = 10 - 1 = 9$
Similarly, number of decision points of y = cyclomatic complexity of module $y - 1 = 10 - 1 = 9$

$\Rightarrow$ cydomatic complexity of integration
= number of decision points of x + number of decision points of $y + 1$
Cyclomatic complexity of integration = 9 + 9 + 1 = 19

61. Topic: Software Engineering

(c) In software maintenance tackling the changes in the hardware or software environment where the software works is adaptive maintenance.

Corrective maintenance: tackles the errors spotted by the users.

Perfective maintenance: implements changes in the existing software or implements new requirements of the user.

Adaptive maintenance: tackles the changes in the hardware or software environment where the software works

Preventive maintenance: takes appropriate measures to avoid problems in future.

62. Topic: Programming and Data Structures

(b) The output of the code will be 10 10 10 10 10. Note that the variable *i* in for loop and variable *i* inside the loop are different the variable *i* defined for the loop goes from 0 to 4 and fails for $i = 5$ and the variable *i* defined inside the for loop is assigned value 10 for every true condition of the for loop. This value of $i = 10$ is printed and then *i* is incremented to 11.
Thus value 10 is printed for $i = 0, 1, 2, 3, 4$ and for $i = 5$. The condition of for loop fails and compiler exist the loop.

63. Topic: Programming and Data Structures

(c) If input is 16-bit integer then the program will prints binary equivalent of num

for input num = 3, output will be 00000000 00000011
for input num = 4, output will be 00000000 00000100
now suppose input is 4 bit wide, let num = 100
$i = 0$; 100 << 0 = 0100 $\rightarrow$ 0100 & 1000 = 0000 $\rightarrow$ prints 0
$i = 1$; 100 << 1 = 1000 $\rightarrow$ 1000 & 1000 = 1000 $\rightarrow$ prints 1
$i = 2$; 100 << 2 = 0100 $\rightarrow$ 0100 & 1000 = 0000 $\rightarrow$ prints 0
$i = 3$; 100 << 3 = 0010 $\rightarrow$ 0010 & 1000 = 000 $\rightarrow$ prints 0

This output is num itself.

64. Topic: Programming and Data Structures

(b) We know that:
tmp = 20 is global function. In main function, tmp is printed. After this function is called, a new local variable tmp = 10 is defined and printed. After this control goes to main (), at print command after function call which will again print global variable tmp = 20. Thus output of program will be: 20 10 20

65. Topic: Software Engineering

(d) Fog index gives the measure of the average length of words and sentence in documents. Higher the value of Fog index more is the average length and sentence in document and it is more difficult to understand the document.

66. Topic: Operating System

(b) Shortest seek time first algorithm seeks according to next shortest distance. Starting from 50 next shortest distance will be 37 since it is 13 tracks away. Time taken = 13 ms

From 37 next shortest distance will be 20 since it is 17 tracks away. Time taken = 17 ms
From 20 next shortest distance will be 19 and time taken = 1 ms
From 19 next shortest distance will be 15 and time taken = 4 ms
From 15 next shortest distance will be 10 and time taken = 5 ms
From 10 next shortest distance will be 7 and time taken = 3 ms
From 7 next shortest distance will be 6 and time taken = 1 ms
From 6 next shortest distance will be 4 and time taken = 2 ms
From 4 next shortest distance will be 2 and time taken = 2 ms
From 2 next shortest distance will be 73 and time taken = 71 ms

Time taken to satisfy all requests = 13 + 17 + 1 + 4 + 5 + 3 + 1 + 2 + 2 + 71 = 119 ms

67. Topic: Databases

(a) Assuming that all nodes are completely full, then every node will have $(n - 1)$ keys.
Given, there are 4 levels including root node. If a new key is inserted then a new node will be created at every level and the tree will be increased to one more level.
So, the maximum number of nodes that could be newly created in the process will be 5.

68. Topic: Operating System

(a) A critical region is a piece of code that accesses shared variables and therefore has to be executed as on atomic action. Thus, in a group of cooperating process, only one process must be executing at a given point of time. So, a critical region is a piece of code which only one process executes at a time.

69. Topic: Programming and Data Structures

(a) Prefix form also known as polish notation requires all operators precede the two operands that they work on So,

$$\begin{aligned}&(a+(b-c)) * ((d-e)/(f+g-h))\\&=(a+(-bc)) * ((-de)/(-+fgh))\\&=(+a-bc) * (1-de-+fgh)\\&=*+a-bc\,/-de-+fgh\end{aligned}$$

Therefore, option (a) is correct.

70. Topic: Programming and Data Structures

(a) Malloc and Calloc are used for dynamic memory allocation. Dynamic memory allocation refers to manual memory allocation. It allows us to obtain more memory when required and release it when not necessary.

Malloc: allocates requested size of bytes and returns a pointer first byte of allocated space.

Calloc: allocates space for an array element, initializes to zero & then return a pointer to memory.

71. Topic: Operating System

(b, d) Each P operation decreases semaphore value by 1 and each V operation increase the semaphore value by 1. If value of semaphore is 10, it will become 7 after
(b) 3P operation $\Rightarrow$ S = S – 3 = 10 – 3 = 7
(d) 2V operation & 5P operations $\Rightarrow$ S = S + 2 – 5

= 10 + 2 – 5
= 7

72. Topic: Operating System

(c) mknod command makes a directory entry and corresponding i-node for a special file. So, the given command 'mknod myfifo b 4 16' will create 'myfifo' special file. It is a special block file with the major device number 4 and minor device number 16. Thus option (c) is correct that the given command will create a block device if the user is root.

73. Topic: Operating System

(a) Jitter means timing error in task. Hard real time OS can predict the deadline and responds at time $t = 0$. Whereas soft real time OS can miss certain deadlines and responds at time $t = 0+$. Therefore, hard real time OS has less jitter than soft real time OS. So, option (a) is true.

74. Topic: Programming and Data Structures

(b) Throwable is a super class of all errors and exceptions in the Java language. Java throw statement can throw the objects that are instances of this class and throwable class and its subclass can only be the argument in a catch clause.

75. Topic: Computer Networks

(d) We know that:
 create a numbered list
 create unordered list.
<dl> creates a description list
<list> creates list items
Therefore, option (d) is correct.

76. Topic: Algorithms

(d) Prim's algorithm solves spanning tree problem. Dijikstra's algorithm solves single source shortest path problem. Bellmon for algorithm solves single source shortest path problem. Floyd–Warshall's algorithm solves all pair shortest path problem. Therefore option (d) is correct.

77. Topic: Theory of Computation

(d) Recursively enumerable languages are closed under union $L \cup P$, intersection $L \cap P$ and closure. But recursively enumerable languages are not closed under set difference. Set difference $L - P$ can be written as L and $\sim P$ and we know that recursively enumerable languages are not closed under complement.
Therefore, L and P are not closed under set difference.

78. Topic: Software Engineering

(b) Cohesion is a measure of internal strength and coupling is a measure of inter-dependency among modules. Therefore, in the context of modular software design cohesion should be high and coupling should be low i.e., internal strength within a module should be high and interdependency among modules should be low. Hence option (b) is correct.

79. Topic: Compiler Design

(d) The output of a lexical analyzer is a stream of tokens Lexical analyzer takes the source code as input and convert it into the stream of tokens as output i.e. it read the source code, one character at a time and translate into a sequence of primitive units called tokens.

80. Topic: Programming and Data Structures

(c) Transitive closure of a binary relation tells whether there exists a path from node i to node j. Which means ordered pair consists of two elements. For a set of n elements the matrix can be found using $n^2(2n-1)(n-1) + n^2(n-1)$ bit operations. By using Warshall's algorithm the time complexity of computation will be $O(n^3)$ bit operations.
Therefore, option (c) is correct.

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Computer Science

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QUESTION PAPER

1. Suppose A is a finite set with n elements. The number of elements and the rank of the largest equivalence relation on A are

(a) $\{n, 1\}$ (b) $\{n, n\}$
(c) $\{n^2, 1\}$ (d) $\{1, n^2\}$

2. Consider the set of integers I. Let D denote "divides with an integer quotient" (e.g., $4D8$ but $4\not{D}7$). Then D is

(a) Reflexive, not symmetric, transitive
(b) Not reflexive, not antisymmetric, transitive
(c) Reflexive, antisymmetric, transitive
(d) Not reflexive, not antisymmetric, not transitive

3. A bag contains 19 red balls and 19 black balls. Two balls are removed at a time repeatedly and discarded if they are of the same colour, but if they are different, black ball is discarded and red ball is returned to the bag. The probability that this process will terminate with one red ball is

(a) 1 (b) 1/21
(c) 0 (d) 0.5

4. If $x = -1$ and $x = 2$ are extreme points of $f(x) = \alpha \log|x| + \beta x^2 + x$ then

(a) $\alpha = -6, \beta = -1/2$ (b) $\alpha = 2, \beta = -1/2$
(c) $\alpha = 2, \beta = 1/2$ (d) $\alpha = -6, \beta = 1/2$

5. Let $f(x) = \log|x|$ and $g(x) = \sin x$. If A is the range of $f(g(x))$ and B is the range of $g(f(x))$ then $A \cap B$ is

(a) $[-1, 0]$ (b) $[-1, 0)$
(c) $[-\infty, 0]$ (d) $[-\infty, 1]$

6. The proposition $(P \Rightarrow Q) \wedge (Q \Rightarrow P)$ is a

(a) tautology (b) contradiction
(c) contingency (d) absurdity

7. If $T(x)$ denotes x is a trigonometric function, $P(x)$ denotes x is a periodic function and $C(x)$ denotes x is a continuous function then the statement "*It is not the case that some trigonometric functions are not periodic*" can be logically represented as

(a) $\neg\exists x[T(x) \wedge \neg P(x)]$ (b) $\neg\exists x[T(x) \vee \neg P(x)]$
(c) $\neg\exists x[\neg T(x) \wedge \neg P(x)]$ (d) $\neg\exists x[T(x) \wedge P(x)]$

8. The number of elements in the power set of $\{\{1,2\},\{2,1,1\},\{2,1,1,2\}\}$ is

(a) 3 (b) 8
(c) 4 (d) 2

9. The function $f : [0, 3] \rightarrow [1, 29]$ defined by $f(x) = 2x^3 - 15x^2 + 36x + 1$ is

(a) injective and surjective
(b) surjective but not injective
(c) injective but not surjective
(d) neither injective nor surjective

10. If vectors $\vec{a} = 2\hat{i} + \lambda\hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} + 3\hat{k}$ are perpendicular to each other, then value of λ is

(a) 2/5 (b) 2
(c) 3 (d) 5/2

11. Consider the schema

Sailors(sid,sname,rating,age) with the following data

Sid	sname	rating	age
22	Dustin	7	45
29	Borg	1	33
31	Pathy	8	55
32	Robert	8	25
58	Raghu	10	17
64	Herald	7	35
71	Vishnu	10	16
74	King	9	35
85	Archer	3	26
84	Bob	3	64
96	Flinch	3	17

For the query

```
SELECT S.rating, AVG(S.age) AS avgage FROM Sailors S
Where S.age >= 18
GROUP BY S.rating
HAVING 1 < (SELECT COUNT(*) FROM Sailors S2
where S.rating = S2.rating)
```

The number of rows returned is

(a) 6 (b) 5
(c) 4 (d) 3

12. Consider a table that describes the customers:

Customers (custid, name, gender, rating)

The *rating* value is an integer in the range 1 to 5 and only two values (*male and female*) are recorded for *gender*. Consider the query "how many male customers have a rating of 5"? The best indexing mechanism appropriate for the query is

(a) Linear hashing (b) Extendible hashing
(c) B+ tree (d) Bit-mapped index

13. Consider the following schema:

Sailors(sid,sname,rating,age)
Boats(bid,bname, colour)
Reserves(sid,bid,day)

Two boats can have the same name but the colour differentiates them.
The two relations

$$\rho(Tempsids, (\pi_{sid,\ bid}\ Reserves)/(\pi_{bid}\ (\sigma_{bname=\text{'Ganga'}}\ Boats))),$$
$$\pi_{\text{sname}}\ (\text{Tempsids} \times \text{Sailors})$$

If/is division operation, the above set of relations represents the query

(a) Names of sailors who have reserved all boats called *Ganga*
(b) Names of sailors who have not reserved any *Ganga* boat
(c) Names of sailors who have reserved at least one *Ganga* boat
(d) Names of sailors who have reserved at most one *Ganga* boat

14. Type IV JDBC driver is a driver

(a) which is written in C++
(b) which requires an intermediate layer
(c) which communicates through Java sockets
(d) which translates JDBC function calls into API not native to DBMS

15. Consider the following table : Faculty (*facName*, *dept*, *office*, *rank*, *dateHired*)

facName	dept	office	rank	dateHired
Ravi	Art	A101	Professor	1975
Murali	Math	M201	Assistant	2000
Narayanan	Art	A101	Associate	1992
Lakshmi	Math	M201	Professor	1982
Mohan	CSC	C101	Professor	1980
Sreeni	Math	M203	Associate	1990
Tanuja	CSC	C101	Instructor	2001
Ganesh	CSC	C105	Associate	1995

(Assume that no faculty member within a single department has same name. Each faculty member has only one office identified in *office*). 3NF refers to third normal form and BNCF refers to Boyce-Codd normal form
Then *Faculty* is

(a) Not in 3NF, in BCNF
(b) In 3NF, not in BCNF
(c) In 3NF, in BCNF
(d) Not in 3NF, not in BCNF

16. Consider the following query:

```
SELECT E.eno, COUNT(*)
FROM Employees E
GROUP BY E.eno
```

If an index on *eno* is available, the query can be answered by scanning only the index if

(a) the index is only hash and clustered
(b) the index is only B+ tree and clustered
(c) index can be hash or B+ tree and clustered or non-clustered
(d) index can be hash or B+ tree and clustered

17. If C is a skew-symmetric matrix of order n and X is $n \times 1$ column matrix, then X^TCX is a

(a) scalar matrix
(b) null matrix
(c) unit matrix
(d) matrix will all elements 1

18. Consider the recurrence equation

$$T(n) = \begin{cases} 2T(n-1), & \text{if } n > 0 \\ 1 & \text{otherwise} \end{cases}$$

Then $T(n)$ is (in big O order)

(a) $O(n)$ (b) $O(2^n)$
(c) $O(1)$ (d) $O(\log n)$

19. Consider the program

```
void function(int n)
{
    int i, j, count=0;
    for (i=n/2; i <= n; i++)
        for (j = 1; j <= n; j = j*2)
            count++;
}
```

The complexity of the program is

(a) $O(\log n)$ (b) $O(n^2)$
(c) $O(n^2 \log n)$ (d) $O(n \log n)$

20. Match the following and choose the correct answer for the order A, B, C, D

A. Strassen matrix multiplication	p. Decrease and Conquer
B. Insertion sort	q. Dynamic Programming
C. Guassian Elimination	r. Divide and Conquer
D. Floyd shortest path algorithm	s. Transform and Conquer

(a) r, s, p, q (b) r, p, s, q
(c) q, s, p, r (d) s, p, q, r

21. For $\Sigma = \{a, b\}$ the regular expression $r = (aa)^*(bb)^*b$ denotes

(a) Set of strings with 2 a's and 2 b's
(b) Set of strings with 2 a's 2 b's followed by b
(c) Set of strings with 2 a's followed by b's which is a multiple of 3
(d) Set of strings with even number of a's followed by odd number of b's

22. Consider the grammar with productions

$$S \to aSb|SS|\varepsilon$$

This grammar is

(a) not context-free, not linear
(b) not context-free, linear
(c) context-free, not linear
(d) context free, linear

23. Identify the language generated by the following grammar

$$S \to AB$$
$$A \to aAb|\varepsilon$$
$$B \to bB|b$$

(a) $\{a^m b^n \mid n \geq m, m > 0\}$
(b) $\{a^m b^n \mid n \geq m, m \geq 0\}$
(c) $\{a^m b^n \mid n > m, m > 0\}$
(d) $\{a^m b^n \mid n > m, m \geq 0\}$

24. Let L_1 be regular language, L_2 be a deterministic context free language and L_3 a recursively enumerable language, but not recursive. Which one of the following statements is false?

(a) $L_3 \cap L_1$ is recursive
(b) $L_1 \cap L_2 \cap L_3$ is recursively enumerable
(c) $L_1 \cup L_2$ is context free
(d) $L_1 \cap L_2$ is context free

25. Let $L = \{a^p \mid p \text{ is a prime}\}$. Then which of the following is true

(a) It is not accepted by a Turing Machine
(b) It is regular but not context free
(c) It is context free but not regular
(d) It is neither regular nor context free, but accepted by a Turing Machine

26. Which of the following are context free?

$$A = \{a^n b^n a^m b^m \mid m, n \geq 0\}$$
$$B = \{a^m b^n a^m b^n \mid m, n \geq 0\}$$
$$C = \{a^m b^n \mid m \neq 2, m, n \geq 0\}$$

(a) A and B only (b) A and C only
(c) B and C only (d) C only

27. Let S be an NP-complete problem. Q and R are other two problems not known to be NP. Q is polynomial time reducible to S and S is polynomial time reducible to R. Which of the following statements is true?

(a) R is NP-complete (b) R is NP-hard
(c) Q is NP-complete (d) Q is NP-hard

28. The number of structurally different possible binary trees with 4 nodes is

(a) 14 (b) 12
(c) 336 (d) 168

29. Using public key cryptography, X adds a digital signature σ to a message M, encrypts $\langle M, \sigma \rangle$ and sends it to Y, where it is decrypted. Which one of the following sequence of keys is used for operations?

(a) Encryption : X's private key followed by Y's private key. Decryption : X's public key followed by Y's public key
(b) Encryption : X's private key followed by Y's public key; Decryption : X's public key followed by Y's private key
(c) Encryption : X's private key followed by Y's public key; Decryption : Y's private key followed by X's public key.
(d) Encryption : X's public key followed by Y's private key; Decryption : Y's public key followed by X's private key.

30. Which of the following are used to generate a message digest by the network security protocols?

(P) SHA-256
(Q) AES
(R) DES
(S) MD5

(a) P and S only (b) P and Q only
(c) R and S only (d) P and R only

31. In the IPv4 addressing format, the number of networks allowed under Class C addresses is

(a) 2^{20} (b) 2^{24}
(c) 2^{14} (d) 2^{21}

32. An Internet Service Provider (ISP) has the following chunk of CIDR-based IP addresses available with it: 245.248.128.0/20. The ISP wants to give half of this chunk of addresses to Organization *A*, and a quarter to Organization *B*, while retaining the remaining with itself. Which of the following is a valid allocation of addresses to *A* and *B*?

(a) 245.248.136.0/21 and 245.248.128.0/22
(b) 245.248.128.0/21 and 245.248.128.0/22
(c) 245.248.132.0/22 and 245.248.132.0/21
(d) 245.248.136.0/24 and 245.248.132.0/21

33. Assume that Source *S* and Destination *D* are connected through an intermediate router *R*. How many times a packet has to visit the network layer and data link layer during a transmission from *S* to *D*?

(a) Network layer – 4 times, Data link layer – 4 times
(b) Network layer – 4 times, Data link layer – 6 times
(c) Network layer – 2 times, Data link layer – 4 times
(d) Network layer – 3 times, Data link layer – 4 times

34. Generally TCP is reliable and UDP is not reliable. DNS which has to be reliable uses UDP because

(a) UDP is slower
(b) DNS servers has to keep connections
(c) DNS requests are generally very small and fit well within UDP segments
(d) None of these

35. Consider the set of activities related to e-mail

A : Send an e-mail from a mail client to mail server
B : Download e-mail headers from mail box and retrieve mails from server to a cache
C : Checking e-mail through a web browser

The application level protocol used for each activity in the same sequence is

(a) SMTP, HTTPS, IMAP (b) SMTP, POP, IMAP
(c) SMTP, IMAP, HTTPS (d) SMTP, IMAP, POP

36. Station *A* uses 32 byte packets to transmit messages to Station *B* using a sliding window protocol. The round trip time delay between *A* and *B* is 40 ms and the bottleneck bandwidth on the path *A* and *B* is 64 kbps. What is the optimal window size that *A* should use?

(a) 5 (b) 10
(c) 40 (d) 80

37. A two way set associative cache memory unit with a capacity of 16 KB is built using a block size of 8 words. The word length is 32 bits. The physical address space is 4 GB. The number of bits in the TAG, SET fields are

(a) 20, 7 (b) 19, 8
(c) 20, 8 (d) 21, 9

38. A CPU has a 32 KB direct mapped cache with 128 byte block size. Suppose A is a 2 dimensional array of size 512×512 with elements that occupy 8 bytes each. Consider the code segment

```
for (i =0; i < 512; i++)
{
   for (j =0; j < 512; j++)
   {
         x += A[i][j];
   }
}
```

Assuming that array is stored in order *A*[0][0], *A*[0][1], *A*[0][2]…, the number of cache misses is

(a) 16384 (b) 512
(c) 2048 (d) 1024

39. A computer with 32 bit word size uses 2s compliment to represent numbers. The range of integers that can be represented by this computer is

(a) -2^{32} to 2^{32} (b) -2^{31} to $2^{32}-1$
(c) -2^{31} to $2^{31}-1$ (d) $-2^{31}-1$ to $2^{32}-1$

40. Let *M* = 11111010 and *N* = 00001010 be two 8 bit two's compliment number. Their product in two's complement is

(a) 11000100 (b) 10011100
(c) 10100101 (d) 11010101

41. For a pipelines CPU with a single ALU, consider the following:

A. The $j+1^{st}$ instruction uses the result of j^{th} instruction as an operand
B. Conditional jump instruction
C. j^{th} and $j+1^{st}$ instructions require ALU at the same time

Which one of the above causes a hazard?

(a) A and B only (b) B and C only
(c) B only (d) A, B, C

42. In designing a computer's cache system, the cache block (or cache line) size is an important parameter. Which one of the following statements is correct in this context?

(a) Smaller block size incurs lower cache miss penalty
(b) Smaller block size implies better spatial locality
(c) Smaller block size implies smaller cache tag
(d) Smaller block size implies lower cache hit time

43. Consider an instruction of the type LW R1, 20(R2) which during execution reads a 32 bit word from memory and stores it in a 32 bit register R1. The effective address of the memory location is obtained by adding a constant 20 and contents of R2. Which one best reflects the source operand

(a) Immediate addressing
(b) Register addressing
(c) Register Indirect addressing
(d) Indexed addressing

44. A sorting technique is called stable if

(a) It takes O(n log n) time
(b) It uses divide and conquer technique
(c) Relative order of occurrence of non-distinct elements is maintained
(d) It takes O(n) space

45. Match the following and choose the correct answer in the order A, B, C

A. Heap construction	p. O(n log n)
B. Hash table construction with linear probing	q. O (n^2)
C. AVL Tree construction	r. O(n)

(Bounds given may or may not be asymptotically tight)

(a) q, r, p (b) p, q, r
(c) q, p, r (d) r, q, p

46. In a compact one dimensional array representation for lower triangular matrix (all elements above diagonal are zero) of size $n \times n$, non-zero elements of each row are stored one after another, starting from first row, the index of $(i, j)^{th}$ element in this new representation is

(a) $i + j$ (b) $j + \frac{i(i-1)}{2}$
(c) $i + j - 1$ (d) $i + \frac{j(j-1)}{2}$

47. Which of the following permutation can be obtained in the same order using a stack assuming that input is the sequence 5, 6, 7, 8, 9 in that order?

(a) 7, 8, 9, 5, 6 (b) 5, 9, 6, 7, 8
(c) 7, 8, 9, 6, 5 (d) 9, 8, 7, 5, 6

48. Quick sort is run on 2 inputs shown below to sort in ascending order

A. 1, 2, 3,…n
B. $n, n - 1, n - 2,\ldots 1$

Let $C1$ and $C2$ be the number of comparisons made for A and B respectively.
Then

(a) $C1 > C2$
(b) $C1 = C2$
(c) $C1 < C2$
(d) Cannot say anything for arbitrary n

49. A binary search tree is used to locate the number 43. Which one of the following probe sequence is not possible?

(a) 61, 52, 14, 17, 40, 43
(b) 10, 65, 31, 48, 37, 43
(c) 81, 61, 52, 14, 41, 43
(d) 17, 77, 27, 66, 18, 43

50. The characters of the string $K\ R\ P\ C\ S\ N\ Y\ T\ J\ M$ are inserted into a hash table of size of size 10 using hash function

$$h(x) = (ord(x) - ord(A) + 1)$$

If linear probing is used to resolve collisions, then the following insertion causes collision

(a) Y (b) C
(c) M (d) P

51. Suppose the numbers 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are inserted in that order into an initially empty binary search tree. The binary search tree uses the reversal ordering on natural numbers i.e. 9 is assumed to be smallest and 0 is assumed to be largest. The in-order traversal of the resultant binary search tree is

(a) 9, 8, 6, 4, 2, 3, 0, 1, 5, 7
(b) 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
(c) 0, 2, 4, 3, 1, 6, 5, 9, 8, 7
(d) 9, 8, 7, 6, 5, 4, 3, 2, 1, 0

52. A priority queue is implemented as a Max-heap. Initially it has 5 elements. The level order traversal of the heap is 10, 8, 5, 3, 2. Two new elements '1' and '7' are inserted into the heap in that order. The level order traversal of the heap after the insertion of the elements is

(a) 10, 8, 7, 5, 3, 2, 1
(b) 10, 8, 7, 2, 3, 1, 5
(c) 10, 8, 7, 1, 2, 3, 5
(d) 10, 8, 7, 3, 2, 1, 5

53. The minimum number of stacks needed to implement a queue is

(a) 3 (b) 1
(c) 2 (d) 4

54. A strictly binary tree with 10 leaves

(a) cannot have more than 19 nodes
(b) has exactly 19 nodes

(c) has exactly 17 nodes
(d) has exactly 20 nodes

55. What is the maximum height of any AVL tree with 7 nodes? Assume that height of tree with single node is 0.

(a) 2 (b) 3
(c) 4 (d) 5

56. Which one of the following property is correct for a red-black tree?

(a) Every simple path from a node to a descendant leaf contains the same number of black nodes
(b) If a node is red, then one children is red and another is black
(c) If a node is red, then both its children are red
(d) Every leaf node (sentinel node) is red

57. The in-order and pre-order traversal of a binary tree are *d b e a f c g* and *a b d e c f g* respectively. The post order traversal of a binary tree is

(a) *e d b g f c a* (b) *e d b f g c a*
(c) *d e b f g c a* (d) *d e f g b c a*

58. A virtual memory system uses FIFO page replacement policy and allocates a fixed number of frames to the process. Consider the following statements

M : Increasing the number of page frames allocated to a process sometimes increases the page fault rate
N : Some programs do not exhibit locality of reference

Which one of the following is true?

(a) Both M and N are true and N is the reason for M
(b) Both M and N are true but N is not the reason for M
(c) Both M and N are false
(d) M is false, but N is true

59. Consider three CPU intensive processes, which require 10, 20, 30 units and arrive at times 0,2,6 respectively. How many context switches are needed if shortest remaining time first is implemented? Context switch at 0 is included but context switch at end is ignored

(a) 1 (b) 2
(c) 3 (d) 4

60. A process executes the following code

for ($i = 0$; $i < n$; i ++) fork();

The total number of child processes created is

(a) n^2 (b) $2^{n+1} - 1$
(c) 2^n (d) $2^n - 1$

61. Consider the following scheduling:

A. Gang scheduling	s. Guaranteed scheduling
B. Rate Monotonic scheduling	t. Thread scheduling
C. Fair share scheduling	u. Real time scheduling

Matching the table in the order A, B, C gives

(a) t, u, s (b) s, t, u
(c) u, s, t (d) u, t, s

62. A system uses FIFO policy for page replacement. It has 4 page frames with no pages loaded to begin with. The system first accesses 50 distinct pages in some order and then accesses the same 50 pages in reverse order. How many page faults will occur?

(a) 96 (b) 100
(c) 97 (d) 92

63. Which of the following is false?

(a) User level threads are not scheduled by the kernel
(b) Context switching between user level threads is faster than context switching between kernel level threads
(c) When a user thread is blocked all other threads of its processes are blocked
(d) Kernel level threads cannot utilize multiprocessor systems by splitting threads on different processors or cores

64. Which of the following is not true with respect to deadlock prevention and deadlock avoidance schemes?

(a) In deadlock prevention, the request for resources is always granted if resulting state is safe
(b) In deadlock avoidance, the request for resources is always granted, if the resulting state is safe
(c) Deadlock avoidance requires knowledge of resource requirements a priori
(d) Deadlock prevention is more restrictive than deadlock avoidance

65. Which one of the following are essential features of object oriented language?

A. Abstraction and encapsulation
B. Strictly-typed
C. Type-safe property coupled with sub-type rule
D. Polymorphism in the presence of inheritance

(a) A and B only
(b) A, D and B only
(c) A and D only
(d) A, C and D only

66. Which languages necessarily need heap allocation in the run time environment?

(a) Those that support recursion
(b) Those that use dynamic scoping
(c) Those that use global variables
(d) Those that allow dynamic data structures

67. Consider the code segment

```
int i, j, x, y, m, n;
n=20;
for (i = 0, i < n; i++)
{
   for (j = 0; j < n; j++)
   {
         if (i % 2)
         {
               x + = ((4*j) + 5*i);
               y += (7 + 4*j);
         }
   }
}
m = x + y;
```

Which one of the following is false?

(a) The code contains loop invariant computation
(b) There is scope of common sub-expression elimination in this code
(c) There is scope of strength reduction in this code
(d) There is scope of dead code elimination in this code

68. Consider the following table:

A. Activation record	p. Linking loader
B. Location counter	q. Garbage collection
C. Reference counts	r. Subroutine call
D. Address relocation	s. Assembler

Matching A, B, C, D in the same order gives:

(a) p, q, r, s (b) q, r, s, p
(c) r, s, q, p (d) r, s, p, q

69. Consider a disk sequence with 100 cylinders. The request to access the cylinder occur in the following sequence:

4, 34, 10, 7, 19, 73, 2, 15, 6, 20

Assuming that the head is currently at cylinder 50, what is the time taken to satisfy all requests if it takes 2 ms to move from one cylinder to adjacent one and shortest seek time first policy is used?

(a) 190 (b) 238
(c) 233 (d) 276

70. A counting semaphore was initialized to 7. Then $20P$ (wait) operations and xV (signal) operations were completed on this semaphore. If the final value of semaphore is 5, then the value x will be

(a) 0 (b) 13
(c) 18 (d) 5

71. A 32 bit *adder* is formed by cascading 4 bit CLA *adder*. The gate delays (latency) for getting the sum bits is

(a) 16 (b) 18
(c) 17 (d) 19

72. We consider the addition of two 2's compliment numbers $b_{n-1}b_{n-2}\ldots b_0$ and $a_{n-1}a_{n-2}\ldots a_0$. A binary adder for adding two unsigned binary numbers is used to add two binary numbers. The sum is denoted by $c_{n-1}c_{n-2}\ldots c_0$. The carry out is denoted by c_{out}. The overflow condition is identified by

(a) $c_{out}(\overline{a_{n-1} \oplus b_{n-1}})$
(b) $\overline{a_{n-1}}b_{n-1}\overline{c_{n-1}} + \overline{a_{n-1}b_{n-1}}c_{n-1}$
(c) $c_{out} \oplus c_{n-1}$
(d) $a_{n-1} \oplus b_{n-1} \oplus c_{n-1}$

73. Consider the function

```
int fun(x: integer)
{
      If x > 100 then fun = x – 10;
      else
      fun = fun(fun(x + 11));
}
```

For the input $x = 95$, the function will return

(a) 89 (b) 90
(c) 91 (d) 92

74. Consider the function

```
int func(int num)
{
   int count = 0;
   while(num)
   {
         count++;
         num >>= 1;
   }
return(count);
}
```

For func(435) the value returned is

(a) 9 (b) 8
(c) 0 (d) 10

75. In IEEE floating point representation, the hexadecimal number 0xC0000000 corresponds to

(a) –3.0 (b) –1.0
(c) –4.0 (d) –2.0

76. Which of the following set of components is sufficient to implement any arbitrary Boolean function?

(a) XOR gates, NOT gates
(b) AND gates, XOR gates and 1
(c) 2 to 1 multiplexer
(d) Three input gates that output $(A \cdot B) + C$ for the inputs A, B, C

77. Consider the following:

A. Condition coverage	p. Black box testing
B. Equivalence class partitioning	q. System testing
C. Volume testing	r. White box testing
D. Beta testing	s. Performance testing

Matching A, B, C, D in the same order gives.

(a) r, p, s, q (b) p, r, q, s
(c) s, r, q, p (d) q, r, s, p

78. Consider the results of a medical experiment that aims to predict whether someone is going to develop myopia based on some physical measurements and heredity. In this case, the input dataset consists of the person's medical characteristics and the target variable is binary: 1 for those who are likely to develop myopia and 0 for those who aren't. This can be best classified as

(a) Regression (b) Decision Tree
(c) Clustering (d) Association Rules

79. Which of the following related to snowflake schema is true?

(a) Each dimension is represented by a single dimensional table
(b) Maintenance efforts are less
(c) Dimension tables are normalized
(d) It is not an extension of star schema

80. Consider the following C function

```
#include <stdio.h>
int main(void)
{
    char c[] = "ICRBCSIT17";
    char *p=c;
    printf("%s", c+2[p] - 6[p] - 1);
    return 0;
}
```

The output of the program is

(a) SI (b) IT
(c) T1 (d) 17

ANSWER KEY

1. (c)	**9.** (b)	**17.** (b)	**25.** (d)	**33.** (d)	**41.** (d)	**49.** (d)	**57.** (c)	**65.** (c)	**73.** (c)
2. (b)	**10.** (d)	**18.** (b)	**26.** (b)	**34.** (c)	**42.** (a)	**50.** (*)	**58.** (b)	**66.** (d)	**74.** (a)
3. (a)	**11.** (d)	**19.** (d)	**27.** (a)	**35.** (c)	**43.** (d)	**51.** (d)	**59.** (b)	**67.** (d)	**75.** (d)
4. (b)	**12.** (d)	**20.** (b)	**28.** (a)	**36.** (b)	**44.** (c)	**52.** (d)	**60.** (d)	**68.** (c)	**76.** (b)
5. (a)	**13.** (a)	**21.** (d)	**29.** (c)	**37.** (b)	**45.** (d)	**53.** (c)	**61.** (a)	**69.** (b)	**77.** (a)
6. (c)	**14.** (c)	**22.** (c)	**30.** (a)	**38.** (a)	**46.** (b)	**54.** (b)	**62.** (a)	**70.** (c)	**78.** (b)
7. (a)	**15.** (b)	**23.** (*)	**31.** (d)	**39.** (c)	**47.** (c)	**55.** (b)	**63.** (d)	**71.** (b)	**79.** (c)
8. (d)	**16.** (c)	**24.** (a)	**32.** (a)	**40.** (a)	**48.** (a)	**56.** (a)	**64.** (a)	**72.** (c)	**80.** (d)

*** None of the options is correct.**

ANSWERS WITH EXPLANATION

1. Topic: Engineering Mathematics

(c) Largest equivalence relation, occurs on A occurs when every element of A is related to every other element of A. Here A is a finite set with n elements therefore total number of elements will be $n \times n = n^2$ (possible ordered pairs) and the rank of largest equivalence relation will be 1. Therefore, correct option is (c) that is, $\{n^2, 1\}$.

2. Topic: Engineering Mathematics

(b) If R is a binary relation on a set A then,
R is reflexive if for all $x \in A$, xRx

R is symmetric if for all $x, y \in A$, if xRy, then yRx

R is transitive if for all $x, y, z \in A$, if xRy and yRz then xRz.

For given relation D, reflexivity is violated for $x = 0$ where $x \in I$. Thus D is not reflexive.
Symmetry relation is violated for $x = 2$ and $y = 1$ for $x, y \in I$
transitive relation is valid for D since for $x, y, z \in I$

$$xDy \text{ and } yDx \Rightarrow xDz$$

Antisymmetric: for all $x \in I$, $R(x, y)$ and $R(y, x)$ then $x = y$ is antisymmetric.
We can easily make a violation as R(–2, 2) and R(2, –2) are not antisymmetric.
Transitivity for all $x \in I$, $R(x, y)$, $R(y, z)$ then $R(x, z)$.
This is always true for the above divides relation. So, the relation is not-reflexive, not-antisymmetric but transitive.

3. Topic: Engineering Mathematics

(a) Given, bag contains 19 red balls say R and 19 black balls say B
Two balls are chosen, if they are of some color that is, RR or BB both balls are discarded.
If balls are of different color RB or BR then Black ball is discarded and red ball is returned to bag. Since red ball is discarded only when both balls are red to only 18 balls among 19 will be discarded. So, the last trial will have either of the following contents in bag.

1. (red, black)
2. (red, red, red)

So, the process will terminate with one red ball in both the cases, thus probability is 1.

4. Topic: Engineering Mathematics

(b) Given:

$$f(x) = \alpha \log |x| + \beta x^2 + x \tag{1}$$

Now differentiate Eq. (1) with respect to x, we have:

$$f'(x) = \frac{\alpha}{x} + 2\beta x + 1$$

at extremum $f'(x) = 0$
So,

$$\frac{\alpha}{x} + 2\beta x + 1 = 0$$
$$\Rightarrow 2\beta x^2 + x + \alpha = 0 \tag{2}$$

It is given that $x = -1$ and $x = 2$ are extreme points put in Eq. (1) we have

$$2\beta - 1 + \alpha = 0 \tag{3}$$
$$8\beta + 2 + \alpha = 0 \tag{4}$$

On solving Eq. (3) and Eq. (4), we have

$$(2\beta - 1 + \alpha) - (8\beta + 2 + \alpha) = 0$$
$$\Rightarrow 2\beta - 1 + \alpha - 8\beta - 2 - \alpha = 0$$
$$\Rightarrow -6\beta - 3 = 0 \Rightarrow \beta = \frac{-1}{2}$$

Put $\beta = \frac{-1}{2}$ in Eq. (3) we have

$$2 \times \frac{-1}{2} - 1 + \alpha = 0$$
$$\Rightarrow -1 - 1 + \alpha = 0 \Rightarrow \alpha = 2$$

Therefore,

$$\alpha = 2 \text{ and } \beta = \frac{-1}{2}$$

5. Topic: Engineering Mathematics

(a) Given:

$$f(x) = \log |x|$$
$$g(x) = \sin x$$

Then,

$$f(g(x)) = \log |g(x)| = \log |\sin x|$$

Now,

$$0 \le |\sin x| \le 1$$

for domain [0, 1] log will give range $(-\infty, 0]$
So,

$$\text{range of } f(g(x)) = (-\infty, 0) = A$$

also,

$$g(f(x)) = \sin(f(x)) = \sin(\log |x|)$$

Now,

$$-\infty \le \log|x| \le \infty$$

For domain $[-\infty, \infty]$ sin function will give range [–1, 1]
Therefore,

$$\text{range of } g(f(x)) = [-1, 1] = B$$

Then, $A \cap B$ is $[-1, 0]$

6. **Topic: Engineering Mathematics**

(c) Given:

$$(P \Rightarrow Q) \wedge (Q \Rightarrow P)$$

Truth table for the above equation is:

P	***Q***	***P→Q***	***Q→P***	***(P ⇒ Q) ∧ (Q ⇒ P)***
T	*T*	*T*	*T*	*T*
T	*F*	*F*	*T*	*F*
F	*T*	*T*	*F*	*F*
F	*F*	*T*	*T*	*T*

Thus, the given proposition is neither a tautology nor contradiction. A proportion which is neither tautology nor contradiction is a contingency.

7. **Topic: Engineering Mathematics**

(a) Given:

$T(x)$ is trigonometric function
$P(x)$ is periodic function
$C(x)$ is continuous function

If we suppose that there is a trigonometric function which is not periodic, then it can be written as:

$$\exists x\, [T(x) \wedge \neg P(x)]$$

Now, for the statement 'It is not the case that some trigonometric functions are not periodic' then it can logically be written as:

$$\neg \exists x [T(x) \wedge \neg P(x)]$$

8. **Topic: Engineering Mathematics**

(d) A power set of a set S is the set of all subsets of S including the empty set and S itself. Since the elements in the set {{1, 2}, {2, 1, 1}, {2, 1, 1, 2}} are all same. Therefore, number of elements in power set will be $\{\{\phi\}, \{2,1,1,2\}\}$ that is two elements.

9. **Topic: Engineering Mathematics**

(b) If there are two sets, set A and set B. Injective means, every member of set A has it's unique matching member in set B. Injective is also called one-to-one.
Surjective means that every member of set B has at least one matching member in set A (maybe more than one). Surjective is also called onto function.
Thus, the function f: [0, 3] → [1, 29] defined by:
$f(x) = 2x^3 - 15x^2 + 36x + 1$ is surjective but not injective because,

$$f'(x) = 6x^2 - 30x + 36 = 6(x^2 - 5x + 6) = 6(x-2)(x-3)$$

So, $f(x)$ increases in (0, 2) and decreases in (2, 3). Hence function is not one-to-one.
Now,

$$f(0) = 1$$
$$f(3) = 54 - 135 + 108 + 1 = 28$$

also, $x = 2$ is point of maximum

$$f(2) = 16 - 60 + 72 + 1 = 29$$

Thus, range of $f(x)$ is [1, 29].
Hence function is surjective.

10. **Topic: Engineering Mathematics**

(d) Given:

$$\vec{a} = 2\hat{i} + \lambda\hat{j} + \hat{k}$$
$$\vec{b} = \hat{i} - 2\hat{j} + 3\hat{k}$$
$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos\theta$$

Since, vectors are perpendicular.

$$\theta = 90$$

So,

$$\cos 90 = 0 \Rightarrow \vec{a} \cdot \vec{b} = 0$$
$$\Rightarrow \vec{a} \cdot \vec{b} = (2\hat{i} + \lambda\hat{j} + \hat{k}) \cdot (\hat{i} - 2\hat{j} + 3\hat{k}) = 0$$
$$\Rightarrow 2 - 2\lambda + 3 = 0 \Rightarrow -2\lambda + 5 = 0$$
$$\Rightarrow \lambda = 5/2$$

11. **Topic: Databases**

(d) Given:

query; SELECT S, rating, AVG(S.age) AS average

FROM Sailors S
Where S.age > = 18
GROUP By S.rating
It selects sailors whose age is greater than 18 and groups then by rating.
HAVING 1 < (SELECT COUNT (*) FROM sailors S2 where S.rating = S2.rating)
It selects rows where rating is equal to rating in row grouped by above rating and finally selects those rows where this count is greater than 1.
Number of rows returned is 3.

12. **Topic: Databases**

(d) Bit mapped index is a special kind of database index that uses bit maps. It is useful for moderate or even high-cardinality data which is accessed in a read-only manner and queries access multiple bitmap indexed columns using AND, OR or XOR operators extensively. It is also useful in data warehousing applications for joining a large fact table to smaller dimensions tables. It can also be used for representing data structure. The best appropriate indexing mechanism here is bit mapped index.

13. Topic: Databases

(a) Given:

Sailors (sid, sname, rating, age)

Boats (bid, bname, colour)

Reserves (Sid, bid, day)

The set relation is given as:

$$\rho(Tempsids, (\pi_{sid,\, bid}\, Reserves)/(\pi_{bid}\, (\sigma_{bname='Ganga'}\, Boats))),$$
$$\pi_{\text{sname}}\, (\text{Tempsids} \times \text{Sailors})$$

The given set of relations represents Names of sailors who have reserved all boats called Ganga.

14. Topic: Programming and Data Structures

(c) JDBC drivers implement defined interfaces in JDBC API for interacting with your database server. In a type IV JDBC driver, a pure java-based driver communicates directly with Vendor's database through socket connection. Thus, type IV JDBC driver is a driver which communicates through Java sockets. It is highest performance driver available for database that is provided by vendor itself.

15. Topic: Databases

(b) In third normal form, all attributes are atomic and there are no repeating groups, all non-key attributes are fully functional dependent on the primary key and there is no transitive functional dependency.

In BCNF or Boyce-Codd normal form, there is no non-trivial functional dependencies of attributes. Here,

facName → dept, office, rank, dak hired

office → dept

facName is primary key therefore faculty is in 3NF. However, office is not a super key therefore Faculty is not in BCNF. Hence option (b) is correct.

16. Topic: Databases

(c) A B+ tree consists of root, internal nodes and leaves. It is a B-tree in which each node contains only keys and not key value pairs.

A clustered index defines the order in which data is physically stored in table.

A non-dustered index doesn't sort the physical data inside the table. In fact, a non-dustered index is stored at one place & table data is stored in another place. Here index 'eno' can be hash or B+ true and clustered or non-clustered.

17. Topic: Engineering Mathematics

(b) Suppose skew-symmetric matrix 'C' is given as

$$C = \begin{bmatrix} 0 & b & c \\ -b & 0 & a \\ -c & -a & 0 \end{bmatrix}$$

and column vector X is

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Then,

$$X^T CX = \begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} 0 & b & c \\ -b & 0 & a \\ -c & -a & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$
$$= \begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} by + cz \\ -bx + az \\ -cx - ay \end{bmatrix}$$
$$= \begin{bmatrix} xby + xcz - bxy + ayz - cxz - ayz \end{bmatrix}$$
$$= \begin{bmatrix} 0 \end{bmatrix}$$

Thus, X^TCX is a null matrix.

18. Topic: Programming and Data Structures

(b) Given:

$$T(n) = \begin{cases} 2T(n-1) & \text{if } n < 0 \\ 1 & \text{otherwise} \end{cases}$$
$$T(n) = 2T(n-1) = 2[2T(n-2)] = 2^2 T(n-2)$$
$$= 2^2[2T(n-3)] = 2^3(T(n-3))$$

or

$$T(n) = 2^k\, [T\,(n-k)]$$

Now, we know $T\,(0) = 1$

$$\Rightarrow n - k = 0 \Rightarrow n = k$$

So,

$$T(n) = 2^k \text{ or } 2^n$$

Therefore, $T(n)$ is $O(2^n)$

19. Topic: Programming and Data Structures

(d) The outer for loop runs $n/2$ times and inner for loop runs log n times both the loops together will run for n log n time. The complexity of program is O($n$log$n$).

20. Topic: Programming and Data Structures

(b) Strassen matrix multiplication is a divide and conquer algorithm that is faster $O(n^{\log 7})$.

Insertion sort is a simple sorting algorithm that sorts array by shifting elements one by one.

Gaussian elimination algorithm is used for solving systems of linear equations. It uses transform and conquer approach to solve set of equations.

Floyd shortest path algorithm is an algorithm for finding shortest path in weighted graph. It uses the principle of dynamic programming.

Thus, Strassen matrix multiplication: divide and conquer insertion sort – decrease and conquer. Gaussian elimination: transform and conquer.
Floyd shortest path algorithm: dynamic programming.

21. Topic: Theory of Computation

(d) Regular expressions are used to denote regular languages. Here $\Sigma = \{a, b\}$,
Σ represents input alphabet and ab is a regular expression with language $\{a, b\}$. Thus, regular expression $r = (aa)^*(bb)^*b$ denotes set of strings with even number of a's followed by odd number of b's.

22. Topic: Theory of Computation

(c) Given:

$$S \to aSb|SS|\varepsilon$$

A grammar is linear if all not terminals are in left side or right side. In the given grammar,

$$S \to asb$$

Non-terminal appears in the middle; therefore, it is non-linear grammar
A context free grammar has rule:

$$S \to \omega$$

where S is a non-terminal and ω is string of terminals and non-terminals.
Thus, the given grammar is context free.
Thus option (c) is correct; the given grammar is context free and nonlinear.

23. Topic: Theory of Computation

(None) Given:

$$S \to AB$$
$$A \to aAb \mid \varepsilon$$
$$B \to bB \mid b$$

The language generated by the given grammar will be: $\{a^m b^n | \, n > m, m \geq 0\}$ none of the options is correct.

24. Topic: Theory of Computation

(a) Given L_1 is a regular language, L_2 is a deterministic context free language and L_3 is recursively enumerable but not recursive language.
Now we know that deterministic context free languages are closed under intersection with regular languages. Every context free language is closed under union. Every regular language is also a context free language, regular language is also recursively enumerable and deterministic context free language is also recursively enumerable and recursively enumerable languages are closed under intersection. Thus statement 1 is false is $L_3 \cap L_1$ is recursive.

25. Topic: Theory of Computation

(d) Given:

$$L = \{a^p/p \text{ is a prime}\}$$

the string generated will be

$$\{aaa, aaaaa, aaaaaaa\ldots\}$$

Since, there is no definite pattern therefore the given language is not regular and hence not context free. However, by using trial division algorithm this language is accepted by Turing machine. Hence L is neither regular nor context free but accepted by a Turing machine.

26. Topic: Theory of Computation

(b) From the given languages A and C are context free while B is not.
Consider $A = \{a^n b^n a^m b^m | m, n \geq 0\}$
for each a push into stack and for each b pop the stack. If stack is empty, string is accepted.
Now, consider $C = \{a^m b^n \mid m \neq 2n, m, n, \geq 0\}$
for each a push into stack and for each b pop the stack twice. If stack is not empty, string is accepted.
Now, consider $B = \{a^m b^n a^m b^n | m, n, \geq 0\}$
here single stack is not sufficient since m and n are not comparable and it is difficult to find, when to push in stack and when to pop. Therefore, B is not context free.

27. Topic: Algorithms

(a) Given, S in NP – complete problem
Q and R are other two problems not known to be NP.
Q is polynomial time reducible to S
S is polynomial time reducible to R
R is NP-Hard. Since S is NP-complete is reducible to R then R is at least as hard as S
Now, intersection of NP-hard and NP is NP complete. Therefore, R is NP-hard and NP therefore R is NP-complete.
Therefore option (a) is correct.

28. Topic: Programming and Data Structures

(a) Structurally different binary trees implies unlabeled binary trees where nodes are not assigned any label.
Total number of such binary trees are given as:

$$\frac{(2n)!}{(n+1)!n!}$$

here $n = 4$
Therefore,

$$\frac{(2\times 4)!}{(4+1)!4!} = \frac{8!}{5!4!} = \frac{8\times 7\times 6\times 5!}{5!\times 4\times 3\times 2\times 1} = 14$$

Thus, number of structurally different possible binary trees with 4 nodes is 14.

29. Topic: Computer Networks

(c) Given, using public key cryptography, X adds a digital signature σ to a message M, encrypts $<M, \sigma>$ and sends it to Y, where it is decrypted.
For these operations the following sequence of keys is used:
Encryption: X's private key followed by Y's public key
Decryption: Y's private key followed by X's public key.
Hence option (c) is correct.

30. Topic: Computer Networks

(a) SHA-256 generates 256-bit message digest.
AES is an encryption standard
DES is also an encryption standard
MD5 is algorithm generates 128-bit message digest
Thus SHA-256 and MD5 are used to generate a message digest by the network security protocols.
Hence option (a) is correct.

31. Topic: Computer Networks

(d) IPv4 addressing format has 32 bits. In classic addresses, first 24 bits are used for Network ID and remaining 8 bits are used for identifying host ID. Now out of the 24 bits, three leading bits are fixed that is 110

$$\Rightarrow 24 - 3 = 21 \text{ bits}$$

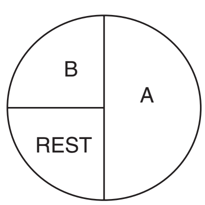
Thus, number of networks allowed under class C addresses is 2^{21}.

32. Topic: Computer Networks

(a) Given:

$$\text{Address allocation to } A = \frac{2^{12}}{2} = 2^{11}$$

$$\text{Address allocation to } B = \frac{2^{12}}{4} = 2^{10}$$



Now setting 12^{th} bit to 1 gives address of A

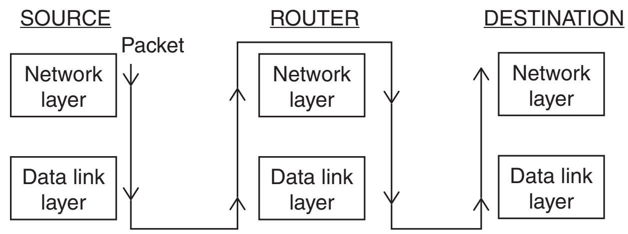
245.248.136.0/21

Now setting 12^{th} and 13^{th} bit to 0 the address of B is

245.248.128.0/22

33. Topic: Computer Networks

(d) Source is connected to destination through an intermediate router R:



Thus, packet visits network layer 3 times and data link layer 4 times.

34. Topic: Computer Networks

(c) TCP is reliable and UDP is not reliable still DNS which has to be reliable uses UDP because UDP is much faster and TCP is slow. DNS requests are generally very small and fit well within UDP segments and reliability can be added on application layer.

35. Topic: Computer Networks

(c) Sending an email from a mail client to mail server uses simple mail transfer protocol or SMTP. SMTP is an internet standard for electronic mail transmission.
Download email headers from mail box and retrieve mails from server to a cache requires Internet message access protocol or IMAP.
Checking e-mail through a web browser requires Hypertext transfer protocol secure or HTTPS. It is a protocol over which data is sent between your browser and the connected website.

36. Topic: Computer Networks

(b) Station A uses 32-byte packets to transmit messages to station B.
round trip time delay between A and B is 40 ms bottle neck bandwidth on the path A and B is 64 kbps
Now, maximum bits transmitted = 64 × 40 = 2560 and frame size = 32 byte = 32 × 8 bits = 256
Therefore,

$$\text{Optimal window size of } A = \frac{2560}{256} = 10$$

37. Topic: Computer Organization and Architecture

(b) Given:

$$\text{Cache memory} = 16 \text{ KB} = 2^{14} \text{ bytes}$$

$$\text{Word length} = 32 \text{ bits} = \frac{32}{8} = 2^2 \text{ bytes}$$

$$\text{Physical address space} = 4 \text{ GB}$$

Now,

Block offset
= number of words in block × number of bytes forward
= 8 × 4 = 32 GB

So,

$$\text{bits} = \log 32 = 5 \text{ bits}$$

Now,

$$\text{Set offset} = \frac{2^{14}}{32 \times 2} = 2^8 \Rightarrow \log 2^8 = 8 \text{ bits}$$

Also

$$\text{Tag bits} = 32 - (8 + 5) = 19 \text{ bits}$$

Therefore, number of bits in TAG field = 19
and number of bits in SET field = 8

38. Topic: Computer Organization and Architecture

(a) Given:

Direct mapped cache = 32 KB
Block size = 128 byte
Size of array $A = 512 \times 512$
Size of each element = 8 bytes

Now,

$$\text{Number of elements in 1 block} = \frac{128}{8} = 16$$

So, block 0 is $A[0][0]$ to $A[0][15]$
and block 1 is $A[0][16]$ to $A[0][31]$
for $i = 0$, $A[0][0]$ not in cache so there is a miss in j loop there is a miss after every 16 elements
Therefore,

$$\text{Total number of misses in } j \text{ loop} = \frac{512}{16} = 32$$

So, total number of misses in code segment is number of i iterations × number of misses in j loop

$$= 512 \times 32 = 16834$$

39. Topic: Digital Logic

(c) Given, computer with 32-bit word size uses 2's compliment to represent numbers. The range of integers that can be represented by this computer is -2^{31} to $2^{31} - 1$.

40. Topic: Digital Logic

(a) Given:

$$M = 1111\ 1010$$
$$N = 0000\ 1010$$

In decimal representation $M = -(0000\ 0110) = -6$
$N = 10$
Therefore,

$$\text{Product } M \times N = 10 \times (-6) = -60$$

In binary representation $M \times N = 11000100$

41. Topic: Computer Organization and Architecture

(d) Given:

A. The j+1[st] instructions uses the result of j[th] instruction as an operand – this will cause data hazard.
B. Conditional jump instruction – this will cause control hazard.
C. j[th] and j+1[st] instructions require ALU at the same time – this will case structural hazard.

Hence all of the above causes a hazard.

42. Topic: Computer Organization and Architecture

(a) In designing a computer's cache system, the cache block (or cache line) size is an important parameter. Smaller block size incurs lower cache miss penalty since during a memory access only a smaller part of nearby address is brought to cache that is spatial locality is reduced.

43. Topic: Computer Organization and Architecture

(d) Indexed addressing best reflects the source operand. In indexed addressing, address field stores starting address and indexed register stores a signed number. It is suitable for iterative operations, array implementation. Register R2 acts as index register and 20 is the base address.

44. Topic: Algorithms

(c) A sorting algorithm is stable if two objects with equal keys appear in the same order in sorted output as they appear in the input array to be sorted. Thus, a sorting technique is called stable if relative order of occurrence of non-distinct elements is maintained.

45. Topic: Programming and Data Structures

(d) Heap is a special case of balanced binary tree data structure where the root node key is compared with its children and arranged accordingly. If b is child of node a then

$$\text{key}(a) > \text{key}(b)$$

Time complexity of heap construction is $O(n)$.
Hash table construction with linear probing is a data structure that associates key with values. Time complexity of hash table construction is $O(n^2)$.
AVL tree construction – AVL tree is a self-balancing tree where the difference between heights of left and right subtree cannot be more than one for all nodes. Time complexity of AVL tree construction is $O(n \log n)$.

46. Topic: Programming and Data Structures

(b) Given:

Number of elements in row 1 = 1
Number of elements in row 2 = 2
Number of elements in row 3 = 3
Number of elements in row 4 = 4
Number of elements in row $i - 1 = i - 1$

Thus, total number of elements up to row $(i - 1)$ is

$$1+2+3+\cdots+i-1=\frac{i(i-1)}{2}$$

Since, i^{th} row contains j elements, index for the $(i, j)^{th}$ element is $j+\frac{i(i-1)}{2}$.

47. Topic: Programming and Data Structures

(c) Given:

Push 5	
Push 6	
Push 7	Pop 7
Push 8	Pop 8
Push 9	Pop 9
	Pop 6
	Pop 5

Therefore, correct option is (c)

48. Topic: Programming and Data Structures

(a) Quick sort uses divide and conquer technique. It selects a value called pivot and partitions the given array around the pivot. All the elements which are smaller than pivot is to the left and elements that are greater than pivot are to the right.

Here,

$$A = 1, 2, 3,\ldots n$$
$$B = n, n-1, n-2, \ldots 1$$

If $C1$ and $C2$ are number of comparisons made for A and B then $C1 > C2$

49. Topic: Programming and Data Structures

(d) A binary search tree is a rooted binary tree whose internal nodes store a key and each node have two distinguished sub-trees called left subtree and right subtree. Keys with value greater than parent node are stored in right tree and keys with value less than parent node are stored in left tree.

In option (d): 17, 77, 27, 66, 18, 43

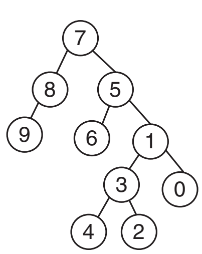
right subtree of node 27 contains 66, 18, 48 however 18 must be to the left sub tree of node 27 this is not possible.

50. Topic: Algorithms

(None) *ord*(A) is not defined and *ord*(x) is also not clearly defined. Solution to this question is not possible.

51. Topic: Programming and Data Structures

(d) We have:

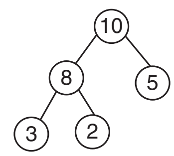


In inorder traversal left subtree is visited first and next root is visited and right subtree is visited in the end. Thus, resultant binary search tree is:

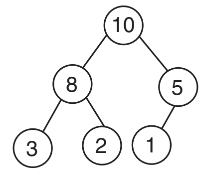
9876543210

52. Topic: Programming and Data Structures

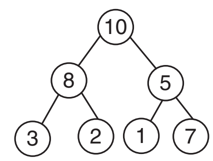
(d) We have:



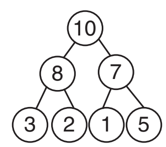
to insert new element 1:



to insert new element 7:



Now, since 5 < 7 then swapping 7 and 5. We have:



Then traversal gives: 10, 8, 7, 3, 2, 1, 5.

53. Topic: Programming and Data Structures

(c) A queue can be implemented using two stacks. All elements are pushed in stack 1 and in de-queue operation, if stack 2 is empty all elements are moved to stack 2 and then popped out of stack 2.

54. Topic: Programming and Data Structures

(b) In a strict binary tree each node can have either 0 children or 2 children. For L number of leaves, number of nodes is $2L - 1$

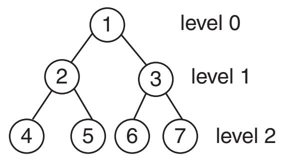
Given:

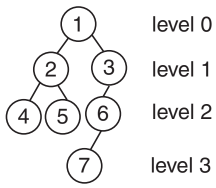
$$\text{Number of leaves} = 10$$
$$\text{Number of nodes} = 2 \times 10 - 1 = 19$$

Therefore, a strict binary tree with 10 leaves has exactly 19 nodes.

55. Topic: Programming and Data Structures

(b) AVL tree is a self-balancing tree where the difference between heights of left and right subtree cannot be more than one for all nodes





If the height of tree with single node is 0 then the maximum height of any AVL tree with 7 nodes can be 3.

56. Topic: Programming and Data Structures

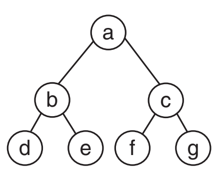
(a) A red-black tree is a balanced binary search tree where every node is colored red or black. If a node is red in color then both its children are black. Every leaf is a NIL node and is colored black. Every simple path from a node to a descendant leaf contains the same number of black nodes. Thus option (a) is correct for a red-black tree.

57. Topic: Programming and Data Structures

(c) In in-order traversal of tree, left subtree is traversed first then the root and finally right subtrees is traversed.
In pre-order traversal of tree, root is visited first, then the left subtree and right subtree is traversed at end.
In post order traversal of tree, left subtree is traversed first then the right subtree and root is visited in the end.
Given:

Inorder traversal is *d b e a f c g*
Preorder traversal is *a b d e c f g*

So, we have following tree:



Then post order traversal is *d e b f g c a*

58. Topic: Operating System

(b) FIFO or first in first out is a simple page replacement algorithm in which operating system keeps track of all pages in the memory in a queue with oldest page in front of queue. When replacement of a page is required, the page in front of queue is removed.
Here, statement *M* is true – this is called Belady's anomaly. When there are more pages, recently requested pages can remain at the bottom of queue longer.
Statement *N* is also true – Locality of reference occurs because of the style of programming. But *N* is not the reason of *M*.

59. Topic: Operating System

(b) In shortest remaining time first scheduling algorithm, the process with least time remaining to complete is executed. Thus, for currently executing processes, the process will run until they are completed or a new process which requires less amount of time is added. Given, 3 processes arrive at times 0, 2 & 6 which require 10, 20 & 30 time units respectively.
At $t = 0$, 1st process arrives. It is the only process available and time unit required = 10
At $t = 2$, 2nd process arrives but since process 1 requires less time therefore it continues to run.
At $t = 6$, 3rd process arrives but since process 1 has shortest remaining time among all therefore process 1 continues.
At $t = 10$, process 1 finished and process 2 is scheduled because it has shortest remaining time.
At $t = 30$, process 2 completes and process 3 is scheduled. Thus, two context switches are needed, process 1 to process 2 and process 2 to process 3.

60. Topic: Operating System

(d) Fork ( ) function call creates a new process called the child process. The child process runs concurrently with process which called the fork ( ) function, this process is called parent process. Following the fork ( ) system call both parent and child processes will execute next instructions following the fork ( ) system call. In the given segment of code, the number of processes doubles at each iteration except for the very first process which will be the parent process. Hence total number of child process created is $2^n - 1$.

61. Topic: Operating System

(a) Gang scheduling is scheduling algorithm for parallel system. It schedules related threads to run simultaneously on different processors.
Rate monotonic scheduling is priority assignment algorithm used in real time operating systems.
Fair share scheduling is a scheduling algorithm in which CPU usage is distributed among system equally.
Therefore,

Gang scheduling – thread scheduling
Rate monotonic scheduling –real time scheduling
Fare share scheduling – guaranteed scheduling

62. Topic: Operating System

(a) FIFO or first in first out is a simple page replacement algorithm in which operating system keeps track of all pages in the memory in a queue with oldest page in front of queue. When replacement of a page is required, the page in front of queue is removed.
Here access to 50 pages will cause 50 page faults when accessed in reverse, first four-page access will not cause page fault while others will cause page faults. Thus, total number of page faults is:

$$(50 + 50 - 4) = 96$$

63. Topic: Operating System

(d) Among the given statements – (d) is false. Kernel level scheduler schedules threads from all process it can distribute threads on different processors or cores. Thus, kernel level threads can utilize multiprocessor systems by splitting threads on different processors or cores.

64. Topic: Operating System

(a) Statement (a) is not true with respect to deadlock prevention and deadlock avoidance schemes. Deadlock prevention scheme handles deadlock by making sure that one of the four necessary conditions do not occur. In deadlock prevention, the request for a resource may not be granted even if the resulting state is safe.

65. Topic: Programming and Data Structures

(c) Essential features of object-oriented language is

1. Inheritance
2. Abstraction
3. Overloading
4. Polymorphism
5. Encapsulation

Hence option (c) is correct.

66. Topic: Programming and Data Structures

(d) The head is memory set aside for dynamic allocation. Heap allocation is needed for dynamic data structures like trees, linked list, etc.
Hence option (d) is correct.

67. Topic: Programming and Data Structures

(d) A code is loop invariant if a fragment of code inside a loop computes same value in all iterations. So option (a) is true.'
The sub expression $4 \times j$ is used twice that can be eliminated. So option (b) is true.
Operations that consume more time and space can be replaced by simple operations that produce same result $4 \times j$ can be replaced by $j << 2$.
So, option (c) is true.
Option (d) is false – there is no scope of dead code elimination in this code.

68. Topic: Operating System

(c) An activation record is created, every time when a subroutine is called.
Assembler uses location counter value to give address to each instruction.
Garbage collector uses reference count to clear memory whose reference count becomes zero.
A linker loader is a loader that can load several compiled codes and link them into single executable. It needs address relocation of the object codes.

activation record – subroutine call
location counter – assembler
reference count – garbage collection
address relocation – linking loader

69. Topic: Operating System

(b) Sequence of request to access the cylinder:

4, 34, 10, 7, 19, 73, 2, 15, 6, 20

head is currently at cylinder 50
time to move from one cylinder to adjacent one = 2 ms
nearest cylinder is 34. Then it continues to move to left reaching all cylinders requested to access up to 2 and then it finally moves to cylinder 73

$$\Rightarrow (50 - 2) + (73 - 2) = 119 \text{ cylinders moved}$$

Therefore, seek time = 119×2 ms = 238 ms.

70. Topic: Operating System

(c) Given:

P operation decrements semaphore value by 1.
V operation increments semaphore value by 1.
Initial value of semaphore = 7
Number of P operations = 20
Number of V operations = x
Final value of semaphore = 5

$$\Rightarrow 7 - 20 + x = 5 \Rightarrow -13 + x = 5 \Rightarrow x = 13 + 5$$
$$x = 18$$

71. Topic: Digital Logic

(b) Given, 32-bit adder is formed by cascading 4-bit CLA added. Thus, there will be 8 CLA adder cascaded.
In each cycle, all address will compute EXOR operations on its 4 bits. Also, can CLA requires 2 cycles to pass on carry out to next CLA.
So,

$$\text{Total gate delays} = 1 + 2 \times 8 + 1 = 18$$

72. Topic: Digital Logic

(c) Given, addition of two 2's compliment numbers

$$b_{n-1}\, b_{n-2} \dots b_0 \text{ and } a_{n-1}\, a_{n-2} \dots a_0$$

To add these binary numbers, a binary adder for adding two unsigned binary numbers is used.

$$\text{Sum} = c_{n-1}\, c_{n-2} \ldots c_0$$
$$\text{Carry out} = c_{\text{out}}$$
$$\text{Overflows condition} = c_{\text{out}} \oplus c_{n-1}$$

73. Topic: Programming and Data Structures

(c) We have:

for $x = 95$: fun (95)
function will return
fun (fun (95 + 11)) = fun (fun (106))
= fun (106 – 10) = fun (96)
= fun (fun (96 + 11)) = fun (fun (107))
= fun (97) = fun (fun (97 + 11)) = fun (fun (108))
= fun (98) = fun (fun (98 + 11)) = fun (fun (109))
= fun (99) = fun (fun (99 + 11)) = fun (fun (110))
= fun (100) = fun (fun (100 + 11) = fun (fun (111))
= fun (101) = 101 – 10 = 91

74. Topic: Programming and Data Structures

(a) Given:

$$\text{num} \gg 1 \equiv \frac{\text{num}}{2}$$
$$\text{Count } 1: \frac{435}{2} = 217$$
$$\text{Count } 2: \frac{217}{2} = 108$$
$$\text{Count } 3: \frac{108}{2} = 54$$
$$\text{Count } 4: \frac{54}{2} = 27$$
$$\text{Count } 5: \frac{27}{2} = 13$$
$$\text{Count } 6: \frac{13}{2} = 6$$
$$\text{Count } 7: \frac{6}{2} = 3$$
$$\text{Count } 8: \frac{3}{2} = 1$$
$$\text{Count } 9: \frac{1}{2} = 0$$

Thus, for fun (435) the value returned = 9

75. Topic: Digital Logic

(d) Given:

$$0 \times C0000000$$
$$= \underset{\text{sign}}{\underset{\downarrow}{1}}\ \underbrace{100\ 00000}_{\substack{\downarrow \\ \text{exponent}}}\ \underbrace{000\ 0000\ 0000\ 0000\ 0000\ 0000}_{\substack{\downarrow \\ \text{mantissa}}}$$

Sign bit is 1 $\Rightarrow$ negative number
Exponent is 10000000 $\Rightarrow$ 128

This is in normalized form therefore original exponent = 128 – 127 = 1

$$\text{Mantissa} = 0$$
$$\text{Number} = (-1)^{\text{sign}} \times 1\cdot(\text{Mantissa})\, 2^{\text{bias exponent-bias}}$$
$$= (-1)^1 \times 1.0 \times 2^{128-127} = -1 \times 2^1 = -2$$

76. Topic: Digital Logic

(b) XOR gate can be used to make a NOT gate $(a \oplus 1 = \bar{a})$
NOT and AND gates are functionally complete.
Thus, AND gates, XOR gates and 1 is sufficient to implement any arbitrary Boolean function.

77. Topic: Algorithms

(a) Condition coverage is a mixed type of white box testing. It tests each Boolean expression with values TRUE and FALSE.
Equivalence class partitioning-divides input test data into each partition at least once of equivalent data from which test cases can be derived.
It is black box testing.

Volume testing: it checks system performance with increasing volumes of data in the database.

Beta testing: it is done at customer site in order to validate usability and functionality. It is a kind of system testing.

78. Topic: Databases

(b) Decision tree is a graph that uses branching method to illustrate every possible outcome of a decision. It comes under supervised learning and is mostly used than regression. Thus, the preferred option which can classify the given scenario is decision tree.

79. Topic: Databases

(c) Snow flake schema is sued to normalize the dimensions tables by removing low cardinality attributes and forming separate tables.
Thus option (c) is correct.

80. Topic: Programming and Data Structures

(d) Given:

$$c = \underset{0}{I}\ \underset{1}{C}\ \underset{2}{R}\ \underset{3}{B}\ \underset{4}{C}\ \underset{5}{S}\ \underset{6}{I}\ \underset{7}{T}\ \underset{8}{1}\ \underset{9}{7}$$
$$2[p] = p[2] = R$$
$$6[p] = p[6] = I$$
$$R - I = 9$$

So,

$$c + 2[p] - 6[p] - 1$$
$$\Rightarrow c + 9 - I = c + 8$$

Therefore, output = 17

Indian Space Research Organisation
Computer Science

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QUESTION PAPER

1. Consider the following program

```
main( )
{
   int x = 1;
   printf("%d", (*char(char*)&x));
}
```

Assuming required header files are included and if the machine in which this program is executed is little-endian, then the output will be

(a) 0 (b) 99999999
(c) 1 (d) unpredictable

2. Consider the following declaration:

```
Structaddr {
            char city[10];
            char street[30];
            int pin;
            };
struct {
        char name[30];
        Int gender;
        struct addr locate;
        } person, *kd = &person;
```

Then `*(kd -> name+2)` can be used instead of

(a) `person.name+2`
(b) `kd -> (name+2)`
(c) `*((*kd).name+2)`
(d) either (a) or (b), but not (c)

3. If a variable can take only integral values from 0 to n, where n is an integer, then the variable can be represented as a bit-field whose width is (the log in the answers are to the base 2, and [log n] means the floor of log n)

(a) $[\log(n)] + 1$ bits
(b) $[\log (n - 1)] + 1$ bits
(c) $[\log (n + 1)] + 1$ bits
(d) None of the above

4. Consider the following C program

```
main()
      {
      fork(); fork(); printf("yes");
      }
```

If we execute this core segment, how many times the string *yes* will be printed?

(a) Only once (b) 2 times
(c) 4 times (d) 8 times

5. Consider the following table in a relational database

Last Name	Rank	Room	Shift
Smith	Manager	234	Morning
Jones	Custodian	33	Afternoon
Smith	Custodian	33	Evening
Doe	Clerical	222	Morning

According to the data shown in the table, which of the following could be a candidate key of the table?

(a) {Last Name} (b) {Room}
(c) {Shift} (d) {Room, Shift}

6. A data driven machine is one that executes an instruction if the needed data is available. The physical ordering of the code listing does not dictate the course of execution. Consider the following pseudo-code

(A) Multiply E by 0.5 to get F
(B) Add A and B to get E
(C) Add B with 0.5 to get D
(D) Add E and F to get G
(E) Add A with 10.5 to get C

Assume A, B, C are already assigned values and the desired output is G. Which of the following sequence of execution is valid?

(a) B, C, D, A, E
(b) C, B, E, A, D
(c) A, B, C, D, E
(d) E, D, C, B, A

7. Assume A and B are non-zero positive integers. The following code segment

```
while (A!=B){
   If (A>B)
         A -= B;
   else
         B -= A;
}
cout<<A;   //printing the value of A
```

(a) Computes the LCM of two numbers
(b) Divides the larger number by the smaller number
(c) Computes the GCD of two numbers
(d) Finds the smaller of two numbers

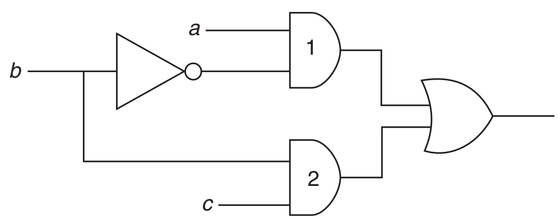
8. A language with string manipulation facilities uses the following operations.

`head(s)` – returns the first character of the string *s*
`tails(s)` – returns all but the first character of the string *s*
`concat(s1, s2)` – concatenates string s1 with s2.

The output of ***concat(head(s), head(tail(tail(s))))***, where *s* is ***acbc*** is

(a) *ab* (b) *ba*
(c) *ac* (d) *aa*

9. In the diagram below, the inverter (NOT gate) and the AND-gates labeled 1 and 2 have delays of 9, 10 and 12 nanoseconds(ns), respectively. Wire delays are negligible. For certain values of *a* and *c*, together with certain transition of *b*, a glitch (spurious output) is generated for a short time, after which the output assumes its correct value. The duration of the glitch is



(a) 7 ns (b) 9 ns
(c) 11 ns (d) 13 ns

10. Which of the following comparisons between static and dynamic type checking is incorrect?

(a) Dynamic type checking slows down the execution
(b) Dynamic type checking offers more flexibility to the programmers
(c) In contrast to static type checking, dynamic type checking may cause failure in runtime due to type errors
(d) Unlike static type checking, dynamic type checking is done during compilation

11. ____________ can detect burst error of length less than or equal to degree of the polynomial and detects burst errors that affect odd number of bits.

(a) Hamming Code (b) CRC
(c) VRC (d) None of the above

12. An array *A* consists of *n* integers in locations *A*[0], *A*[1]…..*A*[*n* – 1]. It is required to shift the elements of the array cyclically to the left by *k* places, where 1 <= *k* <= (*n* – 1). An incomplete algorithm for doing this in linear time, without using another array is given below. Complete the algorithm by filling in the blanks. Assume all the variables are suitably declared.

```
min = n; i = 0;
while (__________________) {
   temp = A[i]; j = i;
   while (________________) {
         A[j] = _______________;
         j = (j + k) mod n;
         If (j < min) then
               min = j;
}
A[(n+i-k)modn]=________________________;
i = _______________;
}
```

(a) `i>min; j! = (n+i) mod n; A[j + k]; temp; i+1;`
(b) `i<min; j! = (n + i) mod n; A[j + k]; temp; i+2;`
(c) `i>min; j! = (n+i+k) mod n; A[j+k]; temp; i+1;`
(d) `i<min; j! = (n+i-k) mod n; A[(j+k) mod n]; temp; i+1;`

13. The difference between a named pipe and a regular file in Unix is that

(a) Unlike a regular file, named pipe is a special file
(b) The data in a pipe is transient, unlike the content of a regular file
(c) Pipes forbid random accessing, while regular files do allow this
(d) All of the above

14. A class of 30 students occupy a classroom containing 5 rows of seats, with 8 seats in each row. If the students seat themselves at random, the probability that the sixth seat in the fifth row will be empty is

(a) $\frac{1}{5}$ (b) $\frac{1}{3}$
(c) $\frac{1}{4}$ (d) $\frac{2}{5}$

15. The domain of the function log(log sin(x)) is

(a) $0 < x < \pi$
(b) $2n\pi < x < (2n + 1)\pi$, for n in N
(c) Empty set
(d) None of the above

16. The following paradigm can be used to find the solution of the problem in minimum time:

Given a set of non-negative integer, and a value K, determine if there is a subset of the given set with sum equal to K.

(a) Divide and Conquer
(b) Dynamic Programming
(c) Greedy Algorithm
(d) Branch and Bound

17. (G, *) is an abelian group. Then

(a) $x = x^{-1}$, for any x belonging to G
(b) $x = x^2$, for any x belonging to G
(c) $(x * y)^2 = x^2 * y^2$, for any x, y belonging to G
(d) G is of finite order

18. Consider the following C code segment:

```
#include <stdio.h>
main()
{
int i, j, x;
scanf("%d", &x)
i = 1, j = 1;
while (i < 10) {
   j = j * i;
   i = i + 1;
   if (i == x) break;
   }
}
```

For the program fragment above, which of the following statements about the variables i and j must be true after execution of this program? [!(exclamation) sign denotes factorial in the answer]

(a) $(j = (x-1)!) \wedge (i \geq x)$
(b) $(j = 9!) \wedge (i = 10)$
(c) $((j = 10!) \wedge (i = 10)) \vee ((j = (x-1)!) \wedge (i = x))$
(d) $((j = 9!) \wedge (i \geq 10)) \vee ((j = (x-1)!) \wedge (i = x))$

19. Given $\sqrt{224_r} = 13_r$ the value of radix r is

(a) 10 (b) 8
(c) 6 (d) 5

20. Determine the number of page faults when references to pages occur in the order – 1, 2, 4, 5, 2, 1, 2, 4. Assume that the main memory can accommodate 3 pages and the main memory already has the pages 1 and 2, with page 1 having brought earlier than page 2. (Assume LRU i.e. Least-Recently-Used algorithm is applied.)

(a) 3 (b) 4
(c) 5 (d) None of the above

21. Consider a system having m resources of the same type. These resources are shared by 3 processes A, B, C which have peak time demands of 3, 4, 6 respectively. The minimum value of m that ensures that deadlock will never occur is

(a) 11 (b) 12
(c) 13 (d) 14

22. A computer has 1000 K of main memory. The jobs arrive and finish in the following sequence.

Job 1 requiring 200 K arrives
Job 2 requiring 350 K arrives
Job 3 requiring 300 K arrives
Job 1 finishes
Job 4 requiring 120 K arrives
Job 5 requiring 150 K arrives
Job 6 requiring 80 K arrives

Among best fit and first fit, which performs better for this sequence?

(a) First fit (b) Best fit
(c) Both perform the same (d) None of the above

23. Disk requests come to a disk driver for cylinders in the order 10, 22, 20, 2 40, 6 and 38, at a time when the disk drive is reading from cylinder 20. The seek time is 6 ms/cylinder. The total seek time, if the disk arm scheduling algorithms is first-come-first-served is

(a) 360 ms (b) 850 ms
(c) 900 ms (d) None of the above

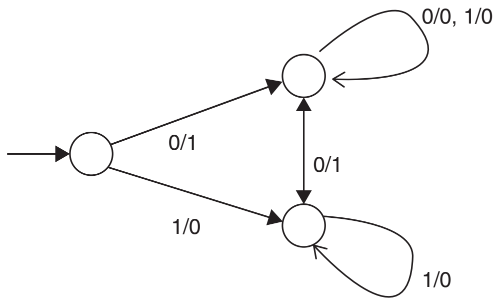
24. Choose the correct statement –

(a) $A = \{a^n b^n \mid n = 1, 2, 3, \ldots\}$ is a regular language
(b) The set B, consisting of all strings made up of only a's and b's having equal number of a's and b's defines a regular language
(c) $L(A^* B) \cap B$ gives the set A
(d) None of the above

25. CFG (Context Free Grammar) is not closed under

(a) Union (b) Complementation
(c) Kleene Star (d) Product

26. The FSM (Finite State Machine) pictured in the figure below



(a) Complements a given bit pattern
(b) Finds 2's complement of a given bit pattern
(c) Increments a given bit pattern by 1
(d) Changes the sign bit

27. A CFG (Context Free Grammar) is said to be in Chomsky Normal Form (CNF), if all the productions are of the form *A* -> *BC* or *A* -> *a*. Let *G* be a CFG in CNF. To derive a string of terminals of length *x*, the number of products to be used is

(a) $2x - 1$ (b) $2x$
(c) $2x + 1$ (d) 2^x

28. Incremental-Compiler is a compiler

(a) which is written in a language that is different from the source language
(b) compiles the whole source code to generate object code afresh
(c) compiles only those portion of source code that have been modified
(d) that runs on one machine but produces object code for another machine

29. DU (Definition-Use)-chains in compiler design

(a) consist of a definition of a variable and all its uses, reachable from that definition
(b) are created using a form of static code analysis
(c) are prerequisite for many compiler optimization including constant propagation and common sub-expression elimination
(d) All of the above

30. Which of the following comment about peep-hole optimization is true?

(a) It is applied to small part of the code and applied repeatedly
(b) It can be used to optimize intermediate code
(c) It can be applied to a portion of the code that is not contiguous
(d) It is applied in symbol table to optimize the memory requirements.

31. A byte addressable computer has a memory capacity of 2^m KB (kbytes) and can perform 2^n operations. An instruction involving 3 operands and one operator needs maximum of

(a) $3m$ bits (b) $3m + n$ bits
(c) $m + n$ bits (d) none of the above

32. A computer uses ternary system instead of the traditional binary system. An *n* bit string in the binary system will occupy

(a) $3 + n$ ternary digits
(b) $2n/3$ ternary digits
(c) $n(\log_2 3)$ ternary digits
(d) $n(\log_3 2)$ ternary digits

33. Which of the following is application of breath-first search on the graph?

(a) Finding diameter of the graph
(b) Finding bipartite graph
(c) Both (a) and (b)
(d) None of the above

34. Microprogram is

(a) the name of a source program in micro computers
(b) set of microinstructions that defines the individual operations in response to a machine-language
(c) a primitive form of macros used in assembly language programming
(d) a very small segment of machine code

35. Given two sorted lists of size *m* and *n* respectively. The number of comparisons needed the worst case by the merge sort algorithm will be

(a) $m \times n$ (b) maximum of *m* and *n*
(c) minimum of *m* and *n* (d) $m + n - 1$

36. A has table with 10 buckets with one slot per bucket is depicted here. The symbols, *S*1 to *S*7 are initially entered using a hashing function with linear probing. The maximum number of comparisons needed in searching an item that is not present is

0	*S*7
1	*S*1
2	
3	*S*4
4	*S*2
5	
6	*S*5
7	
8	*S*6
9	*S*3

(a) 4 (b) 5
(c) 6 (d) 3

37. The running time of an algorithm is given by

$$T(n) = T(n-1) + T(n-2) - T(n-3), \text{ if } n > 3$$
$$= n, \text{ otherwise}$$

Then what should be the relation between $T(1)$, $T(2)$ and $T(3)$, so that the order of the algorithm is constant?

(a) $T(1) = T(2) = T(3)$
(b) $T(1) + T(3) = 2*T(2)$
(c) $T(1) - T(3) = T(2)$
(d) $T(1) + T(2) = T(3)$

38. The number of edges in a regular graph of degree d and n vertices is

(a) maximum of n and d (b) $n + d$
(c) nd (d) $nd/2$

39. Perform window to viewport transformation for the point (20, 15). Assume that ($X_{\text{w min}}$, $Y_{\text{w min}}$) is (0, 0); ($X_{\text{w max}}$, $Y_{\text{w max}}$) is (100, 100); ($X_{\text{v min}}$, $Y_{\text{v min}}$) is (5, 5); ($X_{\text{v max}}$, $Y_{\text{v max}}$) is (20, 20). The value of x and y in the viewport is

(a) $x = 4, y = 4$ (b) $x = 3, y = 3$
(c) $x = 8, y = 7.25$ (d) $x = 3, y = 4$

40. Given relations $R(w, x)$ and $S(y, z)$, the result of

```
SELECT DISTINCT w, x
FROM R, S
```

Is guaranteed to be same as R, if

(a) R has no duplicates and S in non-empty
(b) R and S have no duplicates
(c) S has no duplicates and R is non-empty
(d) R and S have the same number of tuples

41. For a database relation $R(a, b, c, d)$ where the domains of a, b, c and d include only atomic values, only the following functional dependencies and those that can be inferred from them hold

```
a -> c
b -> d
```

The relation is in

(a) First normal form but not in second normal form
(b) Second normal form but not in third normal form
(c) Third normal form
(d) None of the above

42. Consider the set of relations given below and the SQL query that follows:

```
Students : (Roll_number, Name, Date_
of_birth)
Courses : (Course_number, Course_
name, Instructor)
Grades : (Roll_number, Course_number,
Grade)
SELECT DISTINCT Name
FROM Students, Courses, Grades
WHERE Students.Roll_number = Grades.
Roll_number
  AND Courses.Instructor = Sriram
  AND Courses.Course_number = Grades.
    Course_number
  AND Grades.Grade = A
```

Which of the following sets is computed by the above query?

(a) Names of Students who have got an A grade in all courses taught by Sriram
(b) Names of Students who have got an A grade in all courses
(c) Names of Students who have got an A grade in at least one of the courses taught by Sriram
(d) None of the above

43. Consider the following C++ program

```
int a (int m)
{return ++m;}
int b(int &m)
{return ++m;}
intc(char &m)
{return ++m;}
void main()
{
   int p = 0, q = 0, r = 0;
   p += a(b(p));
   q += b(a(q));
   r += a(c(r));
   cout <<p << q << r;
}
```

Assuming the required header files are already included, the above program

(a) results in compilation error
(b) print 123
(c) print 111
(d) print 322

44. Station A uses 32 byte packets to transmit messages to Station B using a sliding window protocol. The round trip delay between A and B is 80 ms and the bottleneck bandwidth on the path between A and B is 128 kbps. What is the optimal window size that A should use?

(a) 20 (b) 40
(c) 160 (d) 320

45. Assuming that for a given network layer implementation, connection establishment overhead is 100 bytes and disconnection overhead is 28 bytes. What would be the minimum size of the packet the transport layer needs to keep up, if it wishes to implement a datagram service above the network layer and needs to keep it overhead to a maintain of 12.5%. (Ignore transport layer overhead)

(a) 512 bytes (b) 768 bytes
(c) 1152 bytes (d) 1024 bytes

46. In cryptography, the following uses transposition ciphers and the keyword is LAYER. Encrypt the following message. (Spaces are omitted during encryption)

WELCOME TO NETWORK SECURITY!

(a) WMEKREETSILTWETCOOCYONRU!
(b) EETSICOOCYWMEKRONRU!LTWET
(c) LTWETONRU!WMEKRCOOCYEETSI
(d) ONRU!COOCYLTWETEETSIWMEKR

47. In a particular program, it is found that 1% of the code accounts for 50% of the execution time. To code a program is C++, it takes 100 man-days. Coding in assembly language is 10 times harder than coding in C++, but runs 5 times faster. Converting an existing C++ program into an assembly language program is 4 times faster.

To completely write the program in C++ and rewrite the 1% code in assembly language, if a project team needs 13 days, the team consists of

(a) 13 programmers
(b) 10 programmers
(c) 8 programmers
(d) 100/13 programmers

48. In unit testing of a module, it is found that for a set of test data, at the maximum 90% of the code alone were tested with the probability of success 0.9. The reliability of the module is

(a) Greater than 0.9 (b) Equal to 0.9
(c) At most 0.81 (d) At least 0.81

49. In a file which contains 1 million records and the order of the tree is 100, then what is the maximum number of nodes to be accessed if B+ tree index is used?

(a) 5 (b) 4
(c) 3 (d) 10

50. A particular disk unit uses a bit string to record the occupancy or vacancy of its tracks, with 0 denoting vacant and 1 for occupied. A 32-bit segment of this string has hexadecimal value D4FE2003. The percentage of occupied tracks for the corresponding part of the disk, to the nearest percentage is

(a) 12 (b) 25
(c) 38 (d) 44

51. Which of the following is dense index?

(a) Primary index
(b) Clusters index
(c) Secondary index
(d) Secondary non key index

52. In E-R model, Y is the dominant entity and X is subordinate entity

(a) If X is deleted, then Y is also deleted
(b) If Y is deleted, then X is also deleted
(c) If Y is deleted, then X is not deleted
(d) None of the above

53. Immunity of the external schemes (or application programs) to changes in the conceptual schema is referred to as:

(a) Physical Data Independence
(b) Logical Data Independence
(c) Both (a) and (b)
(d) None of the above

54. The set of attributes X will be fully functionally dependent on the set of attributes Y if the following conditions are satisfied.

(a) X is functionally dependent on Y
(b) X is not functionally dependent on any subset of Y
(c) Both (a) and (b)
(d) None of these

55. Let us assume that transaction $T1$ has arrived before transaction $T2$. Consider the schedule

$$S = r1(A); r2(B); w2(A); w1(B)$$

Which of the following is true?

(a) Allowed under basic timestamp protocol.
(b) Not allowed under basic timestamp protocols because $T1$ is rolled back.
(c) Not allowed under basic timestamp protocols because T2 is rolled back.
(d) None of these

56. The time complexity of computing the transitive closure of binary relation on a set of n elements is known to be

(a) $O(n)$
(b) $O(n*\log(n))$
(c) $O(n^{\frac{3}{2}})$
(d) $O(n^3)$

57. Given a binary-max heap. The elements are stored in an arrays as 25, 14, 16, 13, 10, 8, 12. What is the content of the array after two delete operations?

(a) 14, 13, 8, 12, 10 (b) 14, 12, 13, 10, 8
(c) 14, 13, 12, 8, 10 (d) 14, 13, 12, 10, 8

58. The Functions Point (FP) metric is

(a) Calculated from user requirement
(b) Calculated from lines of code
(c) Calculated from software complexity assessment
(d) None of the above

59. The lower degree of cohesion is kind of

(a) Logical Cohesion
(b) Coincidental Cohesion
(c) Procedural Cohesion
(d) Communicational Cohesion

60. What is the output of the following program?

```
main () {
int x = 2, y = 5;
if (x < y) return (x = x + y);
   else printf("z1");
printf("z2");
}
```

(a) z2 (b) z1z2
(c) Compilation error (d) None of these

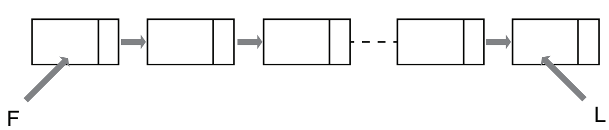
61. The operating system of a computer may periodically collect all the free memory space to form contiguous block of free space. This is called

(a) Concatenation
(b) Garbage collection
(c) Collision
(d) Dynamic memory allocation

62. Any set of Boolean operators that is sufficient to represent all Boolean expressions is said to be complete. Which of the following is not complete?

(a) {AND, OR} (b) {AND, NOT}
(c) {NOT, OR} (d) {NOR}

63. Consider a singly linked list of the form where F is a pointer to the first element in the linked list and L is the pointer to the last element in the list. The time of which of the following operations depends on the length of the list?



(a) Delete the last element of the list
(b) Delete the first element of the list
(c) Add an element after the last element of the list
(d) Interchange the first two elements of the list

64. A particular BNF definition for a "word" is given by the following rules.

```
<word>       : : = <letter> | <letter> <charpair> | <letter><intpair>
<charpair>   : : = <letter><letter> | <charpair><letter><letter>
<intpair>    : : = <integer><integer> | <intpair><integer><integer>
<letter>     : : = a | b | c |.... | y | z
<integer>    : : = 0 | 1 | 2 |.... | 9
```

Which of the following lexical entries can be derived from <word>?

I. pick II. picks III. c44

(a) I, II and III (b) I and II only
(c) I and III only (d) II and III only

65. Of the following, which best characterizes computers that use memory-mapped I/O?

(a) The computer provides special instructions for manipulating I/O ports
(b) I/O ports are placed at addresses on the bus and are accessed just like other memory locations
(c) To perform I/O operations, it is sufficient to place the data in an address register and call channel to perform the operation
(d) I/O can be performed only when memory management hardware is turned on

66. Of the following sorting algorithms, which has a running time that is least dependent on the initial ordering of the input?

(a) Merge Sort (b) Insertion Sort
(c) Selection Sort (d) Quick Sort

67. Processes $P1$ and $P2$ have a producer-consumer relationship, communicating by the use of a set of shared buffers.

```
P1: repeat
     Obtain an empty buffer
     Fill it
     Return a full buffer
    forever
P2: repeat
     Obtain a full buffer
     Empty it
     Return an empty buffer
    forever
```

Increasing the number of buffers is likely to do which of the following?

I. Increase the rate at which requests are satisfied (throughput)
II. Decrease the likelihood of deadlock

III. Increase the ease of achieving a correct implementation

(a) III only (b) II only
(c) I only (d) II and III only

68. In multi-programmed systems, it is advantageous if some programs such as editors and compilers can be shared by several users.

Which of the following must be true of multi-programmed systems in order that a single copy of a program can be shared by several users?

I. The program is a macro
II. The program is recursive
III. The program is reentrant

(a) I only (b) II only
(c) III only (d) I, II and III

69. Let *P* be a procedure that for some inputs calls itself (i.e. is recursive). If *P* is guaranteed to terminate, which of the following statement(s) must be true?

I. *P* has a local variable
II. *P* has an execution path where it does not call itself
III. *P* either refers to a global variable or has at least one parameter

(a) I only (b) II only
(c) III only (d) II and III only

70. Consider the following C program

```
#include <stdio.h>
main()
{
float sum = 0.0, j = 1.0, i = 2.0;
while (i/j > 0.001) {
   j = j + 1;
   sum = sum + i/j;
   printf("%f \n", sum);
}
}
```

How many lines of output does this program produce?

(a) 0 – 9 lines of output
(b) 10 – 19 lines out output
(c) 20 – 29 lines of output
(d) More than 29 lines of output

71. A particular parallel program computation requires 100 s when executed on a single processor. If 40% of this computation is inherently sequential (i.e. will not benefit from additional processors), then theoretically best ossible elapsed times of this program running with 2 and 4 processors, respectively, are

(a) 20 s and 10 s (b) 30 s and 15 s
(c) 50 s and 25 s (d) 70 s and 55 s

72. Consider the following C code segment

```
int f(int x)
{
   if (x < 1) return 1;
   else return (f(x – 1) + g(x));
}
int g (int x)
{
   if (x < 2) return 2;
   else return (f(x – 1) + g(x/2));
}
```

Of the following, which best describes the growth of *f*(*x*) as a function of *x*?

(a) Linear (b) Exponential
(c) Quadratic (d) Cubic

73. For a multi-processor architecture, In which protocol a write transaction is forwarded to only those processors that are known to possess a copy of newly altered cache line?

(a) Snoopy bus protocol
(b) Cache coherency protocol
(c) Directory based protocol
(d) None of the above

74. Avalanche effect in cryptography

(a) Is desirable property of cryptographic algorithm
(b) Is undesirable property of cryptographic algorithm
(c) Has no effect on encryption algorithm
(d) None of the above

75. In neural network, the network capacity is defined as

(a) The traffic carry capacity of the network
(b) The total number of nodes in the network
(c) The number of patterns that can be stored and recalled in a network
(d) None of the above

76. Cloaking is a search engine optimization (SEO) technique. During cloaking

(a) Content presented to search engine spider is different from that presented to user's browser
(b) Content present to search engine spider and browser is same
(c) Contents of user's requested website are changed
(d) None of the above

77. What is one advantage of setting up a DMZ (Demilitarized Zone) with two firewalls?

(a) You can control where traffic goes in the three networks
(b) You can do stateful packet filtering
(c) You can do load balancing
(d) Improve network performance

78. Which one of the following algorithm is not used in asymmetric key cryptography?

(a) RSA Algorithm
(b) Diffie-Hellman Algorithm
(c) Electronic Code Book Algorithm
(d) None of the above

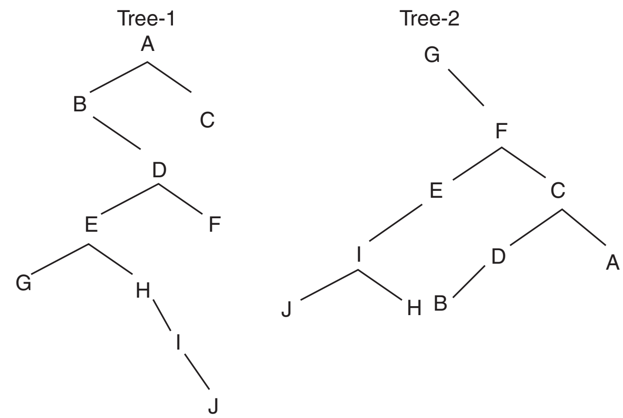
79. A doubly linked list is declared as

```
struct Node {
   int Value;
   struct Node *Fwd;
   struct Node *Bwd;
};
```

Where *Fwd* and *Bwd* represent forward and backward link to the adjacent elements of the list. Which of the following segments of code deletes the node pointed to by *X* from the doubly linked list, if it is assumed that *o* points to neither the first nor the last node of the list?

(a) `X -> Bwd -> Fwd = X -> Fwd; X -> Fwd -> Bwd = X -> Bwd;`
(b) `X -> Bwd.Fwd = X -> Fwd; X.Fwd -> Bwd = X -> Bwd;`
(c) `X.Bwd -> Fwd = X.Bwd; X -> Fwd.Bwd = X.Bwd;`
(d) `X -> Bwd -> Fwd = X -> Bwd; X -> Fwd -> Bwd = X -> Fwd;`

80. If Tree-1 and Tree-2 are the trees indicated below:



Which traversals of Tree-1 and Tree-2, respectively, will produce the same sequence?

(a) Preorder, postorder
(b) Postorder, inorder
(c) Postorder, preorder
(d) Inorder, preorder

ANSWER KEY

1. (c)	**9.** (a)	**17.** (c)	**25.** (b)	**33.** (c)	**41.** (a)	**49.** (b)	**57.** (c)	**65.** (b)	**73.** (c)
2. (c)	**10.** (d)	**18.** (d)	**26.** (c)	**34.** (b)	**42.** (c)	**50.** (d)	**58.** (c)	**66.** (a, c)	**74.** (a)
3. (a)	**11.** (b)	**19.** (d)	**27.** (a)	**35.** (d)	**43.** (a)	**51.** (c)	**59.** (b)	**67.** (c)	**75.** (c)
4. (c)	**12.** (d)	**20.** (b)	**28.** (c)	**36.** (b)	**44.** (b)	**52.** (b)	**60.** (d)	**68.** (c)	**76.** (a)
5. (d)	**13.** (d)	**21.** (a)	**29.** (d)	**37.** (a)	**45.** (d)	**53.** (b)	**61.** (b)	**69.** (d)	**77.** (a)
6. (b)	**14.** (c)	**22.** (a)	**30.** (a)	**38.** (d)	**46.** (b)	**54.** (c)	**62.** (a)	**70.** (b)	**78.** (c)
7. (c)	**15.** (c)	**23.** (d)	**31.** (d)	**39.** (c)	**47.** (c)	**55.** (b)	**63.** (a)	**71.** (d)	**79.** (a)
8. (a)	**16.** (b)	**24.** (d)	**32.** (d)	**40.** (a)	**48.** (c)	**56.** (d)	**64.** (d)	**72.** (b)	**80.** (*)

*** None of the options are correct.**

ANSWERS WITH EXPLANATION

1. Topic: Programming and Data Structures

(c) The little-endian describes the order in which sequence of bytes are stored. In this, least significant value in the sequence (little end) is stored first.
The output of the following program will be 1.

2. Topic: Programming and Data Structures

(c) Given declaration gives us: `*(kd → name + 2)` points to the third character of array 'name[ ]'.
Option (a): `person.name + 2`, points to address of the location.
Option (b): `kd → (name + 2)`, points to address of the location.
Option (c): `*((*kd).name + 2)`, points to the third character of array 'name[ ]'.
Therefore, option (c) `*((*kd).name + 2)` can be used instead of `*(kd → name + 2)`.

3. Topic: Programming and Data Structures

(a) Given, variable can take integral values from 0 to n where n is an integer.
The width of the bit field must be less than or equal to the bit width of the specified type.
Since log 0 is infinity, so the variable is taken from interval $(0, n]$. Also for a number of form 2^n has $(n + 1)$ bits.
Therefore, any number taken from interval $(0, n]$ should have $[\log n] + 1$ bits.

4. Topic: Programming and Data Structures

(c) Fork function creates a new process called the child process of the caller. After the creation of child process, both the caller and child process execute next instruction following the fork ( ).
Here in main ( ) function – when fork ( ) is first time called, a child process is created.
When second fork ( ) function is called – the original caller will create a child process and the child process created during first fork ( ) call will also create a child process. Therefore, after second fork ( ) call 4 process will be running and thus string '*yes*' will be printed 4 times.

5. Topic: Databases

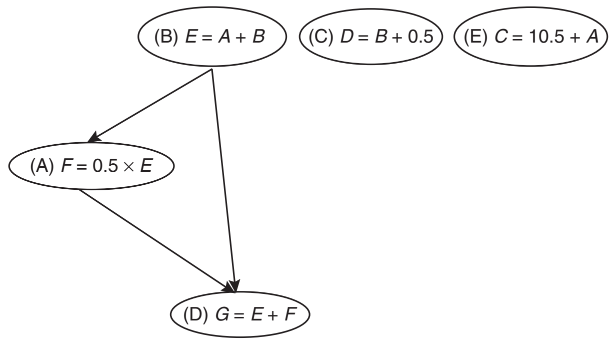
(d) A candidate key of relational database is the minimal set of attribute which can uniquely identify a tuple. The value of candidate key is unique and non-null for every tuple.
{Last Name} cannot be the candidate key because Smith is a repeated value. Therefore, the candidate key is not unique. {Room} is also having a repeated value, therefore, it cannot be the candidate key.
{Shift} also has a repeated value. Therefore, it cannot be the candidate key.
Therefore, only {Room, Shift} is the candidate key.

6. Topic: Databases

(b) Given pseudo-code instructions are:

(A) $F = 0.5 \times E$ (B) $E = A + B$
(C) $D = B + 0.5$ (D) $G = E + F$
(E) $C = 10.5 + A$

The data dependencies of instructions are as follows:



So, instruction (A) cannot be executed before (B) and instruction (D) cannot be executed before (B) and (A). Therefore, valid sequence of execution is C, B, E, A, D.

7. Topic: Programming and Data Structures

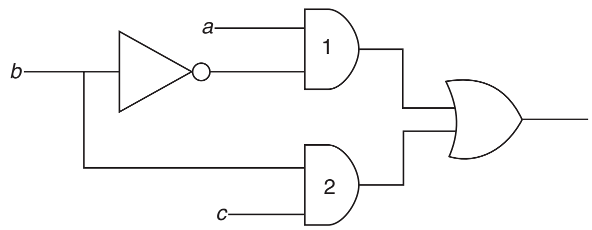
(c) The given code segment is Euclidean algorithm. The code segment subtracts smaller number from larger until both are not equal. This gives GCD of the two numbers.

8. Topic: Programming and Data Structures

(a) Given: `head(s)` – returns the first character of string s.
`tail(s)` – returns all but the first character of the string s.
`concat(s1, s2)` – concatenates string $s1$ with $s2$.
$s = acbc$.
Now, `tail(s)` $= cbc$
`tail(tail(s))` $= bc$
`head(tail(tail(s)))` $= b$
`head(s)` $= a$
`concat (head(s), head(tail(tail(s)))` $= ab$

9. Topic: Digital Logic

(a) Given: Delay of inverter = 9 ns
Delay of AND gate 1 = 10 ns
Delay of AND gate 2 = 12 ns



Output of AND gate 1 will be available to input of OR gate after 9 + 10 = 19 ns
Output of AND gate 2 will be available to input of OR gate after 12 ns
Duration of glitch = 19 – 12 = 7 ns

10. Topic: Programming and Data Structures

(d) Option (d) is incorrect. Static type checking is done during compilation, while dynamic type checking is done during runtime.

11. Topic: Computer Networks

(b) CRC can detect burst error of length less than or equal to degree of the polynomials and detect burnt errors that affect odd number of bits.

12. Topic: Programming and Data Structures

(d) The complete algorithm is

```
min = n, i = 0;
while (i < min) {
   temp = A[i]; j = 1;
   while (j! = (n + i - k) mod n){
          A[j] = A[j + k] mod n];
          j = (j + k) mod n;
          if (j < min) then
          min = j;
          }
   A[(n + i - k) mod n] = temp;
   i = i + 1;
}
```

13. Topic: Operating System

(d) A regular file is an ordinary file, which contains ASCII text, executable program binaries, program data, etc.

A named pipe is a tool that allows two or more system processes to communicate with each other using a file that acts as a pipe between them.

Difference between a named pipe and a regular file in Unix is that unlike a regular file, named pipe is a special file. The data in a pipe is transient, unlike the content of a regular file. Pipes forbid random accessing, while regular files do allows this. Therefore, all of the above are correct.

14. Topic: Engineering Mathematics

(c) Total seats in class room = 5 × 8 = 40

So, probability of a student taking a particular seat $= \dfrac{1}{40}$

Now, probability of 30 students to take a particular seat $= \dfrac{30}{40}$

Therefore, probability of the seat being empty $= 1 - \dfrac{30}{40} = \dfrac{1}{4}$

15. Topic: Engineering Mathematics

(c) Given: log(log sin(x))
We know that sin x can take values between −1 and +1 and log of a number is defined for positive values only.
Therefore log (sin(x)) is defined for (0, 1] and the possible values of log sin(x) are between $[-\infty, 0]$.
Thus, log (log(sin(x)) is not defined which implies domain of the function is an empty set.

16. Topic: Algorithms

(b) Given a set of non-negative integer, and a value K, determine if there is a subset of the given set with sum equal to K, Dynamic Programming can be used to find the solution in minimum time.
Dynamic Programming is a powerful technique that allows to solve problems in time $O(n^2)$ or $O(n^3)$. The key idea of Dynamic Programming is to avoid repeated work by remembering partial result.

17. Topic: Engineering Mathematics

(c) If (G, *) is an abelian group, then

(i) For all $a, b \in G$, $a * b$ also belongs to G
(ii) For all $a, b, c \in G$, $(a * b) * c = a * (b * c)$
(iii) There exists $e \in G$ such that for $a \in G$, $e * a = a * e = a$
(iv) For $a, b \in G$, $a * b = b * a = e$
(v) For $a, b \in G$, $a * b = b * a$

Therefore, option (c) is correct, that is, $(x * y)^2 = x^2 * y^2$ for any $x, y \in G$.

18. Topic: Programming and Data Structures

(d) On execution of program, we get:
For $i >= 10$, loop will run till 9 because of condition in while loop is ($i < 10$).
Then, value of j is 9!
For $i < 10$, loop will terminate when $i = x$
Then, value of j will be $(x – 1)!$
Therefore, statement of option (d) is true, that is, $((j = 9!) \wedge (i \geq 10)) \vee ((j = (x-1)!) \wedge (i = x))$

19. Topic: Engineering Mathematics

(d) Given: $\sqrt{224_r} = 13_r$

$$\Rightarrow 224_r = (13_r)^2$$
$$\Rightarrow 2 \times r^2 + 2 \times r^1 + 4 \times r^0 = (1 \times r^1 + 3 \times r^0)^2$$
$$\Rightarrow 2r^2 + 2r + 4 = (r+3)^2\ 2r^2 + 2r + 4 = r^2 + 9 + 6r$$
$$\Rightarrow 2r^2 + 2r + 4 - r^2 - 9 - 6r = 0 \Rightarrow r^2 - 4r - 5 = 0$$
$$\Rightarrow r^2 - 5r + r - 5 = 0\ r(r-5) + 1(r-5) = 0$$
$$\Rightarrow (r+1)(r-5) = 0 \Rightarrow r = -1 \text{ and } r = 5$$

Since r cannot be negative, therefore, $r = 5$ or value of radix $r = 5$.

20. Topic: Operating System

(**b**) Given is the references to pages that occur in the following order – 1, 2, 4, 5, 2, 1, 2, 4.
We know that main memory can accommodate 3 pages and page 1 and 2 are already in the main memory.
According to LRU, that is, least recently used algorithm, pages that have been used heavily in last few instructions will probably be heavily used again in the next few instructions.

Page 1:	Frame status 1, 2
Page 2:	Frame status 1, 2
Page 4:	Frame status 1, 2, 4 (Page fault)
Page 5:	Page 1 replaced by 5 – Frame status 5, 2, 4 (Page fault)
Page 2:	Frame status 5, 2, 4
Page 1:	Page 4 replaced by 1 – Frame status 5, 2, 1 (Page fault)
Page 2:	Frame status 5, 2, 1
Page 4:	Page 5 replaced by 4 – Frame status 4, 2, 1 (Page fault)

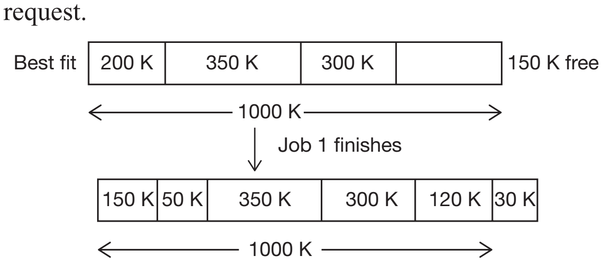
Therefore, number of page faults are 4.

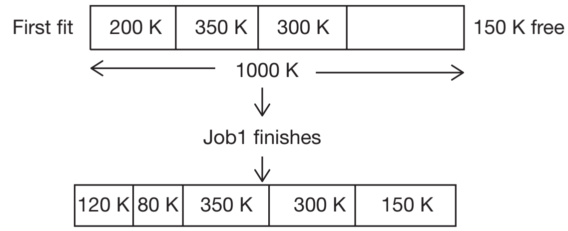
21. Topic: Operating System

(a) Deadlock is a situation which occurs when a set of process are blocked, because each process holds a resource and waiting for another resource that is being used by another process.
Here, the system has m resources and number of processes are 3(A, B, C) having peak time demands = 3, 4, 6 respectively.
Deadlock will never occur if total request resources of i^{th} process are less than total number of processes and resources available.
Here, peak time demand = 3 + 4 + 6 = 13
So, $13 < 3 + m \Rightarrow m > 10$
Therefore, minimum value of m that ensures deadlock never occurs is 11.

22. Topic: Operating System

(a) In best fit, the process is placed in the smallest block of memory in which it will fit.
In first fit, the process is allocated to block of memory encountered first that is large enough to satisfy the request.





Therefore, first fit performs better for this sequence.

23. Topic: Operating System

(d) First come first served scheduling algorithm is a simple scheduling algorithm where process are queued and executed in the order that they arrive in the ready queue.
Given: Disk request order – 10, 22, 20, 2, 40, 6, 38
Header is at cylinder 20 and set time is 6 m/cylinder.
Head movements = (20 – 10) + (22 – 10) + (22 – 2) + (40 – 2) + (40 – 6) + (38 – 6) = 146
Total seek time = 146 × 6 = 876 ms

24. Topic: Theory of Computation

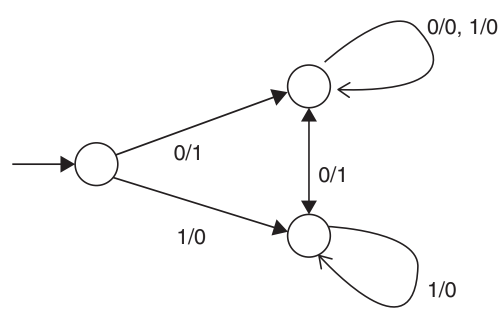
(d) Option (a): $A = \{a^n b^n | n = 1, 2, 3, \ldots\}$ is not regular but a deterministic context free language. So, this statement is incorrect.
Option (b): Set B consisting of all strings made up of only a's and b's having equal number of a's and b's defines a deterministic context free language. So, it is incorrect.
Option (c): $L(A^* B) \cap B$ gives the set B not set A. So, this option is also incorrect.
Therefore, option (d) is correct.

25. Topic: Theory of Computation

(b) Context free grammar (CFG) is a set of recursive rewriting rules. It is used to generate patterns of strings. Therefore, context free grammar (CFG) is not closed under complementation.

26. Topic: Theory of Computation

(c) Given:



A finite state machine is a mathematical model of computation. It is an abstract machine that can be in exactly one of a finite number of states at any given time.
Here, the given machine increments a given bit pattern by 1.

27. Topic: Theory of Computation

(a) A context free grammar is in Chomsky normal form if every rule is of the form

$$A \to BC$$
$$A \to a$$

where a is any terminal and A, B and C are variables except that B and C may not be the start variables. To get a string of n length, CNF takes $2n - 1$ steps. Therefore, to derive a string of terminals of length x, the number of products to be used is $2x - 1$.

28. Topic: Compiler Design

(c) An incremental compiler is a compiler that compiles only those portion of source code that have been modified and updates any corresponding output files that may already exist from previous compilations. By effectively building upon previously compiler output files. Hence option (c) is correct.

29. Topic: Compiler Design

(d) Definition use (DU-chains) is a data structure that consists of a use U of a variable and all the definitions D of that variable. DU is a compiler design that consists of a definition of a variable and all its uses, reachable from that definition. These chains are created using a form of static code analysis. There are prerequisite for many compiler optimization, including constant propagation and common sub-expression elimination. Thus, all of the given options are correct.

30. Topic: Compiler Design

(**a**) Peep-hole optimization is a kind of optimization performed over a very small set of instructions in a segment of generated code. This set is called a "peep-hole" or "window". It is used to improve performance, reduce memory footprint and reduce code size. Hence option (a) is true that it is applied to small part of the code and applied repeatedly.

31. Topic: Operating System

(d) Given:

$$\text{Memory capacity} = 2^m \text{ kB} = 2^m.2^{10}\text{ B}$$

So, number of bits to define operand = $m + 10$
and number of operations perform = 2^n
Now, number of bits to define 1 operation = n bits
Therefore, number of bits required by instruction having 3 operands and 1 operator is

$$3 \times (m + 10) + n = 3m + n + 30$$

So, none of the given options is correct.

32. Topic: Computer Organization and Architecture

(d) Let m be the number of digits occupied by ternary system. Therefore, maximum value represented in ternary system using m digits is equal to maximum value represented in binary system using n digits.
Now, maximum value represented in ternary system using m digits = $3^m - 1$
Maximum value represented in binary system using n digits = $2^n - 1$
Therefore,

$$3^m - 1 = 2^n - 1$$

Now, taking $\log_3$ on both sides

$$\log_3(3^m - 1) = \log_3(2^n - 1)$$
$$\Rightarrow m\log_3 3 = n\log_3 2 \; m = n\log_3 2$$

33. Topic: Algorithms

(c) Breadth-first search is a traversing algorithm. In this, the graph is traversed breadth-wise by moving horizontally and visiting all nodes of the current layer and then moving to the next layer. Applications of breadth-first search on the graph are finding diameter of the graph and finding bipartite graph.

34. Topic: Computer Organization and Architecture

(b) A microprogram consists of a sequence of instructions in a microprogramming language. These instructions specify micro-operations. It defines the individual operations in response to a machine-language instruction.

35. Topic: Algorithms

(d) Merge sort is a divide and conquer algorithm. In this, the input list is divided in two halves and then sorts them individually and merges the two sorted halves. The time complexity of merge sort is $O(n \log n)$ for worst, average and best cases.
For the given two sorted lists of size m and n, the number of comparisons needed the worst case by the merge sort will be $(n + m) - 1$.

36. Topic: Algorithms

(b) In worst case, number of comparisons required in searching an item that is not present is

Size of largest cluster +1

A hash table is a data structure that stores keys/or value pairs. While searching a hash table, when an empty slot is found, the probe stops. In the given hash table, the largest cluster is ($S6$, $S3$, $S7$, $S1$); therefore, the size of cluster is 4. Thus, maximum number of comparisons need in searching an item that is not present in 4 + 1 = 5.

37. Topic: Algorithms

(a) Run time of an algorithm is given as:

$$T(n) = \begin{cases} T(n-1) + T(n-2) - T(n-3), & \text{If } n > 3 \\ n, & \text{otherwise} \end{cases}$$

Let the order of algorithm for which the given run time relation is applicable be constant.
Using $T(n) = T(n-1)$ in the given relation, we get

$$T(n) = T(n) + T(n-2) - T(n-3)$$
$$\Rightarrow T(n-2) = T(n-3)$$

Solving like this further, we have

$$T(1) = T(2) = T(3)$$

38. Topic: Programming and Data Structures

(d) A regular graph is a graph in which all vertices have same degree.
Let d be the degree of each vertex and n be the number of vertices.
Then, sum of degrees = nd
Now, sum of degrees = 2 × Number of edges

Therefore, number of edges $= \frac{\text{Sum of degrees}}{2} = \frac{nd}{2}$

39. Topic: Computer Graphics

(c) Given, $(X_{\text{w min}}, Y_{\text{w min}}) = (0, 0)$
$(X_{\text{w max}}, Y_{\text{w max}}) = (100, 100)$
$(X_{\text{v min}}, Y_{\text{v min}}) = (5, 5)$
$(X_{\text{v max}}, Y_{\text{v max}}) = (20, 20)$
$(X_{\text{w}}, Y_{\text{w}}) = (20, 15)$

We know, $\frac{(X_{\text{v}} - X_{\text{v min}})}{(X_{\text{v max}} - X_{\text{v min}})} = \frac{(X_{\text{w}} - X_{\text{w min}})}{(X_{\text{w max}} - X_{\text{w min}})}$

Therefore, $\frac{X_{\text{v}} - 5}{20-5} = \frac{20-0}{100-0} \Rightarrow \frac{X_{\text{v}} - 5}{15} = \frac{20}{100}$

$$\Rightarrow X_{\text{v}} = \frac{20}{100} \times 15 + 5 = 8$$

And $\frac{Y_{\text{v}} - Y_{\text{v min}}}{Y_{\text{v max}} - X_{\text{v min}}} = \frac{Y_{\text{w}} - Y_{\text{w min}}}{Y_{\text{w max}} - Y_{\text{w min}}}$

Therefore, $\frac{Y_{\text{v}} - 5}{20-5} = \frac{15-0}{100-0} \Rightarrow \frac{Y_{\text{v}} - 5}{15} = \frac{15}{100}$

$$Y_{\text{v}} = \frac{15}{100} \times 15 + 5 = 7.25$$

40. Topic: Databases

(a) SELECT DISTINCT w, x
FROM R, S

This query selects all attributes of R and S.
The result can be equal to R if R has no duplicates and S is non-empty. If R has duplicates, then due to distinct keyword, the duplicates will be eliminated in final result. If S is empty, then $R \times S$ will be empty. Therefore S must be non-empty.

41. Topic: Databases

(a) A relation is in first normal form, if it has atomic values and all values are of same domain and unique.
A relation is in second normal form, if it is in first normal form and it does not have partial dependencies.
A relation is in third normal form, if it is in second normal form and it does not have transitive dependency. Here, the given relation is in first normal form, but it is not in second normal form.

42. Topic: Databases

(c) The SQL query is

```
SELECT DISTINCT Name
FROM Students, Courses, Grades
WHERE Students.Roll_Number = Grades.
Roll-number
AND Courses.Instructor = Sriram
AND Courses.Course-number = Grades.
Course-number
AND Grades.Grades A
```

Course.instructor – Sriram gives the course taught by Sriram. Grades.grade A gives students that can get A grade atleast. Therefore, option (c) is correct.

43. Topic: Programming and Data Structures

(a) The given program results in compilation error. The statement `r += a(c(r))` will result in compilation error.

44. Topic: Computer Networks

(b) Packet used to transmit message = 32 bytes
Round trip delay = 80 ms
Bottleneck bandwidth = 128 kbps
Window size is given by

$$\text{Window size} = \frac{\text{Round trip delay} \times \text{Bottleneck bandwidth}}{\text{Pocket used to transmit message}}$$

$$= \frac{80 \times 10^{-3} \times 128 \times 10^{3}}{32 \times 8} = 40$$

45. Topic: Computer Networks

(d) Given: connection overhead = 100 bytes and disconnection overhead = 28 bytes
So, total overhead = 100 + 28 = 128 bytes
Assume size of packet in transport layer = p
Now, to keep overhead to a minimum of 12.5%. Then,

$$12.5\% \text{ of } p = 128 \Rightarrow \frac{12.5}{100} \times p = 128$$

$$\Rightarrow p = \frac{128 \times 100}{12.5} = 1024 \text{ bytes}$$

46. Topic: Computer Networks

(b) Transposition cipher is an encryption method by which positions held by units of plain text are shifted according to a regular system so that the cipher text constitutes a permutation of the plain text.
Given, the keyword is LAYER $\Rightarrow$ 31524

Write the plain text in rows of width equal to length of keyword and read it by columns in order to define by key

3	1	5	2	4
↓	↓	↓	↓	↓
W	E	L	C	O
M	E	T	O	N
E	T	W	O	R
K	S	E	L	U
R	I	T	Y	!

⇓

EETSI COOCY WMEKR ONRU! LTWET

47. Topic: Programming and Data Structures

(c) Given:
To code a C++ program, 100 man days are required.
So, in 1 man day, 1% code is done.
Coding in assembly language is 10 times harder than coding in C++.
So, time consumed in coding in assembly language is 10 times more; also, coding in assembly language seems 5 times faster.
Converting an existing C++ program to assembly language is 4 times faster.
Therefore, to complete a code in assembly language, $\frac{100\times 10}{4} = 250$ man days are required.

That is, 1% of code is completed in $250\times\frac{1}{100} = 2.5$ man days
Thus, time consumed in writing 99% code in C++ and 1% code in assembly language is

$$99 \times 1 + 2.5 \times 1 = 101.5 \text{ man days}$$

Given, project team needs 13 days therefore,

Number of programmers in team $= \frac{101.5}{13} = 8$

48. Topic: Software Engineering

(c) Given, unit testing covers 90% of the code probability of success = 0.9.
Therefore, reliability of module = 90% × 0.9

$$= \frac{90}{100}\times 0.9 = 0.9\times 0.9 = 0.81$$

49. Topic: Databases

(b) A B+ tree is a B tree in which each node contains only key and not key-value pairs. Here it is given that number of records in a file is 1 million and order of tree is 100.
Now for a B+ tree of nodes n with k search keys, the path cannot be longer than $[\log_{n/2}(k)]$.

$$\Rightarrow \log_{50}(10^6) = 4$$

Therefore, maximum number of nodes to be accessed is 4.

50. Topic: Computer Organization and Architecture

(d) Given:
Bit string is D4FE2003, converting to binary representation

1101 0100 1111 1110 0010 0000 0000 0011

So, number of vacant tracks = 18 (number of 0's in the string) and number of occupied tracks = 14 (number of 1's in the string).

Therefore, percentage of occupied tracks $= \frac{14}{32}\times 100$
$= 43.75\% \sim 44\%$

51. Topic: Databases

(c) A dense index contains search key value and a pointer to actual record. In dense index, an index record appears for every search key value in file. A secondary index is a dense index because in secondary index, an index is maintained for every key value.

52. Topic: Databases

(b) Given: Y is dominant entity (strong)
X is subordinate entity (weak)

A strong entity connects weak entity using identifying relationship. Any change in dominant entity will be reflected in subordinate entity. Therefore, if Y is deleted X will also be deleted.

53. Topic: Databases

(b) Immunity of external schemas to change in conceptual schema means that if there are any changes in one level of database design, then it will not affect other part of database design (external schemas). In physical data independence, any change in physical storage schema will not affect conceptual schema.
In logical data independence, any change in conceptual schema will not affect external schema. So, the correct option is (b).

54. Topic: Databases

(c) A set of attributes X will be fully functionally dependent on the set of attributes Y if following conditions are satisfied.
(i) X is functionally dependent on Y
(ii) X is not functionally dependent on any subset of Y.

Hence option (c) is correct.

55. Topic: Databases

(b) Given schedule

$$S = r1(A);\ r2(B);\ w2(A);\ w1(B)$$

According to basic timestamp ordering protocol, conflicting operations should be executed in the timestamp order. In the given schedule, $T2$ is violating this protocol; therefore, a rollback is required. Therefore, this schedule is not allowed index basic timestamp protocol because $T1$ is rolled back.

56. Topic: Algorithms

(d) The time complexity of computing the transitive closure of binary relation on a set of n elements can be found using Warshall's algorithm for transitive closure computing. It is an efficient method of finding adjacency matrix of the transitive closure of relation. The time complexity is known to be $O(n^3)$.

57. Topic: Programming and Data Structures

(c) Given, elements are stored in an array as 25, 14, 16, 13, 10, 8, 12 in binary-max heap.
A binary heap is a complete binary tree. In binary-max heap, the key at root must be maximum among all keys present in binary heap.
Initial array: 25, 14, 16, 13, 10, 8, 12
after 1st deletion : 16, 14, 12, 13, 10, 8
after 2nd deletion : 14, 13, 12, 8, 10
Therefore, content of the array after two deletion operations is 14, 13, 12, 8, 10

58. Topic: Software Engineering

(c) A function point is a unit of measurement. It is used to calculate the functional size measurement of software. They are derived using an empirical relationship based on direct measures of software's information domain and assessments of software complexity. Therefore, option (c) is correct.

59. Topic: Software Engineering

(b) Lowest level of cohesion is coincidental cohesion. It is exhibited if the task performed are totally unrelated.
Next level of cohesion is logical cohesion. It is exhibited if the task performed are conceptually related.
Temporal cohesion: It is exhibited if the task performed are invoked at or near the same time.
Procedural cohesion: It is exhibited if the task performed are steps in the same application domain process.
Informational cohesion: It is exhibited if the task performed operates on the same information or data.
Hence the lowest degree of cohesion is kind of coincidental cohesion.

60. Topic: Programming and Data Structures

(d) Given: $x = 2, y = 5$
Here $x < y$. Therefore, $x = x + y$ is computed and returned so output is none of the options given.

61. Topic: Operating System

(b) The operating system of a computer may periodically collect all the free memory space to form contiguous block of free space. This is called Garbage collection. It is the job of a garbage collector to collect all free memory space at one place.

62. Topic: Digital Logic

(a) Any set of Boolean operators that is sufficient to represent all Boolean expression is said to be complete set of Boolean operators if it contains NAND or NOR gate.
{AND, OR} is not complete set.
{AND, NOT} is complete because NAND gate can be implemented using AND and NOT gates.
{NOT, OR} is complete because NOR gate can be implemented using NOT and OR gates.
{NOR} is complete.

63. Topic: Programming and Data Structures

(a) Option (a): To delete the last element of the list we need to traverse $(n – 1)$ elements therefore the time taken depends on the length of the list.
Option (b): To delete the first element of the list, move the pointer F from the first element to the next element. So, this operation does not depend on length of the list.
Option (c): To add an element after the last element of the list, move the pointer L to the new element. So, this operation does not depend on length of the list.
Option (d): Interchanging the first two elements of the list is also independent of the length of the list.
So, the correct option is (a).

64. Topic: Databases

(d) Given <word> :: = <letter> | <letter> <charpair> | <letter> <intpair> |
(i) Now <word> :: = <letter> <charpair>
Use <charpair> :: = <charpair> <letter> <letter>
Therefore, <word>::=<letter><chairpair> <letter> <letter>
Use <charpair> :: = <letter> <letter>
Therefore,
<word> :: = <letter> <letter> <letter> <letter> <letter>
↓ ↓ ↓ ↓ ↓
p *i* *c* *k* *s*
(ii) Now <word> :: = <letter> <intpair>
Use <intpair> :: = <integer> <integer>
Therefore, <word> :: = <letter> <integer> <integer
↓ ↓ ↓
c 4 4
Therefore, picks and c44 can be derived.

65. Topic: Computer Organization and Architecture

(b) Memory-mapped I/O is a method of performing input/outputs between central processing unit and peripheral devices in a computer. Memory-mapped I/O uses same address space to address both memory and I/O devices. Here, I/O ports are placed at addresses on the bus and are accessed just like other memory locations best characterized computers that use memory mapped I/O.

66. Topic: Algorithms

(a), (c) Option (a): Time complexity of merge sort: O(n log n). It is independent of the ordering of the array.
Option (b): Time complexity of insertion sort: O(n) if array is sorted and O(n^2) if array is not sorted. Therefore it depends on the ordering of the array.
Option (c): Time complexity of selection sort: O(n^2) and is independent of ordering of the array.
Option (d): Time complexity of quick sort: O(n^2) for sorted array and O(n log n) for unsorted array. Therefore it depends on the ordering of the array.

67. Topic: Operating System

(c) Given, processes $P1$ and $P2$ have a product–consumer relationship, communicating by the use of a set of shared buffers. By increasing the number of buffers, the throughput will increase. It will increase the rate at which requests are satisfied.

68. Topic: Operating System

(c) Given, in multi-programmed system, it is advantageous if some programs such as editors and compilers can be shared by several users. So, in order to have a single copy of a program shared by several users, the program must be reentrant.
Reentrant describes a computer program that is written so that the same copy in memory can be shared by multiple users.

69. Topic: Programming and Data Structures

(d) Given, P is a procedure that is recursive and it has guaranteed to terminate.
If P has a local variable, then each time P is called, a new copy of local variables will be created and it is not possible to track the number of times P is called.
Therefore in order to guarantee that P terminates, there has to be an execution path with P does not call itself and P either refers to a global variable or has at least one parameter. Hence, option (d) is correct.

70. Topic: Programming and Data Structures

(**b**) When the given program executes, we have:
While loop continues till $i/j > 0.001$
Initially, $i = 2.0$
$j = 1.0$

$$\frac{i}{j} > 0.001 \Rightarrow \frac{i}{j} \times 1000 > 1 \Rightarrow 1000 \times i > j$$

Therefore, j should be greater than 2000 for $i = 2.0$
Now, j increases as 2^n.
So, $2^n > 2000$
For $n = 11$ we have $2^n = 2048$
Therefore, for $n = 12$, the condition fails and exists in the loop.
So, the program produces 10 – 19 lines of output.
Hence, option (b) is correct.

71. Topic: Operating System

(**d**) Parallel program when executed on single processor requires 100 s.
40% computation is inherently sequential which implies 40 s.
So, remaining computation time = 100 s – 40 s = 60 s
For 2 processors: This computation is divided into 30 s – 30 s.
Therefore, total computation time = 40 + 30 = 70 s.
For 4 processors: This computation time is divided in 15 s – 15 s – 15 s – 15 s.
Therefore total computation time = 40 + 15 = 55 s.

72. Topic: Programming and Data Structures

(**b**) Given:

$$\begin{aligned} f(x) &= f(x-1) + g(x) \\ &= f(x-1) + f(x-1) + g(x/2) \\ &= 2f(x-1) + f(x/2-1) + g(x/4) \end{aligned}$$

So, time to execute $f(x)$ is

$$\begin{aligned} T(x) &= 2T(x-1) + T\left(\frac{x}{2}-1\right) + T\left(\frac{x}{2^2}-1\right) + T\left(\frac{x}{2^3}-1\right) \\ &\quad + \cdots + T\left(\frac{x}{2^{\log x}}-1\right) + 0(1) \\ &\approx 2T(x-1) + 0(1) \end{aligned}$$

Therefore, growth of $f(x)$ is exponential.

73. Topic: Computer Organization and Architecture

(c) For a multiprocessor architecture, directory-based protocol is used to forward a write transaction to only those processors that are known to possess a copy of newly altered cache line. Directory-based protocol deals with cache coherence in systems containing more processors that can be accommodated on a single bus.

74. Topic: Computer Networks

(a) Avalanche effect is a desirable property of the cryptographic algorithm. It ensures that a person cannot easily predict a message based on the changes in the hash value through statistical analysis. If an input is changed slightly, the output changes significantly.

75. Topic: Computer Networks

(c) In neural network, the network capacity is defined as the number of patterns that can be stored and recalled in a network.

76. Topic: Computer Networks

(a) Cloaking is a search engine optimization technique in which the content presented to the search engine spider is different from that presented to the user's browser. This is done by delivering context based on the IP addresses or user-agent HTTP header of the user requesting the page. When a user is identified as a search engine spider, a server-side script delivers a different version of the web-page, one that contains content not present on the visible page or that is present but not searchable.

77. Topic: Computer Networks

(a) Demilitarized Zone (DMZ) is a physical or logical sub-network that separates an internal local area network from other untrusted networks, usually the Internet. A demilitarized zone segments a network. DMZ can be setup using firewalls. The first firewall allows traffic destined to DMZ only and second firewall allows traffic from DMZ to internal network. The advantage of doing this is that you can control where traffic goes in the three networks.

78. Topic: Computer Networks

(c) An asymmetric cryptography uses public and private keys to encrypt and decrypt data. RSA algorithm and Diffie-Hellman algorithm are used in asymmetric key cryptography, while electronic code book algorithm is used in symmetric key cryptography.

79. Topic: Programming and Data Structures

(a) Given, a doubly linked list

```
struct   node {
   int value;
   struct Node * Fwd;
   struct Node * Bwd;
   };
```

Fwd → forward link
Bwd → backward link
To delete node pointed to by *X*, following segment of code is used.

```
X → Bwd → Fwd = X → Fwd;
X → Fwd → Bwd = X → Bwd;
```

80. Topic: Programming and Data Structures

(None) In inorder traversal: left node of a tree is visited first, then the root node and finally the right node of the tree is visited.
In preorder traversal: root node is visited first, then the left subtree and in the last the right subtree.
In postorder traversal: left subtree is visited first, then the right subtree and finally the root node.

Tree1: Postorder → GJIHEFDBCA
Preorder → ABDEGHIJFC
Inorder → BGEHIJDFAC

Tree2: Postorder → JHIEBDACFG
Preorder → GFEIJHCDBA
Inorder → GJIHEFBDCA

Indian Space Research Organisation Computer Science

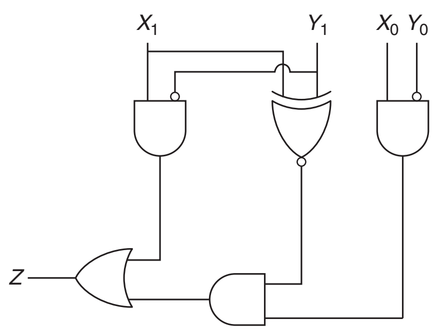
2020

QUESTION PAPER

1. Regression testing in primarily related to
   (a) Functional testing
   (b) Development testing
   (c) Data flow testing
   (d) Maintenance testing

2. Of the following sort algorithms, which has execution time that is least dependent on initial ordering of the input?
   (a) Insertion sort (b) Quick sort
   (c) Merge sort (d) Selection sort

3. The following circuit compares two 2-bit binary numbers, X and Y represented by X_1X_0 and Y_1Y_0 respectively. (X_0 and Y_0 represent Least Significant Bits)



   Under what condition Z will be 1?
   (a) $X > Y$ (b) $X < Y$
   (c) $X = Y$ (d) $X != Y$

4. What is the availability of the software with following reliability figures
   Mean Time Between Failures (MTBF) is 20 days
   Mean Time To Repair (MTTR) is 20 hours
   (a) 90% (b) 96%
   (c) 24% (d) 50%

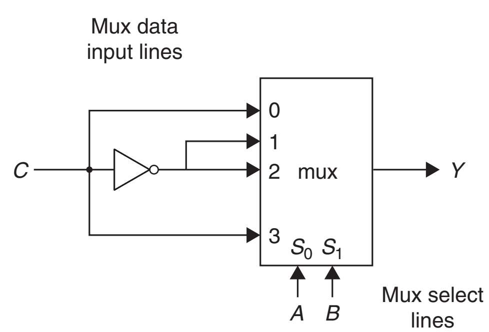
5. What is the defect rate for Six sigma?
   (a) 1.0 defect per million lines of code
   (b) 1.4 defects per million lines of code
   (c) 3.0 defects per million lines of code
   (d) 3.4 defects per million lines of code

6. Consider a 5-segment pipeline with a clock cycle time 20 ns in each sub operation. Find out the approximation speed-up ratio between pipelined and non-pipelined system to execute 100 instructions. (if an average, every five cycles, a bubble due to data hazard has to be introduced in the pipeline)
   (a) 5 (b) 4.03
   (c) 4.81 (d) 4.17

7. Consider a 32-bit processor which supports 70 instructions. Each instruction is 32 bit long and has 4 fields namely opcode, two register identifiers and an immediate operand of unsigned integer type. Maximum value of the immediate operand that can be supported by the processor is 8191. How many registers the processor has?
   (a) 32 (b) 64
   (c) 128 (d) 16

8. In a 8-bit ripple carry adder using identical full address, each full adder takes 34 ns for computing sum. If the time taken for 8-bit addition is 90 ns, find time taken by each full adder to find carry.
   (a) 6 ns (b) 7 ns
   (c) 10 ns (d) 8 ns

9. Following Multiplexer circuit is equivalent to



   (a) Sum equation of full adder
   (b) Carry equation of full adder
   (c) Borrow equation for full subtractor
   (d) Difference equation of a full subtractor

10. A stack is implemented with an array of '$A[0\ldots N-1]$' and a variable 'pos'. The push and pop operations are defined by the following code.

```
push (x)
    A[pos] ← x
    pos ← pos - 1
end push
pop ( )
    pos ← pos + 1
    return A[pos]
end pop
```

Which of the following will initialize an empty stack with capacity N for the above implementation?

(a) pos $\leftarrow -1$ (b) pos $\leftarrow 0$
(c) pos $\leftarrow 1$ (d) pos $\leftarrow N-1$

11. Given that

$B(a)$ means "a is a bear"
$F(a)$ means "a is a fish" and
$E(a, b)$ means "a eats b"
Then what is the best meaning of

$$\forall x F[(x) \to \forall y(E(y,x) \to b(y))]$$

(a) Every fish is eaten by some bear
(b) Bears eat only fish
(c) Every bear eats fish
(d) Only bears eat fish

12. Following declaration of an array of struct, assumes size of byte, short, int and long are 1, 2, 3 and 4 respectively. Alignment rule stipulates that n-byte field must be located at an address divisible by n. The fields in a struct are not rearranged, padding is used to ensure alignment. All elements of array should be same size.

```
Struct complx
    Short s
    Byte b
    Long l
    Int i
End complx
Complx  C[10]
```

Assuming C is located at an address divisible by 8, what is the total size of C, in bytes?

(a) 150 (b) 160
(c) 200 (d) 240

13. The immediate addressing mode can be used for

1. Loading internal registers with initial values
2. Perform arithmetic or logical operation on data contained in instructions

Which of the following is true?

(a) Only 1
(b) Only 2
(c) Both 1 and 2
(d) Immediate mode refers to data is cache

14. Statements associated with registers of a CPU are given. Identify the false statement.

(a) The program counter holds the memory address of the instruction in execution.
(b) Only opcode is transferred to the control unit.
(c) An instruction in the instruction register consists of the opcode and the operand.
(d) The value of the program counter is incremented by 1 once its value has been read to the memory address register.

15. Which of the following affects the processing power assuming they do not influence each other.

1. Data bus capability
2. Addressing scheme
3. Clock speed

(a) 3 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

16. Convert the pre-fix expression to in-fix

$$- * + A\ B\ C * - D\ E + F\ G$$

(a) (A – B) * C + (D * E) – (F + G)
(b) (A + B) * C – (D – E) * (F – G)
(c) (A + B – C) * (D – E) * (F + G)
(d) (A + B) * C – (D * E) – (F + G)

17. G is an undirected graph with vertex set $\{v_1, v_2, v_3, v_4, v_5, v_6, v_7\}$ and edge set $\{v_1v_2, v_1v_3, v_1v_4, v_2v_4, v_2v_5, v_3v_4, v_4v_5, v_4v_6, v_5v_6, v_6v_7\}$. A breadth first search of the graph is performed with v_1 as the root node. Which of the following is a tree edge?

(a) v_2v_4 (b) v_1v_4
(c) v_4v_5 (d) v_3v_4

18. If the array A contains the items 10, 4, 7, 23, 67, 12 and 5 in the order, what will be the resultant array A after third pass of insertion sort?

(a) 67, 12, 10, 5, 4, 7, 23
(b) 4, 7, 10, 23, 67, 12, 5
(c) 4, 5, 7, 67, 10, 12, 23
(d) 10, 7, 4, 67, 23, 12, 5

19. Huffman tree is constructed for the following data: {A, B, C, D, E} with frequency {0.17, 0.11, 0.24, 0.33 and 0.15} respectively, 100 00 01101 is decoded as

(a) BACE (d) CADE
(c) BAD (d) CADD

20. Given the grammar

```
S → T * S | T
T → U + T | U
U → a | b
```

Which of the following statements is wrong?

(a) Grammar is not ambiguous
(b) Priority of + over * is ensured

(c) Right to left evaluation of * and + happens
(d) None of these

21. What is the complexity of the following code?

```
sum = 0;
     for (i = 1, i <= n; i* = 2)
     for (i = 1; j <= n; j++)
          sum ++;
```

Which of the following is not a valid string?

(a) $O(n^2)$ (b) $O(n \log n)$
(c) $O(n)$ (d) $O(n \log n \log n)$

22. In the following procedure

```
Integer procedure P(X, Y);
Integer X, Y;
value x;
begin
      K = 5;
      L = 8;
      P = x + y;
end
```

X is called by value and Y is called by name. If the procedure were invoked by the following program fragment

```
K = 0;
L = 0;
Z = P(K, L)
```

then the value Z would be set equal to

(a) 5 (b) 8
(c) 13 (d) 0

23. Consider product of three matrices M_1, M_2 and M_3 having w rows and x columns, x rows and y columns, y rows and z columns. Under what condition will it take less time to compute the product as $(M_1M_2)M_3$ than to compute $M_1(M_2M_3)$?

(a) Always take the same time
(b) $(1/x+1/z) < (1/w+1/y)$
(c) $x > y$
(d) $(w+x) > (y+z)$

24. A new flipflop with inputs X and Y, has the following property

Inputs			
X	***Y***	**Current State**	**Next State**
0	0	Q	1
0	1	Q	$\bar{Q}$
1	1	Q	0
1	0	Q	Q

Which of the following expresses the next state in terms of X, Y, current state?

(a) $(\bar{X} \wedge \bar{Q}) \vee (\bar{Y} \wedge Q)$ (b) $(\bar{X} \wedge Q) \vee (\bar{Y} \wedge \bar{Q})$
(c) $(X \wedge \bar{Q}) \vee (Y \wedge Q)$ (d) $(X \wedge \bar{Q}) \vee (\bar{Y} \wedge Q)$

25. What is the output of the following 'C' code assuming it runs on a byte addressed little endian machine?

```
#include <stdio.h>
int main( )
{
int x; char *ptr;
x = 622,100,101;
printf("%d", (*(char *)&x) * (x % 3));
return 0;
}
```

(a) 622 (b) 311
(c) 22 (d) 110

26. What is the output in a 32 bit machine with 32 bit compiler?

```
#include<stdio.h>
rer(int **ptr2,int **ptr1)
{
    int* ii;
    ii=*ptr2;
    *ptr2=*ptr1;
    *ptr1=ii;
    **ptr1 *= **ptr2;
    **ptr2 += **ptr1;
}
void main( )
{
    int var1=5, var2=10;
    int *ptr1=&var1, *ptr2=&var2;
    rer(&ptr1,&ptr2);
    printf("%d %d", var2,var1);
}
```

(a) 60 70 (b) 50 50
(c) 50 60 (d) 60 50

27. Which of the following is an efficient method of cache updating?

(a) Snoopy writes (b) Write through
(c) Write within (d) Buffered write

28. In a columnar transposition cipher, the plain text is "the tomato is a plant in the night shade family", keyword is "TOMATO". The cipher text is

(a) "TINESAX / EOAHTFX / HTLTHEY / MAIIAIX / TAPNGDL / OSTANHMX"
(b) "TINESAX / EOAHTFX / MAIIAIX / HTLTHEY / TAPNGDL / OSTNHMX"

(c) "TINESAX / EOAHTFX / HTLTHEY / MAIIAIX / OSTNHMX / TAPNGDL"
(d) "EOAHTFX / TINESAX / HTLTHEY / MAIIAIX / TAPNGDL / OSTNHMX"

29. Avalanche effect in cryptography refers
(a) large changes in cipher text when the keyboard is changed minimally
(b) large changes in cipher text when the plain text is changed
(c) large Impact of keyboard change to length of the cipher text
(d) none of the above

30. A magnetic disk has 100 cylinders, each with 10 tracks of 10 sectors. If each sector contains 128 bytes, what is the maximum capacity of the disk in kilobytes?
(a) 1,280,000 (b) 1280
(c) 1250 (d) 128,000

31. How many total bits are required for a direct-mapped cache with 128 KB of data and 1 word block size, assuming a 32-bit address and 1 word size of 4 bytes?
(a) 2 Mbits (b) 1.7 Mbits
(c) 2.5 Mbits (d) 1.5 Mbits

32. Properties of 'DELETE' and TRUNCATE' commands indicate that
(a) after the execution of 'TRUNCATE' operation, COMMIT, and ROLLBACK statements cannot be performed to retrieve the lost data, while 'DELETE' allow it.
(b) after the execution of 'DELETE' and 'TRUNCATE' operation retrieval is easily possible for the lost data.
(c) after the execution of 'DELETE' operation, COMMIT and ROLLBACK statements can be performed to retrieve the lost data, while TRUNCATE do not allow it.
(d) after the execution of 'DELETE' and 'TRUNCATE' operation no retrieval is possible for the lost data.

33. Remote Procedure Calls are used for
(a) communication between two processes remotely different from each other on the same system
(b) communication between two processes on the same system
(c) communication between two processes on separate systems
(d) none of the above

34. Consider the following recursive C function that takes two arguments

```
unsigned int rer (unsigned int n,
unsigned int r) {
if (n > 0) return (n%r + rer (n/r,
r));
}
```

What is the return value of the function `rer` when it is called as `rer (513, 2)`?
(a) 9 (b) 8
(c) 5 (d) 2

35. A given grammar is called ambiguous if
(a) two or more productions have the same non-terminal on the left hand side
(b) a derivation tree has more than one associated sentence
(c) there is a sentence with more than one derivation tree corresponding to it
(d) brackets are not present in the grammar

36. What is the output of the code given below?

```
#include<stdio.h>
int main( )
{
char name[ ]="satellites";
int len;
int size;
len = strlen(name);
size = sizeof(name);
printf("%d", len * size);
return 0;
}
```

(a) 100 (b) 110
(c) 40 (d) 44

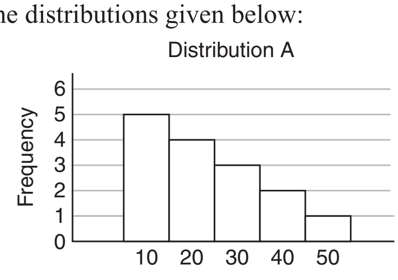
37. Checksum field in TCP header is
(a) ones complement of sum of header and data in bytes
(b) ones complement of sum of header, data and pseudo header in 16 bit words
(c) dropped from IPv6 header format
(d) better than md5 or sh1 methods

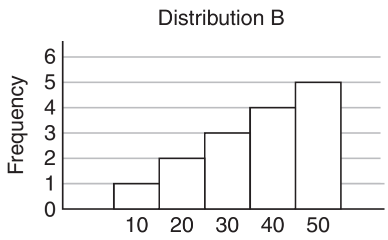
38. If $x + 2y = 30$, then

$$\left(\frac{2y}{5}+\frac{x}{3}\right)+\left(\frac{x}{5}+\frac{2y}{3}\right)$$

will be equal to
(a) 8 (b) 16
(c) 18 (d) 20

39. For the distributions given below:





Which of the following is correct for the above distributions?

(a) Standard deviation of A is significantly lower than standard deviation of B.
(b) Standard deviation of A is slightly lower than standard deviation of B.
(c) Standard deviation of A is same as standard deviation of B.
(d) Standard deviation of A is significantly higher than standard deviation of B.

40. The hardware implementation which provides mutual exclusion is

(a) Semaphores
(b) Test and set instruction
(c) Both options
(d) None of the options

41. Which of the following algorithms defines time quantum?

(a) shortest job scheduling algorithm
(b) round robin scheduling algorithm
(c) priority scheduling algorithm
(d) multilevel queue scheduling algorithm

42. Dispatch latency is defined as

(a) the speed of dispatching a process from running to the ready state.
(b) the time of dispatching a process from running to ready state and keeping the CPU idle.
(c) the time to stop one process and start running another one.
(d) none of these

43. An aid to determine the deadlock occurrence is

(a) resource allocation graph
(b) starvation graph
(c) inversion graph
(d) none of the above

44. Consider the following page reference string.

1 2 3 4 2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 6

What are the minimum number of frames required to get a single page fault for the above sequence assuming LRU replacement strategy?

(a) 7 (b) 4
(c) 6 (d) 5

45. Three CPU-bond tasks, with execution times of 15, 12 and 5 time units respectively arrive at time 0, t and 8, respectively. If the operating system implements a shortest remaining time first scheduling algorithm, what should be the value of t to have 4 context switches? Ignore the context switches at time 0 and at the end.

(a) $0 < t < 3$ (b) $t = 0$
(c) $t <= 3$ (d) $3 < t < 8$

46. The post-order traversal of a binary tree is ACEDBHIGF. The pre-order traversal is

(a) ABCDEFGHI (b) FBADCEGIH
(c) FABCDEGHI (d) ABDCEFGIH

47. In linear hashing, if blocking factor *bfr*, loading factor *i* and file buckets *N* are known, the number of records will be

(a) $cr = i + bfr + N$ (b) $r = i - bfr - N$
(c) $r = i + bfr - N$ (d) $r = i * bfr * N$

48. What does compaction refer to

(a) a technique for overcoming internal fragmentation.
(b) a paging technique.
(c) a technique for overcoming external fragmentation.
(d) a technique for compressing the data.

49. The operating system and the other processes are protected from being modified by an already running process because

(a) they run at different time instants and not in parallel.
(b) they are in different logical addresses.
(c) they use a protection algorithm in the scheduler.
(d) every address generated by the CPU is being checked against the relocation and limit parameters.

50. A grammar is defined as

$A \rightarrow BC$
$B \rightarrow x \mid Bx$
$C \rightarrow B \mid D$
$D \rightarrow y \mid Ey$
$E \rightarrow z$

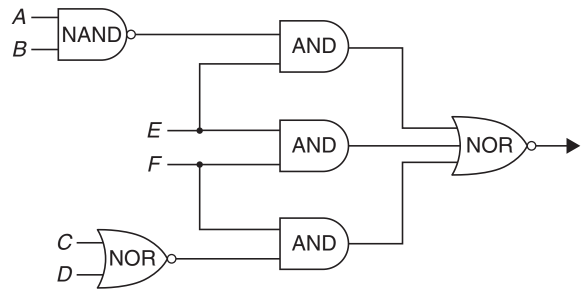
The non-terminal alphabet of the grammar is

(a) $\{A, B, C, D, E\}$ (b) $\{B, C, D, E\}$
(c) $\{A, B, C, D, E, x, y, z\}$ (d) $\{x, y, z\}$

51. If $A = \{x, y, z\}$ and $B = \{u, v, w, x\}$, and the universe is $\{s, t, u, v, w, x, y, z\}$. Then $(A \cup \bar{B}) \cap (A \cap B)$ is equal to

(a) $\{u, v, w, x\}$ (b) $\{\ \}$
(c) $\{u, v, w, x, y, z\}$ (d) $\{u, v, w\}$

52. Consider the following circuit



The function by the network above is

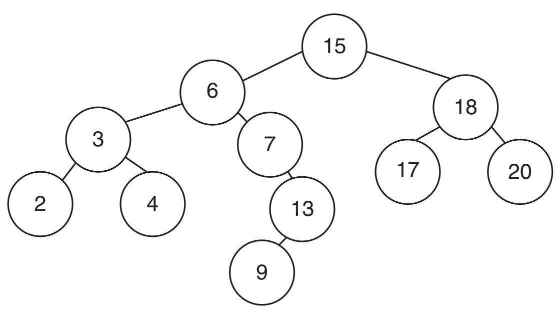
(a) $\overline{AB}E + EF + \overline{CD}F$

(b) $(\overline{E} + AB\overline{F})(C + D + \overline{F})$

(c) $(\overline{AB} + E)(\overline{E} + \overline{F})(C + D + \overline{F})$

(d) $(A + B)\overline{E} + \overline{EF} + CD\overline{F}$

53. What is the in-order successor of 15 in the given binary search tree?



(a) 18 (b) 6
(c) 17 (d) 20

54. The minimum height of an AVL tree with n nodes is

(a) Ceil($\log_2(n + 1)$) (b) $1.44\log_2 n$
(c) Floor($\log_2(n + 1)$) (d) $1.64\log_2 n$

55. The master theorem

(a) assumes the subproblems are unequal sizes.
(b) can be used if the subproblems are of equal size.
(c) cannot be used for divide and conquer algorithms.
(d) cannot be used for asymptotic complexity analysis.

56. Raymonds tree based algorithm ensures

(a) no starvation, but deadlock may occur in rare cases.
(b) no deadlock, but starvation may occur.
(c) neither deadlock nor starvation can occur.
(d) deadlock may occur in cases where the process is already starved.

57. Consider the following pseudo-code

```
I = 0, J = 0, K = 8;
while (I < K - 1) // while-I
    {
        J = J + 1;
        while (J < K)          //while-2
           {
                 if (x[I] < x[J])
          {
          temp = x[I];
          x[I] = x[J];
          x[J] = temp;
          }
           } //end of while-2
           I = I + 1;
    } //end of while-1
```

The cyclomatic complexity of the above is

(a) 3 (b) 2
(c) 4 (d) 1

58. In a class definition with 10 methods, to make the class maximally cohesive, number of direct and indirect connections required among the methods are

(a) 90, 0 (b) 45, 0
(c) 10, 10 (d) 45, 45

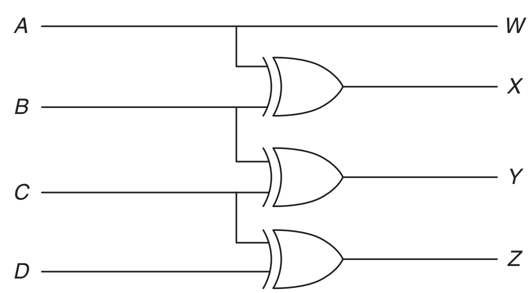
59. Of the following, which best approximates the ratio of the number of nonterminal nodes in the total number of nodes in a complex K-ary tree of depth N?

(a) $1/N$ (b) $N - 1/N$
(c) $1/K$ (d) $K - 1/K$

60. Minimum number of NAND gates required to implement the following binary equation $Y = (\overline{A} + \overline{B})(C + D)$

(a) 4 (b) 5
(c) 3 (d) 6

61. If $ABCD$ is a 4-bit binary number, then what is code generated by the following circuit?



(a) BCD code (b) Gray code
(c) 8421 code (d) Excess-3 code

62. The number of tokens in the following C code segment is

```
switch(inputvalue)
{
case 1: b = c* d; break;
default: b = b++; break;
}
```

(a) 27 (b) 29
(c) 26 (d) 24

63. In a two-pass assembler, resolution of subroutine calls and inclusion of labels in the symbol table is done during

(a) second pass
(b) first pass and second pass respectively
(c) second pass and first pass respectively
(d) first pass

64. One instruction tries to write on operand before it is written by previous instruction. This may lead to a dependency called

(a) True dependency (b) Anti-dependency
(c) Output dependency (d) Control hazard

65. If every non-key attribute functionally dependent on the primary key, then the relation will be in

(a) First normal form
(b) Second normal form
(c) Third normal form
(d) Fourth normal form

66. The SQL query

```
SELECT columns
FROM TableA
RIGHT OUTER JOIN TableB
ON A.columnName = B.columnName
WHERE A.columnName IS NULL
```

returns the following:

(a) All rows in TableB, which meet equality condition above and none from TableA, which meet the condition.
(b) All rows in TableA, which meet equality condition above and none from TableB, which meet the condition.
(c) All rows in TableB, which meet equality condition.
(d) All rows in TableA, which meet equality condition.

67. To send same bit sequence, NRZ encoding require

(a) Same clock frequency as Manchester encoding
(b) Half the clock frequency as Manchester encoding
(c) Twice the clock frequency as Manchester encoding
(d) A clock frequency which depend on number of zeros and ones in the bit sequence

68. The persist timer is used in TCP to

(a) To detect crashes from the other end of the connection
(b) To enable retransmission
(c) To avoid deadlock condition
(d) To timeout FIN_Wait1 condition

69. An array of 2 two byte integers is stored in big endian machine in byte addresses as shown below. What will be its storage pattern in little endian machine?

Address	Data
0×104	78
0×103	56
0×102	34
0×101	12

(a) 0×104 12
0×103 56
0×102 34
0×101 78

(b) 0×104 12
0×103 34
0×102 56
0×101 78

(c) 0×104 56
0×103 78
0×102 12
0×101 34

(d) 0×104 56
0×103 12
0×102 78
0×101 34

70. A non-pipelined CPU has 12 general purpose registers ($R0$, $R1$, $R2$,….$R12$). Following operations are supported

ADD Ra, Rb, Rr — Add Ra to Rb and store the result in Rr
MUL Ra, Rb, Rr — Multiply Ra to Rb and store the result in Rr

MUL operations takes two clock cycles, ADD takes one clock cycle.
Calculate minimum number of clock cycles required to compute the value of the expression $XY + XYZ + YZ$. The variables X, Y, Z are initially available in registers $R0$, $R1$ and $R2$ and contents of these registers must not be modified.

(a) 5 (b) 6
(c) 7 (d) 8

71. Context free languages are closed under

(a) union, intersection
(b) union, kleene closure
(c) intersection, complement
(d) complement, kleene closure

72. Which of the following is true?

(a) Every subset of a regular set is regular
(b) Every finite subset of non-regular set is regular
(c) The union of two non-regular set is not regular
(d) Infinite union of finite set is regular

73. The language which is generated by the grammar $S \to aSa \mid bSb \mid a \mid b$ over the alphabet $\{a, b\}$ is the set of

(a) Strings that begin and end with the same symbol
(b) All odd and even length palindromes
(c) All odd length palindromes
(d) All even length palindromes

74. Which of the following classes of languages can validate an IPv4 address in dotted decimal format? It is to be ensured that the decimal values lie between 0 and 255.

(a) RE and higher
(b) CFG and higher
(c) CSG and higher
(d) Recursively enumerable language

75. Minimum number of states required in DFA accepting binary strings not ending in "101" is

(a) 3 (b) 4
(c) 5 (d) 6

76. Which of the following is a type of a out-of-order execution, with the reordering done by a compiler

(a) loop unrolling
(b) dead code elimination
(c) strength reduction
(d) software pipelining

77. A stack organized computer is characterized by instructions with

(a) indirect addressing (b) direct addressing
(c) zero addressing (d) index addressing

78. A computer which issues instructions in order, has only 2 registers and 3 opcodes ADD, SUB and MOV. Consider 2 different implementations of the following basic block:

Case 1	Case 2
$t_1 = a + b;$	$t_2 = c + d;$
$t_2 = c + d;$	$t_3 = c - t_2;$
$t_3 = e - t_2;$	$t_1 = a + b;$
$t_4 = t_1 - t_2$	$t_4 = t_1 - t_2;$

Assume that all operands are initially in memory. Final value of computation also has to reside in memory. Which one is better in terms of memory accesses and by how many MOV instructions?

(a) Case 2, 2 (b) Case 2, 3
(c) Case 1, 2 (d) Case 1, 3

79. Which one indicates a technique of building cross compilers?

(a) Beta cross (b) Canadian cross
(c) Mexican cross (d) X-cross

80. Consider a 2-dimensional array x with 10 rows and 4 columns, with each element storing a value equivalent to the product of row number and column number. The array is stored in row major format. If the first element $x[0][0]$ occupies the memory location with address 1000 and each element occupies only one memory location, which all locations (in decimal) will be holding a value of 10?

(a) 1018, 1019 (b) 1022, 1041
(c) 1013, 1014 (d) 1000, 1399

ANSWER KEY

1. (d)	**2.** (c)	**3.** (a)	**4.** (b)	**5.** (d)	**6.** (b)	**7.** (b)	**8.** (d)	**9.** (a), (d)	**10.** (d)
11. (d)	**12.** (*)	**13.** (c)	**14.** (a)	**15.** (d)	**16.** (*)	**17.** (b)	**18.** (b)	**19.** (a)	**20.** (d)
21. (*)	**22.** (b)	**23.** (b)	**24.** (a)	**25.** (c)	**26.** (c)	**27.** (a)	**28.** (a)	**29.** (b)	**30.** (c)
31. (d)	**32.** (a), (c)	**33.** (c)	**34.** (d)	**35.** (c)	**36.** (b)	**37.** (b), (c)	**38.** (b)	**39.** (c)	**40.** (b)
41. (b)	**42.** (c)	**43.** (a)	**44.** (c)	**45.** (a)	**46.** (b)	**47.** (d)	**48.** (c)	**49.** (d)	**50.** (a)
51. (*)	**52.** (b)	**53.** (c)	**54.** (c)	**55.** (b)	**56.** (b)	**57.** (c)	**58.** (b)	**59.** (a)	**60.** (c)
61. (b)	**62.** (c)	**63.** (c)	**64.** (c)	**65.** (c)	**66.** (a)	**67.** (b)	**68.** (c)	**69.** (c)	**70.** (b)
71. (b)	**72.** (b)	**73.** (c)	**74.** (a)	**75.** (b)	**76.** (d)	**77.** (c)	**78.** (a)	**79.** (b)	**80.** (*)

* None of the options is correct.

ANSWERS WITH EXPLANATION

1. Topic: Software Engineering

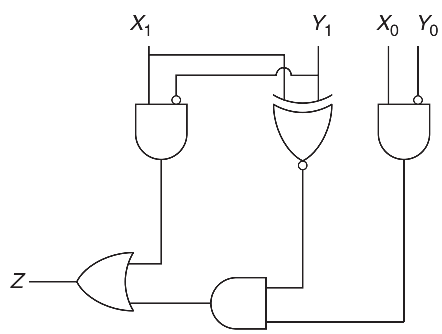
(d) Regression testing is defined as a type of software testing to confirm that a recent program or code change has not adversely affected existing features. Regression testing is primarily related to maintenance testing.

2. Topic: Algorithms

(c) Merge sort is dividing and conquer algorithm. It divides input array in two halves and calls itself for the two halves and then merges the two sorted halves. So, merge sort is least dependent on initial ordering of the input.

3. Topic: Digital Logic

(a) The following circuit compares two 2-bit binary numbers X and Y, that is, X_1X_0 and Y_1Y_0.



The expression of output Z is given by

$$Z = X_1\bar{Y}_1 + (X_1 \odot Y_1)X_0\bar{Y}_0$$

We know output of OR gate is 1 if either of the input is 1 or both are 1.

If $X_1 = 1$ and $Y_1 = 0$ then $X > Y$

If $X_1 = Y_1$, $X_0 = 1$ and $Y_0 = 0$ then $X > Y$

So, $Z = 1$ when $X > Y$

4. Topic: Software Engineering

(b) Given that, mean time between failures (MTBF) = 20 days and mean time of repair (MTTR) = 20 hr. Availability is given by

$$\text{Availability} = \frac{\text{Total uptime}}{(\text{Total uptime} + \text{Total downtime})} \times 100$$

$$= \frac{\text{MTBF}}{(\text{MTBF} + \text{MTTR})} \times 100$$

$$= \frac{20\,\text{days}}{(20\,\text{days} + 20\,\text{hr})} \times 100$$

$$= \frac{(20 \times 24)\,\text{hr}}{[(20 \times 24)\,\text{hr} + 20\,\text{hr}]} \times 100$$

$$= \frac{20 \times 24 \times 100}{(20 \times 24) + 20} = \frac{480 \times 100}{480 + 20} = \frac{480 \times 100}{500}$$

$$= 96\%$$

So, availability of the software is 96%.

5. Topic: Operating System

(d) Six sigma performance produces a defect free product. It calculates the number of defects that the process delivers. Defect rate for six sigma is 3.4 defects per million lines of code.

6. Topic: Computer Organization and Architecture

(b) Given that, number of instructions = 100, stages in pipeline = 5 and clock cycle per stage = 20 ns.

We know that,

T_{pipeline} = (time for pipelined execution without stalls) + (overhead due to stalls)

$T_{\text{without pipeline}}$ = (number of instructions) × (stages in pipeline) × (clock cycle per stage)

Time for pipelined execution without stalls = $(99 + 5) \times 20 = 104 \times 20 = 2080$

Overhead due to stalls = $20 \times 20 = 400$

So, $T_{\text{without pipeline}} = 100 \times 5 \times 20 = 10000$

and $T_{\text{pipeline}} = 2080 + 400 = 2480$

Speed-up ratio is given by

$$\frac{T_{\text{without pipeline}}}{T_{\text{pipeline}}} = \frac{10000}{2480} = 4.03$$

7. Topic: Operating System

(b) Given that, 32-bit processor supports 70 instructions, each instruction length = 32 bit long and has 4 fields and maximum value of immediate operand that can be supported = 8191.

Number of bits needed for opcode = ceil (log 70) = $\log_2(70) = 7$

Number of bits needed for immediate operand = ceil (log 8191) = $\log_2(8191) = 13$

Number of bits needed for each register identifiers

$$= \frac{32 - (7 + 13)}{2} = \frac{12}{2} = 6$$

So, total number of registers = $2^6 = 64$

8. Topic: Digital Logic

(d) Given that, number of bits $n = 8$, total delay $t_{tot} = 90$ ns and delay in computing sum $t_{sum} = 34$ ns. Time taken by 8-bit addition by 8-bit ripple carry adder is equal to total delay take for addition, that is,

$$t_{tot} = t_{sum} + (n-1)t_{carry}$$

$$\Rightarrow 90\text{ ns} = 32\text{ ns} + (8-1)\times t_{carry}$$

$$\Rightarrow 90\text{ ns} = 34\text{ ns} + (7\times t_{carry})$$

$$\Rightarrow t_{carry} = \frac{90\text{ ns} - 34\text{ ns}}{7} = \frac{56}{7} = 8\text{ ns}$$

9. Topic: Digital Logic

(a) Output of given multiplexer is

A	*B*	*C*	*Y*
0	0	0	0
1	0	1	1
0	1	0	1
1	1	1	0
0	0	0	1
1	0	1	0
0	1	0	0
1	1	1	1

So, the expression for output is

$$Y = A \oplus B \oplus C$$

Thus, multiplex circuit is equivalent to sum equation of full adder.

10. Topic: Programming and Data Structures

(d) Given that, array $A[0\ldots N-1]$ and a variable 'pos' From the given code, with each push command, pos is decreasing by 1.

So, for an array of length $N - 1$, the empty indices will decrease by 1 after each push command is executed.

Therefore, the above implementation is for an empty stack with capacity N and pos being initialized to $N - 1$, that is,

$$\text{pos} \leftarrow N - 1$$

11. Topic: Databases

(d) Given that, $B(a) \Rightarrow$ "*a* is a bear", $F(a) \Rightarrow$ "*a* is a fish" and $E(a, b) \Rightarrow$ "*a* eats *b*".

Then meaning of $\forall x F[(x) \rightarrow \forall y(E(y,x) \rightarrow B(y))]$ is only "bears eat fish", where *y* is a bear and *x* is a fish.

12. Topic: Programming and Data Structures

(*) Given, array of struct

```
Struct complex
      Short s
      Byte b
      Long l
      Int i
End complex
Complex C[10]
```

We know that, size of byte = 1, size of short = 2, size of int = 3 and size of long = 4.

C is located at an address divisible by 8, assuming base address to be 0.

S		*b*	*l*				*i*		
0	1	2	3	4	5	6	7	8	9

So, data type will be 12 byte after padding. Since, array is 10 elements long. Therefore, total size of *C* = 120 bytes

13. Topic: Computer Organization and Architecture

(c) The immediate addressing mode can be used for both loading internal registers with initial values and performing arithmetic or logical operation on data contained in instructions. So both (1) and (2) are true.

14. Topic: Computer Organization and Architecture

(a) The program counter holds the memory address of next instruction and not the current instruction. So, statement in option (a) is false.

15. Topic: Operating System

(d) Data bus capability, addressing scheme and clock speed all affects the processing power assuming they do not influence each other.

16. Topic: Programming and Data Structures

(*) An expression is called infix if the operator appears in between the operands in expression. Algorithm for prefix to infix is

1. Read prefix expression from right to left.
2. If symbol is operand then push it onto stack.
3. If symbol is operator, pop two operands from the stack and create a string by concatenating the two operands and the operator between them and push the resultant string back to stack.
4. Repeat above steps until end of prefix expression.

So infix expression of prefix expression is

$$- * ABC * - DE + FG$$

$$AB - C * (D - E) * (F + G)$$

Here an operand between A and B is missing. So, it cannot be converted into infix and none of the options is correct.

17. Topic: Algorithms

(b) Given that, vertex set $\{v_1, v_2, v_3, v_4, v_5, v_6, v_7\}$ and edge set $\{v_1v_2, v_1v_3, v_1v_4, v_2v_4, v_2v_5, v_3v_4, v_4v_5, v_4v_6, v_5v_6, v_6v_7\}$

Breadth first search is an algorithm for searching graph. It starts at the root of graph and explores all neighbors at present depth and then moves on to nodes at next level.

So, a breadth first search of graph with v_1 as root node will have v_1v_4 as the tree edge.

18. Topic: Algorithms

(b) Given an array $A = \{10, 4, 7, 23, 67, 12 \text{ and } 5\}$

In the insertion sort, the array is sorted by placing each element at its suitable place in each iteration.

First iteration: $\{\underline{4, 10}, 7, 23, 67, 12, 5\}$

Second iteration: $\{\underline{4, 7, 10}, 23, 67, 12, 5\}$

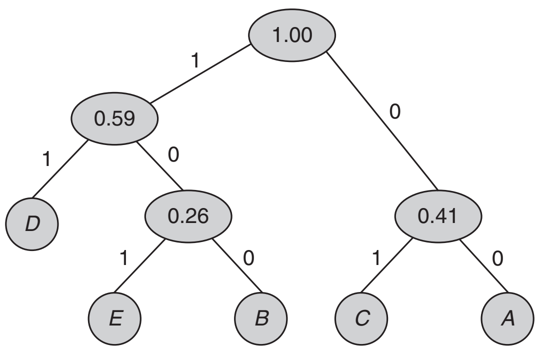
Third iteration: $\{\underline{4, 7, 10, 23}, 67, 12, 5\}$

So, resultant array A after third pass will be {4, 7, 10, 23, 67, 12, 5}

19. Topic: Algorithms

(a) Given data {A, B, C, D, E}, with frequency {0.7, 0.11, 0.24, 0.33, 0.15} and coded data 100 00 01101.

So, A = 0.7, B = 0.11, C = 0.24, D = 0.33 and E = 0.15



Huffman code: A = 00, B = 100, C = 01, D = 11 and E = 101

So, 100 00 01 101 is decoded as BACE

20. Topic: Theory of Computation

(d) Given, grammar

```
S → T * S/T
T → U + T/U
U → a/b
```

We have following points from this grammar:

(i) This grammar is not ambiguous.

(ii) Priority of + over * is ensured.

(iii) Right to left evaluation of * and + happens.

So, none of the statements is wrong.

21. Topic: Programming and Data Structures

(*) Given code,

```
sum = 0;
for (i = 1, i < = n; i* = 2)
       for (i = 1; j < = n; j++)
       sum ++;
```

Time complexity of loop is O(log n) if loop variable is multiplied by a constant amount.

So, none of the options is correct.

22. Topic: Programming and Data Structures

(b) Given, procedure

```
integer procedure P(X, Y)
integer X, Y;
value X;
begin
        K = 5;
        L = 8;
        P = X + Y;
End
```

Here X is called by value, Y is called name and

```
K = 0;
L = 0;
Z = P(K, L)
```

Then, $L = 8$ and $K = 0$, therefore, $Z = 8$

23. Topic: Algorithms

(b) Given, matrices M_1 with w rows and x columns, M_2 with x rows and y columns and M_3 with y rows and z columns. So, $M_1 = w * x$, $M_2 = x * y$ and $M_3 = y * z$.

To compute $(M_1M_2)M_3$, we have time = $wxy + wyz$

To compute $M_1(M_2M_3)$, we have time = $xyz + wxz$

So, time to compute $(M_1M_2)M_3$ is less than $M_1(M_2M_3)$ if

$$wxy + wyz < xyz + wxz$$

$$\Rightarrow \frac{wxy}{wxyz} + \frac{wyz}{wxyz} < \frac{xyz}{wxyz} + \frac{wxz}{wxyz}$$

$$\Rightarrow \frac{1}{x} + \frac{1}{z} < \frac{1}{w} + \frac{1}{y}$$

24. Topic: Digital Logic

(a) Given flipflop truth table is

Inputs			
X	Y	Current state	Next state
0	0	Q	1
0	1	Q	$\bar{Q}$
1	0	Q	0
1	1	Q	Q

So, next state in terms of X, Y current state is expressed as

$$(\bar{X}\wedge\bar{Q})\vee(\bar{Y}\wedge Q)$$

25. Topic: Programming and Data Structures

(d) Given that, $x = 622, 100, 101$
Since, comma (,) is left to right associative operator, so, x will store value 622.
Now, binary value of 622 is 0000001001101110.
Here 01101110 is lower byte pointed by character pointer.
So, `(*(char*)&x)` will give 01101110.
The decimal of 01101110 is 110 and `(x%3)`= 622%3 = 1.
Therefore, `(*(char*)&x)*(x%3)` gives
110 * 1 = 110

26. Topic: Programming and Data Structures

(d) From given program we have var 1 = 5, var 2 = 10, ptr 1 = address of var 1 and ptr 2 = address of var 2.
After function call `rer(&ptr1,&ptr2)`, we get ptr 2 = address of ptr 2 and ptr 1 = address of ptr 1
So, ptr 1 = ptr 1 * ptr 2 = 5 × 10 = 50 and ptr 2 = ptr 2 + ptr 1 = 10 + 50 = 60
Then, var 1 = 50 and var 2 = 60.
Therefore, output = 60 50

27. Topic: Computer Organization and Architecture

(a) Cache is special high speed memory used to speed up and synchronizing with high speed. Snoopy writes is an efficient method of cache updating. In snoopy writes, the cache controller monitors the operations of the other bus masters.

28. Topic: Computer Networks

(a) given, plain text: "the tomato is a plant in the night shade family" and keyword: TOMATO
Cipher text is encrypted text. In columnar transposition cipher, the given plain text is written in rows the number of rows depend on keywords "TOMATO", that is number of rows are six then reads the cipher text in columns one by one according to keywords in alphabetical order.

T	**O**	**M**	**A**	**T**	**O**
T	H	E	T	O	M
A	T	O	I	S	A
P	L	A	N	T	I
N	T	H	E	N	I
G	H	T	S	H	A
D	E	F	A	M	I
L	Y	X	X	X	X

So, the cipher text is in manner A/M/O/O/T/T alphabetically.
TINESAX/EOAHTFX/HTLTHEY/MAIIAIX/TAPNGDL/OSTANHMX

29. Topic: Computer Networks

(b) Avalanche effect is cryptography refers to large changes in cipher text when the plain text is changed.

30. Topic: Operating System

(c) Given that, magnetic disk has 100 cylinders, 10 tracks and 20 sectors. Each sector contains 128 bytes.
So, maximum capacity is given by
Number of cylinders × Number of tracks × Number of sectors × Number of bytes in each sector = 100 cylinders × 10 tracks × 10 sectors × 128 bytes

$$= 1280000 \text{ bytes}$$
$$= \frac{1280000}{1024} \text{ kilo bytes}$$
$$= 1250 \text{ kilo bytes}$$

31. Topic: Computer Organization and Architecture

(d) Given that, block size = 1 word = 4 B, cache data size = 128 kB = 128 × 8 kbits = 1024 kbits = 2^{17}

$$\text{Number of lines } = \frac{\text{Cache size}}{\text{Line size}} = \frac{128\,\text{kB}}{4\,\text{B}} = 32\,\text{K}$$

Number of cache lines = 32 × $\log_2$(128 kB) = 32 × 17 = 544 kbits
Total bits = number of cache lines + cache data size
= 544 + 1024 = 1564 kbits
Therefore, total bits $\simeq 1.5\,\text{Mbits}$

32. Topic: Databases

(a), (c) Properties of 'DELETE' and 'TRUNCATE' commands indicate that after the execution of 'TRUNCATE' operation, COMMIT, and ROLL BACK statements cannot be performed to retrieve the lost data, while 'DELETE' allow it.
After the execution of 'DELETE' operation, COMMIT and ROLL BACK statements can be performed to retrieve the lost data, while TRUNCATE do not allow it.
So both option (a) and (c) are correct.

33. Topic: Operating System

(c) Remote produce calls are used for communication between two processes on separate systems.

34. Topic: Programming and Data Structures

(d) Given a program

```
unsigned int rer (unsigned int n,
unsigned int r)
{  if(n  <  0)return(n%r  +  rer
(n/r,r));
}
```

Now return value of function rer when it is called as rer(513,2) is

$513 \% 2 = 1$
$256 \% 2 = 0$
$128 \% 2 = 0$
$64 \% 2 = 0$
$32 \% 2 = 0$
$16 \% 2 = 0$
$8 \% 2 = 0$
$4 \% 2 = 0$
$2 \% 2 = 0$
$1 \% 2 = 1$

So, 1 + 0 + 0 + 0 + 0 + 0 + 0 + 0 + 0 + 1 = 2

Therefore, return value = 2

35. Topic: Compiler Design

(c) A given grammar is called ambiguous if string generated has more than one phase tree, that is, there is a sentence with more than one derivation tree corresponding to it.

36. Topic: Programming and Data Structures

(b) Given program execute as

```
len=strlen(name)
```

name [ ] = “satellites”

Len = strlen (satellites\0)

$\Rightarrow$ len = 10

```
size=sizeof(name)
```

size = sizeof(satellites\0)

$\Rightarrow$ size = 11

So, len * size = $10 \times 11 = 110$

Therefore, output of code is 110.

37. Topic: Computer Networks

(b) Checksum field in TCP header is ones complement of sum of header, data and pseudo header in 16-bit words.

38. Topic: Engineering Mathematics

(b) Given that $x + 2y = 30$ (1)

Then $\left(\frac{2y}{5}+\frac{x}{3}\right)+\left(\frac{x}{5}+\frac{2y}{3}\right)$

$$= \frac{6y+5x}{15}+\frac{3x+10y}{15} = \frac{6y+5x+3x+10y}{15}$$

$$= \frac{8x+16y}{15} = \frac{8(x+2y)}{15}$$

$$\left(\frac{2y}{5}+\frac{x}{3}\right)+\left(\frac{x}{5}+\frac{2y}{3}\right) = \frac{8\times 30}{15} = 16 \text{ (Using Eq. (1))}$$

39. Topic: Engineering Mathematics

(c) Given that, in distribution A: frequencies = 5, 4, 3, 2, 1

Mean of the frequencies is, $\bar{x} = \frac{15}{5} = 3$

Standard deviation

$$= \sqrt{\frac{[(5-3)^2+(4-3)^2+(3-3)^2+(2-3)^2+(1-3)^2]}{4}}$$

$$= \sqrt{\frac{4+1+0+1+4}{4}} = \sqrt{\frac{10}{4}}$$

Now, in distribution B: frequencies = 1, 2, 3, 4, 5

Mean of the frequencies is, $\bar{x} = \frac{15}{5} = 3$

Standard deviation

$$= \sqrt{\frac{(1-3)^2+(2-3)^2+(3-3)^2+(4-3)^2+(5-3)^2}{4}}$$

$$= \sqrt{\frac{4+1+0+1+4}{4}} = \sqrt{\frac{10}{4}}$$

so standard deviation of A is same as standard deviation of B.

40. Topic: Computer Organization and Architecture

(b) The hardware implementation which provides mutual exclusion is test and set instruction. It is a special assembly language instruction that does two operations autonomously.

41. Topic: Operating System

(b) Round robin scheduling algorithm defines time quantum. It allocates CPU to each process in the system for a fixed time slot called quantum. Once a process is executed for a given time period that process is preempted and an other process executes for given time period.

42. Topic: Operating System

(c) Dispatch latency is defined as the time to stop one process and starts running another one by the dispatcher.

43. Topic: Operating System

(a) Dead lock occurrence is determined using resource allocation graph. Resource allocation graph is a diagrammatic representation of processes and resources in the system.

44. Topic: Operating System

(c) Given, page reference string: 1 2 3 4 2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 6

In LRU or least recently used replacement strategy, the data is read sequentially into the cache replacing a least recently used buffer. The buffer is placed on most recently used end of data buffer chain. The buffer moves towards LRU end as more pages are read into the cache.

	1	2	3	4	2	1	5	6	2	1	2	3	7	6	3	2	1	2	3	6
				4	4	4	4	6	6	6	6	6	7	7	7	7	1	1	1	1
			3	3	3	3	5	5	5	5	5	3	3	3	3	3	3	3	3	3
		2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
	1	1	1	1	1	1	1	1	1	1	1	1	1	6	6	6	6	6	6	6
Faults	1^{st}	2^{nd}	3^{rd}	4^{th}	0	0	5^{th}	6^{th}	0	0	0	7^{th}	0	8^{th}	0	0	9^{th}	0	0	0

Since, first four faults are in same frame. Therefore, minimum number of frames required to get a single page fault will be 6.

45. Topic: Operating System

(a) Given, three CPU – bond tasks

Execution time of task 1 = 15 time units

Execution time of task 2 = 12 time units

Execution time of task 3 = 5 time units

Arrival time of tasks are 0, t and 8.

In shortest remaining time first scheduling algorithm, the smallest amount of time remaining unit completion is selected to execute.

If $t = 1$ or $t = 2$ there are 4 context switches and for $t = 0$ and $t \geq 3$ context switches $\neq 4$.

Therefore, to have 4 context switches, value of t should lie in between $0 < t < 3$.

46. Topic: Programming and Data Structures

(b) Post-order traversal of binary tree is ACEDBHIGF.

In in-order traversal, left node is visited first then root node and right node is visited in the last.

In pre-order traversal, root node is visited first, left node is visited next and right node is visited in the last.

In post-order traversal, left node is visited first, right node is visited next and root node is visited in the last.

In pre-order traversal, the binary tree is FBADCEGIH.

47. Topic: Algorithms

(d) Given that, blocking factor = bfr, loading factor = i, file buckets = N

Linear hashing is a dynamic data structure which implements a hash table and grows or shrinks one bucket at a time.

Then number of records = $i * bfr * N$

48. Topic: Operating System

(c) Compaction refers to a technique for overcoming external fragmentation which occurs when memory is divided into variable size partitions based on the size of processes.

49. Topic: Operating System

(d) The operating system and the other processes are protected from being modified by an already running process because every address generated by the CPU is being checked against the relocation and limit parameters.

50. Topic: Theory of Computation

(a) Given grammar

$A \to BC$

$B \to x \mid Bx$

$C \to B \mid D$

$D \to y \mid Ey$

$E \to z$

Here A, B, C, D and E all derive something in grammar therefore non-terminal alphabet of the grammar is $\{A, B, C, D, E\}$.

51. Topic: Engineering Mathematics

(*) Given that, $A = \{x, y, z\}$, $B = \{u, v, w, x\}$ and $U = \{s, t, u, v, w, x, y, z\}$.

So, $\bar{B} = \{s, t, y, z\}$

Then, $A \cup \bar{B} = \{s, t, x, y, z\}$

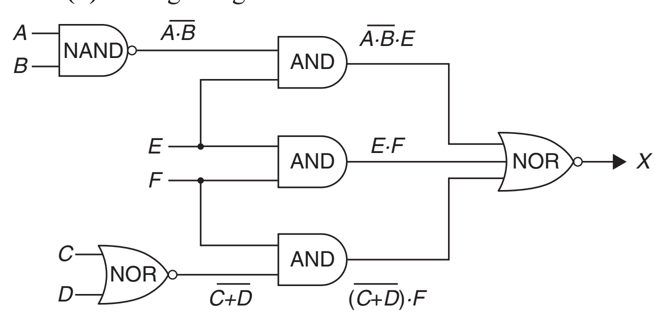
and $A \cap B = \{x\}$

Therefore, $(A \cup \bar{B}) \cap (A \cap B) = \{x\}$

None of the given options is correct.

52. Topic: Digital Logic

(b) The given gate circuit is



Expression of output is given by

$$X = \overline{\overline{A \cdot B} \cdot E + E \cdot F + (\overline{C + D}) \cdot F}$$

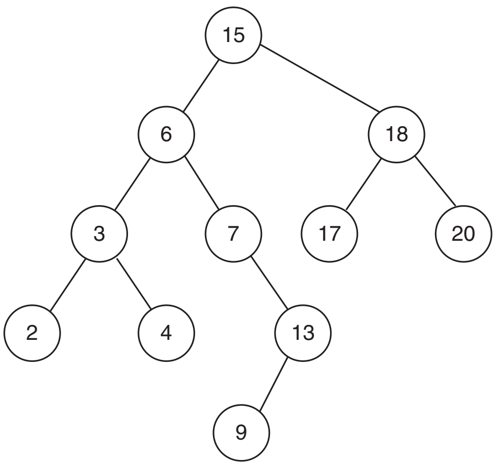
$$= \overline{\overline{A \cdot B} \cdot E} \cdot \overline{E \cdot F} \cdot \overline{(\overline{C + D}) \cdot F}$$

$$= (A \cdot B + \bar{E}) \cdot (\bar{E} + \bar{F}) \cdot ((C + D) + \bar{F})$$

$$= (AB\overline{E} + \overline{E}\overline{E} + AB\overline{F} + \overline{E}\overline{F})(C + D + \overline{F})$$
$$= (AB\overline{E} + \overline{E} + AB\overline{F} + \overline{E}\overline{F})(C + D + \overline{F})$$
$$= (AB\overline{F} + \overline{E}(AB + 1 + \overline{F}))(C + D + \overline{F})$$
$$= (AB\overline{F} + \overline{E})(C + D + \overline{F})$$

53. Topic: Programming and Data Structures

(c) In in-order traversal left node is visited first then the root node and right node is visited in the last.



So, order is as follows: 2, 3, 4, 6, 9, 13, 7, 15, 17, 18, 20
Therefore, successor of 15 is 17

54. Topic: Theory of Computation

(c) If there are n nodes in AVL tree, minimum height of an AVL tress is Floor($\log_2(n + 1)$).

55. Topic: Algorithms

(b) The master theorem assumes the subproblems are of equal sizes. So, option (a) is false.
The master theorem can be used if the subproblems are of equal size. So option (b) is true.
The master theorem can be used for divide and conquer algorithms. So option (c) is false.
The master theorem can be used for asymptotic complexity analysis. So option (d) is false.

56. Topic: Algorithms

(b) Raymond's tree based algorithm is lock based algorithm. In this algorithm all sites are arranged as a directed tree such that the edges of tree are assigned direction towards the site that holds the token.
Raymond's tree based algorithm ensures no deadlock, but starvation may occur.

57. Topic: Programming and Data Structures

(c) Cyclomatic complexity indicates complexity of a program. It measures number of linearly independent paths through the source code. It is given by

$$C = E - N + 2P$$

where, E is number of edges, N is number of nodes and P is number of connected components.
In the given program,
number of while loops = 2. Complexity of while loop is 1
number of if statements = 1. Complexity of if statement is 2.
Therefore, cyclomatic complexity = $2 \times 1 + 1 \times 2 = 2 + 2 = 4$

58. Topic: Computer Networks

(b) In a class definition with 10 methods, we have to make class maximally cohesive with number of direct and indirect connections.
To do this, every method should be related to every other method. Therefore,

$$\text{Number of methods } = \frac{n(n-1)}{2} = \frac{10 \times 9}{2} = 45$$

59. Topic: Programming and Data Structures

(c) Ratio of number of non terminal nodes in total number of nodes in complex K-ary tree of depth N is

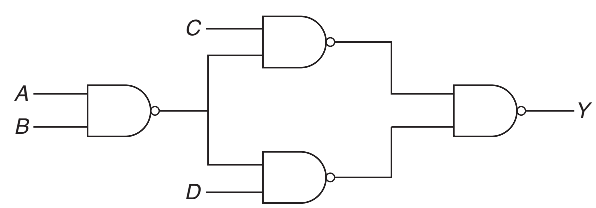
$$\frac{1 + K + K^2 + \cdots + K^{N-2}}{1 + K + K^2 + \cdots + K^{N-1}} = \frac{K^{N-1} - 1}{K^N - 1} \sim \frac{1}{K}$$

60. Topic: Digital Logic

(a) Given binary equation

$$Y = (\overline{A} + \overline{B})(C + D)$$
$$= (\overline{AB})(C + D) = (\overline{AB})C + (\overline{AB})D$$

Minimum number of NAND gets required to implement in this equation is 4.



61. Topic: Digital Logic

(b) From given logic gate circuit, we have following outputs
$W = A$, $X = A \oplus B$, $Y = B \oplus C$ and $Z = C \oplus D$
The truth table of input output of the given logic gate circuit is

Input				Output			
A	**B**	**C**	**D**	**W**	**X**	**Y**	**Z**
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0

Thus the successive values differ by one bit. So, code generated is a gray code.

62. Topic: Programming and Data Structures

(c) The number of tokens in the given C code are 26

1. switch
2. (
3. input value
4.)
5. {
6. case
7. 1
8. :
9. b
10. =
11. c
12. *
13. d
14. ;
15. break
16. ;
17. default
18. :
19. b
20. =
21. b
22. ++
23. ;
24. break
25. ;
26. }

63. Topic: Compiler Design

(c) In a two-pass assembler, first pass generates the symbol table and second pass generates machine code. So, the resolution of subroutine calls and inclusion of labels in the symbol table is done during second pass and first pass respectively.

64. Topic: Computer Organization and Architecture

(c) When one instruction tries to write an operand before it is written by previous instruction, it may lead to a dependency called output dependency. When two instructions written on operand one after other, this leads to write after write dependency (WAW) which is called output dependency.

65. Topic: Databases

(b) If every non-key attribute is functionally dependent on the primary key, then the relation will be in second normal form.

66. Topic: Databases

(*) The given SQL query returns all rows in TABLE B, which do not meet equality condition and none from TABLE A which meet the condition.
So, none of the options is correct.

67. Topic: Computer Networks

(c) In Manchester encoding the frequency of data transmission is twice that of NRZ thus to send same bit sequence NRZ encoding requires twice the clock frequency as Manchester encoding.

68. Topic: Computer Networks

(c) The persist timer is used in TCP to avoid deadlock condition.

69. Topic: Computer Organization and Architecture

(c) We know that in little endian machine, LSB is stored first and in big endian machine, MSB is stored first.
Data stored in big endian machine is given by
0×104 78
0×103 56
0×102 34
0×101 12
Data stored in little endian machine is given by
0×104 56
0×103 78
0×102 12
0×101 34

70. Topic: Computer Organization and Architecture

(d) Given expression

$$XY + XYZ + YZ$$

X, Y and Z are initially available in registers $R0$, $R1$ and $R2$.
To perform the given operation, 3 multiplication and 2 addition operations are required.
MUL operation clock cycles $= 3 \times 2 = 6$
ADD operation clock cycles $= 2 \times 1 = 2$
So, minimum number of clock cycles required $= 6 + 2 = 8$

71. Topic: Theory of Computation

(b) Context free language is a language generated by a context free grammar and is used to generate most arithmetic expressions. Context free languages are closed under union and kleene closure.

72. Topic: Engineering Mathematics

(b) Every finite subset of non-regular set is regular.

73. Topic: Theory of Computation

(c) the language generated by the grammar

$$S \rightarrow aSa \mid bsb \mid a \mid b$$

over alphabet $\{a, b\}$ is

$$\{a, b, aaa, aba, bab, bbb, aaaaa, aabaa, \ldots\}$$

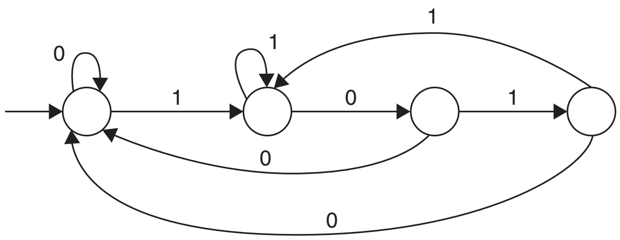
That is, all odd length palindromes.

74. Topic: Computer Networks

(a) To have decimal values between 0 and 255 we have to build finite automata. So, RE and higher classes of languages can validate an IPv4 address given in dotted decimal format.

75. Topic: Theory of Computation

(b) Minimum number of states required in DFA accepting binary strings not ending is "101" is 4.



76. Topic: Compiler Design

(a) Out-of-order execution is a technique where such cycles are used that might get wasted in certain type of costly delay. Loop unrolling is an out-of-order execution technique where reordering is done by a compiler.

77. Topic: Programming and Data Structures

(c) A stack organized computer is characterized by instructions with zero addressing.

78. Topic: Operating System

(*) Computer has 2 registers and 3 opcodes (ADD, SUB, MOV)

Case 1	Case 2
$t_1 = 1 + b;$	$t_2 = c + d;$
$t_2 = c + d;$	$t_3 = c + t_2;$
$t_3 = e - t_2;$	$t_1 = a + b;$
$t_4 = t_1 - t_2;$	$t_4 = t_1 - t_2;$

Instructions	Instructions
MOV R1, a	MOV R1, c
ADD R1, b	ADD R1, d
MOV R2, c	MOV R2, R1
ADD R2, d	SUB R1, e[e – t_2]
SUB R2, e[e – R2]	MOV R1, a
SUB R1, R2 [$t_1 - t_2$]	ADD R1, b
MOV R1, mom	SUB R1, R2
	MOV R1, mom

So, Case 1 is better than Case 2 in terms of memory access by one operation.

79. Topic: Compiler Design

(b) Cross compiler is compiler capable of creating executable code for a platform other than the one on which compiler is running. Canadian cross is a technique of building cross compilers.

80. Topic: Programming and Data Structures

(b) Given, 2 dimensional array x with 10 row and 4 columns.
First element $x[0][0]$ occupy the memory location with address 1000 and each element occupies only one memory location.
Address of $[L_r \ldots n][L_c \ldots m]$th element in row major order is given by

$$B + W[n(i - L_r) + (j - L_c)]$$

where, B is base address, W is element size, n is number of rows, L_r is the first row number and L_c is the first column number.
For the value of 10 we have two cases 10×1 and 5×2.
Case I: 10×1, $i = 10$ and $j = 1$, we get

$$1000 + 1[(10 - 0) \times 4 + (1 - 0)] = 1041$$

Case II: 5×2, $i = 5$ and $j = 2$, we get

$$1000 + 1[(5 - 0) \times 4 + (2 - 0)] = 1022$$

Practice Tests

Indian Space Research Organisation
Computer Science

Practice Test – 1

QUESTION PAPER

1. Three houses are available in a locality. Three persons apply for the houses. Each applies for one house without consulting others. The probability that all the three apply for the same house is

(a) 2/9 (b) 1/9
(c) 8/9 (d) 7/9

2. An unbiased die with faces marked 1, 2, 3, 4, 5 and 6 is rolled four times. Out of four face values obtained, the probability that the minimum face value is not less than 2 and the maximum face value is not greater than 5 is

(a) 16/81 (b) 1/81
(c) 80/81 (d) 65/81

3. The value of $\sin\theta + \cos\theta$ will be greatest when $\theta =$ _____.

(a) 45 (b) 30
(c) 90 (d) 60

4. How long is an IPv6 address?

(a) 32 bits (b) 128 bytes
(c) 64 bits (d) 128 bits

5. What flavor of Network Address Translation can be used to have one IP address allow many users to connect to the global Internet?

(a) NAT (b) Static
(c) Dynamic (d) PAT

6. What are the two main types of access control lists (ACLs)?

(i) Standard (ii) IEEE (iii) Extended (iv) Specialized

(a) (i) and (iii) (b) (ii) and (iv)
(c) (iii) and (iv) (d) (i) and (ii)

7. What is a stub network?

(a) A network with more than one exit point.
(b) A network with more than one exit and entry point.
(c) A network with only one entry and no exit point
(d) A network that has only one entry and exit point.

8. Which of the following languages is more suited to a structured program?

(a) PL/1 (b) FORTRAN
(c) BASIC (d) PASCAL

9. Which of the following is the 1's complement of 10?

(a) 01 (b) 110
(c) 11 (d) 10

10. The time required for the fetching and execution of one simple machine instruction is

(a) delay time (b) CPU cycle
(c) real time (d) seek time

11. A single packet on a data link is known as

(a) Path (b) Frame
(c) Block (d) Group

12. ASCII stands for

(a) American Standard Code for Information Interchange
(b) All-purpose Scientific Code for Information Interchange
(c) American Security Code for Information Interchange
(d) American Scientific Code for Information Interchange

13. A program that converts computer data into some code system other than the normal one is known as

(a) encoder (b) simulation
(c) emulator (d) coding

14. Which of the following are the two main components of the CPU?

(a) control unit and registers
(b) registers and main memory
(c) control unit and ALU
(d) ALU and bus

15. The function of CPU is

(a) to provide a hard copy
(b) to read, interpret and process the information and instruction
(c) to communicate with the operator
(d) to provide external storage of text

16. When an input electrical signal $A = 10100$ is applied to a NOT gate, its output signal is

(a) 01011 (b) 10001
(c) 10101 (d) 00101

17. Your router has the following IP address on Ethernet: 172.16.2.1/23. Which of the following can be valid host IDs on the LAN interface attached to the router?

(i) 172.16.1.100 (ii) 172.16.1.198 (iii) 172.16.2.255 (iv) 172.16.3.0

(a) (i) only (b) (ii) and (iii) only
(c) (iii) and (iv) only (d) None of these

18. Frames from one LAN can be transmitted to another LAN via the device

(a) router (b) bridge
(c) repeater (d) modem

19. Which of the following is not a logical data-base structure?

(a) Tree (b) Relational
(c) Network (d) Chain

20. Conversion of decimal number $(61)_{10}$ to its binary number equivalent is

(a) $(110011)_2$ (b) $(11001110)_2$
(c) $(111101)_2$ (d) $(11111)_2$

21. What logic function is obtained by adding an inverter to the inputs of an AND gate?

(a) OR (b) NAND
(c) XOR (d) NOR

22. A medium for transferring data between two locations is called

(a) network
(b) communication channel
(c) modem
(d) bus

23. What is the name of the technique in which the operating system of a computer executes several programs concurrently by switching back and forth between them?

(a) Partitioning (b) Multitasking
(c) Windowing (d) Paging

24. The slowest transmission speeds are those of

(a) twisted-pair wire (b) coaxial cable
(c) fibre-optic cable (d) microwaves

25. A computer program consists of

(a) system flowchart
(b) program flowchart
(c) algorithms written in computer's language
(d) discrete logical steps.

26. Which is the computer memory that does not forget?

(a) ROM (b) RAM
(c) NVRAM (d) All of these

27. A combinational logic circuit which is used when it is desired to send data from two or more source through a single transmission line is known as

(a) encoder (b) decoder
(c) multiplexer (d) demultiplexer

28. Which of the following services use TCP?

(i) DHCP (ii) SMTP (iii) HTTP (iv) TFTP (v) FTP

(a) (i) and (ii) (b) (ii), (iii) and (v)
(c) (i), (ii) and (iv) (d) (i), (iii) and (iv)

29. Which of the following memories must be refreshed many times per second?

(a) Static RAM (b) Dynamic RAM
(c) EPROM (d) ROM

30. An adder in which the bits of the operands are added one after another is

(a) half-adder (b) full-adder
(c) serial adder (d) all of these

31. The register which contains the instruction that is to be executed is known as

(a) index register
(b) instruction register
(c) memory address register
(d) memory data register

32. Which of the following is used to hold ROM, RAM, CPU and expansion cards?

(a) Computer bus (b) Motherboard
(c) Cache memory (d) All of these

33. A hard disk is divided into tracks which are further subdivided into

(a) clusters (b) sectors
(c) vectors (d) heads

34. Subtract $(1010)_2$ from $(1101)_2$ using 1's complement

(a) $(1100)_2$ (b) $(0011)_2$
(c) $(1001)_2$ (d) $(0101)_2$

35. Two pipes A and B can fill a cistern in $37\frac{1}{2}$ min and 45 min, respectively. Both pipes are opened together. The cistern will be filled in just half an hour if the B is turned off after ______.

(a) 5 min (b) 9 min
(c) 10 min (d) 15 min

36. A port can be ______.

(a) strictly for input (b) strictly for output
(c) bidirectional (d) all of the above

37. Ether LAN uses

(a) polar encoding
(b) differential manchester encoding
(c) manchester encoding
(d) NRZ

38. Two dice are tossed. The probability that the total score is a prime number is

(a) 1/6 (b) 5/12
(c) 1/2 (d) 7/9

39. Four bits are used for packed sequence numbering in a sliding window protocol used in a computer network. What is the maximum window size?

(a) 4 (b) 8
(c) 15 (d) 16

40. Which of the following technology can give high speed RAM?

(a) TTL (b) CMOS
(c) ECL (d) NMOS

41. Which memory is difficult to interface with processor?

(a) Static memory (b) Dynamic memory
(c) ROM (d) None of these

42. Convert the decimal number, 187 to 8-bit binary.

(a) $(10111011)_2$ (b) $(11011101)_2$
(c) $(10111101)_2$ (d) $(10111100)_2$

43. Which method bypasses the CPU for certain types of data transfer?

(a) Software interrupts
(b) Interrupt-driven I/O
(c) Polled I/O
(d) Direct memory access (DMA)

44. From a group of 7 men and 6 women, five persons are to be selected to form a committee so that at least 3 men are there on the committee. In how many ways can it be done?

(a) 564 (b) 645
(c) 735 (d) 756

45. An Enterprise Resource Planning application is an example of $a(n)$ _________.

(a) Single-user database application
(b) Multiuser database application
(c) E-commerce database application
(d) Data mining database application

46. Which protocol reduces administrative overhead in a switched network by allowing the configuration of a new VLAN to be distributed to all the switches in a domain?

(a) STP (b) VTP
(c) DHCP (d) ISL

47. The full adder circuit adds ________ digit at a time.

(a) 1 (b) 2
(c) 3 (d) 4

48. Which of the following statements are true regarding the command *ip route* 172.16.4.0 255.255.255.0 192.168.4.2?

1. The command is used to establish a static route.
2. The default administrative distance is used.
3. The command is used to configure the default route.
4. The subnet mask for the source address is 255.255255.0.

(a) 1 and 2 (b) 2 and 4
(c) 3 and 4 (d) All of these

49. Process is

(a) contents of main memory.
(b) a job in secondary memory.
(c) a program in execution.
(d) program in high level language kept on disk.

50. A combinational logic circuit which is used when it is desired to send data from two or more sources through a single transmission line is known as

(a) encoder (b) decoder
(c) multiplexer (d) demultiplexer

51. The process of jointly establishing communication is called _________.

(a) DMA
(b) bidirectional addressing
(c) multiplexing
(d) handshaking

52. Which type of gate can be used to add two bits?

(a) Ex-OR (b) Ex-NOR
(c) Ex-NAND (d) NOR

53. To make antenna more directional, either its size must be increased or

(a) the number of its feed horns must be increased
(b) the frequency of its transmission must be increased
(c) its effective isotropic radiated power (eirp) must be increased
(d) its footprint must be increased

54. In a directional coupler

(a) isolation (dB) equals coupling plus directivity
(b) coupling (dB) equals isolation plus directivity
(c) directivity (dB) equals isolation plus coupling
(d) isolation (dB) equals (coupling) (directivity)

55. Operating system

(a) links a program with the subroutines it references
(b) provides a layered, user-friendly interface
(c) enables a programmer to draw a flowchart
(d) none of these

56. A transponder is a satellite equipment which

(a) receives a signal from earth station and amplifies
(b) changes the frequency of the received signal
(c) retransmits the received signal.
(d) does all of the above-mentioned functions

57. Tracking of extra-terrestrial objects requires

(a) high transmitting power
(b) very sensitive receiver
(c) fully steerable antenna
(d) all of these

58. The parallel transmission of digital data

(a) is much slower than the serial transmission of data.
(b) requires only one signal line between sender and receiver.
(c) requires as many signal lines between sender and receiver as there are data bits.
(d) is less expensive than the serial method of data transmission.

59. The hexadecimal number $A0$ has the decimal value equivalent to

(a) 80 (b) 256
(c) 100 (d) 160

60. A wrist grounding strap contains which of the following?

(a) Surge protector (b) Capacitor
(c) Voltmeter (d) Resistor

61. CSMA (Carrier Sense Multiple Access) is

(a) A method of determining which device has access to the transmission medium at any time
(b) A method access control technique for multiple-access transmission media.
(c) A very common bit-oriented data link protocol issued by ISO.
(d) Network access standard for connecting stations to a circuit-switched network

62. The method for updating the main memory as soon as a word is removed from the Cache is called

(a) write-through (b) write-back
(c) protected write (d) cache-write

63. EPROM contents can be erased by exposing it to

(a) ultraviolet rays.
(b) infrared rays.
(c) burst of microwaves.
(d) intense heat radiations.

64. In computers, subtraction is generally carried out by

(a) 9's complement (b) 10's complement
(c) 1's complement (d) 2's complement

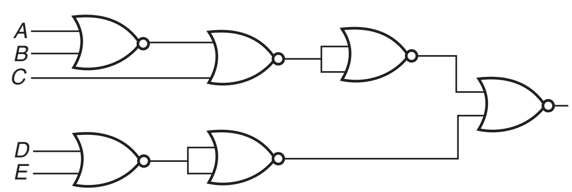
65. A JFET

(a) is a current-controlled device
(b) has a low input resistance
(c) is a voltage-controlled device
(d) is always forward-biased

66. The Boolean expression $Y = \left(A+\bar{C}+\bar{A}C\right)\bar{B}$ is given by

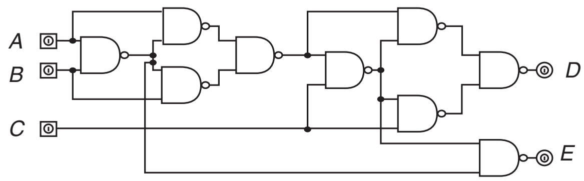
(a) $A\bar{C}$ (b) $B\bar{C}$
(c) $\bar{B}$ (d) AB

67. The circuit shown in the following figure realizes the function



(a) $(A+B\cdot\bar{C})\cdot(\bar{D}\cdot\bar{E})$ (b) $\left(\overline{A+B}+C\right)\left(\overline{DE}\right)$
(c) $\left(A+\overline{B+C}\right)\left(\overline{DE}\right)$ (d) $\left(A+B+\bar{C}\right)\left(\bar{D}\bar{E}\right)$

68. The circuit shown in the given figure is a



(a) full adder (b) full subtracter
(c) shift register (d) decade counter

69. Eigen vectors of $\begin{bmatrix} 1 & \sin\theta \\ \sin\theta & 1 \end{bmatrix}$ are

(a) $\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ (b) $\frac{-1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$
(c) $\frac{1}{\sqrt{2}}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ (d) $\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

70. Given $P(A) = 1/4$, $P(B) = 1/3$ and $P(A \cup B) = 1/2$. Value of $P(A/B)$ is

(a) 1/4 (b) 1/3
(c) 1/6 (d) 1/7

71. How is a *JK* flip–flop made to toggle?

(a) $J = 0, K = 0$ (b) $J = 1, K = 0$
(c) $J = 0, K = 1$ (d) $J = 1, K = 1$

72. The number of flip–flops used in decade counter is

(a) 3 (b) 2
(c) 4 (d) None of these

73. The decimal equivalent of the hexadecimal number *E*5 is

(a) 279 (b) 229
(c) 427 (d) 3000

74. Which part of the computer is used for calculating and comparing?

(a) Disk unit (b) Control unit
(c) ALU (d) Modem

75. A CPU's processing power is measured in

(a) IPS (b) CIPS
(c) MIPS (d) nano-seconds

76. Which computer memory is volatile?

(a) RAM (b) ROM
(c) EPROM (d) PROM

77. The computer code for the interchange of information between terminals is

(a) ASCII (b) BCD
(c) EBCDIC (d) All of these

78. Which of the following is not a sequential storage device?

(a) Magnetic disk (b) Magnetic tape
(c) Paper tape (d) All of these

79. The data recording area between the blank gaps on magnetic tape is called a

(a) Record (b) Block
(c) Field (d) Database

80. On a systems flowchart, the online manual keeping of input data is identified by using the

(a) online storage symbol
(b) online keyboard symbol
(c) keeping operation symbols
(d) manual operation symbols

Indian Space Research Organisation
Computer Science

Practice Test – 2

QUESTION PAPER

1. An example of a hierarchical data structure is

(a) array (b) link list
(c) tree (d) ring

2. Which of the following system software does the job of merging the records from two files into one?

(a) security software
(b) utility program
(c) networking software
(d) documentation system

3. In SQL, which command is used to enable/disable a database trigger?

(a) ALTER TRIGGER
(b) ALTER DATABASE
(c) ALTER TABLE
(d) MODIFY TRIGGER

4. Which of the following TCP/IP protocol is used for transferring electronic mail messages from one machine to another?

(a) FTP (b) SNMP
(c) SMTP (d) RPC

5. Consider $T(n) = 6T(n/2) + n^3$; then $T(n)$ is equal to

(a) $(n/2)$ (b) $O(n^3 \log n)$
(c) $(n^2 \log n)$ (d) $O(n^3)$

6. The principle of locality is used in

(a) interrupt (b) registers
(c) DMA (d) cache memory

7. Which of the following is a regular expression?

(a) $L = \{0^m 1^n \mid m, n \geq 0\}$ (b) $L = \{0^n 1^n \mid n \geq 0\}$
(c) $L = \{xx \mid x \in \{0,1\}^*\}$ (d) $L = \{1^{n^2} 0^n \mid n \geq 0\}$

8. Given the following expression grammar:

$$E \rightarrow E * F \mid F + E \mid F$$
$$F \rightarrow F - F \mid id$$

Which of the following is true?

(a) * has higher precedence than +
(b) – has higher precedence than *
(c) + and – have same precedence
(d) + has higher precedence than *

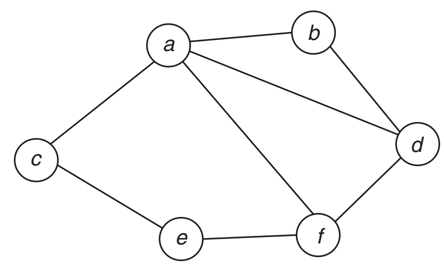
9. In modern operating systems such as Windows 2000, all the processor-dependent code is isolated in a dynamic link library called

(a) NTFS file system
(b) hardware abstraction layer
(c) microkernel
(d) CORBA process manager

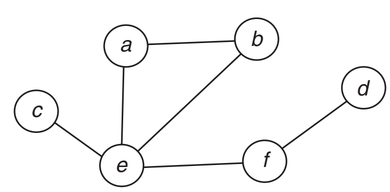
10. If $f(x) = \begin{cases} 4x - 5, & \text{if } x \leq 2 \\ x - \lambda, & \text{if } x > 2 \end{cases}$, then find λ for which $\lim_{x \to 2} f(x)$ exists.

(a) 1 (b) –1
(c) 5 (d) 2

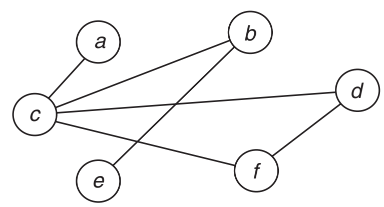
11. Which one of the following options is the complement of the given graph?

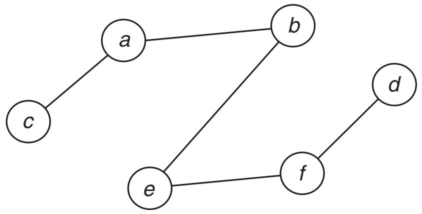


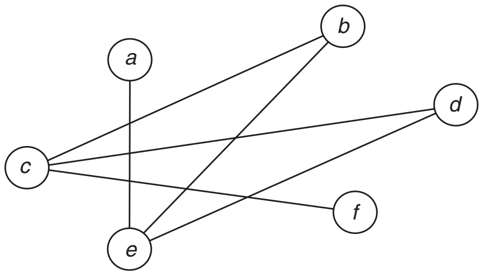
(a)



(b)



(c) 

(d) 

12. Virtual memory

(a) is a method of memory allocation by which the program is subdivided into equal portions, or pages and core is subdivided into equal portions or blocks
(b) consists of those addresses that may be generated by a processor during execution of a computation
(c) is a method of allocating processor time
(d) allows multiple programs to reside in separate areas of core at the time

13. The modify operation is likely to be done after

(a) Delete (b) Look-up
(c) Insert (d) All of these

14. The slowest transmission speeds are those of

(a) twisted-pair wire (b) coaxial cable
(c) fiber-optic cable (d) microwaves

15. Consider the following statements:

I. An algorithm is the number of steps to be performed to solve a problem
II. An algorithm is the number of steps and their implementation in any language to a given problem to solve a problem
III. To solve a given problem there may be more than one algorithm

Which of the following is true?

(a) I is correct
(b) II is correct
(c) I and III are correct
(d) I and II are correct

16. Which memory unit has lowest access time?

(a) Cache (b) Registers
(c) Optical disk (d) Main memory

17. The class of context-free language is not closed under

(a) union (b) star
(c) repeated concatenation (d) intersection

18. The number of tokens in the following C statement is

```
printf("i=%d, & i =%x", i & i);
```

(a) 13 (b) 6
(c) 10 (d) 11

19. The most important feature of spiral model is

(a) requirement analysis
(b) risk management
(c) quality management
(d) configuration management

20. Discuss the verification of Lagrange's mean value theorem for $f(x) = 2\sin x + \sin 2x$ on $[0, \pi]$.

(a) Not applicable
(b) Applicable for $c = \pm\pi/3$
(c) Applicable for $c = -\pi/3$
(d) Applicable for $c = \pi/3$

21. Find the number of edges in the graph K_6.

(a) 12 (b) 15
(c) 16 (d) 30

22. Which is a permanent database in the general model of compiler?

(a) Literal table (b) Identifier table
(c) Terminal table (d) Source code

23. Which of the following is not a logical data-base structure?

(a) Tree (b) Relational
(c) Network (d) Chain

24. How many digits of the Network User Address are known as the DNIC (Data Network Identification Code)?

(a) First three (b) First four
(c) First five (d) First seven

25. Compilers require efficient searching strategies, for which they rely on

(a) Hash tables (b) String matching
(c) Binary search tables (d) Binary search trees

26. During DMA transfer, the DMA controller takes over the buses to manage the transfer

(a) directly from CPU to memory
(b) directly from memory to CPU
(c) directly between the memory and registers
(d) directly between the I/O device and memory

27. Which of the following is true for regular sets $A = (1101 + 1)^*$ and $B = ((1101)^*1^*)^*$?

(a) $A \subset B$
(b) $B \subset A$
(c) $A = B$
(d) A and B are incomparable

28. The identifications of common sub-expression and replacement of run-time computations by compile time computations is

(a) local optimization
(b) global optimization
(c) constant folding
(d) common fielding

29. Cyclomatic complexity of a flow graph G with n vertices and e edges is

(a) $V(G) = e + n - 2$ (b) $V(G) = e - n + 2$
(c) $V(G) = e + n + 2$ (d) $V(G) = e - n - 2$

30. The order and degree of the differential equation $y + \frac{dy}{dx} = \frac{1}{4}\int y \cdot dx$ are

(a) order = 2 and degree = 1
(b) order = 1 and degree = 2
(c) order = 1 and degree = 1
(d) order = 2 and degree = 2

31. Two dice are thrown and the sum of the numbers which come up on the dice is noted. Consider the following events:

A = the sum is even
B = the sum is multiple of 3
C = the sum is less than 4
D = the sum is greater than 11
Which pairs of these events are mutually exclusive?

(a) A and B (b) B and C
(c) C and D (d) A and D

32. Operating system functions may include

(a) input/output control (b) virtual storage
(c) multi-programming (d) all of these

33. Which of the following tools is not used in modelling the new system?

(a) Decision tables (b) Data dictionary
(c) Data-flow diagrams (d) All of these

34. The most flexibility in how devices are wired together is provided by

(a) bus networks
(b) ring networks
(c) star networks
(d) T-switched networks

35. Consider a binary search tree having n elements. The time required to search a given element will be:

(a) $O(n)$ (b) $O(\log n)$
(c) $O(n^2)$ (d) $O(n \log n)$

36. Booth's algorithm is used for the arithmetic operation of

(a) addition (b) subtraction
(c) multiplication (d) division

37. The language generated by the following grammar is

$$S \rightarrow aSa \mid bSb \mid \varepsilon$$

(a) $a^m b^n$ $n \geq 0, m \geq 0$
(b) $a^n b^m$ $n \geq 1, m \geq 1$
(c) Odd length palindrome
(d) Even length palindrome

38. The term thrashing is used to define

(a) a reduce page I/O
(b) a decreased degree of multiprogramming
(c) an excessive page I/O
(d) improvement(s) in the system performance

39. Testing of software with actual data and in actual environment is called

(a) alpha testing (b) beta testing
(c) regression testing (d) none of these

40. Find the solution to $(x+1)\frac{dy}{dx} = 2xy$.

(a) $\log x = \log(y + 1) + C$
(b) $\log y = 1/x + 1$
(c) $\log y = 2[x - \log|x + 1|] + C$
(d) $y/x = C$

41. Out of all the words that can be formed from the letters of KING, what is the probability that a word chosen at random will start with K.

(a) 1/4 (b) 1/2
(c) 3/4 (d) 2/5

42. Moving process from main memory to disk is called

(a) scheduling (b) caching
(c) swapping (d) spooling

43. Difference between Decision Tables and Decision Tree is

(a) value to end user
(b) form of representation
(c) one shows the logic while other shows the process
(d) all of these

44. An FM signal contains intelligence in

(a) its frequency variations
(b) its amplitude variations
(c) both amplitude and frequency variations
(d) none of these

45. Variables of function call are allocated in

(a) registers and stack (b) cache and heap
(c) stack and heap (d) registers and heap

46. The reason for improvement in CPU performance during pipelining is

(a) reduced memory access time.
(b) increased clock speed.
(c) introduction of parallelism.
(d) increase in cache memory.

47. Number of states in DFA accepting the following language is

$$L = \{a^n b^{2m} \mid n, m \geq 1\}$$

(a) 2 (b) n
(c) m (d) 5

48. Page fault occurs when

(a) the page is not in cache memory.
(b) the page is in main memory.
(c) the page is not in main memory.
(d) the page has an address, which cannot be loaded.

49. Object Request Broker (ORB) is

I. A software program that runs on the client as well as on the application server.
II. A software program that runs on the client side only.
III. A software program that runs on the application server, where most of the components reside.

(a) I, II and III (b) I and II
(c) II and III (d) I only

50. If $w = \log z$, then what is the value of dw/dz?

(a) $1/z$ (b) z
(c) z^2 (d) $1/z^2$

51. In order to allow only one process to enter its critical section, binary semaphore are initialized to

(a) 0 (b) 1
(c) 2 (d) 3

52. In which addressing mode the contents of a register specified in the instruction are first decremented, and these contents are used as the effective address of the operands?

(a) Index addressing (b) Indirect addressing
(c) Auto increment (d) Auto decrement

53. The reservation system of Indian Railways is an example of

(a) transaction processing system
(b) interactive decision support system
(c) management controls system
(d) expert system

54. In PCM system,

(a) a large bandwidth is required
(b) the biggest disadvantage is its incompatibility with TDM
(c) companding is used to overcome quantising noise
(d) none of these

55. Predict the output of:

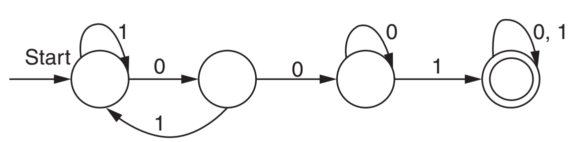
```
void main()
{
   float x = 1.1;
   double y = 1.1;
if(x==y)
printf("I see You");
else
printf("I hate You");
}
```

(a) I see You
(b) I hate You
(c) Cannot compare double and float
(d) Run time error

56. Use of cache memory enhances

(a) I/O access time.
(b) memory access time.
(c) effective memory access time.
(d) secondary storage access time.

57. Consider the following DFS automaton M.



Let S denote the seven bits binary strings in which the first, fourth and last bits are 1. The number of strings in S that are accepted by M is

(a) 6 (b) 5
(c) 4 (d) 7

58. If m is the number of resources and n is the number of processes in the system, then Banker's algorithm although a general algorithm may require operations.

(a) $n \times m$ (b) $(m \times m) \times n$
(c) $m \times (n \times n)$ (d) $(n \times m) \times m$

59. Key process areas of CMM level 4 are also classified by a process which is

(a) CMM level 2 (b) CMM level 3
(c) CMM level 5 (d) All of the above

60. Find the Taylor's expression of $f(z) = \frac{1}{(z+1)^2}$ about the point $z = -i$.

(a) $\sum_{n=2}^{\infty}(-1)^n \frac{(z-i)^n}{(1+i)^{n+1}}$

(b) $\frac{i}{2}\left[\sum_{n=2}^{\infty}(-1)^n \frac{(n+1)(z-i)^n}{(1-i)^n}\right]$

(c) $\frac{i}{2}\left[\sum_{n=2}^{\infty}(-1)^n \frac{(z-i)^n}{(1+i)^{n+1}}\right]$

(d) $\frac{i}{2}\left[1+\sum_{n=2}^{\infty}(-1)^n \frac{(n+1)(z+i)^n}{(1+i)^n}\right]$

61. The outstanding invoice file should be stared on a Pardon Access Storage Device if

(a) invoice data entry is on-line
(b) payment recording is done in a batch rode
(c) inquiries concerning payable are to be answered on-line
(d) last record points to the first record

62. The network layer, in reference to the OSI model, provide

(a) data link procedures that provide for the exchange of data via frames that can be sent and received
(b) the interface between the X.25 network and packet mode device
(c) the virtual circuit interface to packet-switched service
(d) all of these

63. If the input to a differentiating circuit is saw-tooth wave, then output will be ________ wave.

(a) square (b) rectangular
(c) triangular (d) saw-tooth

64. Global variable conflicts due to multiple file occurrence is resolved during

(a) load time
(b) execution time
(c) link time
(d) parsing phase of compiler

65. A correct output is achieved from a master-slave *JK* flip–flop only if its inputs are stable while the

(a) clock is LOW
(b) lave is ready to receive the input from master
(c) master gives a reset signal to slave
(d) clock is HIGH

66. Which of the following derivations does a top-down parser use while parsing an input string? The input is assumed to be scanned in left to right order.

(a) Leftmost derivation
(b) Topmost derivation
(c) Rightmost derivation
(d) Leftmost derivation traced out in reverse

67. Mutual exclusion problem occurs

(a) between two disjoint processes that do not interact
(b) among processes that share resources
(c) among processes that do not use the same resource
(d) between two processes that uses different resources of different machines

68. Validation means

(a) are we building the product right
(b) are we building the right product
(c) certification of fields
(d) None of these

69. A 32-bit microprocessor has word length equal to

(a) 1 byte (b) 2 byte
(c) 4 byte (d) 8 byte

70. For what value of x, the matrix $A = \begin{bmatrix} 1 & -2 & 3 \\ 1 & 2 & 1 \\ x & 2 & -3 \end{bmatrix}$ is singular?

(a) 1 (b) –1
(c) 2 (b) –3

71. Which of the following statements is true?

(a) An unambiguous grammar need not have same leftmost and rightmost derivation.
(b) An LL(1) parser is not a top-down parser.
(c) LALR is less powerful than SLR.
(d) An ambiguous grammar can be LR(k) for any k.

72. Which of the following is an important characteristic of LAN?

(a) Application independent interfaces
(b) Unlimited expansion
(c) Low cost access for low bandwidth channels
(d) Parallel transmission

73. Which gate acts as universal gate?

(a) AND (b) NAND
(c) OR (d) EX-OR

74.

```
#define f(a,b) a+b
#define g(a,b)a*b
main()
{
   int m;
   m=2*f(3,g(4,5));
   printf("\n m is %d", m);
}
```

What is the value of m?

(a) 70 (b) 50
(c) 46 (d) 69

75. An inverter circuit can be realized with how many NAND gates?

(a) 1 (b) 2
(c) 3 (d) Not possible

76. The process of assigning load addresses to various parts of the program and adjusting the code and data in the program to reflect the assigned addresses is called

(a) address assembly (b) parsing
(c) relocation (d) indexing

77. Linux operating system uses

(a) i-Node scheduling
(b) fair pre-emptive scheduling
(c) RR scheduling
(d) highest penalty ratio next

78. Find a, b, c and d such that $\begin{bmatrix} a-b & 2c+d \\ 2a-b & 2a+d \end{bmatrix} = \begin{bmatrix} 5 & 3 \\ 12 & 15 \end{bmatrix}$?

(a) $a = 7, b = 2, c = 1$ and $d = 1$
(b) $a = 6, b = 1, c = 2$ and $d = 2$
(c) $a = 8, b = 3, c = 3$ and $d = 3$
(d) $a = 9, b = 4, c = 1$ and $d = 1$

79. A program to find the factorial of a given number is written (a) using recursion and (b) without using recursion. Which of these will result in stack overflow for a given number?

(a) Only first program with recursion
(b) Both may result for the same number
(c) Only second program
(d) None of these

80. A modulus-12 ring counter requires a minimum of ________.

(a) 10 flip–flops (b) 12 flip–flops
(c) 8 flip–flops (d) 6 flip–flops

Indian Space Research Organisation
Computer Science

Practice Test – 3

QUESTION PAPER

1. Match the following:

IC Family	GATE
TTL	NOR
CMOS	NOT
ECL	NAND

(a) TTL:NAND; ECL:NOR; CMOS:NOT
(b) TTL:NAND; ECL:NOT; CMOS:NOR
(c) TTL:NOT; ECL:NAND; CMOS:NOR
(d) TTL:NOR; ECL:NOT; CMOS:NAND

2. Which of the following satisfies the commutative law but not the associative law?

(a) NOR (b) OR
(c) XOR (d) NAND

3. What is the control unit's function in the CPU?

(a) To decode program instructions
(b) To transfer data to primary storage
(c) To perform logical operations
(d) To store arithmetic operations

4. _______ is often used to prove the correctness of a recursive function.

(a) Associativity (b) Commutativity
(c) Mathematical induction (d) Factorial

5. Which of the following CFG will generate the language $L = \{a^m b^n c^p d^q \mid m + n = p + q\}$

(a) $S \to aSd \mid A \mid B$
$A \to aAc \mid C$
$B \to bBd \mid D$
$C \to bCc \mid \varepsilon$

(b) $S \to aSd \mid A \mid B$
$A \to aAc \mid D$
$B \to bBd \mid C$
$C \to CBc \mid \varepsilon$

(c) $S \to aSd \mid A \mid B$
$A \to aAB \mid C$
$B \to bBd \mid C$
$C \to bCc \mid \varepsilon$

(d) $S \to aSd \mid A \mid B$
$A \to aAc \mid C$
$B \to bBd \mid C$
$C \to bCc$

6. Let us suppose, in a programming language, an identifier is permitted to be a letter followed by any number of letters or digits. If L and D denote the sets of letters and digits, respectively, which of the following expressions defines an identifier?

(a) $(L.D)^*$ (b) $L(L + D)$
(c) $L + (L.D)^*$ (d) $L(L.D)^*$

7. Which of the following features will characterize an OS as multiprogrammed OS?

I. More than one program may be loaded into main memory at the same time.
II. If a program waits for certain event, another program is immediately scheduled.
III. If the execution of a program terminates, another program is immediately scheduled.

(a) I only
(b) I and II only
(c) I and III only
(d) I, II and III only

8. Superficially the term "object-oriented", means that, we organize software as a

(a) collection of continuous objects that incorporates both data structure and behaviour
(b) collection of discrete objects that incorporates both discrete structure and behaviour
(c) collection of discrete objects that incorporates both data structure and behaviour
(d) collection of objects that incorporates both discrete data structure and behaviour

9. Which class of IP addresses is used for multicasting?

(a) Class E (b) Class C
(c) Class A (d) Class D

10. Which one of the following options is not the eigenvector of matrix $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$?

(a) $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ (b) $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

(c) $\begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$ (d) $\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$

11. A school awarded 38 prizes in football, 15 in basketball and 20 in cricket. If these prizes went to a total of 58 students and only 3 students got prizes in all three sports, how many received medals in exactly two of the three sports?

(a) 18 (b) 15
(c) 12 (d) 9

12. Match the following:

IC	Family	Characteristics
X.	TTL	High fan-out
Y.	ECL	Low propagation delay
Z.	CMOS	High power dissipation

Code:	**X**	**Y**	**Z**
(a)	3	2	1
(b)	3	1	2
(c)	2	1	3
(d)	1	2	3

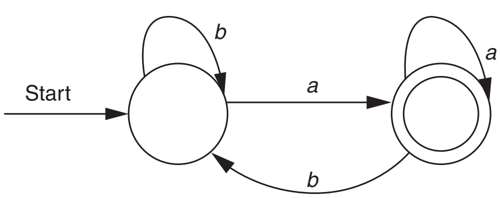
13. CPU fetches the data and instructions from _______.

(a) ROM (b) control unit
(c) RAM (d) coprocessors chip

14. In a doubly linked list, the number of pointers affected for an insertion operation will be

(a) 4
(b) 0
(c) 1
(d) Depending upon the nodes of the doubly linked list.

15. Consider the FA shown in the figure below, which language is accepted by the FA



(a) $b + (a + b)^*b$ (b) $(b + a + b)^*a$
(c) $b + a^*b$ (d) $b + a^*b^*$

16. Consider LR(1) collections of grammar (G), in state I_0 (that is, first state) what is the follow set of LR(1) item, $T \rightarrow .TF$?

(a) $\$| +$ (b) $\$| + |a|b$
(c) $\$| + |a$ (d) $+ |a|b$

17. The P and V operations on counting semaphores, where s is a counting semaphore are defined as follows:

$P(s)$:$s = s - 1$
if $s < 0$ then wait;
$V(s)$:$s = s + 1$
if $s = 0$ then wake up a process waiting on s.

Assume that Pb and Vb are the wait and signal operation on binary semaphore. Two binary semaphores Xb and Yb are used to implement the semaphore operation $P(s)$ and $V(s)$ as follows:

```
P(s): Pb(Xb);
s=s-1
if(s<0)
{
Vb(Xb);
Pb(Yb);
}
else Vb(Xb);
V(s): Pb(Yb);
s= s +1
if(s<=0)
Vb(Yb);
```

The initial value of Xb and Yb are, respectively,

(a) 0 and 0 (b) 0 and 1
(c) 1 and 0 (d) 1 and 1

18. Which one of the following is not a key process area in CMM level 5?

(a) Defect prevention
(b) Process change management
(c) Software product engineering
(d) Technology change management

19. The options field of IPv4 is used for

(a) Time stamping, strict source routing
(b) loose source routing, strict source routing
(c) time stamping only
(d) time stamping, loose source routing, strict source routing

20. Two dice are thrown simultaneously. What is the probability of getting a total of at least 10?

(a) 1/6 (b) 2/3
(c) 5/36 (d) 3/7

21. Which one of the following is NOT logically equivalent to $\neg\exists x(\forall y(\alpha) \wedge \forall z(\beta))$?

(a) $\forall x(\exists y(\neg\alpha) \rightarrow \forall z(\beta))$
(b) $\forall x(\forall z(\beta) \rightarrow \exists y(\neg\alpha))$

(c) $\forall x(\forall y(\alpha) \to \exists z(\neg\beta))$

(d) $\forall x(\exists y(\neg\alpha) \to \exists z(\neg\beta))$

22. Which of the following statements is false?

A. The 2's complement of 0 is 0.
B. In 2's complement, the leftmost bit cannot be used to express a quantity.
C. For an n-bit word (2's complement) which includes the sign bit, there are $2n - 1$ positive integers, $2n + 1$ negative integers and one 0 for a total of $2n$ unique states.
D. In 2's complement, the significant information is contained in the 1's of positive numbers and 0's of the negative numbers.

(a) A (b) B
(c) C (d) D

23. Which of the following affects the processing power of the CPU?

(a) Data bus, addressing schemes
(b) Clock speed, addressing schemes
(c) Clock speed, data bus
(d) Clock speed, data bus, addressing schemes

24. The queue data structure is to realised by using stack. The number of stacks needed would be

(a) It cannot be implemented
(b) 2 stacks
(c) 4 stacks
(d) 1 stacks

25. Let $L = L_1 \cap L_2$, where L_1 and L_2 are languages defined below

$L_1 = \{a^m b^m c a^n b^m \mid m, n \geq 0\}$ and $L_2 = \{a^i b^j c^k \mid i, j, k \geq 0\}$.
Then L is

(a) regular but not recursive
(b) context-free
(c) context-free but not regular
(d) recursively enumerable but not context-free

26. Which grammar rules violate the requirement of the operator grammar? A, B, C are variables and a, b, c are terminals.

I. $A \to BC$ II. $A \to CcBb$
III. $A \to BaC$ IV. $A \to \varepsilon$

(a) I only (b) I and II
(c) I and III (d) I and IV

27. In which stage of database design, all the necessary fields and their types of a database are listed?

(a) Data definition
(b) Data field definition
(c) E-R diagram
(d) User definition

28. Which of the following system components is responsible for ensuring that the system is working to fulfil its objective?

(a) Input (b) Output
(c) Feedback (d) Control

29. In a fully connected mesh network with d devices and c connections, there are ________ physical channels to link all devices.

(a) $d*(d - 1)/2$ (b) $d*(d + 1)/2*c$
(c) $(2*d + 2*c)/2$ (d) $2*(d + 1) + c$

30. Chances of solving a question by A, B and C are 1/2, 1/3 and 1/4, respectively. What is the probability that the question will be solved?

(a) 1/4 (b) 2/3
(c) 1/2 (d) 3/4

31. Consider the following logical inferences:

I_1: If it rains, then the cricket match will not be played.
The cricket match was played.
Inference: There was no rain.
I_2: If it rains, then the cricket match will not be played.
It did not rain.
Inference: The cricket match was played.
Which of the following is TRUE?

(a) Both I1 and I2 are correct inferences.
(b) I1 is correct but I2 is not a correct inference.
(c) I1 is not correct but I2 is a correct inference.
(d) Both I1 and I2 are not correct inferences.

32. The dual of any Boolean expression is obtained by

(a) interchanging POS and SOP terms
(b) interchanging 0's and 1's
(c) interchanging Boolean sums and Boolean products and also interchanging 0's and 1's
(d) interchanging Boolean sums and Boolean products

33. The contents of the flag register after execution of the following program by 8085 microprocessor will be:

Program
```
SUB A
MVI B, (01)H
DCR B
HLT
```

(a) $(54)_H$ (b) $(00)_H$
(c) $(01)_H$ (d) $(45)_H$

34. The time complexity of searching an element in a linked list of length n will be

(a) $O(n)$ (b) $O(n^2 - 1)$
(c) $O(\log n)$ (d) $O(\log n^2)$

35. Which of the following is/are correct?

I. A language is context-free if and only if it is accepted by the PDA.
II. PDA is a finite automata with push-down stack.

(a) Only I is true
(b) Only II is true
(c) Both I and II are true
(d) Both I and II are false

36. Consider the following grammar:

$$S \to (L)|a$$
$$L \to SL'$$
$$L' \to \varepsilon \mid + SL'$$

In the predictive parser table M of the grammar, the entries $M[S, (]$ and $M[L',)]$, respectively,

(a) $S \to (L)$ and $L' \to ,SL'$ (b) $S \to a$ and $L' \to \varepsilon$
(c) $S \to (L)$ and $L' \to \varepsilon$ (d) $S \to a$ and $L' \to ,SL'$

37. The maximum length of an attribute of type text is

(a) 127 (b) 255
(c) 256 (d) It is a variable

38. Modifying the software to match changes in the ever-changing environment is called

(a) adaptive maintenance
(b) corrective maintenance
(c) perfective maintenance
(d) preventive maintenance

39. The firewall is used for

(a) sensing the size of the packet
(b) isolating intranet from extranet
(c) screening packets to/from the network and filtering of network traffic
(d) isolating Internet from virtual LAN

40. If A and B are two events such that $P(A) = 0.3$, $P(B) = 0.6$ and $P(B/A) = 0.5$, then what is the value of $P(A \cup B)$.

(a) 0.50 (b) 0.75
(c) 0.67 (d) 0.33

41. A function $f(x)$ is defined for five different values of (x) at $x = 1, 2, 3, 4$ and 5. It is desired to determine $f'(x)$ at $x = 3$. One of the following formulas would preferably be used for computation:

(a) Stirling's formula
(b) Newton's backward formula
(c) Newton's forward formula
(d) None of these

42. A sequential circuit gives an output depending upon

(a) its present state inputs only.
(b) its present-1 state input only.
(c) both present- and past-state input.
(d) the size of the memory.

43. An N-bit carry look-ahead adder, where N is a multiple of 4, employs ICS 74181 (40-bit ALU) and 74182 (4-bit carry look-ahead generator).

The minimum addition time using the best architecture for adder is

(a) proportional to N
(b) proportional to $\log N$
(c) a constant
(d) None of these

44. A hashing algorithm uses linear probing; the average search time will be less if the load factor

(a) is less than one (b) equals one
(c) is greater than one (d) equals two

45. How many strings of length less than 4 contains the language described by the regular expression

$$(x + y)^*y(a + ab)^*?$$

(a) 3 (b) 9
(c) 10 (d) 11

46. Windows 2000 operating system include the following process states:

(a) Ready, running and waiting
(b) Ready, standby, running, waiting, transition and terminated
(c) Ready, running, waiting, transition, sleep and terminated
(d) Standby, running, transition, sleep and terminated

47. When a transaction has completed all its operations and is waiting for commit or rollback action, the state of transaction is

(a) partially commit
(b) ready commit
(c) half commit
(d) commit and rollback

48. Software measurement is useful to

(a) indicate quality of the product
(b) track progress
(c) form a baseline for estimation and prediction
(d) all of these

49. LDAP stands for
- (a) lightweight digital audio protocol
- (b) lightweight directory access protocol
- (c) large domain access protocol
- (d) large data audio protocol

50. What is the variance of the number of heads in a simultaneous toss of three coins?
- (a) 3/2 (b) 1/2
- (c) 3/5 (d) 3/4

51. By applying the principle of temporal locality, processes are likely to reference pages that _____.
- (a) have been referenced recently.
- (b) are located at address near recently referenced pages in memory.
- (c) have been preloaded into memory.
- (d) have to be reloaded into memory.

52. Priority is provided by _______ for access to memory by various I/O channels and processors.
- (a) a register
- (b) a counter
- (c) the processor scheduler
- (d) a controller

53. In tree construction, which one will be suitable and efficient data structure?
- (a) Queue (b) Linked list
- (c) Heap (d) String

54. If a node has K children in B tree, then the node contains exactly ______ keys.
- (a) K^2 (b) $K-1$
- (c) $K+1$ (d) $\sqrt{K}$

55. Consider the following laws:

$$L_1 : (L^*)^* = L^*$$
$$L_2 : \varnothing^* = \varnothing$$
$$L_3 : \varepsilon^* = \varepsilon$$
$$L_4 : L^+ = L^* + \varepsilon$$
$$L_5 : L^* = \varepsilon + L^+$$

Which of the above laws hold for regular expression?
- (a) L_1, L_2 and L_5 (b) L_1, L_3 and L_5
- (c) L_2, L_3 and L_4 (d) L_1, L_4 and L_5

56. A memory management algorithm, realizing virtual memory, partially swaps out a process. This is similar to which kind of CPU scheduling?
- (a) Short-term scheduling
- (b) Long-term scheduling
- (c) Medium-term scheduling
- (d) Mutual exclusion

57. Which of the following statement is not false?
- (a) Time stamp protocols avoid deadlock.
- (b) Locking technique is used to avoid deadlock.
- (c) Two Phase locking does not provide serializability.
- (d) Two Phase deals with input and storing phase.

58. The minimum size of an Ethernet frame is
- (a) 18 bytes (b) 64 bytes
- (c) 46 bytes (d) 56 bytes

59. A digital signature used to provide security, makes use of
- (a) digitally scanned signatures
- (b) a unique ASCII code number of the sender
- (c) private key encryption
- (d) public key encryption

60. If it is desired to find derivative of a function for a given data near the beginning of the table, one would preferably employ
- (a) Stirling's formula
- (b) Newton's backward formula
- (c) Newton's forward formula
- (d) None of these

61. Which of the following is a correct statement related to $L2$ cache memory?
- (a) The level 1 cache is always faster than the level 2 cache.
- (b) The level 2 cache is used to mitigate the dynamic slowdown every time a level 1 cache miss occurs.
- (c) Level 2 cache comes as on board only.
- (d) In modern day computer, the level 2 cache is considered an internal cache.

62. Which of the following is not true about spanning tree?
- (a) It is a tree derived from a graph
- (b) All the nodes of a network appear on the tree only once
- (c) Spanning tree cannot have at most two edges repeated
- (d) Spanning tree cannot be minimum and maximum

63. Big-O estimate for the factorial function, that is, $n!$, is given by
- (a) $O(n!)$ (b) $O(n^n)$
- (c) $O(n \times n!)$ (d) $O(\sqrt{n})$

64. In a compiler the module that checks every character of the source text is called
- (a) the machine-dependant optimizer
- (b) the code checker
- (c) the lexical analyzer
- (d) the syntax analyzer

65. The working set model is used in memory management to implement the concept of

(a) segmentation (b) principle of locality
(c) paging (d) thrashing

66. Data integrity control makes use of ___________ for maintaining the integrity of database.

(a) specific alphabets (uppercase and lowercase alphabets)
(b) passwords
(c) relational algebra
(d) storing on a backup hard disk

67. Using a 7-bit sequence number, what is the maximum size (in bits) of the sender and receiver window using stop-and-wait protocols?

(a) 1 and 1 (b) 1 and 7
(c) 7 and 1 (d) 7 and 7

68. The <b> tag makes the enclosed text bold. The other alternative is to use

(a) <strong> (b) <dark>
(c) <big> (d) <highblack>

69. The searching technique in which less number of comparisons are done is

(a) binary searching
(b) linear search searching
(c) hashing
(d) Woline's search

70. What are the values of x and y, if $A = \begin{bmatrix} 3 & 0 & 0 \\ 5 & x & 0 \\ 3 & 6 & y \end{bmatrix}$ and eigenvalues of A are 3, 4 and 1?

(a) $x = 2, y = 2,$ (b) $x = 3, y = 2,$
(c) $x = 2, y = 3,$ (d) $x = 4, y = 1$

71. Function points can be calculated by

(a) UFP × CAF (b) UFP × FAC
(c) UFP × Cost (d) UFP × Productivity

72. Which of the following has compilation error in C?

(a) int *n* = 17; (b) char ch = 99;
(c) float *f* = (float) 99.32; (d) #include<stdio.*h*>

73. Let A and B be two $n \times n$ matrices. The efficient algorithm to multiply the two matrices has the time complexity $O(n^x)$, where x is _______.

(a) 1.35 (b) 2.53
(c) 2.81 (d) 3.45

74. Match the following.

Group 1	Group 2
A. Lexical analysis	P. Directed acyclic graphs
B. Code optimization	Q. Syntax trees
C. Code generation	R. PDA
D. Abelian groups	S. Finite automaton

(a) A-S, B-P, C-R, D-Q
(b) A-Q, B-R, C-S, D-P
(c) A-S, B-Q, C-R, D-P
(d) A-P, B-S, C-R, D-Q

75. Which scheduling algorithm gives a minimum average waiting time?

(a) RR (b) SJF
(c) FCFS (d) Priority

76. Data warehouse provides

(a) transaction responsiveness
(b) storage, functionality responsiveness to queries
(c) demand and supply responsiveness
(d) storage of transactions

77. Vulnerable time in CSMA is

(a) It is double the transmission time as it includes the sensing time
(b) half of average frame transmission time as collisions are less
(c) 2 × (the average frame transmission time)
(d) None of these

78. Who describes and controls the making of various Web standards?

(a) The World Wide Web Consortium
(b) IEEE and IETE
(c) Administrator and Mozilla
(d) IEEE

79. Which of the following strings can definitely be said to be token without looking at the next input character while compiling a Pascal program?

I. Begin
II. Goto
III. <>

(a) 1 (b) 2
(c) 3 (d) All of these

80. If the disk is rotating at 3600 rpm, determine the effective data transfer rate which is defined as the number of bytes transferred per second between disk and memory. (Given size of track = 512 bytes)

(a) 20 kbps (b) 30 kbps
(c) 40 kbps (d) 50 kbps

Answer Key

PRACTICE TEST – 1

1.	(b)	**9.**	(a)	**17.**	(c)	**25.**	(c)	**33.**	(b)	**41.**	(b)	**49.**	(c)	**57.**	(d)	**65.**	(c)	**73.**	(b)
2.	(a)	**10.**	(b)	**18.**	(b)	**26.**	(a)	**34.**	(b)	**42.**	(a)	**50.**	(c)	**58.**	(c)	**66.**	(c)	**74.**	(c)
3.	(a)	**11.**	(b)	**19.**	(d)	**27.**	(c)	**35.**	(b)	**43.**	(d)	**51.**	(d)	**59.**	(d)	**67.**	(a)	**75.**	(c)
4.	(d)	**12.**	(a)	**20.**	(c)	**28.**	(b)	**36.**	(d)	**44.**	(d)	**52.**	(a)	**60.**	(d)	**68.**	(a)	**76.**	(a)
5.	(d)	**13.**	(a)	**21.**	(d)	**29.**	(b)	**37.**	(c)	**45.**	(b)	**53.**	(b)	**61.**	(b)	**69.**	(a)	**77.**	(a)
6.	(a)	**14.**	(c)	**22.**	(b)	**30.**	(c)	**38.**	(b)	**46.**	(b)	**54.**	(a)	**62.**	(b)	**70.**	(a)	**78.**	(a)
7.	(d)	**15.**	(b)	**23.**	(c)	**31.**	(a)	**39.**	(c)	**47.**	(b)	**55.**	(b)	**63.**	(a)	**71.**	(d)	**79.**	(b)
8.	(d)	**16.**	(a)	**24.**	(a)	**32.**	(b)	**40.**	(c)	**48.**	(a)	**56.**	(d)	**64.**	(d)	**72.**	(c)	**80.**	(b)

PRACTICE TEST – 2

1.	(c)	**9.**	(b)	**17.**	(d)	**25.**	(a)	**33.**	(d)	**41.**	(a)	**49.**	(d)	**57.**	(c)	**65.**	(d)	**73.**	(b)
2.	(b)	**10.**	(b)	**18.**	(c)	**26.**	(d)	**34.**	(a)	**42.**	(c)	**50.**	(a)	**58.**	(c)	**66.**	(a)	**74.**	(c)
3.	(a)	**11.**	(d)	**19.**	(b)	**27.**	(c)	**35.**	(a)	**43.**	(b)	**51.**	(b)	**59.**	(c)	**67.**	(b)	**75.**	(a)
4.	(c)	**12.**	(b)	**20.**	(d)	**28.**	(c)	**36.**	(c)	**44.**	(a)	**52.**	(d)	**60.**	(d)	**68.**	(b)	**76.**	(c)
5.	(d)	**13.**	(b)	**21.**	(b)	**29.**	(b)	**37.**	(d)	**45.**	(c)	**53.**	(a)	**61.**	(c)	**69.**	(c)	**77.**	(b)
6.	(d)	**14.**	(a)	**22.**	(c)	**30.**	(a)	**38.**	(c)	**46.**	(c)	**54.**	(c)	**62.**	(c)	**70.**	(b)	**78.**	(a)
7.	(a)	**15.**	(c)	**23.**	(d)	**31.**	(c)	**39.**	(b)	**47.**	(d)	**55.**	(b)	**63.**	(b)	**71.**	(a)	**79.**	(a)
8.	(b)	**16.**	(b)	**24.**	(b)	**32.**	(d)	**40.**	(c)	**48.**	(c)	**56.**	(c)	**64.**	(c)	**72.**	(a)	**80.**	(b)

PRACTICE TEST – 3

1.	(a)	**9.**	(a)	**17.**	(c)	**25.**	(c)	**33.**	(a)	**41.**	(a)	**49.**	(b)	**57.**	(a)	**65.**	(b)	**73.**	(c)
2.	(d)	**10.**	(d)	**18.**	(c)	**26.**	(d)	**34.**	(a)	**42.**	(c)	**50.**	(d)	**58.**	(b)	**66.**	(a)	**74.**	(a)
3.	(a)	**11.**	(d)	**19.**	(d)	**27.**	(a)	**35.**	(c)	**43.**	(c)	**51.**	(a)	**59.**	(d)	**67.**	(a)	**75.**	(b)
4.	(c)	**12.**	(a)	**20.**	(a)	**28.**	(d)	**36.**	(c)	**44.**	(a)	**52.**	(d)	**60.**	(c)	**68.**	(a)	**76.**	(a)
5.	(c)	**13.**	(c)	**21.**	(d)	**29.**	(a)	**37.**	(b)	**45.**	(d)	**53.**	(b)	**61.**	(a)	**69.**	(c)	**77.**	(d)
6.	(b)	**14.**	(a)	**22.**	(c)	**30.**	(d)	**38.**	(a)	**46.**	(b)	**54.**	(b)	**62.**	(d)	**70.**	(d)	**78.**	(a)
7.	(d)	**15.**	(b)	**23.**	(d)	**31.**	(b)	**39.**	(c)	**47.**	(a)	**55.**	(b)	**63.**	(b)	**71.**	(a)	**79.**	(c)
8.	(c)	**16.**	(b)	**24.**	(b)	**32.**	(c)	**40.**	(b)	**48.**	(d)	**56.**	(c)	**64.**	(c)	**72.**	(d)	**80.**	(b)

Answers with Explanation

PRACTICE TEST – 1

1. **Topic: Engineering Mathematics**

   (b) Given:

   Total number of houses available = 3

   Probability of applying for a particular house = $\frac{1}{3}$

   Probability that all apply for some house is

$$\left(\frac{1}{3}\times\frac{1}{3}\times\frac{1}{3}\right)\times 3 = \frac{1}{9}$$

2. **Topic: Engineering Mathematics**

   (a) We have:

$$\text{Total outcomes} = 6$$
$$\text{Favorable outcomes} = 2, 3, 4, 5$$

   Therefore, probability that the minimum face value is not less than 2 and the maximum face value is not greater than 5 is:

$$\frac{4}{6} = \frac{2}{3}$$

   Since, the dice is rolled 4 times.
   So, required probability is:

$$\left(\frac{2}{3}\right)^4 = \frac{16}{81}$$

3. **Topic: Engineering Mathematics**

   (a) We have:

   For $\theta = 45°$, $\sin 45° + \cos 45° = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2} = 1.414$

   For $\theta = 30°$, $\sin 30° + \cos 30° = \frac{1}{2} + \frac{\sqrt{3}}{2} = \frac{1+\sqrt{3}}{2} = 1.366$

   For $\theta = 90°$, $\sin 90° + \cos 90° = 1 + 0 = 1$

   For $\theta = 60°$, $\sin 60° + \cos 60° = \frac{\sqrt{3}}{2} + \frac{1}{2} = \frac{\sqrt{3}+1}{2} = 1.366$

   Therefore, for $\theta = 45°$, $\sin\theta + \cos\theta$ will be greatest.

4. **Topic: Computer Networks**

   (d) IPv6 address is expressed by eight 4 digit hexadecimal numbers, which represent 16 bits each.

   Therefore, the total length of the address is

$$8 \times 16 = 128 \text{ bits}$$

5. **Topic: Computer Networks**

   (d) Port Address Translation (PAT) or NAT overloading, and maps multiple private IP addresses to a single public IP address. Multiple addresses can be mapped to a single address because each private address is tracked by a port number. PAT uses unique source port numbers on the inside global IP address to distinguish between translations. The port number is 16-bit integers. The total number of internal addresses that can be translated to one external address could theoretically be as high as 65,536 per IP address. Realistically, the number of ports that can be assigned a single IP address is around 4000. PAT attempts to preserve the original source port. If this source port is already used, PAT assigns the first available port number starting from the beginning of the appropriate port group 0-511, 512-1023, or 1024-65535. When there are no more ports available and there is more than one external IP address configured, PAT moves to the next IP address to try to allocate the original source port again. This process continues until it runs out of available ports and external IP addresses.

6. **Topic: Computer Networks**

   (a) An access control list (ACL), with respect to a computer file system, is a list of permissions attached to an object. An ACL specifies which users or system processes are granted access to objects, as well as what operations are allowed on given objects. Each entry in a typical ACL specifies a subject and an operation.

7. **Topic: Computer Networks**

   (d) A stub network, or pocket network, is a somewhat casual term describing a computer network, or part of an internetwork, with no knowledge of other networks, that will typically send much or all of its non-local traffic out via a single path, with the network aware only of a default route to non-local destinations.

8. **Topic: Programming and Data Structures**

   (d) Pascal is an imperative and procedural programming language; it was developed on the pattern of the ALGOL

60 language. Wirth had already developed several improvements to this language as part of the ALGOL X proposals, but these were not accepted and Pascal was developed separately and released in 1970. A derivative known as Object Pascal designed for object-oriented programming was developed in 1985; this was used by Apple Computer and Borland in the late 1980s and later developed into Delphi on the Microsoft Windows platform. Extensions to the Pascal concepts led to the Pascal-like languages Modula-2 and Oberon.

9. **Topic: Digital Logic**

   (a) 1's complement is obtained by converting all 1 to 0 and all 0 to 1.
   Therefore, 1's complement of 10 is 01.

10. **Topic: Operating System**

    (b) An instruction cycle also known as CPU cycle is the basic operational process of a computer. It is the process by which a computer retrieves a program instruction from its memory, determines what actions the instruction dictates, and carries out those actions. This cycle is repeated continuously by a computer's central processing unit (CPU), from boot-up to when the computer is shut down.

11. **Topic: Computer Networks**

    (b) A frame is a part of a web page or browser window which displays content independent of its container, with the ability to load content independently. The HTML or media elements that go in a frame may or may not come from the same web site as the other elements of content on display.
    In HTML, a frameset is a group of named frames to which web pages and media can be directed; an iframe provides for a frame to be placed inside the body of a document.

12. **Topic: Programming and Data Structures**

    (a) ASCII abbreviated from American Standard Code for Information Interchange, is a character encoding standard for electronic communication. ASCII codes represent text in computers, telecommunications equipment, and other devices. Most modern character-encoding schemes are based on ASCII, although they support many additional characters.
    ASCII is the traditional name for the encoding system; the Internet Assigned Numbers Authority (IANA) prefers the updated name US-ASCII, which clarifies that this system was developed in the US and based on the typographical symbols predominantly in use there.

13. **Topic: Digital Logic**

    (a) An encoder is a device, circuit, transducer, software program, algorithm or person that converts information from one format or code to another, for the purposes of standardization, speed or compression.

14. **Topic: Computer Organization and Architecture**

    (c) The control unit (CU) is a component of a computer's central processing unit (CPU) that directs the operation of the processor. It tells the computer's memory, arithmetic/logic unit and input and output devices how to respond to a program's instructions.
    It directs the operation of the other units by providing timing and control signals. Most computer resources are managed by the CU. It directs the flow of data between the CPU and the other devices. John von Neumann included the control unit as part of the von Neumann architecture. In modern computer designs, the control unit is typically an internal part of the CPU with its overall role and operation unchanged since its introduction.

15. **Topic: Computer Organization and Architecture**

    (b) The fundamental operation of most CPUs, regardless of the physical form they take, is to execute a sequence of stored instructions that is called a program. The instructions to be executed are kept in some kind of computer memory. Nearly all CPUs follow the fetch, decode and execute steps in their operation, which are collectively known as the instruction cycle.
    After the execution of an instruction, the entire process repeats, with the next instruction cycle normally fetching the next-in-sequence instruction because of the incremented value in the program counter. If a jump instruction was executed, the program counter will be modified to contain the address of the instruction that was jumped to and program execution continues normally. In more complex CPUs, multiple instructions can be fetched, decoded, and executed simultaneously. This section describes what is generally referred to as the "classic RISC pipeline", which is quite common among the simple CPUs used in many electronic devices (often called microcontroller). It largely ignores the important role of CPU cache, and therefore the access stage of the pipeline.
    Some instructions manipulate the program counter rather than producing result data directly; such instructions are generally called "jumps" and facilitate program behavior like loops, conditional program execution (through the use of a conditional jump), and existence of functions. In some processors, some other instructions change the state of bits in a "flags" register. These flags can be used to influence how a program behaves, since they often indicate the outcome of various operations. For example, in such processors a "compare" instruction evaluates two values and sets or clears bits in the flags register to indicate which one is greater or whether they are equal; one of these flags could then be used by a later jump instruction to determine program flow.

16. **Topic: Digital Logic**

    (a) Truth table of NOT gate is as shown below:

Input	Output
0	1
1	0

Therefore, for input electrical signal $A = 10100$, output is as listed below:

A	**Output**
1	0
0	1
1	0
0	1
0	1

An inverter circuit outputs a voltage representing the opposite logic-level to its input. Its main function is to invert the input signal applied. If the applied input is low then the output becomes high and vice versa. Inverters can be constructed using a single NMOS transistor or a single PMOS transistor coupled with a resistor. Since this 'resistive-drain' approach uses only a single type of transistor, it can be fabricated at low cost. However, because current flows through the resistor in one of the two states, the resistive-drain configuration is disadvantaged for power consumption and processing speed. Alternatively, inverters can be constructed using two complementary transistors in a CMOS configuration. This configuration greatly reduces power consumption since one of the transistors is always off in both logic states. Processing speed can also be improved due to the relatively low resistance compared to the NMOS-only or PMOS-only type devices. Inverters can also be constructed with bipolar junction transistors (BJT) in either a resistor–transistor logic (RTL) or a transistor–transistor logic (TTL) configuration.

17. Topic: Computer Networks

(c) Given:

IP address = 172.16.2.1/23 $\Rightarrow$ 255.255.254.0

Thus third octet is of block size = 2.
Router's interface is in 2.0 subnet and the broadcast address is 3.255 because the next subnet is 4.0.
The valid host range is 2.1 through 3.254.
The router is using the first valid host address in the range.

18. Topic: Computer Networks

(b) A network bridge is a computer networking device that creates a single aggregate network from multiple communication networks or network segments. This function is called network bridging. Bridging is distinct from routing, as routing allows multiple different networks to communicate independently while remaining separate whilst bridging connects two separate networks as if they are only one network (hence, the name "bridging"). In the OSI model, bridging is performed in the first two layers, below the network layer (layer 3). If one or more segments of the bridged network are wireless, the device is known as a wireless bridge and the function as wireless bridging.

19. Topic: Databases

(d) Logical data models represent the abstract structure of a domain of information. They are often diagrammatic in nature and are most typically used in business processes that seek to capture things of importance to an organization and how they relate to one another. Once validated and approved, the logical data model can become the basis of a physical data model and form the design of a database. Logical data models should be based on the structures identified in a preceding conceptual data model, since this describes the semantics of the information context, which the logical model should also reflect. Even so, since the logical data model anticipates implementation on a specific computing system, the content of the logical data model is adjusted to achieve certain efficiencies.
The term 'Logical Data Model' is sometimes used as a synonym of 'domain model' or as an alternative to the domain model. While the two concepts are closely related, and have overlapping goals, a domain model is more focused on capturing the concepts in the problem domain rather than the structure of the data associated with that domain.

20. Topic: Digital Logic

(c) Given:

$$(61)_{10}$$

Now, convert $(61)_{10}$ decimal numbers into binary number, we get:

$$\begin{array}{r|r|l} 2 & 61 & 1 \\ 2 & 30 & 0 \\ 2 & 15 & 1 \\ 2 & 7 & 1 \\ 2 & 3 & 1 \\ & 1 & \end{array}$$

Therefore,

$$(61)_{10} = (111101)_2$$

21. Topic: Digital Logic

(d) Let two inputs A and B.
Now, inverting the inputs, we get:

$$\overline{A} \text{ and } \overline{B}$$

Now, taking AND of both inverting inputs, we get:

$$\overline{A} \cdot \overline{B} = \overline{(A + B)}$$

The logic function obtained is NOR.

22. Topic: Operating System

(b) A communication channel or simply channel refers either to a physical transmission medium such as a wire, or to a logical connection over a multiplexed medium such as a radio channel in telecommunications and computer networking. A channel is used to convey an information signal, for example a digital bit stream, from one or several senders (or transmitters) to one or several receivers. A channel has a certain capacity for transmitting information, often measured by its bandwidth in Hz or its data rate in bits per second.

Communicating data from one location to another requires some form of pathway or medium. These pathways, called communication channels, use two types of media: cable (twisted-pair wire, cable, and fiber-optic cable) and broadcast (microwave, satellite, radio, and infrared). Cable or wire line media use physical wires of cables to transmit data and information. Twisted-pair wire and coaxial cables are made of copper, and fiber-optic cable is made of glass.

23. Topic: Operating System

(c) A windowing system (or window system) is software that manages separately different parts of display screens. It is a type of graphical user interface (GUI) which implements the WIMP (windows, icons, menus, and pointer) paradigm for a user interface.

Each currently running application is assigned a usually resizable and usually rectangular shaped surface of the display to present its graphical user interface to the user; these windows may overlap each other, as opposed to a tiling interface where they are not allowed to overlap. Usually a window decoration is drawn around each window. The programming of both the window decoration and of available widgets inside of the window, which are graphical elements for direct user interaction, such as sliders, buttons, etc., is eased and simplified through the use of widget toolkits.

24. Topic: Computer Networks

(a) Ethernet over twisted pair technologies uses twisted-pair cables for the physical layer of an Ethernet computer network. They are a subset of all Ethernet physical layers.

Early Ethernet had used various grades of coaxial cable, but in 1984, StarLAN showed the potential of simple unshielded twisted pair. This led to the development of 10BASE-T and its successors 100BASE-TX, 1000BASE-T and 10GBASE-T, supporting speeds of 10, 100 Mbit/s and 1 and 10 Gbit/s respectively.

25. Topic: Programming and Data Structures

(c) An algorithm is an unambiguous specification of how to solve a class of problems. Algorithms can perform calculation, data processing and automated reasoning tasks. An algorithm is an effective method that can be expressed within a finite amount of space and time and in a well-defined formal language for calculating a function. Starting from an initial state and initial input (perhaps empty), the instructions describe a computation that, when executed, proceeds through a finite number of well-defined successive states, eventually producing "output" and terminating at a final ending state. The transition from one state to the next is not necessarily deterministic; some algorithms, known as randomized algorithms, incorporate random input.

26. Topic: Computer Organization and Architecture

(a) Read-only memory (ROM) is a type of non-volatile memory used in computers and other electronic devices. Data stored in ROM can only be modified slowly, with difficulty, or not at all, so it is mainly used to store firmware (software that is closely tied to specific hardware, and unlikely to need frequent updates) or application software in plug-in cartridges.

Strictly, read-only memory refers to memory that is hard-wired, such as diode matrix and the later mask ROM (MROM), which cannot be changed after manufacture. Although discrete circuits can be altered in principle, integrated circuits (ICs) cannot, and are useless if the data is bad or requires an update. That such memory can never be changed is a disadvantage in many applications, as bugs and security issues cannot be fixed, and new features cannot be added.

27. Topic: Digital Logic

(c) A multiplexer (or mux) is a device that selects one of several analog or digital input signals and forwards the selected input into a single line. A multiplexer of 2^n inputs has n select lines, which are used to select which input line to send to the output. Multiplexers are mainly used to increase the amount of data that can be sent over the network within a certain amount of time and bandwidth. A multiplexer is also called a data selector. Multiplexers can also be used to implement Boolean functions of multiple variables.

28. Topic: Computer Networks

(b) SMTP, HTTP, FTP are service use TCP.

29. Topic: Computer Organization and Architecture

(b) Dynamic random-access memory (DRAM) is a type of random access semiconductor memory that stores each bit of data in a separate tiny capacitor within an integrated circuit. The capacitor can either be charged or discharged; these two states are taken to represent the two values of a bit, conventionally called 0 and 1.

30. Topic: Digital Logic

(c) The serial binary adder or bit-serial adder is a digital circuit that performs binary addition bit by bit. The serial full adder has three single-bit inputs for the numbers to be added and the carry in. There are two single-bit outputs for the sum and carry out. The carry-in signal is the previously calculated carry-out signal. The addition is performed by adding each bit, lowest to highest, one per clock cycle.

31. Topic: Computer Organization and Architecture

(a) An index register in a computer's CPU is a processor register used for modifying operand addresses during the run of a program, typically for doing vector/array operations.

The contents of an index register is added to (in some cases subtracted from) an immediate address (one that is part of the instruction itself) to form the "effective" address of the actual data (operand). Special instructions are typically provided to test the index register and, if the test fails, increments the index register by an immediate constant and branches, typically to the start of the loop. Some instruction sets allow more than one index register to be used; in that case additional instruction fields specify which index registers to use. While normally processors that allow an instruction to specify multiple index registers add the contents together, IBM had a line of computers in which the contents were or'd together.

32. Topic: Computer Organization and Architecture

(b) A motherboard (sometimes alternatively known as the mainboard, system board, baseboard, planar board or logic board, or colloquially, a mobo) is the main printed circuit board (PCB) found in general purpose microcomputers and other expandable systems. It holds and allows communication between many of the crucial electronic components of a system, such as the central processing unit (CPU) and memory, and provides connectors for other peripherals. Unlike a backplane, a motherboard usually contains significant sub-systems such as the central processor, the chipset's input/output and memory controllers, interface connectors, and other components integrated for general purpose use.

33. Topic: Computer Organization and Architecture

(b) A division of a storage medium on a hard drive or diskette that is a wedge shaped section of one of the circular tracks. Each arc is a sector that typically holds 512 bytes of data and is given a sector number for interleaving purposes, so the term sector may refer to the entire single arc. The size of sectors can be customized to maximize the storage area. For example, if a user stores smaller files, decreasing the sector size allows more files to fill the space without any leftover room. The illustration gives an example of what the sector would look like on a disk platter.

34. Topic: Digital Logic

(b) Given:

$$(1101)_2 \tag{1}$$

Now, 1's complement of $(1101)_2$ is

$$(0010)_2 \tag{2}$$

Now, Add Eq. (1) with Eq. (2):

$$\begin{array}{r} 1101 \\ +0010 \\ \hline 1100 \end{array}$$

Now, converting all 1 to 0 and all 0 to 1 of the above result, we get

$$(0011)_2$$

35. Topic: Engineering Mathematics

(b) Pipe A can fill a tank in $37\frac{1}{2}$ min $= \frac{75}{2}$ min

Part filled by pipe A in 1 min $= \frac{2}{75}$

Pipe B can fill a tank in 45 min

Part filled by pipe B in 1 min $= \frac{1}{45}$

Part filled by Pipe A and B in 1 min $= \frac{2}{75} + \frac{1}{45} = \frac{6+5}{225} = \frac{11}{225}$

Assume that B is turned off after x min. That is, for x min, both pipe A and B were open.

Part filled in x min by Pipe A and $B = x \times \frac{11}{225} = \frac{11x}{225}$

Now, the cistern must be filled in $(30 - x)$ min by pipe A alone.

Part filled in $(30 - x)$ min by pipe A

$$(30-x)\times\frac{2}{75} = \frac{2(30-x)}{75}$$

$$\frac{11x}{225} + \frac{2(30-x)}{75} = 1 \Rightarrow \frac{11x}{225} + \frac{2(30-x)}{75} = 1$$

$$\Rightarrow 11x + 6(30-x) = 225 \Rightarrow 11x + 180 - 6x = 225$$

$$\Rightarrow 5x = 45 \Rightarrow x = 9$$

36. Topic: Computer Organization and Architecture

(d) In computer hardware, a port serves as an interface between the computer and other computers or peripheral devices. In computer terms, a port generally refers to the part of connection available for connection between one computer to peripherals like input and output ones. Computer ports have many uses, to connect a monitor, webcam, speakers, or other peripheral devices.

37. Topic: Computer Networks

(c) Ether LAN uses Manchester encoding because using Manchester encoding has a nice advantage, self-clocking (lower error rate and a more reliable transmission). This is because rather at looking at the +5 V to 0 V to encode a bit, it will depend on the direction of a transmission how a bit is encoded.

38. Topic: Engineering Mathematics

(b) We have the total number of events as

$$n(S) = (6 \times 6) = 36$$

Let E be the event that the sum is a prime number.
Then,

$$E = \{(1, 1), (1, 2), (1, 4), (1, 6), (2, 1), (2, 3), (2, 5), (3, 2), (3, 4), (4, 1), (4, 3), (5, 2), (5, 6), (6, 1), (6, 5)\}$$

So,

$$n(E) = 15$$

Therefore, the required probability is

$$P(E) = \frac{n(E)}{n(S)} = \frac{15}{36} = \frac{5}{12}$$

39. Topic: Computer Networks

(c) Given:

Sender window size is $n - 1$

and

Receiver window size is 1

The maximum size is $2^n - 1$
Here, $n = 4$

$$2^4 - 1 = 16 - 1 = 15$$

40. Topic: Operating System

(c) Many of the fastest computing applications are ECL logic based machines, making BiCMOS an attractive choice for densities above the practical limits of bipolar ECL memory, Internally synchronous memories may employ dynamic logic techniques for speed.

41. Topic: Operating System

(b) Dynamic memory refreshes periodically so due to refreshment it is difficult to interface it.

42. Topic: Digital Logic

(a) Given:

$$(187)_{10}$$

Now, convert decimal to binary:

$$\begin{array}{r|c|l} 2 & 187 & 1 \\ 2 & 93 & 1 \\ 2 & 46 & 0 \\ 2 & 23 & 1 \\ 2 & 11 & 1 \\ 2 & 5 & 1 \\ 2 & 2 & 0 \\ & 1 & \end{array}$$

Therefore,

$$(187)_{10} = (10111011)_2$$

43. Topic: Computer Organization and Architecture

(d) Direct memory access (DMA) is a feature of computer systems that allows certain hardware subsystems to access main system memory (random-access memory), independent of the central processing unit (CPU).

44. Topic: Engineering Mathematics

(d) We may have (3 men and 2 women) or (4 men and 1 woman) or (5 men only).
Therefore, the required number of ways is

$$({}^7C_3 \times {}^6C_2) + ({}^7C_4 \times {}^6C_1) + ({}^7C_5)$$
$$= \left(\frac{7!}{3! \times 4!} \times \frac{6!}{2! \times 4!}\right) + \left(\frac{7!}{4! \times 3!} \times \frac{6!}{1! \times 5!}\right) + \left(\frac{7!}{5! \times 2!}\right)$$
$$= 525 + 210 + 21 = 756$$

45. Topic: Software Engineering

(b) Due to large amount of data management, most ERP systems are multi-user. In this situation, the data are both integrated and shared. A database is integrated when the same information is not recorded in two places. For example, both the Library department and the Account department of the college database may need student addresses. Even though both departments may access different portions of the database, the students' addresses should only reside in one place.

46. Topic: Computer Networks

(b) Virtual Trunk Protocol (VTP) is used to pass a VLAN database to any or all switches in the switched network. The three VTP modes are server, client, and transparent.

47. Topic: Digital Logic

(b) Full Adder is the adder which adds three inputs and produces two outputs. The first two inputs are A and B and the third input is an input carry as C-IN. The output carry is designated as C-OUT and the normal output is designated as S which is SUM. A full adder logic is designed in such a manner that can take eight inputs together to create a byte-wide adder and cascade the carry bit from one adder to the another.

48. Topic: Computer Networks

(a) The mask is the mask used on the remote network, not the source network. Since there is no number at the end of the static route, it is using the default administrative distance of 1.

49. Topic: Programming and Data Structures

(c) A computer program is a passive collection of instructions while a process is the actual execution of those instructions. Several processes may be associated with the same program; for example, opening up several instances of the same program often means more than one process is being executed.

50. Topic: Digital Logic

(c) A multiplexer (or mux) is a device that selects one of several analog or digital input signals and forwards the selected input into a single line.

51. Topic: Digital Logic

(d) The process of jointly establishing communication is called handshaking

52. Topic: Digital Logic

(a) A basic Binary Adder circuit can be made from standard AND and Ex-OR gates allowing us to "add" together two single bit binary numbers, A and B. The addition of these two digits produces an output called the SUM of the addition and a second output called the CARRY or Carry-out, (COUT) bit according to the rules for binary addition. One of the main uses for the Binary Adder is in arithmetic and counting circuits.

53. Topic: Computer Networks

(b) To make an antenna more directional, it must have a small or narrow beam width. Since beam width is directional to "lambda" or wavelength, that means it is inversely proportional to the frequency. Therefore, increasing its frequency will give it a narrow beam width thus, making the antenna more directional.

54. Topic: Computer Networks

(a) Directional couplers are four-port circuits where one port is isolated from the input port. Directional couplers are passive reciprocal networks,

55. Topic: Operating System

(b) Operating system provides qa layered, user-friendly Linking of subroutine is the function of linker not OS.
Enabling the programmer to draw a flowchart is the function of application software's.

56. Topic: Computer Networks

(d) Transponder function: Transmit & receive signal for up-link and down-link frequency respectively.

57. Topic: Computer Networks

(d) For extra-terrestrial materials all the three are required.

58. Topic: Computer Networks

(c) Parallel transmission of digital data means hole digital input transfers at a time. The number of input lines required is same as the number of data lines.
Example: 8-bit parallel in parallel/serial out shift register 8 input lines are connected to all 8-D flip–flops.
But for serial in parallel/serial out has only one input data line is enough.

59. Topic: Digital Logic

(d) Given:

$$(A0)_{16}$$

Now, convert hexadecimal to deciamal number

$$(A0)_{16} = A\times16^1 + 0\times16^0 = 10\times16^1 + 0\times16^0 = (160)_{10}$$

Therefore,

$$(A0)_{16} = (160)_{10}$$

60. Topic: Computer Organization and Architecture

(d) A wrist grounding strap contains resistor. An anti-static wrist strap, ESD wrist strap, or ground bracelet is an antistatic device used to safely ground a person working on electronic equipment, to prevent the buildup of static electricity on their body, which can result in electrostatic discharge (ESD).

61. Topic: Computer Networks

(b) CSMA is a method access control technique for multiple-access.
Carrier-sense multiple access (CSMA) is a media access control (MAC) protocol in which a node verifies the absence of other traffic before transmitting on a shared transmission medium, such as an electrical bus or a band of the electromagnetic spectrum.

62. Topic: Computer Networks

(b) The method for updating the man memory as soon as a word is removed from the cache is called write back.

63. Topic: Computer Organization and Architecture

(a) EPROM contents can be erased by exposing it to Ultraviolet rays The Ultraviolet light passes through a window in the IC package to the EPROM chip where it releases stored charges. Thus the stored contents are erased.

64. Topic: Computer Organization and Architecture

(d) By using 2's complement we can subtract in computers.

65. Topic: Computer Organization and Architecture

(c) JFET is a voltage-controlled device in which current flows from the SOURCE terminal (equivalent to the emitter in a bipolar transistor) to the DRAIN (equivalent to the collector). A voltage applied between the source terminal and a GATE terminal(equivalent to the base) is used to control the source–drain current.

66. Topic: Digital Logic

(c) Given:

$$Y = (A + \overline{C} + \overline{A}\overline{C})\overline{B}$$

Using distributive law $A + (B \cdot C) = (A + B) \cdot (C + A)$, we have

$$Y = (A + \overline{A} \cdot C + \overline{C}) \cdot \overline{B} = ((\overline{A} + \overline{A}) \cdot (A + C) + \overline{C}) \cdot \overline{B}$$

Now, $A + \overline{A} = 0$ identity. Using this, we get

$$Y = (1 \cdot (A + C) + \overline{C}) \cdot \overline{B}$$

Using $1 \cdot A = A$, we have

$$Y = (A + C + \bar{C}) \cdot \bar{B}$$

Now, again using $C + \bar{C} = 1$, we have

$$Y = (A + 1) \cdot \bar{B}$$

Using $A + 1 = 1$ identity, we have

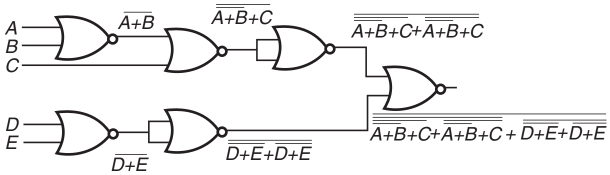
$$Y = 1 \cdot \bar{B} = \bar{B} \qquad \text{(Since } 1 \cdot A = A\text{)}$$

Therefore,

$$Y = \bar{B}$$

67. Topic: Digital Logic

(a) Output of NOR gate with inputs A and B is $\overline{A+B}$:



Solving the resultant

$$Y = \overline{\overline{(\overline{\overline{A+B}+C} + \overline{\overline{A+B}+C})} + (\overline{\overline{D+E}} + \overline{\overline{D+E}})},$$

using De Morgan's theorem $\overline{A+B} = \bar{A} \cdot \bar{B}$, we have

$$Y = \overline{\overline{(\overline{\bar{A} \cdot \bar{B} + C} + \overline{\bar{A} \cdot \bar{B} + C})} + (\overline{\bar{D} \cdot \bar{E} + \bar{D} \cdot \bar{E}})}$$

$$= \overline{\overline{(\overline{\bar{A} \cdot \bar{B}} \cdot \bar{C} \cdot \overline{\bar{A} \cdot \bar{B}} \cdot \bar{C})} + (\overline{\bar{D} \cdot \bar{E}} \cdot \overline{\bar{D} \cdot \bar{E}})}$$

Using De Morgan's theorem $\overline{A \cdot B} = \bar{A} + \bar{B}$, we get

$$Y = \overline{\overline{(\bar{\bar{A}} + \bar{\bar{B}} \cdot \bar{C} \cdot \bar{\bar{A}} + \bar{\bar{B}} \cdot \bar{C})} + (\bar{\bar{D}} + \bar{\bar{E}} \cdot \bar{\bar{D}} + \bar{\bar{E}})}$$

Using double negation identity, that is, $\bar{\bar{A}} = A$, we have

$$Y = \overline{\overline{(A + B \cdot \bar{C} \cdot A + B \cdot \bar{C})} + (D + E \cdot D + E)}$$

Using De Morgan's theorem,

$$Y = \overline{(\bar{A} \cdot \bar{B} + \bar{\bar{C}} + \bar{A} \cdot \bar{B} + \bar{\bar{C}}) + (D + E \cdot D + E)}$$

$$= (\bar{\bar{A}} + \bar{\bar{B}} \cdot \bar{C} \cdot \bar{\bar{A}} + \bar{\bar{B}} \cdot \bar{C}) \cdot (\bar{D} \cdot \bar{E} + \bar{D} \cdot \bar{E}) \qquad \text{(Since } \bar{\bar{C}} = C\text{)}$$

$$= (A + B \cdot \bar{C} \cdot A + B \cdot \bar{C}) \cdot (\bar{D} \cdot \bar{E} + \bar{D} \cdot \bar{E})$$

$$= (A(1 + B \cdot \bar{C}) + B \cdot \bar{C}) \cdot (\bar{D} \cdot \bar{E} + \bar{D} \cdot \bar{E})$$

$$= (A \cdot 1 + B \cdot \bar{C}) \cdot (\bar{D} \cdot \bar{E} + \bar{D} \cdot \bar{E}) \qquad (\text{Since } 1 + A = 1)$$

$$= (A + B \cdot \bar{C}) \cdot (\bar{D} \cdot \bar{E} + \bar{D} \cdot \bar{E}) \qquad \text{(Since } A \cdot 1 = A\text{)}$$

$$= (A + B \cdot \bar{C}) \cdot (\bar{D} \cdot \bar{E}(1 + 1))$$

$$= (A + B \cdot \bar{C}) \cdot (\bar{D} \cdot \bar{E})$$

$$Y = (A + B \cdot \bar{C}) \cdot (\bar{D} \cdot \bar{E})$$

68. Topic: Digital Logic

(a) The final Boolean expression for given circuit is:

A	B	C	D	E
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

The given Boolean expression is equivalent to full adder Boolean expression.

69. Topic: Engineering Mathematics

(a) Let λ be the eigen value.
Then, $|A - I\lambda| = 0$

$$\begin{vmatrix} 1-\lambda & \sin\theta \\ \sin\theta & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda)^2 - \sin^2\theta = 0 \Rightarrow (1-\lambda)^2 = \sin^2\theta$$

$$\Rightarrow (1-\lambda) = \sin\theta \Rightarrow \lambda = 1 + \sin\theta$$

Now, we have, $(A - I\lambda)X = 0$

$$\begin{bmatrix} -\sin\theta & \sin\theta \\ \sin\theta & -\sin\theta \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\Rightarrow x_1 = x_2$$

Therefore, the eigen vector of given matrix is $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

Now, the normalised eigen vector is $\frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

70. Topic: Engineering Mathematics

(a) Given:

$$P(A) = \frac{1}{4}, P(B) = \frac{1}{3} \text{ and } P(A \cup B) = \frac{1}{2}$$

We know that

$$P(A/B) = \frac{P(A \cdot B)}{P(B)} = \frac{P(A) + P(B) - P(A \cup B)}{P(B)}$$

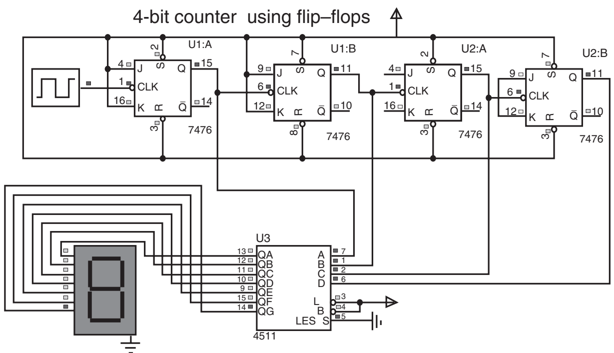
$$P(A/B) = \frac{\left(\frac{1}{4} + \frac{1}{3} - \frac{1}{2}\right)}{\frac{1}{3}} = \frac{1}{4}$$

71. Topic: Digital Logic

(d) When $J = K = 1$ then the race condition is occurs that means both output wants to be HIGH. Hence, there is toggle condition is occurs.

72. Topic: Digital Logic

(c) 4 flip–flops are required to count up to 9 starting from 0 as shown in figure below.



73. Topic: Digital Logic

(b) Given:

$$(E5)_{16}$$

Now, convert hexadecimal number into Decimal number. We get:

$$(E5)_{16} = E \times 16^1 + 5 \times 16^0$$
$$= 14 \times 16^1 + 5 \times 16^0 = 224 + 5 = (229)_{10}$$

Therefore,

$$(E5)_{16} = (229)_{10}$$

74. Topic: Computer Organization and Architecture

(c) An arithmetic logic unit (ALU) is a digital circuit used to perform arithmetic and logic operations. It represents the fundamental building block of the central processing unit (CPU) of a computer. Modern CPUs contain very powerful and complex ALUs.

75. Topic: Computer Organization and Architecture

(c) The number of machine code instructions a computer can process while executing a "standard" program is measured in MIPS (Million Instructions Per Second). Specifically, MIPS is a method of measuring the raw speed of a computer's processor and is defined as the number of machine instructions (in millions) that a processor can execute in 1 s.

76. Topic: Computer Organization and Architecture

(a) Most RAM (random access memory) used for primary storage in personal computers is volatile memory. RAM is much faster to read from and write to than the other kinds of storage in a computer, such as the hard disk or removable media.

77. Topic: Digital Logic

(a) ASCII is a code for representing 128 English characters as numbers, with each letter assigned a number from 0 to 127. For example, the ASCII code for uppercase M is 77. Most computers use ASCII codes to represent text, which makes it possible to transfer data from one computer to another.

78. Topic: Computer Organization and Architecture

(a) Magnetic sequential access memory is typically used for secondary storage in general-purpose computers due to their higher density at lower cost compared to RAM, as well as resistance to wear and non-volatility. Magnetic tape is the only type of sequential access memory still in use; historically, drum memory has also been used.

79. Topic: Computer Organization and Architecture

(b) In a typical format, data is written to tape in blocks with inter-block gaps between them, and each block is written in a single operation with the tape running continuously during the write. However, since the rate at which data is written or read to the tape drive is not deterministic, a tape drive usually has to cope with a difference between the rate at which data goes on and off the tape and the rate at which data is supplied or demanded by its host.

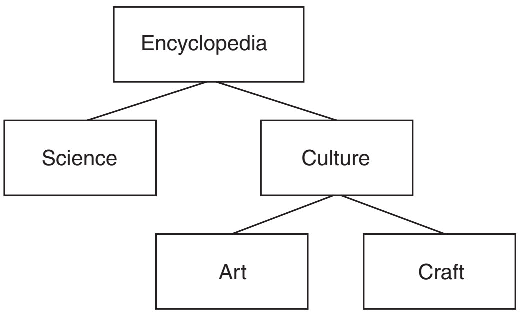
80. Topic: Computer Organization and Architecture

(b) On a systems flowchart, the online manual keeping of input data is identified by using the online keyboard symbol.

PRACTICE TEST – 2

1. **Topic: Programming and Data Structures**

(c) A tree structure or tree diagram is a way of representing the hierarchical nature of a structure in a graphical form. It is named a "tree structure" because the classic representation resembles a tree, even though the chart is generally upside down compared to an actual tree, with the "root" at the top and the "leaves" at the bottom.
A tree structure showing the possible hierarchical originations of an encyclopedia is depicted as follows:



2. **Topic: Operating System**

(b) A program that performs a very specific task, usually related to managing system resources. Operating systems contain a number of utilities for managing disk drives, printers, and other devices.
Utilities differ from applications mostly in terms of size, complexity and function. For example, word processors, spreadsheet programs, and database applications are considered applications because they are large programs that perform a variety of functions not directly related to managing computer resources.

3. **Topic: Databases**

(a) Disabling Triggers: Example The sample schema hr has a trigger named update_job_history created on the employees table. The trigger fires whenever an UPDATE statement changes an employee's job_id. The trigger inserts into the job_history table a row that contains the employee's ID, begin and end date of the last job, and the job ID and department.
When this trigger is created, the database enables it automatically. You can subsequently disable the trigger with this statement:
ALTER TRIGGER update_job_history DISABLE;
When the trigger is disabled, the database does not fire the trigger when an UPDATE statement changes an employee's job.
Enabling Triggers: Example after disabling the trigger, you can subsequently enable it with this statement:
ALTER TRIGGER update_job_history ENABLE;
After you re-enable the trigger, the database fires the trigger whenever an UPDATE statement changes an employee's job. If an employee's job is updated while the trigger is disabled, then the database does not automatically fire the trigger for this employee until another transaction changes the job_idagain.

4. **Topic: Computer Networks**

(c) SMTP (Simple Mail Transfer Protocol) is a TCP/IP protocol used in sending and receiving e-mail. However, since it is limited in its ability to queue messages at the receiving end, it is usually used with one of two other protocols, POP3 or IMAP that let the user save messages in a server mailbox and download them periodically from the server.

5. **Topic: Algorithms**

(d) Given

$$T(n) = 6T(n/2) + n^3$$

By using Master Theorem.

$$a = 6,\ b = 2,\ f(n) = n^3$$
$$g(n) = (n^{\log_b a}) = n^{\log_2 6} = O(n^{2.58})$$

$f(n) = \Omega(n^{\log_2 6+\varepsilon})$, $f(n)$ is larger than $g(n)$, but is not a larger polynomial.
By case 3, $T(n) = O(f(n)) = O(n)$

6. **Topic: Computer Organization and Architecture**

(d) It is used in cache memory to help the program access small amounts of address space at any instant.

7. **Topic: Theory of Computation**

(a) In option (a), the given expression will generate any number of 0's followed by any number of 1's, which can be accepted by FA. So, it is a regular expression.
In option (b), the given expression will generate number of 0's followed by number of 1's (both should be same in number), which will be accepted by PDA. So, it is not a regular expression.
In option (c), the given expression will generate any string "*x*" twice, which will be accepted by PDA. So it is not a regular expression.
In option (d), the given expression will generate twice the n number of 1's followed by n number of 0's, which cannot be accepted by FA. So, it is not a regular expression.

8. **Topic: Complier Design**

(b) *id* has higher precedence than any other symbol.

9. Topic: Operating System

(b) The hardware abstraction layer (HAL) provides logical link between NT-based operating systems and physical hardware of the computer.

10. Topic: Engineering Mathematics

(b) We have

$$f(x) = \begin{cases} 4x-5, & \text{if } x \le 2 \\ x-\lambda, & \text{if } x > 2 \end{cases}$$

Now, $\lim\limits_{x\to 2^-} f(x) = \lim\limits_{h\to 0} f(2-h)$

$$= \lim_{h\to 0} 4(2-h)-5 = \lim_{h\to 0}(3-4h) = 0$$

and

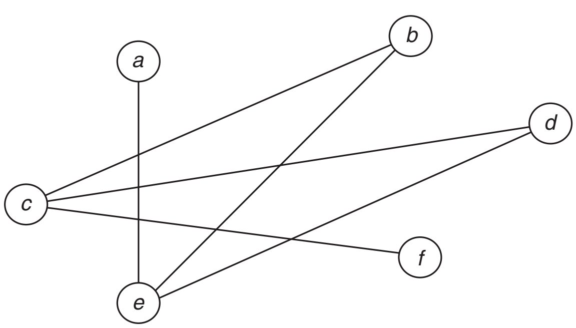
$$\lim_{x\to 2^+} f(x) = \lim_{h\to 0} f(2+h)$$
$$= \lim_{h\to 0} f(2+h) - \lambda = (2-\lambda)$$

Now, if limit exists, then

$$\lim_{x\to 2^-} f(x) = \lim_{x\to 2^+} f(x)$$
$$\Rightarrow 3 = 2-\lambda \Rightarrow \lambda = -1$$

11. Topics: Engineering Mathematics

(d) We have to keep in mind that the two vertices are adjacent in the complement only when they are not originally adjacent.



12. Topic: Operating System

(b) Virtual memory consists of those addresses that may be generated by a processor during execution of a computation.

13. Topic: Databases

(b) A data search performed within a predefined table of values (array, matrix, etc.) or within a data file.

14. Topic: Computer Networks

(a) Twisted pair cabling is a type of wiring in which two conductors of a single circuit are twisted together for the purposes of canceling out electromagnetic interference (EMI) from external sources; for instance, electromagnetic radiation from unshielded twisted pair (UTP) cables, and crosstalk between neighboring pairs. It was invented by Alexander Graham Bell.

15. Topic: Algorithms

(c) I. An algorithm is the number of steps to be performed to solve a problem. Statement is true.

II. An algorithm is the number of steps and their implementation in any language to a given problem to solve a problem. Statement is false.

III. To solve a given problem there may be more than one algorithm. Statement is true.

Both statements I and III are correct.

16. Topic: Computer Organization and Architecture

(b) Registers are used for processing and manipulating data and for holding memory addresses that are available to the machine-code programmer. So, they have lowest access time.

17. Topic: Theory of Computation

(d) Context-free language is not close under intersection. Please refer table 5.6 (closure property of languages) in text.

18. Topic: Complier Design

(c) The tokens in C tokens include identifiers, keywords, constants, operators, string literals and other separators. Thus, there are 10 tokens in the given printf statement.

19. Topic: Software Engineering

(b) Risk management helps in developing a cost effective project where risks are analyzed and risk strategies are decided upon.

20. Topic: Engineering Mathematics

(d) Since $\sin x$ and $\sin 2x$ are continuous and differentiable everywhere, the function $f(x)$ is continuous on $[0,\pi]$ and differentiate on $[0,\pi]$. Hence, Lagrange's mean value theorem is applicable.

Thus, there exists at least one $c \in [0,\pi]$ such that

$$f'(c) = \frac{f(\pi)-f(0)}{\pi - 0}$$

Now,

$$f(x) = 2\sin x + \sin 2x$$
$$f'(x) = 2\cos x + 2\cos 2x$$
$$f(0) = 0 \text{ and } f(\pi) = 2\sin\pi + \sin\pi = 0$$

Hence,

$$f'(x) = \frac{f(x)-f(0)}{\pi - 0} \Rightarrow 2\cos x + 2\cos 2x = \frac{0}{\pi}$$
$$\Rightarrow \cos x + \cos 2x = 0 \Rightarrow \cos 2x = -\cos x$$
$$\Rightarrow \cos 2x = \cos(\pi - x)$$
$$\Rightarrow 2x = \pi - x \Rightarrow x = \frac{\pi}{3}$$

Thus, $c = \frac{\pi}{3} \in (0,\pi)$ such that $f'(c) = \frac{f(\pi)-f(0)}{\pi - 0}$.

21. Topic: Engineering Mathematics

(b) The number of edges in a complete graph is given by

$$m = \frac{n(n-1)}{2}$$

where n is number of vertices. For K_6:

$$m = \frac{6(6-1)}{2} = 15$$

22. Topic: Complier Design

(c) Terminal Table a permanent table which lists all key words and special symbols of the language in symbolic form

23. Topic: Databases

(d) The logical units of database space allocation are data blocks, extents, segments, and tablespaces. At a physical level, the data is stored in data files on disk. The data in the data files is stored in operating system blocks.
Tree: A tree is a widely used abstract data type (ADT)—or data structure implementing this ADT—that simulates a hierarchical tree structure, with a root value and sub trees of children with a parent node, represented as a set of linked nodes.
Relational: A relational database is a database model that stores data in tables. ... Each table in a relational database contains rows (records) and columns (fields). In computer science terminology, rows are sometimes called “tuples,” columns may be referred to as “attributes,” and the tables themselves may be called “relations.”
Network: A computer network is a set of computers connected together for the purpose of sharing resources. The most common resource shared today is connection to the Internet. Other shared resources can include a printer or a file server.

24. Topic: Computer Networks

(b) Data Network Identification Code (DNIC) is the CCITT International X.121 format, the first four digits of the international data number, three digits to identify the country (one to identify a zone and two to identify the country within the zone) and one to identify the PDN (allowing only ten in each country) and the 1-digit network code.

25. Topic: Algorithms

(a) Hash tables have less complexity, so they are used by the compiler for efficient searching.

26. Topic: Computer Organization and Architecture

(d) DMA controller manages transfer between I/O device and memory.

27. Topic: Theory of Computation

(c) Both the regular expressions will generate the same set of strings, so both the regular sets are equal.

28. Topic: Complier Design

(c) Constant folding is the process of evaluating the constant expression at compile time.

29. Topic: Software Engineering

(b) Cyclomatic complexity is a software metric developed by Thomas J. McCabe. It is used to indicate the complexity of a program. It directly measures the number of linearly independent paths through a program’s source code. It is also known as conditional complexity. It is defined as

$$V(G) = e - n + 2$$

30. Topic: Engineering Mathematics

(a) We have

$$y + \frac{dy}{dx} = \frac{1}{4}\int y \cdot dx$$

Now, differentiating with respect to x, we get.

$$\Rightarrow \frac{dy}{dx} + \frac{d^2y}{dx^2} = \frac{1}{4}y$$

Differential equation is of order 2 and degree 1.

31. Topic: Engineering Mathematics

(c) We have

$$S = \begin{Bmatrix} (1,1),(1,2),(1,3),(1,4),(1,5),(1,6), \\ (2,1),(2,2),(2,3),(2,4),(2,5),(2,6), \\ (3,1),(3,2),(3,3),(3,4),(3,5),(3,6), \\ (4,1),(4,2),(4,3),(4,4),(4,5),(4,6), \\ (5,1),(5,2),(5,3),(5,4),(5,5),(5,6), \\ (6,1),(6,2),(6,3),(6,4),(6,5),(6,6) \end{Bmatrix}$$

$n(S) = \text{Total number of events} = 36$

Now, A = the sum is even

$$A = \begin{Bmatrix} (1,1),(1,3),(1,5),(2,2),(2,4),(2,6), \\ (3,1),(3,3),(3,5),(4,2),(4,4),(4,6), \\ (5,1),(5,3),(5,5),(6,2),(6,4),(6,6) \end{Bmatrix}$$

$$n(A) = 18$$

B = the sum is multiple of 3

$$B = \begin{Bmatrix} (1,2),(2,1),(1,5),(5,1),(2,4),(4,2), \\ (3,3),(3,6),(6,3),(4,5),(5,4),(6,6) \end{Bmatrix}$$

$$n(B) = 12$$

C = the sum is less than 4

$$C = \{(1,1),(1,2),(2,1)\}$$

$$n(C) = 3$$

D = the sum is greater than 11

$$D = \{(6,6)\}$$

$$n(D) = 1$$

We observe that

$$A \cap B = \{(1,5),(2,4),(3,3),(4,2),(5,1),(6,6)\} \neq \phi$$

Thus, A and B are not mutually exclusive events.

$$C \cap D = \phi$$

Also,

$$A \cap D = \{(6,6)\} \neq \phi$$

Hence, C and D are mutually exclusive events.

32. Topic: Operating System

(d) Input/Output Control System is any of several packages on early IBM entry-level and mainframe computers that provided low level access to records on peripheral equipment.

In computer science, storage virtualization is "the process of presenting a logical view of the physical storage resources to" a host computer system, "Treating all storage media in the enterprise as a single pool of storage."

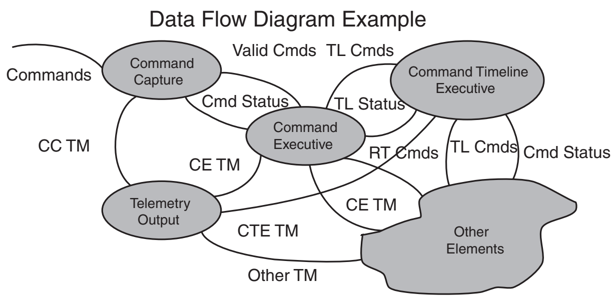
Multi-programming is a rudimentary form of parallel processing in which several programs are run at the same time on a uniprocessor. Since there is only one processor, there can be no true simultaneous execution of different programs. Instead, the operating system executes part of one program, then part of another, and so on. To the user it appears that all programs are executing at the same time.

33. Topic: Databases

(d) Decision Table: A list of decisions and their criteria. Designed as a matrix, it lists criteria (inputs) and the results (outputs) of all possible combinations of the criteria. It can be placed into a program to direct its processing.

Data Dictionary: A data dictionary is a collection of descriptions of the data objects or items in a data model for the benefit of programmers and others who need to refer to them. A first step in analyzing a system of objects with which users interact is to identify each object and its relationship to other objects.

Data-flow diagram: A data flow diagram (DFD) is a graphical representation of the "flow" of data through an information system, modelling its process aspects. A DFD is often used as a preliminary step to create an overview of the system without going into great detail, which can later be elaborated. DFDs can also be used for the visualization of data processing (structured design).



34. Topic: Computer Networks

(a) Bus network is an arrangement in a local area network (LAN) in which each node (workstation or other device) is connected to a main cable or link called the bus. The illustration shows a bus network with five nodes.

35. Topic: Algorithms

(a) The time required for search operation in a binary search depends on the height of the tree. In a complete binary tree with n nodes, search operation takes O($\log_2 n$) in worst case scenario. But if all the nodes of tree are in linear chain, then the same operations will take O(n) in worst-case scenario. So, option (a) is correct.

36. Topic: Computer Organization and Architecture

(c) It is a multiplication algorithm that multiplies two signed binary numbers in 2's complement notation.

37. Topic: Theory of Computation

(d) In the given grammar, variable S will generate a pair of symbol 'a' or symbol 'b' with variable S in the middle of them, and this is in loop so it can be generated any number of times. In the last variable, S will be replaced by ε. Some of the strings which can be generated by the given grammar are {*aa*, *bb*, *abba*, *aabbaa*, *baab*, etc}. So, the given grammar will generate even length palindrome.

38. Topic: Operating System

(c) Thrashing means that the system more busy in dealing with page faults instead of doing some constructive work. Thus, it leads to excessive page I/O.

39. Topic: Software Engineering

(b) Users or an independent test team at the developer's site performs alpha testing. Alpha testing is often employed as a form of internal acceptance testing, before the software goes to beta testing.
Beta testing is performed after alpha testing for user acceptance testing.
Regression testing focuses on finding defects after a major code change has occurred.

40. Topic: Engineering Mathematics

(c) We have

$$(x+1)\frac{dy}{dx} = 2xy$$
$$\Rightarrow (x+1)dy = 2xydx$$
$$\Rightarrow \frac{dy}{y} = \frac{2x}{x+1}dx, \text{ if } x \neq -1$$
$$\Rightarrow \int \frac{dy}{y} = 2\int \frac{x}{x+1}dx$$
$$\Rightarrow \int \frac{1}{y}dy = 2\int \frac{x+1-1}{x+1}dx$$
$$\Rightarrow \int \frac{1}{y}dy = 2\int \left[1-\frac{1}{x+1}\right]dx$$
$$\log y = 2[x - \log|x+1|] + C$$

41. Topic: Engineering Mathematics

(a) We know that

$$P(E) = \frac{n(E)}{n(S)}$$

Where E is given event and S is total events. Thus,

$$n(E) = \text{words starting with K} = 3!$$
$$n(S) = \text{total number of words} = 4!$$

Hence,

$$P(E) = \frac{n(E)}{n(S)} = \frac{3!}{4!} = \frac{1}{4}$$

42. Topic: Operating System

(c) *Swapping* is a mechanism in which a process can be *swapped* temporarily out of *main memory* (or move) to secondary storage (disk) and make that memory available to other processes. At some later time, the system swaps back the process from the secondary storage to *main memory*.

43. Topic: Databases

(b) Decision Tree: A decision tree is a graphical representation of possible solutions to a decision based on certain conditions. It's called a decision tree because it starts with a single box (or root), which then branches off into a number of solutions, just like a tree.
Decision Tables: Program embedded decision tables. Decision tables can be, and often are, embedded within computer programs and used to "drive" the logic of the program. ... Multiple conditions can be coded for in similar manner to encapsulate the entire program logic in the form of an "executable" decision table or control table.

44. Topic: Computer Networks

(a) Frequency modulation uses the information signal, $V_m(t)$ to vary the carrier frequency within some small range about its original value. Here are the three signals in mathematical form:

- Information: $V_m(t) = V_m \cos(\omega_m t)$
- Carrier: $V(t) = V_c \cos[\omega_c t + \theta(t)]$
- FM: $V_{FM}(t) = V_c \cos[\omega_c t + \beta \cos \omega_m(t))]$

45. Topic: Programming and Data Structures

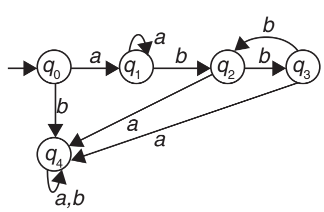
(c) The variables of function call can be stored in stack and heap.

46. Topic: Computer Organization and Architecture

(c) Instruction-level parallelism is implemented within a single processor to allow faster CPU throughput.

47. Topic: Theory Computation

(d) DFA for given language $L = a^n b^{2m} | n, m \geq 1$ is



q_0 {Starting state}
q_3 {Final state}

48. Topic: Operating System

(c) If page is not found in page table, it generates the page fault.

49. Topic: Software Engineering

(d) ORB manages interaction between clients and servers.

50. Topic: Engineering Mathematics

(a) We have

$$w = \log z$$

Let $w = u + iv = \log(x + iy)$

$$= \frac{1}{2}\log(x^2 + y^2) + i\tan^{-1}\frac{y}{x}$$

Thus,

$$u = \frac{1}{2}\log(x^2 + y^2), v = i\tan^{-1}\frac{y}{x}$$

Therefore,

$$\frac{\partial u}{\partial x} = \frac{x}{x^2 + y^2}$$

$$\frac{\partial u}{\partial y} = \frac{y}{x^2 + y^2}$$

$$\frac{\partial v}{\partial x} = \frac{-y}{x^2 + y^2}$$

$$\frac{\partial v}{\partial y} = \frac{x}{x^2 + y^2}$$

$$\frac{dw}{dz} = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = \frac{x}{x^2 + y^2} + i\left[\frac{-y}{x^2 + y^2}\right]$$

$$= \frac{x - iy}{(x + iy)(x - iy)} = \frac{1}{(x + iy)} = \frac{1}{z} \qquad (z \neq 0)$$

51. Topic: Operating System

(b) Binary semaphores have two methods associated with it (up, down/lock, unlock)
Binary semaphores can take only 2 values (0/1). They are used to acquire locks. When a resource is available, the process in charge set the semaphore to 1 else 0.

52. Topic: Operating System

(d) Auto increment Mode = after operand addressing, the contents of the register is incremented.
Decrement Mode = before operand addressing, the contents of the register is decrement.
These contents are then used as the effective address of the operand.

53. Topic: Databases

(a) A *transaction process system* (TPS) is an information *processing system* for business *transactions* involving the collection, modification and retrieval of all *transaction* data. Characteristics of a TPS include performance, reliability and consistency. TPS is also known as *transaction processing* or real-time *processing*.

54. Topic: Digital Logic

(c) PCM system: In a PCM stream, the amplitude of the analog signal is sampled regularly at uniform intervals, and each sample is quantized to the nearest value within a range of digital steps. Linear pulse-code modulation (LPCM) is a specific type of PCM where the quantization levels are linearly uniform.
Companding: The process involves decreasing the number of bits used to record the strongest (loudest) signals. In the digital file format, companding improves the signal-to-noise ratio at reduced bit rates. For example, a 16-bit PCM signal may be converted to an eight-bit ".wav" or ".au" file.

55. Topic: Programming and Data Structures

(b) The stored value may or may not be exact; the precision of the value depends upon the number of bytes. Double stores value 1.1 with more precision than float because double takes 8 bytes, whereas float takes 4 bytes.

56. Topic: Computer Organization and Architecture

(c) Cache memory compensates the speed mismatch between processor and main memory access time.

57. Topic: Theory of Computation

(c) Strings which will be accepted by machine M can be found by fixing 1 at places first, fourth and last. We will have following four strings:

1001001 1001011 1001111 1111001

58. Topic: Operating System

(c) Complexity of Safe Sequence using Banker's Algorithm is

$$m \times n^2$$

where n is the number of processes, and m is the number of resources.

59. Topic: Software Engineering

(c) There are five maturity levels in CMM model–(1) Initial, (2) Managed, (3) Defined, (4) Quantitatively Managed and (5) Optimizing. The organization at level 2 includes level 1. So, level 4 is included in level 5.

60. Topic: Engineering Mathematics

(d) We have

$$f(z) = \frac{1}{(z+1)^2}$$

Now, we need to find the Taylor's expansion of $f(z)$ about $z = -i$, that is in power of $z + i$.
Putting $z + i = t$, we get

$$f(z) = \frac{1}{(t-i+1)^2} = (1-i)^{-2}\left[1+\frac{t}{(1-i)}\right]^{-2}$$

$$= \frac{i}{2}\left[1-\frac{2t}{1-i}+\frac{3t^2}{(1-t)^2}-\frac{4t^3}{(1-t)^3}+\cdots\right]$$

[expanding by binomial theorem]

$$\frac{i}{2}\left[1+\sum_{n=2}^{\infty}(-1)^n\frac{(n+1)(z+i)^n}{(1+i)^n}\right]$$

61. Topic: Computer Organisation and Architechure

(c) The outstanding invoice file should be stared on a Pardon Access storage device if inquiries concerning payable are to be answered on-line.

62. Topic: Computer Networks

(c) The network layer, in reference to the OSI model, provides the virtual circuit interface to packet-switched service.

Network Layer: The network Layer controls the operation of the subnet. The main aim of this layer is to deliver packets from source to destination across multiple links (networks). If two computers (system) are connected on the same link, then there is no need for a network layer. It routes the signal through different channels to the other end and acts as a network controller.

It also divides the outgoing messages into packets and to assemble incoming packets into messages for higher levels.

OSI Model: OSI (Open Systems Interconnection) is reference model for how applications can communicate over a network. A reference model is a conceptual framework for understanding relationships. The purpose of the OSI reference model is to guide vendors and developers so the digital communication products and software programs they create will interoperate, and to facilitate clear comparisons among communications tools. Most vendors involved in telecommunications make an attempt to describe their products and services in relation to the OSI model.

63. Topic: Digital Logic

(b) If the input to a differentiating circuit is saw-tooth wave then output will be rectangular wave.

64. Topic: Programming and Data Structures

(c) A global variable provides access from multiple files. Due to merging of one file code to another code, conflict occurs in the variable names and this cause an error at link time.

65. Topic: Digital Logic

(d) The clock is high, only the flip-flops are stable when the inputs are stable.

66. Topic: Complier Design

(a) Top–down parser uses leftmost derivation for constructing parse tree.

67. Topic: Operating System

(b) In this problem, if process P_i is executing in its critical section, then no other process can execute their critical sections.

68. Topic: Software Engineering

(b) Verification means are we building the product right, whereas validation means whether we are building the right product.

69. Topic: Computer Organization and Architecture

(c) Data structures containing such different sized words refer to them as WORD (16 bits/2 bytes), DWORD (32 bits/4 bytes) and QWORD (64 bits/8 bytes) respectively.

70. Topic: Engineering Mathematics

(b) The matrix A is singular if

$$A = 0 \Rightarrow \begin{bmatrix} 1 & -2 & 3 \\ 1 & 2 & 1 \\ x & 2 & -3 \end{bmatrix} = 0$$

$$\Rightarrow 1(-6-2)-1(6-6)+x(-2-6)=0$$

$$\Rightarrow -8-8x=0$$

$$\Rightarrow x=-1$$

71. Topic: Complier Design

(a) Only option (a) is true. LL(1) parser is a top–down parser; LALR is more powerful.

72. Topic: Computer Networks

(a) API-an application programming interface (API) is a set of subroutine definitions, protocols, and tools for building application software. In general terms, it is a set of clearly defined methods of communication between various software components. A good API makes it easier to develop a computer program by providing all the building blocks, which are then put together by the programmer.

73. Topic: Digital Logic

(b) In digital electronics, a NAND gate (negative-AND) is a logic gate which produces an output which is false only if all its inputs are true; thus its output is complement to that of the AND gate. A LOW (0) output results only if both the inputs to the gate are HIGH (1); if one or both inputs are LOW (0), a HIGH (1) output results. It is made using transistors and junction diodes. By De Morgan's theorem, AB=A+B, and thus a NAND gate is equivalent to inverters followed by an OR gate.

74. Topic: Programming and Data Structures

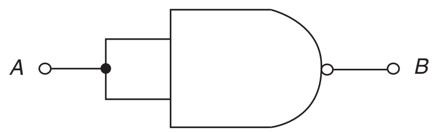
(c) We have

$$m=2*f(3,g(4,5));$$

The function g(4,5) will return 20 by performing multiplication operations. Then function f (3,20) will return 23 by performing addition operation. In the last output of function f (3,20) is multiplied with 2. So, the final answer will be 46.

75. Topic: Digital Logic

(a) An inverter or logic NOT gate can be made using standard NAND gate by connecting together ALL their inputs to a common input signal, for example,



Two-input NAND gate equivalent

76. Topic: Complier Design

(c) The process adjusting the code and data in the program to reflect the assigned addresses is called relocation.

77. Topic: Operating System

(b) It uses fair pre-emptive scheduling.

78. Topic: Engineering Mathematics

(a) We know that the corresponding elements of two equal matrices are equal. Therefore,

$$a - b = 5 \quad (1)$$

$$2a - b = 12 \quad (2)$$

$$2c + d = 3 \quad (3)$$

$$2a + d = 15 \quad (4)$$

Subtracting Eq. (1) from Eq. (2), we get

$$a = 7$$

$$7 - b = 5 \Rightarrow b = 2$$

Substituting the value of a in Eq. (4), we get

$$14 + d = 15 \Rightarrow d = 1$$

Substituting the value of d in Eq. (3), we get

$$2c + 1 = 3 \Rightarrow c = 1$$

Hence, $a = 7$, $b = 2$, $c = 1$ and $d = 1$.

79. Topic: Programming and Data Structures

(a) When the factorial program is written using recursion only then will it use stack, so stack overflow will occur in recursive program only.

80. Topic: Digital Logic

(b) In the ring counter, n flip-flops are used for the mod-n. 12 flip–flops because ring counter uses states n, so here modulus is 12.

PRACTICE TEST – 3

1. **Topic: Digital Logic**
   (a) The NAND function is actually the simplest, most natural mode of operation for this TTL design. The basic gates are OR/NOR gates for ECL Construction. Inverters can be constructed using two complementary transistors in a CMOS configuration.

2. **Topic: Digital Logic**
   (d) Let $\uparrow$ signify the NAND operation. Then there exist propositions p, q, r such that
   $$p\uparrow(q\uparrow r)\ (p\uparrow q)\uparrow r$$
   This implies NAND is not associative.
   $$p\uparrow q = q\uparrow p$$
   This implies NAND is commutative.

3. **Topic: Computer Organization and Architecture**
   (a) Control unit controls several units of CPU and helps decode program instructions.

4. **Topic: Programming and Data Structures**
   (c) Mathematical induction is the method through which correctness of recursive function can be proved.

5. **Topic: Theory of Computation**
   (c) Strings generated by L should have sum of number of 'a' and 'b' equal to total number of 'c' and 'd'. Only in option (c), the grammar can generate such string which has number of $(a + b)$ equal to number of $(c + d)$.

6. **Topic: Compiler Design**
   (b) An identifier expression cannot have character such as *.

7. **Topic: Operating System**
   (d) Multiprogrammed OS performs all the given three tasks.

8. **Topic: Software Engineering**
   (c) This is the definition of object-oriented according to software engineering.

9. **Topic: Computer Networks**
   (a) Class D is used for multicasting.
   Class D Leading Bits – 1110; Value Range – 224—239; Starting Address – 224.0.0.0; Ending Address – 239.255.255.255

10. **Topic: Engineering Mathematics**
    (d) We have
    $$A=\begin{bmatrix}1&1&0\\0&2&2\\0&0&3\end{bmatrix}$$
    The characteristic equation is given by
    $$|A-\lambda I|=0$$
    $$\begin{vmatrix}1-\lambda&1&0\\0&2-\lambda&2\\0&0&3-\lambda\end{vmatrix}=0$$
    $$\Rightarrow(1-\lambda)[(2-\lambda)(3-\lambda)-0]$$
    $$-0[(3-\lambda)(2-\lambda)-0]+0[2-0]=0$$
    $$\Rightarrow(1-\lambda)(2-\lambda)(3-\lambda)=0$$
    $$\Rightarrow\lambda=1,2,3$$
    Using $|A-\lambda I|\,X=0$ and putting $\lambda=1$, we get
    $$\begin{bmatrix}0&1&0\\0&1&2\\0&0&2\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$$
    Hence, $x = k$, $y = 0$ and $z = 0$
    Therefore, the eigenvector can be of the form $\begin{bmatrix}k\\0\\0\end{bmatrix}$
    Similarly we do for $\lambda=2$, we get
    $$x = k,\ y = k \text{ and } z = 0$$
    Therefore, the eigenvector can be of the form $\begin{bmatrix}k\\k\\0\end{bmatrix}$
    Similarly we do for $\lambda=3$, we get
    $$x = 2k,\ y = k \text{ and } z = 2k$$
    Therefore, the eigenvector can be of the form $\begin{bmatrix}2k\\k\\2k\end{bmatrix}$
    Hence, the eigenvector which is not of the matrix A is $\begin{bmatrix}1\\2\\0\end{bmatrix}$.

11. **Topic: Engineering Mathematics**
    (d) Suppose A, B and C denote the set of students who got prizes in football, basketball and cricket, respectively.
    Then, we have
    $$n(A)=38,\ n(B)=15,\ n(C)=20,\ n(A\cup B\cup C)=58$$
    and
    $$n(A\cap B\cap C)=3$$

Also, we know that

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$

$$\Rightarrow 59 = 38 + 15 + 20 - n(A \cap B) - n(A \cap C) - n(B \cap C) + 3$$

$$\Rightarrow n(A \cap B) + n(A \cap C) + n(B \cap C) = 18$$

Now, the number of men who received medals in exactly two sports is

$$n(A \cap B) + n(A \cap C) + n(B \cap C) - 3n(A \cap B \cap C)$$
$$= 18 - (3 \times 3) = 9$$

12. Topic: Digital Logic

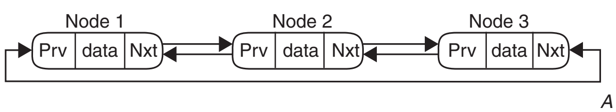
(a) The power dissipation is dependent on the power supply voltage, frequency, output load, and input rise time. Power dissipation for TTL gates is usually 10 mW per gate. ECL is the fastest logic gate and has least propagation delay of 2 ns. CMOS gates have high fan-outs.

13. Topic: Computer Organization and Architecture

(c) The CPU fetches data and instructions from RAM and executes it.

14. Topic: Programming and Data Structures

(a) Let a node X be inserted after node 1, then both pointer of node X (Prv and Nxt) will be affected and node 1's Nxt pointer will point to node X and node 2's Prv pointer will also point to node *X*. So, total four pointers will be affected.



15. Topic: Theory of Computation

(b) Given that FA will accept any string which has the last symbol '*a*'. So, only option (b) expression $(b + a + b)^*a$ can generate all combinations of strings which end with symbol '*a*'.

16. Topic: Compiler Design

(b) I_0:

$E' \rightarrow .E, \$$
$E \rightarrow .E + T, \$| +$
$E \rightarrow .T, \$| +$
$T \rightarrow .TF, \$| + |a|b$
$T \rightarrow .F, \$| + |a|b$
$F \rightarrow .F^*, \$| + |a|b|^*$
$F \rightarrow .a, \$| + |a|b|^*$
$F \rightarrow .b, \$| + |a|b|^*$

17. Topic: Operating System

(c) From the give code, we get $P(s)$ and $V(s)$ decrements and increment the value of semaphore, respectively. So, from the given conditions, we conclude that to avoid mutual exclusion the value of *Xb* should be 1 and that of *Yb* should be 0.

18. Topic: Software Engineering

(c) The key process areas at level 5 covers the following issues–Defect Prevention, Technology Change Management and Process Change Management. Software Product Engineering has been addressed in level 3.

19. Topic: Computer Networks

(d) It may contain values for various options, such as strict source routing, security, record route, time stamp, etc.

20. Topic: Engineering Mathematics

(a) We have

$$S = \left\{\begin{matrix} (1,1),(1,2),(1,3),(1,4),(1,5),(1,6), \\ (2,1),(2,2),(2,3),(2,4),(2,5),(2,6), \\ (3,1),(3,2),(3,3),(3,4),(3,5),(3,6), \\ (4,1),(4,2),(4,3),(4,4),(4,5),(4,6), \\ (5,1),(5,2),(5,3),(5,4),(5,5),(5,6), \\ (6,1),(6,2),(6,3),(6,4),(6,5),(6,6) \end{matrix}\right\}$$

Total outcomes = 36
Let *A* be the event of getting a total of at least 10. Then,

$$A = \{(6, 4), (4, 6), (5, 5), (6, 5), (5, 6), (6, 6)\}$$

Therefore,

Favorable outcomes = 6

Hence,

$$\text{Required probability} = \frac{6}{36} = \frac{1}{6}$$

21. Topic: Engineering Mathematics

(d) We have

$$\neg\exists x\left(\forall y(\alpha) \wedge \forall z(\beta)\right)$$

Using identity $\neg(p \wedge q) \equiv p \rightarrow \neg q$, we get

$$\neg\exists x\left(\forall y(\alpha) \wedge \forall z(\beta)\right) \equiv \forall \times\left[\forall y(\alpha) \rightarrow \exists z(\neg\beta)\right]$$

Using identity $p \Rightarrow q \equiv \neg q \Rightarrow \neg p$, we get

$$\neg\exists x\left(\forall y(\alpha) \wedge \forall z(\beta)\right) \equiv \forall \times\left[\forall z(\beta) \rightarrow \exists y(\neg\alpha)\right]$$

Hence, the correct option is (d).

22. Topic: Digital Logic

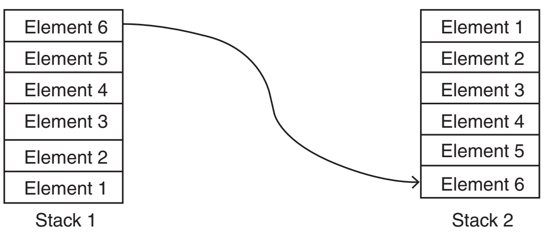
(c) For an n-bit word which includes the sign bit, there are $2^{n-1} - 1$ positive integers, 2^{n-1} negative integers and one 0, for a total of 2^n unique states.

23. Topic: Computer Organization and Architecture

(d) CPU processing speed is affected by clock cycle, data bus for routing data and addressing modes.

24. Topic: Programming and Data Structures

(b) Let there are 6 elements (1, 2, 3, 4, 5 and 6). First we will push them in stack 1 and then pop from stack 1 and push in stack 2. The same behaviour will be observed when we will pop from stack 2.



25. Topic: Theory of Computation

(c) Language L_1 is CFL and language L_2 is regular. So, the result of $L_1 \cap L_2$ will be context free.

26. Topic: Compiler Design

(d) Operator grammar has two properties:
First, there should not be any null production. Second, there should not be any production which has two consecutive variables in the right-hand side of production.

27. Topic: Databases

(a) In data definition stage, all the fields and their types are listed.

28. Topic: Software Engineering

(d) The components of the control system ensure smooth functioning of the system to achieve its objective.

29. Topic: Computer Networks

(a) The total number of wired links required to establish a fully connected mesh network of d nodes can be calculated as $c = d(d - 1)/2$.

30. Topic: Engineering Mathematics

(d) The probability that A can solve the question is $\frac{1}{2}$.
The probability that A cannot solve the question is

$$1 - \frac{1}{2} = \frac{1}{2}$$

Similarly, the probabilities that B and C cannot solve the question are

$$1 - \frac{1}{3} = \frac{2}{3} \text{ and } 1 - \frac{1}{4} = \frac{3}{4}$$

The probabilities that A, B and C cannot solve the question is

$$\left(\frac{1}{2}\right)\left(\frac{2}{3}\right)\left(\frac{3}{4}\right) = \frac{1}{4}$$

Hence, the probability that at least one student will solve the question is

$$1 - \frac{1}{4} = \frac{3}{4}$$

31. Topic: Engineering Mathematics

(b) For I_1, since we know that the cricket match was played, we can infer that there was no rain. Hence, inference 1 is correct.
For I_2, since we know that it did not rain. However, this only confirms that the match can be played and tells nothing about whether the match was played or not. Hence, inference 2 is incorrect.
Therefore, I_1 is correct and I_2 is incorrect inference.

32. Topic: Digital Logic

(c) The dual of a Boolean expression is obtained by interchanging Boolean sums and products and interchanging 0's and 1's. For example,
The dual of $x(y + 0) = x + (y \times 1)$.

33. Topic: Computer Organization and Architecture

(a) On execution of program, we have

SUB A	Subtract contents of memory location whose contents are stored in A
MVI B, $(01)_H$	Move B with $(01)_H$
DCR B	Decrement B gives $(54)_H$
HLT	

34. Topic: Algorithms

(a) The worst-case time complexity for searching an element in a linked list is O(n), so option (a) is correct. If in the question it is not mentioned about (best or worst or average) time complexity then we always calculate as worst time complexity.

35. Topic: Theory of Computation

(c) Both I and II are true.

36. Topic: Compiler Design

(c) Parsing table

	A	**,**	**(**	**)**	**$**
S	$S \to a$		$S \to (L)$		
L	$L \to SL'$		$L \to SL'$		
L'		$L' \to ,SL'$		$L' \to \varepsilon$	

37. Topic: Databases

(b) The maximum length of an attribute of type text is 255

38. Topic: Software Engineering

(a) Corrective maintenance refers to modifications initiated by defects in the software.
Perfective maintenance refers to improving processing efficiency or performance, or restructuring the software to improve changeability.
Preventive maintenance refers to prevention of breakdowns.

39. Topic: Web Technologies

(c) Firewall is used to prevent unauthorized access of private networks through packet screening from outside user, especially Internet.

40. Topic: Engineering Mathematics

(b) We have

$$P(A \cap B) = P(A)P\left(\frac{B}{A}\right) = 0.3 \times 0.5 = 0.15$$

Therefore,

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} \Rightarrow P\left(\frac{A}{B}\right) = \frac{0.15}{0.6} = \frac{1}{4}$$

Now,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$\Rightarrow P(A \cup B) = 0.3 + 0.6 - 0.15 = 0.75$$

41. Topic: Engineering Mathematics

(a) Since derivative is desired to be computed at the centre of the given data table, Stirling's formula will be used.

42. Topic: Digital Logic

(c) Sequential logic is a logic circuit in which the output depends not only on the present value of its input signals but on their past history. This is in contrast to combinational logic, where the output depends only on the present input.

43. Topic: Computer Organization and Architecture

(c) The addition time for N-bit carry lookahead adder is always a constant.

44. Topic: Algorithms

(a) As load time is the ratio of number of records present to the total number of records, so is load time is less than 1, the probability of collisions decreases and hence, search time reduces.

45. Topic: Theory of Computation

(d) From the regular expression $(x + y)^*y(a + ab)^*$, it can be seen that it contains 11 strings as follows:

$\{xy, yy, y, ya, yab, xya, yya, yaa, xxy, yyy, xyy\}$

46. Topic: Operating System

(b) It includes ready, standby, running, waiting, transition and terminated states.

47. Topic: Databases

(a) The state of that transaction is partially commit.

48. Topic: Software Engineering

(d) Software measurement is useful to indicate quality of the product, track progress, assess productivity and form a baseline for estimation and prediction.

49. Topic: Web Technologies

(b) The LDAP stands for Lightweight Directory Access Protocol which is a directory service protocol based on client-server model.

50. Topic: Engineering Mathematics

(d) Suppose X is the number of heads in a simultaneous toss of three coins.
Hence, X can take value 0, 1, 2, 3.
Now,

$$P(x = 0) = P(TTT) = \frac{1}{8}$$
$$P(x = 1) = P(HTT, THT, TTH) = \frac{3}{8}$$
$$P(x = 2) = P(HHT, HTH, THH) = \frac{3}{8}$$
$$P(x = 3) = P(HHH) = \frac{1}{8}$$

Hence, the probability distribution of X is given by

X:	0	1	2	3
$p(X)$:	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Now, let us construct the table as follows:

x_i	p_i	$p_i x_i$	$p_i x_i^2$
0	1/8	0	0
1	3/8	3/8	3/8
2	3/8	6/8	12/8
3	1/8	3/8	9/8

From the table, we have

$$\sum p_i x_i = \frac{3}{8} \text{ and } \sum p_i x_i^2 = 3$$

Now,

$$Var(x) = \sum p_i x_i^2 - \left(\sum p_i x_i\right)^2 = 3 - \left(\frac{3}{2}\right)^2 = \frac{3}{4}$$

51. Topic: Computer Organization and Architecture

(a) Temporal locality refers to reuse of resources referenced within a short time frame.

52. Topic: Computer Organization and Architecture

(d) Controller sets priority for memory access by various I/O devices and processes.

53. Topic: Programming and Data Structures

(b) Linked list is the best suitable and efficient data structure for constructing a tree because an item can be inserted and deleted from linked list with less cost.

54. Topic: Algorithms

(b) The number of keys = $K - 1$.

55. Topic: Theory of Computation

(b) L_1: both the expression generates any string made up of input symbol of L. it also include ε.
L_2: $\varnothing$* can generate {$\varnothing$ and ε}
L_3: ε * can only generate ε.
L_4: L^+ contain all the string made up of input symbol of L excluding ε.
L_5: L^* contain all the string made up of input symbol of L including ε.
Son only L_1, L_3 and L_5 are applicable.

56. Topic: Operating System

(c) Medium-term scheduling is one in which decision is taken to add the number of processes that are present either partially or fully in the main memory. This is similar to CPU scheduling. In shortterm scheduling the available process is executed by the processor. In long-term scheduling, the process is added to the pool of processes to be executed.

57. Topic: Databases

(a) Time stamp protocol is a deadlock free protocol.

58. Topic: Computer Networks

(b) We have

$$\begin{aligned} \text{Ethernet header} = 18\text{ Bytes } &\big[\text{Dest. Mac }(6) \\ &+\text{Source Mac }(6) \\ &+\text{Length }(2)+\text{CRC }(4)\big] \end{aligned}$$

$$\text{Minimum Data Portion} = 46 \text{ Bytes and Minimum}$$
$$\text{Ethernet Frame Size} = 64 \text{ Bytes}$$

59. Topic: Web Technologies

(d) Digital signature is an asymmetric cryptography technique which uses public key encryption mechanism to provide integerity, authentication and non-repudiation.

60. Topic: Engineering Mathematics

(c) Newton's forward formula is employed to find derivative of a function for a given data near the beginning of the table.

61. Topic: Computer Organization and Architecture

(a) $L2$ level of cache is placed between the $L1$ and RAM. The $L1$ cache is always the fastest.

62. Topic: Programming and Data Structures

(d) A spanning tree is a subgraph of a given graph G, which covers all the vertices of G and should be a tree. Spanning tree could be minimum or maximum.

63. Topic: Algorithms

(b) Factorial function

$$f(n) = n! = n(n-1)\times(n \quad 2)\ldots 1$$
$$n! = n(n-1)(n-2)\ldots 1 \le n \times n \ldots \times n = (n^n)$$

So, Big O Estimation of $f(n!) = O(n^n)$

64. Topic: Compiler Design

(c) Lexical analyzer reads the source program character by character at a time and unites them into a stream of tokens.

65. Topic: Operating System

(b) The working set model is used in memory management to implement the concept of Principle of locality.

66. Topic: Databases

(a) It promotes and enforces some integrity rules for the reducing data redundancy and increasing data consistency.

67. Topic: Computer Networks

(a) Stop-and-wait protocol uses 1 bit sequence.

68. Topic: Web Technologies

(a) <strong> tag is used to define important text by bold the text.
<dark> – there is no tag in html named dark.
<big> tag is used to make text bigger than the normal text.
<highblack> – there is no tag in html named highblack.

69. Topic: Algorithms

(c)

Search Complexity	Average Case	Worst Case
Binary	$O(\log n)$	$O(n)$
Hash table	$O(1)$	$O(n)$
Linear search	$O(n)$	$O(n)$

Thus, hash table takes less search time.

70. Topic: Engineering Mathematics

(d) We have

$$A = \begin{bmatrix} 3 & 0 & 0 \\ 5 & x & 0 \\ 3 & 6 & y \end{bmatrix}$$

Now, eigenvalues of A are 1, 3 and 4.

Sum of eigenvalues = Sum of diagonal

Therefore,

$$3+x+y=1+3+4 \Rightarrow x+y=5$$

Also, product of eigenvalues = $|A|$
Therefore,

$$3(xy-0)-5(0-0)+3(0-0)=1\times 3\times 4$$
$$\Rightarrow xy=4$$

Now,

$$y=5-x \Rightarrow (5-x)x=4$$
$$\Rightarrow 5x-x^2=4 \Rightarrow x^2-5x+4=0$$
$$\Rightarrow (x-4)(x-1)=0 \Rightarrow x=1,4$$
$$\Rightarrow y=4,1$$

Hence, from the given options

$$x=4, y=1$$

71. Topic: Software Engineering

(a) Function Point is given by

FP = UFP × CAF

UFP stands for Unadjusted Function Point, while CAF stands for Complexity Adjustment Factor.

72. Topic: Programming and Data Structures

(d) The syntax for inclusion of header file is #include <stdio.*h*>. There should be space between #include and <stdio.*h*>.

73. Topic: Algorithms

(c) Strassen's surprising algorithm can multiply two n × n matrices in $O(n^{\log 7})$ time, which is equal to $O(n^{2.81})$ time.

74. Topic: Compiler Design

(a) A. Lexical analysis – S. Finite automaton
B. Code optimization – Directed acyclic graphs
C. Code generation – PDA
D. Abelian groups – Syntax trees
Hence, A-S, B-P, C-R, D-Q

75. Topic: Operating System

(b) Shortest job first (SJF) algorithm.

76. Topic: Databases

(a) It provides data storage and responsiveness to queries.

77. Topic: Computer Networks

(d) Vulnerable time for CSMA is the propagation time T_p needed for a signal to propagate from one end of the medium to the other.

78. Topic: Web Technologies

(a) The World Wide Web Consortium (W3C) is the international standard organization for managing and controlling web standard. IEEE and IETE are professional association societies for advancement of technology.

79. Topic: Compiler Design

(c) Begin and Goto cannot be properly grouped as keywords without considering the delimiter. But <> can be considered as Boolean operator, not matter what follows the symbol.

80. Topic: Operating System

(b) WE have

$$3600 \text{ rotation} \rightarrow 60 \text{ s}$$
$$1 \text{ rotation} \rightarrow 60/3600 = 1/60 \text{ s}$$
$$\text{Transfer rate} = (60\times \text{track size}) \text{ per second}$$
$$= 60\times 512 \text{ bytes s}^{-1}$$
$$= 30 \text{ Kbps}$$

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